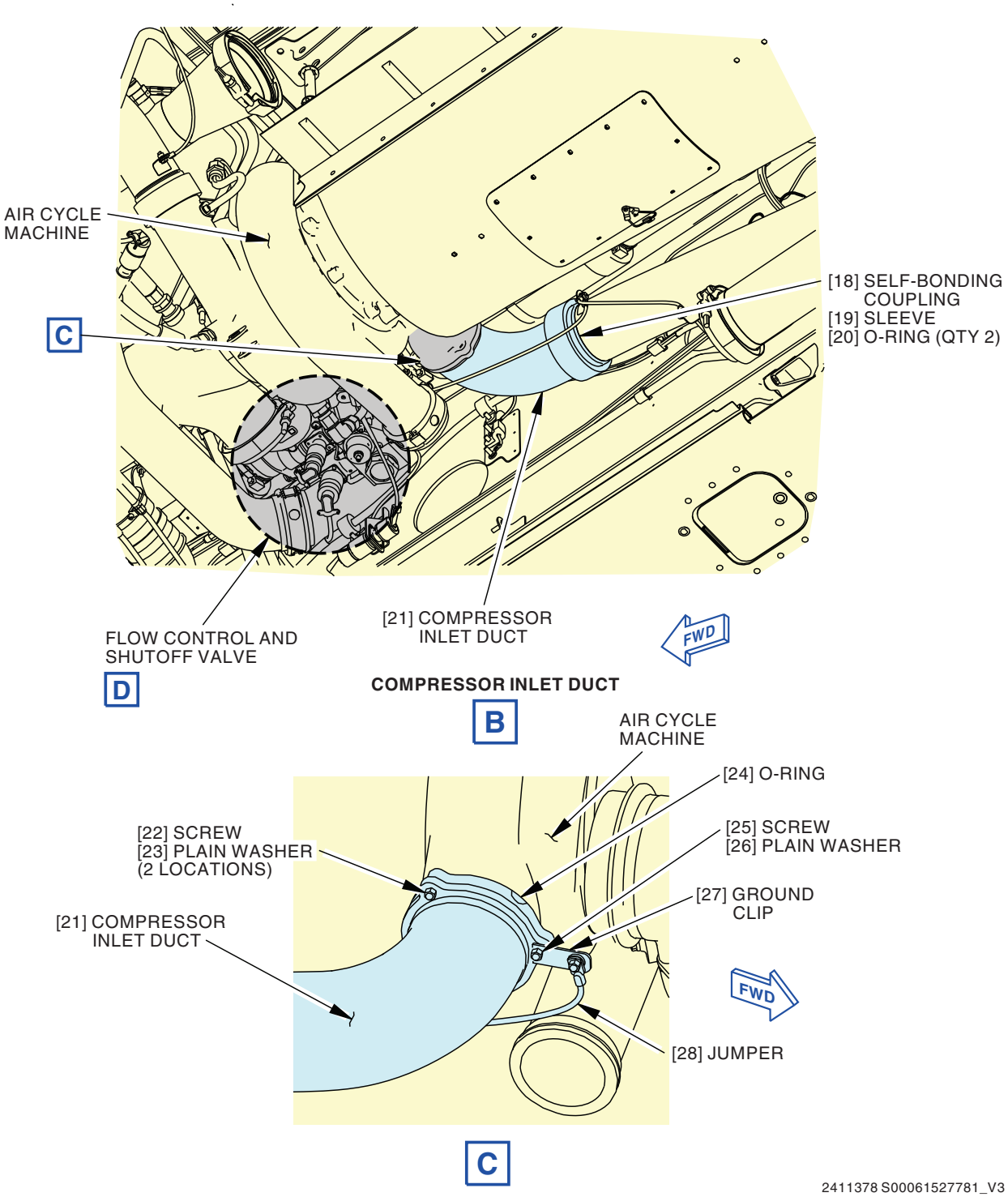
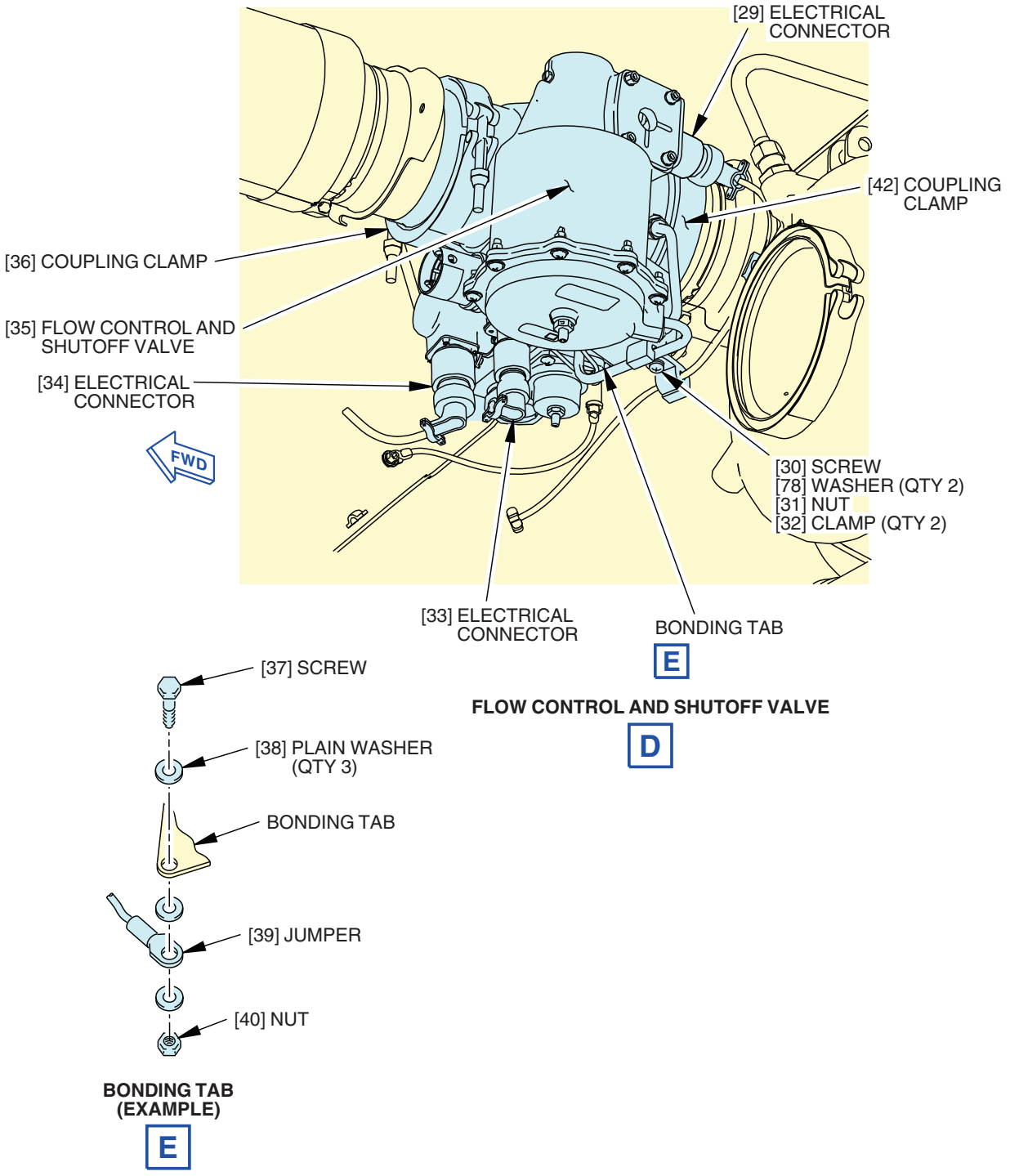


Left Flow Control and Shutoff Valve Installation
Figure 401/21-51-01-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Flow Control and Shutoff Valve Installation
Figure 401/21-51-01-990-801 (Sheet 3 of 3)

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TASK 21-51-01-400-801

3. Left Flow Control and Shutoff Valve Installation

(Figure 401)

A. General

(1) The high pressure water separator mix muff is referred to as HPWS mix muff in this task.

B. References

Reference	Title
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
21-51-00-700-802	Pack Flow Control Valve Operational Test (P/B 501)
21-51-17-400-801	High Pressure Water Separator Mix Muff Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

D. Consumable Materials

Reference	Description	Specification
C00852	Compound - Antiseize, Molybdenum Disulfide-Petrolatum	MIL-PRF-83483
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00602	Lubricant - Fluorinated	BMS3-25
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	
D50294	Compound - Antiseize, Pure Nickel Special - Never Seez	BAC5008
D50320	Lubricant - Moly-50 Anti-Seize	MIL-PRF-83483

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	O-ring	21-51-03-01-100	PTF ALL
14	O-ring	21-51-03-02-055	PTF ALL
17	Packing	21-51-03-02-040	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
20	O-ring	21-51-10-01-068	PTF ALL
24	O-ring	21-51-03-02-025	PTF ALL
35	Flow control and shutoff valve	21-51-01-10-050	PTF ALL

F. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left

G. Access Panels

Number	Name/Location
192CL	ECS Access Door

H. Prepare for the Installation

SUBTASK 21-51-01-010-013

(1) Remove the covers from the duct openings.

SUBTASK 21-51-01-640-001

(2) Apply Never Seez, D50294, to the T-bolt threads on the coupling clamps [36].

I. Left Flow Control and Shutoff Valve Installation

SUBTASK 21-51-01-980-001

(1) Put the flow control and shutoff valve [35] in its position.

SUBTASK 21-51-01-420-014

(2) Loosely install the coupling clamps [36].

SUBTASK 21-51-01-110-001

(3) Clean the fay surfaces of the steel bonding tab on the flow control and shutoff valve [35], the blue terminal on the jumper [39], and the plain washers [38] (SWPM 20-20-00, Paragraph 2).

SUBTASK 21-51-01-420-015

(4) Install the screw [37], plain washers [38], jumper [39], and nut [40] to the bonding tab on the flow control and shutoff valve [35].

SUBTASK 21-51-01-420-021

(5) Tighten the nut [40] to 31.5 ±3.5 in-lb (3.6 ±0.4 N·m).

SUBTASK 21-51-01-760-001

(6) Measure the electrical bonding resistance between the flow control and shutoff valve [35] and structure (SWPM 20-20-00).

(a) Use an non-intrinsically safe bonding meter, COM-614 .

(b) Make sure that the overall bonding resistance is 0.010 ohms (10.0 ohms) or less.

SUBTASK 21-51-01-420-022

(7) Connect the electrical connector [29], electrical connector [33], and electrical connector [34] to the flow control and shutoff valve [35].

SUBTASK 21-51-01-420-023

(8) Find the running torque of the locknut on each clamp trunnion on the coupling clamp [36], as follows:

NOTE: The running torque is the torque necessary to turn the locknut on the trunnion. The running torque must be 6.5 ±3.5 in-lb (0.7 ±0.4 N·m).

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(a) With an applicable torque wrench, measure the torque that is necessary to turn the locknut on the trunnion.

(b) Record the torque value.

SUBTASK 21-51-01-420-024

(9) Tighten the locknuts on the coupling clamp [36] and coupling clamp [42], as follows:

(a) Tighten the locknuts to 5 in-lb (0.6 N·m) above the running torque.

NOTE: If the running torque was measured at 8 in-lb (0.9 N·m), the torque required will be 13 in-lb (1.5 N·m).

(b) Align the valve flanges with the adjacent duct flanges.

(c) Tighten the locknut on the coupling clamp [36] to 60 +5 / -0 in-lb (6.8 +0.6 / -0.0 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.310 in. (7.874 mm).

(d) Tighten the locknut on the coupling clamp [42] to 60 +5 / -0 in-lb (6.8 +0.6 / -0.0 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-51-01-420-037

(10) Install the screw [30], washers [78], nut [31], and clamps [32] to attach the wirings to the flow control and shutoff valve [35].

SUBTASK 21-51-01-410-020

(11) Install the compressor inlet duct [21].

(a) Apply a light coating of Molykote 44 grease, medium, D50104, on a new O-ring [24].

(b) Install the new O-ring [24] on the flange of the compressor inlet duct [21].

(c) Apply compound, C00852, on the threads of the screw [25] and two screws [22].

(d) Align the compressor inlet duct [21] with the air cycle machine.

(e) Install the screw [25], plain washer [26], and ground clip [27] into the compressor inlet duct [21].

(f) Install the screws [22] and plain washers [23] into the compressor inlet duct [21].

(g) Apply a light coating of lubricant, D00602, on new O-rings [20].

(h) Install the new O-rings [20] on the compressor inlet duct [21] and adjacent duct.

(i) Move the sleeve [19] over the duct ends.

(j) Install the self-bonding coupling [18].

SUBTASK 21-51-01-410-012

(12) Install the HPWS mix muff (TASK 21-51-17-400-801).

SUBTASK 21-51-01-410-021

(13) Install the mix muff duct [5].

(a) Apply Never Seez, D50294, to the T-bolt threads on the two coupling clamps [4].

(b) Align the mix muff duct [5] in its position.

(c) Install the coupling clamps [4].

(d) Tighten the nuts on the coupling clamps [4] to 60 +5 / -0 in-lb (6.8 +0.6 / -0.0 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-51-01-110-002

(14) Clean the fay surfaces of the titanium bonding tab on the mix muff duct [5], the blue terminal on the bonding jumper [9] and bonding jumper [10], and the three washers [7] (SWPM 20-20-00, Paragraph 2).

SUBTASK 21-51-01-420-025

(15) Install the screw [6], washers [7], bonding jumper [9], bonding jumper [10], and nut [8] on the bonding tab on the mix muff duct [5].

SUBTASK 21-51-01-420-036

(16) Tighten the nut [8] to 36.5 ±3.5 in-lb (4.1 ±0.4 N·m).

SUBTASK 21-51-01-760-002

(17) Measure the bonding resistance between the bonding tab on the mix muff duct [5] and structure (SWPM 20-20-00).

(a) Use an non-intrinsically safe bonding meter, COM-614 .

(b) Make sure that the overall bonding resistance is 0.0010 ohms (1.0 milliohms) or less.

SUBTASK 21-51-01-410-022

(18) Prepare the compressor outlet duct [11] for installation, as follows:

(a) Apply a light coat of Molykote 44 grease, medium, D50104, to a new packing [17] and two new O-rings [14].

(b) Install the new O-ring [14] on the flange of the compressor outlet duct [11].

(c) Install the new O-ring [14] on the flange of the air cycle machine.

(d) Install the new packing [17] on the flange of the heat exchanger.

(e) Apply compound, D50320, to the T-bolt threads on the clamp coupling [16].

SUBTASK 21-51-01-410-013

(19) Install the compressor outlet duct [11].

(a) Align the compressor outlet duct [11] into its position.

(b) Loosely install the clamp coupling [16] between the heat exchanger and compressor outlet duct [11].

(c) Slide the sleeve [13] between the air cycle machine and compressor outlet duct [11].

(d) Install the coupling [12] over the outlet duct connection.

(e) Tighten the nut on the clamp coupling [16] to 47.5 ±2.5 in-lb (5.4 ±0.3 N·m).

(f) With hand force, gently push the compressor outlet duct in the direction of the adjacent mix muff duct until the compressor outlet duct reaches the maximum flexed position.

(g) Make sure that there is a minimum clearance of 0.20 in. (5.1 mm) to the adjacent mix muff duct when the compressor outlet duct is in the flexed position.

(h) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the flexhose [2].

(i) Apply a light coating of lubricant, D00602, on new O-ring [3].

(j) Install the new O-ring [3] on the compressor outlet duct [11].

(k) Connect the flexhose [2] to the compressor outlet duct [11] (TASK 20-10-51-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-01-410-023

(20) Install the electrical connector [1] to the ram air control sensor.

SUBTASK 21-51-01-420-026

(21) Install the electrical connector [15] to the compressor discharge overheat switch.

J. Left Flow Control and Shutoff Valve Installation Test

SUBTASK 21-51-01-860-030

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-51-01-860-012

(2) On the P5-10 Bleed Air Control Module, remove the DO-NOT-OPERATE tags from these switches:

(a) L PACK and R PACK switches

(b) BLEED 1 and 2 switches

(c) BLEED APU switch.

SUBTASK 21-51-01-710-003

(3) Do an operational check of the flow control and shutoff valve [35] (TASK 21-51-00-700-802).

K. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-01-410-014

(1) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

——— END OF TASK ———

TASK 21-51-01-000-802

4. Right Flow Control and Shutoff Valve Removal

(Figure 402)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-51-000-801	Flareless Tubing Assembly Removal (P/B 401)
21-51-04-000-801	Air Cycle Machine Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-01-860-010

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove pressure from the pneumatic system (TASK 36-00-00-860-806).

SUBTASK 21-51-01-860-011

(2) On the P5-10 Bleed Air Control Module, set the switches and attach DO NOT OPERATE tags, STD-858, as follows:

(a) Set the L PACK and R PACK switches to the OFF position.

(b) Set the BLEED 1 and 2 switches to the OFF position.

(c) Set the BLEED APU switch to the OFF position.

SUBTASK 21-51-01-010-008

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-51-01-010-009

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To open the right Environmental Control System (ECS) access doors do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-01-010-010

(5) It is necessary, remove the air cycle machine (TASK 21-51-04-000-801).

SUBTASK 21-51-01-020-021

(6) Remove the mix muff duct [51].

(a) Remove the cap screw [74], three plain washers [75], jumper [76], and nut [77] from the bonding tab.

(b) Hold the mix muff duct [51] while you remove the coupling clamps [53].

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(c) Remove the mix muff duct [51].

SUBTASK 21-51-01-010-011

(7) Remove the compressor inlet duct [52].

(a) Remove the self-bonding coupling [56].

NOTE: Lift the three latch pawls at the same time to release the clamp.

(b) Move the sleeve [55] to the compressor inlet duct [52].

(c) Remove the compressor inlet duct [52].

(d) Remove and discard the packings [54] or seals [41].

SUBTASK 21-51-01-010-012

(8) Disconnect the sense line [64], as follows:

(a) Remove the nut [60], washer [61], and loop clamp [59] from the clip [63].

(b) Disconnect the B-nut [57] and B-nut [58] (TASK 20-10-51-000-801).

(c) Remove the sense line [64].

F. Right Flow Control and Shutoff Valve Removal

SUBTASK 21-51-01-020-022

(1) Remove the electrical connector [66], electrical connector [67], and electrical connector [69] from the flow control and shutoff valve [68].

SUBTASK 21-51-01-020-025

(2) Remove the cap screw [70], three washers [71], jumper [72], and nut [73] from the bonding tab on the flow control and shutoff valve [68].

SUBTASK 21-51-01-020-027

(3) Remove the flow control and shutoff valve [68] from the ducts, as follows:

(a) Support the flow control and shutoff valve [68].

(b) Loosen the T-bolts on the coupling clamp [65] and coupling clamp [79].

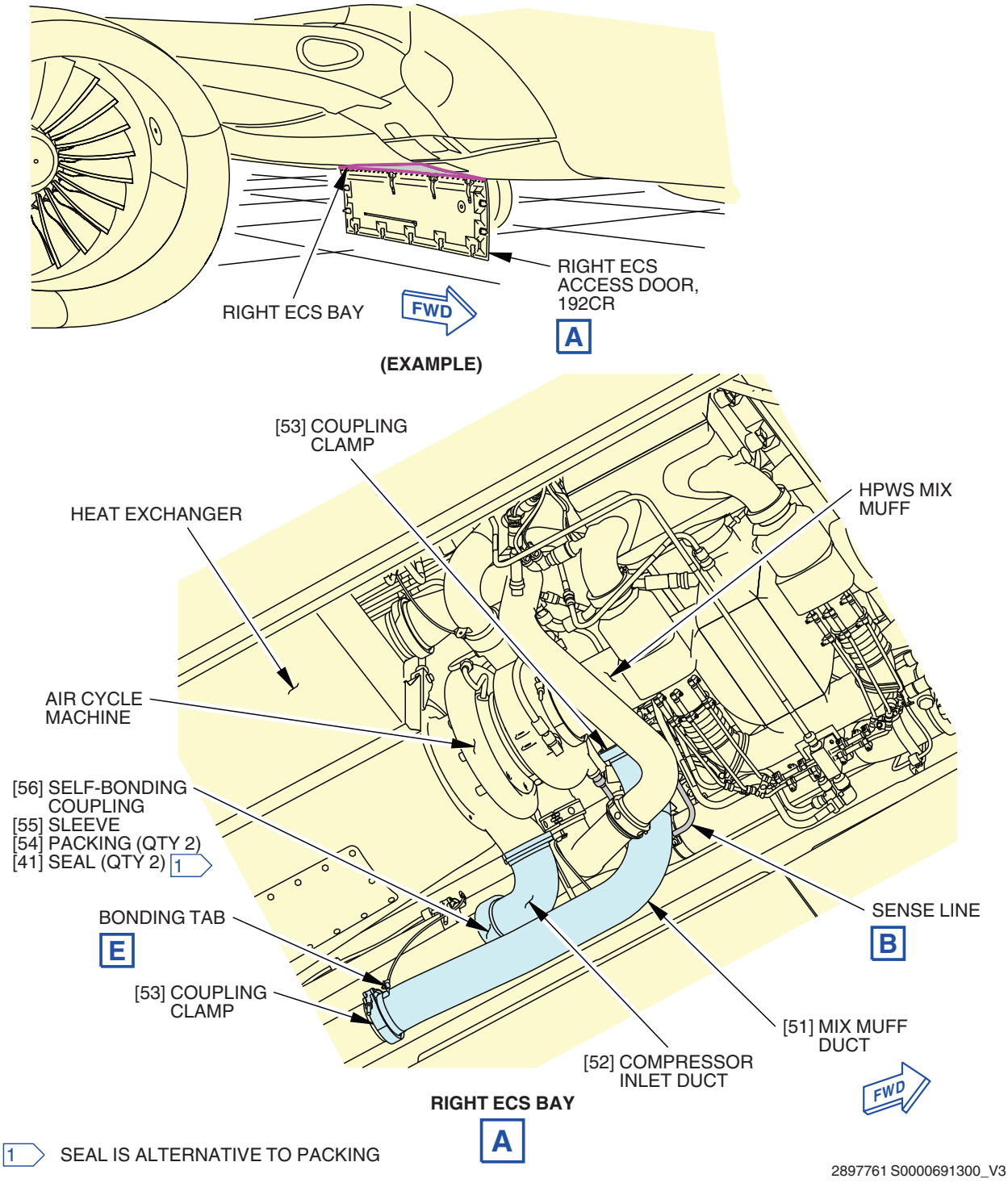
(c) Remove the flow control and shutoff valve [68].

SUBTASK 21-51-01-410-011

(4) Put covers on the duct openings to keep out unwanted materials.

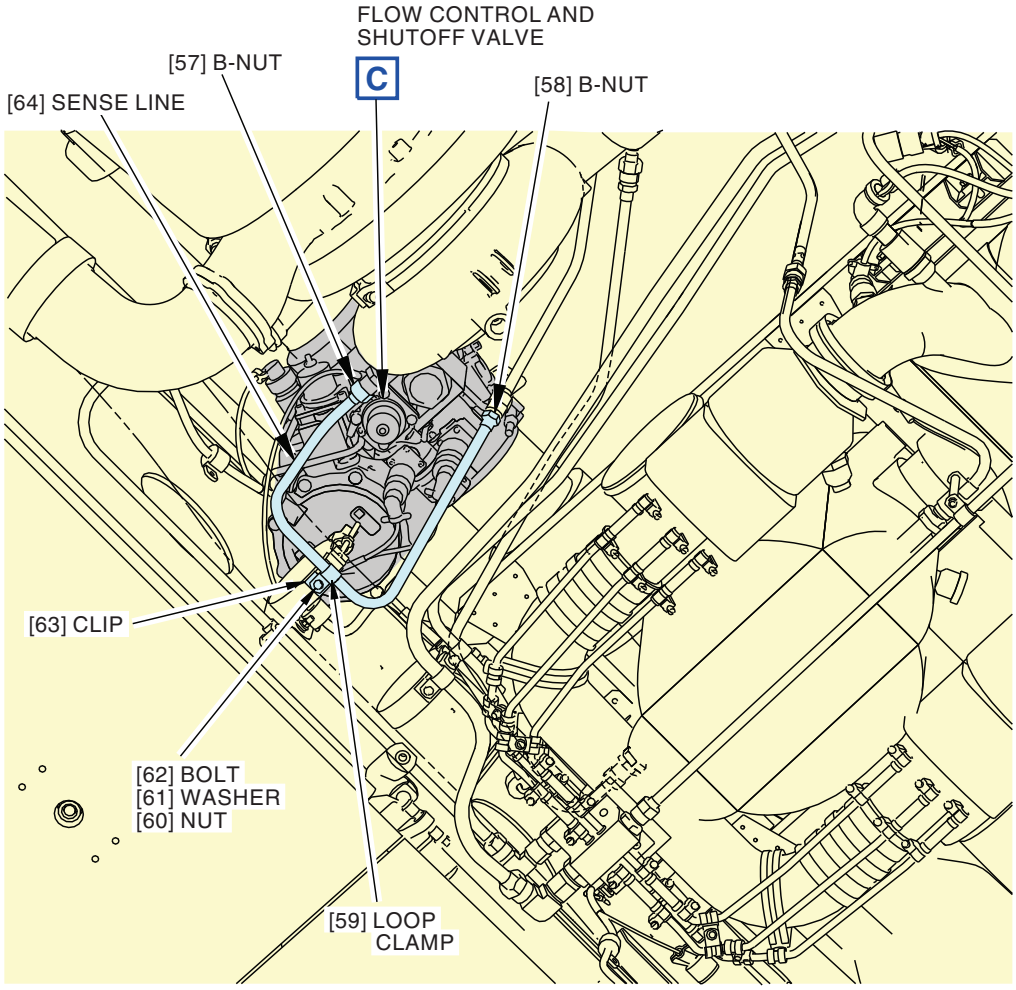
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



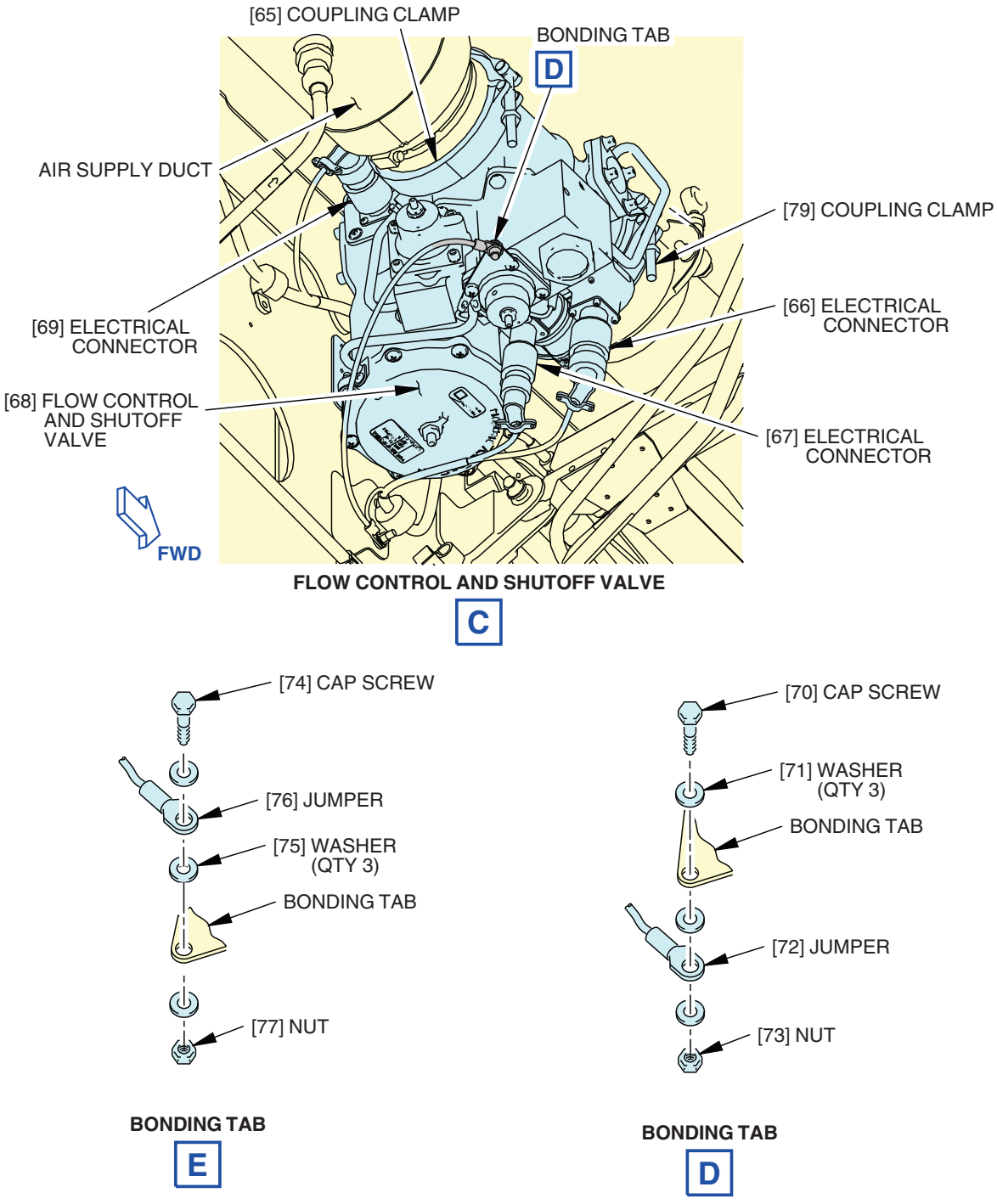
Right Flow Control and Shutoff Valve Installation
Figure 402/21-51-01-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right Flow Control and Shutoff Valve Installation
Figure 402/21-51-01-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Right Flow Control and Shutoff Valve Installation
Figure 402/21-51-01-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-01-400-802

5. Right Flow Control and Shutoff Valve Installation

(Figure 402)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
21-51-00-700-802	Pack Flow Control Valve Operational Test (P/B 501)
21-51-04-400-801	Air Cycle Machine Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Consumable Materials

Reference	Description	Specification
D00602	Lubricant - Fluorinated	BMS3-25
D50294	Compound - Antiseize, Pure Nickel Special - Never Seez	BAC5008

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
41	Seal	21-51-10-02-072	PTF ALL
54	Packing	21-51-10-02-068	PTF ALL
68	Flow control and shutoff valve	21-51-01-10-050	PTF ALL

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Installation

SUBTASK 21-51-01-010-016

(1) Remove the covers from the duct openings.

SUBTASK 21-51-01-640-002

(2) Apply Never Seez, D50294, to the T-bolt threads on the coupling clamps [65].

SUBTASK 21-51-01-980-003

(3) Put the flow control and shutoff valve [68] in its position.

H. Right Flow Control and Shutoff Valve Installation

SUBTASK 21-51-01-420-017

(1) Loosely install the coupling clamps [65] on the ends of the flow control and shutoff valve [68].

SUBTASK 21-51-01-110-003

(2) Clean the fay surfaces of the steel bonding tab on the flow control and shutoff valve [68], blue terminal on the jumper [72], and washer [71] (SWPM 20-20-00, Paragraph 2).

SUBTASK 21-51-01-420-027

(3) Install the cap screw [70], three washers [71], jumper [72], and nut [73] to the bonding tab on the flow control and shutoff valve [68].

SUBTASK 21-51-01-420-028

(4) Tighten the nut [73] to 31.5 ±3.5 in-lb (3.6 ±0.4 N·m).

SUBTASK 21-51-01-760-003

(5) Measure the bonding resistance between the flow control and shutoff valve [68] and structure with an non-intrinsically safe bonding meter, COM-614 (SWPM 20-20-00, Paragraph 21).

(a) Make sure that the overall bonding resistance is 0.010 ohms (10.0 milliohms) or less.

SUBTASK 21-51-01-420-029

(6) Connect the electrical connector [66], electrical connector [67], and electrical connector [69] to the flow control and shutoff valve [68].

SUBTASK 21-51-01-210-001

(7) Make sure that there is a minimum of 0.10 in. (2.5 mm) clearance between the flow control and shutoff valve [68] and the keel beam.

(a) If it is necessary, turn the flow control and shutoff valve [68] clockwise to get clearance.

SUBTASK 21-51-01-420-030

(8) Find the running torque of the locknut on each trunnion on the coupling clamp [65], as follows:

NOTE: The running torque is the torque necessary to turn the locknut on the trunnion. The running torque must be 6.5 ±3.5 in-lb (0.7 ±0.4 N·m).

(a) With an applicable torque wrench, measure the torque that is necessary to turn the locknut on the trunnion.

(b) Record the torque value.

SUBTASK 21-51-01-420-031

(9) Tighten the locknuts on the coupling clamp [65] and coupling clamp [79], as follows:

(a) Tighten the locknuts to 5 in-lb (1 N·m) above the running torque.

NOTE: If the running torque was measured at 8 in-lb (1 N·m), the torque required will be 13 in-lb (1 N·m).

(b) Align the valve flanges with the adjacent duct flanges.

(c) Tighten the locknut on the coupling clamp [65] to 60 +5 / -0 in-lb (6.8 +0.6 / -0.0 N·m).

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1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

(d) Tighten the locknut on the coupling clamp [79] to 60 +5 / -0 in-lb (6.8 +0.6 / -0.0 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.310 in. (7.874 mm).

SUBTASK 21-51-01-420-032

(10) Connect the sense line [64], as follows:

(a) Put the sense lines into position.

(b) Connect the B-nut [57] and B-nut [58] (TASK 20-10-51-400-801).

(c) Put the loop clamp [59] on the sense line [64].

(d) Align the loop clamp [59] with the clip [63].

(e) Install the washer [61] and nut [60] on the bolt [62] that attaches the loop clamp [59] to the clip [63].

SUBTASK 21-51-01-420-033

(11) Install the compressor inlet duct [52].

(a) Apply a light coating of lubricant, D00602, on the new packings [54] or new seals [41].

NOTE: It is acceptable to use seals as alternative to packings.

(b) Install the packings [54] or seals [41] on the flanges of the compressor inlet duct [52] and adjacent duct.

(c) Put the compressor inlet duct [52] in its position.

(d) Move the sleeve [55] over the duct ends.

(e) Install the self-bonding coupling [56].

SUBTASK 21-51-01-410-016

(12) If removed, install the air cycle machine (TASK 21-51-04-400-801).

SUBTASK 21-51-01-020-037

(13) Install the mix muff duct [51].

(a) Apply Never Seez, D50294, to the T-bolt threads on the two coupling clamps [53].

(b) Align the mix muff duct [51] in its position.

(c) Install the coupling clamps [53].

(d) Tighten the nuts on the coupling clamps [53] to 60 +5 / -0 in-lb (6.8 +0.6 / -0.0 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

(e) With hand force, gently push the compressor outlet duct in the direction of the adjacent mix muff duct until the compressor outlet duct reaches the maximum flexed position.

(f) Make sure that there is a minimum clearance of 0.20 in. (5.1 mm) to the adjacent mix muff duct when the compressor outlet duct is in the flexed position.

SUBTASK 21-51-01-110-004

(14) Clean the fay surfaces of the titanium bonding tab on the mix muff duct [51], blue terminal on the jumper [76], and plain washers [75] (SWPM 20-20-00, Paragraph 2).

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SUBTASK 21-51-01-420-034

(15) Install the cap screw [74], three plain washers [75], jumper [76], and nut [77] on the bonding tab on the mix muff duct [51].

SUBTASK 21-51-01-420-035

(16) Tighten the nut [77] to 36.5 ±3.5 in-lb (4.1 ±0.4 N·m).

SUBTASK 21-51-01-760-004

(17) Measure the bonding resistance between the bonding tab on the mix muff duct [51] and structure with an non-intrinsically safe bonding meter, COM-614 (SWPM 20-20-00, Paragraph 21).

(a) Make sure that the overall bonding resistance is 0.0010 ohms (1.0 milliohms) or less.

I. Flow Control and Shutoff Valve Installation Test

SUBTASK 21-51-01-010-019

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-51-01-860-021

(2) On the P5-10 Bleed Air Control Module, remove the DO-NOT-OPERATE tags from these switches:

(a) L PACK and R PACK switches

(b) BLEED 1 and 2 switches

(c) BLEED APU switch.

SUBTASK 21-51-01-710-004

(3) Do an operational test of the flow control and shutoff valve [68] (TASK 21-51-00-700-802).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-01-410-017

(1) To close the right Environmental Control System (ECS) access doors do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CONDITIONING ACCESSORY UNIT (ACAU) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the air conditioning accessory unit (ACAU).

(2) An installation of the air conditioning accessory unit (ACAU).

(3) There are two air conditioning accessory units (ACAU) installed in the electronic equipment center on the E4-1 shelf.

TASK 21-51-02-000-801

2. Air Conditioning Accessory Unit (ACAU) Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
24-22-00-860-802	Remove Electrical Power (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

SUBTASK 21-51-02-860-001

(1) Remove electrical power. To remove electrical power, do this task: Remove Electrical Power, TASK 24-22-00-860-802

SUBTASK 21-51-02-010-001

(2) Open this access panel:

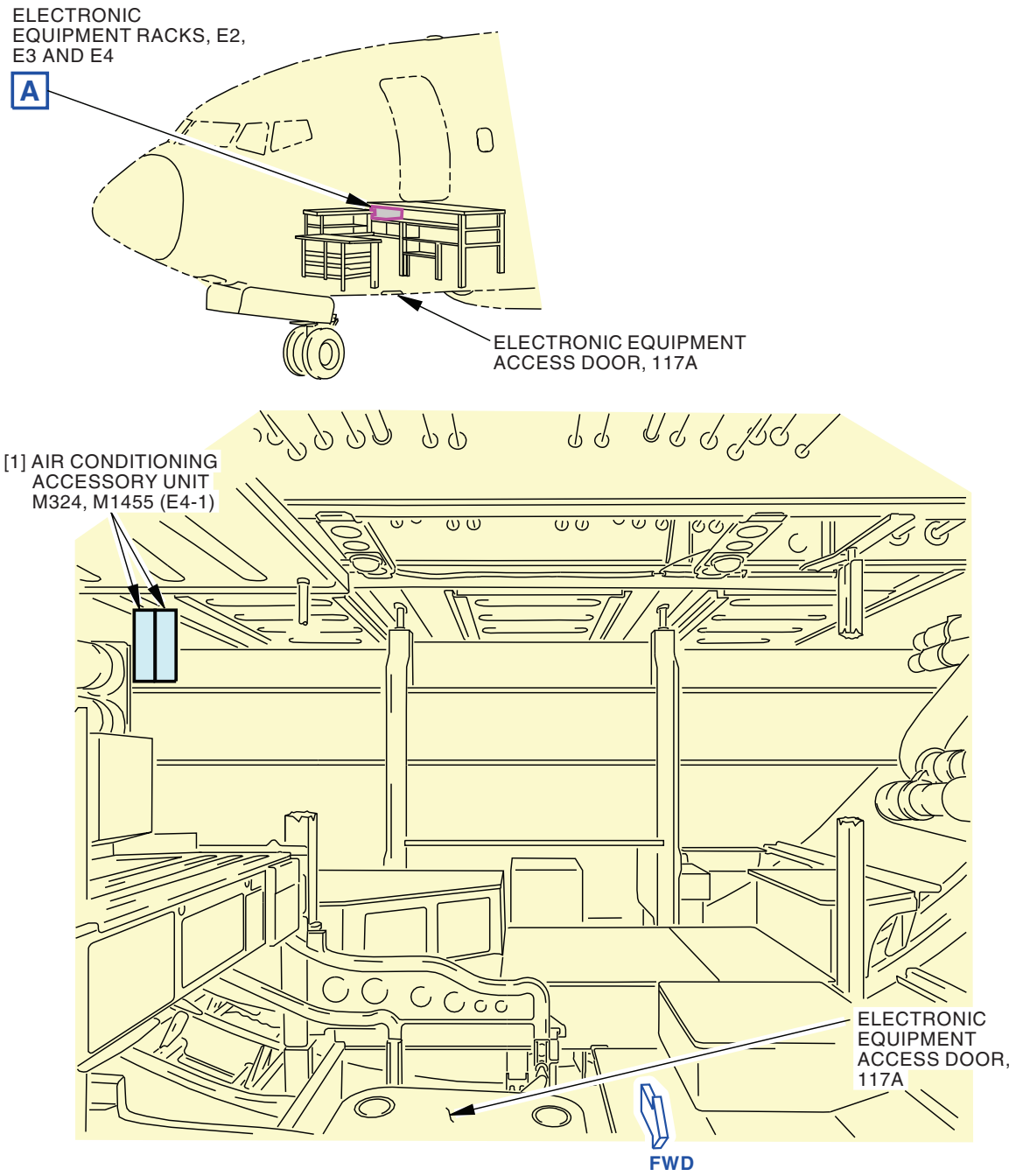
Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-51-02-020-001

(3) Remove the ACAU [1]. To remove the ACAU [1], do this task: E/E Box Removal, TASK 20-10-07-000-801.

——— END OF TASK ———

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ELECTRONIC EQUIPMENT RACKS, E2, E3 AND E4

A

2411383 S00061527792_V1

Air Conditioning Accessory Unit (ACAU) Installation
Figure 401/21-51-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-02-400-801

3. Air Conditioning Accessory Unit (ACAU) Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-07-400-801	E/E Box Installation (P/B 401)
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
192	Lower Wing-To-Body Fairing - Under Wing Box

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

D. Air Conditioning Accessory Unit (ACAU) Installation

SUBTASK 21-51-02-420-001

(1) Install the ACAU [1]. To install the ACAU [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

E. Air Conditioning Accessory Unit (ACAU) Installation Test

SUBTASK 21-51-02-710-001

(1) Do this test of the ACAU [1]:

NOTE: If you replaced the Air Conditioning Accessory Unit (ACAU M324 or M1455) because it had a defect, it is recommended that the complete adjustment/test procedure for the affected system be performed. For example, if an ACAU was replaced to correct a defect in the ram air system, then it is recommended that a test of the ram air system be performed. If the ACAU is being replaced for any other reason you may do just the test that follows:

(a) If you installed the M324 ACAU [1], do this test of the left pack overheat relay:

1) Make sure that the L and R PACK switches on the P5-10 Air Conditioning Panel are set to OFF.

2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

3) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Disconnect electrical connector D532 from the left compressor discharge overheat switch, S2.

5) Install a jumper wire between pins 1 and 2 of connector D532 on the ships' wiring.

6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

7) Make sure that the left PACK light on the P5-10 Air Conditioning Panel comes on.

8) Make sure that the AIR COND and MASTER CAUTION lights on the P7 glareshield panel come on.

9) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

10) Remove the jumper wire that was installed between pins 1 and 2 of connector D532 on the ships' wiring.

11) Re-connect electrical connector D532 to the left compressor discharge overheat switch, S2.

12) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

13) Depress the TRIP RESET switch on the P5-10 Air Conditioning Panel.

14) Make sure the left PACK light on the P5-10 Air Conditioning Panel goes off.

15) If the MASTER CAUTION or AIR COND annunciator lights are on, push and release one of the two MASTER CAUTION lights.

16) Make sure the AIR COND and MASTER CAUTION lights on the P7 glareshield go off.

17) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) If you installed the M1455 ACAU [1], do this test of the right pack overheat relay:

1) Make sure that the L and R PACK switches on the P5-10 Air Conditioning Panel are set to OFF.

2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

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CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

3) To open the right ECS access door do these steps:

(TASK 06-41-00-800-801)

a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

4) Disconnect electrical connector D538 from the right compressor discharge overheat switch, S5.

5) Install a jumper wire between pins 1 and 2 of connector D538 on the ships' wiring.

6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

7) Make sure the right PACK light on the P5-10 Air Conditioning Panel comes on.

8) Make sure that the AIR COND and MASTER CAUTION lights on the P7 glareshield panel come on.

9) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

10) Remove the jumper wire that was installed between pins 1 and 2 of connector D538 on the ships' wiring.

11) Reconnect electrical connector D538 to the right compressor discharge overheat switch, S5.

12) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

13) Depress the TRIP RESET switch on the P5-10 Air Conditioning Panel.

14) Make sure the right PACK light on the P5-10 Air Conditioning Panel goes off.

15) If the MASTER CAUTION or AIR COND annunciator lights are on, push and release one of the two MASTER CAUTION lights.

16) Make sure the AIR COND and MASTER CAUTION lights on the P7 glareshield go off.

17) To close the right ECS access door, do these steps:

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(TASK 06-41-00-800-801)

a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-02-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-51-02-860-016

(2) Supply the electrical power, do this task: Supply Electrical Power, TASK 24-22-00-860-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CONDITIONING ACCESSORY UNIT - ADJUSTMENT/TEST

1. General

A. This procedure has one task. The task is an operational test of the Air Conditioning Accessory Unit (M324 or M1455 relays module) after installation.

B. There are two air conditioning accessory units (ACAU) installed in the electronic equipment center on the E4-1 shelf.

TASK 21-51-02-710-801

2. Air Conditioning Accessory Unit Operational Test

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-25-00-710-801	Recirculation System - Operational Test (P/B 501)
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

D. Prepare for the Test

SUBTASK 21-51-02-410-002

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Air Conditioning Accessory Unit (ACAU) Relay Module Test

SUBTASK 21-51-02-710-002

(1) Do these checks for the post-installation test of the M324 ACAU No. 1:

(a) Do a test of the left pack overheat relay:

1) Set the L PACK and R PACK switches, on the P5-10 Air Conditioning Panel, to the OFF position.

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2) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

3) Open this access door:

Number	Name/Location
192CL	ECS Access Door

4) Disconnect electrical connector, D532, from the left compressor discharge overheat switch, S2.

5) Install a jumper wire between pins 1 and 2 of electrical connector, D532.

6) Make sure that the left PACK light, on the P5-10 Air Conditioning Panel, comes on.

7) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, come on.

8) Remove the jumper wire that was installed between pins 1 and 2 of electrical connector, D532.

9) Reconnect electrical connector, D532, to the left compressor discharge overheat switch, S2.

10) Push the TRIP RESET switch, on the P5-10 Air Conditioning Panel.

11) Make sure that the left PACK light, on the P5-10 Air Conditioning Panel, goes off.

12) If the MASTER CAUTION or AIR COND annunciator lights are on, push and release one of the two MASTER CAUTION lights.

13) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, go off.

14) Close this access door:

Number	Name/Location
192CL	ECS Access Door

(b) Do a test of the left ram air system relays:

1) Make sure that the deflector door for the left ram air inlet is extended.

2) Make sure that the airplane is in the ground mode (TASK 32-09-00-860-802).

3) Open this access door:

Number	Name/Location
192CL	ECS Access Door

4) Disconnect electrical connector, D40844, near the left pack valve.

NOTE: A disconnected electrical connector, D40844, simulates that the left pack valve is open.

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

5) Extend the trailing edge flaps to Flaps Position 1 (TASK 27-51-00-860-803).

6) Make sure that the deflector door for the left ram air system is fully extended.

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7) Put the airplane in the air mode (TASK 32-09-00-860-801).

8) Make sure that these results occur:

a) Make sure that the left ram air inlet deflector door retracts fully.

b) Make sure that the left RAM DOOR FULL OPEN light remains on.

WARNING

MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

9) Retract the trailing edge flaps to the fully up position (TASK 27-51-00-860-804).

10) Make sure that these results occur:

a) Make sure that the left ram air inlet modulation panels stay near the closed position.

b) Make sure that the left RAM DOOR FULL OPEN light goes off.

11) Return the airplane to the ground mode (TASK 32-09-00-860-802).

12) Make sure that these results occur:

a) Make sure that the left ram air inlet modulation panels move to the fully open position.

b) Make sure that the left ram air inlet deflector door moves to the extended position.

13) Reconnect the electrical connector, D40844, near the left pack valve.

14) Close this access door:

Number	Name/Location
192CL	ECS Access Door

(c) Do a test of the cabin air recirculation fan relays:

1) Do an operational test of the recirculation system (TASK 21-25-00-710-801).

(d) Do a test of the flight deck zone overheat relay:

NOTE: The flight deck zone overheat relay is in the Air Conditioning Accessory Unit, M324.

1) Make sure that the L and R PACK switches, on the P5-10 Air Conditioning Panel, are set to the OFF position.

2) Get access to the duct overheat switch for the control cabin as follows:

a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

b) Find the duct overheat switch, S931, to the left of the E2 rack.

c) Disconnect electrical connector, D3892, from the control cabin duct overheat switch, S931.

d) Install a jumper wire between pins 1 and 2 of electrical connector, D3892.

e) Make sure that the CONT CAB ZONE TEMP light, on the P5-17 Temperature Control Panel, comes on.

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f) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, come on.

g) Remove the jumper wire that was installed between pins 1 and 2 of electrical connector, D3892.

h) Reconnect electrical connector, D3892, to the control cabin duct overheat switch, S931.

i) Depress the TRIP RESET switch, on the P5-10 Air Conditioning Panel.

j) Make sure that the CONT CAB ZONE TEMP light, on the P5-17 Temperature Control Panel, goes off.

k) Push and release one of the two MASTER CAUTION lights, on the P7 glareshield panel.

l) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, go off.

m) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

(e) Do a test of the aft passenger cabin zone overheat relay:

1) Set the L PACK and R PACK switches, on the P5-10 Air Conditioning Panel, to the OFF position.

2) Get access to the duct overheat switch, S1077, for the aft passenger cabin duct as follows:

a) Remove the applicable ceiling panel that is forward of the overwing escape hatches (TASK 25-21-45-000-801).

3) Disconnect electrical connector, D11836, from the aft passenger cabin duct overheat switch, S1077.

4) Install a jumper wire between pins 1 and 2 of electrical connector, D11836.

5) Make sure that the AFT CAB ZONE TEMP light, on the P5-17 Temperature Control Panel, comes on.

6) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, come on.

7) Remove the jumper wire that was installed between pins 1 and 2 of electrical connector, D11836.

8) Reconnect electrical connector, D11836, to the aft passenger cabin duct overheat switch, S1077.

9) Depress the TRIP RESET switch, on the P5-10 Air Conditioning Panel.

10) Make sure that the AFT CAB ZONE TEMP light, on the P5-17 Temperature Control Panel, goes off.

11) Push and release one of the two MASTER CAUTION lights, on the P7 glareshield panel.

12) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, go off.

13) Install the applicable ceiling panel that was removed to gain access (TASK 25-21-45-400-801).

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SUBTASK 21-51-02-730-001

(2) Do these checks for the post-installation test of the M1455 ACAU No. 2:

(a) Do a test of the right pack overheat relay:

1) Set the L PACK and R PACK switches, on the P5-10 Air Conditioning Panel, to the OFF position.

2) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

3) To open the right ECS access door do these steps:

(TASK 06-41-00-800-801)

a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

4) Disconnect electrical connector, D538, from the right compressor discharge overheat switch, S5.

5) Install a jumper wire between pins 1 and 2 of electrical connector, D538.

6) Make sure that the right PACK light, on the P5-10 Air Conditioning Panel, comes on.

7) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, come on.

8) Remove the jumper wire that was installed between pins 1 and 2 of electrical connector, D538.

9) Reconnect electrical connector, D538, to the right compressor discharge overheat switch, S5.

10) Push the TRIP RESET switch, on the P5-10 Air Conditioning Panel.

11) Make sure that the right PACK light, on the P5-10 Air Conditioning Panel, goes off.

12) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, go off.

13) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

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b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(b) Do a test of the right ram air system relays:

1) Make sure that the deflector door for the right ram air inlet is extended.

2) Make sure that the airplane is in the ground mode (TASK 32-09-00-860-802).

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

3) To open the right ECS access door do these steps:

(TASK 06-41-00-800-801)

a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

4) Disconnect electrical connector, D40846, near the right pack valve.

NOTE: A disconnected electrical connector, D40846, simulates that the right pack valve is open.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

5) Extend the trailing edge flaps to Flaps Position 1 (TASK 27-51-00-860-803).

6) Make sure that the deflector door for the right ram air system is fully extended.

7) Put the airplane in the air mode (TASK 32-09-00-860-801).

8) Make sure that these results occur:

a) Make sure that the right ram air inlet deflector door retracts fully.

b) Make sure that the right ram air inlet modulation panels move toward the closed position.

WARNING MAKE SURE PERSONS AND EQUIPMENT ARE CLEAR OF THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES. THE LEADING EDGE AND TRAILING EDGE CONTROL SURFACES CAN EXTEND AND RETRACT QUICKLY. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

9) Retract the trailing edge flaps to the fully up position (TASK 27-51-00-860-804).

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10) Make sure that these results occur:
    a) Make sure that the right ram air inlet modulation panels stay near the closed position.
11) Return the airplane to the ground mode (TASK 32-09-00-860-802).
12) Make sure that these results occur:
    a) Make sure that the right ram air inlet modulation panels move to the fully open position.
    b) Make sure that the right ram air inlet deflector door moves to the extended position.
13) Reconnect electrical connector D40846 near the right pack valve.
14) To close the right ECS access door, do these steps:

    (TASK 06-41-00-800-801)

    a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

    b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

    NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Do a test of the cabin air recirculation fan relays:
1) Do an operational test of the recirculation system (TASK 21-25-00-710-801).

(d) Do a test of the passenger zone overheat relay:
1) Set the L PACK and R PACK switches, on the P5-10 Air Conditioning Panel, re set to the OFF position.
2) Get access to the forward cabin duct overheat switch:
    a) Remove the applicable ceiling panel that is forward of the overwing escape hatches (TASK 25-21-45-000-801).
3) Disconnect electrical connector, D11834, from the forward cabin duct overheat switch, S1076.
4) Install a jumper cable between pins 1 and 2 of electrical connector, D11834.
5) Make sure that the FWD CAB ZONE TEMP light, on the P5-17 Temperature Control Panel, comes on.
6) Make sure the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, comes on.
7) Remove the jumper cable that was installed between pins 1 and 2 of electrical connector, D11834.
8) Reconnect electrical connector, D11834, to the forward cabin duct overheat switch, S1076.
9) Depress the TRIP RESET switch, on the P5-10 Air Conditioning Panel.
10) Make sure that the FWD CAB ZONE TEMP light, on the P5-17 Temperature Control Panel, goes off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

11) Push and release one of the two MASTER CAUTION lights, on the P7 glareshield panel.

12) Make sure the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, go off.

13) Install the ceiling panel that was removed for access to the forward cabin duct overheat switch, S1076 (TASK 25-21-45-400-801).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-02-410-003

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 21-51-02-710-802

3. Pack Overheat Shutoff Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

D. Prepare for the Test

SUBTASK 21-51-02-860-009

(1) Make sure that the switches, on the P5-10 air conditioning module, set to these positions:

(a) L PACK to OFF

(b) R PACK to OFF

(c) BLEED 1 to OFF

(d) BLEED 2 to OFF

(e) ISOLATION VALVE to OPEN

(f) L RECIRC FAN to AUTO

(g) R RECIRC FAN to AUTO

SUBTASK 21-51-02-710-003

(2) Set the CONT CAB, FWD CAB, and AFT CAB temperature selector switches, on the P5-17 temperature control module, to the OFF position.

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SUBTASK 21-51-02-010-002

(3) Make sure that this access panel is open:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-02-010-003

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To make sure that the right ECS access door is open do these steps:

(TASK 06-41-00-800-801)

(a) Make sure that this access panel is open:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Make sure that this access panel is open:

Number	Name/Location
192CR	ECS Access Door

E. Pack Overheat Shutoff Operational Test

SUBTASK 21-51-02-710-005

(1) Do a test of the pack overheat relays:

(a) Disconnect these electrical connectors from the pack overheat switches:

1) Electrical connector, D532, from the left pack compressor discharge overheat switch, S2.
2) Electrical connector, D10056, from the left pack air cycle machine overheat switch, S938.
3) Electrical connector, D538, from the right pack compressor discharge overheat switch, S5.
4) Electrical connector, D10058, from the right pack air cycle machine overheat switch, S939.

(b) Install a jumper wire between pins 1 and 2 of electrical connector, D532, for the left compressor discharge overheat switch, S2.

(c) Make sure that the AIR COND and MASTER CAUTION lights, on the P7 glareshield panel, and the left PACK light, on the P5-10 air conditioning panel, come on.

(d) Push either MASTER CAUTION light to reset the light.

(e) Make sure that the left PACK lights stays on and the AIR COND annunciator goes off.

(f) Push either of the MASTER CAUTION annunciator lights.

(g) Make sure that the MASTER CAUTION and the AIR COND annunciator lights come on and the left PACK light stays on.

(h) Remove the jumper wire from electrical connector, D532.

(i) Push the TRIP RESET switch, on the P5-10 air conditioning panel.

(j) Make sure that the left PACK light goes off.

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(k) Install a jumper wire between pins 1 and 2 of electrical connector, D10056, for the left air cycle machine overheat switch, S938.

(l) Make sure that the left PACK light comes on.

(m) Remove the jumper wire between pins 1 and 2 of electrical connector, D10056.

(n) Push the TRIP RESET switch, on the P5-10 air conditioning panel.

(o) Make sure that the left PACK light goes off.

(p) Install a jumper wire between pins 1 and 2 of electrical connector, D538, for the right compressor discharge overheat switch, S5.

(q) Make sure that the right PACK light come on.

(r) Push either MASTER CAUTION light to cancel the Master Caution.

(s) Make sure that the right PACK light stays on.

(t) Push either MASTER CAUTION annunciator light.

(u) Make sure that the MASTER CAUTION and AIR COND annunciator lights come on and the R PACK light stays on.

(v) Remove the jumper wire from electrical connector, D538.

(w) Push the TRIP RESET switch, on the P5-10 air conditioning panel.

(x) Make sure that the right PACK light goes off.

(y) Install a jumper wire between pins 1 and 2 of electrical connector, D10058, for the right air cycle machine overheat switch, S939.

(z) Make sure that the right PACK light comes on.

(aa) Remove the jumper wire from electrical connector, D10058.

(ab) Push the TRIP RESET switch, on the P5-10 air conditioning panel.

(ac) Make sure that the right PACK light goes off.

SUBTASK 21-51-02-710-004

(2) Connect these electrical connectors to the pack overheat switches:

(a) Electrical connector, D532, to the left pack compressor discharge overheat switch, S2.

(b) Electrical connector, D10056, to the left pack air cycle machine overheat switch, S938.

(c) Electrical connector, D538, to the right pack compressor discharge overheat switch, S5.

(d) Electrical connector, D10058, to the right pack air cycle machine overheat switch, S939.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-02-410-006

(1) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-02-410-007

(2) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

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(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-02-860-012

(3) Make sure that the R PACK and L PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

SUBTASK 21-51-02-860-013

(4) Make sure that the BLEED 1 and BLEED 2 switches, on the P5-10 air conditioning panel, are in the OFF position.

SUBTASK 21-51-02-860-014

(5) Make sure that the ISOLATION VALVE switch, on the P5-10 air conditioning panel, is in the OPEN position.

SUBTASK 21-51-02-860-015

(6) Make sure that the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, are in the AUTO position.

——— END OF TASK ———

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HEAT EXCHANGER AND PLENUM/DIFFUSER ASSEMBLY - CLEARING DEBRIS/BLOCKAGES

1. General

A. This procedure has this task:

(1) A blowing out of debris/blockages with nitrogen gas/compressed air from the heat exchanger.

TASK 21-51-03-160-805

2. Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air

(Figure 201, Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Boeing recommends performing this clearing procedure using nitrogen gas/compressed air to effectively clear blocked heat exchangers.

(2) This task includes instructions to blow out debris or contaminant blockages trapped within the cool air side heat exchanger grills.

(a) This is an effective method for clearing debris/blockages, especially for aircraft operating in challenging environments (cottonwood/bug season, etc.).

(b) This method does not clean/break down contaminants (dust, dirt, airborne particles, pollen, etc.) baked onto the heat exchanger grills over time.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-10099	Cleaning Wand Eqpt - Heat Exchanger, Pneumatic Part #: B21005-10 Supplier: 81205 Opt Part #: B21005-1 Supplier: 81205
SPL-14922	Cleaning Wand Eqpt - Heat Exchanger Part #: C21010-40 Supplier: 81205 Opt Part #: C21010-1 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-1290	Source - Nitrogen, 0-200 PSIG
STD-6452	Air Source - Regulated, Dry Filtered, 0-200psig

D. Consumable Materials

Reference	Description	Specification
A50026	Adhesive - One-Component, Fluorosilicone, Momentive Performance Materials FRV1106 (Formally GE Silicones)	BAC5010 Type 84
A50035	Sealant - Dow Corning Q4-2817 Fluorosilicone Sealant	
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A
G51935	Air - Clean, Dry	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Gasket	21-51-24-04A-190	PTF ALL
		21-51-24-04B-190	PTF ALL
5	Gasket	21-51-03-02-118	PTF ALL
		21-51-03-03-145	PTF ALL

F. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

H. Prepare for the Blowing Out

SUBTASK 21-51-03-860-036

(1) Set the L PACK or R PACK switches, on the P5–10 air conditioning panel, to the OFF position.

(a) Install the DO NOT OPERATE tags, STD-858, on the switches.

SUBTASK 21-51-03-860-037

(2) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel, to the CLOSE position.

SUBTASK 21-51-03-010-022

(3) To get access to the left pack heat exchangers, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

WARNING

OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(b) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-03-010-023

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the right pack heat exchangers, do these steps:

(TASK 06-41-00-800-801)

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-03-020-080

(5) Remove the bolts [4] and washers [3] that attach the ram air inlet duct access panel [2] to the ram air inlet duct.

SUBTASK 21-51-03-010-024

(6) Remove the ram air inlet duct access panel [2].

NOTE: Removal of the access panel is necessary so that the debris and blockages will not accumulate in the ram air inlet duct.

NOTE: There is a gasket bonded to the access panel. The gasket can stay in place and reused if it is not damaged.

SUBTASK 21-51-03-020-081

(7) Remove the bolts [8] and washers [7] that attach each of the plenum/diffuser access panels [6] and gaskets [5] to the plenum/diffuser assemblies for the primary and secondary heat exchangers.

NOTE: Each access panel is installed with a gasket. The gaskets can be reused if there is no damage.

SUBTASK 21-51-03-020-082

(8) If necessary, remove the bonding jumpers [11] from the plenum/diffuser access panels [6].

NOTE: The bonding jumper removal is not necessary if the task can be completed with the access panel moved to the side.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(a) Remove the sealant that covers the jumper terminals and the head of the bolts [9] on the plenum/diffuser access panels [6] (SWPM 20-20-00 and SWPM 20-20-10).

(b) Remove the bolts [9], washers [10], washers [12], and nuts [13] that attach the bonding jumpers [11] to the plenum/diffuser access panels [6].

SUBTASK 21-51-03-020-083

(9) Remove the plenum/diffuser access panels [6] and gaskets [5] from the plenum/diffuser.

I. Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air

SUBTASK 21-51-03-160-011

(1) To blow out debris/blockages with the nitrogen gas/compressed air in the heat exchanger grills with the cleaning wand, SPL-14922, do the following steps:

WARNING COMPRESSED GAS CAN BE DANGEROUS. EYE PROTECTION IS MANDATORY. APPLY DRY NITROGEN CAREFULLY IN DIRECTION AWAY FROM EYES AND FACE AND OTHER PERSONNEL TO AVOID POSSIBLE INJURY.

(a) Blow out contaminant blockages with compressed clean dry air, G51935, or nitrogen, G00018 (Figure 202).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Connect one end of the cleaning wand, SPL-14922, air hose to either the 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, outlet.

   NOTE: The cleaning wand, SPL-14922, includes an air hose, air wand, water hose, water wand, and soap injector.

2) Connect other end of the cleaning wand, SPL-14922, air hose to the cleaning wand, SPL-14922, air wand inlet.

   NOTE: The adjustable neck of the cleaning wand, SPL-14922, can be lengthened or shortened by its number of beads. It is acceptable to adjust the length by adding or removing the beads as necessary.

3) Adjust the regulator on compressed clean dry air, G51935, or nitrogen, G00018, to supply a recommended pressure of 100 psi (689 kPa) maximum to the cleaning wand, SPL-14922.

CAUTION DO NOT TOUCH THE COOLING FINS OF THE HEAT EXCHANGERS WITH THE TIP OF THE CLEANING WAND. DAMAGE TO THE COOLING FINS OF THE HEAT EXCHANGERS CAN OCCUR.

4) Insert the cleaning wand, SPL-14922, through the plenum/diffuser access openings and blow clean dry air, G51935, or nitrogen, G00018, through the cooling fins of the primary and secondary heat exchangers.

   a) Make sure that the cooling fins of the heat exchanger are not damaged when the cleaning wand, SPL-14922, was inserted through the plenum/diffuser access panel openings.

   b) Squeeze the handle on the cleaning wand, SPL-14922, air wand to blow the clean dry air, G51935, or nitrogen, G00018, through the heat exchangers until all debris is blown out of the cooling fins.

      <1> Make sure that the pressure gauge on the regulator does not indicate more than 100 psi (689 kPa) when the cleaning wand, SPL-14922, operates.

      <2> Blow the clean dry air, G51935, or nitrogen, G00018, through all the areas of the heat exchangers.

      <3> Make sure that dirt, sand, gravel or plant fiber is no longer being blown out on the opposite side (inlet side) of the heat exchanger.

      <4> Continue blowing the clean dry air, G51935, or nitrogen, G00018, through the heat exchangers until debris is no longer detected being blown out.

   c) Make sure that all loose dirt, sand, gravel, or plant fiber is removed from the ram air inlet that was blown out of the heat exchangers.

5) Remove the cleaning wand, SPL-14922, from the inside of the plenum/diffuser access opening.

6) Decrease the compressed clean dry air, G51935, or nitrogen source, STD-1290, pressure to 0 psi (0 kPa).

7) Remove the cleaning wand, SPL-14922.

SUBTASK 21-51-03-160-012

(2) To blow out heat exchanger debris/blockage with nitrogen gas/compressed air with the heating exchanger cleaning wand, SPL-10099, do the following steps:

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WARNING COMPRESSED GAS CAN BE DANGEROUS. EYE PROTECTION IS MANDATORY. APPLY DRY NITROGEN CAREFULLY IN DIRECTION AWAY FROM EYES AND FACE AND OTHER PERSONNEL TO AVOID POSSIBLE INJURY.

(a) Attach the heating exchanger cleaning wand, SPL-10099, to the regulator on a clean dry air, G51935, bottle or nitrogen, G00018, bottle.

(b) Adjust the regulator on a clean dry air, G51935, bottle or nitrogen, G00018, bottle to supply a maximum of 80 psi (552 kPa) to the heating exchanger cleaning wand, SPL-10099.

NOTE: If a clean dry air, G51935, bottle is used, the moisture contaminant or water concentration in the air of 0.00002 grams maximum is not required.

CAUTION DO NOT TOUCH THE COOLING FINS OF THE HEAT EXCHANGERS WITH THE TIP OF THE CLEANING WAND. DAMAGE TO THE COOLING FINS OF THE HEAT EXCHANGERS CAN OCCUR.

(c) Insert the heating exchanger cleaning wand, SPL-10099, through the plenum/diffuser access openings and blow clean dry air, G51935, or nitrogen, G00018, through the cooling fins of the primary and secondary heat exchangers.

1) Make sure that the cooling fins of the heat exchanger are not damaged when the heating exchanger cleaning wand, SPL-10099, was inserted through the plenum/diffuser access panel openings.

2) Blow the clean dry air, G51935, or nitrogen, G00018, through the heat exchangers until all debris is blown out of the cooling fins.

a) Make sure that the pressure gauge on the heating exchanger cleaning wand, SPL-10099, does not indicate more than 80 psi (552 kPa) when the wand operates.

b) Blow the clean dry air, G51935, or nitrogen, G00018, through all the areas of the heat exchangers.

c) Make sure that dirt, sand, gravel or plant fiber is no longer being blown out on the opposite side (inlet side) of the heat exchanger.

d) Continue to blow the clean dry air, G51935, or nitrogen, G00018, through the heat exchangers until debris is no longer detected being blown out.

3) Make sure that all loose dirt, sand, gravel, or plant fiber is removed from the ram air inlet that was blown out of the heat exchangers.

(d) Remove the heating exchanger cleaning wand, SPL-10099, from the inside of the plenum/diffuser access opening and decrease the clean dry air, G51935, or nitrogen, G00018, pressure to 0 psi (0 kPa).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-03-211-004

(1) Examine the gasket [1] and gaskets [5] for damage.

(a) Replace the gasket [1] or gaskets [5] if there is damage or deterioration.

SUBTASK 21-51-03-420-016

(2) If removed, install the bonding jumpers [11] to the plenum/diffuser access panels [6].

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(a) Make sure that the bonding jumpers [11], bolts [9], washers [10], washers [12], nuts [13], and plenum/diffuser access panels [6] are clean.

1) If installed, make sure that the nutplates are clean.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(b) Put the bonding jumpers [11] in their positions on the plenum/diffuser access panels [6].

(c) Install the bolts [9], washers [10], washers [12], and nuts [13].

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(d) Tighten the bolts [9] to 22 in-lb (2.49 N·m) - 26 in-lb (2.94 N·m).

(e) Measure the electrical bonding resistance between the bolts [9] and anti-static coating of plenum/diffuser access panels [6].

1) Use a digital/analog multimeter, COM-1793.

2) Make sure that the electrical bonding resistance is 0.3 Mohm (300 000 ohm) or less.

(f) Apply a coat of Dow Corning Q4-2817 sealant, A50035, or FRV1106 adhesive, A50026, to cover the jumper terminals and the heads of the bolts [9].

NOTE: The sealant is optional if the plenum was built before 2022/2023, and did not originally leave Honeywell production with the sealant applied.

SUBTASK 21-51-03-410-024

(3) Install the plenum/diffuser access panels [6] and gaskets [5] with the bolts [8] and washers [7].

(a) Tighten the bolts [8] to 22 in-lb (2.5 N·m) - 26 in-lb (2.9 N·m).

SUBTASK 21-51-03-410-025

(4) Install the ram air inlet duct access panel [2] and gasket [1] with the bolts [4] and washers [3].

SUBTASK 21-51-03-080-010

(5) Remove the DO NOT OPERATE tag, STD-858, from the L PACK and R PACK switches.

(a) Make sure that the R PACK and L PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

SUBTASK 21-51-03-860-039

(6) Operate the air conditioning pack for five minutes (TASK 21-00-00-800-803).

NOTE: This will remove all water in the ram air ducts and the air cycle machine bearings.

SUBTASK 21-51-03-860-040

(7) Stop the air conditioning pack (TASK 21-00-00-800-804).

SUBTASK 21-51-03-860-041

(8) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-03-410-026

(9) If the left pack heat exchangers are blown out, do these steps:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Install this access panel:

(TASK 53-51-21-400-802)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-03-410-027

(10) If the right pack heat exchangers are blown out, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Install this access panel:

(TASK 53-51-21-400-802)

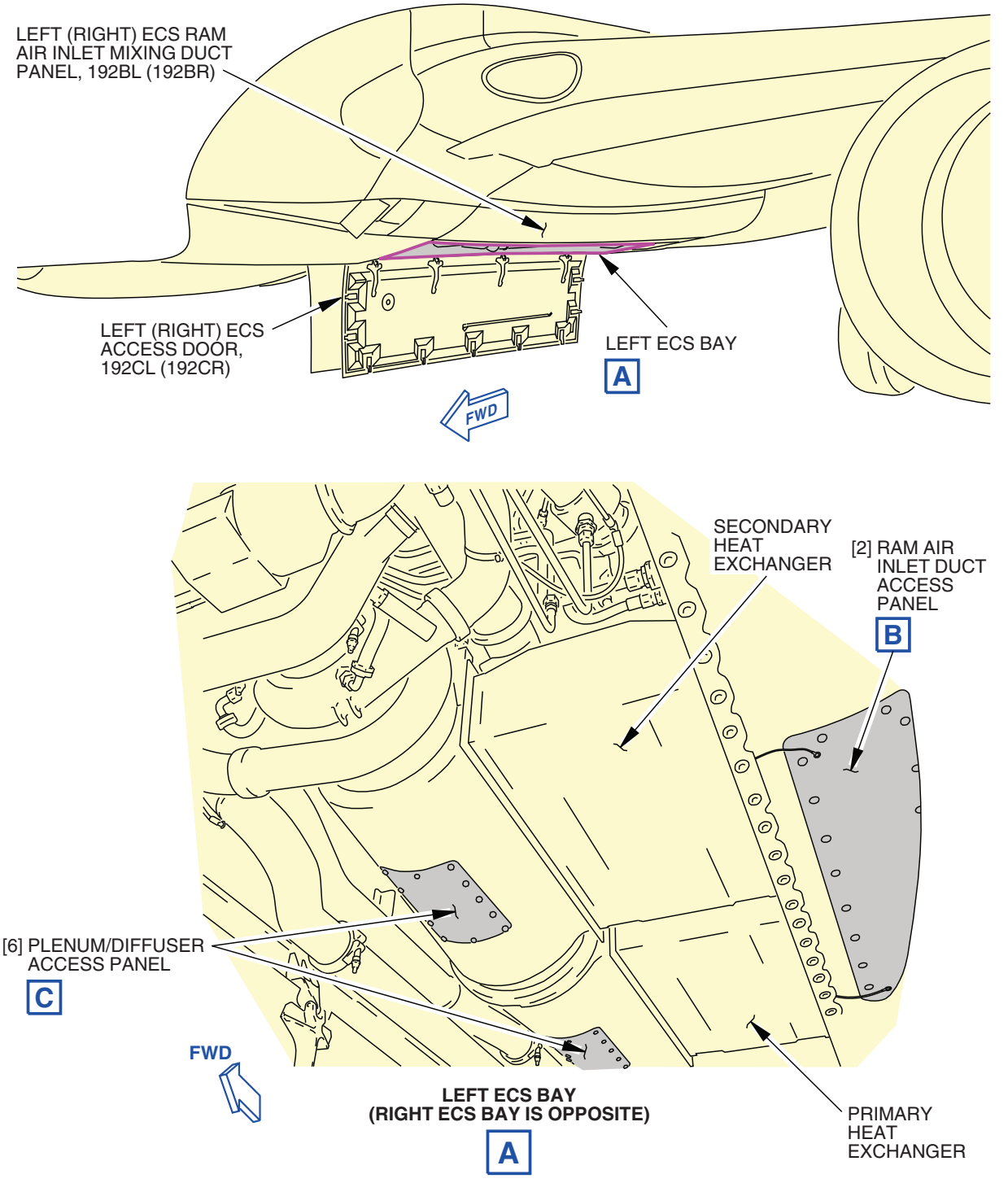
Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-03-860-038

(11) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel, to the OPEN position.

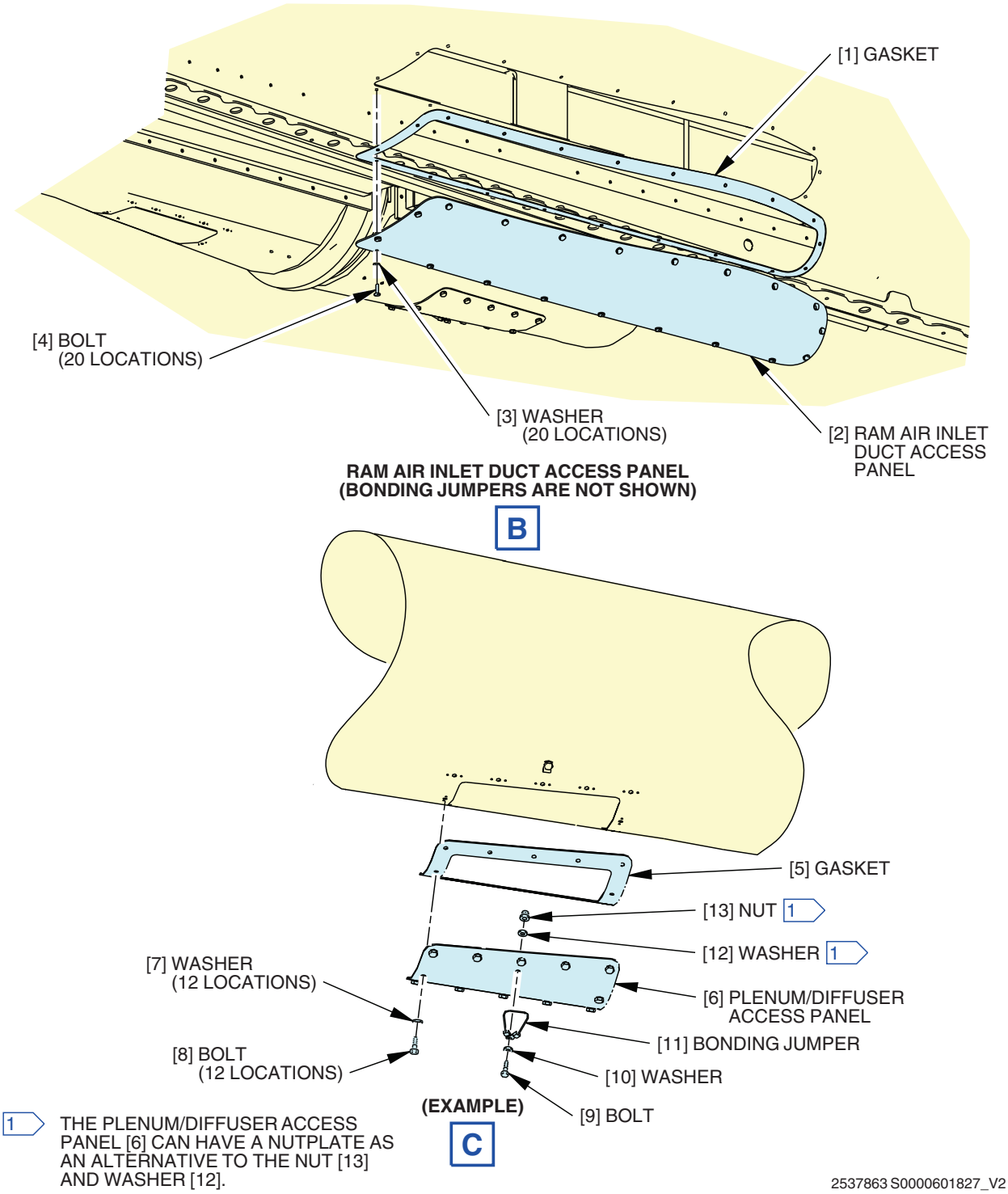
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



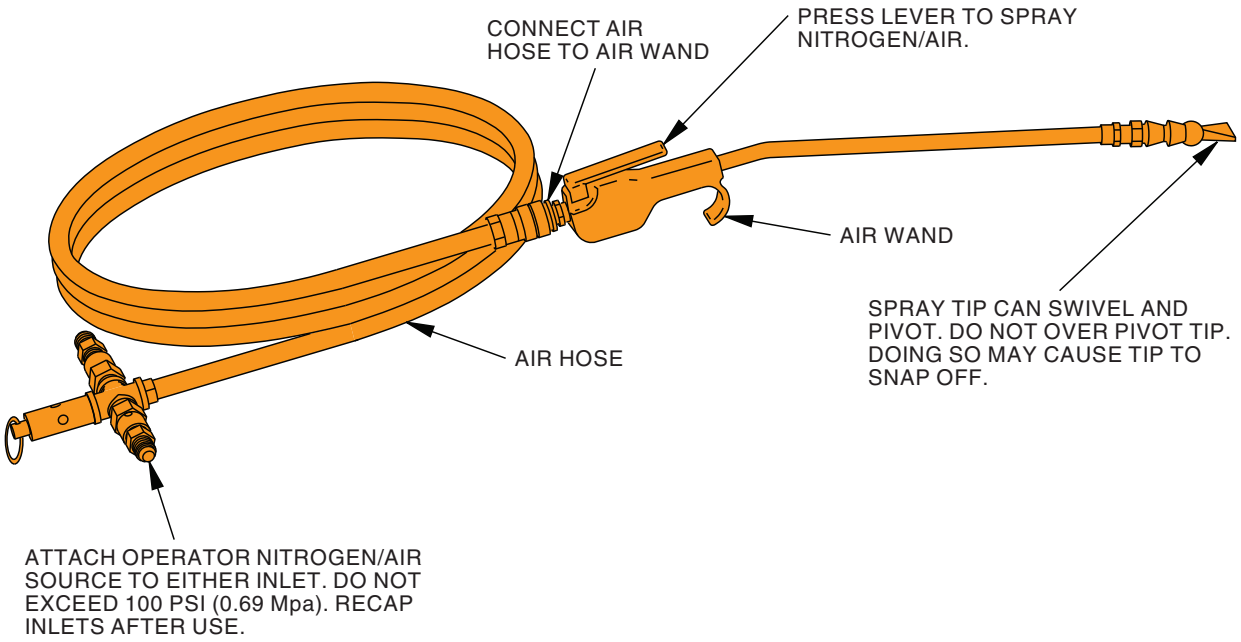
Heat Exchanger Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air
Figure 201/21-51-03-990-815 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Heat Exchanger Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air
Figure 201/21-51-03-990-815 (Sheet 2 of 2)

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Air Wand and Air/Nitrogen Hose
Figure 202/21-51-03-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HEAT EXCHANGER AND PLENUM/DIFFUSER ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the primary and secondary heat exchangers.

NOTE: The plenum/diffuser assembly is attached to the heat exchangers.

(2) An installation of the primary and secondary heat exchangers.

TASK 21-51-03-000-803

2. Heat Exchanger and Plenum/Diffuser Assembly - Removal

(Figure 401, Figure 402, Figure 403)

A. General

(1) It is recommended that you remove the primary heat exchanger, the secondary heat exchanger, the primary plenum/diffuser, and the secondary plenum/diffuser as one unit. As an option, you may choose to remove and install a heat exchanger separately, or with the air cycle machine as an assembly, if you determine this best suits your maintenance needs.

(2) The heat exchangers, the plenum/diffusers and air cycle machine together weigh approximately 125 lb (57 kg). If the adapter, SPL-1608, is not used, a minimum of three persons will be necessary to complete this task.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-04-000-801	Air Cycle Machine Removal (P/B 401)
21-51-24-000-801	Ram Air Ducts Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
47-11-01-000-802	Air Separation Module Assembly - Removal (P/B 401)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1570	Fixture - Lift, Engine Accessory Part #: A71015-157 Supplier: 81205 Opt Part #: A71015-108 Supplier: 81205
SPL-1608	Jack Adapter - Installation/Removal, Air Conditioning Pack Part #: C21005-70 Supplier: 81205 Opt Part #: C21005-62 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G50262	Wiper - Cleaning	BMS15-5

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Prepare for the Removal

SUBTASK 21-51-03-040-001

(1) Make sure that the pneumatic power is off (TASK 36-00-00-860-806).

SUBTASK 21-51-03-860-001

(2) Do these steps, on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position.

1) Attach a DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position.

1) Attach a DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position.

1) Attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-03-860-002

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-03-860-032

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

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SUBTASK 21-51-03-860-028

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-03-010-001

(6) To remove the heat exchangers for the left pack, do these steps:

(TASK 06-41-00-800-801)

WARNING

OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-03-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(7) To remove the heat exchangers for the right pack, do these steps:

(TASK 06-41-00-800-801)

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Heat Exchanger and Plenum/Diffuser Assembly Removal

SUBTASK 21-51-03-020-001

(1) Disconnect the flexible hose [71] from the spud on the top of the heat exchanger inlet duct [15] as follows:

(a) Loosen the hose clamps [70].

(b) Slide the flexible hose [71] aft toward the Nitrogen Generation System (NGS) heat exchanger until the flexible hose [71] is off the spud on the heat exchanger inlet duct [15].

SUBTASK 21-51-03-020-002

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(2) If it is necessary, remove the outer seal [12] for the Environmental Control System (ECS) access door:

(a) Remove the bolts [10] and nutplates.

(b) Remove the retainer [13] and seal [12].

SUBTASK 21-51-03-020-003

(3) Remove the ram air exhaust elbow [2] from the primary plenum/diffuser [8], do these steps:

(a) Loosen the aft clamp [4].

(b) Move the aft end of the flexible duct [114] forward off of the ram air exhaust duct [107].

(c) If it is necessary remove the bolts [106] and ram air exhaust duct [107] (TASK 21-51-24-000-801).

(d) Remove the bolts [6] and washers [7].

(e) Remove the bolt [115], washer [116], bonding jumper [117], and washer [118].

(f) Remove the ram air exhaust elbow [2].

(g) Remove the gasket [3].

SUBTASK 21-51-03-020-004

(4) Do these steps to remove the heat exchanger inlet duct [15]:

(a) If it is necessary to get more access in the left ram air duct compartment, remove the Nitrogen-Enriched Air Distribution System (NEADS) tube (TASK 47-11-01-000-802).

NOTE: It is possible to disconnect the ends and loosen the two clamps for the applicable section of the NEADS tube assembly. This is to rotate the NEADS tube outwards to get more access.

(b) Remove the bolts [111] and washers [112] that hold each bonding jumpers [113] to the inlet duct.

NOTE: These bolts are longer to accommodate the bonding jumper. Keep them separate from the other bolts for the installation.

(c) Loosen the clamp [14].

(d) Move the flexible hose off the heat exchanger inlet duct [15].

(e) Remove the bolts [16], washers [17], bolt [123], and washer [124].

(f) Move the heat exchanger inlet duct [15] down to disengage the upper flange from the retainer on the lower wing beam.

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SUBTASK 21-51-03-010-003

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(5) To remove the compressor outlet duct [29], do these steps:

(a) Disconnect the flex hose [105] from the compressor outlet duct [29].

(b) Remove the electrical connector [18] from the ram air control sensor.

(c) Remove the electrical connector [22] from the compressor discharge overheat switch.

(d) Remove the clamp [24] at the air cycle machine [39].

1) Lift the three latch pawls at the same time to release the clamp [24].

(e) Move the sleeve [25] to the compressor outlet duct [29].

(f) Remove the coupling [19] at the secondary heat exchanger [58].

(g) Remove the compressor outlet duct [29].

(h) Remove and discard the packing [20] at the secondary heat exchanger [58].

(i) Remove and discard the packings [26].

SUBTASK 21-51-03-020-009

(6) Remove the clamp [21] from the outboard duct of the secondary heat exchanger [58].

SUBTASK 21-51-03-020-011

(7) Do this step to disconnect the tie rod assemblies [89] from the forward and aft end of each of the two heat exchangers:

(a) Remove the bolt [54], washer [55], and bushing [56] from each of the four tie rod assemblies [89].

SUBTASK 21-51-03-020-012

(8) Do these steps to remove the duct [108]:

(a) Remove the clamp [109] and clamp [110].

(b) Remove the duct [108].

SUBTASK 21-51-03-020-013

(9) Remove the clamp [1] from primary heat exchanger [9] outlet and bleed air inlet ducts.

SUBTASK 21-51-03-020-014

(10) If the hoist adapter, SPL-1608, is to be used, do these steps to install the adapter:

(a) Install the adapter, SPL-1608, on the engine accessory lift fixture, SPL-1570.

(b) Put the engine accessory lift fixture, SPL-1570, in its position under the heat exchangers.

(c) Make sure that the frame of the adapter, SPL-1608, is horizontal.

1) If it is necessary, use the square drive socket [104] to adjust the angle of the frame at the bottom of the adapter, SPL-1608.

(d) Lift the engine accessory lift fixture, SPL-1570, until it touches the heat exchangers.

1) Make sure that the straps [101] are not caught between the adapter, SPL-1608, and the heat exchangers.

(e) If it is necessary, adjust the sliding tray [103] to make sure that both heat exchangers are firmly seated the in the trays.

1) Loosen the nuts [102].

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2) Move the sliding tray [103] forward or aft.

3) Make sure that the heat exchangers are between the end pieces of the trays and the flanges of the heat exchangers are not on the tabs.

4) Tighten the nuts [102].

(f) Put the straps [101] around the heat exchangers.

(g) Tighten the straps [101].

SUBTASK 21-51-03-020-015

WARNING IF YOU DO NOT USE THE HOIST ADAPTER, HAVE A MINIMUM OF TWO PERSONS HELP YOU REMOVE THE HEAT EXCHANGERS. THE HEAT EXCHANGERS WEIGH APPROXIMATELY 125 POUNDS, AND IF THEY FALL, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Do these steps to disconnect the heat exchangers from the lower wing beam:

(a) Remove the bolts [59] and washers [60] that attach the primary heat exchanger [9] and secondary heat exchanger [58] to the lower wing beam.

(b) Remove the bolts [57] that attach the primary heat exchanger [9] and secondary heat exchanger [58] to the lower wing beam.

SUBTASK 21-51-03-020-074

(12) Do these steps to disconnect the secondary plenum/diffuser [23] from the air cycle machine [39]:

(a) Remove the bolts [37] and washers [38] that connect the air cycle machine [39] to the secondary plenum/diffuser [23].

(b) Carefully pull the secondary plenum/diffuser [23] from the air cycle machine [39].

(c) Remove and discard the gasket [36] on the air cycle machine [39].

SUBTASK 21-51-03-020-075

CAUTION KEEP THE HEAT EXCHANGERS LEVEL IN RELATION TO EACH OTHER. IF THE HEAT EXCHANGERS BEND AT THE FLEXIBLE DUCT, DAMAGE TO THE DIFFUSERS AND THE FLEXIBLE DUCT CAN OCCUR.

(13) Carefully lower the heat exchangers from the ECS bay.

SUBTASK 21-51-03-480-001

(14) Put covers on the duct openings to keep out unwanted materials.

SUBTASK 21-51-03-020-017

(15) Do these steps to separate the primary plenum/diffuser [8] from the secondary plenum/diffuser [23]:

(a) Remove the clamps [61] that attach the flexible ducts [62] to the plenum/diffusers.

(b) Carefully move one heat exchanger away from the other.

(c) Move the flexible duct [62] off the plenum/diffuser.

SUBTASK 21-51-03-020-019

(16) Do these steps to remove the secondary heat exchanger [58] from the secondary plenum/diffuser [23] or the primary heat exchanger [9] from the primary plenum/diffuser [8]:

(a) Remove the bolts [63] and washers [64] that attach the clip assembly [65] to the heat exchanger.

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(b) Remove the clip assembly [65].

(c) Remove the bolts [63] and washers [64] that attach the clip assembly [66] to the heat exchanger.

(d) Remove the clip assembly [66].

(e) Remove the bolts [67] and washers [68] that attach the plenum/diffuser to the heat exchanger.

(f) Remove the primary plenum/diffuser [8] from the primary heat exchanger [9] or the secondary plenum/diffuser [23] from the secondary heat exchanger [58].

(g) Remove and discard the gasket [69] from the primary heat exchanger [9] or secondary heat exchanger [58].

(h) Remove the shim [69A] from the primary heat exchanger [9].

SUBTASK 21-51-03-020-020

(17) If necessary, remove the primary diffuser assembly [82] from the primary plenum, do these steps (Figure 402):

(a) Remove the screws [81].

(b) Remove the bolts [83] and washers [84].

(c) If the new primary diffuser assembly [82] will be installed, remove the bolts [90], washers [91], and bracket [92].

1) Use wiper, G50262, moist in alcohol, B00130, to remove remaining adhesive from the primary plenum and bracket [92].

2) Make sure that the bracket [92] is not damaged.

(d) Make a record of the orientation of the diffuser assembly [82] with respect to its position in the plenum so that it can be reinstalled in the same orientation as follows:

1) The smaller diameter end of the diffuser is installed forward.

2) Make a record of the side of the diffuser that is up and which side is down with respect to the installed position on the aircraft.

NOTE: The primary diffuser has a fan bypass check valve which is installed in the down position. The primary diffuser is aft of the secondary diffuser.

(e) Remove the diffuser assembly [82] from the primary plenum.

SUBTASK 21-51-03-020-021

(18) If necessary, remove the secondary diffuser [85] from the secondary plenum, do these steps (Figure 402):

(a) Remove the bolts [83] and washers [84].

(b) Make a record of the orientation of the secondary diffuser with respect to its position in the plenum as follows:

NOTE: This is necessary so that you can reinstall the diffuser in the same orientation.

1) The smaller diameter end of the diffuser is installed forward.

2) Make a record of the side of the diffuser that is up and which side is down with respect to the installed position on the aircraft.

(c) Remove the secondary diffuser [85] from the secondary plenum.

SUBTASK 21-51-03-212-001

(19) Inspect the diffuser assembly [82] from the primary plenum/diffuser as follows:

(a) Make sure that the fan bypass check valve door is installed.

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(b) Make sure that the fan bypass check valve door operates normally.

(c) If the fan bypass check valve door is not installed or does not operate normally, replace the diffuser assembly [82] with a serviceable unit.

I. Inspection of the Air Cycle Machine

SUBTASK 21-51-03-211-001

NOTE: It is recommended to do an inspection of the Air Cycle Machine (ACM) when you get access to the fan wheel. This inspection makes sure that the component operates correctly and helps to prevent a non-scheduled removal. If you remove the heat exchanger and plenum/diffuser assembly to clean the off-wing heat exchanger and do not have ECS related problems, then the ACM inspection is not necessary.

NOTE: If operators choose to remove the heat exchangers, plenum/diffusers, and air cycle machine as one assembly, refer to Air Cycle Machine Removal, TASK 21-51-04-000-801.

(1) If the air cycle machine [39] was removed from the secondary plenum/diffuser [23], do these steps to inspect the impeller blade [74] of the air cycle machine [39]:

(a) Make sure that the chord length (width) of each impeller blade tip is a minimum of 0.38 in. (9.65 mm).

(b) Do a visual check that no impeller blades [74] are broken or cracked.

(c) Replace the air cycle machine [39] with a new or serviceable unit if the inspection is unsatisfactory.

SUBTASK 21-51-03-700-001

(2) Do these steps to do a check of the torque necessary to turn the rotating assembly of the air cycle machine [39]:

(a) Put the air cycle machine [39] in a horizontal position.

(b) Put a torque wrench on nut [73].

NOTE: A torque wrench with a gage that shows the torque applied is recommended.

1) You will have to turn nut [72] if the air cycle machine [39] was not removed from the secondary plenum/diffuser [23].

CAUTION

TURN THE ROTATING COMPONENTS OF THE ACM ONLY IN THE DIRECTION THAT THEY TURN WHEN IN OPERATION. IF YOU TURN THE ACM ROTATING COMPONENTS IN THE INCORRECT DIRECTION, YOU CAN CAUSE DAMAGE TO THE FOIL AIR BEARINGS.

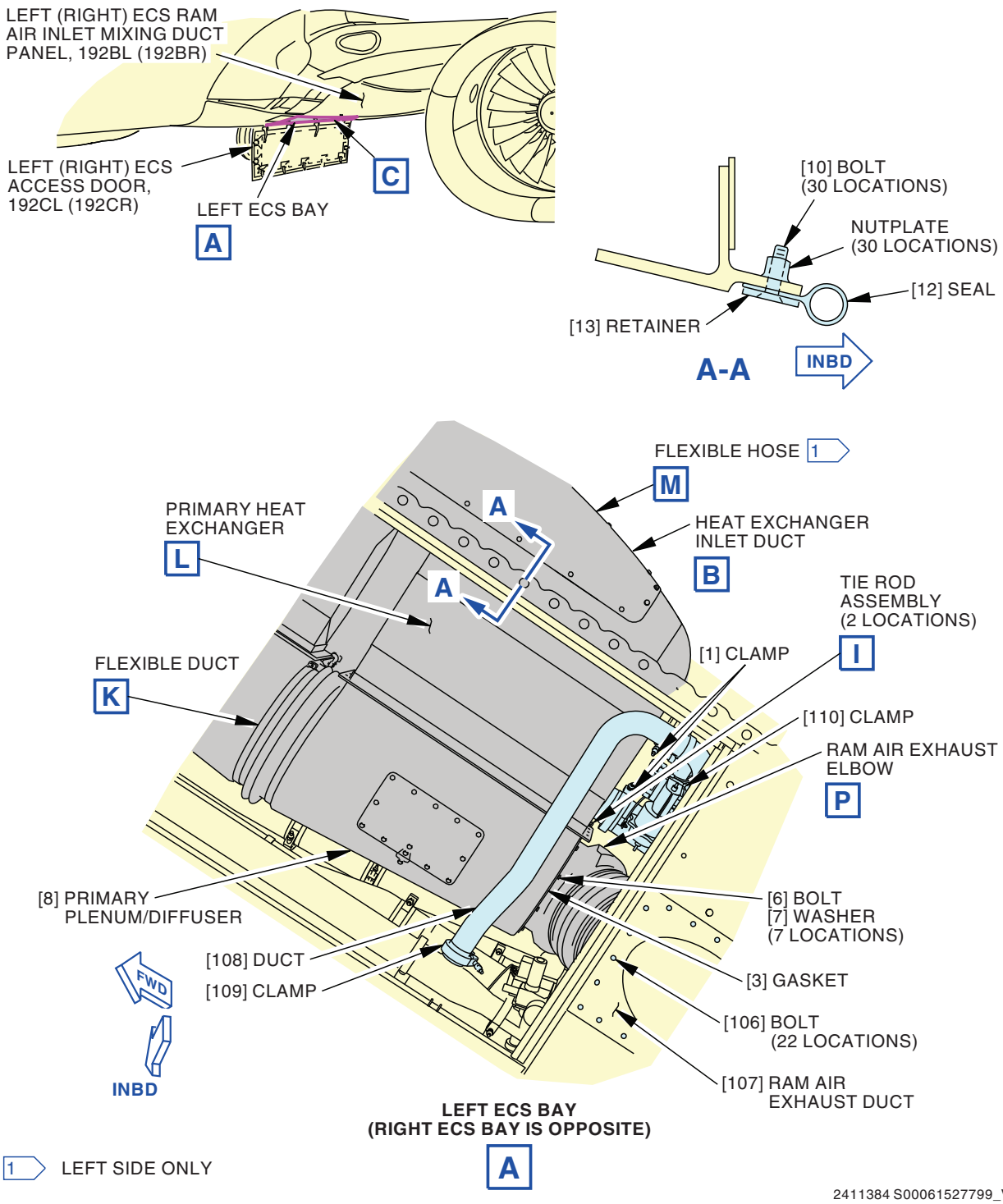
(c) Make sure that you turn nut [72] clockwise or nut [73] counterclockwise while you turn the rotating assembly.

(d) Record the torque necessary to turn the rotating assembly.

(e) If the torque was more than 20.0 in-lb (2.3 N·m), replace the air cycle machine [39] with a new or serviceable unit.

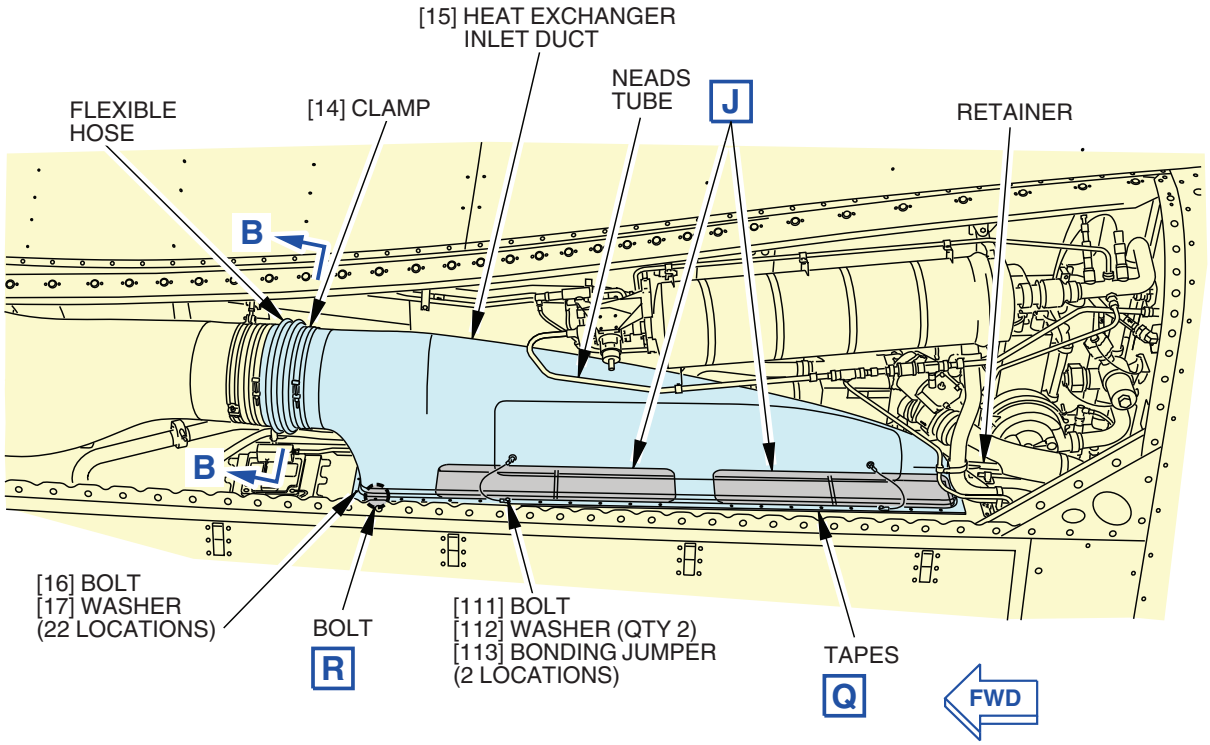
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



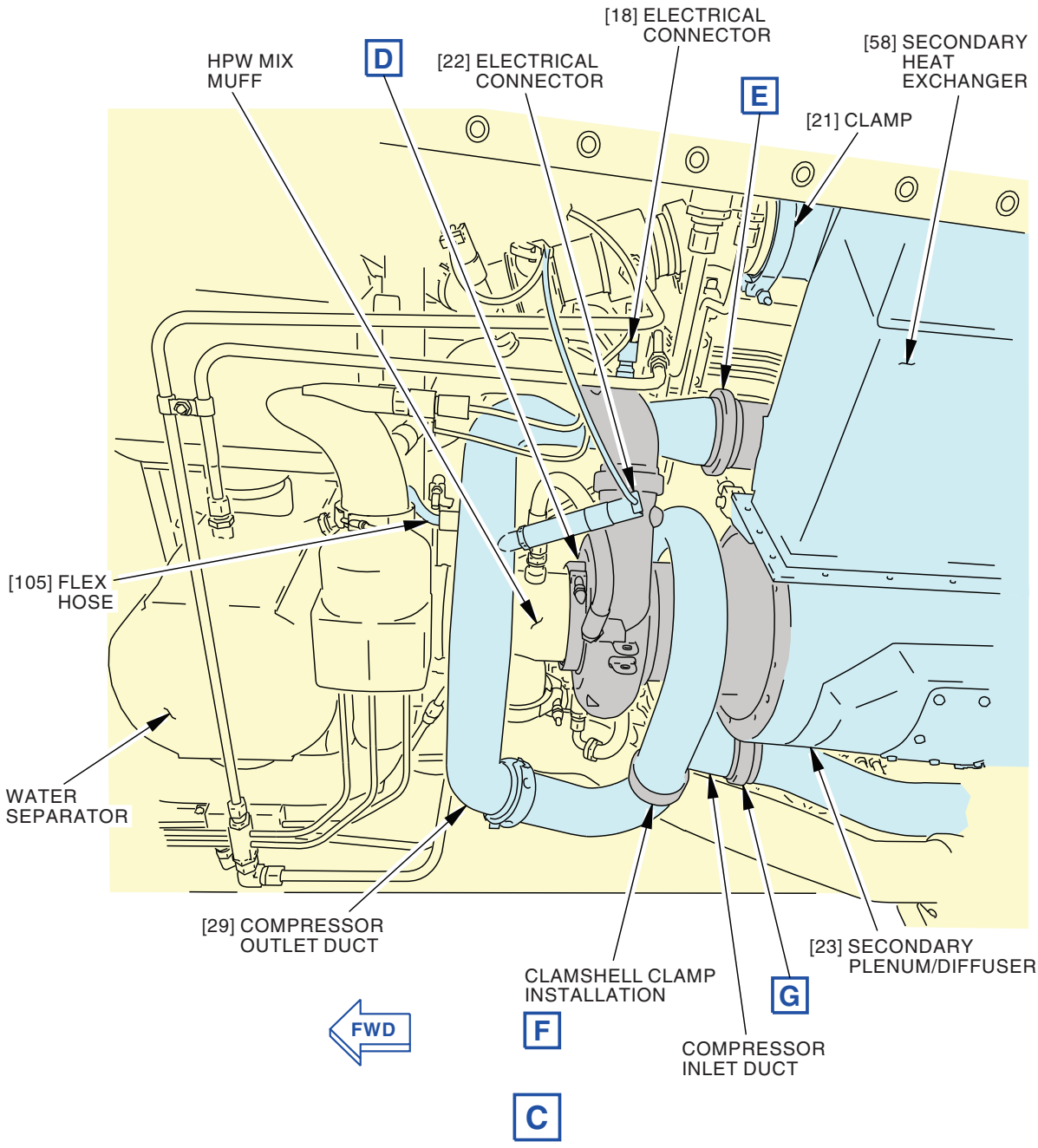
Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 1 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 2 of 13)

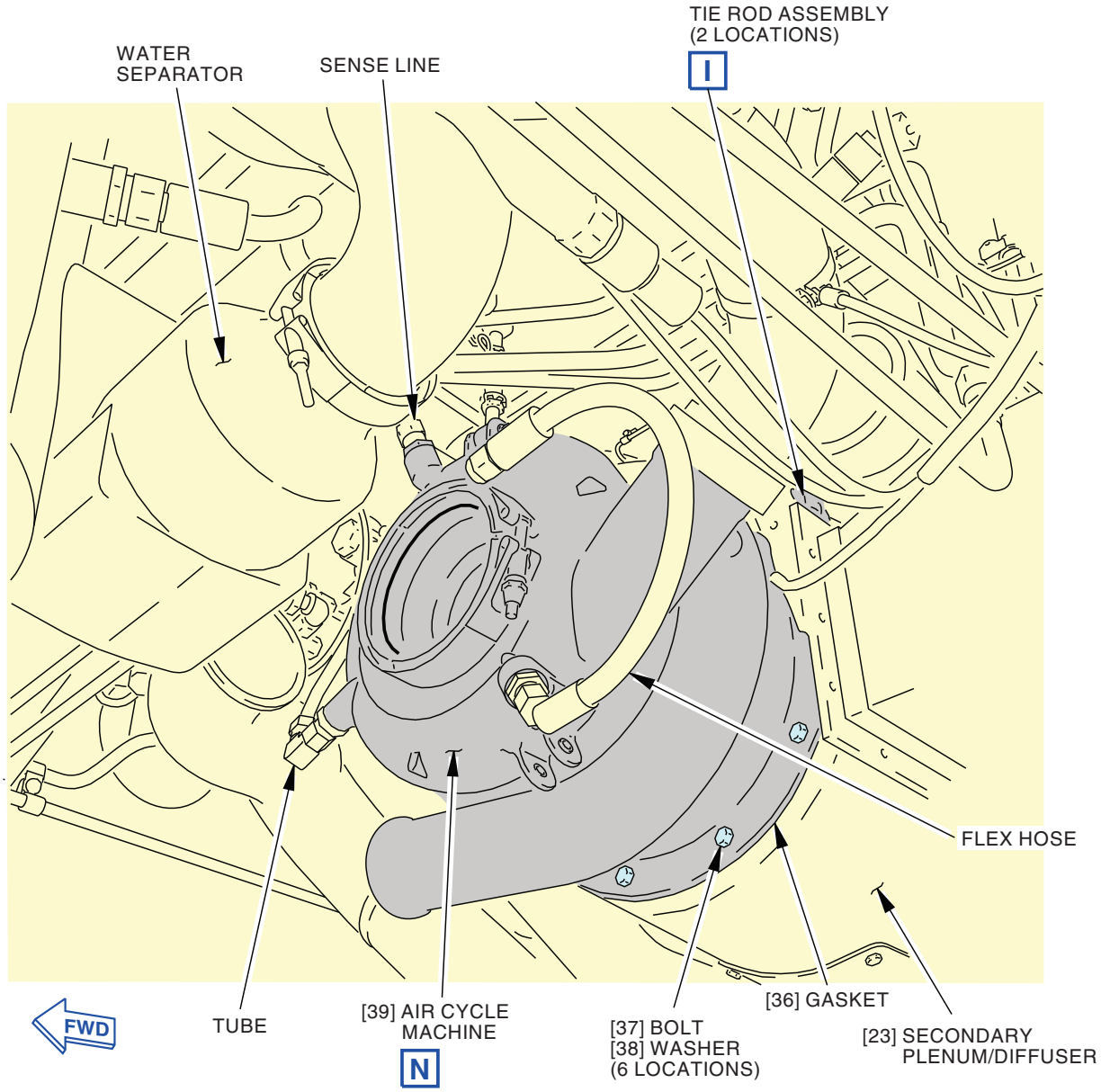
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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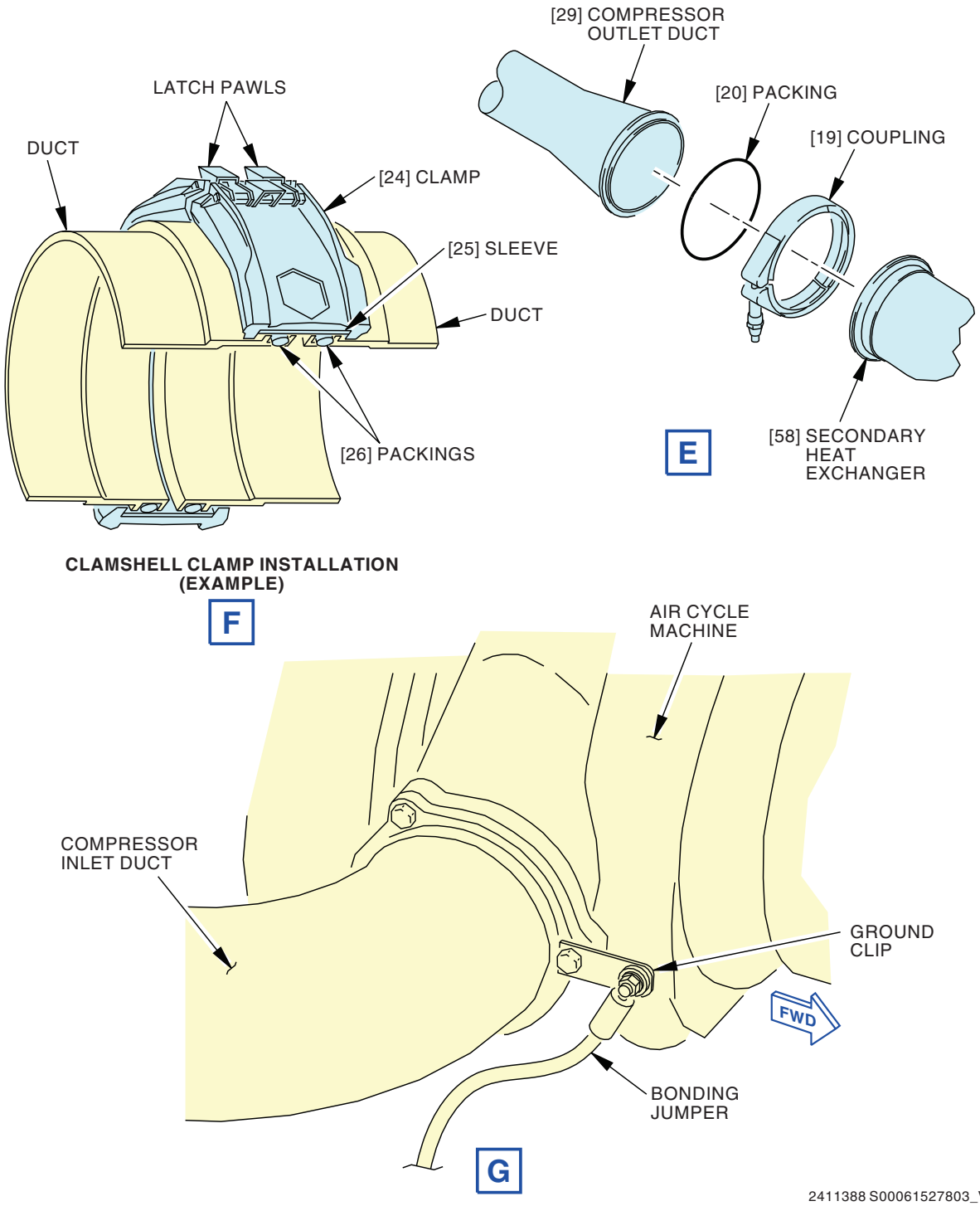
Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 3 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



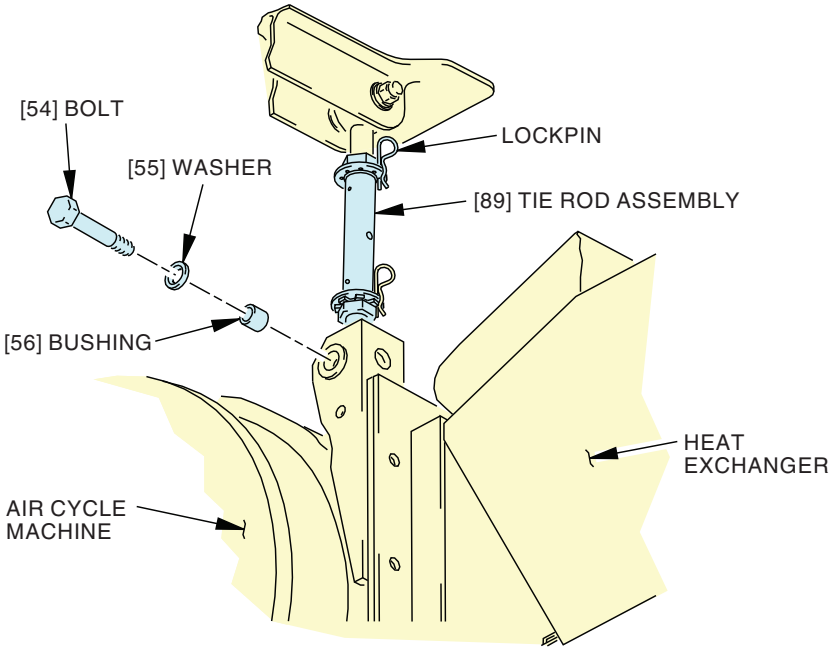
Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 4 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 5 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



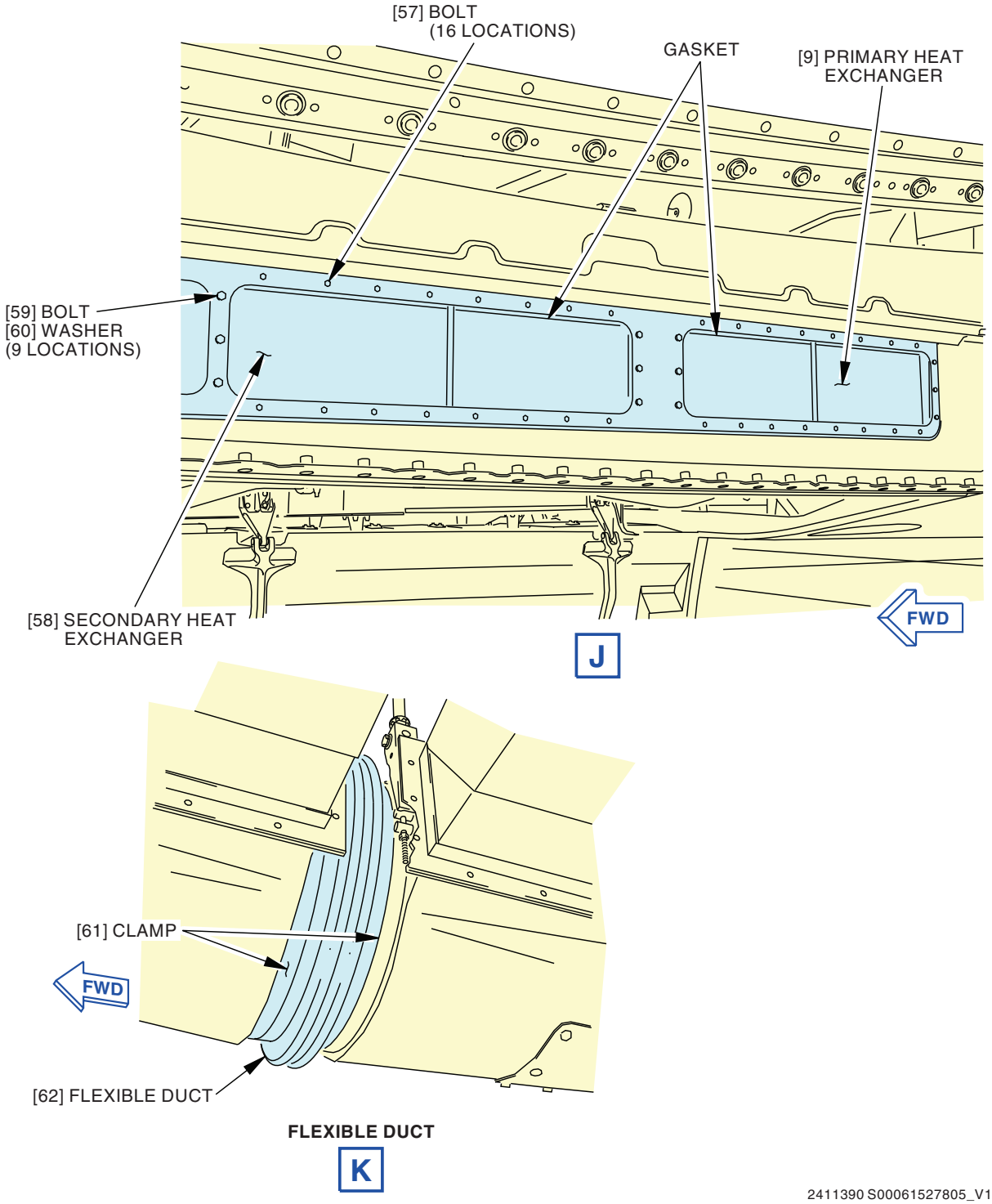
**TIE ROD ASSEMBLY
(EXAMPLE)**

I

2896782 S0000690905_V1

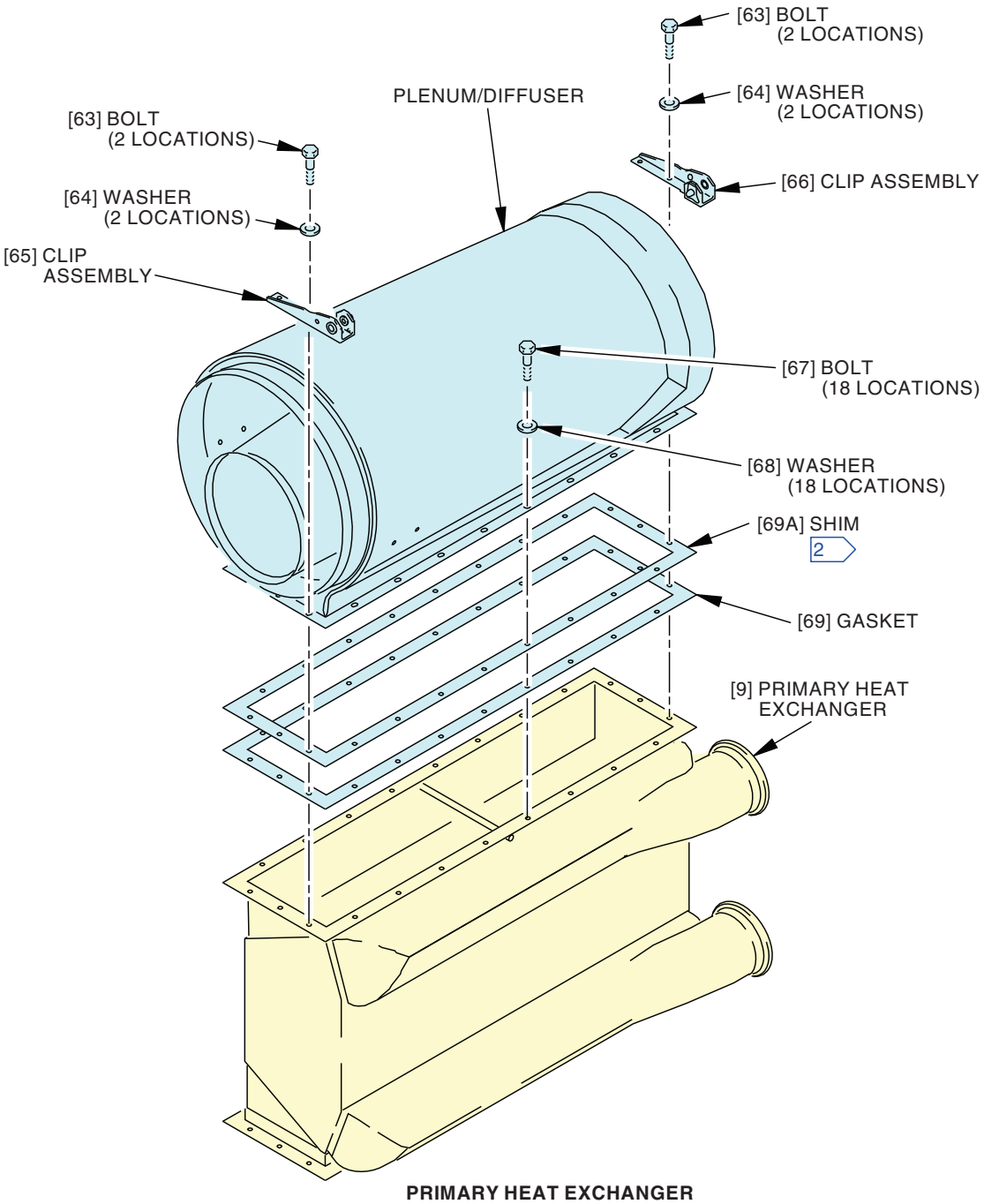
**Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 6 of 13)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 7 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PRIMARY HEAT EXCHANGER
(SECONDARY HEAT EXCHANGER IS EQUIVALENT)

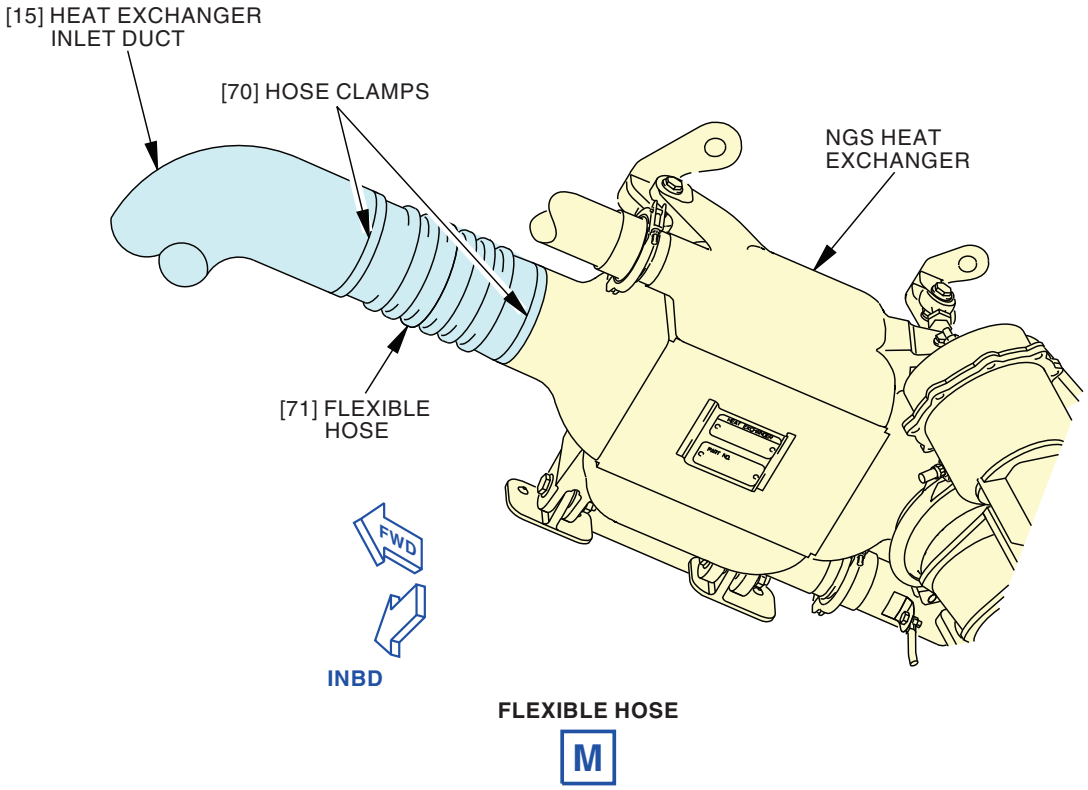
2 FOR PRIMARY HEAT EXCHANGER ONLY

L

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Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 8 of 13)

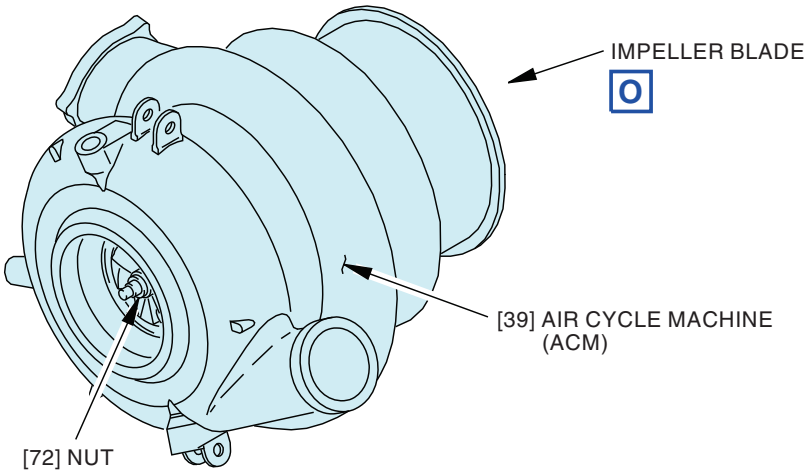
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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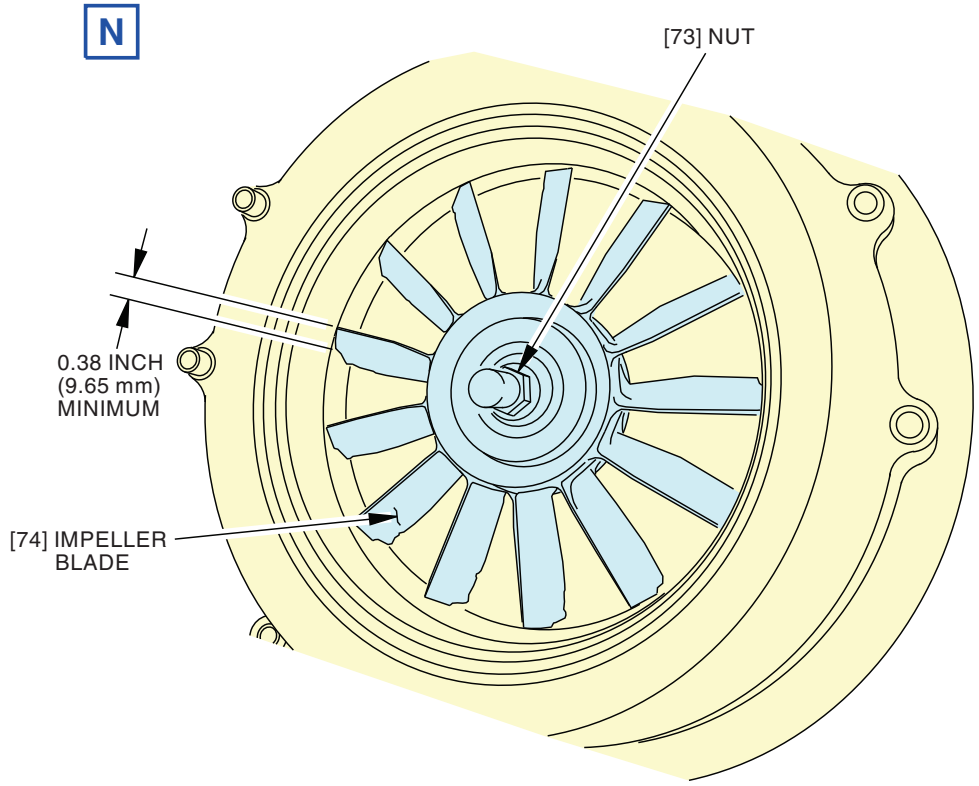
Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 9 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AIR CYCLE MACHINE (ACM)

N



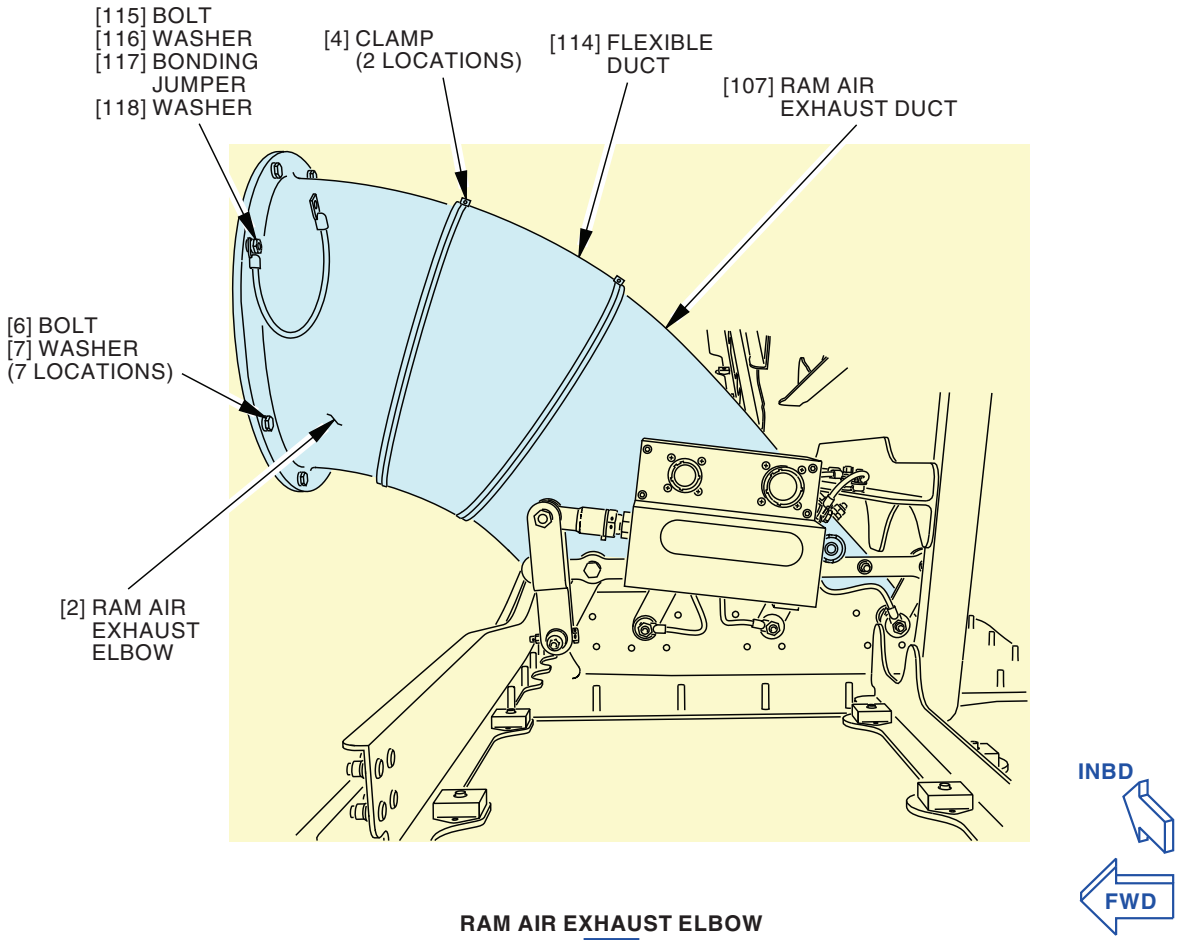
IMPELLER BLADE

O

2411393 S00061527808_V1

Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 10 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



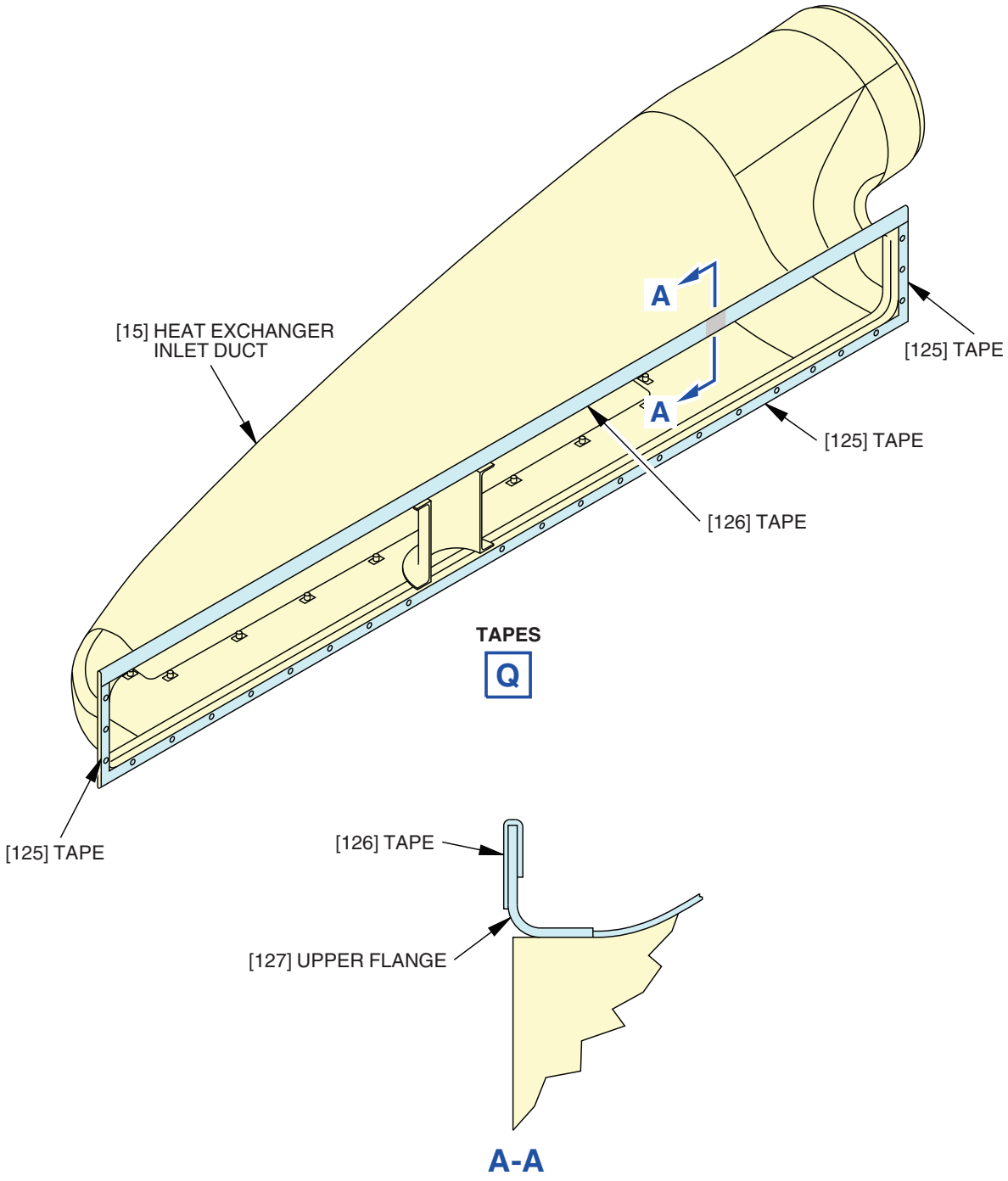
RAM AIR EXHAUST ELBOW

P

2920082 S0000702934_V1

Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 11 of 13)

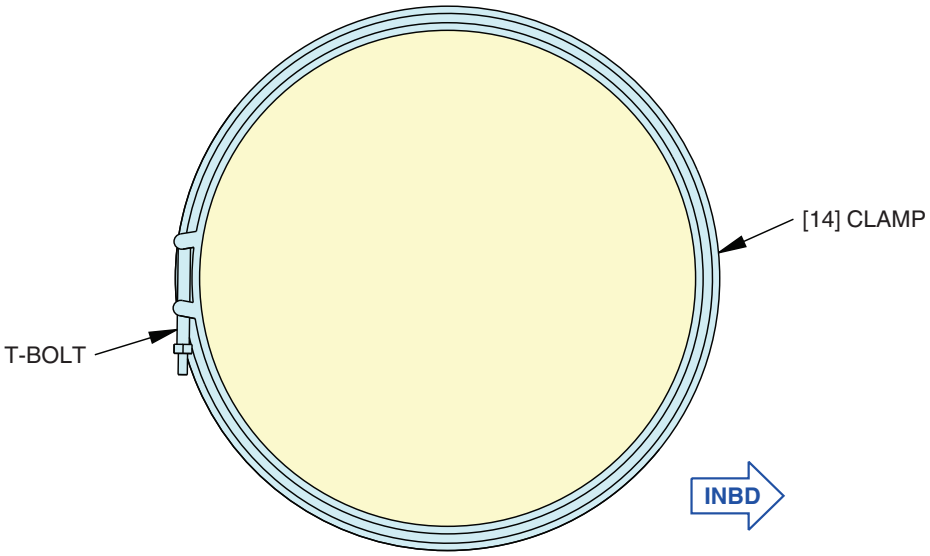
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3083157 S0000838321_V2

Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 12 of 13)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



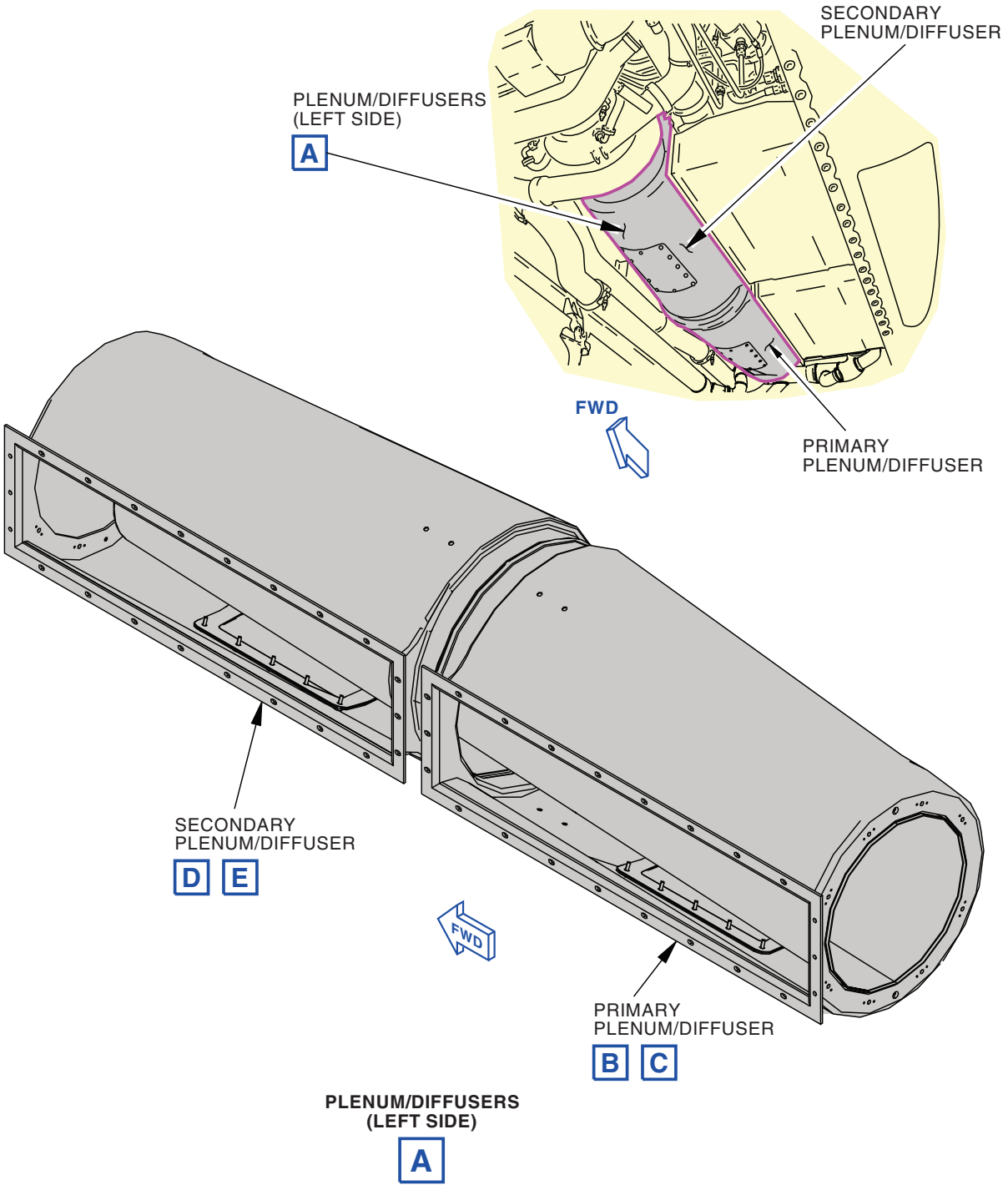
**(CLAMP T-BOLT ORIENTATION IN THE LEFT ECS BAY IS SHOWN,
CLAMP T-BOLT ORIENTATION IN THE RIGHT ECS BAY IS OPPOSITE)**

B-B

3099602 S0000845720_V1

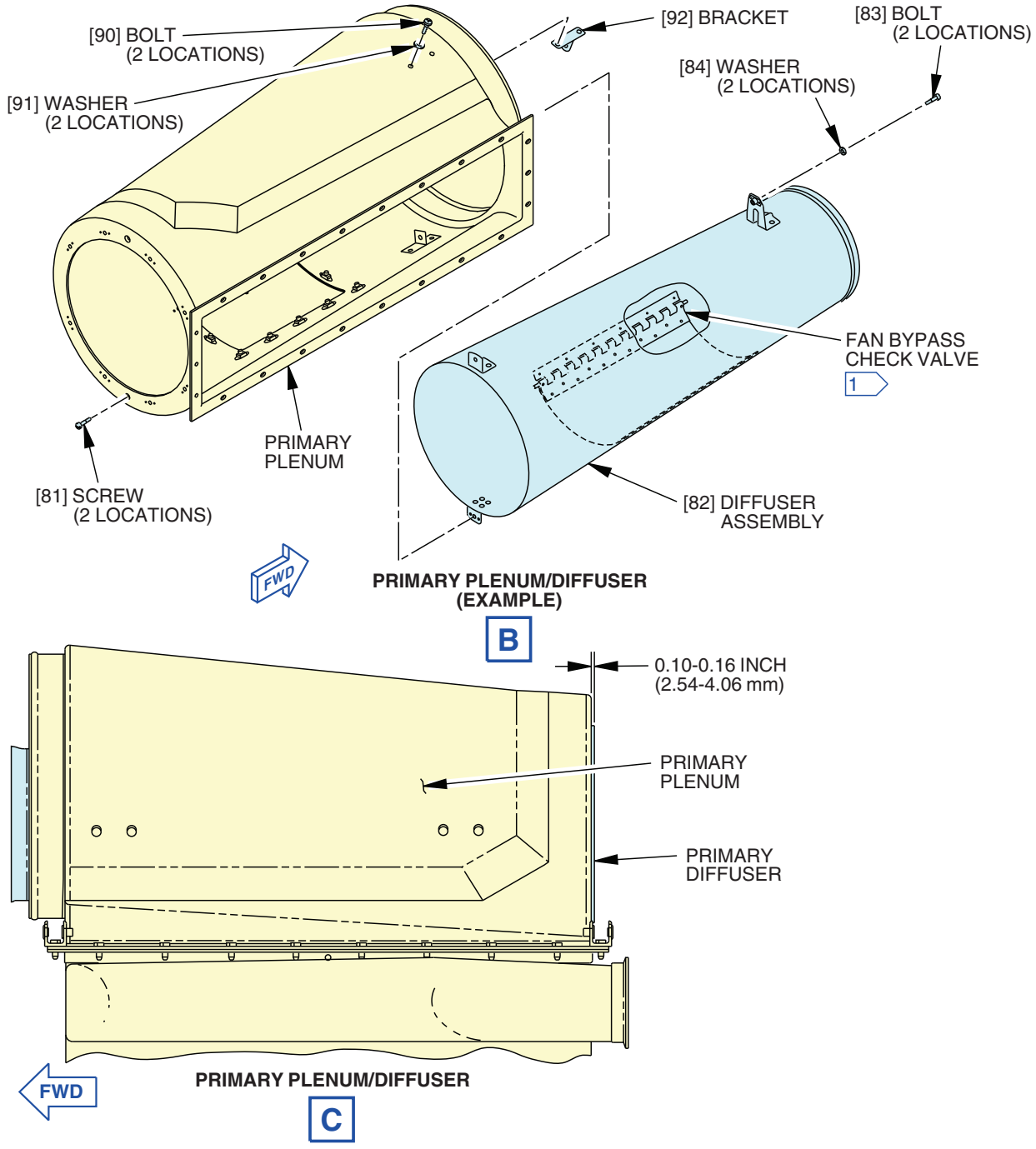
**Heat Exchanger and Plenum/Diffuser Assembly Installation
Figure 401/21-51-03-990-801 (Sheet 13 of 13)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Diffuser Assembly Installation
Figure 402/21-51-03-990-802 (Sheet 1 of 3)

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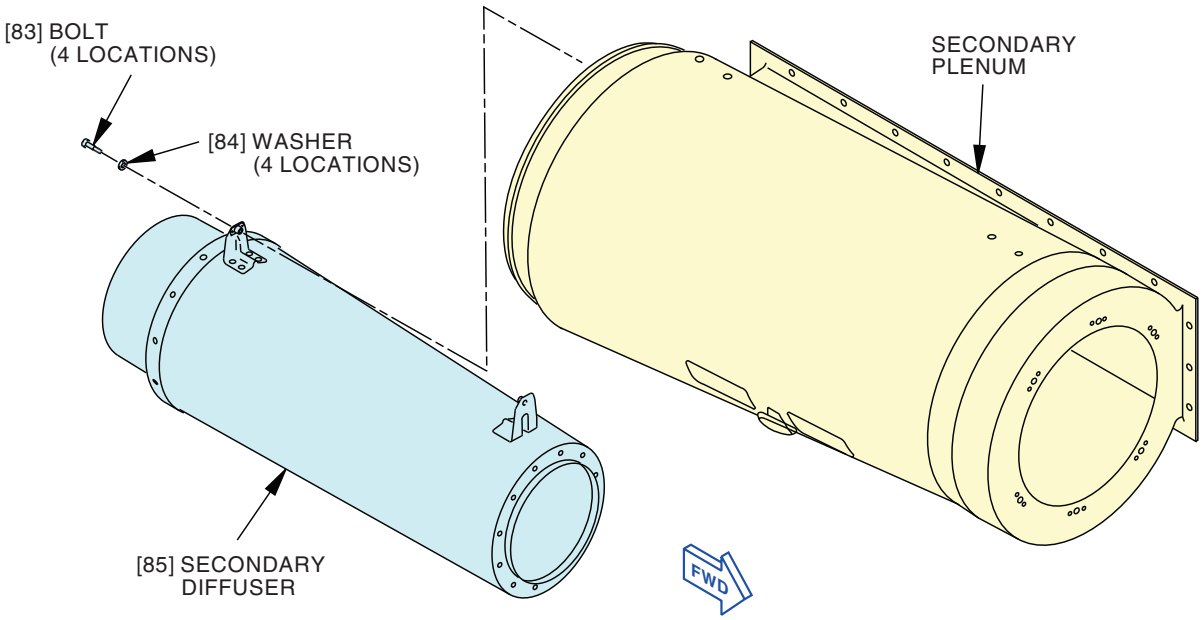


1 INSTALL PRIMARY PLENUM/DIFFUSER SO THAT FAN BYPASS CHECK VALVE IS AT BOTTOM OF DIFFUSER WHEN PLENUM/DIFFUSER IS INSTALLED ON PRIMARY HEAT EXCHANGER.

2411395 S00061527810_V3

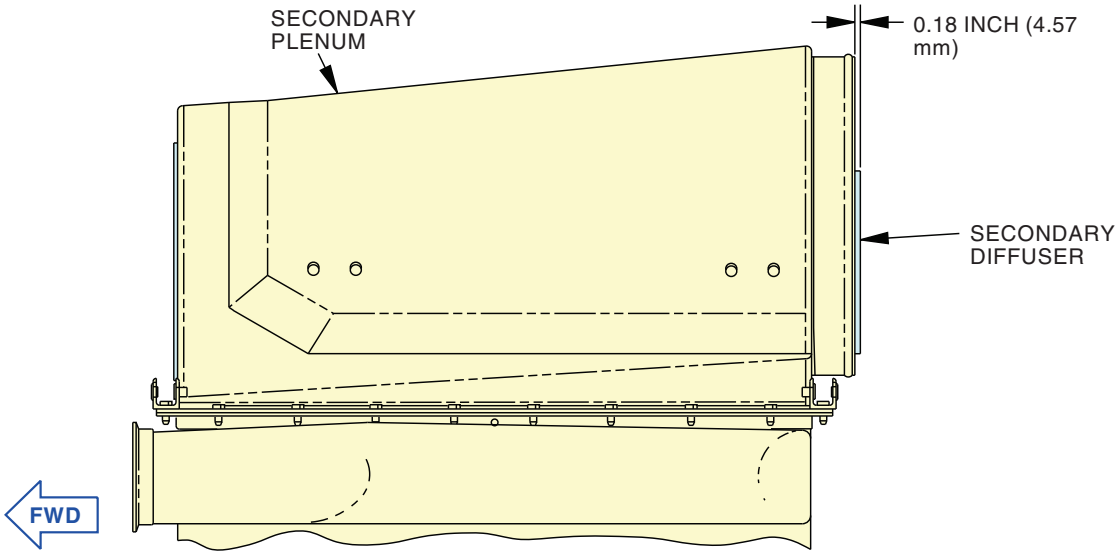
Diffuser Assembly Installation
Figure 402/21-51-03-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SECONDARY PLENUM/DIFFUSER
(EXAMPLE)

D



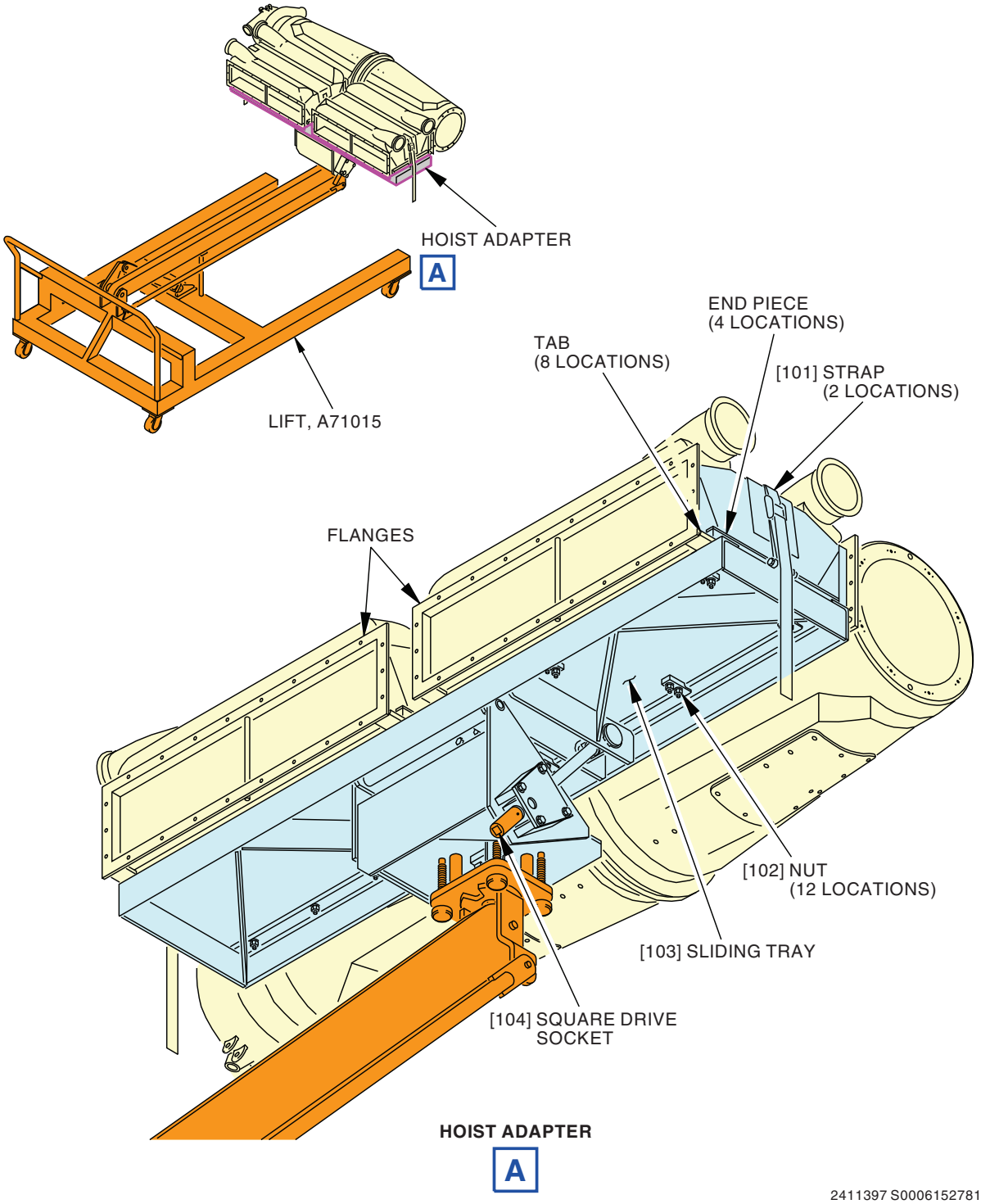
SECONDARY PLENUM/DIFFUSER

E

2411396 S00061527811_V1

Diffuser Assembly Installation
Figure 402/21-51-03-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411397 S00061527812_V1

Heat Exchanger Tool Installation
Figure 403/21-51-03-990-803

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TASK 21-51-03-400-802

3. Heat Exchanger and Plenum/Diffuser Assembly - Installation

(Figure 401, Figure 402, Figure 403)

A. General

(1) This task gives the instructions for the installation of the heat exchanger and the plenum/diffuser assembly.

(2) It is recommended that you install the primary heat exchanger, secondary heat exchanger, primary plenum/diffuser, and secondary plenum/diffuser as an assembly. As an option, you may choose to remove and install a heat exchanger separately, or with the air cycle machine as an assembly, if you determine this best suits your maintenance needs.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
47-00-00-790-801	Leak Check of the Nitrogen Generation System (P/B 601)
47-00-00-800-801	Nitrogen Generation System Ground Operation (P/B 201)
47-11-01-400-801	Air Separation Module Assembly - Installation (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)
SWPM 20-00-00	Standard Wiring Practices Manual
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536

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(Continued)

Reference	Description
SPL-1570	Fixture - Lift, Engine Accessory Part #: A71015-157 Supplier: 81205 Opt Part #: A71015-108 Supplier: 81205
SPL-1608	Jack Adapter - Installation/Removal, Air Conditioning Pack Part #: C21005-70 Supplier: 81205 Opt Part #: C21005-62 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A50035	Sealant - Dow Corning Q4-2817 Fluorosilicone Sealant	
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
C00766	Primer - Nonchromated Primer For Composites	BMS10-103 Type I
C00767	Coating - Anti-Static Coating	BMS10-21 Type III
C00852	Compound - Antiseize, Molybdenum Disulfide-Petrolatum	MIL-PRF-83483
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00649	Lubricant - O-Ring - Krytox GPL206	
D50063	Grease - Perfluoropolyether, fuel and oxygen resistant - Krytox 240AC	MIL-PRF- 27617 Type III
D50320	Lubricant - Moly-50 Anti-Seize	MIL-PRF-83483
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50637	Wiper - MicroCare Fiber Wipes	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Primary plenum/diffuser	21-51-03-03-072	PTF ALL
9	Primary heat exchanger	21-51-03-01-245	PTF ALL
		21-51-03-01-250	PTF ALL
		21-51-03-03-200	PTF ALL
20	Packing	21-51-03-02-040	PTF ALL
		21-51-03-02-200	PTF ALL
23	Secondary plenum/diffuser	21-51-03-02-090	PTF ALL
		21-51-03-02-280	PTF ALL
26	Packing	21-51-03-02-055	PTF ALL
		21-51-03-02-210	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
36	Gasket	21-51-03-02-075	PTF ALL
		21-51-03-02-230	PTF ALL
58	Secondary heat exchanger	21-51-03-02-138	PTF ALL
		21-51-03-02-400	PTF ALL
69	Gasket	21-51-03-02-134	PTF ALL
		21-51-03-02-395	PTF ALL
		21-51-03-03-195	PTF ALL
69A	Shim	21-51-03-03-196	PTF ALL
82	Diffuser assembly	21-51-03-03-325	PTF ALL
125	Tape	21-51-24-04A-275	PTF ALL
		21-51-24-04B-275	PTF ALL
126	Tape	21-51-24-04A-280	PTF ALL
		21-51-24-04B-280	PTF ALL

F. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

G. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

H. Plenum and Diffuser Assemble

SUBTASK 21-51-03-200-002

(1) Make sure that the identification (ID) plates are installed on the primary and secondary plenums, diffusers and heat exchangers.

NOTE: The ID plates on the primary and secondary plenum assemblies contain the heat exchange unit information. The configuration of the equipment is shown by the part number, dash number, series number, and change number stamped on the identification plate. It is necessary to order and install ID plates if the new plenum assemblies are supplied without ID plates.

(a) If the new assemblies will be installed, make sure that all applicable data from the original identification plates are transferred.

SUBTASK 21-51-03-420-001

(2) If it is necessary, do these steps to install the primary diffuser assembly [82] in the primary plenum (Figure 402).

NOTE: The diffuser assembly with the fan bypass check valve is installed in the primary plenum/diffuser assembly.

(a) If the bracket [92] is removed, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Apply a smooth coat of Dow Corning Q4-2817 sealant, A50035, to the mating surfaces of the bracket [92] and primary plenum.

2) Lubricate the threads of the bolts [90] with compound, C00852.

3) Loosely install the bolts [90] and washers [91] to hold the bracket [92] in its place.

CAUTION

MAKE SURE THAT THE FAN BYPASS CHECK VALVE IS INSTALLED AT THE BOTTOM OF THE DIFFUSER. THE AIR CONDITIONING PACK WILL OVERHEAT IF THE FAN BYPASS CHECK VALVE IS NOT INSTALLED AT THE BOTTOM OF THE DIFFUSER.

(b) Put the diffuser assembly [82] in its position in the primary plenum with the smaller diameter end of the diffuser assembly forward and with the fan bypass check valve at the bottom with respect to the installed position on the aircraft.

1) Make sure that the smaller diameter end is forward when you install the primary diffuser assembly [82].

(c) Lubricate the threads of the screws [81] and bolts [83] with compound, C00852.

(d) Align the forward diffuser brackets with the forward plenum brackets and loosely install the bolts [83] and washers [84].

(e) If the bolts [83] have a self-locking feature, determine the run-on torque as follows:

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

1) Use a 1/4-inch drive torque wrench to measure the torque required to turn the bolts [83] before the washers [84] contacts the bracket.

(f) Loosely install the screws [81] through the holes in the plenum into the aft brackets on the diffuser assembly [82].

(g) If the screws [81] has a self-locking feature, determine the run-on torque as follows:

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

1) Use a 1/4-inch drive torque wrench to measure the torque required to turn the screw [81] before the screw [81] head contacts the plenum.

(h) Position the diffuser assembly [82] to the dimension (View C, Figure 402).

(i) Make sure that the fan bypass check valve is installed at the bottom of the diffuser.

(j) Loosely install the bolts [90] through the holes in the plenum into the bracket [92].

1) Find the run-on torque as the bolts [90] are installed in the nutplate self-locking feature.

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

a) Use the torque wrench to measure the torque required to turn the bolt [90] before the washer [91] contacts the plenum.

(k) Tighten the bolts [83], bolts [90], and screws [81] to 22 in-lb (2.5 N·m) - 26 in-lb (2.9 N·m) plus the value of the run-on torque of the fastener.

SUBTASK 21-51-03-420-002

(3) If it is necessary, do these steps to install the secondary diffuser [85] in the secondary plenum:

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(a) Put the secondary diffuser [85] in the large end of the secondary plenum with the smaller diameter end of the diffuser assembly forward with respect to the installed position on the aircraft.

1) Make sure that the smaller diameter end is forward when you install the secondary diffuser [85].

(b) Lubricate the threads of the bolts [83] with compound, C00852.

(c) Align the secondary diffuser brackets with the secondary plenum brackets and loosely install the bolts [83] and washers [84].

(d) If the bolts [83] has a self-locking feature, determine the run-on torque as follows:

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

1) Use a 1/4-inch drive torque wrench to measure the torque required to turn the bolts [83] before the washers [84] contacts the bracket.

(e) Position the secondary diffuser [85] to the dimension (View E, Figure 402).

(f) Tighten the bolts [83] to 22 in-lb (2.5 N·m) - 26 in-lb (2.9 N·m) plus the value of the run-on torque of the bolt.

I. Heat Exchanger and Plenum/Diffuser Assemble

SUBTASK 21-51-03-420-012

(1) Do these steps to attach the secondary heat exchanger [58] to the secondary plenum/diffuser [23] or primary heat exchanger [9] to the primary plenum/diffuser [8]:

(a) Do these steps to attach the heat exchanger to the plenum:

1) Put a new gasket [69] in its position on the primary heat exchanger [9] or secondary heat exchanger [58].

2) For the primary heat exchanger [9], put the shim [69A] in its position between the plenum/diffuser and gasket [69].

3) Put the primary plenum/diffuser [8] on the primary heat exchanger [9] or secondary plenum/diffuser [23] on the secondary heat exchanger [58].

4) Install the clip assembly [65] using the bolts [63] and washers [64] as follows:

a) Lubricate the threads of the bolts [63] with compound, C00852.

b) Align the clip assembly [65] with the primary heat exchanger [9].

<1> Find the run-on torque as you install the bolts [63] in the nutplate self-locking feature.

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

<2> Use a torque wrench to measure the torque required to turn the bolt [63] before the washer [64] contacts the clip assembly [65].

c) Tighten the bolts [63] to 22 in-lb (2.5 N·m) - 26 in-lb (2.9 N·m) plus the value of the run-on torque of the bolt.

5) Install the clip assembly [66] using the bolts [63] and washers [64] as follows:

a) Lubricate the threads of the bolts [63] with compound, C00852.

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b) Find the run-on torque as you install the bolts [63] in the nutplate self-locking feature.

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

c) Use a torque wrench to measure the torque required to turn the bolt [63] before the washer [64] contacts the clip assembly [66].

d) Tighten the bolts [63] to 22 in-lb (2.5 N·m) - 26 in-lb (2.9 N·m) plus the value of the run-on torque of the bolt.

6) Install the bolts [67] and washers [68] as follows:

a) Lubricate the threads of the bolts [67] with compound, C00852.

b) Align the primary plenum/diffuser [8] with the primary heat exchanger [9].

<1> Find the run-on torque as you install the bolts [67] in the nutplate self-locking feature.

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

<2> Use a torque wrench to measure the torque required to turn the bolt [67] before the washer [68] contacts the primary plenum.

c) Tighten the bolts [67] to 22 in-lb (2.5 N·m) - 26 in-lb (2.9 N·m) plus the value of the run-on torque of the bolt.

SUBTASK 21-51-03-020-066

(2) Do these steps to join the primary plenum/diffuser [8] to the secondary plenum/diffuser [23]:

(a) Put the flexible duct [62] in its position on the end of the secondary plenum/diffuser [23].

(b) Carefully move the heat exchangers together while you align the diffusers and plenum ducts to minimize misalignment.

(c) Move the ends of the flexible duct [62] over the ends of the plenums.

(d) Install the clamps [61] that attach the flexible duct [62] to the plenum/diffusers.

1) Tighten the clamps [61] until they can not be rotated around the hose under moderate hand pressure.

2) Tighten the clamp nuts an additional four turns.

NOTE: There shall be a minimum of three threads protruding from the end of the clamp nut when installation is complete.

J. Heat Exchanger and Plenum/Diffuser Installation

SUBTASK 21-51-03-080-001

(1) Remove the covers from the duct openings.

SUBTASK 21-51-03-020-025

(2) If the hoist adapter, SPL-1608, is to be used, do these steps:

(a) Put the heat exchangers in their position on the adapter, SPL-1608.

(b) Put the straps [101] around the heat exchangers.

(c) Tighten the straps [101].

(d) Put the engine accessory lift fixture, SPL-1570, in its position below the Environmental Control System (ECS) bay.

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SUBTASK 21-51-03-020-026

WARNING IF YOU DO NOT USE THE HOIST ADAPTER, HAVE A MINIMUM OF TWO PERSONS HELP YOU INSTALL THE HEAT EXCHANGERS. THE HEAT EXCHANGERS WEIGH APPROXIMATELY 125 POUNDS, AND IF THEY FALL, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these steps to attach the heat exchangers to the lower wing beam:

(a) Examine the gasket that is bonded to the lower wing beam.

1) If the gasket is damaged replace the gasket.

NOTE: A new gasket can be made from raw material to replace the damaged gasket.

CAUTION KEEP THE HEAT EXCHANGERS LEVEL IN RELATION TO EACH OTHER. IF THE HEAT EXCHANGERS BEND AT THE FLEXIBLE DUCT, DAMAGE TO THE DIFFUSERS AND THE FLEXIBLE DUCT CAN OCCUR.

(b) Carefully raise the primary heat exchanger [9] and secondary heat exchanger [58] to their position in the ECS bay.

1) Install a new gasket [36] on the air cycle machine [39].

2) Align the holes in the gasket [36] with the holes on the plenum.

(c) Install the bolts [57] that attach the primary heat exchanger [9] and secondary heat exchanger [58] to the lower wing beam.

(d) Install the bolts [59] and washers [60].

(e) Apply a light layer of compound, C00852, to the threads of the bolts [37].

(f) Install the bolts [37] and washers [38] that connect the air cycle machine [39] to the secondary plenum/diffuser [23].

1) Tighten the bolts [37] to 40 in-lb (4.5 N·m) - 45 in-lb (5.1 N·m).

SUBTASK 21-51-03-020-027

(4) If the hoist adapter, SPL-1608, is being used, do these steps:

(a) Release the straps [101] on the adapter, SPL-1608.

(b) Lower the engine accessory lift fixture, SPL-1570, and remove it from the ECS bay.

SUBTASK 21-51-03-820-001

(5) Do these steps to install the tie rod assemblies [89] on the forward and aft ends of the primary heat exchanger [9] and secondary heat exchanger [58]:

(a) Install the bolt [54], washer [55], and bushing [56] on each of the four tie rod assemblies.

(b) Adjust the tie rod assemblies to align the duct connections on the heat exchangers and the air cycle machine with the ducts in the ECS bay.

(c) Install the lock pins on the tie rods.

SUBTASK 21-51-03-420-004

(6) Do these steps to install the duct [108]:

(a) Put the duct [108] in its position.

(b) Install the clamp [109] and clamp [110].

1) Tighten the clamp [109] and clamp [110] to 55 in-lb (6.2 N·m) - 60 in-lb (6.8 N·m).

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SUBTASK 21-51-03-020-028

(7) Install the clamps [1] for the primary heat exchanger [9] outlet and bleed air inlet ducts.

(a) Tighten the nut on the clamps [1] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

SUBTASK 21-51-03-020-029

(8) Install the clamp [21] on the outboard duct of the secondary heat exchanger [58].

(a) Tighten the nut on the clamp [21] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

SUBTASK 21-51-03-410-001

(9) To install the compressor outlet duct [29], do these steps:

(a) Apply a light coat of Krytox GPL206 lubricant, D00649, or Krytox 240AC perfluoropolyether grease, D50063, to the new packing [20] and new packings [26].

(b) Install the packings [26] on the duct ends.

(c) Install the packing [20] at the secondary heat exchanger [58].

(d) Put the compressor outlet duct [29] into its position.

(e) Apply compound, D50320, to the bolt threads of the coupling [19].

(f) Install the coupling [19] at the secondary heat exchanger [58].

NOTE: Do not tighten the clamp fully.

(g) Move the sleeve [25] over the outlet duct connection.

(h) Install the clamp [24] over the outlet duct connection.

(i) Tighten the nut on the coupling [19] to 45 in-lb (5.1 N·m) - 50 in-lb (5.6 N·m).

(j) With hand force, carefully push the compressor outlet duct [29] in the direction of the adjacent mix muff duct until the compressor outlet duct [29] is at the position of maximum flex.

(k) Make sure that there is a minimum clearance of 0.20 in. (5.1 mm) to the adjacent mix muff duct when the compressor outlet duct [29] is at the position of maximum flex.

(l) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the flex hose [105].

(m) Connect the flex hose [105] to the duct.

(n) Install the electrical connector [18] for the ram air control sensor.

(o) Install the electrical connector [22] for the compressor discharge overheat switch.

SUBTASK 21-51-03-020-033

(10) Install the ram air exhaust elbow [2] to the primary plenum/diffuser [8], do these steps (TASK 21-51-24-400-801):

(a) Examine the gasket [3].

1) If it is necessary, replace the gasket [3].

(b) Put the ram air exhaust elbow [2] and flexible duct [114] in position.

(c) Install the bolts [6] and washers [7].

(d) Install the bonding jumper [117], bolt [115], washer [116], and washer [118] with the countersink side of the washer [116] under the head of the bolt [115].

1) Measure the bonding resistance between the bonding jumper [117] terminal and anti-static coating on the ram air exhaust elbow [2] (SWPM 20-00-00).

a) Use an digital/analog multimeter, COM-1793.

b) Make sure that the bonding resistance is 300,000 ohms or less.

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2) Apply primer, C00766, on the exposed anti-static primer area around the ground stud.

(e) If it is removed, install the ram air exhaust duct [107].

(f) Install the flexible duct [114] with clamps [4].

SUBTASK 21-51-03-020-034

(11) Do these steps to install the heat exchanger inlet duct [15]:

(a) Examine the tape [125] and tape [126] on the flange of the heat exchanger inlet duct [15] for these condition:

1) Missing or damaged sections

2) Presence of oil or grease

3) Dirt or debris.

(b) Remove any damaged or unserviceable section of tape.

WARNING

KEEP THE SOLVENTS AWAY FROM SOURCES OF HEAT, FIRE OR SPARKS. IF YOU DO NOT, AN EXPLOSION CAN OCCUR. DO NOT PERMIT THE SOLVENTS TO TOUCH YOUR SKIN, EYES OR CLOTHING. DO NOT BREATHE THE VAPORS. WEAR PROTECTIVE CLOTHING AND EYE PROTECTION. MAKE SURE THERE IS A GOOD FLOW OF CLEAN AIR. IN A CONFINED SPACE, USE MECHANICAL VENTILATION OR RESPIRATORY PROTECTION. IF YOU DO NOT OBEY, INJURY TO PERSONS CAN OCCUR.

1) Clean the section of the duct flange where tape is to be installed as follows:

a) Remove any fragments of tape from the flange surface where the new tape will be applied.

b) Clean the flange surface with cotton wiper, G00034, wet with solvent, B00083, until surface is clean.

c) Dry the cleaned surfaces with cotton wiper, G00034, until all traces of the solvent are removed.

2) Install a new section of the tape [125] on the duct flange.

a) Cut the tape [125] to the required length.

b) Install the tape [125] on the flange surface.

<1> Do not stretch or wrinkle the tape [125] as you place it on the flange.

c) Cut holes in the tape [125] at the bolt locations.

3) Install a new section of the tape [126] on the duct flange.

a) Cut the tape [126] to the required length.

b) Install the tape [126] on the flange surface.

<1> Do not stretch or wrinkle the tape [126] as you place it on the flange.

<2> Make sure that you install the tape [126] on the upper flange [127] so that it wraps over the top of the flange and onto the back surface of the flange.

(c) Put the heat exchanger inlet duct [15] in its position with the upper flange under the retainer on the lower wing beam.

(d) Install the bolts [16] and washers [17].

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(e) To install the bolt [123] and washer [124], do these steps:

1) Use cotton wiper, G00034, moist in solvent, B00148, to clean the mating surfaces of the heat exchanger inlet duct [15], bolt [123], and washer [124].

2) Apply coating, C00767, continuously from the inside of the countersink onto the heat exchanger inlet duct [15] surface.

a) Make sure that the new anti-static coating overlapping the original anti-static coating by a minimum of 0.10 in. (2.54 mm).

b) Let coating, C00767, air dry for a minimum of 20 minutes.

3) Install the bolt [123] and washer [124].

4) Measure the electrical bonding resistance between the bolt [123] and anti-static coating of the heat exchanger inlet duct [15] (SWPM 20-20-00).

a) Use a digital/analog multimeter, COM-1793.

b) Make sure that the electrical bonding resistance is 300,000 ohm or less.

c) The measurement point on the duct surface must be more than 0.25 in. (6.35 mm) away from the bolt [123].

d) If it is necessary, remove a small area of the surface finish to get access to the anti-static coating of the heat exchanger inlet duct [15].

e) If the resistance measurement is not satisfactory, disassemble the bonding connection, clean the mating surfaces, and repeat the bonding connection installation.

f) If it is necessary, put a thin coating of primer, C00766, on the surface of the heat exchanger inlet duct [15] after the measurement is completed.

(f) Move the flexible hose on to heat exchanger inlet duct [15].

(g) Tighten the clamp [14].

1) Make sure that the clamp [14] is not installed on the duct bead.

2) Keep a 0.12 in. (3.0 mm) minimum distance between the clamp and edge of flexible hose.

3) Make sure that the T-bolt of the clamp [14] points in the outboard direction (View B-B, Figure 401).

(h) Connect the flexible hose [71] from the Nitrogen Generation System (NGS) heat exchanger to the heat exchanger inlet duct [15].

1) Slide the flexible hose [71] over the spud on the top of the heat exchanger inlet duct [15].

2) Tighten the hose clamps [70] to 13 in-lb (1.5 N·m) - 18 in-lb (2.0 N·m).

(i) If applicable, clean the mating surfaces of the bolts [111], washers [112], and the bonding jumpers [113].

1) Use solvent, B00148, or solvent, B00062, to clean the mating surfaces.

2) Remove all traces of the solvent with MicroCare fiber wipes, G50637.

(j) If applicable, install the bolts [111], washers [112], and the bonding jumpers [113] that hold the heat exchanger inlet duct to the airplane structure.

1) Make sure that the bolts [111] are the same length as the ones that were removed.

2) Install one washer [112] between the bonding jumper [113] and the bolt head.

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a) Make sure that the washer [112] is installed with the countersink next to the bolt head.

3) Install one washer [112] between the inlet duct and the bonding jumper [113].

(k) Measure the electrical bonding resistance between the terminal of the bonding jumper [113] and anti-static coating of the heat exchanger inlet duct [15] (SWPM 20-20-00).

1) Use a digital/analog multimeter, COM-1793.

2) Make sure that the electrical bonding resistance is 300,000 ohms or less.

a) The measurement point on the duct surface should be more than 0.25 in. (6.35 mm) away from the bolt [111].

b) If it is necessary, remove a small area of the surface finish to gain access to the anti-static coating of the heat exchanger inlet duct [15].

c) If the resistance measurement is not satisfactory, then disassemble the bonding connection, clean the mating surfaces, and repeat the bonding connection installation.

3) If it is necessary, put a thin coating of primer, C00766, on the surface of the heat exchanger inlet duct [15] after the measurement is completed.

(l) If it was removed, install or connect the Nitrogen-Enriched Air Distribution System (NEADS) tube (TASK 47-11-01-400-801).

1) Do this task: Nitrogen Generation System Ground Operation, TASK 47-00-00-800-801.

2) Do this task: Leak Check of the Nitrogen Generation System, TASK 47-00-00-790-801.

SUBTASK 21-51-03-020-035

(12) If it was removed, install the outer seal [12] for the ECS access door:

(a) Put the seal [12] and retainer [13] in their position on the lower wing beam.

(b) Install the bolts [10] and nutplates.

SUBTASK 21-51-03-860-003

(13) Remove the DO NOT OPERATE tags from these switches, on the P5-10 air conditioning panel:

(a) The L PACK and R PACK switches

(b) The BLEED 1 and BLEED 2 switches

(c) The BLEED APU switch.

SUBTASK 21-51-03-860-004

(14) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

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SUBTASK 21-51-03-860-029

(15) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-03-860-031

(16) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

K. Heat Exchanger and Plenum/Diffuser Installation Test

SUBTASK 21-51-03-860-006

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-03-860-007

(2) Put the applicable L PACK or R PACK switch, on the P5-10 air conditioning panel, to the AUTO position.

SUBTASK 21-51-03-790-001

(3) Do a soap bubble test of all the applicable duct joints.

NOTE: No leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches to the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the applicable clamps.

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps to the applicable installation torque value:

a) Tighten the clamp [109] and clamp [110] to 55 in-lb (6.2 N·m) - 60 in-lb (6.8 N·m).

b) Tighten the nut on clamp [1], clamp [21], and clamp [40] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

c) Tighten the nut on coupling [19] to 45 in-lb (5.1 N·m) - 50 in-lb (5.6 N·m).

d) Tighten the hose clamps [70] to 13 in-lb (1.5 N·m) - 18 in-lb (2.0 N·m).

5) Put the L PACK and R PACK switches to the AUTO position.

6) Make sure that the leak stops.

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L. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-03-860-008

(1) Put the applicable L PACK or R PACK switches to the OFF position.

SUBTASK 21-51-03-860-009

(2) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-03-410-002

(3) If the heat exchangers for the left pack were removed, do these steps:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-03-410-003

(4) If the heat exchangers for the right pack were removed, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

——— END OF TASK ———

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FAN BYPASS CHECK VALVE DOOR - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) An operational check of the fan bypass check valve door.

TASK 21-51-03-710-802

2. Fan Bypass Check Valve Door - Operational Check

(Figure 601)

A. General

(1) This procedure has instructions to make sure that the fan bypass check valve door is operational.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-03-000-803	Heat Exchanger and Plenum/Diffuser Assembly - Removal (P/B 401)
21-51-03-400-802	Heat Exchanger and Plenum/Diffuser Assembly - Installation (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2
STD-858	Tag - DO NOT OPERATE

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D. Consumable Materials

Reference	Description	Specification
A50035	Sealant - Dow Corning Q4-2817 Fluorosilicone Sealant	
A50613	Adhesive - Sealant, Silicone RTV, Noncorrosive, High Strength, Gray	MIL-A-46146 Group II Type I
B50420	Solvent - Cleaner, Desoclean 45	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Gasket	21-51-03-03-145	PTF ALL

F. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

G. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

H. Prepare for the Fan Bypass Check Valve Door Operational Check

SUBTASK 21-51-03-040-003

(1) If necessary remove pneumatic pressure, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-03-860-024

(2) Do these steps, on the P5-10 air conditioning panel (found on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach the DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-03-860-025

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

SUBTASK 21-51-03-010-016

(4) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-03-010-025

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To open the right ECS access doors do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

I. Fan Bypass Check Valve Door Operational Check

SUBTASK 21-51-03-010-018

(1) Do the steps that follow to remove the plenum/diffuser access panel [3]:

(a) Remove the bolts [1] and washers [2] from the plenum/diffuser access panel [3] and gasket [4].

(b) If necessary, remove the bonding jumper [10] from the plenum/diffuser access panel [3].

NOTE: The bonding jumper removal is not necessary if the task can be completed with the access panel moved to the side.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

1) Remove the sealant that covers the jumper terminal and the head of the bolt [7] on the plenum/diffuser access panel [3] (SWPM 20-20-00 and SWPM 20-20-10).

2) Remove the bolt [7], washer [8], washer [12], and nut [13] that attaches the bonding jumper [10] to the plenum/diffuser access panel [3].

(c) Remove the plenum/diffuser access panel [3] from the plenum diffuser.

SUBTASK 21-51-03-020-084

(2) If the gasket [4] is damaged, do these steps to remove it:

(a) Use a sealant removal tool handle, COM-4746, to remove the gasket [4] from the plenum/diffuser access panel [3].

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WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

(b) Use solvent, B50420, to remove remaining adhesive from the plenum/diffuser access panel [3].

SUBTASK 21-51-03-200-001

(3) Make sure that the fan bypass check valve door [5] is installed and closed.

NOTE: Gravity force keeps the door closed when the system is off.

SUBTASK 21-51-03-710-001

(4) Make sure that the fan bypass check valve door [5] moves freely, do the steps that follow:

(a) Lightly push the fan bypass check valve door [5] up until it hits the fan bypass check valve door stop.

(b) Operate the door through 3 cycles and examine the door for free movement.

SUBTASK 21-51-03-100-005

(5) If you see dirt or contamination, clean the dirt or contamination with soap and water where the fan bypass check valve door [5] and primary diffuser assembly [6] touch.

SUBTASK 21-51-03-400-001

(6) If the fan bypass check valve door [5] is not installed or does not operate correctly, replace the primary diffuser assembly [6] with a serviceable unit, do these tasks:

- Heat Exchanger and Plenum/Diffuser Assembly - Removal, TASK 21-51-03-000-803
- Heat Exchanger and Plenum/Diffuser Assembly - Installation, TASK 21-51-03-400-802.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-03-420-014

(1) If removed, install the bonding jumper [10] to the plenum/diffuser access panel [3].

(a) Make sure that the bonding jumper [10], bolt [7], washer [8], washer [12], nut [13], and plenum/diffuser access panel [3] are clean.

1) If installed, make sure that the nutplate is clean.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(b) Put the bonding jumper [10] in its position on the plenum/diffuser access panel [3].

(c) Install the bolt [7], washer [8], washer [12], and nut [13].

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(d) Tighten the bolt [7] to 22 in-lb (2.49 N·m) - 26 in-lb (2.94 N·m).

(e) Measure the electrical bonding resistance between the bolt [7] and the plenum/diffuser access panel [3].

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.3 Mohm (300 000 ohm) or less.

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(f) Apply a coat of Dow Corning Q4-2817 sealant, A50035, to cover the jumper terminal and the head of the bolt [7].

NOTE: The sealant is optional if the plenum was built before 2022/2023, and did not originally leave Honeywell production with the sealant applied.

SUBTASK 21-51-03-420-015

(2) Do the steps that follow to install the plenum/diffuser access panel [3]:

(a) If the gasket [4] was removed, do these steps:

WARNING DO NOT GET ADHESIVE IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM ADHESIVE. PUT ON GOGGLES, AND GLOVES WHEN YOU USE ADHESIVE. ADHESIVE IS POISONOUS AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING KEEP ADHESIVE AWAY FROM THE FLAME AND HEAT. ADHESIVE IS FLAMMABLE AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

1) Apply a 1.00 in. (25.40 mm) wide strip of gray adhesive sealant, A50613, to the inner surface of the plenum/diffuser access panel [3].
2) Put the new gasket [4] on the adhesive.
3) Let the adhesive cure.
4) If the gasket [4] is over the plenum/diffuser access panel [3] edges, cut the outer edges of the gasket [4] to the contour of the plenum/diffuser access panel [3].
5) Use the holes in the plenum/diffuser access panel [3] to locate the 0.247 in. (6.27 mm) - 0.258 in. (6.55 mm) diameter holes in the gasket [4].

(b) Put the plenum/diffuser access panel [3] in its position.

(c) Install the washers [2] and bolts [1].

(d) Tighten the bolts [1] to 22 in-lb (2.5 N·m) - 26 in-lb (2.9 N·m).

SUBTASK 21-51-03-410-018

(3) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-03-410-028

(4) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

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SUBTASK 21-51-03-860-026

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-03-860-027

(6) Do these steps, on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

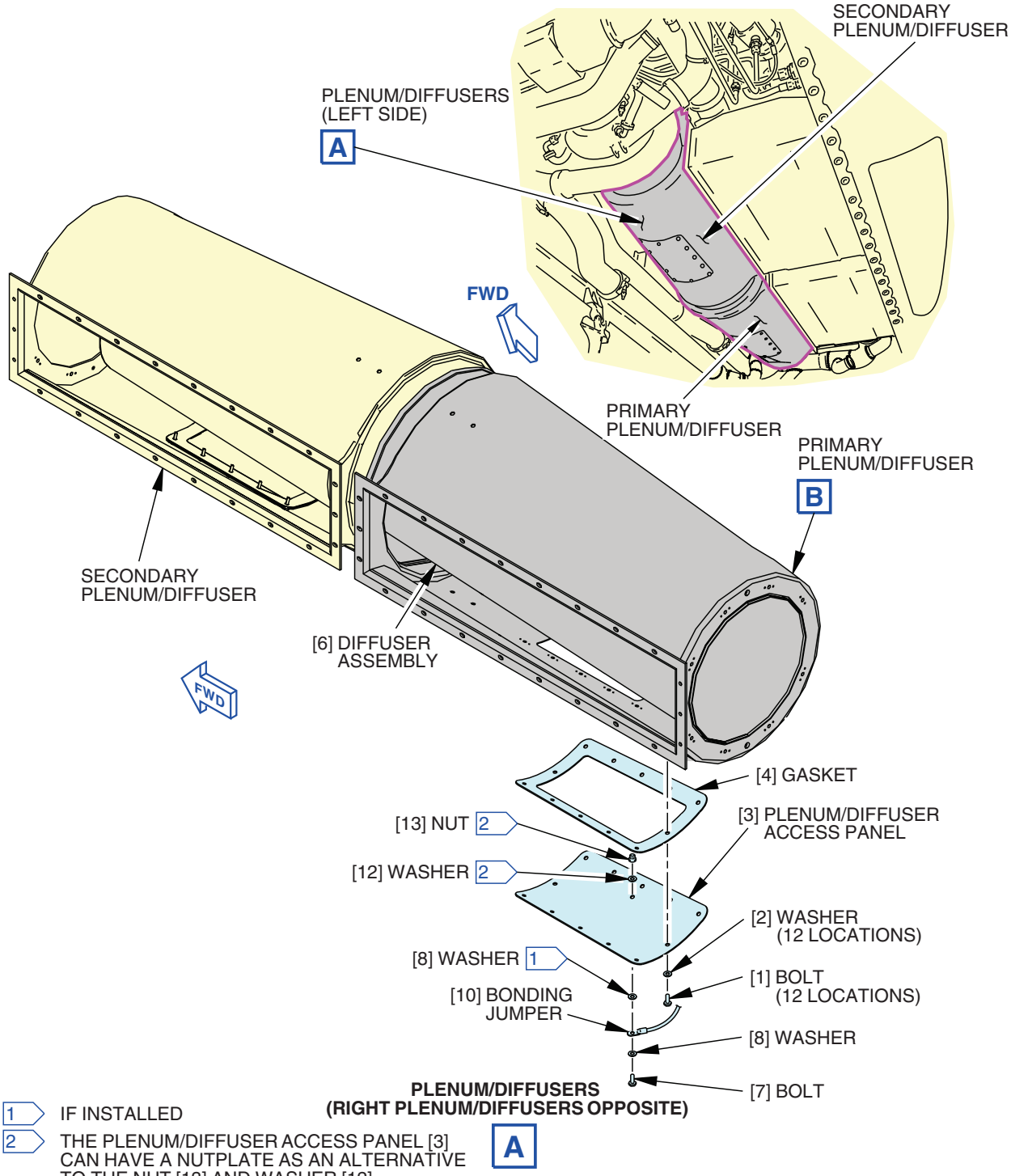
(a) Remove the DO NOT OPERATE tags, STD-858, from the L and R Pack switches.

(b) Remove the DO NOT OPERATE tags, STD-858, from the BLEED 1 and 2 switches.

(c) Remove the DO NOT OPERATE tag, STD-858, from the BLEED APU switch.

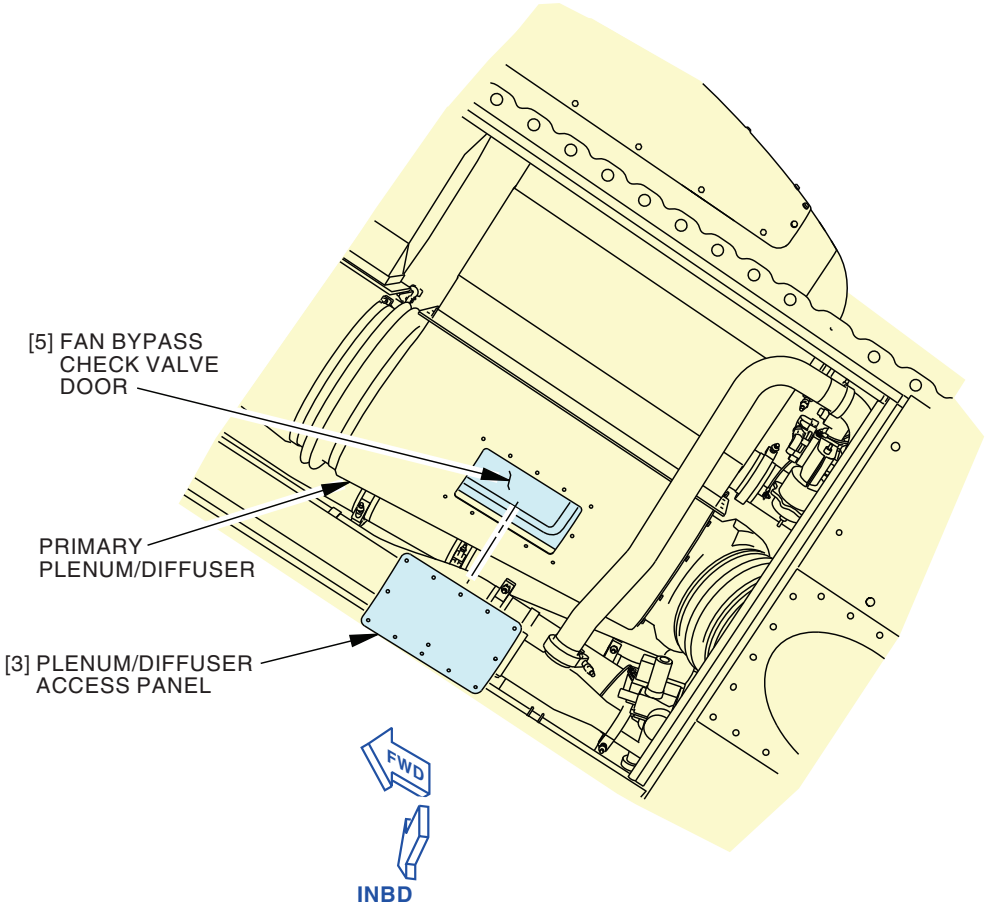
——— END OF TASK ———

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Fan Bypass Check Valve Door Operational Check
Figure 601/21-51-03-990-812 (Sheet 1 of 2)

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Fan Bypass Check Valve Door Operational Check
Figure 601/21-51-03-990-812 (Sheet 2 of 2)

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HEAT EXCHANGER AND PLENUM/DIFFUSER ASSEMBLY - CLEANING/PAINTING

1. General

A. This procedure has these tasks to clean the primary and secondary heat exchangers:

NOTE: All of these procedures can be accomplished solely to meet the scheduled maintenance task requirement (i.e. each procedure is an alternate to the other).

(1) A heat exchanger and plenum/diffuser assembly on-wing cleaning procedure with backflush equipment that uses air and water.

(2) Heat exchangers on-wing cleaning with Soap-Water-Air.

(3) Alternatively to on-wing cleaning, replace heat exchanger with new or shop-cleaned (per vendor CMM) heat exchanger.

TASK 21-51-03-000-804

2. Heat Exchanger and Plenum/Diffuser Assembly Cleaning

(Figure 701)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Boeing notes that this procedure for cleaning with backflush equipment is not effective for clearing blocked heat exchangers and less effective for dirty heat exchangers. To effectively clear blocked heat exchangers, do this task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, TASK 21-51-03-160-805.

(2) You may use either a ground air source or Auxiliary Power Unit (APU) bleed air to do this procedure.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
21-51-03-160-805	Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air (P/B 201)
21-51-24-000-801	Ram Air Ducts Removal (P/B 401)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
52-48-42-000-802	Environmental Control Systems (ECS) High Pressure Door Removal (P/B 401)
52-48-42-400-802	Environmental Control System (ECS) High Pressure Door Installation (P/B 401)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-1607	Equipment - Backflush, A/C Pack Heat Exchanger Part #: C21003-160 Supplier: 81205 Part #: C21003-161 Supplier: 81205 Opt Part #: C21003-133 Supplier: 81205 Opt Part #: C21003-134 Supplier: 81205 Opt Part #: C21003-155 Supplier: 81205 Opt Part #: C21003-156 Supplier: 81205 Opt Part #: C21003-80 Supplier: 81205 Opt Part #: C21003-81 Supplier: 81205 Opt Part #: C21003-82 Supplier: 81205
SPL-2462	Deactivator - Check Valve Part #: C21006-7 Supplier: 81205 Opt Part #: C21006-1 Supplier: 81205 Opt Part #: PF80-012-500 Supplier: 053H3
SPL-13921	Backflush Equipment - Bleed Air Hose Part #: C21003-166 Supplier: 81205 Opt Part #: C21003-135 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-3926	Water Source - Cold, Regulated, 0 to 60 PSIG

D. Consumable Materials

Reference	Description	Specification
A50035	Sealant - Dow Corning Q4-2817 Fluorosilicone Sealant	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Gasket	21-51-24-04A-190	PTF ALL
		21-51-24-04B-190	PTF ALL

F. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

H. Prepare for the Cleaning

SUBTASK 21-51-03-040-002

(1) Set the L PACK and R PACK switches, on the P5-10 air conditioning panel, to the OFF position.

(a) Put DO NOT OPERATE tags, STD-858, on the switches.

SUBTASK 21-51-03-860-011

(2) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel, to the OPEN position.

SUBTASK 21-51-03-010-004

(3) To get access to the heat exchangers for the left pack, do these steps:

(TASK 06-41-00-800-801)

WARNING OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-03-010-005

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the heat exchangers for the right pack, do these steps:

(TASK 06-41-00-800-801)

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-03-010-006

(5) Remove the bolts [4] and washers [3] that attach the ram air inlet duct access panel [2] to the ram air inlet duct.

SUBTASK 21-51-03-010-007

(6) Remove the ram air inlet duct access panel [2].

NOTE: Removal of the access panel is necessary so that the cleaning water will not accumulate in the ram air inlet duct.

NOTE: There is a gasket bonded to the access panel. The gasket can stay in place and reused if it is not damaged.

SUBTASK 21-51-03-010-008

(7) Remove the bolts [8] and washers [7] that attach each of the two plenum/diffuser access panels [6] and gaskets [5] to the plenum/diffuser assemblies for the primary and secondary heat exchangers.

NOTE: Each access panel is installed with a gasket. The gaskets can be reused if there is no damage.

SUBTASK 21-51-03-020-036

(8) If necessary, remove the bonding jumpers [11] from the plenum/diffuser access panels [6].

NOTE: The bonding jumper removal is not necessary if the task can be completed with the access panel moved to the side.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(a) Remove the sealant that covers the jumper terminals and the head of the bolts [9] on the plenum/diffuser access panels [6] (SWPM 20-20-00 and SWPM 20-20-10).

(b) Remove the bolts [9], washers [10], washers [12], and nuts [13] that attach the bonding jumpers [11] to the plenum/diffuser access panels [6].

SUBTASK 21-51-03-010-009

(9) Remove the plenum/diffuser access panels [6] from the plenum/diffuser.

SUBTASK 21-51-03-480-002

(10) Install a backflusher, SPL-1607, heat exchanger cleaning heads, do these steps:

(a) Install the backflush units (cleaning heads) as follows:

1) Use the applicable backflush unit to clean the left pack primary heat exchanger or the right pack secondary heat exchanger.

2) Use the applicable backflush unit to clean the left pack secondary heat exchanger or the right pack primary heat exchanger.

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CAUTION

DO NOT APPLY FORCE TO THE PLENUM/DIFFUSER ASSEMBLY WHEN YOU INSTALL THE CLEANING HEADS OF THE HEAT EXCHANGER. YOU CAN CAUSE DAMAGE TO THE PLENUM/DIFFUSER ASSEMBLY.

(b) Carefully insert the supply tubes of the cleaning head into the access holes in the plenum/diffuser assembly so that the ends of the supply tubes point toward the exit of the heat exchanger.

CAUTION

DO NOT TIGHTEN THE SCREWS ON THE CLEANING HEAD TOO MUCH. TOO MUCH TORQUE WILL CAUSE DAMAGE TO THE PLENUM/DIFFUSER.

(c) Carefully tighten the screws on the cleaning head so that the rubber seals of the cleaning head just make good contact with the plenum/diffuser.

(d) Do the above steps again to install the cleaning head for the other heat exchanger.

SUBTASK 21-51-03-480-003

(11) If the backflusher, SPL-1607, has a water supply valve and air supply valve, make sure that they are closed.

NOTE: Not all configurations of the backflush equipment have the water supply valve and air supply valve.

SUBTASK 21-51-03-480-004

(12) Install plastic sheeting to seal off the plenum/diffuser as follows:

(a) Remove the two clamps that attach the flexible duct between the elbow duct at the aft end of the plenum/diffuser and the ram air exhaust duct (TASK 21-51-24-000-801).

(b) Remove the flexible duct.

(c) Install a piece of plastic sheeting to cover the opening of the elbow duct that is attached to the aft end of the plenum/diffuser.

(d) Secure the plastic sheeting to the end of the elbow duct with the clamps that are used to install the flexible duct.

SUBTASK 21-51-03-940-001

(13) Connect the water line of the 0 to 60 PSIG regulated cold water source, STD-3926, to the water supply hose of the backflush equipment.

(a) Do not start the flow of water at this time.

SUBTASK 21-51-03-941-001

(14) Do these steps if you use the airplane APU pneumatic air to back flush the heat exchangers:

(a) Make sure that this access panel is open:

(TASK 52-48-42-000-802)

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Look at the dual duct pressure indicator, on the P5-10 air conditioning panel, to make sure that there is no pressure in the pneumatic system.

1) If there is pressure in the pneumatic system, remove the pressure as follows:

a) Do this step: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do the two steps that follow at the same time to prepare to supply pneumatic air to the backflush equipment:

WARNING WEAR GLOVES THAT WILL GIVE YOU PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. THE GROUND AIR CONNECTOR CAN BE VERY HOT IF THE PACKS HAVE BEEN OPERATED IMMEDIATELY BEFORE THIS PROCEDURE. YOU CAN BADLY BURN YOUR HANDS IF YOU TOUCH A HOT GROUND AIR CONNECTOR.

1) Install the deactivator, SPL-2462, in the ground pneumatic connector to hold the check valve open.

2) Connect the bleed air hose, SPL-13921, supplied with the backflush equipment to the ground pneumatic connector.

NOTE: The air hose connector holds the deactivator, SPL-2462, in position.

(d) Connect the other end of the air hose that you just connected to the ground pneumatic connector to the air inlet fitting on the backflush equipment air hose.

SUBTASK 21-51-03-941-002

(15) If you use a ground pneumatic air source to back flush the heat exchangers, do this step:

(a) Connect the ground pneumatic air supply line to the air inlet fitting on the backflush equipment.

I. Heat Exchanger and Plenum/Diffuser Assembly Cleaning

SUBTASK 21-51-03-020-037

CAUTION DO NOT USE SOAP OR A DETERGENT SOLUTION TO CLEAN THE HEAT EXCHANGERS. USE COLD WATER ONLY. SOAP OR A DETERGENT SOLUTION CAN CAUSE DAMAGE TO THE AIR CYCLE MACHINE.

(1) Use the backflusher, SPL-1607, to clean the heat exchanger.

NOTE: You must do these steps for each of the heat exchangers.

(a) Connect the air hose to the cleaning head.

(b) Connect the water supply hose to the cleaning head.

(c) If you use the airplane pneumatic system air for the air source, do this task: Supply Pressure to the Pneumatic System with the APU, TASK 36-00-00-860-803.

(d) If you use a ground pneumatic air for the air source, do the following step:

CAUTION DO NOT SUPPLY MORE THAN 30 PSIG OF GROUND AIR PRESSURE TO THE HEAT EXCHANGERS. TOO MUCH PRESSURE CAN CAUSE DAMAGE TO THE HEAT EXCHANGERS OR THE RAM AIR EXIT DUCTS.

1) Start the ground air source and adjust the pressure regulator to a maximum of 30 psig (2 kg/cm^2).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT YOU CORRECTLY ATTACH THE HOSE ASSEMBLY BEFORE YOU SUPPLY PRESSURE. WHEN YOU PUT PRESSURE ON THE HOSE, THE ASSEMBLY CAN MOVE QUICKLY. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(e) If installed, slowly open the air supply valve on the manifold of the cleaning tool.

(f) Slowly open the water supply valve on the 0 to 60 PSIG regulated cold water source, STD-3926.

(g) If installed, slowly open the water supply valve on the manifold of the cleaning tool.

(h) Clean the heat exchanger for 5 minutes.

(i) Close the water supply valve on the 0 to 60 PSIG regulated cold water source, STD-3926, and continue to supply air for an additional 2 minutes.

(j) If you use the airplane pneumatic system air, set the APU BLEED switch, on the P5-10 panel, to the OFF position.

(k) If you use a ground pneumatic air for the air source, set the valve on the ground pneumatic cart to the OFF position.

WARNING PUT ON GLOVES FOR PROTECTION FROM THE HOT WATER IN THE HOSES. HOT WATER CAN BURN YOU.

(l) Disconnect the air hose from the cleaning head.

WARNING PUT ON GLOVES FOR PROTECTION FROM THE HOT WATER IN THE HOSES. HOT WATER CAN BURN YOU.

(m) Disconnect the water supply hose from the cleaning head.

(n) Do the steps above, as necessary, for the other heat exchangers.

SUBTASK 21-51-03-020-038

(2) If not already done, close the valve at the water source.

SUBTASK 21-51-03-030-001

(3) Disconnect the water line from the 0 to 60 PSIG regulated cold water source, STD-3926, and the water supply hose of the backflush equipment.

SUBTASK 21-51-03-880-004

(4) If you used a ground air supply for the air source, do these steps:

(a) Stop the operation of the ground air source.

(b) Disconnect the ground air supply line from the air hose inlet fitting.

SUBTASK 21-51-03-880-001

(5) If you used the airplane pneumatic system air for the air source, do these steps:

(a) If not already done, set the APU BLEED switch, on the P5-10 panel, to the OFF position.

(b) If installed, open the air supply valve on the backflusher equipment.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(c) Disconnect the bleed air hose, SPL-13921, from the ground pneumatic connector and from the air hose inlet fitting.

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(d) Remove the check valve deactivator from the ground pneumatic connector.

(e) Close this access panel:

(TASK 52-48-42-400-802)

Number	Name/Location
192DR	ECS High Pressure Access Door

SUBTASK 21-51-03-020-039

(6) Remove the heat exchanger cleaning heads, do these steps:

(a) If not already done, disconnect the air hose from the cleaning head.

(b) If not already done, disconnect the water supply hose from the cleaning head.

(c) Loosen the screws on the cleaning head.

(d) Carefully remove the cleaning head.

SUBTASK 21-51-03-211-002

(7) Examine the gasket [1] and gaskets [5] for damage.

(a) Replace the gasket if there is damage or deterioration.

SUBTASK 21-51-03-410-004

(8) If removed, install the bonding jumpers [11] to the plenum/diffuser access panels [6].

(a) Make sure that the bonding jumpers [11], washers [10], bolts [9], washers [12], nuts [13], and plenum/diffuser access panels [6] are clean.

1) If installed, make sure that the nutplates are clean.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(b) Put the bonding jumpers [11] in their positions on the plenum/diffuser access panels [6].

(c) Install the washers [10], bolts [9], washers [12], and nuts [13].

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(d) Tighten the bolts [9] to 22 in-lb (2.49 N·m) - 26 in-lb (2.94 N·m).

(e) Measure the electrical bonding resistance between the bolts [9] and the plenum/diffuser access panels [6].

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.3 Mohm (300 000 ohm) or less.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Apply a coat of Dow Corning Q4-2817 sealant, A50035, to cover the jumper terminals and the heads of the bolts [9].

NOTE: The sealant is optional if the plenum was built before 2022/2023, and did not originally leave Honeywell production with the sealant applied.

SUBTASK 21-51-03-420-007

(9) Install the plenum/diffuser access panels [6] and gaskets [5] with the bolts [8] and washers [7].

(a) Tighten the bolts [8] to 22 in-lb (2 N·m) - 26 in-lb (3 N·m).

SUBTASK 21-51-03-420-008

(10) Install the ram air inlet duct access panel [2] and gasket [1] with the bolts [4] and washers [3].

SUBTASK 21-51-03-080-002

(11) Remove the plastic sheeting from the elbow duct at the aft end of the plenum/diffuser as follows:

(a) Remove the clamp that secures the plastic sheeting to the elbow duct.

(b) Remove the plastic sheeting.

(c) Install the flexible duct between the elbow duct at the aft end of the plenum/diffuser and the ram air exhaust duct (TASK 21-51-24-400-801).

(d) Install the clamps to secure the flexible duct.

(e) Torque the clamps to 13 in-lb (1.5 N·m) - 17 in-lb (1.9 N·m).

SUBTASK 21-51-03-080-009

(12) Remove the DO NOT OPERATE tag, STD-858, from the L PACK and R PACK switches.

(a) Make sure that the R PACK and L PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

SUBTASK 21-51-03-880-002

(13) Operate the air conditioning pack for five minutes (TASK 21-00-00-800-803).

NOTE: This will remove all water in the ram air ducts and the air cycle machine bearings.

SUBTASK 21-51-03-880-003

(14) Stop the air conditioning pack (TASK 21-00-00-800-804).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-03-860-012

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-03-100-001

(2) If you cleaned the left pack heat exchangers, do these steps:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-03-100-002

(3) If you cleaned the right pack heat exchangers, do these steps:

(TASK 06-41-00-800-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

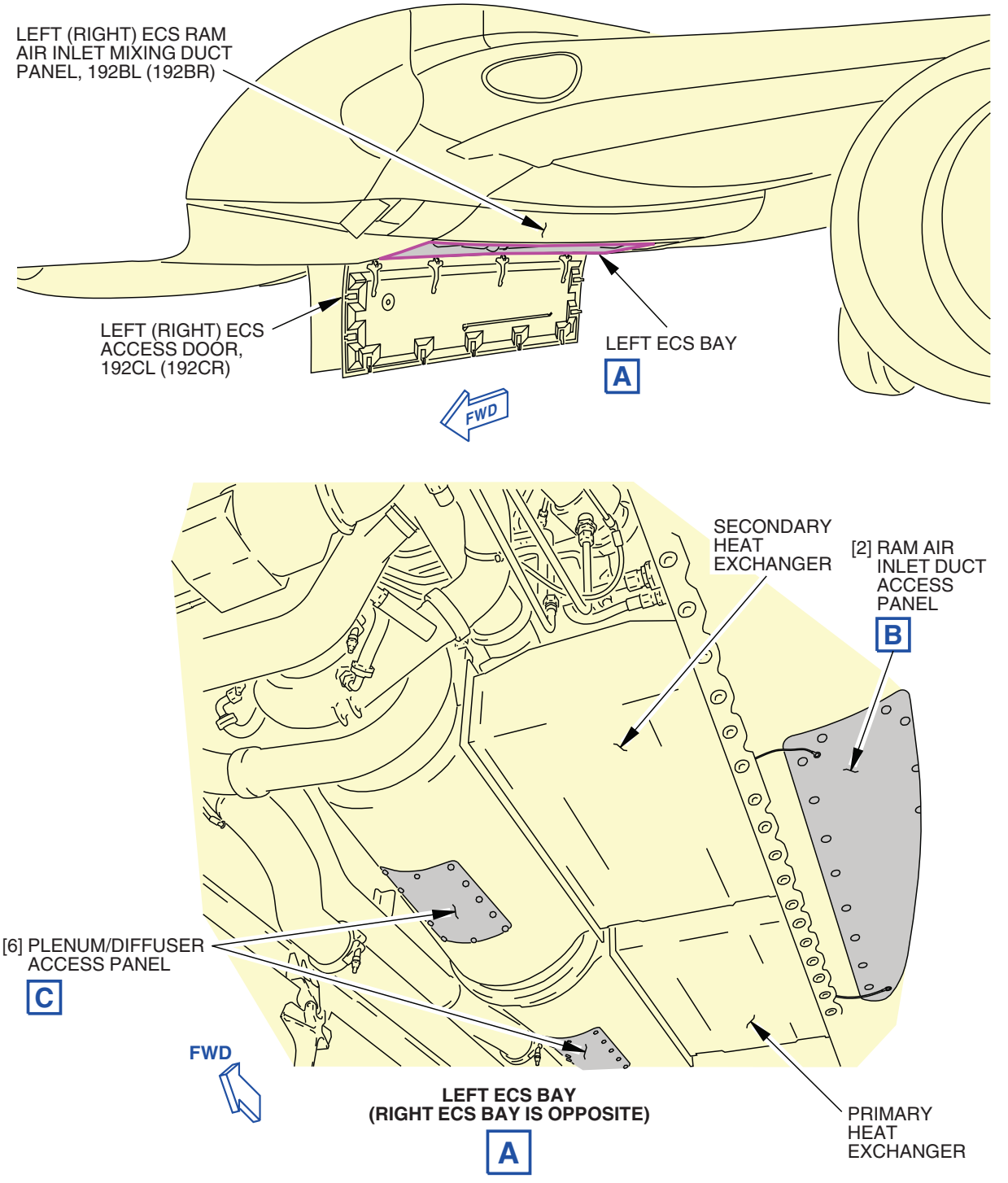
(c) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

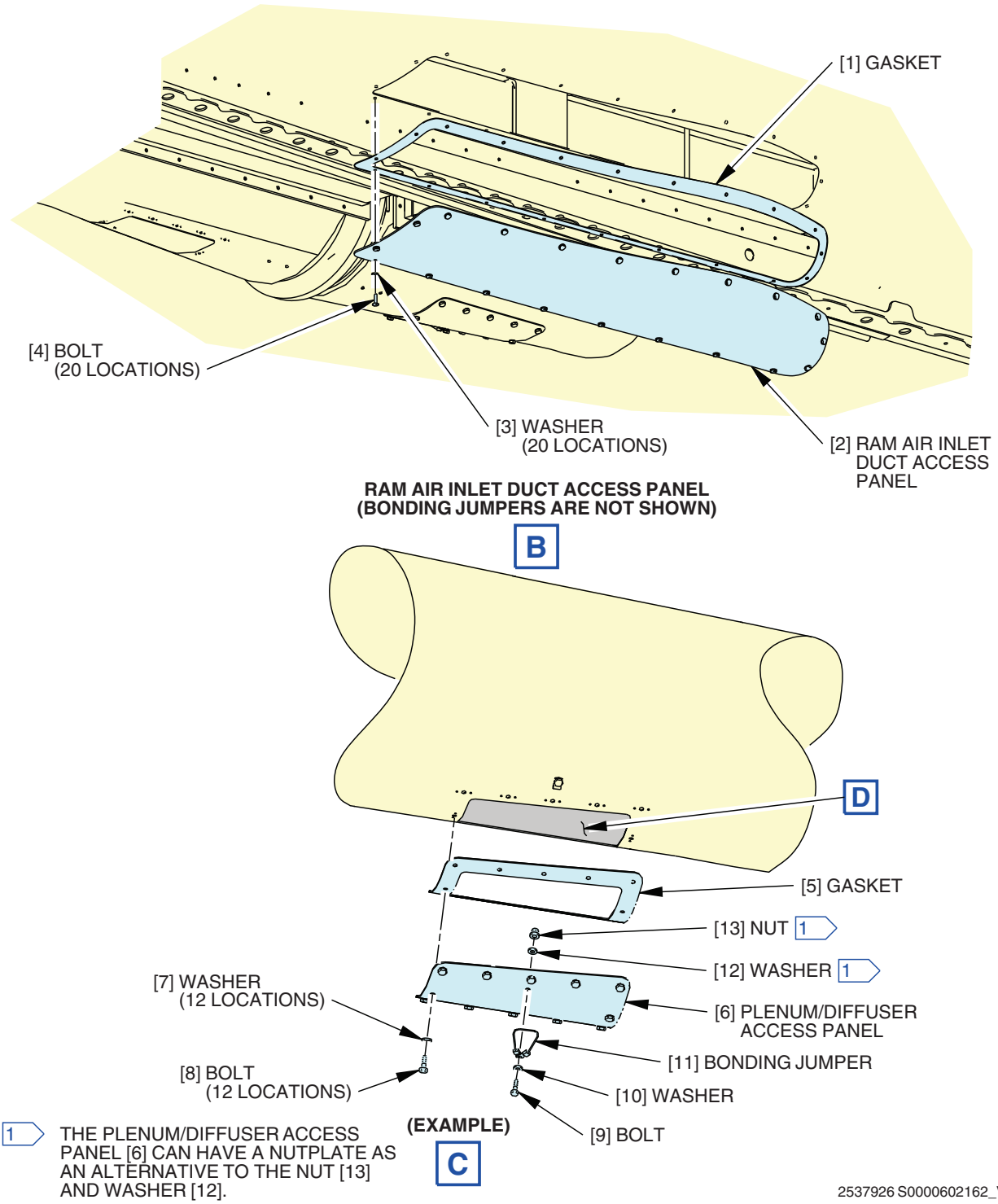
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



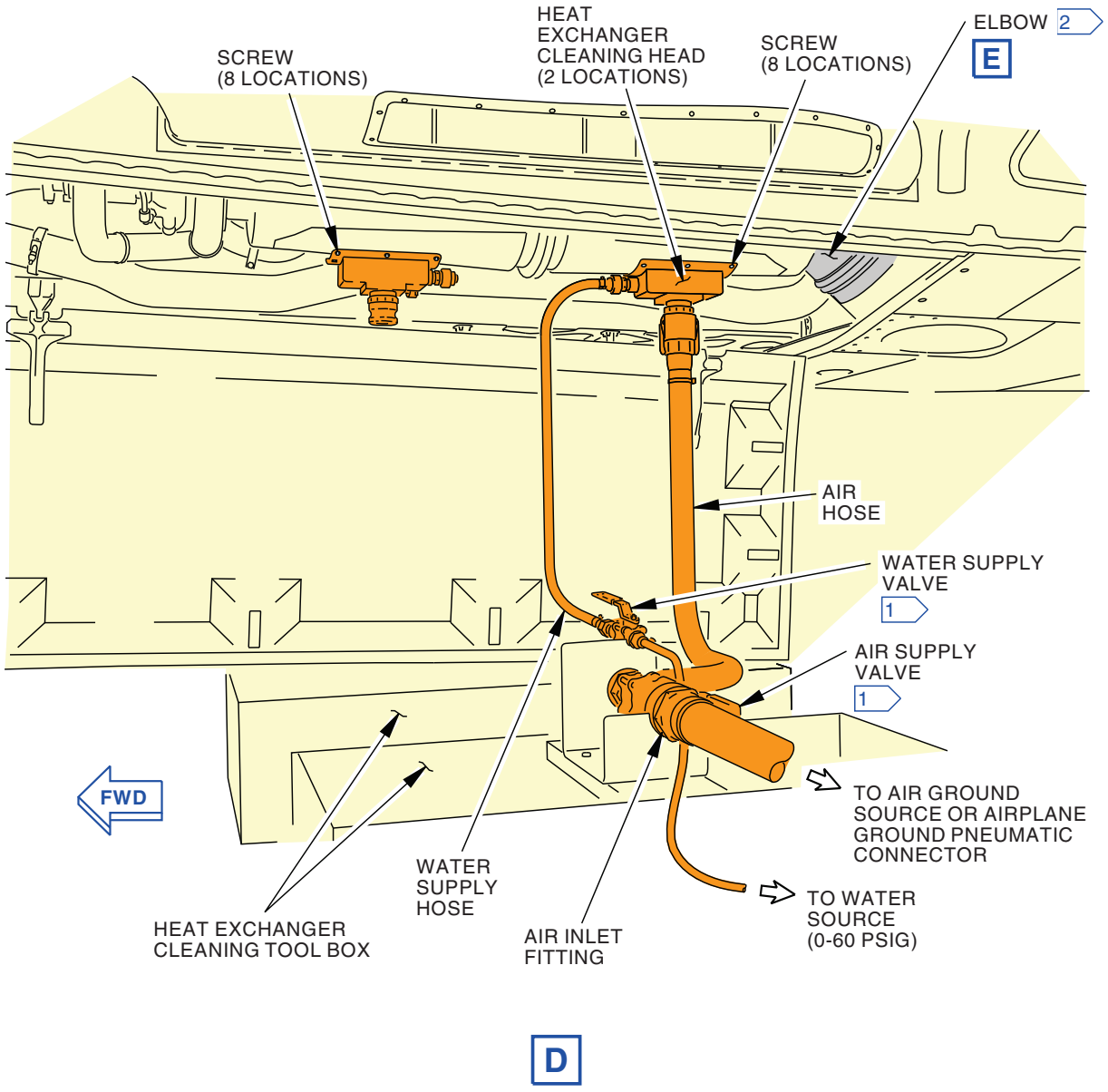
Heat Exchanger and Plenum/Diffuser Assembly Cleaning
Figure 701/21-51-03-990-804 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Heat Exchanger and Plenum/Diffuser Assembly Cleaning
Figure 701/21-51-03-990-804 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



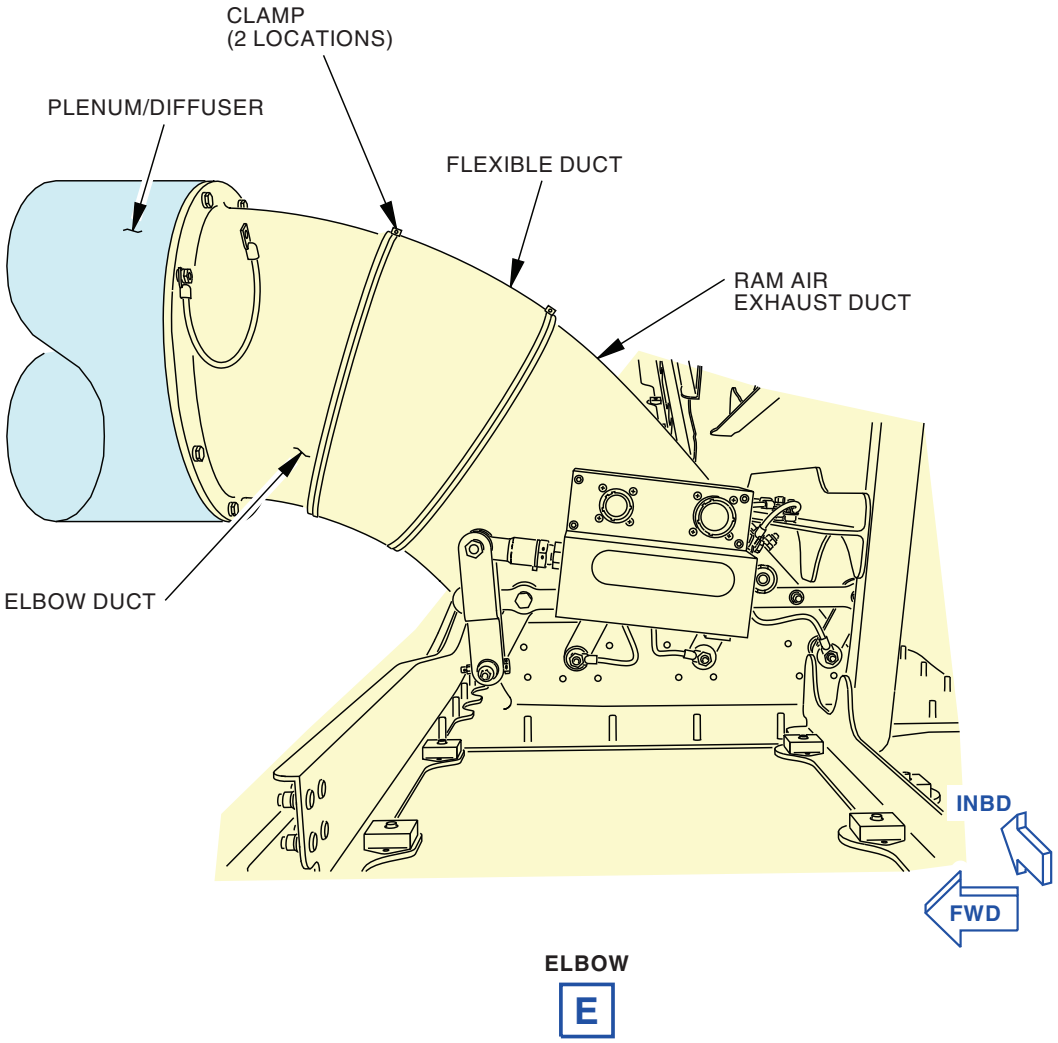
1 IF INSTALLED

2 AIRPLANES WITH EXHAUST LOUVERS

2411399 S00061527817_V1

Heat Exchanger and Plenum/Diffuser Assembly Cleaning
Figure 701/21-51-03-990-804 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Heat Exchanger and Plenum/Diffuser Assembly Cleaning
Figure 701/21-51-03-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-03-160-803

3. Heat Exchanger - Cleaning with Soap-Water-Air

(Figure 702, Figure 703, Figure 704)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Boeing notes that this procedure for cleaning with soap-water-air is not effective for clearing blocked heat exchangers and less effective for dirty heat exchangers. To effectively clear blocked heat exchangers, do this task: Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air, TASK 21-51-03-160-805.

(2) This procedure can improve heat exchanger performance between scheduled maintenance intervals when internal heat exchanger contamination is present (resulting in decreased performance) due to local environment conditions.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
21-51-03-160-805	Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-14922	Cleaning Wand Eqpt - Heat Exchanger Part #: C21010-40 Supplier: 81205 Opt Part #: C21010-1 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

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(Continued)

Reference	Description
STD-1290	Source - Nitrogen, 0-200 PSIG
STD-6452	Air Source - Regulated, Dry Filtered, 0-200psig

D. Consumable Materials

Reference	Description	Specification
A50035	Sealant - Dow Corning Q4-2817 Fluorosilicone Sealant	
B50026	Soap - Clear Liquid Dishwashing	
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A
G50256	Water, Regular	
G50321	Air - Clean, Dry	BAC5402 Table I

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Gasket	21-51-24-04A-190	PTF ALL
		21-51-24-04B-190	PTF ALL
5	Gasket	21-51-03-02-118	PTF ALL
		21-51-03-03-145	PTF ALL

F. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

G. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

H. Prepare for the Cleaning

SUBTASK 21-51-03-860-016

(1) Set the L PACK or R PACK switches, on the P5–10 air conditioning panel, to the OFF position.

(a) Put DO NOT OPERATE tags, STD-858, on the switches.

SUBTASK 21-51-03-860-017

(2) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel, to the CLOSE position.

SUBTASK 21-51-03-010-010

(3) To get access to the heat exchangers for the left pack, do these steps:
(TASK 06-41-00-800-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-03-010-011

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the heat exchangers for the right pack, do these steps:

(TASK 06-41-00-800-801)

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-03-020-048

(5) Remove the bolts [4] and washers [3] that attach the ram air inlet duct access panel [2] and gasket [1] to the ram air inlet duct.

SUBTASK 21-51-03-010-012

(6) Remove the ram air inlet duct access panel [2] and gasket [1].

NOTE: Removal of the access panel is necessary so that the cleaning water will not accumulate in the ram air inlet duct.

SUBTASK 21-51-03-020-049

(7) Remove the bolts [8] and washers [7] that attach each of the two plenum/diffuser access panels [6] and gaskets [5] to the plenum/diffuser assemblies for the primary and secondary heat exchangers.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-03-020-050

(8) If necessary, remove the bonding jumpers [11] from the plenum/diffuser access panels [6].

NOTE: The bonding jumper removal is not necessary if the task can be completed with the access panel moved to the side.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(a) Remove the sealant that covers the jumper terminals and the head of the bolts [9] on the plenum/diffuser access panels [6] (SWPM 20-20-00 and SWPM 20-20-10).

(b) Remove the bolts [9], washers [10], washers [12], and nuts [13] that attach the bonding jumpers [11] to the plenum/diffuser access panels [6].

SUBTASK 21-51-03-020-051

(9) Remove the plenum/diffuser access panels [6] from the plenum/diffuser.

SUBTASK 21-51-03-160-003

(10) Make sure that the regulator connected to the 0-200psig dry filtered regulated air source, STD-6452, or the nitrogen source, STD-1290, have a pressure gauge capable of adjusting pressure outlet from 0 psi (0 kPa) to 200 psi (1379 kPa).

I. Heat Exchanger Cleaning with Soap-Water-Air

SUBTASK 21-51-03-160-004

WARNING MAKE SURE THAT YOU WEAR A FACE MASK, EYE PROTECTION, AND WATER RESISTANT CLOTHING DURING THE OPERATION. PREVENT SOAP AND HEAT EXCHANGER CONTAMINATION ON THE SKIN, EYES, AND HAIR. IF YOU DO NOT OBEY INSTRUCTIONS, INJURY TO PERSONNEL CAN OCCUR.

(1) Clean the heat exchangers with soap, water, and compressed air as follows:

WARNING COMPRESSED GAS CAN BE DANGEROUS. EYE PROTECTION IS MANDATORY. APPLY DRY NITROGEN CAREFULLY IN DIRECTION AWAY FROM EYES AND FACE AND OTHER PERSONNEL TO AVOID POSSIBLE INJURY.

(a) Soak with soap, B50026, and water, G50256, to remove dust/fine particle residue (Figure 703).

1) Connect the cleaning wand, SPL-14922, soap injector to the cleaning wand, SPL-14922, water wand water inlet.

a) Make sure that the lever on the cleaning wand, SPL-14922, water wand is OFF (CLOSE).

2) Put the cleaning wand, SPL-14922, soap injector tube inlet into the soap container.

3) Connect one end of the cleaning wand, SPL-14922, water hose to the cleaning wand, SPL-14922, soap injector.

a) Make sure that the lever on the cleaning wand, SPL-14922, soap injector is OFF (CLOSE).

4) Connect other end of the cleaning wand, SPL-14922, water hose to the water cart.

5) If using clean dry air, G50321, (Optional – Not Required):

a) Make sure that one end of the cleaning wand, SPL-14922, air hose connected to either the 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, outlet.

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b) Connect other end of the cleaning wand, SPL-14922, air hose to the cleaning wand, SPL-14922, water wand air inlet.

c) Adjust the regulator on the 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, to supply 50 psi (345 kPa) maximum to the cleaning wand, SPL-14922.

6) Insert the cleaning wand, SPL-14922, through the plenum/diffuser access openings and spray the soap, B50026, and water, G50256, through the cooling fins of the primary and secondary heat exchangers.

a) Put the cleaning wand, SPL-14922, through the plenum/diffuser access panel openings.

b) Make sure that you do not damage the cooling fins of the heat exchangers when you put the cleaning wand, SPL-14922, through the plenum/diffuser access panel openings.

c) Do not touch the cooling fins of the heat exchangers with the tip of the cleaning wand, SPL-14922, when you spray the soap, B50026, and water, G50256, through the heat exchangers.

d) Move the lever on the cleaning wand, SPL-14922, water wand to ON (OPEN).

e) Move the lever on the cleaning wand, SPL-14922, soap injector to ON (OPEN).

NOTE: Switching the order will most likely cause user to get wet.

f) Spray the soap, B50026, and water, G50256, through the heat exchangers to soak and to remove dust and small particles.

<1> Squeeze the handle on the cleaning wand, SPL-14922, air wand to blow the clean dry air, G50321, or nitrogen, G00018, with soap, B50026, and water, G50256, if clean dry air, G50321, or nitrogen, G00018, is needed.

<2> Spray through all the areas of the heat exchangers.

7) Remove the cleaning wand, SPL-14922, from the plenum/diffuser access opening.

8) Decrease the 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, pressure to zero psi (if clean dry air, G50321, or nitrogen, G00018, is used, but not required).

9) Move the lever on the cleaning wand, SPL-14922, soap injector to OFF (CLOSE).

10) Move the lever on the cleaning wand, SPL-14922, water wand to OFF (CLOSE).

11) Remove the cleaning wand, SPL-14922, water hose from the cleaning wand, SPL-14922, soap injector.

12) Remove the cleaning wand, SPL-14922, soap injector.

(b) Rinse with high pressure clean water, G50256 (Figure 704).

1) Make sure that the cleaning wand, SPL-14922, air hose is connected to the cleaning wand, SPL-14922, water wand air inlet.

2) Connect the cleaning wand, SPL-14922, water hose to the cleaning wand, SPL-14922, water wand water inlet.

3) Adjust the regulator on the 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, to supply 50 psi (345 kPa) maximum to the cleaning wand, SPL-14922.

NOTE: Vaporized clean dry air, G50321, is made with use of pressure over 100 psi.

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4) Put the cleaning wand, SPL-14922, through the plenum/diffuser access opening.

5) Clean with strong pressure clean water, G50256, through the cooling fins of the primary and secondary heat exchangers as follows:

a) Make sure that you do not damage the cooling fins of the heat exchangers when you put the cleaning wand, SPL-14922, through the plenum/diffuser access panel openings.

b) Make sure that you do not touch the cooling fins of the heat exchangers with the tip of the cleaning wand, SPL-14922, when you wash the heat exchangers with high pressure water.

c) Put the lever on the cleaning wand, SPL-14922, water wand to ON (OPEN) for water, G50256, to start flowing.

d) Squeeze the handle on the cleaning wand, SPL-14922, water wand to clean dry air, G50321, or nitrogen, G00018, to start flowing.

e) Wash through the heat exchangers with the strong pressure clean water, G50256, to clean the soap, B50026.

<1> Clean all the areas of the heat exchangers.

6) Remove the cleaning wand, SPL-14922, from the plenum/diffuser access opening.

7) Decrease the 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, pressure to zero psi.

8) Put the lever on the cleaning wand, SPL-14922, water wand to OFF (CLOSE).

9) Remove the cleaning wand, SPL-14922, water hose and cleaning wand, SPL-14922, water wand air inlet.

(c) Blow off residual water, G50256, to protect the Air Cycle Machine (ACM) (Figure 704).

1) Connect the cleaning wand, SPL-14922, air hose to the cleaning wand, SPL-14922, air inlet.

2) Adjust the regulator on 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, to supply 100 psi (689 kPa) maximum to the cleaning wand, SPL-14922.

3) Put the cleaning wand, SPL-14922, through the plenum/diffuser openings.

4) Blow clean dry air, G50321, or nitrogen, G00018, through the cooling fins of the primary and secondary heat exchangers to dry of remaining water, G50256.

a) Put the cleaning wand, SPL-14922, through the plenum/diffuser access panel.

<1> Make sure that you do not damage the cooling fins of the heat exchangers when you put the cleaning wand, SPL-14922, through the plenum/diffuser access panel.

b) Blow clean dry air, G50321, or nitrogen, G00018, through the heat exchangers until all water, G50256, is blown out of the cooling fins as follows:

<1> Do not touch the cooling fins of the heat exchangers with the cleaning wand, SPL-14922, when you blow clean dry air, G50321, or nitrogen, G00018, through the heat exchangers.

<2> Make sure that the pressure gage on the regulator does not show more than 100 psi when you operate the cleaning wand, SPL-14922.

<3> Blow the clean dry air, G50321, or nitrogen, G00018, through all the areas of the heat exchangers.

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<4> Make sure that water, G50256, is fully removed from the heat exchanger.

<5> Blow clean dry air, G50321, or nitrogen, G00018, through the heat exchangers until water, G50256, is not being blown out.

5) Remove the cleaning wand, SPL-14922, from the plenum/diffuser access opening.

6) Decrease the 0-200psig dry filtered regulated air source, STD-6452, or nitrogen source, STD-1290, pressure to zero psi.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-03-211-005

(1) Examine the gasket [1] and gaskets [5] for damage.

(a) Replace the gasket [1] or gaskets [5] if there is damage or deterioration.

SUBTASK 21-51-03-410-011

(2) If removed, install the bonding jumpers [11] to the plenum/diffuser access panels [6].

(a) Make sure that the bonding jumpers [11], washers [10], bolts [9], washers [12], nuts [13], and plenum/diffuser access panels [6] are clean.

1) If installed, make sure that the nutplates are clean.

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(b) Put the bonding jumpers [11] in their positions on the plenum/diffuser access panels [6].

(c) Install the washers [10], bolts [9], washers [12], and nuts [13].

NOTE: The access panel can have a nutplate as an alternative to the nut and washer.

(d) Tighten the bolts [9] to 22 in-lb (2.49 N·m) - 26 in-lb (2.94 N·m).

(e) Measure the electrical bonding resistance between the bolts [9] and the plenum/diffuser access panels [6].

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.3 Mohm (300 000 ohm) or less.

(f) Apply a coat of Dow Corning Q4-2817 sealant, A50035, to cover the jumper terminals and the heads of the bolts [9].

NOTE: The sealant is optional if the plenum was built before 2022/2023, and did not originally leave Honeywell production with the sealant applied.

SUBTASK 21-51-03-410-010

(3) Install the plenum/diffuser access panels [6] and the gaskets [5] with the bolts [8] and washers [7].

(a) Tighten the bolts [8] to 22 in-lb (2 N·m) - 26 in-lb (3 N·m).

SUBTASK 21-51-03-410-029

(4) Install the ram air inlet duct access panel [2] and the gasket [1] with the bolts [4] and washers [3].

SUBTASK 21-51-03-080-006

(5) Remove the DO NOT OPERATE tag, STD-858, from the L PACK and R PACK switches.

(a) Make sure that the R PACK and L PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

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SUBTASK 21-51-03-080-011

(6) Operate the air conditioning pack for five minutes (TASK 21-00-00-800-803).

NOTE: This will remove all water in the ram air ducts and the air cycle machine bearings.

SUBTASK 21-51-03-880-005

(7) Stop the air conditioning pack (TASK 21-00-00-800-804).

SUBTASK 21-51-03-880-006

(8) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-03-410-012

(9) If you cleaned the left pack heat exchangers, do these steps:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-03-410-013

(10) If you cleaned the right pack heat exchangers, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Install this access panel:

(TASK 53-51-21-400-802)

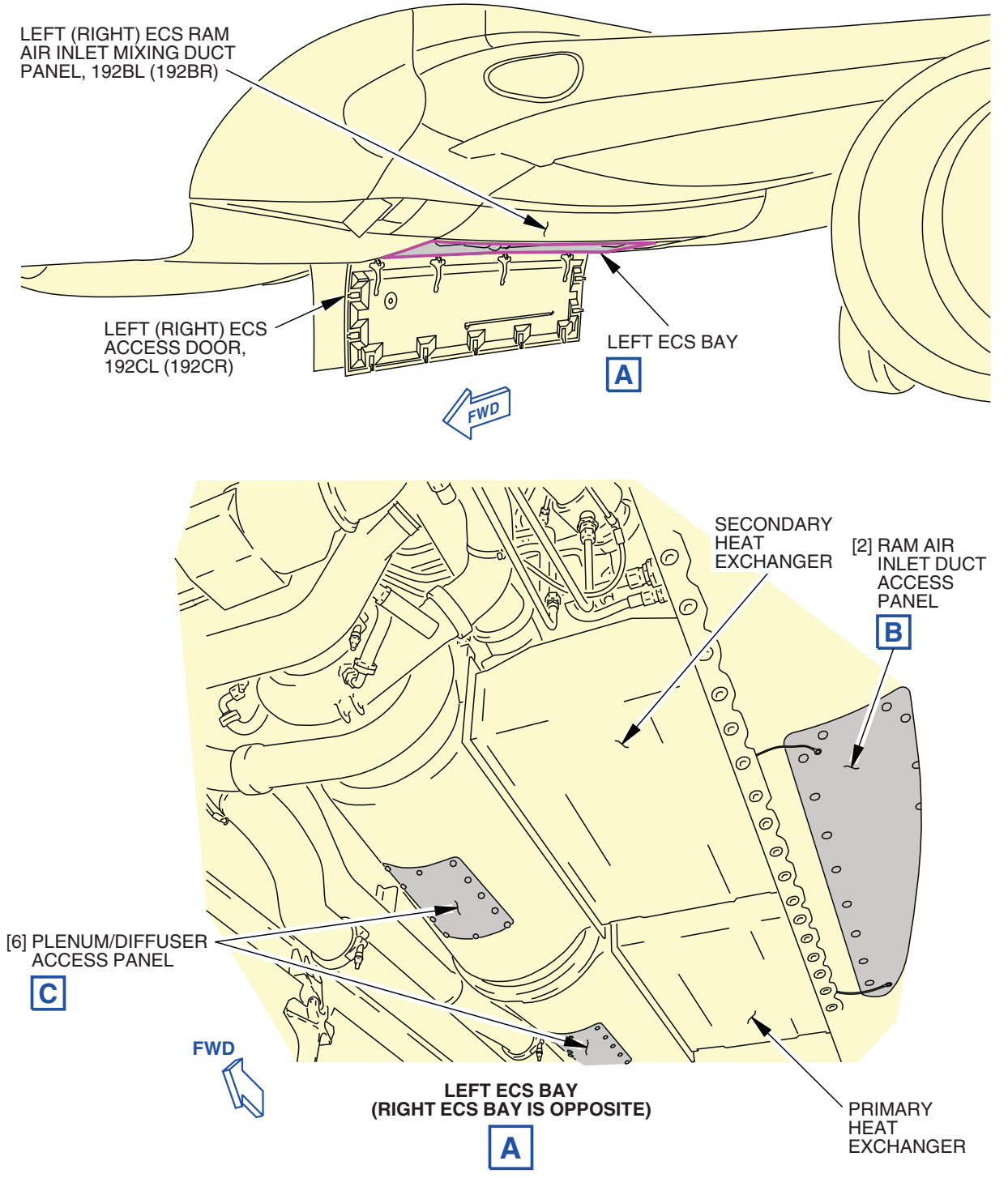
Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-03-860-023

(11) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel, to the OPEN position.

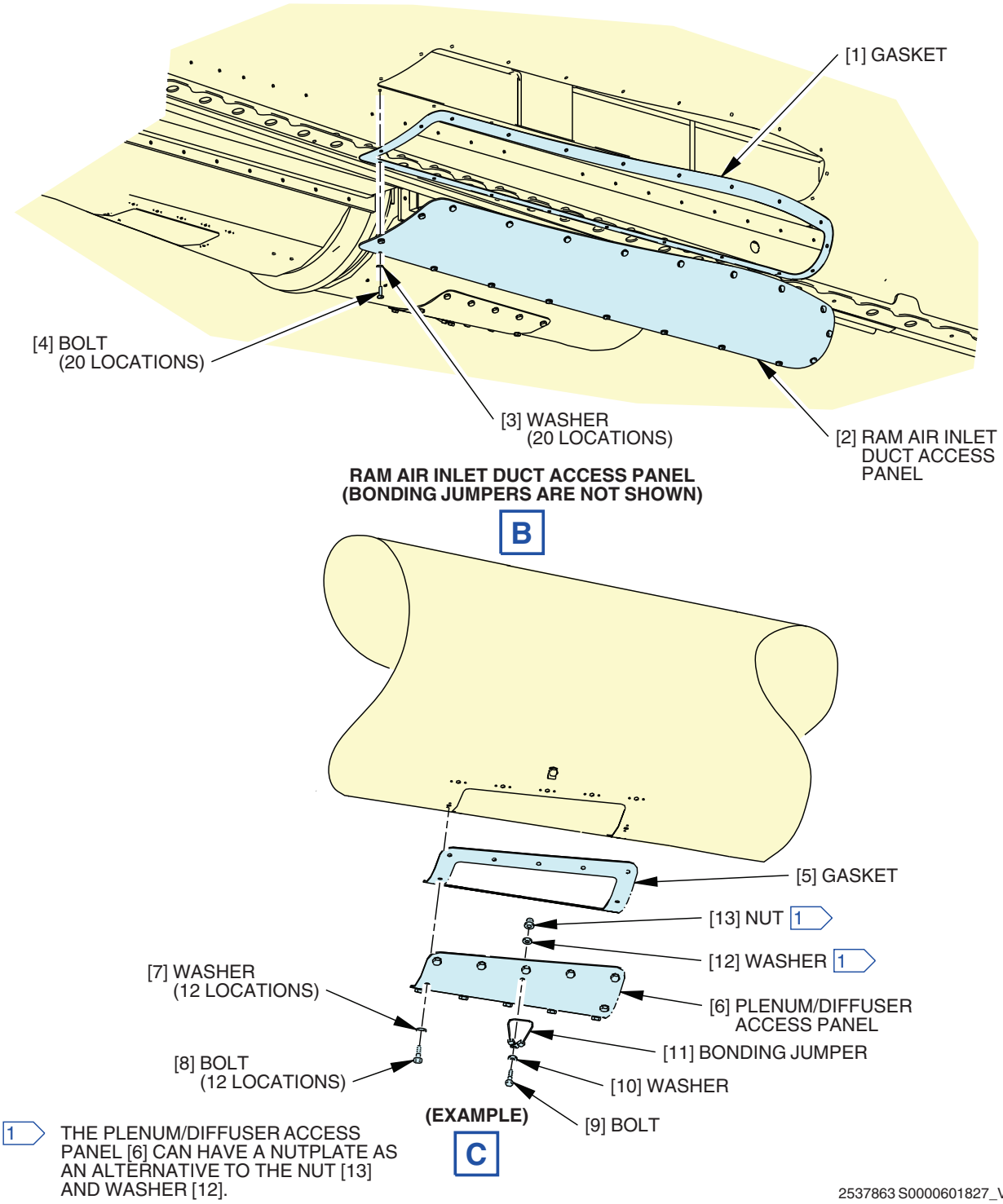
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



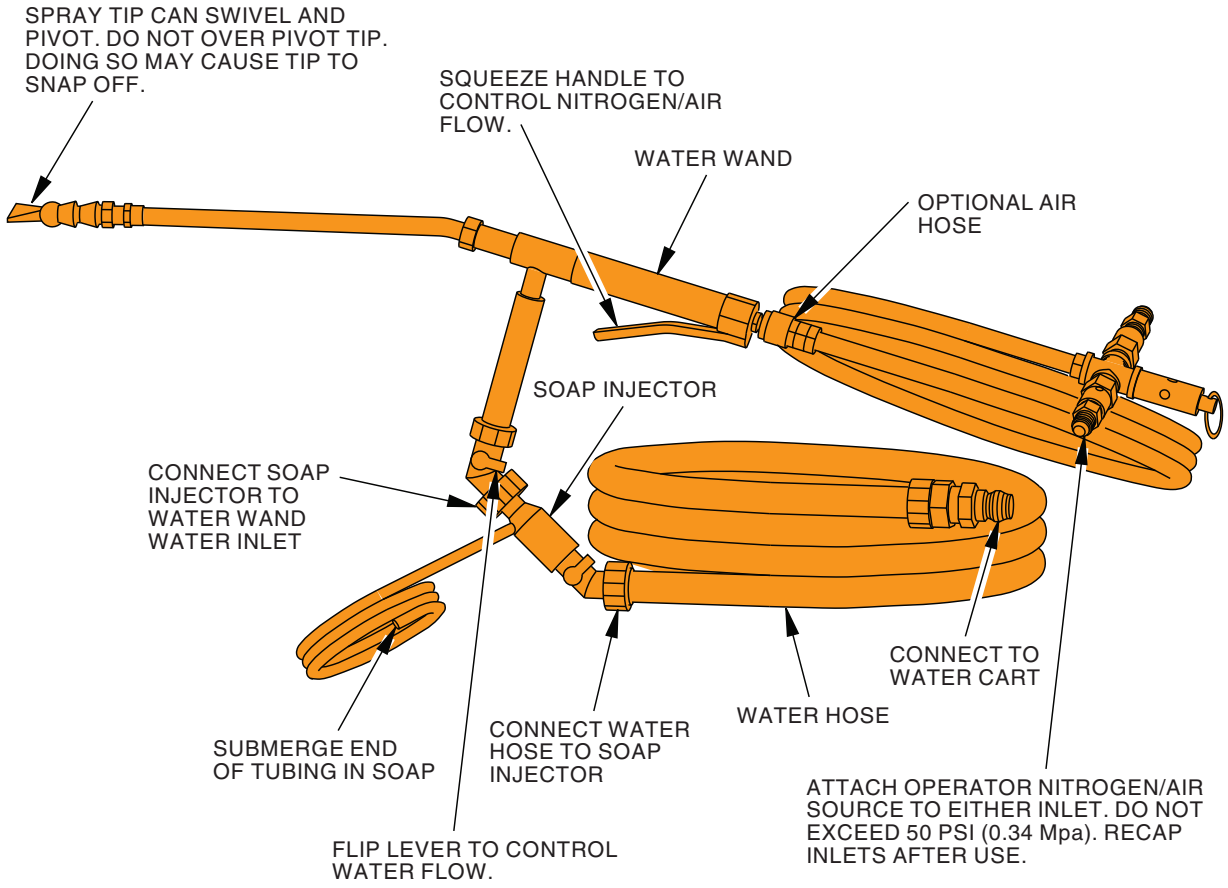
Heat Exchanger Cleaning
Figure 702/21-51-03-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Heat Exchanger Cleaning
Figure 702/21-51-03-990-806 (Sheet 2 of 2)

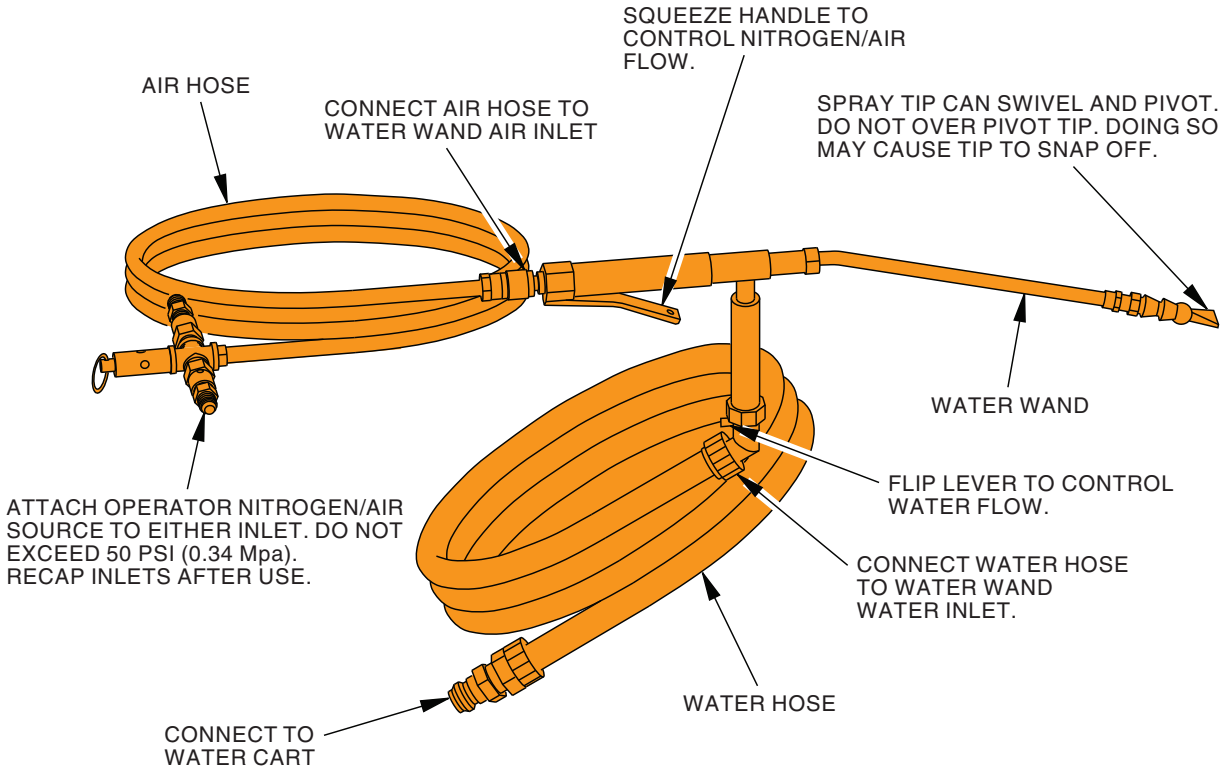
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2522146 S0000594061_V2

Soap Injector with Water Wand and Hose
Figure 703/21-51-03-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2522266 S0000594062_V2

Water Wand, Water Hose and Air/Nitrogen Hose
Figure 704/21-51-03-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CYCLE MACHINE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the air cycle machine

(2) An installation of the air cycle machine.

B. There is an ACM for each pack. The ACMs are located in the forward area of the air conditioning bays.

C. The ACM, together with the heat exchangers, converts high temperature, high pressure airflow to low temperature, low pressure airflow.

D. The high pressure water system mix muff must be removed before the removal of the air cycle machine.

TASK 21-51-04-000-801

2. Air Cycle Machine Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the Air Cycle Machine (ACM).

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-17-000-801	High Pressure Water Separator Mix Muff Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for the Removal

SUBTASK 21-51-04-860-002

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that there is no pressure in the pneumatic system.

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(a) If it is necessary, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-04-860-003

(2) Do these steps on the P5-10 air conditioning panel:

(a) Set the L and R PACK switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach the DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-04-860-010

(3) For left pack air cycle machine [1], do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-04-860-014

(4) For right pack air cycle machine [1], do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-04-010-001

(5) When the air cycle machine for the left pack will be removed, open this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-04-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(6) When the air cycle machine for the right pack will be removed, open these access doors in this sequence:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

G. Air Cycle Machine Removal

SUBTASK 21-51-04-010-003

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Remove the compressor outlet duct [12].

(a) Disconnect the flex hose from the outlet duct [12].

(b) Remove the electrical connector [3] from the ram air control sensor.

(c) Remove the electrical connector [6] from the compressor discharge overheat switch.

(d) Remove the clamshell clamps [7] at the air cycle machine [1].

1) Lift the latch pawls at the same time to release the clamps [7].

(e) Move the sleeves [8] to the outlet duct [12].

(f) Remove the coupling [4] at the heat exchanger.

(g) Remove the compressor outlet duct [12].

(h) Remove and discard the packing [5] at the heat exchanger.

(i) Remove and discard the packings [9].

SUBTASK 21-51-04-860-004

(2) Remove the HPWS Mix Muff [2], do this task: High Pressure Water Separator Mix Muff Removal, TASK 21-51-17-000-801.

SUBTASK 21-51-04-210-001

(3) Check the flexible duct [13] that is connected to the high pressure water separator condenser for any of the conditions listed below:

(a) Fraying

(b) discoloration

(c) rips

(d) splits

(e) tears

(f) rupture defects.

(g) If any of the above conditions are found, remove the flexible duct [13] from the condenser as follows:

1) Loosen the clamps [38] that attaches the flexible duct [13] to the water separator condenser.

2) Remove the flexible duct [13] from the water separator condenser.

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3) Discard the flexible duct [13].

SUBTASK 21-51-04-020-001

(4) Disconnect the hoses and sense line from the air cycle machine [1].

(a) Remove the flex hose [21] from the air cycle machine [1].

(b) Disconnect the sense line [25] from the air cycle machine [1].

(c) Disconnect the sense line [14] from the air cycle machine [1].

(d) If the air cycle machine [1] will be replaced, remove the unions [15] and discard the O-rings [16].

NOTE: Keep the unions for installation on the new air cycle machine.

SUBTASK 21-51-04-020-002

(5) Remove the air cycle machine [1].

(a) Remove the clamshell clamps [18].

1) Lift the latch pawls at the same time to release the clamps [18].

(b) Slide the sleeves [19] away from the air cycle machine [1] and onto the duct [17].

(c) Remove and discard the packings [20].

(d) Disconnect the compressor inlet duct.

1) Remove the screw [33] and washer [39] that hold the ground clip [34] to the air cycle machine [1].

2) Move the bonding jumper [35] away from the air cycle machine [1].

a) Do not remove the ground clip [34] from the bonding jumper [35].

3) Remove the screws [31] and washers [39] that hold the duct to the air cycle machine [1].

4) Remove and discard the O-ring [32].

(e) Remove the bolts [23] and washers [24] that hold the air cycle machine [1] to the secondary heat exchanger.

WARNING HAVE A SECOND PERSON HOLD THE ACM WHEN YOU REMOVE THE NUT. THE ACM CAN FALL WHEN THE NUT IS REMOVED. IF THE ACM FALLS, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(f) Remove the nut [30] and washer [29] from the bolt [26] that holds the air cycle machine [1] to the tie rod.

WARNING HAVE A SECOND PERSON HOLD THE ACM WHEN YOU REMOVE THE BOLT. THE ACM CAN FALL WHEN THE BOLT IS REMOVED. IF THE ACM FALLS, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(g) Remove the bolt [26] and washer [27] from the tie rod.

(h) Carefully move forward then lower the air cycle machine [1] to remove it from the air conditioning bay.

(i) Remove the bushing [28] from the tie rod.

(j) Remove the gasket [22].

1) Discard the gasket [22].

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SUBTASK 21-51-04-620-001

(6) Put covers on the duct openings to prevent the entry of unwanted materials.

SUBTASK 21-51-04-700-001

(7) Check the torque necessary to turn the rotating assembly of the air cycle machine [1] as follows:

(a) Put the air cycle machine [1] in a horizontal position.

(b) Put a torque wrench on the nut [37].

NOTE: A torque wrench with a gage that shows the torque applied is recommended.

CAUTION

TURN THE ROTATING COMPONENTS OF THE ACM ONLY IN THE DIRECTION THAT THEY TURN WHEN IN OPERATION. IF YOU TURN THE ACM ROTATING COMPONENTS IN THE INCORRECT DIRECTION, YOU CAN CAUSE DAMAGE TO THE FOIL AIR BEARINGS.

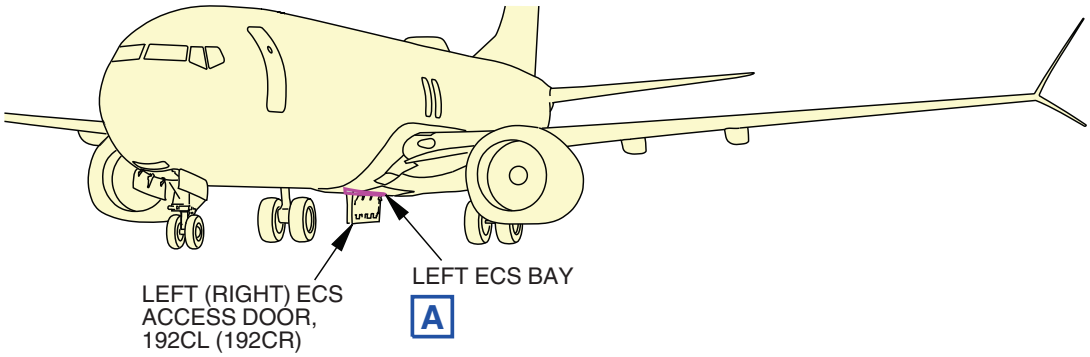
(c) Make sure that you turn the nut [37] counterclockwise while you turn the rotating assembly.

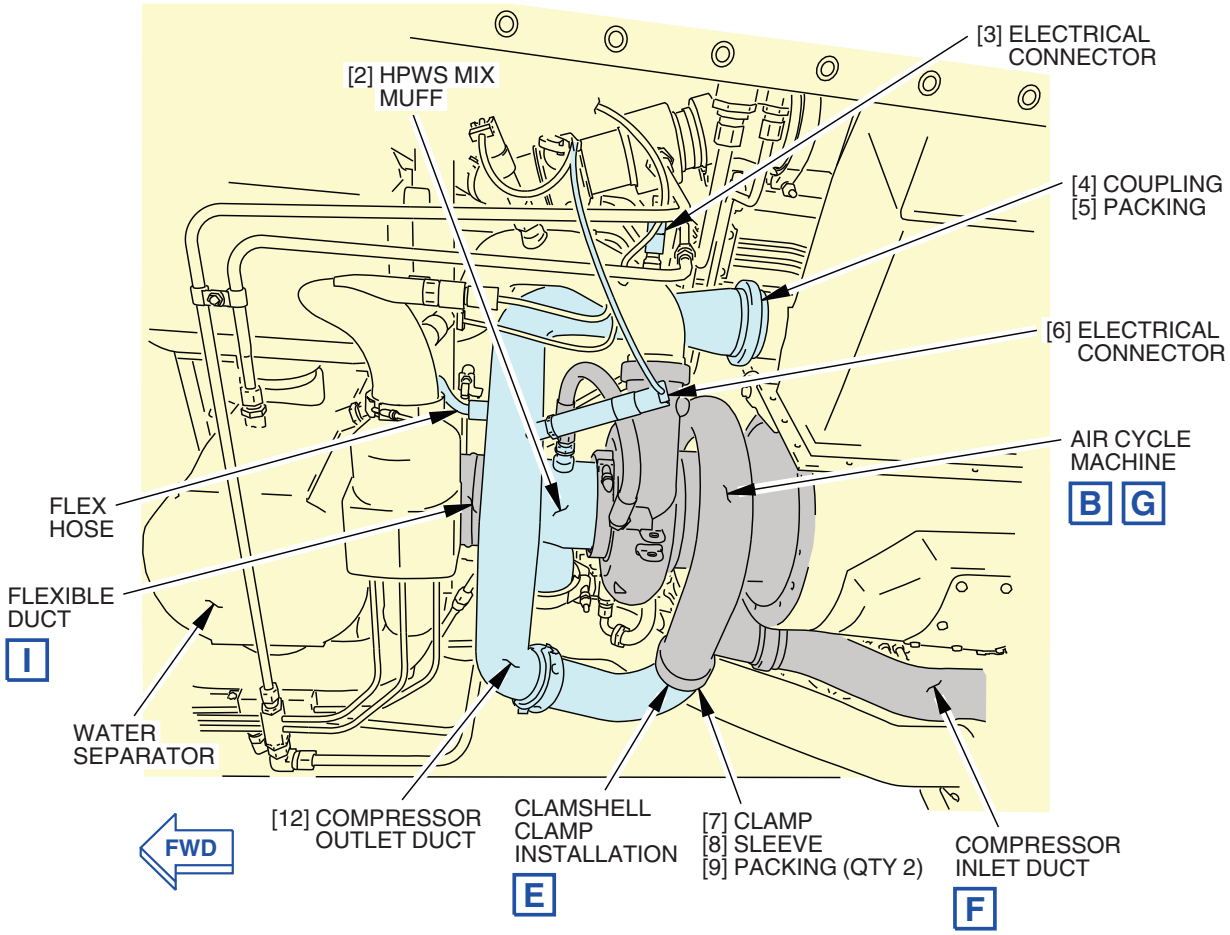
(d) Record the torque necessary to turn the rotating assembly.

(e) If the torque was more than 20.0 in-lb (2.3 N·m), replace the air cycle machine [1] with a new or serviceable unit.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





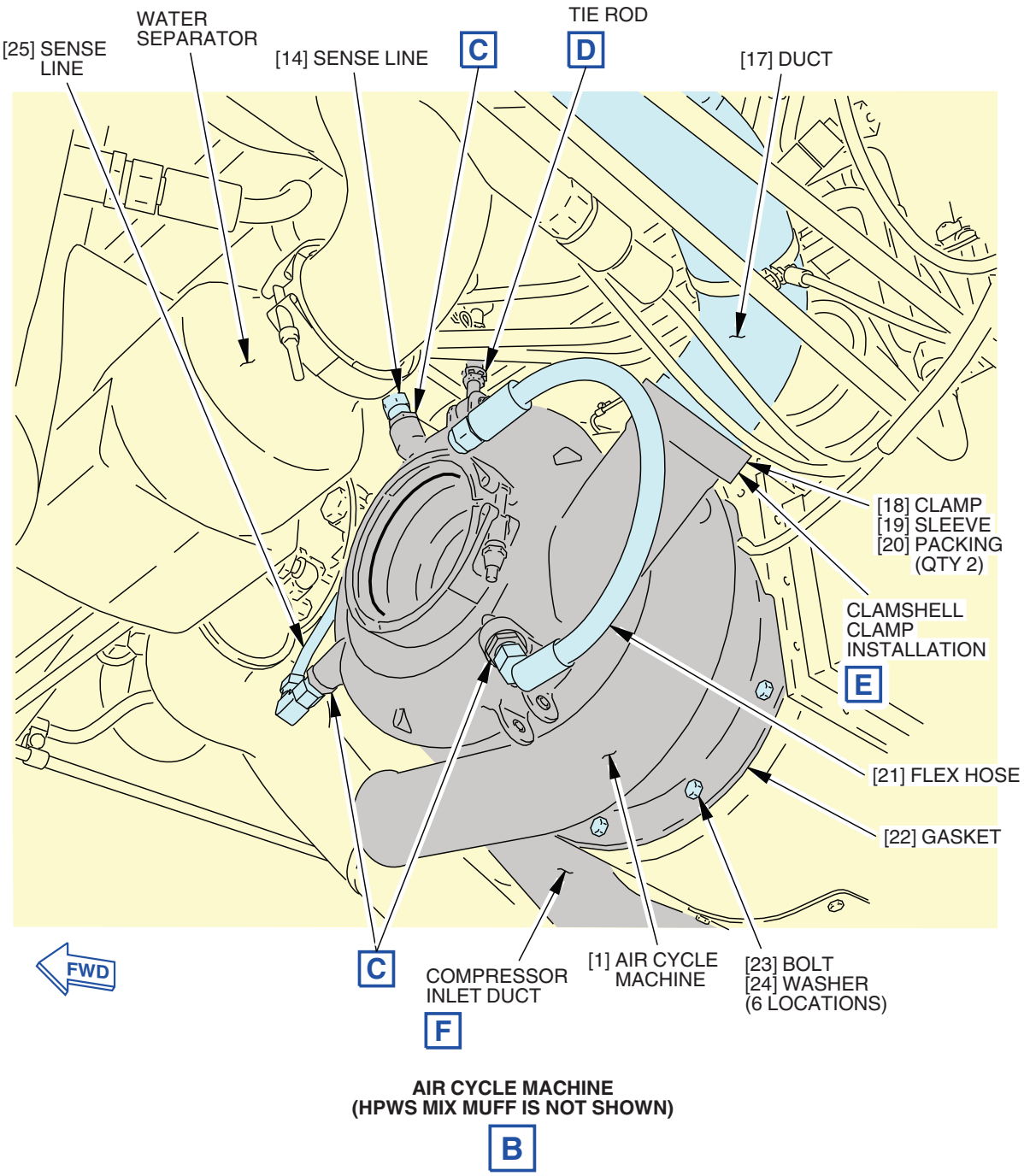
LEFT ECS BAY
(RIGHT ECS BAY IS EQUIVALENT)

A

2411402 S00061527825_V2

Air Cycle Machine Installation
Figure 401/21-51-04-990-801 (Sheet 1 of 6)

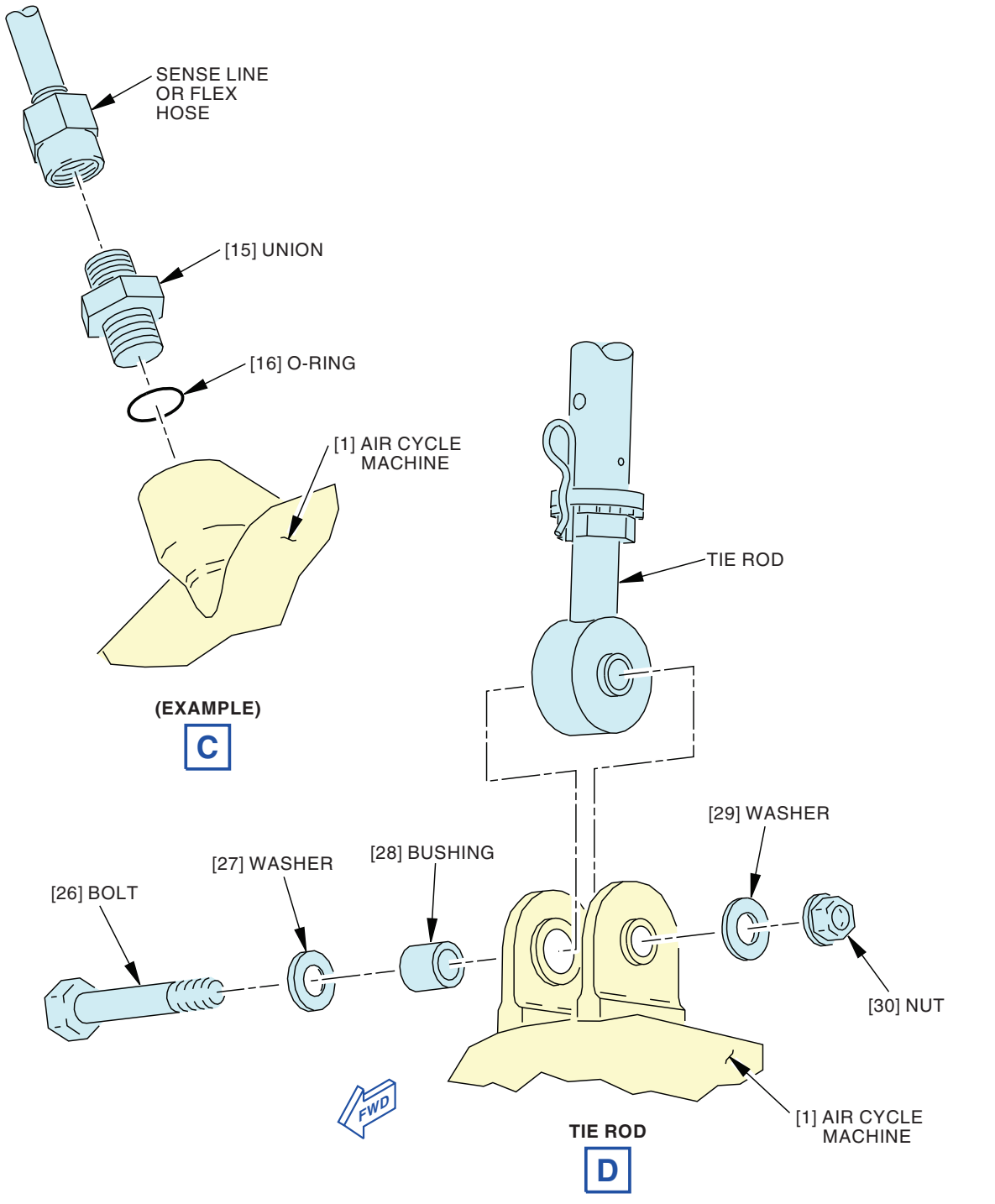
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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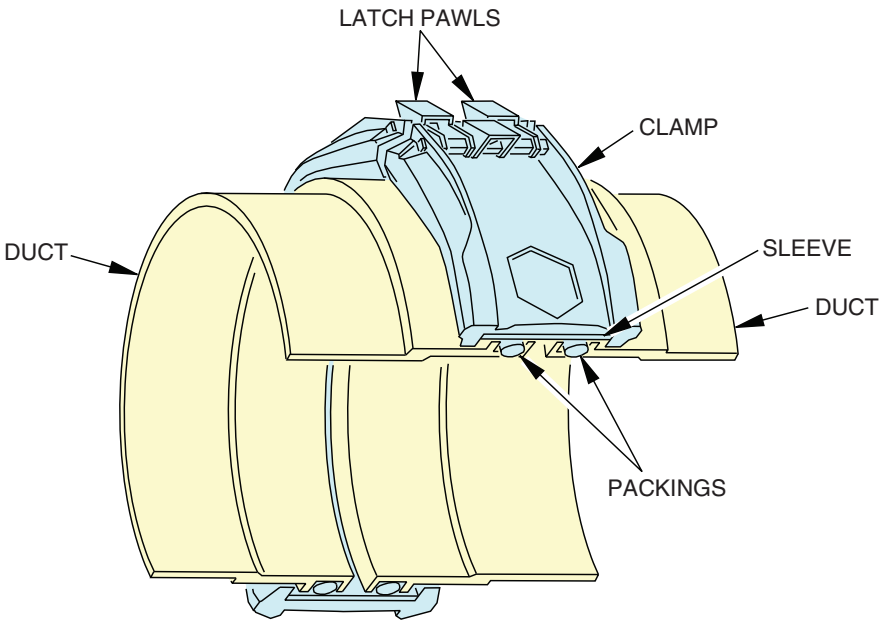
Air Cycle Machine Installation
Figure 401/21-51-04-990-801 (Sheet 2 of 6)

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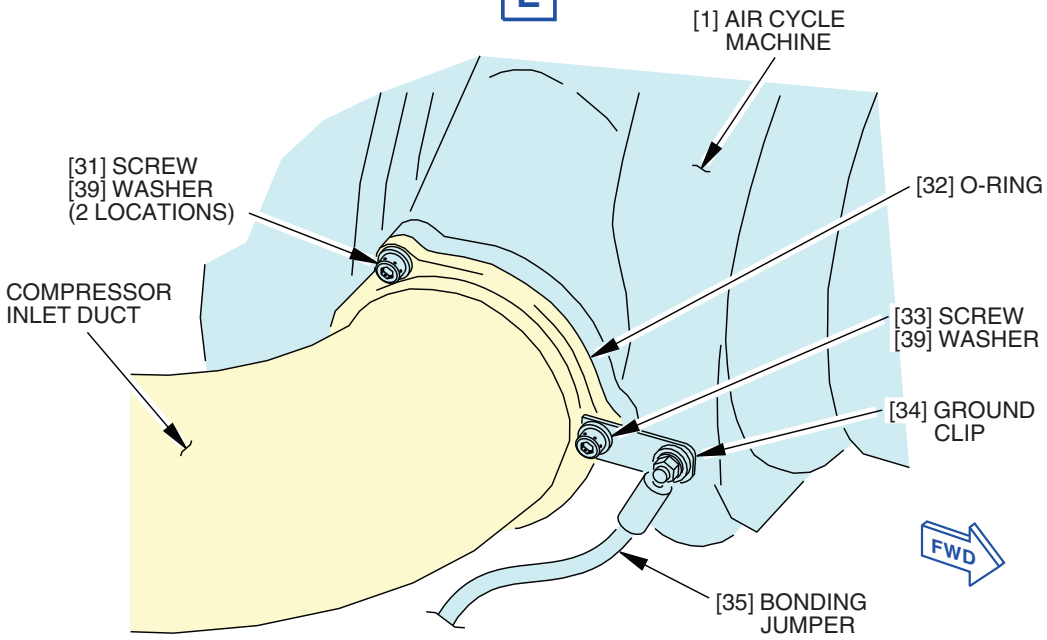
Air Cycle Machine Installation
Figure 401/21-51-04-990-801 (Sheet 3 of 6)

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CLAMSHELL CLAMP INSTALLATION (EXAMPLE)

E



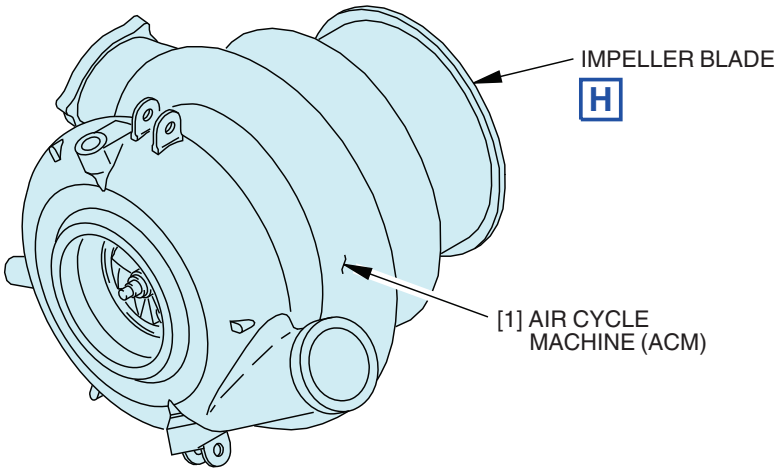
COMPRESSOR INLET DUCT

F

2411405 S00061527828_V2

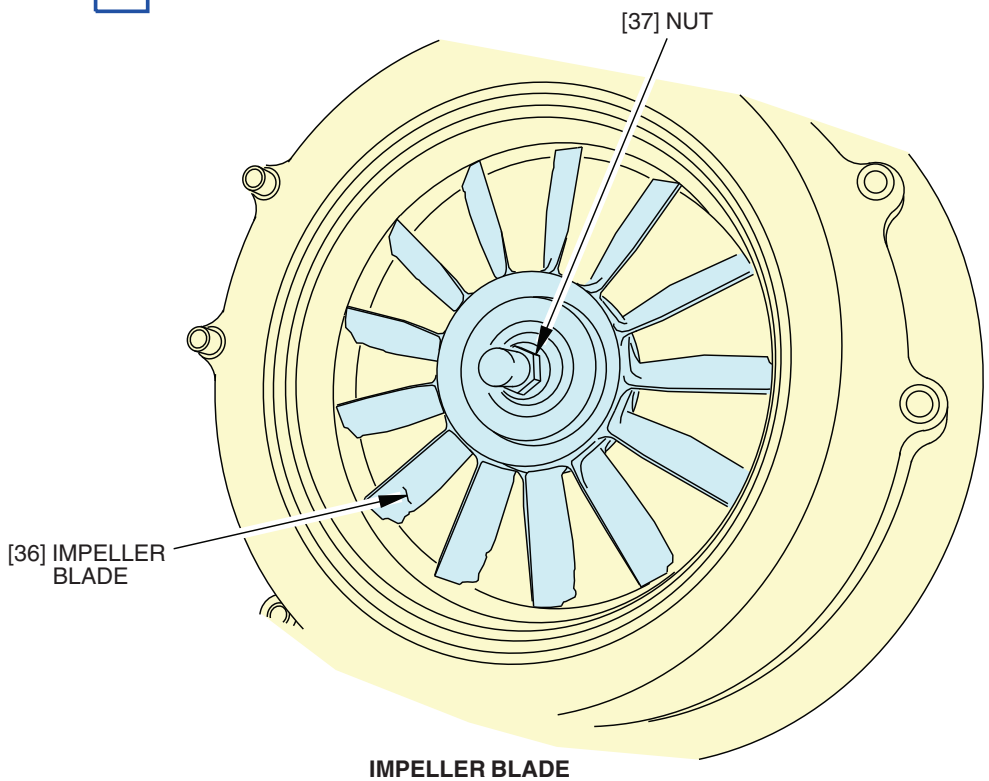
Air Cycle Machine Installation
Figure 401/21-51-04-990-801 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AIR CYCLE MACHINE (ACM)

G



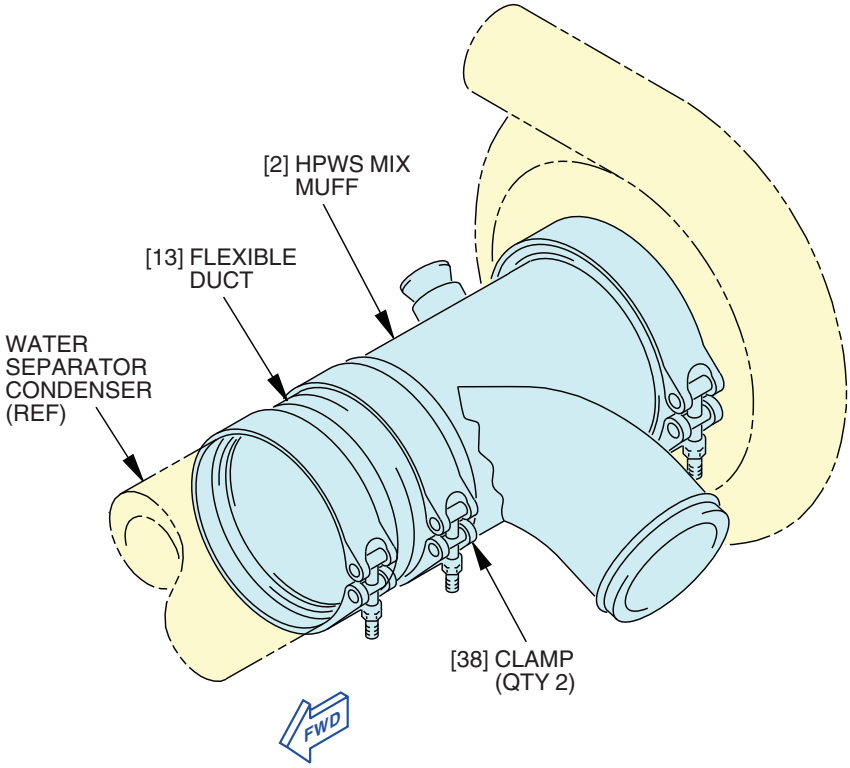
IMPELLER BLADE

H

2411406 S00061527829_V2

Air Cycle Machine Installation
Figure 401/21-51-04-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLEXIBLE DUCT
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

I

2411407 S00061527830_V4

Air Cycle Machine Installation
Figure 401/21-51-04-990-801 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-04-400-801

3. Air Cycle Machine Installation

(Figure 401)

A. General

(1) This task gives instructions to install the Air Cycle Machine (ACM).

(2) Do the steps in the cooling pack water removal/ice protection system inspection. If the water removal paths are blocked, ice can form in the ACM. The ice can cause failure of the ACM.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
21-51-03-000-803	Heat Exchanger and Plenum/Diffuser Assembly - Removal (P/B 401)
21-51-12-000-801	Condenser Removal (P/B 401)
21-51-13-000-801	Reheater Removal (P/B 401)
21-51-14-000-801	Water Extractor Removal (P/B 401)
21-51-15-000-801	Water Extractor Duct Removal (P/B 401)
21-51-16-000-801	Water Spray Nozzle Removal (P/B 401)
21-51-17-400-801	High Pressure Water Separator Mix Muff Installation (P/B 401)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
C00852	Compound - Antiseize, Molybdenum Disulfide-Petrolatum	MIL-PRF-83483
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00504	Grease - Petrolatum	VV-P-236
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	
D50320	Lubricant - Moly-50 Anti-Seize	MIL-PRF-83483
G01061	Water - Distilled	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Air cycle machine	21-51-04-01-025	PTF ALL
		21-51-04-01-510	PTF ALL
5	Packing	21-51-03-02-040	PTF ALL
		21-51-03-02-200	PTF ALL
9	Packing	21-51-03-02-055	PTF ALL
		21-51-03-02-210	PTF ALL
13	Flexible duct	21-51-08-05-130	PTF ALL
		21-51-17-02-020	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
16	O-ring	21-51-53-01-185	PTF ALL
		21-51-53-02-196	PTF ALL
20	Packing	21-51-41-01-035	PTF ALL
		21-51-41-02-090	PTF ALL
22	Gasket	21-51-03-02-075	PTF ALL
		21-51-03-02-230	PTF ALL
32	O-ring	21-51-03-02-025	PTF ALL
		21-51-03-02-185	PTF ALL

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Cooling Pack Water Removal/Ice Protection System Inspection

SUBTASK 21-51-04-200-001

(1) If the cause of the air cycle machine [1] failure is unknown, then do all of the steps that follow.

SUBTASK 21-51-04-200-002

(2) If the cause of the air cycle machine [1] failure is known, then do applicable steps that follow.

SUBTASK 21-51-04-630-002

(3) Do the steps that follow if the air cycle machine [1] was replaced due to bearing failure or damage to the compressor:

(a) Disconnect the water drain lines from the bottom of the water extractors (TASK 21-51-14-000-801).

1) Use compressed air to remove unwanted material from the water drain lines.

2) Reconnect the water drain lines to the bottom of the water extractors (TASK 21-51-14-000-801).

(b) Disconnect the water line and the air line which are attached to the water spray nozzle (TASK 21-51-16-000-801).

1) Use compressed air to remove unwanted material from the water and air lines.

2) Reconnect the water line and the air line which are attached to the water spray nozzle (TASK 21-51-16-000-801).

(c) Remove the flex hose [21] and sense line [25].

1) Use compressed air to remove unwanted material from the flex hose [21] and sense line [25].

2) Reconnect the sense line [25].

NOTE: The flex hose will be installed after the ACM is installed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-04-630-003

(4) If the air cycle machine [1] was replaced due to damage to the compressor, do these steps:

(a) Inspect the components that follow for damage or unwanted materials:

1) Heat Exchangers (TASK 21-51-03-000-803)

2) Water Extractor Duct (TASK 21-51-15-000-801)

3) Water Extractors (TASK 21-51-14-000-801)

4) Reheater (TASK 21-51-13-000-801)

5) Condensor (TASK 21-51-12-000-801)

6) Water Spray Nozzle (TASK 21-51-16-000-801).

(b) Replace the components that have damage or unwanted materials.

SUBTASK 21-51-04-630-004

(5) If the air cycle machine [1] was replaced due to damage to the impeller blade [36], do these steps:

(a) Remove the plenum/diffuser access panels, do these steps:

1) Remove the bolts and washers that attach the plenum/diffuser access panels to the plenum diffuser assemblies for the primary and secondary heat exchangers.

2) Remove the bolt, washers, and nut that attach the bonding jumper to the forward access panel.

3) Remove the plenum/diffuser access panels and the gaskets.

(b) Inspect the plenum for unwanted materials.

(c) Install the plenum/diffuser access panels, do these steps:

1) Install the plenum/diffuser access panels and the gaskets with the bolts and washers.

2) Put the bonding jumper in its position and install the bolts, washers, and nut.

H. Air Cycle Machine Installation

SUBTASK 21-51-04-860-011

(1) For left pack air cycle machine [1], do this step:

(a) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-04-860-015

(2) For right pack air cycle machine [1], do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-04-630-005

(3) Remove the duct covers.

SUBTASK 21-51-04-210-002

(4) Make sure that there are no unwanted materials in the ducts.

SUBTASK 21-51-04-420-001

(5) Install the air cycle machine [1].

(a) Install a new gasket [22].

NOTE: The holes in the gasket must be aligned with the bolt holes in the ACM and the secondary heat exchanger.

(b) Install the new O-ring [32] for the compressor inlet duct.

1) If a lubricant is required to aide in working the O-ring [32] into its installed position, use distilled water, G01061, to wet the surface of the O-ring [32].

2) If the use of the distilled water, G01061, is not adequate to install the O-ring [32] or if a lubricant is needed to hold the O-ring [32] in place for the installation, apply a light coat of grease, D00504, or Molykote 44 grease, medium, D50104, to the O-ring [32].

(c) Install the bushing [28] in the tie rod.

(d) Put the air cycle machine [1] into its position.

(e) Put the washer [27] on the shank of the bolt [26].

NOTE: The countersunk side of the washer must be adjacent to the bolt head.

WARNING HAVE A SECOND PERSON HOLD THE ACM WHEN YOU INSTALL THE BOLT. THE ACM CAN FALL WHEN THE BOLT IS LOOSE. IF THE ACM FALLS, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(f) Push the bolt [26] with the washer [27] through the tie rod and bushing [28].

WARNING HAVE A SECOND PERSON HOLD THE ACM WHEN YOU INSTALL THE NUT. THE ACM CAN FALL WHEN THE NUT IS LOOSE. IF THE ACM FALLS, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(g) Install the washer [29] and nut [30] on the bolt [26].

1) Do not tighten the nut [30] fully.

(h) Apply a light coat of compound, C00852, to the threads of the bolts [23].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Install the bolts [23] and washers [24] that hold the air cycle machine [1] to the secondary heat exchanger.

1) Do not tighten the bolts [23] fully.

(j) Connect compressor inlet duct to the air cycle machine [1].

1) Apply a light coat of compound, C00852, to the threads of the screws [31] and screw [33].

2) Put the washer [39] onto the screw [33].

3) Put the ground clip [34] onto the screw [33].

a) Do not remove the ground clip [34] from the bonding jumper [35].

4) Install the ground clip [34], washer [39], and screw [33] on the air cycle machine [1] compressor inlet flange.

a) Do not tighten the screw [33] fully.

5) Install the screws [31] and washers [39] on theair cycle machine [1] compressor inlet flange.

a) Do not tighten the screws [31] fully.

(k) Apply a light coat of distilled water, G01061, or grease, D00504, or Molykote 44 grease, medium, D50104, on the new packings [20].

(l) Install the packings [20] for the turbine inlet duct.

(m) Move the sleeves [19] on to the air cycle machine [1] and duct connections.

(n) Install the clamshell clamps [18] over the duct connections.

(o) Do these steps to tighten the bolts [23]:

1) Find the run-on torque as you install the bolts [23] in the nutplate self-locking feature.

NOTE: Run-on torque is the torque required to overcome the nutplate self-locking feature.

2) Use a torque wrench to measure the torque required to turn the bolt [23] before the washer [24] contacts the air cycle machine [1].

3) Tighten the bolts [23] to 40.0 in-lb (4.5 N·m) - 45.0 in-lb (5.1 N·m) plus the value of the run-on torque.

(p) Tighten the screws [31] and screw [33] to 22.0 in-lb (2.5 N·m) - 26.0 in-lb (2.9 N·m).

(q) Tighten the nut [30] to 50.0 in-lb (5.6 N·m) - 65.0 in-lb (7.3 N·m).

SUBTASK 21-51-04-420-002

(6) Connect the sense lines and hose to the air cycle machine [1].

(a) If you install a new air cycle machine [1], do these steps:

1) Put a new O-ring [16] on each of the unions [15].

2) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the three unions [15].

3) Install the union [15] onto the air cycle machine [1].

a) Tighten the union [15] to 170 ±9 in-lb (19 ±1 N·m).

(b) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the sense line [14].

(c) Connect the sense line [14] to the air cycle machine [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Tighten the sense line [14] to 170 ±9 in-lb (19 ±1 N·m).

(d) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the flex hose [21].

(e) Install the flex hose [21] on the air cycle machine [1].

1) Tighten the flex hose [21] to 170 ±9 in-lb (19 ±1 N·m).

(f) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the sense line [25].

(g) Connect the sense line [25] to the air cycle machine [1].

1) Tighten the sense line [25] to 170 ±9 in-lb (19 ±1 N·m).

SUBTASK 21-51-04-420-003

(7) If the flexible duct [13] was discarded after inspection, install a new flexible duct [13] as follows:

(a) Loosely position the clamps [38] onto the new flexible duct [13].

(b) Install the new flexible duct [13] onto the water separator condenser.

(c) Position the forward clamp [38] to retain the new flexible duct [13] onto the water separator condenser.

(d) Tighten the forward clamp [38] to 13.0 in-lb (1.5 N·m) - 17.0 in-lb (1.9 N·m).

NOTE: The aft clamp is connected in the installation of the High Pressure Water Separator (HPWS) mix muff in the next step.

SUBTASK 21-51-04-860-005

(8) Do this task: High Pressure Water Separator Mix Muff Installation, TASK 21-51-17-400-801.

SUBTASK 21-51-04-410-001

(9) Install the compressor outlet duct [12].

(a) Apply a light coat of distilled water, G01061, or grease, D00504, or Molykote 44 grease, medium, D50104, to the new packing [5] and new packings [9].

(b) Install the packings [9] on the duct ends.

(c) Install the packing [5] at the heat exchanger.

(d) Put the compressor outlet duct [12] into its position.

(e) Apply compound, D50320 to the bolt threads of the coupling [4].

(f) Install the coupling [4] at the heat exchanger.

1) Do not tighten the coupling [4] fully.

(g) Move the sleeves [8] over the outlet duct connection.

(h) Install the clamshell clamps [7] over the outlet duct connection.

(i) Tighten the nut on the coupling [4] to 45.0 in-lb (5.1 N·m) - 50.0 in-lb (5.6 N·m).

(j) With hand force, gently push the compressor outlet duct in the direction of the adjacent mix muff duct until the compressor outlet duct reaches the maximum flexed position.

(k) Make sure that there is a minimum clearance of 0.20 in. (5.1 mm) to the adjacent mix muff duct when the compressor outlet duct is in the flexed position.

(l) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the flex hose.

(m) Connect the flex hose to the outlet duct [12].

(n) Install the electrical connector [3] for the ram air control sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(o) Install the electrical connector [6] for the compressor discharge overheat switch.

I. Air Cycle Machine Installation Test

SUBTASK 21-51-04-860-013

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-04-860-006

(2) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-04-080-001

(3) Remove the DO NOT OPERATE tags from the L PACK and R PACK switches, on the P5-10 air conditioning panel.

SUBTASK 21-51-04-860-007

(4) Put the applicable L PACK or R PACK switch, on the P5-10 air conditioning panel, to the AUTO position.

SUBTASK 21-51-04-790-001

(5) Do a soap bubble test of all the duct joints at the mix muff and at the air cycle machine [1].

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches, on the P5-10 air conditioning panel, to the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

2) Loosen the clamps.

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps.

5) Put the L PACK and R PACK switches to the AUTO position.

6) Make sure that the leak has been repaired.

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SUBTASK 21-51-04-740-001

(6) Do an IBIT test on the left and right integrated air system controllers (M2957 and M2939).

NOTE: This will clear any fault introduced during maintenance when power is interrupted.

(a) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-04-860-008

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-04-410-004

(2) If the air cycle machine for the right pack was removed, close these access doors in this sequence:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-04-410-002

(3) If the air cycle machine for the left pack was removed, close this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-04-860-009

(4) Put the L PACK and R PACK switches, on the P5-10 air conditioning panel, to the OFF position.

SUBTASK 21-51-04-080-002

(5) Remove the DO NOT OPERATE tags from the BLEED 1 and 2 switches.

SUBTASK 21-51-04-080-003

(6) Remove the DO NOT OPERATE tag from the BLEED APU switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONDITIONED AIR CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the conditioned air check valve

(2) An installation of the conditioned air check valve

(3) A removal of the left pack conditioned air check valve sense line

(4) An installation of the left pack conditioned air check valve sense line.

(5) There is a conditioned air check valve for each pack. The valves are found in the forward end of the air conditioning bays.

TASK 21-51-07-000-801

2. Conditioned Air Check Valve Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-07-860-002

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-07-860-003

(2) Do these steps on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach DO NOT OPERATE tag, STD-858.

(b) Set the BLEED 1 and BLEED 2 switches to the OFF position and attach DO NOT OPERATE tag, STD-858.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Set the BLEED APU switch to the OFF position and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-07-010-001

(3) For the left conditioned air check valve, open this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-07-010-002

(4) For the right conditioned air check valve, open these access doors in this sequence:

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

F. Conditioned Air Check Valve Removal

SUBTASK 21-51-07-020-001

(1) To disconnect the sense line from the conditioned air check valve [1], do these steps:

(a) Loosen the b-nut to disconnect the sense line.

(b) Remove the union [5].

(c) Remove and discard the o-ring [6].

SUBTASK 21-51-07-020-002

(2) Remove the conditioned air check valve [1] as follows:

(a) Make a record of the position of the conditioned air check valve [1] to make sure it is installed in the same position.

(b) Remove the clamp assembly [2] from the conditioned air check valve [1] as follows:

1) Remove the washers [11] and nuts [12].

2) Remove the brackets [13].

3) Loosen the nuts [4] on the clamps [14] of the clamp assembly [2] so that the clamps [14] are free to slide.

4) Slide the flexible duct [3] aft until it is free of the conditioned air check valve [1].

(c) Remove the bolts [10], washers [8], and nuts [9], at 14 locations, that hold the conditioned air check valve [1] to the bulkhead penetration duct assembly [15].

(d) Remove the conditioned air check valve [1].

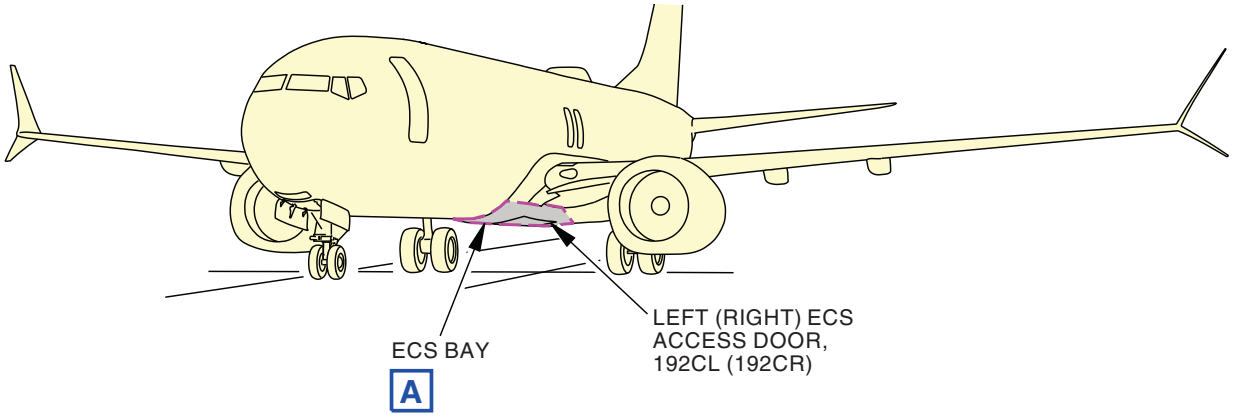
(e) Leave the flexible duct [3] on the condensor outlet duct [16].

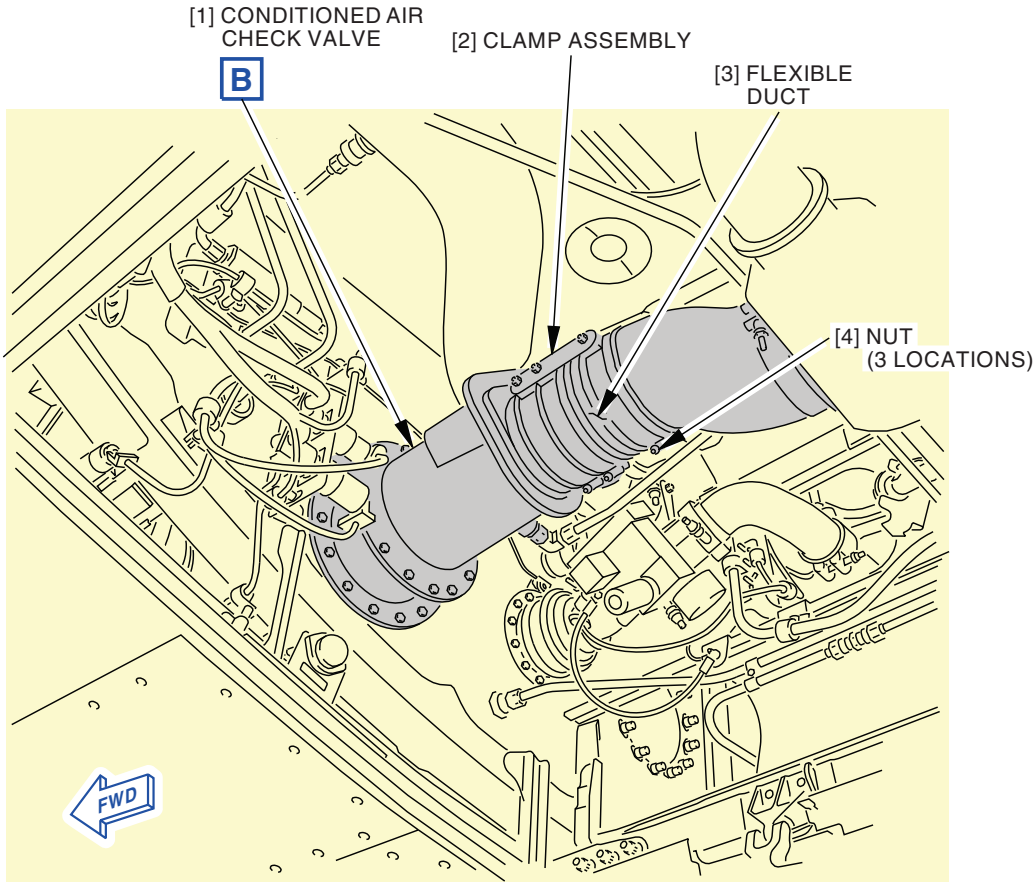
SUBTASK 21-51-07-620-001

(3) Put covers on the duct openings to prevent the entry of unwanted materials.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





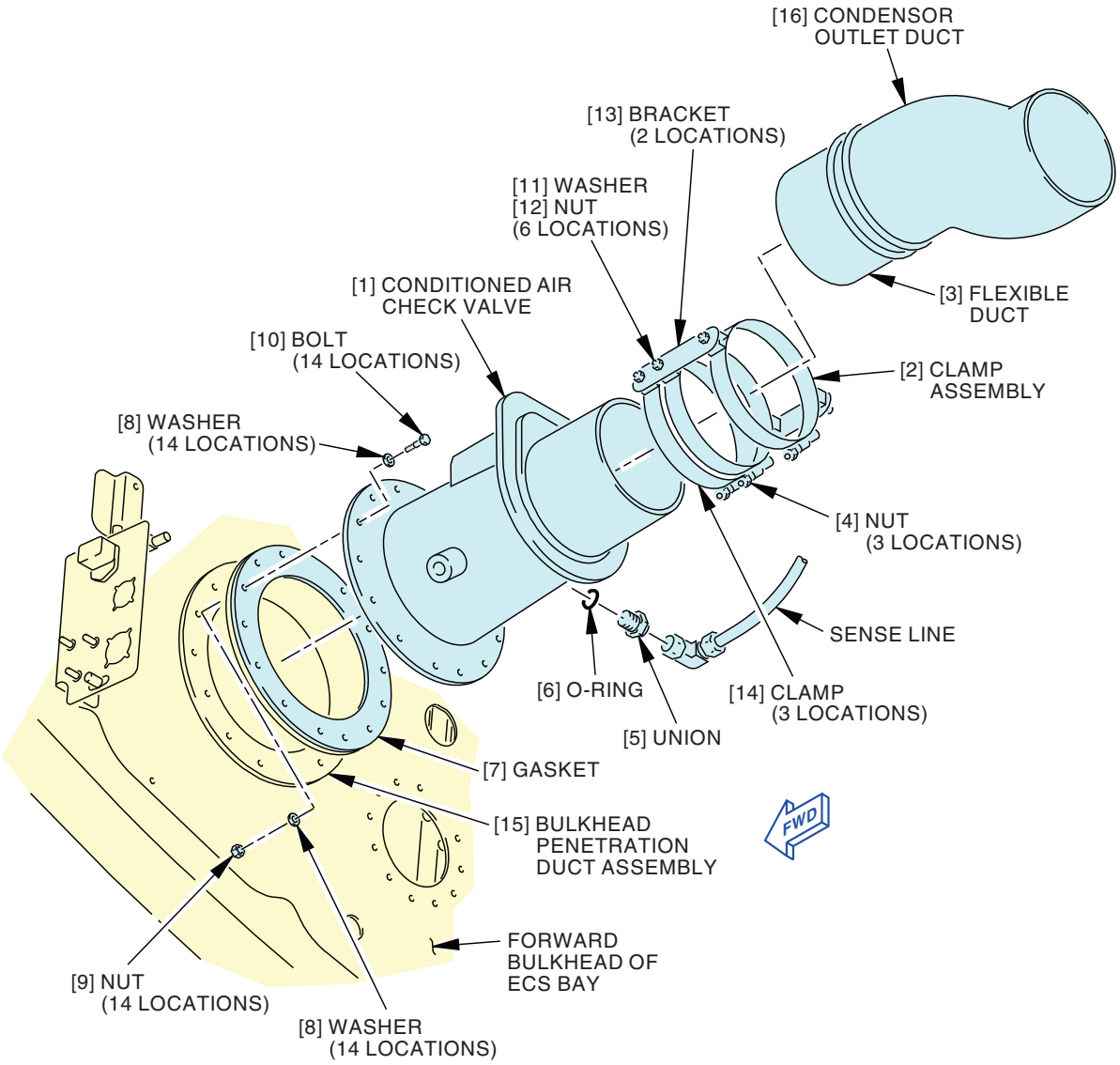
LEFT ECS BAY
(RIGHT ECS BAY IS EQUIVALENT)

A

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Conditioned Air Check Valve Installation
Figure 401/21-51-07-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONDITIONED AIR CHECK VALVE

B

2411409 S00061527836_V1

Conditioned Air Check Valve Installation
Figure 401/21-51-07-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-07-400-801

3. Conditioned Air Check Valve Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Conditioned air check valve	21-51-07-01-050	PTF ALL
		21-51-07-02-050	PTF ALL
6	O-ring	21-51-07-01-045	PTF ALL
		21-51-07-02-075	PTF ALL
7	Gasket	21-51-07-01-017	PTF ALL
		21-51-07-02-040	PTF ALL

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Conditioned Air Check Valve Installation

SUBTASK 21-51-07-630-001

(1) Remove the duct covers.

SUBTASK 21-51-07-420-001

(2) Install the conditioned air check valve [1] as follows:

(a) Examine the gasket [7] on the flange of the bulkhead penetration duct assembly [15].

(b) Replace the gasket [7] if it is damaged or worn.

(c) Put the conditioned air check valve [1] in its position.

1) Make sure that the check valve is installed in the same orientation as when it was removed.

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(d) Install the bolts [10], washers [8], and nuts [9], at 14 locations, that attach the conditioned air check valve [1] to the bulkhead penetration duct assembly [15].

(e) Slide the forward end of the flexible duct [3] forward onto the aft end of the conditioned air check valve [1].

(f) Position the individual clamps [14] so that these conditions are met:

1) Make sure that the clamps are not installed on the duct bead.

2) Keep a 0.125 in. (3.175 mm) minimum distance between the clamps [14] and the edge of the flexible duct [3].

3) Make sure that the two forward clamps [14] are located forward of the bead on the aft end of the conditioned air check valve [1].

4) Make sure that the aft clamp [14] is located aft of the bead on the forward end of the condensor outlet duct [16].

5) Make sure that the forward clamp [14] is located inside (aft) of the bead on the flexible duct [3].

6) Make sure that the aft clamp [14] is located inside (forward) of the bead on the flexible duct [3].

7) Install the brackets [13] that connect the clamps [14] together.

a) If necessary, reposition the clamps [14] axially so that the brackets [13] can be installed.

b) Make sure that the clamps [14] are positioned so that the position requirements are still met.

c) Install the washers [11] and nuts [12] that retain the brackets [13] to the clamps [14].

d) Tighten the nuts [12] to 63 in-lb (7.1 N·m) - 67 in-lb (7.6 N·m).

(g) Install the nuts [4] on the clamps [14].

(h) Determine the run-on torque value of the nuts on the T-bolts for each nut and T-bolt combination as follows:

1) Rotate the nut [4] onto the T-bolt until it is approximately one turn from seating and at least 3 T-bolt threads are protruding from the top of the nut.

2) Measure the torque value required to turn the nut [4] the last revolution before it seats.

3) Make a record of this torque value.

NOTE: The torque value should be between 2 in-lb (0.2 N·m) - 18 in-lb (2 N·m).

(i) Tighten the nuts [4] to 12 in-lb (1.4 N·m) - 14 in-lb (1.6 N·m) plus the run-on torque value determined for each nut [4].

1) For example, if the measured run-on torque value for a nut was 5 in-lb (0.6 N·m), then add 5 in-lb (0.6 N·m) to 12 in-lb (1.4 N·m) - 14 in-lb (1.6 N·m) to determine a final torque value of 17 in-lb (1.9 N·m) - 19 in-lb (2.1 N·m).

SUBTASK 21-51-07-020-003

(3) Do these steps to install the sense line:

(a) Install a new o-ring [6] on the union [5].

(b) Apply a layer of Pure Nickel Special compound, D00006, to the threads of the union [5].

(c) Install the union [5] in the conditioned air check valve [1].

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(d) Connect the sense line to the union [5].

G. Conditioned Air Check Valve Installation Test

SUBTASK 21-51-07-860-004

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-07-860-005

(2) Remove the DO NOT OPERATE tag from these switches on the P5-10 bleed air control module:

(a) BLEED APU

(b) BLEED 1 and BLEED 2

(c) L PACK and R PACK.

SUBTASK 21-51-07-860-006

(3) Put the applicable L PACK or R PACK switch, on the P5-10 bleed air control module, to the AUTO position.

SUBTASK 21-51-07-790-003

(4) Make sure that there are no concentrated air leaks at the ends of the conditioned air check valve [1].

NOTE: A concentrated leak is a leak where you can feel or hear airflow at a distance of less than 12 in. (30 cm) from the source.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-07-010-003

(1) If the left conditioned air check valve was removed, close this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-07-010-004

(2) If the right conditioned air check valve was removed, close these access doors in this sequence:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-07-860-007

(3) Put these switches, on the P5-10 bleed air control module, to the OFF position:

(a) L PACK

(b) R PACK.

SUBTASK 21-51-07-860-010

(4) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-07-000-802

4. Left Pack Conditioned Air Check Valve Sense Line Removal

(Figure 402)

A. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door

E. Prepare for the Removal

SUBTASK 21-51-07-860-011

(1) Make sure that the pneumatic system is off (TASK 36-00-00-860-806).

SUBTASK 21-51-07-860-012

(2) Do these steps, on the P5-10 Bleed Air Control Module (located on the P5 Forward Overhead Panel):

(a) Set the L PACK switch to the OFF position.

1) Attach a DO NOT OPERATE tag, STD-858.

(b) Set the BLEED 1 switch to the OFF position.

1) Attach a DO NOT OPERATE tag, STD-858.

(c) Set the BLEED APU switch to the OFF position.

1) Attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-07-010-005

(3) For the left conditioned air check valve, open this access panel:

Number	Name/Location
192CL	ECS Access Door

F. Left Pack Conditioned Air Check Valve Sense Line Removal

SUBTASK 21-51-07-020-005

(1) Remove the sense line [17], do these steps:

(a) Loosen the coupling nuts [24] to disconnect the sense line [17] from the conditioned air check valve and sense line [18].

(b) Remove the bonding jumper [20] as follows:

1) Remove the nut [19], washers [21], and screw [22] that attach the one end of the bonding jumper [20] to the bonding tab bracket.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the clamp [23] that attach the sense line [17] to the bonding tab bracket.

(d) Remove the sense line [17].

SUBTASK 21-51-07-020-006

(2) Remove the sense line [18], do these steps:

(a) Loosen the coupling nut [31] to disconnect the sense line [18].

(b) Remove the nut [28], washer [29], screw [30], and spacer [27].

1) Remove the clamp [26].

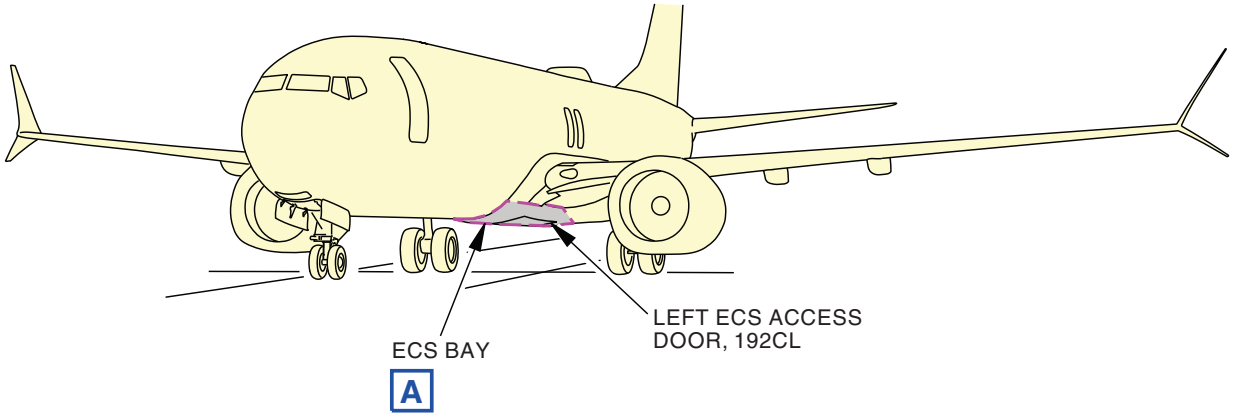
(c) Remove the sense line [18].

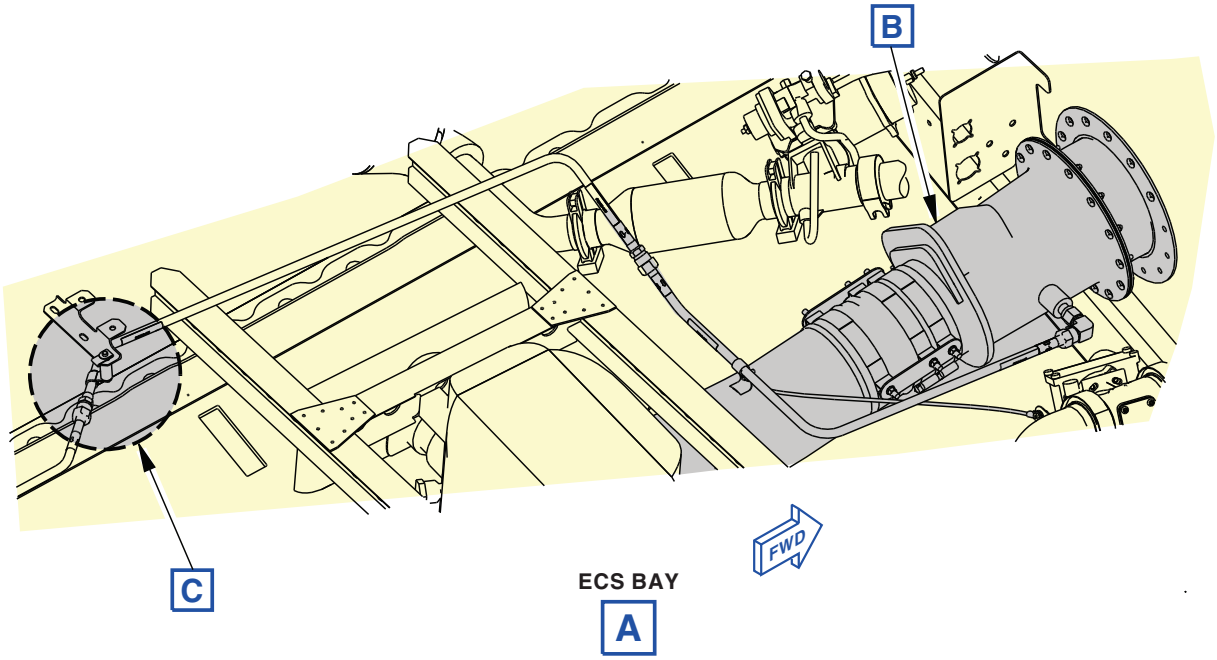
SUBTASK 21-51-07-620-002

(3) Put the covers on the sense line openings to prevent the entry of unwanted materials.

——— END OF TASK ———

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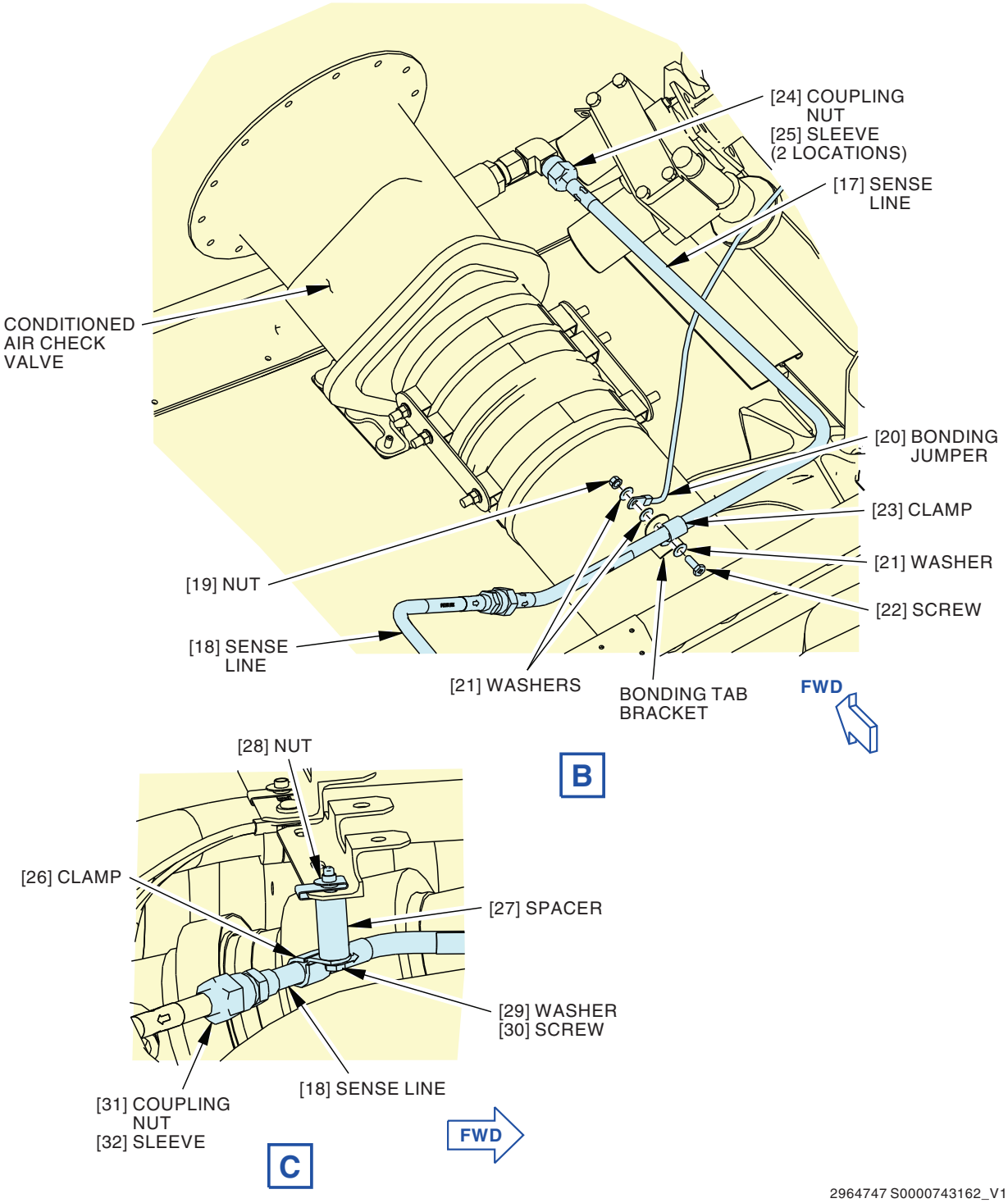




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Left Pack Conditioned Air Check Valve Sense Line Installation
Figure 402/21-51-07-990-802 (Sheet 1 of 2)

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Left Pack Conditioned Air Check Valve Sense Line Installation
Figure 402/21-51-07-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-07-400-802

5. Left Pack Conditioned Air Check Valve Sense Line Installation

(Figure 402)

A. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G50262	Wiper - Cleaning	BMS15-5

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door

F. Left Pack Conditioned Air Check Valve Sense Line Installation

SUBTASK 21-51-07-620-003

(1) Remove the covers from the sense line openings.

SUBTASK 21-51-07-420-002

(2) Install the sense line [18], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT YOU USE TWO WRENCHES TO TIGHTEN TUBE COUPLING NUTS. ONE WRENCH TO HOLD THE FITTING AND THE SECOND WRENCH TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(a) Tighten the coupling nut [31] to 161.5 in-lb (18.2 N·m) - 178.5 in-lb (20.2 N·m).

(b) Install the clamp [26].

1) Install the nut [28], washer [29], screw [30], and spacer [27].

SUBTASK 21-51-07-420-003

(3) Install the sense line [17], do these steps:

CAUTION

MAKE SURE THAT YOU USE TWO WRENCHES TO TIGHTEN TUBE COUPLING NUTS. ONE WRENCH TO HOLD THE FITTING AND THE SECOND WRENCH TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(a) Tighten the coupling nuts [24] to 161.5 in-lb (18.2 N·m) - 178.5 in-lb (20.2 N·m).

(b) Do these steps to install the bonding jumper [20] to the bonding tab bracket:

1) Clean the contact surfaces with a wiper, G50262, soaked with alcohol, B00130.

a) Rub dry with a clean and dry wiper, G50262.

b) Continue to clean and dry the surface until the dry wiper, G50262, stays clean.

2) Install the clamp [23] to the bonding tab bracket.

3) Install the nut [19], washers [21], screw [22], and one end of the bonding jumper [20].

4) Tighten the nut [19] to 33 in-lb (3.7 N·m) - 40 in-lb (4.5 N·m).

(c) Measure the electrical bonding resistance between the bonding jumper [20] and airplane skin (SWPM 20-20-00).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

G. Left Pack Conditioned Air Check Valve Sense Line Installation Test

SUBTASK 21-51-07-860-013

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-07-020-007

(2) Remove the DO NOT OPERATE tag from L PACK switch, on the P5-10 Bleed Air Control Module (located on the P5 Forward Overhead Panel).

SUBTASK 21-51-07-860-014

(3) Put the applicable L PACK switch, on the P5-10 Bleed Air Control Module, to the AUTO position.

SUBTASK 21-51-07-790-002

(4) Make sure that there are no concentrated air leaks at the joints at the ends of the sense line [17] and sense line [18].

NOTE: A concentrated leak is a leak where you can feel or hear airflow at a distance of less than 12 in. (30 cm) from the source.

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(a) If there is leakage, do these steps:

1) Put the L PACK switch, on the P5-10 Bleed Air Control Module, to the OFF position.

CAUTION MAKE SURE THAT YOU USE TWO WRENCHES TO LOOSEN TUBE COUPLING NUTS. USE ONE WRENCH TO HOLD THE FITTING. USE THE OTHER WRENCH TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

2) Loosen the coupling nuts [24] and coupling nut [31].

3) Make sure that the sense line [17] and sense line [18] are aligned.

CAUTION MAKE SURE THAT YOU USE TWO WRENCHES TO TIGHTEN TUBE COUPLING NUTS. ONE WRENCH TO HOLD THE FITTING AND THE SECOND WRENCH TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

4) Tighten the coupling nuts [24] and coupling nut [31] to 161.5 in-lb (18.2 N·m) - 178.5 in-lb (20.2 N·m).

5) Put the L PACK switch, on the P5-10 Bleed Air Control Module, to the AUTO position.

6) Make sure that the leak has been repaired.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-07-410-001

(1) For the left conditioned air check valve, close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-07-860-015

(2) Remove the DO NOT OPERATE tags from these switches, on the P5-10 Bleed Air Control Module:

(a) BLEED APU

(b) BLEED 1.

SUBTASK 21-51-07-860-016

(3) Put L PACK switch, on the P5-10 Bleed Air Control Module, to the OFF position.

SUBTASK 21-51-07-860-017

(4) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIGH PRESSURE WATER SEPARATOR (HPWS) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the High Pressure Water Separator (HPWS)

(2) An installation of the HPWS.

B. The HPWS contains: a condenser, a reheater, two water extractors, a manifold assembly, and a frame assembly.

C. The HPWS is referred to as the water separator in this procedure.

TASK 21-51-08-000-801

2. High Pressure Water Separator Removal

(Figure 401)

A. General

(1) This task provides steps to remove the High Pressure Water Separator (HPWS).

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for the Removal

SUBTASK 21-51-08-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-08-860-002

(2) Do these steps on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach DO NOT OPERATE tags, STD-858.

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(c) Set the BLEED APU switch to the OFF position and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-08-860-008

(3) For the left high pressure water separator [1], open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-08-860-009

(4) For the right high pressure water separator [1], open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-08-860-010

(5) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

SUBTASK 21-51-08-010-001

(6) To get access to the HPWS for the left cooling pack, open this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-08-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(7) To get access to the HPWS for the right cooling pack, open these access doors in this sequence:

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

G. High Pressure Water Separator Removal

SUBTASK 21-51-08-010-003

(1) For the water separator [1] at the left cooling pack do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the screw [3], washers [4], nut [5], and clamps [36].

SUBTASK 21-51-08-020-005

(2) If it is necessary, remove the screw [37], washers [45], spacer [38], nut [47], and clamps [36] from the sense line [2] and water extractor duct drain line [6].

SUBTASK 21-51-08-010-020

(3) Loosen the B-nut of the sense line [2] to disconnect the sense line [2] from the union [28].

SUBTASK 21-51-08-010-004

(4) Remove the sense line [2].

SUBTASK 21-51-08-010-016

(5) If it is necessary, remove the union [28] and packing [29] from the water separator [1].

(a) Discard the packing [29].

SUBTASK 21-51-08-010-021

(6) Loosen the B-nut of the water extractor duct drain line [6] to disconnect the water extractor duct drain line [6] from the union [30].

SUBTASK 21-51-08-010-005

(7) Remove the water extractor duct drain line [6].

SUBTASK 21-51-08-010-017

(8) If it is necessary, remove the union [30] and packing [31] from the water separator [1].

(a) Discard the packing [31].

SUBTASK 21-51-08-010-006

(9) Do these steps to disconnect the lines for the water spray nozzle from the manifold:

(a) Loosen the B-nut of the air line elbow [39] to disconnect the air line elbow [39] from the union [32].

(b) Loosen the B-nut of the water line elbow [40] to disconnect the water line elbow [40] from the union [41].

SUBTASK 21-51-08-010-018

(10) If it is necessary, remove the union [32], union [41], packing [33], and packing [42] from the water separator [1].

(a) Discard the packing [33] and packing [42].

SUBTASK 21-51-08-010-007

(11) To remove the coupling assembly from the condenser, do these steps:

(a) Remove the coupling [7].

(b) Move the sleeve [8] and ring [9] onto the duct.

(c) Remove the retainer [10].

SUBTASK 21-51-08-010-008

(12) For the water separator [1] at the left cooling pack, do this step:

(a) Remove the clamps [11] from the flexible duct [12].

SUBTASK 21-51-08-010-023

(13) For the water separator [1] at the right cooling pack, do this step:

(a) Remove the clamp [11] from the flexible duct [12].

SUBTASK 21-51-08-010-009

(14) Remove the flexible duct [12].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-08-010-010

(15) Loosen the B-nut of the condensor outlet sense line [25] to disconnect the condensor outlet sense line [25] from the union [34].

SUBTASK 21-51-08-010-019

(16) If it is necessary, remove the union [34] and packing [35] from the water separator [1].

(a) Discard the packing [35].

SUBTASK 21-51-08-010-011

(17) Loosen the B-nut of the de-icing hose [26] to disconnect the de-icing hose [26] from the union [43].

SUBTASK 21-51-08-010-022

(18) If it is necessary, remove the union [43] and packing [44] from the water separator [1].

(a) Discard the packing [44].

SUBTASK 21-51-08-020-006

(19) Disconnect the electrical connectors [27] from the pack temperature sensor and pack temperature bulb.

SUBTASK 21-51-08-020-003

(20) Do these steps to disconnect the duct [48] and duct [49] from the reheater:

(a) Remove the bolts [13], washers [14], washers [15], and nuts [16].

(b) Remove and discard the O-rings [17] from the duct [48] and duct [49].

SUBTASK 21-51-08-020-004

WARNING

BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. INJURIES TO PERSONS CAN OCCUR.

(21) To remove the water separator [1] from the airplane, do these steps:

(a) For each of the two forward attach points, do the steps that follow:

1) Remove the nut [21], washer [20], countersunk washer [19], and bolt [18].

2) Remove the bushing [22] from the bracket.

(b) For each of the two aft attach points, do the steps that follow:

1) Remove the nut [21], washer [20], countersunk washer [19], and bolt [23].

2) Remove the two bushings [24] from the bracket.

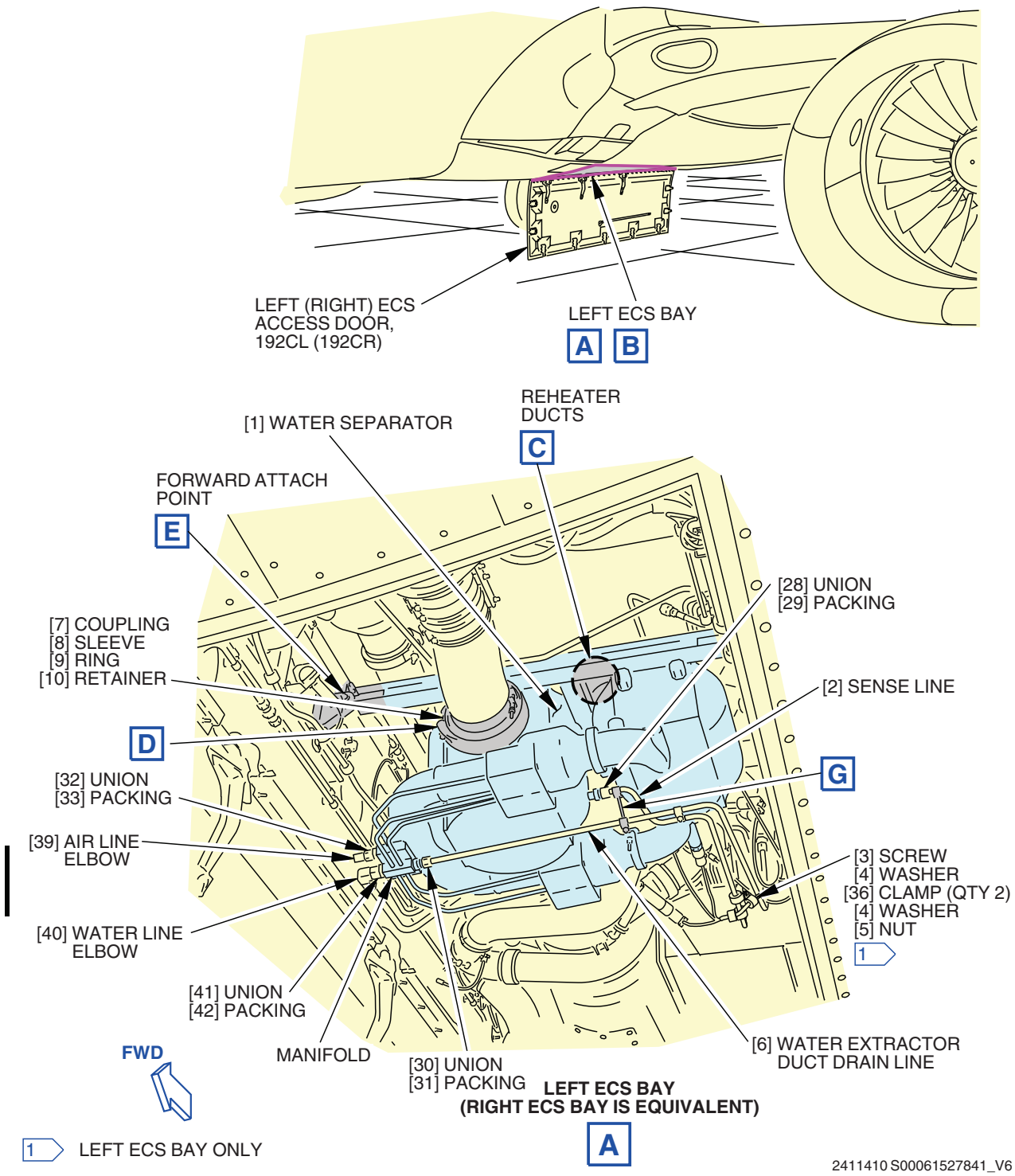
(c) Carefully lower the water separator [1] from the Environmental Control System (ECS) bay.

SUBTASK 21-51-08-480-001

(22) Put covers on the ducts and lines to keep out unwanted materials.

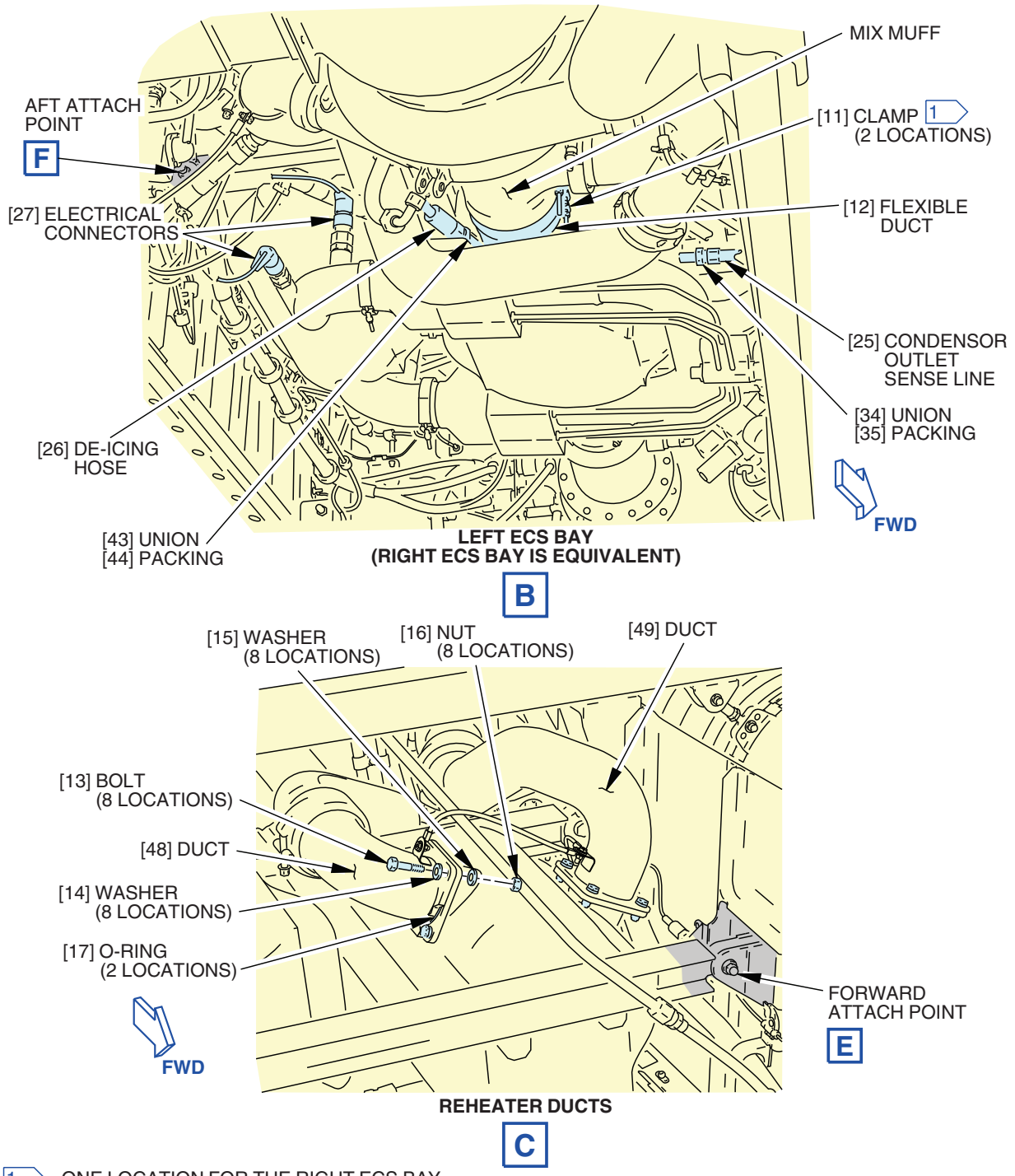
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Water Separator Installation
Figure 401/21-51-08-990-801 (Sheet 1 of 3)

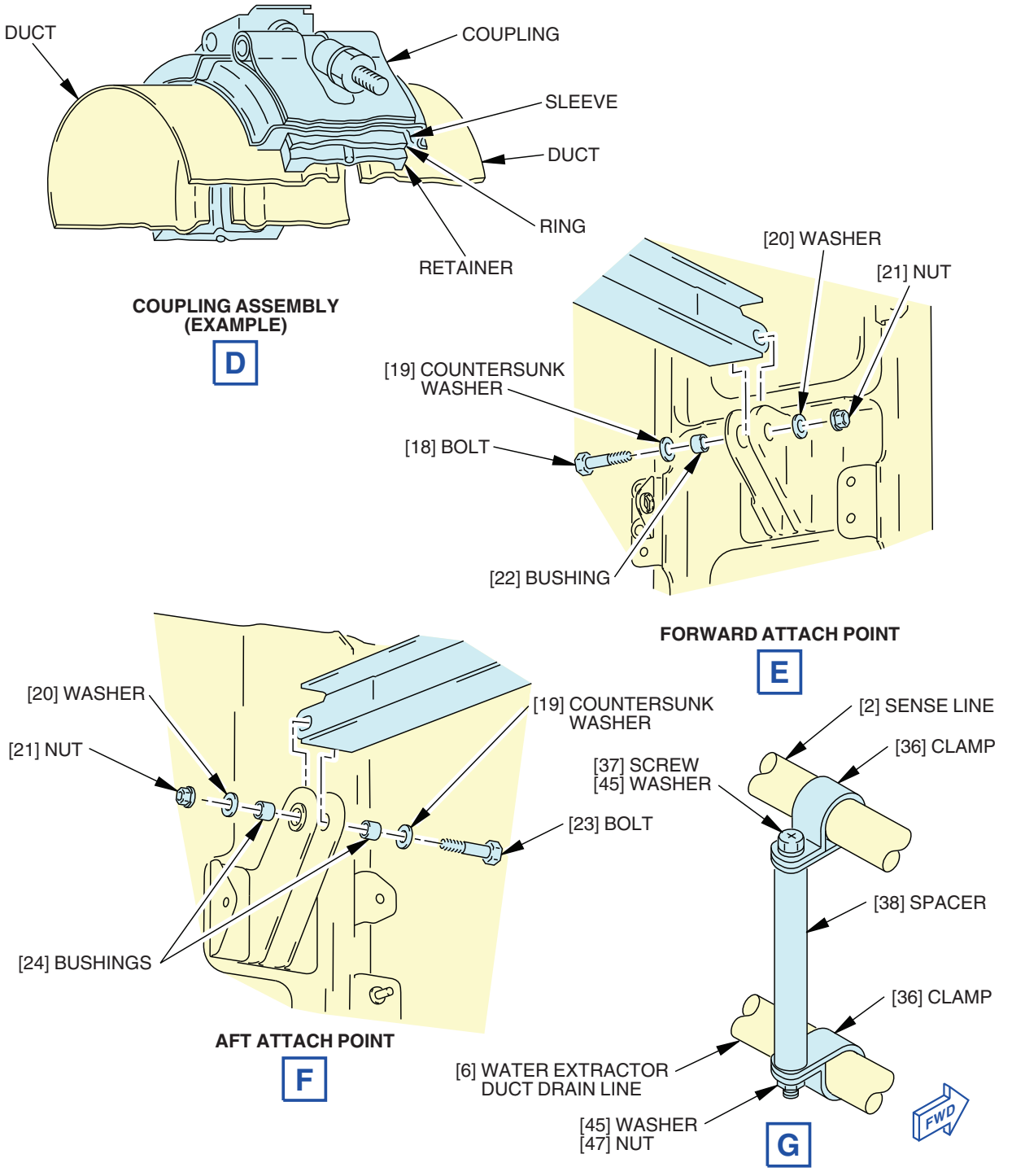
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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High Pressure Water Separator Installation
Figure 401/21-51-08-990-801 (Sheet 2 of 3)

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High Pressure Water Separator Installation
Figure 401/21-51-08-990-801 (Sheet 3 of 3)

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TASK 21-51-08-400-801

3. High Pressure Water Separator Installation

(Figure 401)

A. General

(1) This task provides steps to install the High Pressure Water Separator (HPWS).

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00548	Lubricant - Fluorolube GR-660	BAC5506
D00602	Lubricant - Fluorinated	BMS3-25
D50063	Grease - Perfluoropolyether, fuel and oxygen resistant - Krytox 240AC	MIL-PRF- 27617 Type III
D50203	Grease - Castrol Braycote 601EF	BMS3-25 Type II
D50233	Grease - Dupont Krytox 240AZ	MIL-PRF-27617 Type I
D50234	Grease - Dupont Krytox 240AB	MIL-PRF-27617 Type II
D50268	Lubricant - Castrol Braycote 815Z	BMS3-25 Type I
D50380	Grease - Braycote Micronic 601EF	BMS3-25 Type III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50411	Torque Stripe - EC-1252 [pink or yellow (as required per BAC5001-10)]	BMS8-45 Type I
G51905	Putty/Lacquer - Tamper-proof, Hydraulic Fluid Resistant, Yellow	BMS8-45 Type III

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Water separator	21-51-08-01-170	PTF ALL
		21-51-08-05-185	PTF ALL
17	O-ring	21-51-08-01-065	PTF ALL
		21-51-08-05-065	PTF ALL
		21-51-41-09-030	PTF ALL
29	Packing	21-51-08-01-150	PTF ALL
		21-51-08-05-165	PTF ALL
31	Packing	21-51-08-01-150	PTF ALL
		21-51-08-05-165	PTF ALL
33	Packing	21-51-08-01-145	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
33 (cont.)		21-51-08-05-160	PTF ALL
35	Packing	21-51-08-01-150	PTF ALL
		21-51-08-05-170	PTF ALL
42	Packing	21-51-08-01-150	PTF ALL
		21-51-08-05-165	PTF ALL
44	Packing	21-51-08-01-155	PTF ALL
		21-51-08-05-170	PTF ALL

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. High Pressure Water Separator Installation

SUBTASK 21-51-08-480-002

(1) Remove the covers from the ducts and lines.

SUBTASK 21-51-08-420-001

WARNING BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. INJURIES TO PERSONS CAN OCCUR.

(2) To install the water separator [1] on the airplane, do these steps:

(a) Carefully position the water separator [1] in the Environmental Control System (ECS) bay.

(b) For each of the two forward attach points do the steps that follow:

1) Install the bushing [22] in the bracket.

2) Install the nut [21], washer [20], countersunk washer [19], and bolt [18].

a) Position the countersunk side of the countersunk washer [19] under the bolt head.

(c) For each of the two aft attach points do the steps that follow:

1) Install the bushings [24] in the bracket.

2) Install the nut [21], washer [20], countersunk washer [19], and bolt [23].

a) Position the countersunk side of the countersunk washer [19] under the bolt head.

(d) For the right water separator [1], make sure that there is a minimum of 0.10 in. (2.54 mm) clearance between the supply duct to the trim air Pressure Regulating and Shutoff Valve (PRSOV) and water collection manifold.

1) If there is not a minimum of 0.10 in. (2.54 mm) clearance, do one of these steps to provide the minimum clearance:

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a) Reposition the supply duct to the trim air PRSOV.

b) Remove material from the water collection manifold.

SUBTASK 21-51-08-420-002

(3) Do these steps to connect the duct [48] and duct [49] to the reheater:

(a) Wipe the new O-rings [17] with clean and dry cotton wiper, G00034, to remove foreign objects.

(b) Apply a thin coat of one of these lubricants to the new O-rings [17]:

1) Braycote Micronic 601EF grease, D50380

2) Braycote 601EF Grease, D50203

3) Braycote 815Z lubricant, D50268

4) Fluorolube GR-660 lubricant, D00548

5) Krytox 240AB grease, D50234

6) Krytox 240AZ grease, D50233

7) Krytox 240AC perfluoropolyether grease, D50063.

(c) Install the O-ring [17] for the duct [48] and duct [49].

(d) Position the duct [48] and duct [49] on their respective locations.

(e) Install the bolts [13], washers [14], washers [15], and nuts [16].

SUBTASK 21-51-08-420-003

(4) Install the flexible duct [12] on the mix muff and the water separator [1].

SUBTASK 21-51-08-420-004

(5) For the water separator [1] at the left cooling pack, do these steps:

(a) Apply Pure Nickel Special compound, D00006, to the T-bolts threads of the clamps [11].

(b) Tighten the clamps [11] to 13 in-lb (1.5 N·m) - 17 in-lb (1.9 N·m).

(c) Apply an approximately 0.125 in. (3.18 mm) torque stripe, G50411, or tamper-proof putty/lacquer, G51905, to the clamps [11].

SUBTASK 21-51-08-420-023

(6) For the water separator [1] at the right cooling pack, do these steps:

(a) Apply Pure Nickel Special compound, D00006, to the T-bolts threads of the clamp [11].

(b) Tighten the clamp [11] to 13 in-lb (1.5 N·m) - 17 in-lb (1.9 N·m).

(c) Apply an approximately 0.125 in. (3.18 mm) torque stripe, G50411, or tamper-proof putty/lacquer, G51905, to the clamp [11].

SUBTASK 21-51-08-420-019

(7) If it is necessary, do these steps to install the union [43]:

(a) If it is installed, remove and discard a plug from the opening on the water separator [1].

(b) Wipe the new packing [44] with clean and dry cotton wiper, G00034, to remove foreign objects.

(c) Lubricate the new packing [44] with lubricant, D00602, or Fluorolube GR-660 lubricant, D00548, or Krytox 240AC perfluoropolyether grease, D50063, or Krytox 240AZ grease, D50233, or Krytox 240AB grease, D50234, or equivalent.

(d) Install the packing [44] on the union [43].

(e) Lubricate the threads of the union [43] with Pure Nickel Special compound, D00006.

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(f) Install the union [43] on the water separator [1].

1) Tighten the union [43] to 280 ±28 in-lb (32 ±3 N·m).

SUBTASK 21-51-08-420-005

(8) Install the B-nut of the de-icing hose [26] on the union [43].

(a) Make sure that the plug is removed.

(b) Tighten the B-nut of the de-icing hose [26] to 280 ±14 in-lb (32 ±2 N·m).

SUBTASK 21-51-08-420-014

(9) If it is necessary, do these steps to install the union [34]:

(a) Wipe the new packing [35] with clean and dry cotton wiper, G00034, to remove foreign objects.

(b) Lubricate the new packing [35] with lubricant, D00602, or Fluorolube GR-660 lubricant, D00548, or Krytox 240AC perfluoropolyether grease, D50063, or Krytox 240AZ grease, D50233, or Krytox 240AB grease, D50234, or equivalent.

(c) Install the packing [35] on the union [34].

(d) Lubricate the threads of the union [34] with Pure Nickel Special compound, D00006.

(e) Install the union [34] on the water separator [1].

1) Tighten the union [34] to 170 ±17 in-lb (19 ±2 N·m).

SUBTASK 21-51-08-420-006

(10) Install the B-nut of the condensor outlet sense line [25] on the union [34].

(a) Tighten the B-nut of the condensor outlet sense line [25] to 170 ±9 in-lb (19 ±1 N·m).

SUBTASK 21-51-08-420-021

(11) Connect the electrical connectors [27] to the pack temperature sensor and pack temperature bulb.

SUBTASK 21-51-08-420-009

(12) To install the coupling assembly on the condenser, do these steps:

(a) Put the retainer [10] over the duct bead and the bead on the condenser.

(b) Move the sleeve [8] and ring [9] over the retainer [10].

(c) Install the coupling [7].

1) Tighten the coupling [7] to 100 ±5 in-lb (11 ±1 N·m).

SUBTASK 21-51-08-420-015

(13) If it is necessary, do these steps to install the union [32] and union [30]:

(a) Wipe the new packing [33] with clean and dry cotton wiper, G00034, to remove foreign objects.

(b) Lubricate the new packing [33] and new packing [31] with lubricant, D00602, or Fluorolube GR-660 lubricant, D00548, or Krytox 240AC perfluoropolyether grease, D50063, or Krytox 240AZ grease, D50233, or Krytox 240AB grease, D50234, or equivalent.

(c) Install the packing [33] on the union [32].

(d) Install the packing [31] on the union [30].

(e) Lubricate the threads of the union [32] and union [30] with Pure Nickel Special compound, D00006.

(f) Install the union [32] and union [30] on the water separator [1].

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1) Tighten the union [30] to 170 ±17 in-lb (19 ±2 N·m).

2) Tighten the union [32] to 140 ±14 in-lb (16 ±2 N·m).

SUBTASK 21-51-08-420-020

(14) If it is necessary, do these steps to install the union [41] on the lower port of the manifold:

(a) Wipe the new packing [42] with clean and dry cotton wiper, G00034, to remove foreign objects.

(b) Lubricate the new packing [42] with lubricant, D00602, or Fluorolube GR-660 lubricant, D00548, or Krytox 240AC perfluoropolyether grease, D50063, or Krytox 240AZ grease, D50233, or Krytox 240AB grease, D50234, or equivalent.

(c) Install the packing [42] on the union [41].

(d) Lubricate the threads of the union [41] with Pure Nickel Special compound, D00006.

(e) Install the union [41] on the water separator [1].

1) Tighten the union [41] to 170 ±17 in-lb (19 ±2 N·m).

SUBTASK 21-51-08-420-010

(15) Do these steps to connect the lines for the water spray nozzle to the manifold:

(a) Install the B-nut of the air line elbow [39] on the union [32].

1) Tighten the B-nut of the air line elbow [39] to 170 ±9 in-lb (19 ±1 N·m).

(b) Install the B-nut of the water line elbow [40] on the union [41].

1) Tighten the B-nut of the water line elbow [40] to 280 ±14 in-lb (32 ±2 N·m).

SUBTASK 21-51-08-210-001

(16) Make sure that there is not a plastic plug in the water overflow port in the water collection manifold:

NOTE: This port must be open in case the water spray nozzle becomes clogged.

(a) Remove the plastic plug if it is installed.

SUBTASK 21-51-08-420-016

(17) If it is necessary, do these steps to install the union [28]:

(a) Wipe the new packing [29] with clean and dry cotton wiper, G00034, to remove foreign objects.

(b) Lubricate the new packing [29] with lubricant, D00602, or Fluorolube GR-660 lubricant, D00548, or Krytox 240AC perfluoropolyether grease, D50063, or Krytox 240AZ grease, D50233, or Krytox 240AB grease, D50234, or equivalent.

(c) Install the packing [29] on the union [28].

(d) Lubricate the threads of the union [28] with Pure Nickel Special compound, D00006.

(e) Install the union [28] on the water separator [1].

1) Tighten the union [28] to 170 ±17 in-lb (19 ±2 N·m).

SUBTASK 21-51-08-420-011

(18) Install the B-nut of the sense line [2] on the union [28].

(a) Tighten the B-nut of the sense line [2] to 170 ±9 in-lb (19 ±1 N·m).

SUBTASK 21-51-08-420-012

(19) Install the B-nut of the water extractor duct drain line [6] on the union [30].

(a) Tighten the B-nut of the water extractor duct drain line [6] to 170 ±9 in-lb (19 ±1 N·m).

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SUBTASK 21-51-08-420-018

(20) If it is necessary, install the clamps [36], screw [37], washers [45], spacer [38], and nut [47] between the sense line [2] and water extractor duct drain line [6].

SUBTASK 21-51-08-420-013

(21) For the water separator [1] on the left cooling pack do this step:

(a) Install the clamps [36], screw [3], washers [4], and nut [5].

H. Post Installation Test

SUBTASK 21-51-08-860-004

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-08-860-011

(2) For the left high pressure water separator [1], remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-08-860-012

(3) For the right high pressure water separator [1], remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-08-860-013

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

SUBTASK 21-51-08-860-005

(5) Do these steps on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Remove the DO NOT OPERATE tags and set the applicable L or R PACK switch to the AUTO position.

(b) Remove the DO NOT OPERATE tags and set the BLEED 1 and 2 switches to the ON position.

(c) Remove the DO NOT OPERATE tag and set the BLEED APU switch to the ON position.

SUBTASK 21-51-08-790-001

(6) Do a soap bubble test of all the applicable duct joints.

NOTE: No leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches in the OFF position.

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WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the clamps.
3) Make sure that the ducts are aligned at the joints.
4) Tighten the clamps.
5) Put the L PACK and R PACK switches in the AUTO position.
6) Make sure that the leak has been repaired.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-08-860-006

(1) Do these steps on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position.

SUBTASK 21-51-08-860-007

(2) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-08-410-001

(3) If the HPWS for the right cooling pack was removed, close these access doors in this sequence:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-08-410-002

(4) If the HPWS for the left cooling pack was removed, close this access door:

Number	Name/Location
192CL	ECS Access Door

——— END OF TASK ———

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TEMPERATURE CONTROL VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the temperature control valve

(2) Installation of temperature control valve.

B. There are two temperature control valves, one for each cooling pack. The temperature control valves are located in the aft area of each air conditioning bay, inboard of the heat exchanger.

TASK 21-51-10-000-801

2. Temperature Control Valve Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-10-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-10-860-002

(2) Put the L PACK and R PACK switches, on the P5-10 panel, in the OFF position and attach DO NOT OPERATE tags, STD-858.

SUBTASK 21-51-10-860-003

(3) For the left temperature control valve [1], do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

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(b) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-10-010-001

(4) For the right temperature control valve [1], do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(b) Open these access panels in the sequence that follows:

(TASK 06-41-00-800-801)

1) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Temperature Control Valve Removal

SUBTASK 21-51-10-020-001

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(1) Disconnect the electrical connector [7] from the temperature control valve [1].

SUBTASK 21-51-10-020-002

(2) Remove the screw [3], washers [4], and nut [5] that hold the bonding jumper [2] to the temperature control valve [1].

SUBTASK 21-51-10-020-003

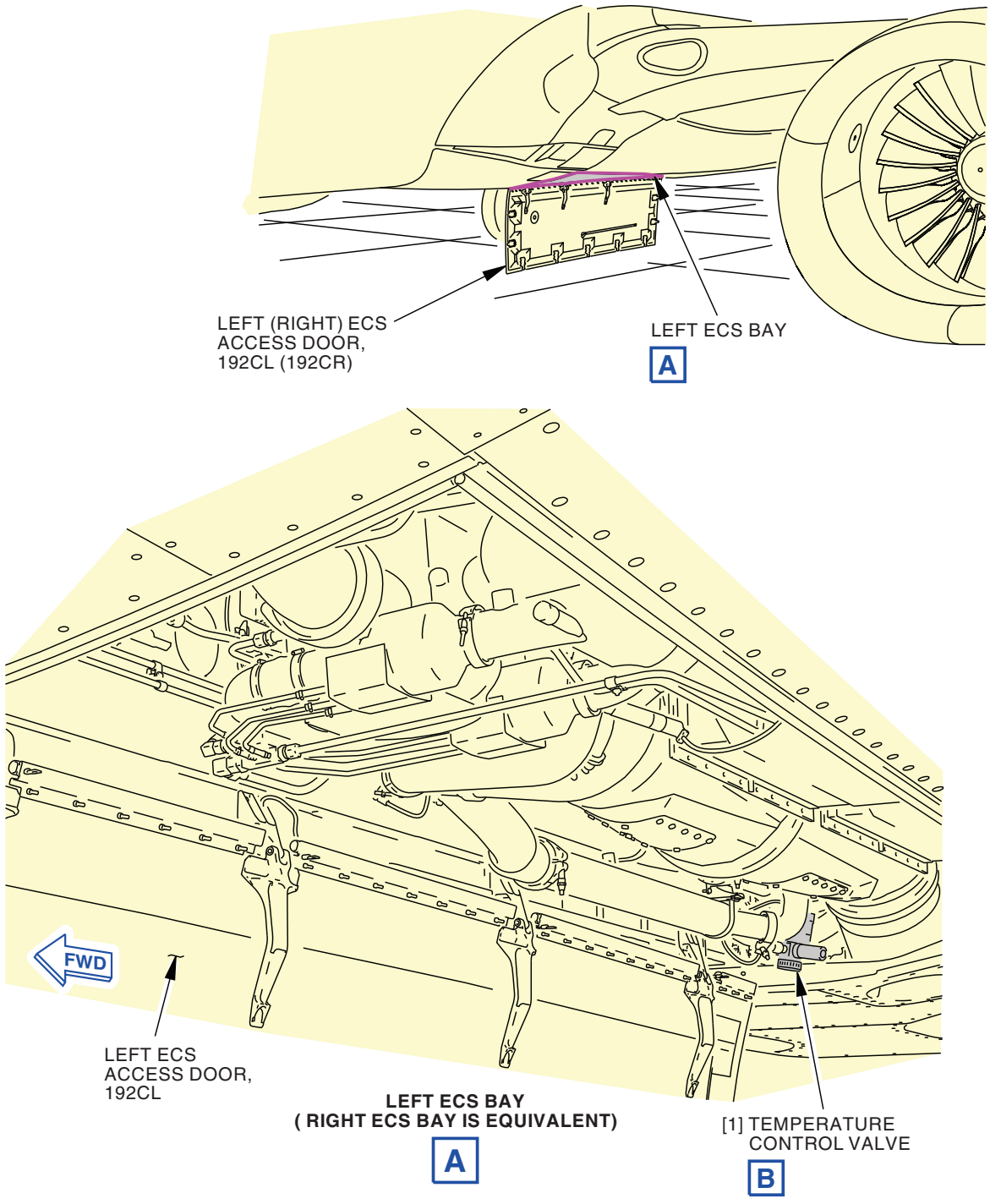
(3) Hold the temperature control valve [1] while you remove the two clamps [6].

SUBTASK 21-51-10-160-001

(4) Carefully remove the temperature control valve [1] from its position between the ducts.

END OF TASK

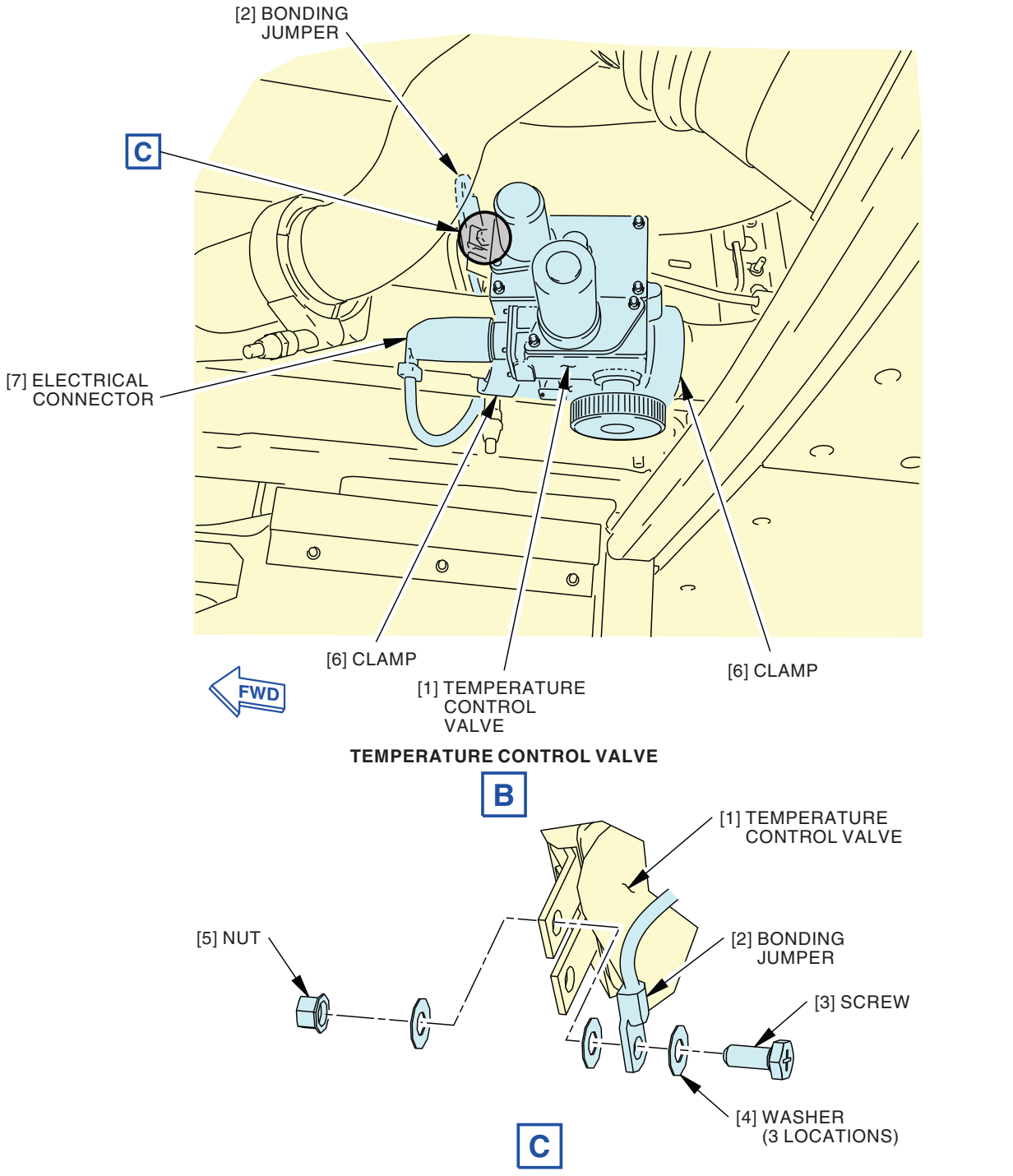
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Temperature Control Valve Installation
Figure 401/21-51-10-990-801 (Sheet 1 of 2)

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Temperature Control Valve Installation
Figure 401/21-51-10-990-801 (Sheet 2 of 2)

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TASK 21-51-10-400-801

3. Temperature Control Valve Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B50262	Alcohol - Isopropyl, grade A (water content 0.1 percent max.)	TT-I-735
G51677	Cloth - 100% Synthetic or Blended Synthetic, Cotton or Cellulose Material	AMS3819 Class 1, 2, or 4, Grade A or B, Form 1

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

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F. Temperature Control Valve Installation

SUBTASK 21-51-10-160-002

(1) Carefully put the temperature control valve [1] into its position between the air ducts.

SUBTASK 21-51-10-020-004

(2) Install the two clamps [6] that hold the temperature control valve [1] to the ducts.

NOTE: Do not fully tighten the clamps.

SUBTASK 21-51-10-420-001

(3) Do these steps to connect the bonding jumper [2] to the temperature control valve [1]:

(a) Clean the terminal of the bonding jumper [2] and washers [4] with cloth, G51677, soaked with a solvent, B00062, or alcohol, B50262.

(b) Put the bonding jumper [2] in its position on the temperature control valve [1].

(c) Install the screw [3], washers [4], and nut [5] that hold the bonding jumper [2] to the temperature control valve [1].

1) Tighten the nut [5] to 35 in-lb (4.0 N·m) - 45 in-lb (5.1 N·m).

(d) Measure the electrical bonding resistance between the temperature control valve [1] and bonding jumper [2].

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.002 ohm (2 milliohm) or less.

SUBTASK 21-51-10-420-002

(4) Install the electrical connector [7] on the temperature control valve [1].

SUBTASK 21-51-10-420-003

(5) Tighten the clamps [6] to 57.5 ±2.5 in-lb (6.5 ±0.3 N·m).

(a) Make sure that the loop bands do not touch.

(b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-51-10-860-004

(6) If the left temperature control valve [1] was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-10-860-005

(7) If the right temperature control valve [1] was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Temperature Control Valve Installation Test

SUBTASK 21-51-10-860-006

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-51-10-730-001

(2) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

SUBTASK 21-51-10-860-007

(3) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-10-860-008

(4) On the P5-10 air conditioning panel, do these steps:

(a) Set the L and R PACK switches to the AUTO position and remove the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the ON position.

(c) Set the BLEED APU switch to the ON position.

SUBTASK 21-51-10-860-009

(5) On the P5-17 cabin temperature panel, do these steps:

(a) Set the TRIM AIR switch to the ON position.

(b) Set the cabin temperature selectors (CONT CAB, FWD CAB, and AFT CAB) to the AUTO position.

SUBTASK 21-51-10-790-001

(6) Do a soap bubble test of the duct joints at the temperature control valve [1].

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches to the OFF position.

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the clamps [6].

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps [6] to 57.5 ±2.5 in-lb (6.5 ±0.3 N·m).

a) Make sure that the loop bands do not touch.

b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

5) Put the L PACK and R PACK switches to the AUTO position.

6) Make sure that the leak has been repaired.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-10-860-010

(1) For the left temperature control valve [1], close this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-10-010-002

(2) For the temperature control valve [1], close these access panels in the sequence that follows: (TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-10-860-011

(3) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-10-860-012

(4) Remove electrical power if it is not necessary (TASK 24-22-00-860-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STANDBY TEMPERATURE CONTROL VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the standby temperature control valve

(2) An installation of standby temperature control valve.

B. There are two standby temperature control valves, one for each cooling pack. The valves are found in the aft, outboard area of each air conditioning bay.

TASK 21-51-11-000-801

2. Standby Temperature Control Valve Removal

(Figure 401)

A. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

E. Prepare for the Removal

SUBTASK 21-51-11-860-002

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-11-860-003

(2) Put the L PACK and R PACK switches, on the P5-10 panel, in the OFF position and attach DO NOT OPERATE tags, STD-858.

SUBTASK 21-51-11-860-004

(3) For the left standby temperature control valve, open these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-11-860-019

(4) For the left standby temperature control valve, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	3	C01613	AIR COND PACK CONT VALVES L ALT

SUBTASK 21-51-11-010-001

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) For the right standby temperature control valve, do these steps:

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Open this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

SUBTASK 21-51-11-860-020

(6) For the right standby temperature control valve, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
D	4	C01614	AIR COND PACK CONT VALVES R ALT

F. Standby Temperature Control Valve Removal

SUBTASK 21-51-11-020-001

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the electrical connector [3] from the standby temperature control valve [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-11-020-002

(2) Do these steps to disconnect the four sense lines from the valve:

(a) Make a record of the sense line connections to make sure they are installed in the same locations.

(b) Remove the screw [15] and washer [16] that hold the acm turbine muffler sense line [4] clamp to the structure.

(c) Remove the screw [5], spacer [22], washers [6], and nut [7] that hold the clamps together.

(d) Remove the sense line for the conditioned air check valve sense line [11].

(e) Remove the sense line for the condensor outlet pressure sense line [10].

(f) Remove the sense line for the acm turbine muffler sense line [4].

(g) Remove the sense line for the condensor inlet pressure sense line [9].

SUBTASK 21-51-11-020-003

(3) Do the applicable step to remove the bonding jumper(s).

(a) On the LH standby temperature control valve [1], remove the screw [14], washers [13], nut [12], bonding jumper [2], and bonding jumper [21].

(b) On the RH standby temperature control valve [1], remove the screw [14], washers [13], nut [12], bonding jumper [2], and bonding jumper [21].

SUBTASK 21-51-11-020-004

(4) Hold the standby temperature control valve [1] while you remove the two clamps [8].

SUBTASK 21-51-11-160-001

(5) Carefully remove the standby temperature control valve [1] from its position between the ducts.

SUBTASK 21-51-11-020-005

(6) Remove the reducer [19] from the conditioned air check valve sense line [11] location.

SUBTASK 21-51-11-020-006

(7) Remove the unions [17] at the remaining three sense line locations.

SUBTASK 21-51-11-020-007

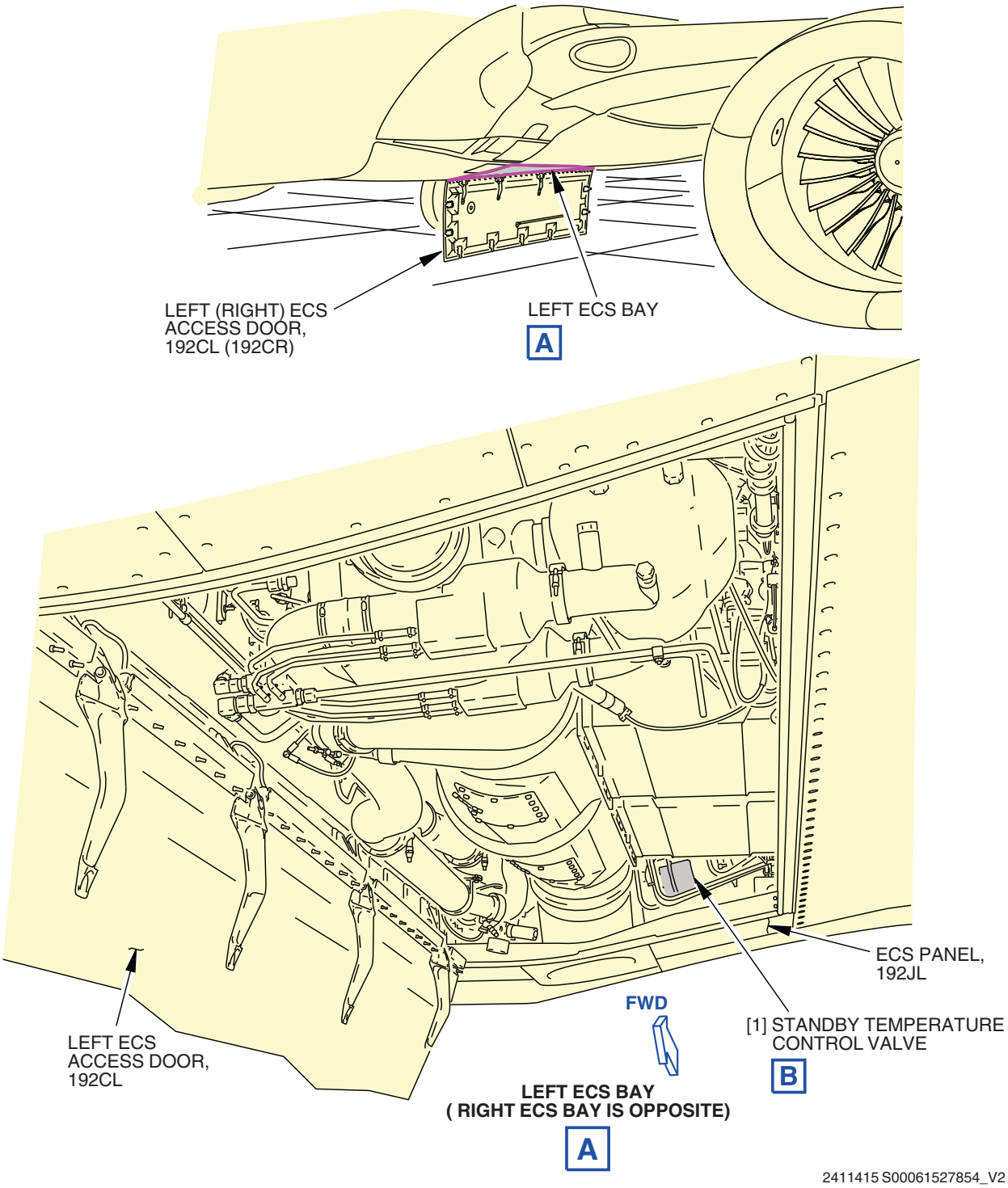
(8) Remove and discard the packing [20] from the conditioned air check valve sense line [11] location.

SUBTASK 21-51-11-020-008

(9) Remove and discard the packings [18] at the remaining three sense line locations.

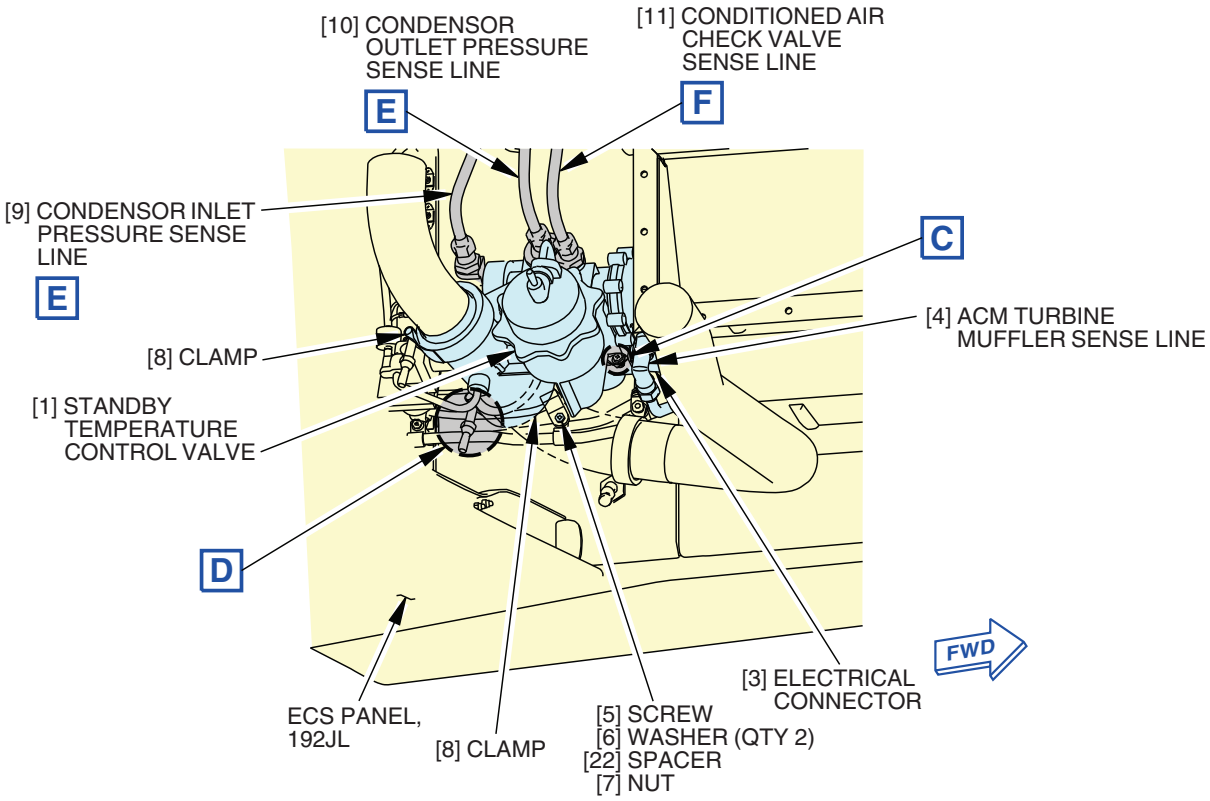
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

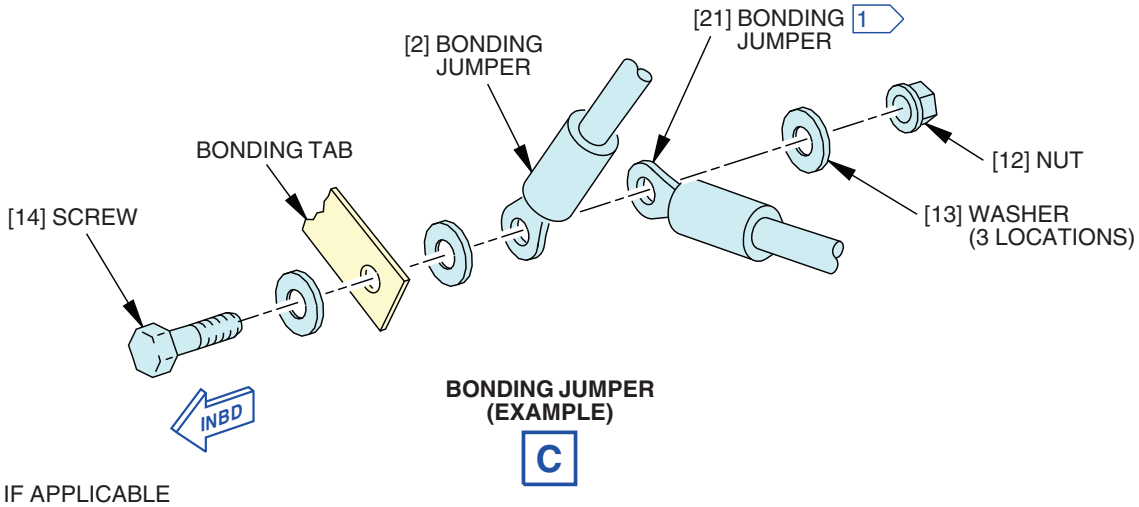


Standby Temperature Control Valve Installation
Figure 401/21-51-11-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STANDBY TEMPERATURE CONTROL VALVE
(LEFT SIDE IS SHOWN, RIGHT SIDE IS ALMOST THE SAME)



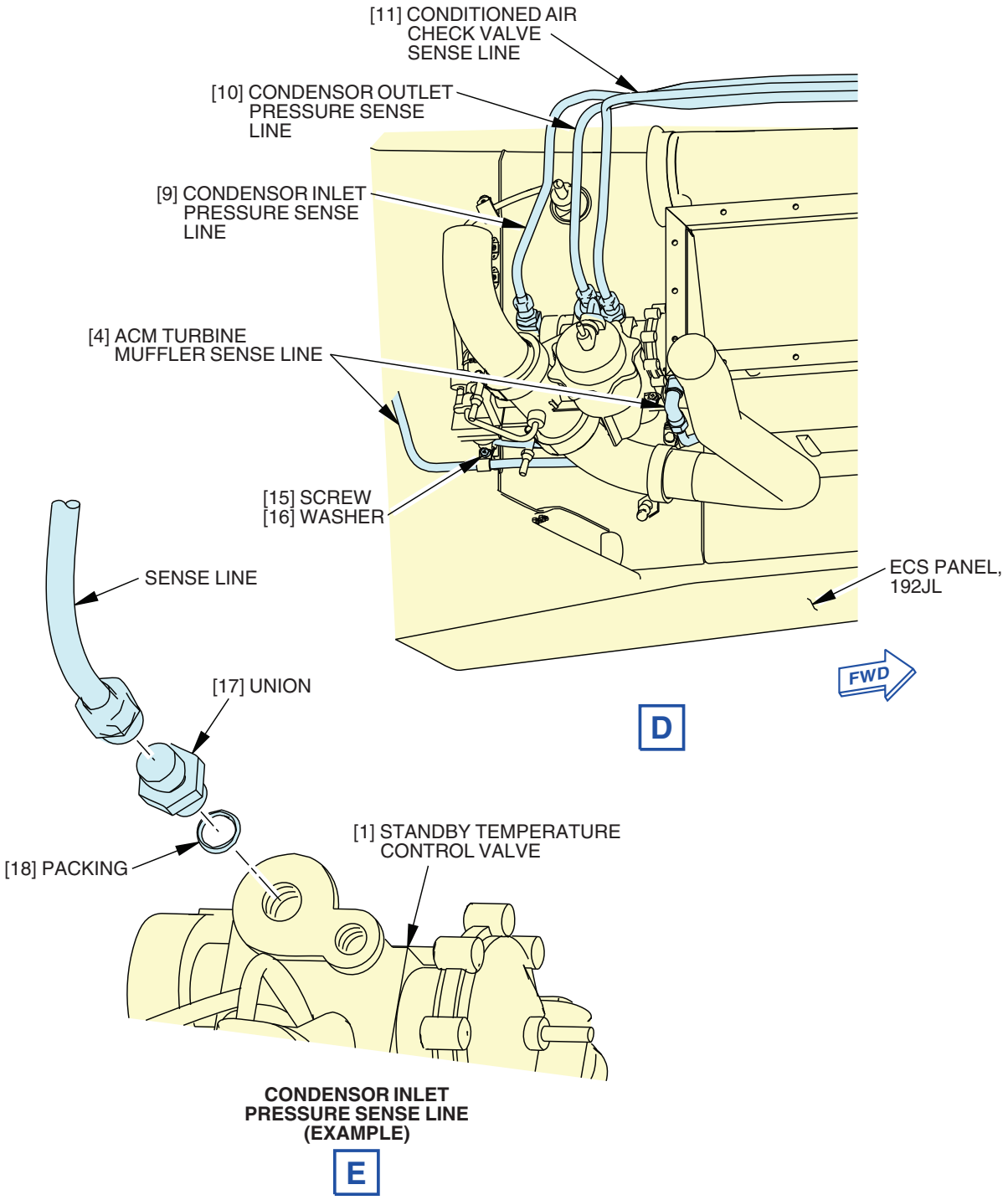
BONDING JUMPER
(EXAMPLE)

1 IF APPLICABLE

2411416 S00061527855_V5

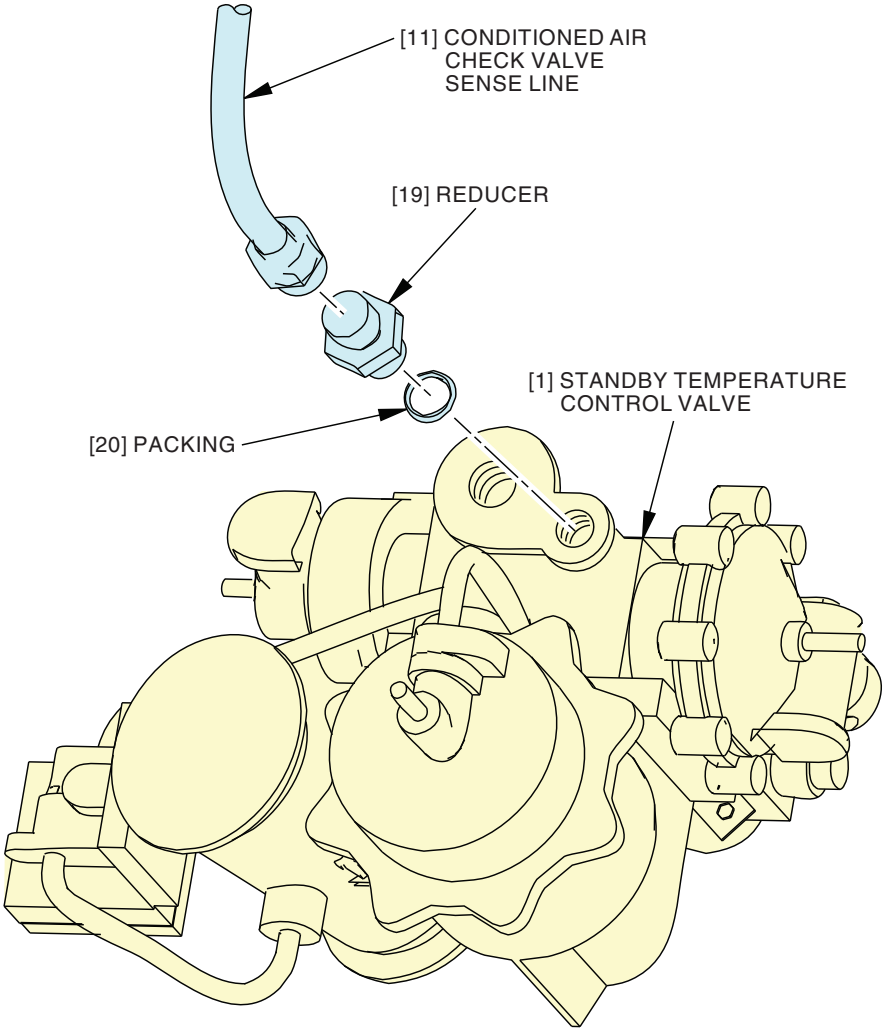
Standby Temperature Control Valve Installation
Figure 401/21-51-11-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Standby Temperature Control Valve Installation
Figure 401/21-51-11-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONDITIONED AIR CHECK VALVE SENSE LINE

F

2411418 S00061527857_V2

Standby Temperature Control Valve Installation
Figure 401/21-51-11-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-11-400-801

3. Standby Temperature Control Valve Installation

(Figure 401)

A. References

Reference	Title
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
18	Packing	21-51-11-01-155	PTF ALL
20	Packing	21-51-11-01-150	PTF ALL

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
192JR	Air Conditioning Panel - Aft

G. Standby Temperature Control Valve Installation

SUBTASK 21-51-11-420-001

(1) Install the reducer [19] on the standby temperature control valve [1].

(a) Do these steps for the reducer at the conditioned air check valve sense line [11] location:

1) Install a new packing [20], on the reducer [19].

2) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the reducer [19].

3) Install the reducer [19], on the standby temperature control valve [1].

SUBTASK 21-51-11-420-002

(2) Install the union [17] on the standby temperature control valve [1].

(a) Do these steps for each of the remaining three unions:

1) Install a new packing [18], on the union [17].

2) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the union [17].

3) Install the union [17] on the standby temperature control valve [1].

SUBTASK 21-51-11-160-002

(3) Carefully put the standby temperature control valve [1] into its position between the air ducts.

SUBTASK 21-51-11-020-009

(4) Install the clamps [8] that hold the valve to the ducts.

(a) Do not fully tighten the clamps [8].

SUBTASK 21-51-11-110-001

(5) Solvent clean the mating surfaces of the bonding jumper(s), bonding tab, and washers [13] (TASK 20-10-34-110-802).

SUBTASK 21-51-11-420-003

(6) Do the applicable step to install the bonding jumper(s).

(a) On the LH standby temperature control valve [1], install the screw [14], washers [13], nut [12], bonding jumper [2], and bonding jumper [21].

(b) On the RH standby temperature control valve [1], install the screw [14], washers [13], nut [12], bonding jumper [2], and bonding jumper [21].

SUBTASK 21-51-11-760-001

(7) Measure the electrical bonding resistance between the bonding tab and the bonding jumper (SWPM 20-20-00).

(a) Use a non-intrinsically safe bonding meter, COM-614.

(b) Make sure that the electrical bonding resistance is 0.002 ohms (2.0 milliohms) or less.

SUBTASK 21-51-11-420-004

(8) Install the electrical connector [3] on the standby temperature control valve [1].

SUBTASK 21-51-11-020-010

(9) To connect the four sense lines to the standby temperature control valve, do these steps:

(a) Make sure that they are installed in the positions they were before the removal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Install the sense line for the condensor inlet pressure sense line [9].

(c) Install the sense line for the acm turbine muffler sense line [4].

(d) Install the sense line for the condensor outlet pressure sense line [10].

(e) Install the sense line for the conditioned air check valve sense line [11].

(f) Align the three sense line clamps and the wire bundle clamp with the spacer plate nut on the structure.

(g) Install the screw [15] and washer [16] that hold the sense line clamp to the structure.

(h) Align the two sense line clamps and the wire bundle clamp.

(i) Install the screw [5], spacer [22], washers [6], and nut [7] that hold the clamps together.

SUBTASK 21-51-11-420-005

(10) Tighten the clamps [8] to 55 in-lb (6.2 N·m) to 60 in-lb (6.8 N·m).

(a) Make sure that the loop bands do not touch.

(b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-51-11-860-026

(11) For the left standby temperature control valve, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	3	C01613	AIR COND PACK CONT VALVES L ALT

SUBTASK 21-51-11-860-025

(12) For the right standby temperature control valve, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
D	4	C01614	AIR COND PACK CONT VALVES R ALT

H. Standby Temperature Control Valve Installation Test

SUBTASK 21-51-11-860-006

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-11-860-007

(2) Do these steps, on the P5-10 air conditioning panel:

(a) Set the PACK switch to the AUTO position and remove the DO-NOT-OPERATE tag.

1) Keep the other PACK switch in the OFF position.

(b) Set the BLEED 1 and 2 switches to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Set the BLEED APU switch to the ON position.

SUBTASK 21-51-11-860-008

(3) Do these steps on the P5-17 Cabin Temperature Panel:

(a) Put the TRIM AIR switch to the ON position.

(b) Put the cabin temperature selectors (CONT CAB, FWD CAB, and AFT CAB) to the AUTO position.

SUBTASK 21-51-11-790-001

(4) Do a soap bubble test of the duct joints and sense line connections of the standby temperature control valve.

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches to the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

2) Loosen the clamps [8].

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps [8] to 55 in-lb (6.2 N·m) to 60 in-lb (6.8 N·m).

a) Make sure that the loop bands do not touch.

b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

5) Put the L PACK and R PACK switches to the AUTO position.

6) Make sure that the leak has been repaired.

SUBTASK 21-51-11-860-016

(5) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-11-740-001

(6) Do a BITE test on the left and right Integrated Air Systems Controllers (M2939 and M2957):

NOTE: This will clear any fault introduced during maintenance when power is interrupted.

(a) To do the BITE test, do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-11-860-018

(1) For the left standby temperature control valve, close these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

SUBTASK 21-51-11-010-002

(2) For the right standby temperature control valve, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

(b) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-11-010-003

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STANDBY TEMPERATURE CONTROL VALVE - ADJUSTMENT/TEST

1. General

A. This procedure has the following task:

(1) Standby Temperature Control Valve Primary Pneumatic Control - Operational Check.

B. There are two standby temperature control valve, one for each cooling pack. The valves are found in the aft, outboard area of each air conditioning bay.

TASK 21-51-11-710-801

2. Standby Temperature Control Valve Primary Pneumatic Control - Operational Check

A. General

(1) The primary function of the Standby Temperature Control Valve (STCV) is to monitor the differential pressure between the inlet and outlet of the High Pressure Water Separator (HPWS)'s Condenser and the Air Cycle Machine (ACM)'s Turbine Outlet and Condition Air Check Valve. This task will make sure that the primary pneumatic control of the STCV is working properly.

B. References

Reference	Title
21-51-11-000-801	Standby Temperature Control Valve Removal (P/B 401)
21-51-11-400-801	Standby Temperature Control Valve Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-21786	Standby Temperature Control Valve (STCV) Test Equipment Part #: C21013-1 Supplier: 81205
STD-1290	Source - Nitrogen, 0-200 PSIG

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Prepare for the Check

SUBTASK 21-51-11-020-013

(1) Remove the STCV. Do this task: Standby Temperature Control Valve Removal, TASK 21-51-11-000-801.

NOTE: It is acceptable to remove the duct in place of the valve.

SUBTASK 21-51-11-200-001

(2) Make sure that the ball valve [8] and needle valve [7] on the STCV test equipment, SPL-21786, are in the “CLOSED” position.

SUBTASK 21-51-11-820-001

(3) Adjust the nitrogen source, STD-1290, to 150±40 psi.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-11-820-002

(4) Connect the nitrogen source, STD-1290, to the jic fitting [6] on the STCV test equipment, SPL-21786.

(a) If necessary, adjust the jic fitting [6] so that the relief valve no longer activates.

SUBTASK 21-51-11-420-006

(5) Connect the trailing probe hose assembly [10] tube end of the STCV test equipment, SPL-21786, to the trailing probe control valve [5] on the STCV.

F. Test the Low Delta Pressure Servo of the Standby Temperature Control Valve (STCV) - Min Open

SUBTASK 21-51-11-420-007

(1) Connect the turbine muff fitting assembly [11] of the STCV test equipment, SPL-21786, to the turbine muff connection [2] on the STCV as follows:

(a) Make sure that the turbine muff fitting assembly [11] is installed on the hose connection of the ss pipe fitting [12].

(b) Install the turbine muff fitting assembly [11] onto the turbine muff connection [2].

(2) Connect the ss pipe fitting [13] of the STCV test equipment, SPL-21786, to the conditioned air check valve connection [1] on the STCV as follows:

(a) Install the ss pipe fitting [13] onto the conditioned air check valve connection [1].

SUBTASK 21-51-11-020-014

(3) Open the ball valve [8] to allow the nitrogen into the system.

SUBTASK 21-51-11-820-003

(4) Make sure that the pressure gauge reading shows 45±5 psi.

(a) If necessary, adjust the pressure regulator [9] until the pressure gauge reading shows 45±5 psi.

SUBTASK 21-51-11-820-004

(5) Adjust the needle valve [7] until the digital manometer reads 24±1 in/H2O.

(a) Make sure that the butterfly plate begins to open.

1) If the plate doesn't operate correctly, then replace the STCV. This is the task: Standby Temperature Control Valve Removal, TASK 21-51-11-000-801.

SUBTASK 21-51-11-410-001

(6) Close the ball valve [8] to stop the nitrogen supply.

G. Test the Low Delta Pressure Servo of the Standby Temperature Control Valve (STCV) - Full Open

SUBTASK 21-51-11-420-008

(1) If not already installed, connect the turbine muff fitting assembly [11] of the STCV test equipment, SPL-21786, to the turbine muff connection [2] on the STCV as follows:

(a) Make sure that the turbine muff fitting assembly [11] is installed on the hose connection of the ss pipe fitting [12].

(b) Install the turbine muff fitting assembly [11] onto the turbine muff connection [2].

(2) If not already installed, connect the ss pipe fitting [13] of the STCV test equipment, SPL-21786, to the conditioned air check valve connection [1] on the STCV as follows:

(a) Install the ss pipe fitting [13] onto the conditioned air check valve connection [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-11-020-015

(3) Open the ball valve [8] to allow the nitrogen into the system.

SUBTASK 21-51-11-820-005

(4) Make sure that the pressure gauge reading shows 45±5 psi.

(a) If necessary, adjust the pressure regulator [9] until the pressure gauge reading shows 45±5 psi.

SUBTASK 21-51-11-820-006

(5) Adjust the needle valve [7] until the digital manometer reads 86±3 in/H2O.

(a) Make sure that the butterfly plate fully opens.

1) If the plate doesn't operate correctly, then replace the STCV. This is the task: Standby Temperature Control Valve Removal, TASK 21-51-11-000-801.

SUBTASK 21-51-11-410-002

(6) Close the ball valve [8] to stop the nitrogen supply.

H. Test the High Delta Pressure Servo of the Standby Temperature Control Valve (STCV) - Min Open

SUBTASK 21-51-11-420-009

(1) Connect the ss pipe fitting [12] of the STCV test equipment, SPL-21786, to the heat exchanger inlet connection [4] on the STCV as follows:

(a) If the turbine muff fitting assembly [11] is installed, remove it from the end of the hose connection so that the ss pipe fitting [12] is accessible.

(b) Install the ss pipe fitting [12] onto the heat exchanger inlet connection [4].

(2) Connect the ss pipe fitting [13] of the STCV test equipment, SPL-21786, to the heat exchanger outlet connection [3] as follows:

(a) Install the ss pipe fitting [13] onto the heat exchanger outlet connection [3].

SUBTASK 21-51-11-020-016

(3) Open the ball valve [8] to allow the nitrogen into the system.

SUBTASK 21-51-11-820-007

(4) Make sure that the pressure gauge reading shows 45±5 psi.

(a) If necessary, adjust the pressure regulator [9] until the pressure gauge reading shows 45±5 psi.

SUBTASK 21-51-11-820-008

(5) Adjust the needle valve [7] until the digital manometer reads 92±8 in/H2O.

(a) Make sure the butterfly plate begins to open.

1) If the plate doesn't operate correctly, then replace the STCV. This is the task: Standby Temperature Control Valve Removal, TASK 21-51-11-000-801.

SUBTASK 21-51-11-410-003

(6) Close the ball valve [8] to stop the nitrogen supply.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-11-020-017

(1) Disconnect the nitrogen source, STD-1290, from the jic fitting [6].

SUBTASK 21-51-11-020-018

(2) Disconnect the STCV test equipment, SPL-21786, from the STCV.

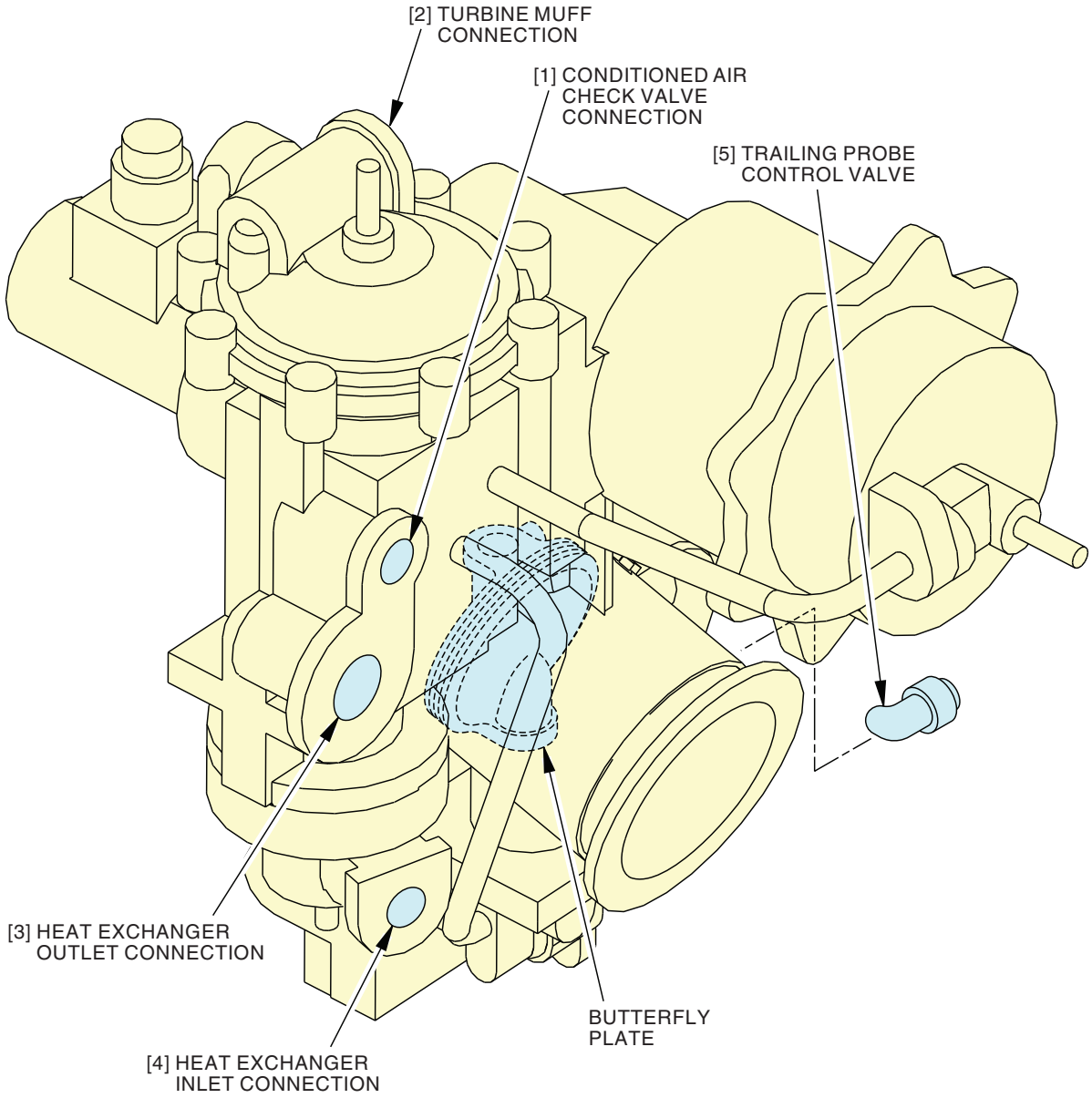
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-11-420-010

(3) Install the STCV. Do this task: Standby Temperature Control Valve Installation, TASK 21-51-11-400-801.

——— END OF TASK ———

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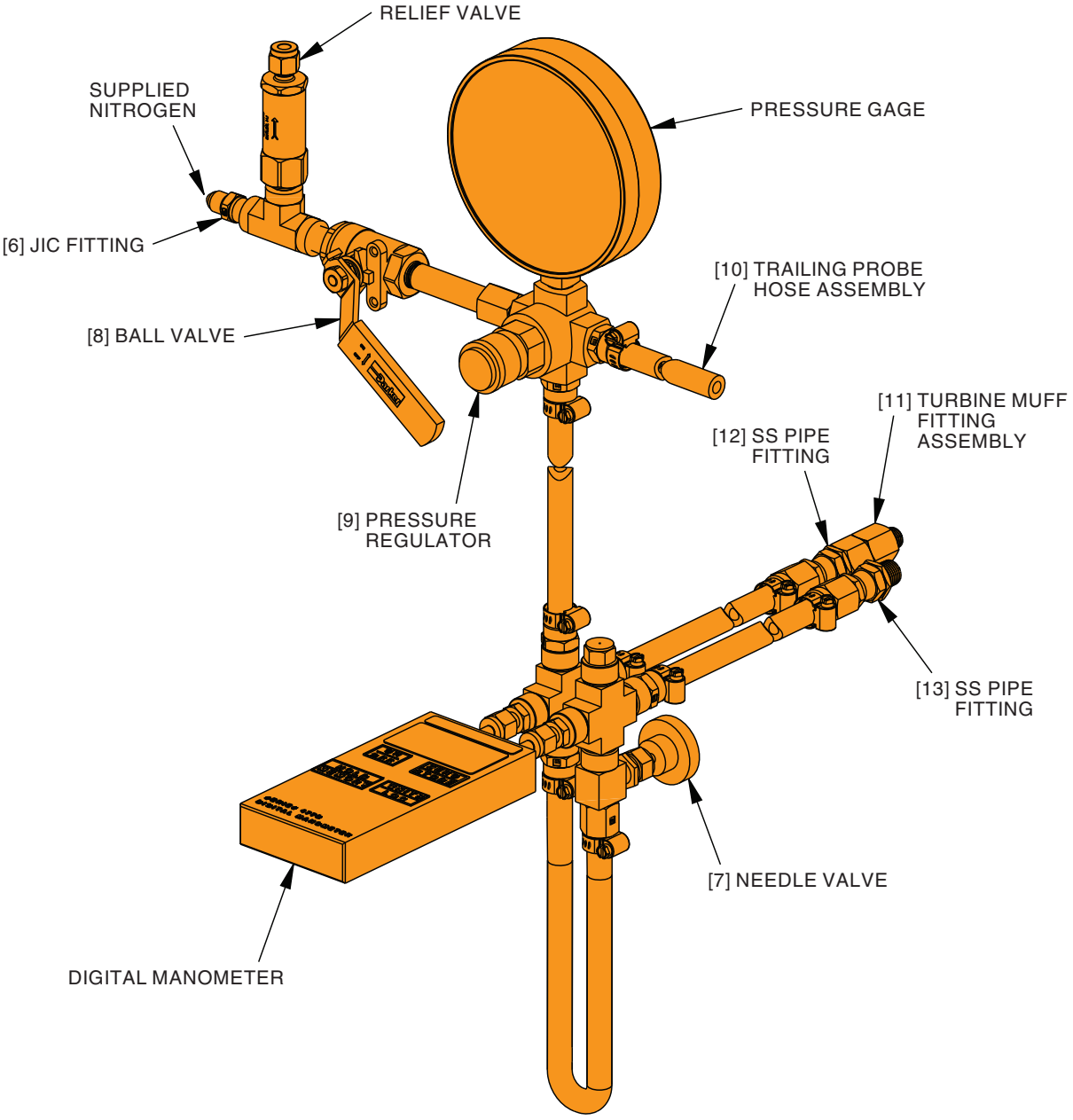


STANDBY TEMPERATURE CONTROL VALVE

3063358 S0000825883_V1

Standby Temperature Control Valve
Figure 501/21-51-11-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C21013 TOOL

3063359 S0000825884_V1

Test Equipment - C21013
Figure 502/21-51-11-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONDENSER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the condenser.

(2) An installation of the condenser.

(3) You must first remove the HPWS from the plane. You can then remove the condenser from the HPWS.

TASK 21-51-12-000-801

2. Condenser Removal

(Figure 401)

A. References

Reference	Title
21-51-08-000-801	High Pressure Water Separator Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

C. Prepare for the Removal

SUBTASK 21-51-12-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-12-020-001

(2) Do this task: High Pressure Water Separator Removal, TASK 21-51-08-000-801.

NOTE: You must first remove the HPWS from the plane. You can then remove the condenser [55] from the HPWS.

D. Condenser Removal

SUBTASK 21-51-12-020-002

(1) To remove the water extractor drain lines and manifold, do these steps:

(a) Remove the clamps [41] that connect the drain lines to the water extractors.

(b) Remove the screws [51], washers [50], and shim [49] that hold the manifold to the condenser [55].

(c) Remove the screws [45], washers [46], and nuts [47] that connect the drain line support clamps [48].

(d) Remove the drain line support clamps [48].

(e) Move the six hoses off the water extractors to complete the removal.

SUBTASK 21-51-12-020-003

(2) To remove the condenser [55] from the frame assembly [54], do these steps:

(a) Remove the clamps [42] that connect the water extractor hoses to the condenser duct.

(b) Remove the bolts [52] and washers [53].

(c) Remove the bolt [44] and washers [43] that connect the condenser [55] to the frame assembly [54].

(d) Remove the condenser [55] from the frame assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-12-020-004

(3) Remove the coupling [59].

(a) Discard the packings [58] on the coupling [59].

SUBTASK 21-51-12-020-005

(4) Remove the plug [56] and packing [57] from the condenser [55].

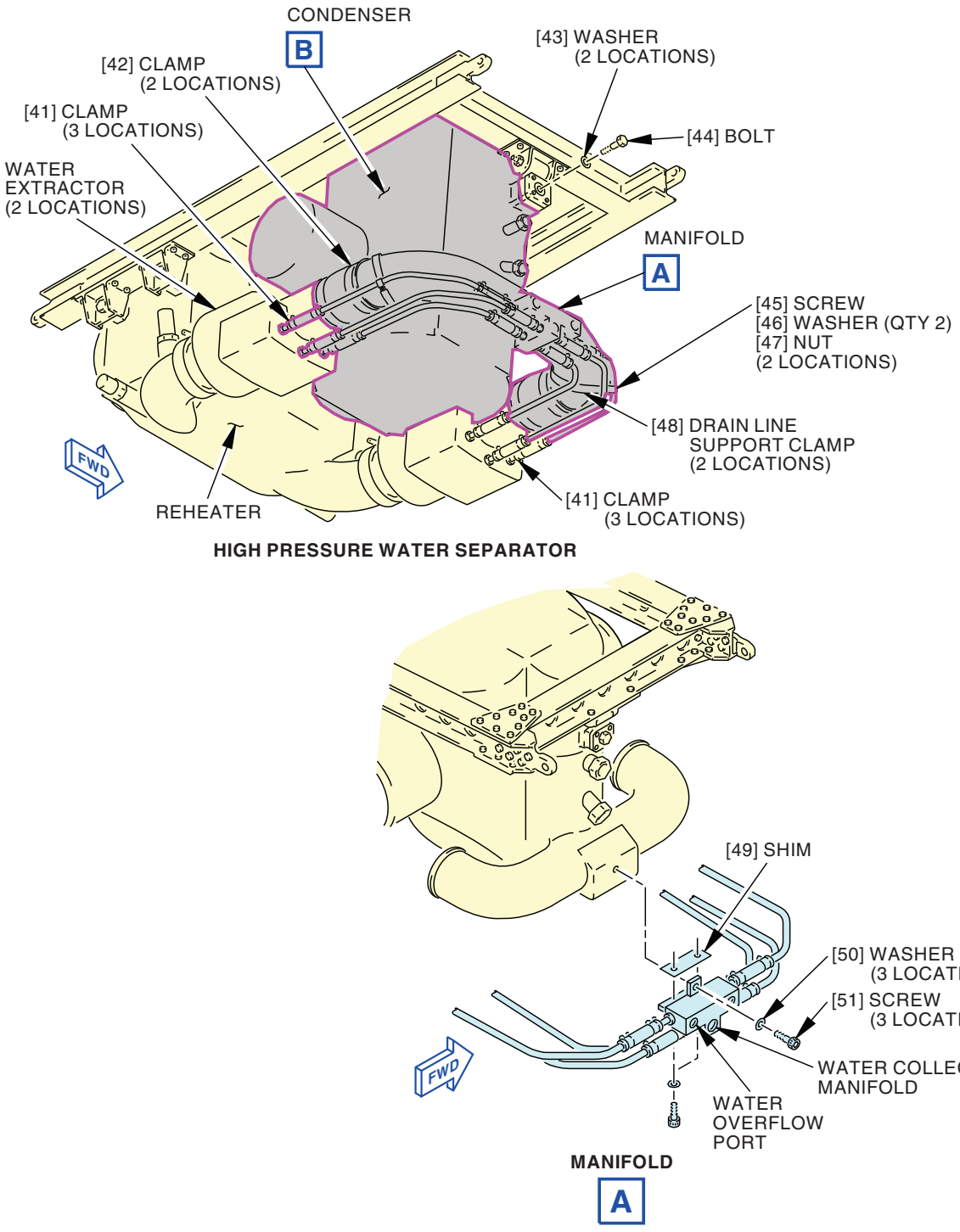
(a) Discard the packing [57].

SUBTASK 21-51-12-480-001

(5) Put covers on the ducts and lines to keep out unwanted materials.

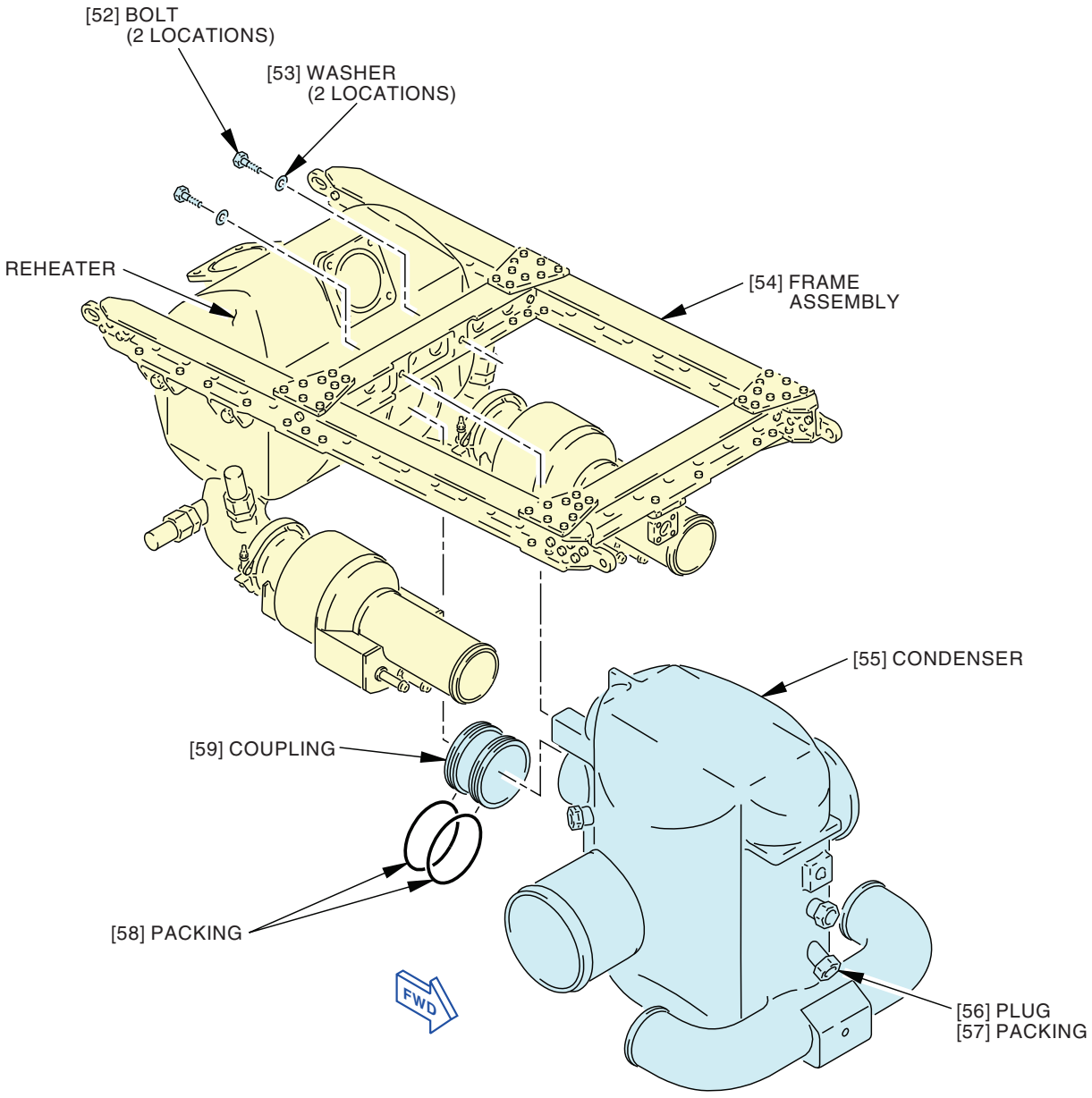
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Condenser Installation
Figure 401/21-51-12-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Condenser Installation
Figure 401/21-51-12-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-12-400-801

3. Condenser Installation

(Figure 401)

A. References

Reference	Title
21-51-08-400-801	High Pressure Water Separator Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D00548	Lubricant - Fluorolube GR-660	BAC5506
D50063	Grease - Perfluoropolyether, fuel and oxygen resistant - Krytox 240AC	MIL-PRF- 27617 Type III
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	
D50203	Grease - Castrol Braycote 601EF	BMS3-25 Type II
D50233	Grease - Dupont Krytox 240AZ	MIL-PRF-27617 Type I
D50234	Grease - Dupont Krytox 240AB	MIL-PRF-27617 Type II
D50268	Lubricant - Castrol Braycote 815Z	BMS3-25 Type I
D50380	Grease - Braycote Micronic 601EF	BMS3-25 Type III
D50389	Grease - Castrol Braycote 1729, Fluorinated Lubricant (use until stock depleted)	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
55	Condenser	21-51-12-01-075	PTF ALL
57	Packing	21-51-08-01-150	PTF ALL
		21-51-08-05-165	PTF ALL
58	Packing	21-51-13-01-030	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Condenser Installation

SUBTASK 21-51-12-480-002

(1) Remove the duct and line covers.

SUBTASK 21-51-12-020-006

(2) To install the plug [56], do these steps:

(a) Apply a thin coat of one of these lubricants to the new packing [57]:

1) Braycote Micronic 601EF grease, D50380
2) Braycote 601EF Grease, D50203
3) Braycote 815Z lubricant, D50268
4) Fluorolube GR-660 lubricant, D00548
5) Braycote 1729 grease, D50389
6) Krytox 240AB grease, D50234
7) Krytox 240AZ grease, D50233

ECCN 9E991 BOEING PROPRIETARY - See title page for details

8) Krytox 240AC perfluoropolyether grease, D50063.

(b) Install the packing [57] on the plug [56].

(c) Install the plug [56] in the condenser [55].

SUBTASK 21-51-12-020-007

(3) To install the coupling [59], do these steps:

(a) Apply a thin coat of Molykote 44 grease, medium, D50104, or an equivalent to the new packings [58].

(b) Install the packings [58] on the coupling [59].

(c) Install the coupling [59] in the duct of the reheater.

SUBTASK 21-51-12-020-008

(4) To install the condenser [55] on the frame assembly, do these steps:

(a) Put the condenser [55] in its position in the frame assembly.

(b) While the condenser [55] is put in the frame assembly, do these steps:

1) Move the hoses on the water extractors over the ducts on the condenser [55].

CAUTION

DO NOT CAUSE DAMAGE TO THE PACKING AT THE COUPLING DURING CONDENSER INSTALLATION. THE DUCT OF THE CONDENSER SETS ON THE COUPLING. DAMAGE TO THE PACKING CAN CAUSE AIR LEAKS.

2) Put the duct on the condenser [55] over the coupling [59].

(c) Install the bolts [52] and washers [53].

(d) Install the bolt [44] and washers [43] that connect the condenser [55] to the frame assembly.

1) Make sure that one of the washers [43] is installed between the condenser [55] and frame assembly.

(e) Install the clamps [42] on the water extractor hoses.

1) Tighten the clamps [42] to 16 in-lb (1.81 N·m) - 18 in-lb (2.03 N·m).

SUBTASK 21-51-12-020-009

(5) To install the water extractor drain lines and manifold, do these steps:

(a) Move the six hoses over the drain line ports of the water extractors.

(b) Put the drain line support clamps [48] on the condenser ducts.

(c) Align the drain line support clamps [48] with the clamp on the water extractor drain line.

(d) Install the screw [45], washers [46], and nut [47] that connect the drain line support clamps [48].

(e) Install the screws [51], washers [50], and shim [49] that hold the manifold to the condenser [55].

1) If a replacement manifold is installed, make sure that the plastic plug removes from the overflow port of the water collection manifold.

(f) Install the clamps [41] that connect the drain line hoses to the water extractors.

SUBTASK 21-51-12-410-001

(6) Install the high pressure water separator (TASK 21-51-08-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

REHEATER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the reheater.

(2) An installation of the reheater.

(3) You must first remove the HPWS from the plane. You can then remove the reheater from the HPWS.

TASK 21-51-13-000-801

2. Reheater Removal

(Figure 401)

A. References

Reference	Title
21-51-08-000-801	High Pressure Water Separator Removal (P/B 401)
21-51-52-000-801	Pack Temperature Sensor Removal (P/B 401)
21-51-60-000-801	Pack Temperature Bulb Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

C. Prepare for the Removal

SUBTASK 21-51-13-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-13-020-001

(2) Do this task: High Pressure Water Separator Removal, TASK 21-51-08-000-801

NOTE: You must first remove the HPWS from the plane. You can then remove the reheater from the HPWS.

D. Reheater Removal

SUBTASK 21-51-13-020-008

(1) Remove the pack temperature bulb (TASK 21-51-60-000-801).

SUBTASK 21-51-13-020-009

(2) Remove the pack temperature sensor (TASK 21-51-52-000-801).

SUBTASK 21-51-13-020-002

(3) Remove the couplings [47] and packings [48] from the water extractors.

(a) Discard the packings [48].

SUBTASK 21-51-13-020-003

(4) To remove the reheater [49] from the frame assembly [44], do these steps:

(a) Remove the bolts [43], washers [41], and bushings [42].

(b) Remove the bolts [50] and washers [41].

(c) Remove the reheater [49] from the frame assembly [44].

SUBTASK 21-51-13-020-004

(5) Remove the coupling [45].

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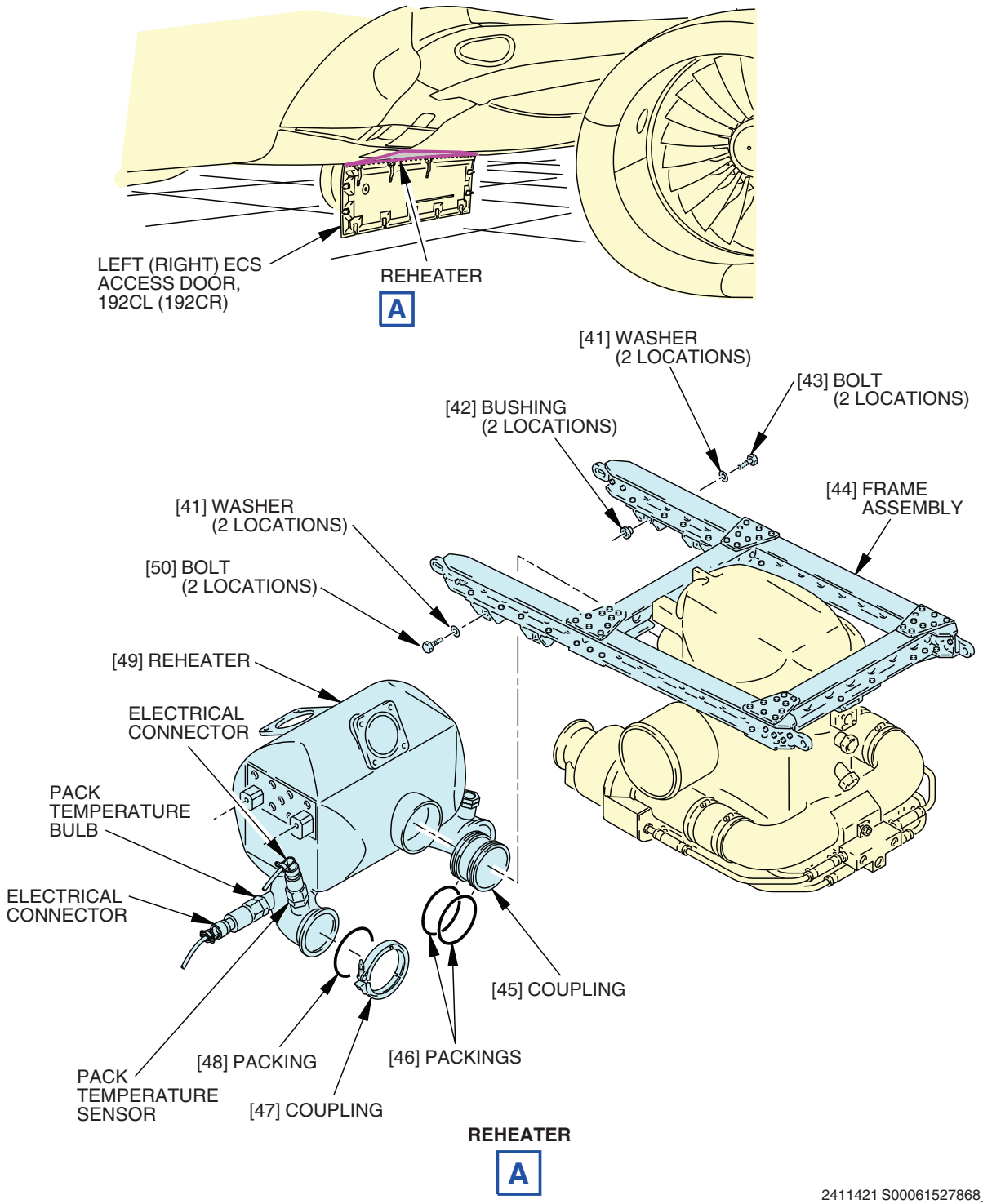
(a) Discard the packings [46] from the coupling [45].

SUBTASK 21-51-13-480-001

(6) Put covers on the ducts and lines to keep out unwanted materials.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Reheater Installation
Figure 401/21-51-13-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-13-400-801

3. Reheater Installation

(Figure 401)

A. References

Reference	Title
21-51-08-400-801	High Pressure Water Separator Installation (P/B 401)
21-51-52-400-801	Pack Temperature Sensor Installation (P/B 401)
21-51-60-400-801	Pack Temperature Bulb Installation (P/B 401)
24-22-00-860-801	Supply Electrical Power (P/B 201)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-804	Supply Pressure to the Pneumatic System with One or Both Engines (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
46	Packing	21-51-13-01-030	PTF ALL
48	Packing	21-51-14-01-100	PTF ALL
49	Reheater	21-51-13-01-035	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Reheater Installation

SUBTASK 21-51-13-480-002

(1) Remove the duct and line covers.

SUBTASK 21-51-13-020-005

(2) Do these steps to install the coupling [45].

(a) Apply a thin coat of Molykote 44 grease, medium, D50104, or an equivalent to the new packings [46].

(b) Install the packings [46] on the coupling [45].

(c) Install the coupling [45] in the condenser.

SUBTASK 21-51-13-020-006

(3) To install the reheater [49] on the frame assembly [44], do these steps:

(a) Put the bushings [42] in their positions in the frame assembly [44].

(b) Put the reheater [49] in its position in the frame assembly [44].

(c) While you put the reheater [49] in the frame assembly [44], do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Be careful not to damage the new packings [46] as you position the duct on the reheater [49] over the coupling [45].

(d) Install the bolts [43] and washers [41].

(e) Install the bolts [50] and washers [41].

SUBTASK 21-51-13-020-007

(4) To install the couplings [47] for the water extractors, do these steps:

(a) Apply a thin coat of Molykote 44 grease, medium, D50104, or an equivalent to the new packings [48].

(b) Put the packings [48] in their positions on the ducts.

(c) Install the couplings [47] on the ducts.

(d) Tighten the couplings [47] to 40 in-lb (4.5 N·m) - 45 in-lb (5.1 N·m).

SUBTASK 21-51-13-020-010

(5) If you install the new reheater [49], remove the plugs and packings from the reheater [49] where you will install the pack temperature sensor and pack temperature bulb.

SUBTASK 21-51-13-420-001

(6) Install the pack temperature sensor (TASK 21-51-52-400-801).

SUBTASK 21-51-13-420-002

(7) Install the pack temperature bulb (TASK 21-51-60-400-801).

SUBTASK 21-51-13-410-001

(8) Install the high pressure water separator (TASK 21-51-08-400-801).

F. Reheater Installation Test

SUBTASK 21-51-13-860-002

WARNING MAKE SURE THAT PERSONNEL ARE CLEAR OF THE RAM AIR INLET AND EXIT DOORS, LINKAGES AND ACTUATORS BEFORE YOU DO THIS TEST. POSSIBLE INJURY TO PERSONNEL CAN OCCUR.

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-51-13-780-001

(2) To prepare for a leak check of the reheater [49] installation, do one of the two steps, (a) or (b), that follow:

(a) Supply pneumatic pressure to the packs with the Auxiliary Power Unit (APU) as follows:

1) Supply pressure to the pneumatic system with the APU (TASK 36-00-00-860-803).

2) Make sure that the aft outflow valve and passenger door are open.

3) On the P5-10 bleed air control module, set the applicable L PACK or R PACK switch to the AUTO position.

(b) Supply pneumatic pressure to the packs with an engine as follows:

1) Supply pressure to the pneumatic system with one or both engines (TASK 36-00-00-860-804).

2) Make sure that the aft outflow valve and passenger door are open.

3) On the P5-10 bleed air control module, set the applicable L PACK or R PACK switch to the AUTO position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-13-790-001

(3) Do a soap bubble test of all the applicable duct joints.

NOTE: No leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches in the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the clamps.

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps.

5) Put the L PACK and R PACK switches in the AUTO position.

6) Make sure that the leak has been repaired.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-13-860-003

(1) Do this step on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position.

SUBTASK 21-51-13-860-004

(2) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER EXTRACTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the water extractor

(2) An installation of the water extractor.

(3) The water extractor is a component of the high-pressure water separator (HPWS). There are two water extractors on the HPWS. There is an HPWS for each of the two packs.

TASK 21-51-14-000-801

2. Water Extractor Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-14-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-14-860-002

(2) Do these steps, on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach the DO NOT OPERATE tag, STD-858.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-14-010-001

(3) To get access to the water extractor [1] for the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-14-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the water extractor [1] for the right cooling pack, open these access panels in this sequence:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Water Extractor Removal

SUBTASK 21-51-14-020-001

(1) To disconnect the drain lines from the water extractor [1], do these steps:

(a) Remove the clamps [6] that hold the hoses of the drain lines to the water extractor [1].

(b) Remove the screw [3], washers [4], and nut [5] that connect the drain line support clamps.

(c) Move the hoses off the water extractor.

SUBTASK 21-51-14-020-002

(2) Remove the clamp [2] that connects the water extractor hose to the condenser duct.

SUBTASK 21-51-14-020-003

(3) Hold the water water extractor [1] while you remove the coupling [7] and packing [8].

(a) Discard the packing [8].

SUBTASK 21-51-14-020-004

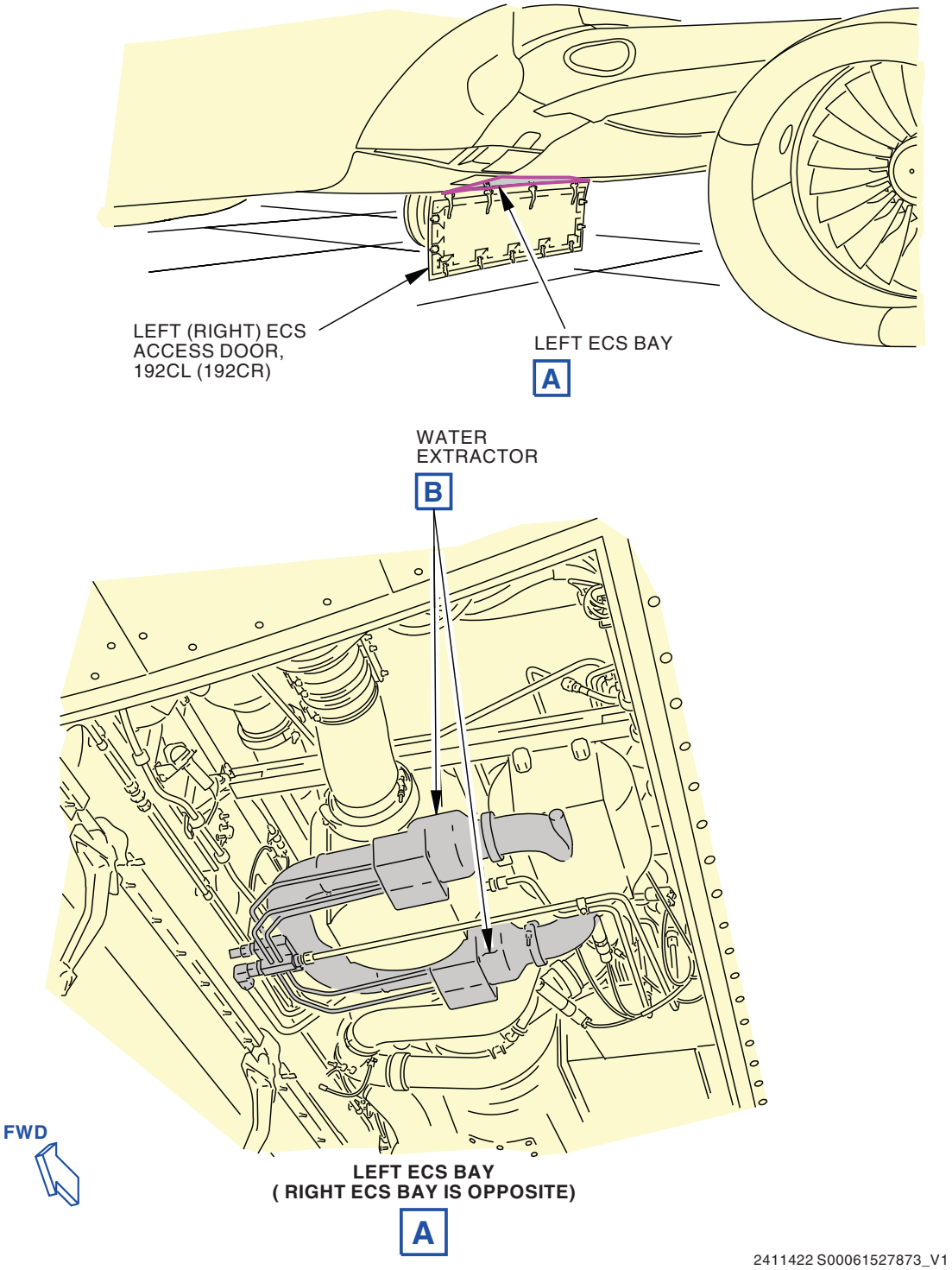
(4) Carefully lower the water extractor [1].

SUBTASK 21-51-14-480-001

(5) Put covers on the ducts and lines to keep out unwanted materials.

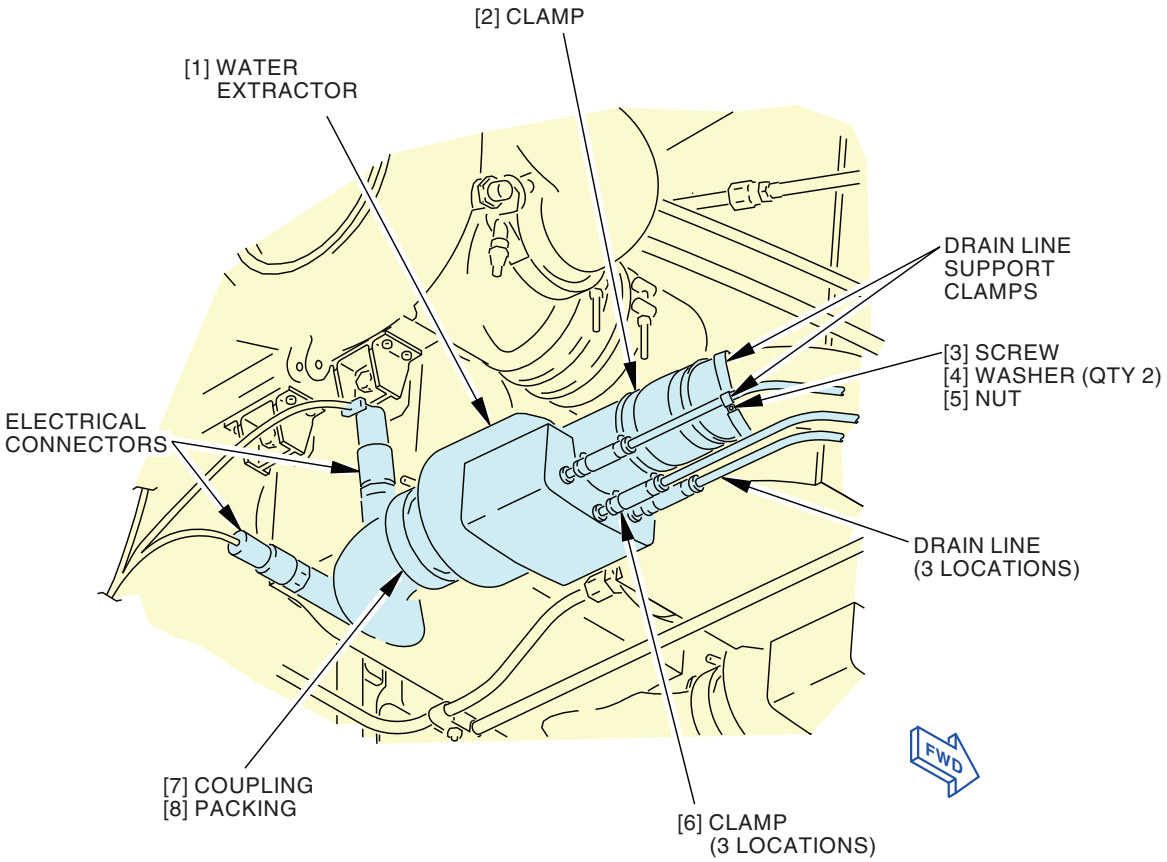
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Extractor Installation
Figure 401/21-51-14-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WATER EXTRACTOR (EXAMPLE)

B

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Water Extractor Installation
Figure 401/21-51-14-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-14-400-801

3. Water Extractor Installation

(Figure 401)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-804	Supply Pressure to the Pneumatic System with One or Both Engines (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Water Extractor Installation

SUBTASK 21-51-14-480-002

(1) Remove the duct and line covers.

SUBTASK 21-51-14-020-005

(2) To install the water extractors, do these steps:

(a) Apply a thin coat of Molykote 44 grease, medium, D50104, or an equivalent to the new packing [8].

(b) Put the packing [8] in its position on the duct.

(c) Move the hose over the end of the water extractor [1].

(d) Hold the water extractor [1] in position.

(e) Loosely install the coupling [7].

(f) Loosely install the clamp [2].

SUBTASK 21-51-14-420-001

(3) Make sure that the water extractor [1] does not touch the condenser.

SUBTASK 21-51-14-420-002

(4) Tighten the coupling [7] to 40 in-lb (4.52 N·m) - 45 in-lb (5.08 N·m).

SUBTASK 21-51-14-420-003

(5) Tighten the clamp [2] to 16 in-lb (1.81 N·m) - 18 in-lb (2.03 N·m).

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SUBTASK 21-51-14-020-006

(6) To install the drain lines on the water extractor [1], do these steps:

(a) Put the hoses on the drain ports of the water extractor [1].

(b) Install the clamps [6] that hold the hoses of the drain lines to the water extractor [1].

(c) Install the screw [3], washers [4], and nut [5] that connect the drain line support clamps.

F. Water Extractor Installation Test

SUBTASK 21-51-14-860-003

WARNING MAKE SURE THAT PERSONNEL ARE CLEAR OF THE RAM AIR INLET AND EXIT DOORS, LINKAGES AND ACTUATORS BEFORE YOU DO THIS TEST. POSSIBLE INJURY TO PERSONNEL CAN OCCUR.

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-51-14-780-001

(2) To prepare for a leak check of the water extractor [1] installation, do one of the two steps, (a) or (b), that follow:

(a) Supply pneumatic pressure to the packs with the Auxiliary Power Unit (APU) as follows:

1) Supply pressure to the pneumatic system with the APU(TASK 36-00-00-860-803).

2) Make sure that the aft outflow valve and a passenger door are open.

3) Set the applicable L PACK or R PACK switch, on the P5-10 bleed air control module, to the AUTO position.

(b) Supply pneumatic pressure to the packs with an engine as follows:

1) Supply pressure to the pneumatic system with one or both engines as follows: (TASK 36-00-00-860-804).

2) Make sure that the aft outflow valve and a passenger door are open.

3) Set the applicable L PACK or R PACK switch, on the P5-10 bleed air control module, to the AUTO position.

SUBTASK 21-51-14-790-001

(3) Do a soap bubble test of all the applicable duct joints.

NOTE: No leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches in the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the clamps.

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps.

5) Put the L PACK and R PACK switches in the AUTO position.

6) Make sure that the leak has been repaired.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-14-860-004

(1) Do these steps, on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position.

SUBTASK 21-51-14-860-005

(2) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-14-410-001

(3) If the water extractor [1] for the right cooling pack was removed, close these access panels in this sequence:

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-14-410-002

(4) If the water extractor [1] for the left cooling pack was removed, close this access panel:

Number	Name/Location
192CL	ECS Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER EXTRACTOR DUCT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the water extractor duct.

(2) An installation of the water extractor duct.

(3) There is a water extractor duct installed for each of the two air conditioning packs.

TASK 21-51-15-000-801

2. Water Extractor Duct Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-15-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-15-860-002

(2) Do these steps, on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach the DO NOT OPERATE tag, STD-858.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-15-010-001

(3) To get access to the water extractor duct [1] for the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-15-010-002

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the water extractor duct [1] for the right cooling pack, open these access panels in this sequence:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Water Extractor Duct Removal

SUBTASK 21-51-15-020-001

(1) To disconnect the water extractor duct drain line [5] for the water extractor duct [1], do these steps:

(a) Remove the screw [2], washer [3], and nut [4] that connect the drain line clamps together.

(b) Loosen the B-nut [13] to disconnect the drain line.

(c) Remove the union [12].

(d) Remove and discard the O-ring [11].

SUBTASK 21-51-15-020-002

(2) Do these steps to remove the coupling assembly:

(a) Remove the clamp [7].

(b) Move the sleeve [8] to the adjacent duct.

(c) Move the ring [10] to the adjacent duct.

(d) Remove the retainer [9].

SUBTASK 21-51-15-020-003

(3) Hold the water extractor duct [1] while you remove the clamp [6].

SUBTASK 21-51-15-020-004

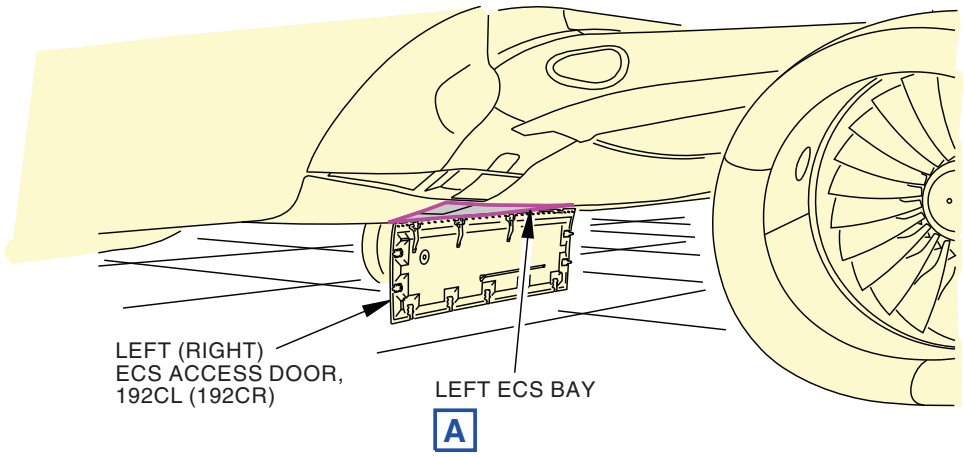
(4) Carefully lower the water extractor duct [1].

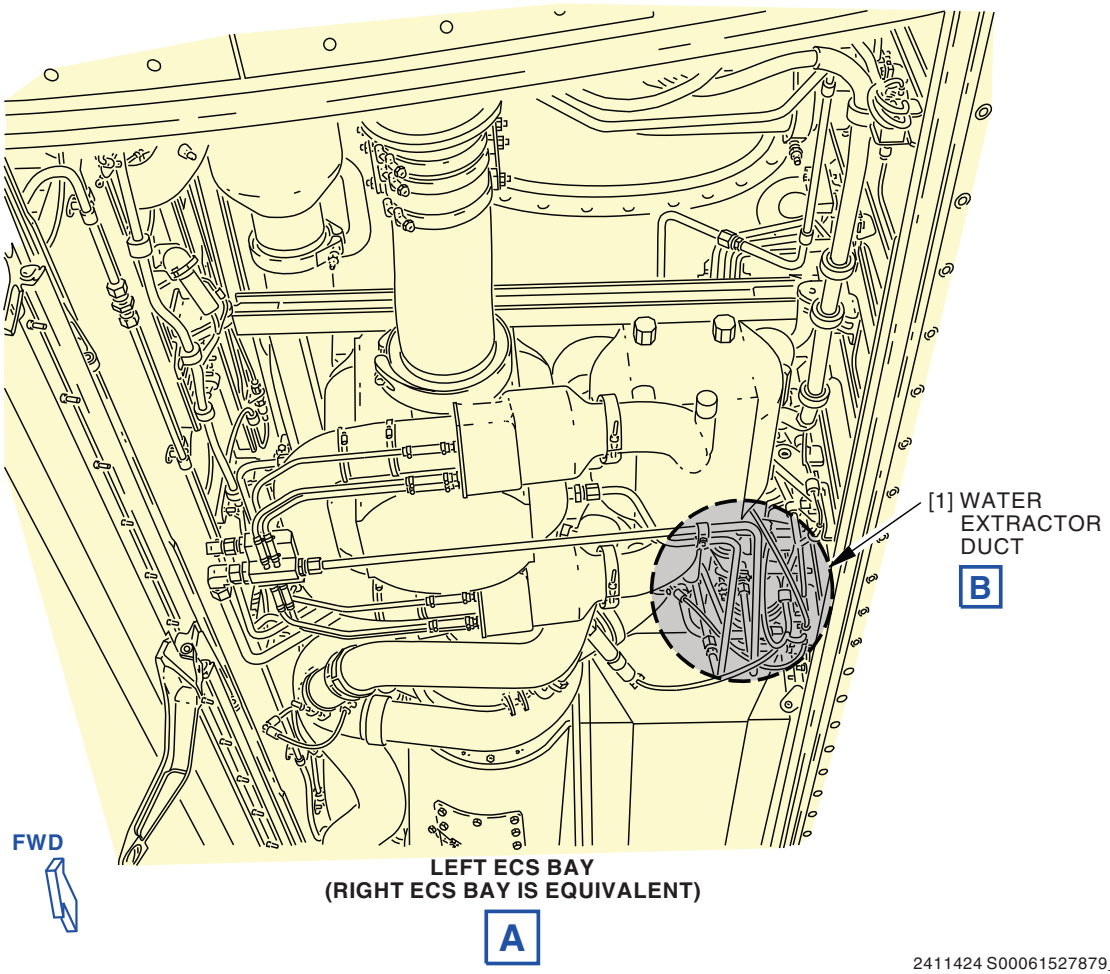
SUBTASK 21-51-15-480-001

(5) Put covers on the ducts and lines to keep out unwanted materials.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

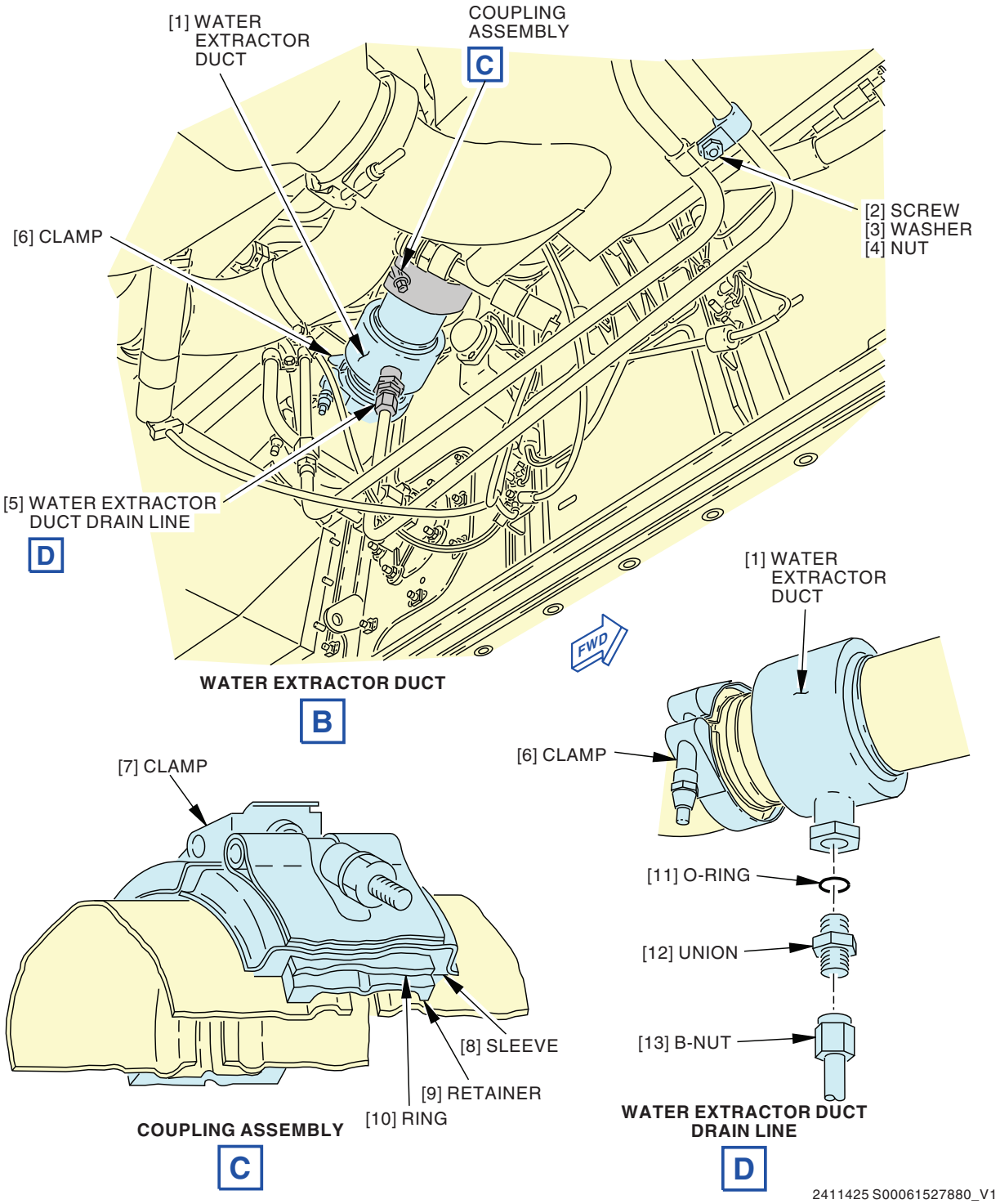




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Water Extractor Duct Installation
Figure 401/21-51-15-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Extractor Duct Installation
Figure 401/21-51-15-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-15-400-801

3. Water Extractor Duct Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Water extractor duct	21-51-08-01-165	PTF ALL
		21-51-08-05-180	PTF ALL
11	O-ring	21-51-08-01-150	PTF ALL
		21-51-08-05-165	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Water Extractor Duct Installation

SUBTASK 21-51-15-480-002

(1) Remove the duct and line covers.

SUBTASK 21-51-15-020-005

(2) To install the water extractor duct [1], do these steps:

(a) Hold the water extractor duct [1] in its position.

1) Make sure that the flow arrow on the water extractor duct [1] points in the forward direction.

(b) Loosely install the clamp [6].

(c) Install the retainer [9] that holds the water extractor duct [1] to the adjacent duct.

(d) Move the ring [10] over the retainer [9].

(e) Move the sleeve [8] over the ring [10].

(f) Install the clamp [7] over the sleeve [8].

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(g) Tighten the clamp [7] to 90 in-lb (10.2 N·m) - 95 in-lb (10.7 N·m).

(h) Tighten the clamp [6] to 55 in-lb (6.2 N·m) - 60 in-lb (6.8 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-51-15-020-006

(3) To connect the water extractor duct drain line [5] to the water extractor duct [1], do these steps:

(a) Install a new O-ring [11] on the union [12].

(b) Apply a thin coating of Pure Nickel Special compound, D00006, to the threads of the union [12].

(c) Install the union [12] on the water extractor duct [1].

(d) Tighten the B-nut [13] to connect the drain line to the duct.

(e) Install the screw [2], washer [3], and nut [4] that hold the drain line clamps together.

G. Water Extractor Duct Installation Test

SUBTASK 21-51-15-860-003

WARNING MAKE SURE THAT PERSONNEL ARE CLEAR OF THE RAM AIR INLET AND EXIT DOORS, LINKAGES AND ACTUATORS BEFORE YOU DO THIS TEST. POSSIBLE INJURY TO PERSONNEL CAN OCCUR.

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-51-15-860-004

(2) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-15-860-005

(3) Do these steps, on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the applicable L or R PACK switch to the AUTO position.

(b) Set the BLEED 1 and 2 switches to the ON position.

(c) Set the BLEED APU switch to the ON position.

SUBTASK 21-51-15-790-001

(4) Do a soap bubble test of all the applicable duct joints.

NOTE: No leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches in the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the clamps.

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps.

a) For the clamp [6], make sure that the loop bands do not touch.

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b) For the clamp [6], make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

5) Put the L PACK and R PACK switches in the AUTO position.

6) Make sure that the leak has been repaired.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-15-860-006

(1) Do this step, on the P5-10 bleed air control module (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position.

SUBTASK 21-51-15-410-001

(2) If the water extractor duct [1] for the right cooling pack was removed, close these access panels in this sequence:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-15-410-002

(3) If the water extractor duct [1] for the left cooling pack was removed, close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-15-860-007

(4) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER SPRAY NOZZLE - REMOVAL/INSTALLATION

TASK 21-51-16-000-801

1. Water Spray Nozzle Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

D. Prepare for the Removal

SUBTASK 21-51-16-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-16-010-001

(2) To get access to the water spray nozzle [1] for the left cooling pack, do these steps:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

WARNING

OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(b) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

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SUBTASK 21-51-16-010-002

(3) To get access to the water spray nozzle [1] for the right cooling pack, do these steps in this specified sequence:

(TASK 06-41-00-800-801)

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-16-010-003

(4) Remove the access panel on the bottom of the ram air inlet duct:

(a) Remove the screws [3] and washers [4] at 20 locations.

(b) Carefully remove the ram air inlet access panel [2].

E. Water Spray Nozzle Removal

SUBTASK 21-51-16-020-001

(1) To disconnect the water line from the water spray nozzle, do these steps:

(a) Loosen the B-nut on the water line.

(b) Remove the union [5] from the water spray nozzle.

(c) Remove and discard the O-ring [6].

SUBTASK 21-51-16-020-002

(2) To disconnect the air line from the water spray nozzle, do these steps:

(a) Loosen the B-nut on the air line.

(b) Remove the union [10].

(c) Remove and discard the O-ring [9].

SUBTASK 21-51-16-020-003

(3) To remove the water spray nozzle [1], do these steps:

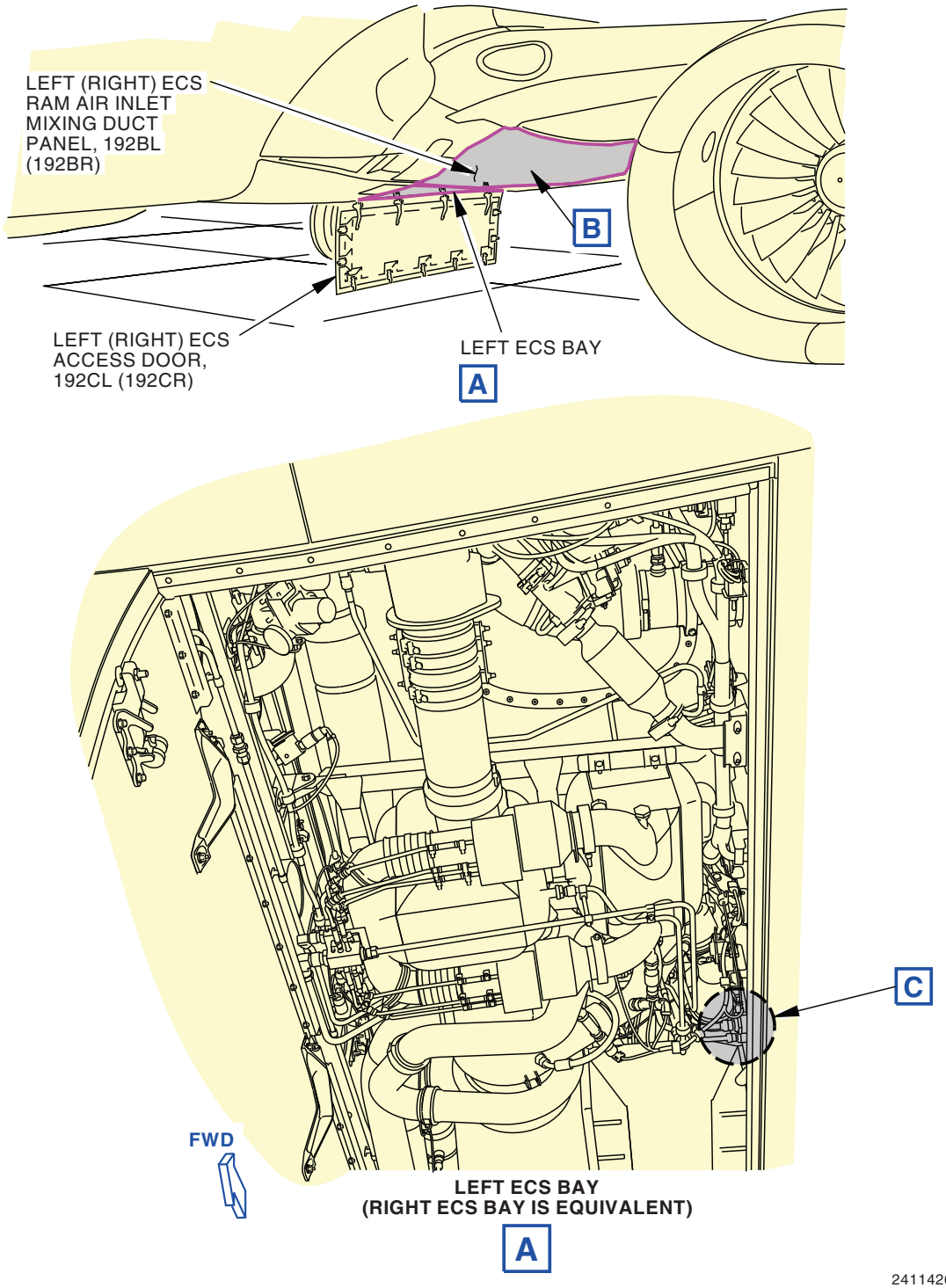
(a) Remove the bolts [11], washers [8] and nuts [7] at five locations.

NOTE: You must reach through the access hole in the ram air inlet to access the bolt and washer.

(b) Remove the water spray nozzle [1].

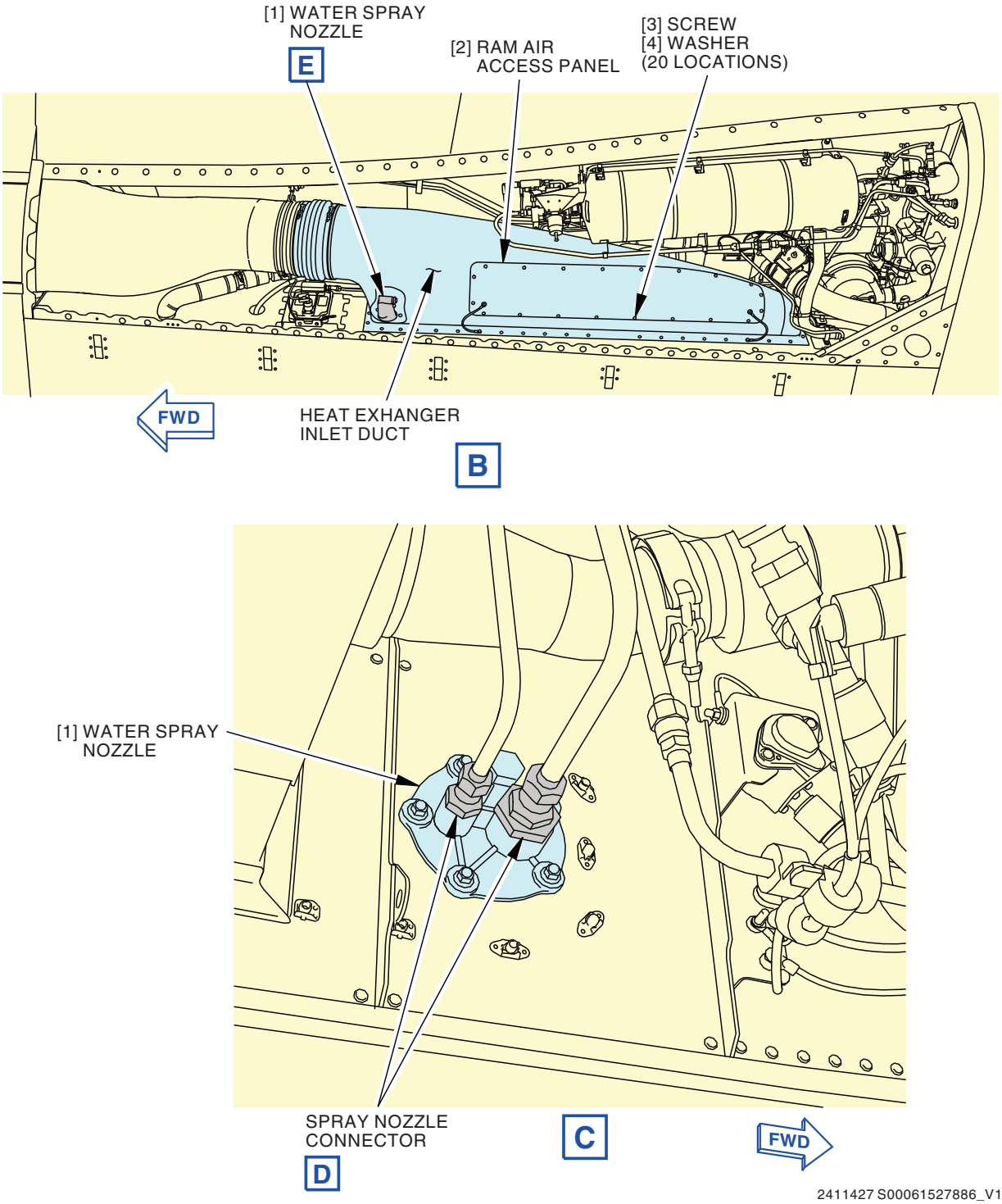
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



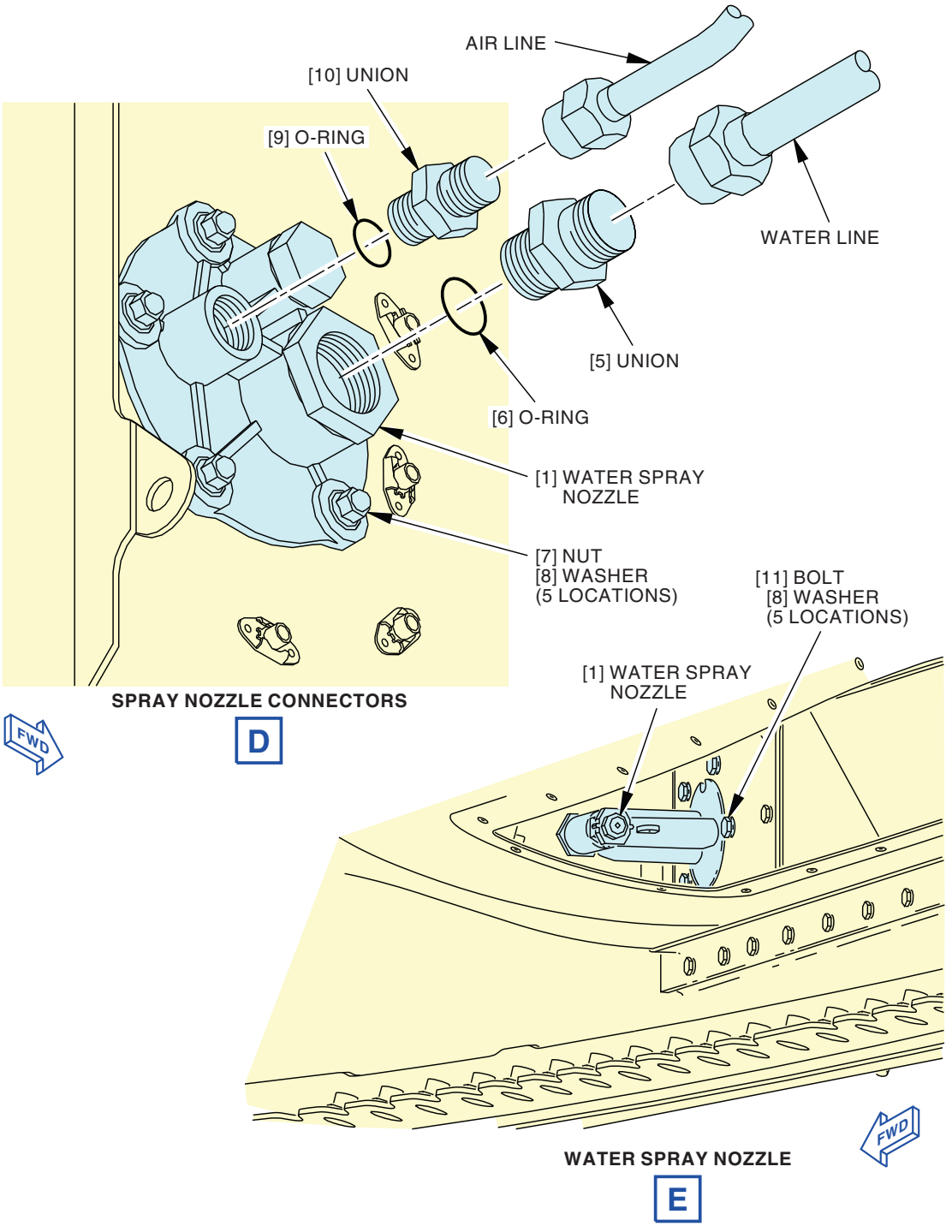
Water Spray Nozzle Installation
Figure 401/21-51-16-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Spray Nozzle Installation
Figure 401/21-51-16-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Spray Nozzle Installation
Figure 401/21-51-16-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-16-400-801

2. Water Spray Nozzle Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Nozzle	21-51-16-01-040	PTF ALL
		21-51-16-02-040	PTF ALL
6	O-ring	21-51-16-01-080	PTF ALL
		21-51-16-02-080	PTF ALL
9	O-ring	21-51-16-01-075	PTF ALL
		21-51-16-02-075	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Water Spray Nozzle Installation

SUBTASK 21-51-16-420-001

(1) To install the water spray nozzle, do these steps:

(a) Put the water spray nozzle [1] in its position.

(b) Install the bolts [11], washers [8], and nuts [7], at five locations.

SUBTASK 21-51-16-020-004

(2) To connect the water line to the water spray nozzle, do these steps:

(a) Install a new O-ring [6] on the union [5].

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(b) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the union [5].

(c) Install the union [5] on the water spray nozzle.

(d) Connect the water line to the water spray nozzle.

SUBTASK 21-51-16-020-005

(3) To connect the air line to the water spray nozzle, do these steps:

(a) Install a new O-ring [9] on the union [10].

(b) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the union [10].

(c) Install the union [10] on the water spray nozzle.

(d) Connect the air line to the water spray nozzle.

G. Water Spray Nozzle - Post Installation Test

SUBTASK 21-51-16-860-002

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-16-860-003

(2) Set the L(R) PACK switches, on the P5-10 overhead panel, to the HIGH position.

SUBTASK 21-51-16-860-004

(3) Set the trim air switch to the ON position.

SUBTASK 21-51-16-860-005

(4) Set the CONT CAB, FWD CAB and AFT CAB zone temperature selectors to the AUTO-C (fully cold) position.

SUBTASK 21-51-16-710-001

(5) Use a flashlight to see that a spray of air and water comes of the L(R) water spray nozzle.

NOTE: The quantity of water is a function of the air humidity and the pack air flow.

SUBTASK 21-51-16-860-006

(6) Set the CONT CAB, FWD CAB and AFT CAB zone temperature selectors to the OFF position.

SUBTASK 21-51-16-860-007

(7) Set the trim air switch to the OFF position.

SUBTASK 21-51-16-860-008

(8) Set the L(R) PACK switches to the OFF position.

SUBTASK 21-51-16-860-009

(9) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-16-410-001

(1) Install the access panel on the ram air inlet duct:

(a) Examine the gasket on the ram air inlet access panel [2].

(b) Replace the gasket if it is damaged or deteriorated.

(c) Hold the ram air inlet access panel [2] in its position while you install the screws [3] and washers [4] at 20 locations.

SUBTASK 21-51-16-410-002

(2) If the water spray nozzle [1] is installed for the left cooling pack, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

(b) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-16-410-003

(3) Do these steps, in the specified sequence, if you installed the water spray nozzle [1] for the right cooling pack:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER SPRAY NOZZLE - INSPECTION/CHECK

TASK 21-51-16-200-801

1. Water Spray Nozzle Check

Figure 601

A. General

(1) This procedure has instructions to do a check of the water spray nozzle and water spray lines of each pack. If these components have blockage, the Air Cycle Machine can get ice.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-16-000-801	Water Spray Nozzle Removal (P/B 401)
21-51-16-400-801	Water Spray Nozzle Installation (P/B 401)
24-22-00-860-801	Supply Electrical Power (P/B 201)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle

D. Consumable Materials

Reference	Description	Specification
B50117	Detergent - General	
B50126	Solvent - Degreasing, General Purpose	MIL-PRF-680
G50256	Water, Regular	

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Prepare for the Check

SUBTASK 21-51-16-010-004

(1) Do these steps to get access to the water spray nozzle for the left cooling pack:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

WARNING

OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(b) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-16-010-005

(2) Do these steps in the specified sequence to get access to the water spray nozzle for the right cooling pack:

(TASK 06-41-00-800-801)

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

(c) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-16-860-010

(3) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

H. Water Spray Nozzle Check

SUBTASK 21-51-16-860-011

(1) Supply pneumatic pressure to the pneumatic system with the APU, do this task: Supply Pressure to the Pneumatic System with the APU, TASK 36-00-00-860-803.

SUBTASK 21-51-16-860-012

(2) Set the L(R) PACK switches, on the P5-10 overhead panel, to the HIGH position.

SUBTASK 21-51-16-860-013

(3) Set the trim air switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-16-860-014

(4) Set the CONT CAB, FWD CAB and AFT CAB zone temperature selectors to the AUTO-C (fully cold) position.

SUBTASK 21-51-16-210-001

(5) Use a flashlight to see that a spray of air and water comes of the L(R) water spray nozzle.

NOTE: The quantity of water is a function of the air humidity and the pack air flow.

(a) Make sure that the water spray pattern from each spray nozzle is approximately the same.

(b) Make sure that the water sprays are approximately circular and are continuous.

(c) If the water sprays are not satisfactory, do the following steps to examine these parts for leakage or blockage:

1) The water collector manifold

2) The water extractor

3) The water extractor duct

4) The water spray nozzle

5) The air and water supply lines from the water extractor to the water spray nozzle.

SUBTASK 21-51-16-860-015

(6) Set the L(R) PACK switches to the OFF position.

SUBTASK 21-51-16-210-002

(7) Check the water collection manifold for leakage or blockage.

(a) Disconnect the water output port line [2] at the water collection manifold.

(b) Set the L(R) PACK switches, on the P5-10 overhead panel, to the HIGH position.

(c) Feel for air or water to come out of the hole in the water collection manifold where you removed the line.

(d) If there is no air or water, examine the water collection manifold for leakage or blockage.

SUBTASK 21-51-16-420-002

(8) Connect the water output port line [2] at the water collection manifold.

SUBTASK 21-51-16-860-016

(9) Set the L(R) PACK switches to the OFF position.

SUBTASK 21-51-16-210-003

(10) Check the water extractor for leakage or blockage.

(a) Disconnect the water drain lines [1] at each water drain nipple of the water extractors.

(b) Set the L(R) PACK switches, on the P5-10 overhead panel, to the HIGH position.

(c) Feel for air or water to come out of the hole in the water extractor where you removed the line.

(d) If there is no air or water, examine the water extractor for leakage or blockage.

SUBTASK 21-51-16-420-003

(11) Connect the water drain lines [1] at each water drain nipple of the water extractors.

SUBTASK 21-51-16-860-017

(12) Set the L(R) PACK switches to the OFF position.

SUBTASK 21-51-16-210-004

(13) Check the water extractor duct for leakage or blockage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the drain line [3] at the water extractor duct.

(b) Set the L(R) PACK switches, on the P5-10 overhead panel, to the HIGH position.

(c) Feel for air or water to come out of the hole in the water extractor duct where you removed the line.

(d) If there is no air or water, examine the water extractor duct for leakage or blockage.

SUBTASK 21-51-16-420-004

(14) Connect the drain line [3] at the water extractor duct.

SUBTASK 21-51-16-110-001

(15) If the water spray nozzle is blocked or clogged, do the steps that follow to clean the nozzle:

(a) Remove the nozzle from the airplane, do this task: Water Spray Nozzle Removal, TASK 21-51-16-000-801.

(b) Soak the nozzle assembly in solvent degreasing solvent, B50126.

(c) Use compressed air to blow out loose particles from the nozzle.

(d) Do steps (b) and (c) again if necessary to clean the inner areas of the nozzle.

(e) Use a soft bristle brush, STD-123, to remove external particles from the nozzle.

(f) Soak the nozzle in general detergent, B50117, and water, G50256, to remove the solvent.

(g) Dry the nozzle with compressed air.

(h) Examine the nozzle to make sure that there is no external damage.

(i) Use a low pressure air source of between 0.5 psi (34.5 mBar) - 1 psi (69 mBar) to blow air through the nozzle.

1) Make sure that air flows freely through the air inlet and water inlet.

(j) Install the nozzle, do this task: Water Spray Nozzle Installation, TASK 21-51-16-400-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-16-860-018

(1) Set the CONT CAB, FWD CAB and AFT CAB zone temperature selectors to the OFF position.

SUBTASK 21-51-16-860-019

(2) Set the L(R) PACK switches to the OFF position.

SUBTASK 21-51-16-860-020

(3) Set the trim air switch to the OFF position.

SUBTASK 21-51-16-860-021

(4) Remove the pneumatic pressure, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-16-410-004

(5) Do these steps if the water spray nozzle is installed for the left cooling pack:

(a) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

Close this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-16-410-005

(6) Do these steps, in the specified sequence, for the right cooling pack:

(TASK 06-41-00-800-801)

(a) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

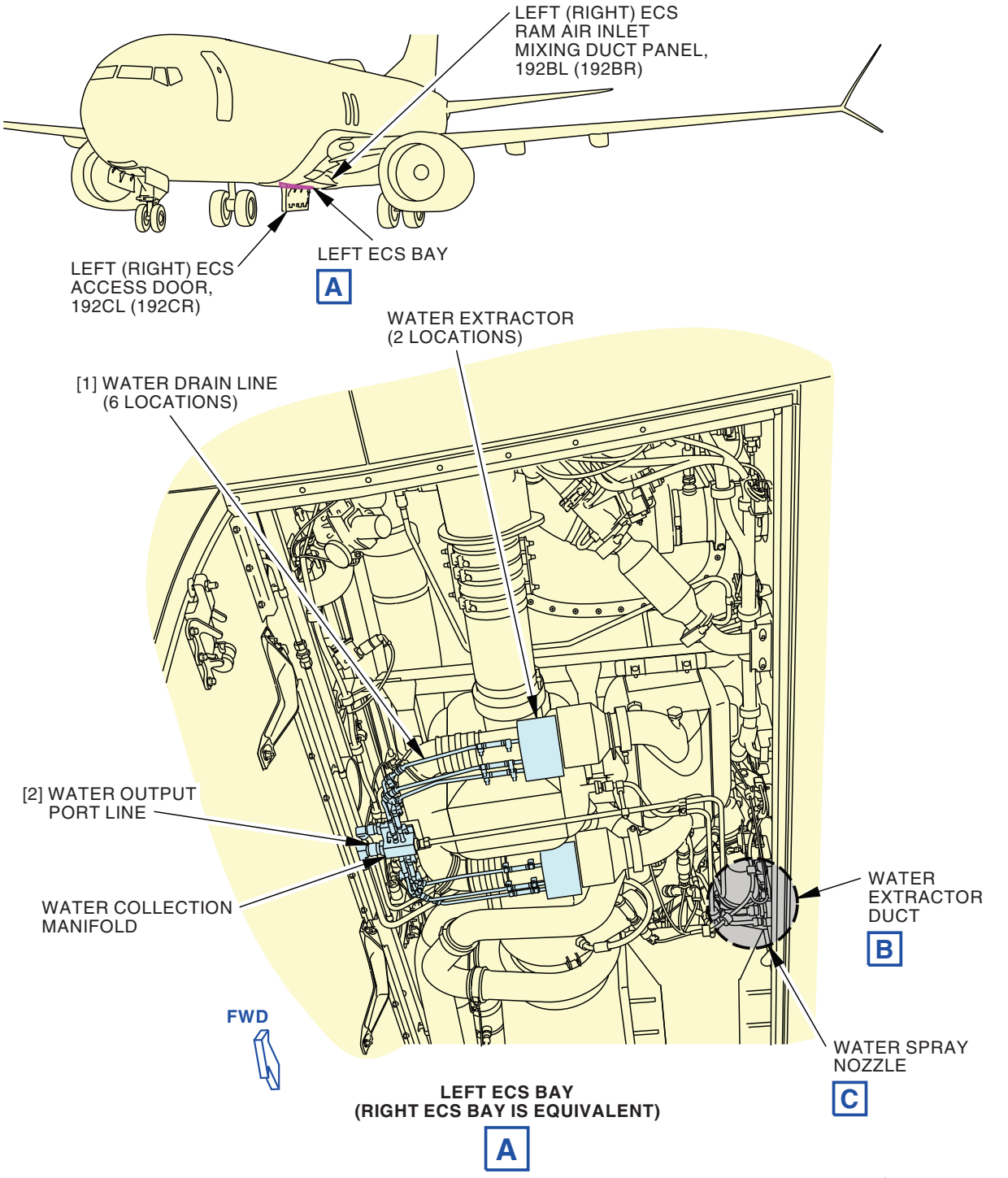
(c) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

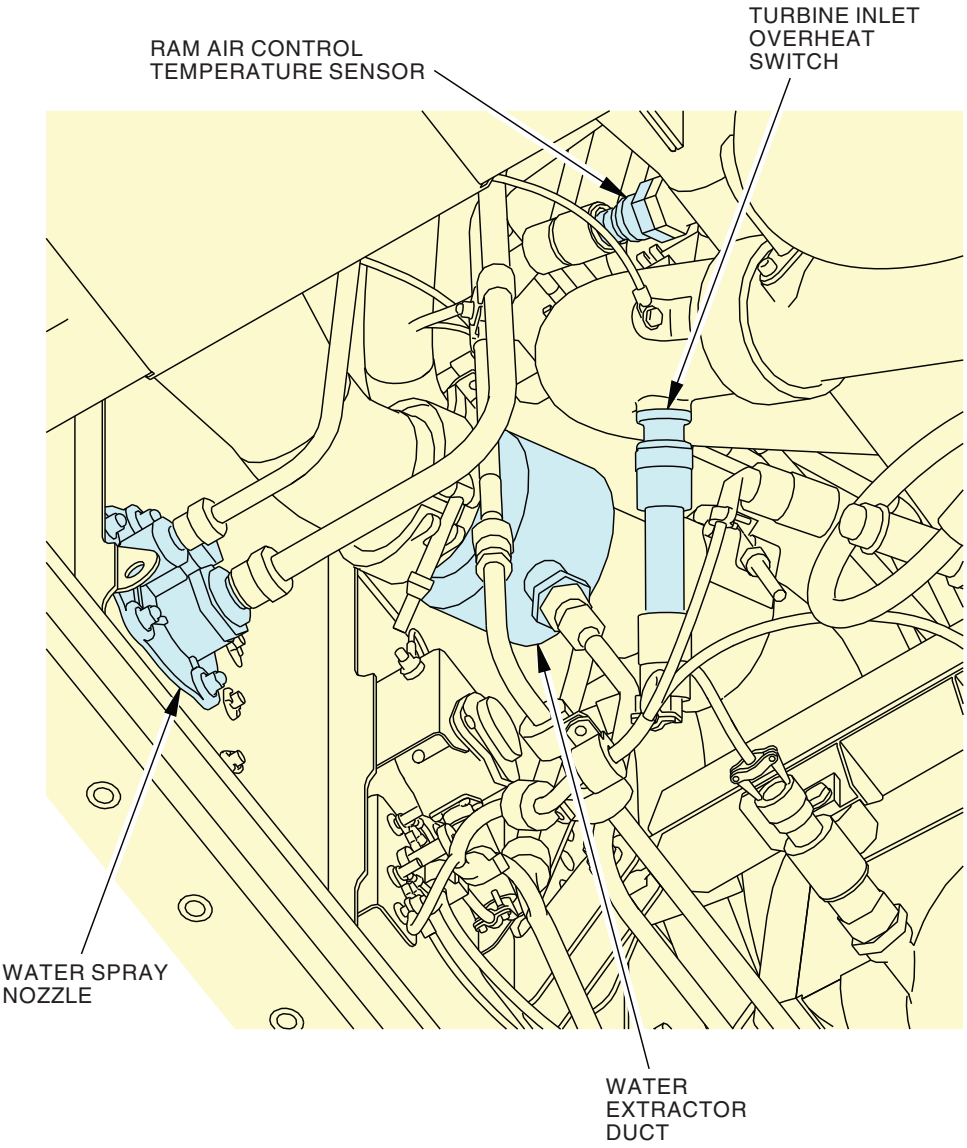
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Spray Nozzle Check
Figure 601/21-51-16-990-802 (Sheet 1 of 3)

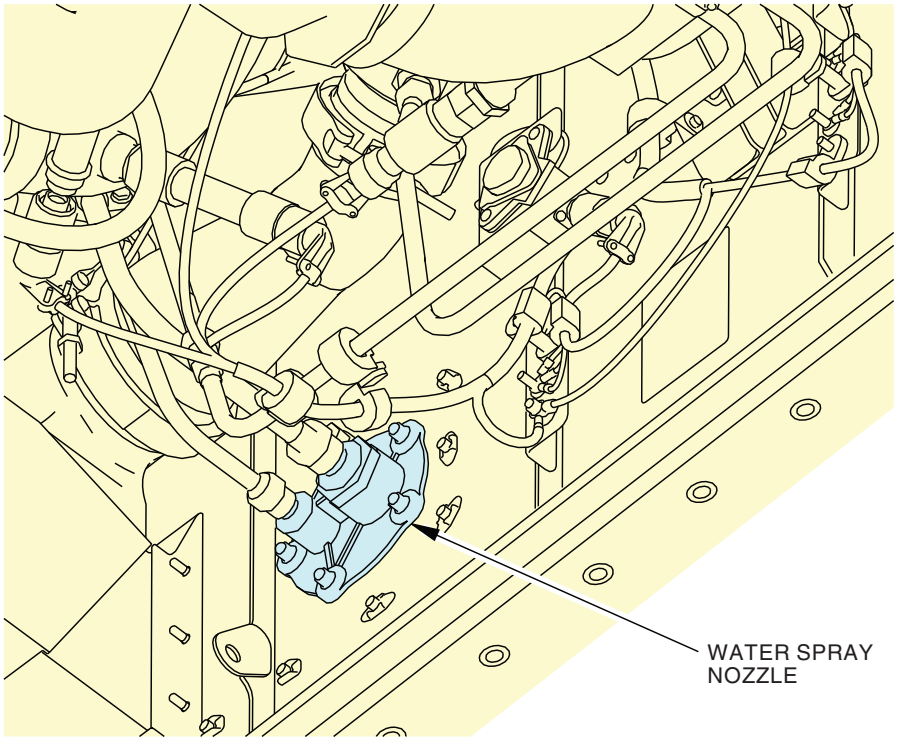
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Water Spray Nozzle Check
Figure 601/21-51-16-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WATER SPRAY NOZZLE

C

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Water Spray Nozzle Check
Figure 601/21-51-16-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIGH PRESSURE WATER SEPARATOR MIX MUFF - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the high pressure water separator mix muff.

(2) An installation of the high pressure water separator mix muff.

TASK 21-51-17-000-801

2. High Pressure Water Separator Mix Muff Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-17-860-003

(1) Set switches on the P5-10 panel on the forward overhead panel to these positions:

(a) L and R PACK switches to OFF position.

1) Attach a DO NOT OPERATE tag, STD-858.

(b) BLEED 1 and BLEED 2 switches to OFF position.

1) Attach the DO NOT OPERATE tag, STD-858.

(c) BLEED APU switch to OFF position.

1) Attach the DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-17-010-001

(2) To get access to the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-17-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(3) To get access to the right cooling pack, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-17-020-001

(4) To get access to the mix muff [16], remove the compressor outlet duct [3], do these steps:

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(a) Disconnect the electrical connector [1] from the compressor outlet overheat switch.

(b) Disconnect the electrical connector [2] from the ram air control temperature sensor.

(c) Remove the flexhose [9] from the compressor outlet duct [3].

(d) Remove the clamshell clamp [6].

(e) Remove the sleeve [7] from the compressor outlet duct [3].

CAUTION

HOLD THE DUCT WHEN YOU LOOSEN THE CLAMP. THE DUCT CAN FALL WHEN THE CLAMP IS LOOSE. THE DUCT CAN BE DAMAGED IF IT FALLS.

(f) Remove the clamp [4] that holds the compressor outlet duct [3] to the heat exchanger.

(g) Remove the compressor outlet duct [3].

(h) Remove and discard the o-rings [8].

(i) Remove and discard the packing [5].

F. Mix Muff Removal

SUBTASK 21-51-17-020-002

(1) To remove the mix muff [16], do these steps:

(a) Disconnect the flexhose [10] from the mix muff [16].

(b) Loosen the clamps [14] that attach the flexible duct [15] to the mix muff [16].

(c) Move the flexible duct [15] off the mix muff [16].

(d) Remove the clamp [13].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

HOLD THE MIX MUFF WHEN YOU LOOSEN THE CLAMP. THE MIX MUFF CAN FALL WHEN THE CLAMP IS LOOSE. THE MIX MUFF CAN BE DAMAGED IF IT FALLS.

(e) Loosen the clamp [11] that attaches the mix muff [16] to the air cycle machine.

(f) Remove the mix muff [16].

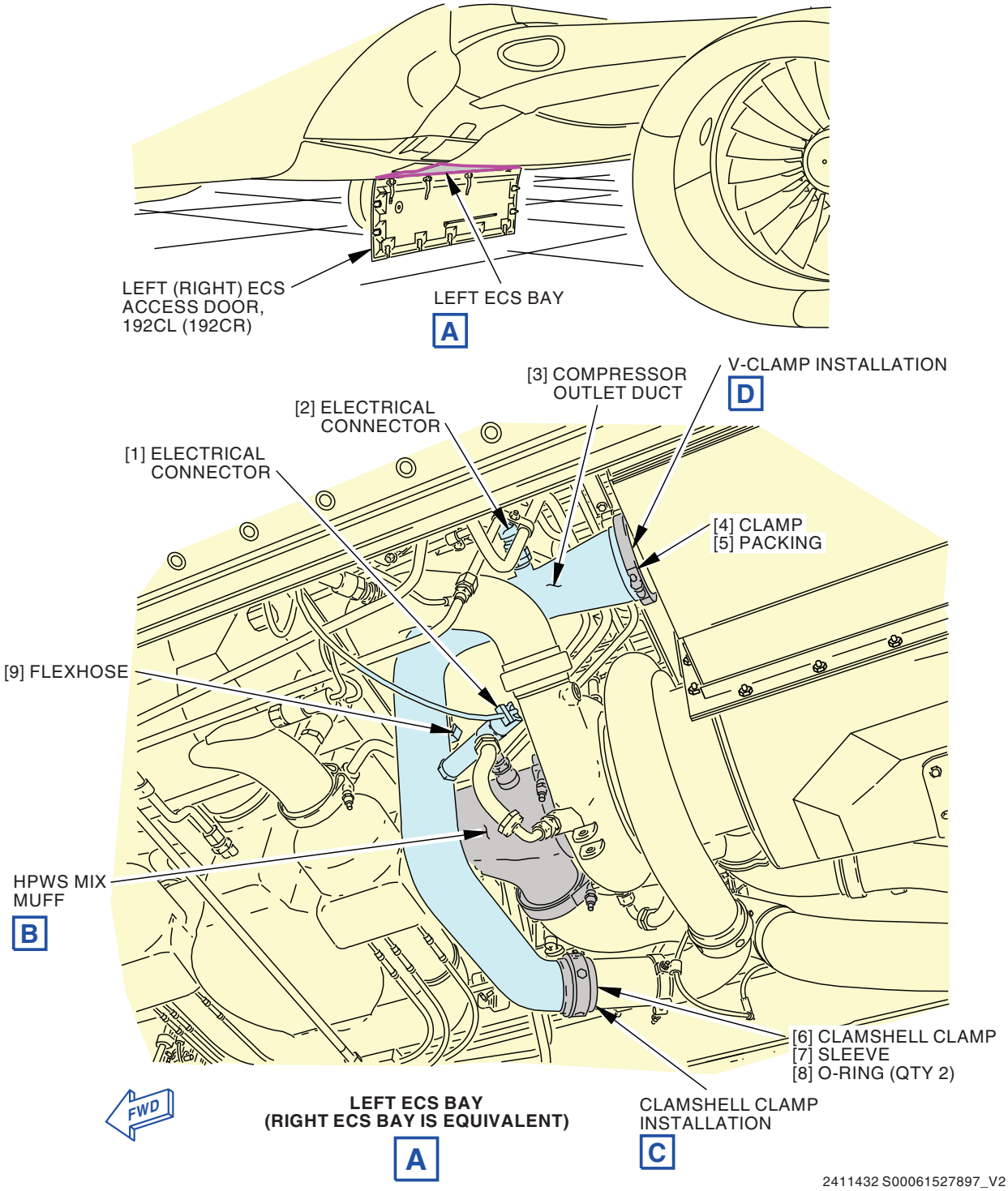
(g) Remove and discard the packing [12].

SUBTASK 21-51-17-620-001

(2) Put covers on the duct openings to prevent the entry of unwanted materials.

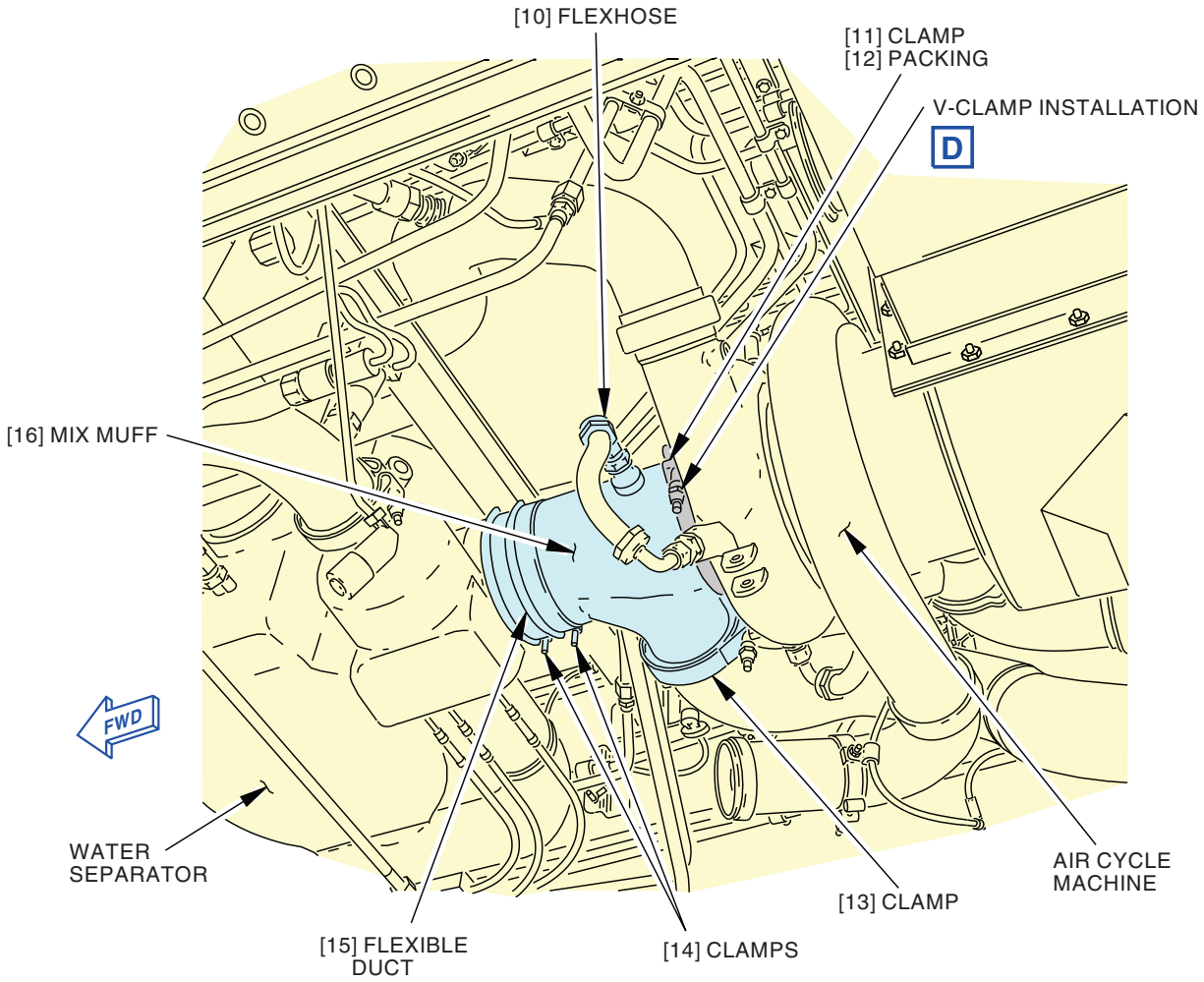
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



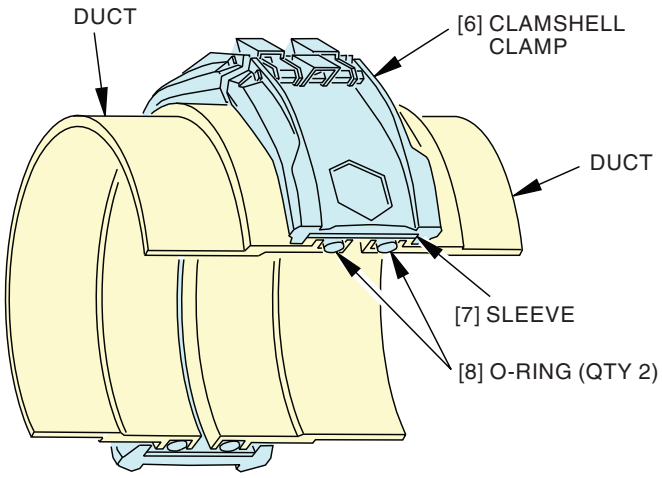
High Pressure Water Separator (HPWS) Mix Muff Installation
Figure 401/21-51-17-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



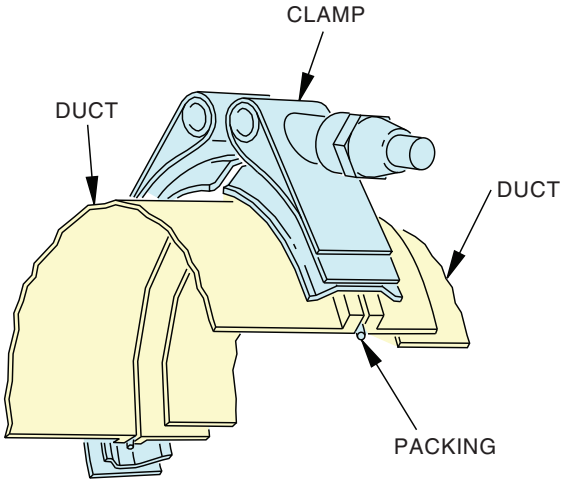
High Pressure Water Separator (HPWS) Mix Muff Installation
Figure 401/21-51-17-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CLAMSHELL CLAMP INSTALLATION

C



V-CLAMP INSTALLATION (EXAMPLE)

D

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High Pressure Water Separator (HPWS) Mix Muff Installation
Figure 401/21-51-17-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-17-400-801

3. High Pressure Water Separator Mix Muff Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00504	Grease - Petrolatum	VV-P-236
D00602	Lubricant - Fluorinated	BMS3-25
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	
D50320	Lubricant - Moly-50 Anti-Seize	MIL-PRF-83483

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Packing	21-51-03-02-040	PTF ALL
		21-51-03-02-200	PTF ALL
8	O-ring	21-51-03-02-055	PTF ALL
		21-51-03-02-210	PTF ALL
12	Packing	21-51-17-02-015	PTF ALL
		21-51-17-08-015	PTF ALL
16	Mix muff	21-51-17-02-040	PTF ALL
		21-51-17-08-040	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Mix Muff Installation

SUBTASK 21-51-17-630-001

(1) Remove the duct covers.

SUBTASK 21-51-17-020-003

(2) To install the mix muff [16], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Lubricate the new packing [12] with lubricant, D00602.

(b) Install the new packing [12].

(c) Put the mix muff [16] into its position between the ducts.

1) Hold the mix muff [16] in position when you install the clamps.

NOTE: The duct connections are not self aligning.

(d) Install the clamp [11] that holds the mix muff [16] to the air cycle machine.

1) Do not fully tighten the clamp [11].

(e) Install the clamp [13] that holds the mix muff [16] to the temperature control duct.

1) Do not fully tighten the clamp [13].

(f) Move the flexible duct [15] on the mix muff [16].

(g) Install the clamps [14] on the flexible duct [15].

(h) Tighten the clamps [14] to 15 ±2 in-lb (1.7 ±0.2 N·m).

(i) Tighten the clamp [11] to 62.5 ±2.5 in-lb (7.1 ±0.3 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.610 in. (15.494 mm).

(j) Tighten the clamp [13] to 62.5 ±2.5 in-lb (7.1 ±0.3 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

(k) Apply a light coating of Pure Nickel Special compound, D00006, to the threads of the flexhose [10].

(l) Install the flexhose [10] on the mix muff [16].

G. Compressor Outlet Duct Installation

SUBTASK 21-51-17-020-004

(1) To install the compressor outlet duct [3], do these steps:

(a) Apply a light coat of grease, D00504, or Molykote 44 grease, medium, D50104, to the packing [5] and o-rings [8].

(b) Install the new packing [5].

(c) Install the new o-rings [8] on the duct ends.

(d) Put the compressor outlet duct [3] into its position.

1) Hold the compressor outlet duct [3] in position when you install the clamps.

NOTE: The duct connections are not self aligning.

(e) Apply compound, D50320, to the bolt threads of the clamp [4].

(f) Install the clamp [4] that holds the compressor outlet duct [3] to the heat exchanger.

1) Do not fully tighten the clamp [4].

(g) Install the sleeve [7] over the duct ends.

(h) Install the clamshell clamp [6].

(i) Tighten the clamp [4] to 47.5 ±2.5 in-lb (5.4 ±0.3 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) With hand force, gently push the compressor outlet duct in the direction of the adjacent mix muff duct until the compressor outlet duct reaches the maximum flexed position.

(k) Make sure that there is a minimum clearance of 0.20 in. (5.1 mm) to the adjacent mix muff duct when the compressor outlet duct is in the flexed position.

(l) Install the electrical connector [2] on the ram air control temperature sensor.

(m) Install the electrical connector [1] on the compressor outlet overheat switch.

(n) Apply a light coating of Pure Nickel Special compound, D00006, to the threads of the flexhose [9].

(o) Reconnect the flexhose [9] to the compressor outlet duct [3].

H. Mix Muff Installation Test

SUBTASK 21-51-17-860-004

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-17-860-005

(2) Put the applicable L PACK or R PACK switch on the P5-10 air conditioning panel to the AUTO position.

SUBTASK 21-51-17-790-001

(3) Do a soap bubble test of the joints at the mix muff and compressor outlet duct.

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches on the P5-10 air conditioning panel in the OFF position.

2) Loosen the clamps.

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps.

a) For the clamp [11] and clamp [13], make sure that the loop bands do not touch.

b) For the clamp [11], make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.610 in. (15.494 mm).

c) For the clamp [13], make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

5) Put the L PACK and R PACK switches on the P5-10 air conditioning panel in the AUTO position.

6) Make sure that the leak has been repaired.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-17-860-006

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-17-860-007

(2) Put the L PACK and R PACK switches on the P5-10 air conditioning panel in the OFF position.

SUBTASK 21-51-17-010-003

(3) If the mix muff for the left cooling pack was removed, close this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-17-010-004

(4) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR INLET ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Ram Air Inlet Actuator Removal.

(2) Ram Air Inlet Actuator Installation.

TASK 21-51-21-000-801

2. Ram Air Inlet Actuator Removal

(Figure 401)

A. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
191FL	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward

E. Prepare for the Removal

SUBTASK 21-51-21-860-010

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM BEFORE YOU DO MAINTENANCE. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-21-860-011

(2) Set these switches on the P5-10 air conditioning panel as follows:

(a) Set the BLEED 1 and BLEED 2 switches to the OFF position.

1) Attach DO NOT OPERATE tags, STD-858.

(b) Set the APU BLEED switch to the OFF position.

1) Attach DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-21-860-001

(3) To remove the ram air inlet actuator [15] for the left pack, do these steps:

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(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-51-21-010-001

(4) Open these access panels to remove the ram air inlet actuator for the left pack:

Number	Name/Location
191FL	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

SUBTASK 21-51-21-860-002

(5) To remove the ram air inlet actuator [15] for the right pack, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-51-21-010-002

(6) Open these access panels to remove the ram air inlet actuator for the right pack:

Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

F. Ram Air Inlet Actuator Removal

SUBTASK 21-51-21-020-001

(1) Remove the electrical connector [29] and electrical connector [30] from the actuator [15].

SUBTASK 21-51-21-020-002

(2) Remove the blue end of the bonding jumper [5] from the actuator [15] as follows:

(a) Remove enough of the sealant to get access to the screw [1] and nut [6].

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(b) Remove the screw [1], washer [3], washer [4], and nut [6] from the ground tab [19] on the actuator [15].

SUBTASK 21-51-21-020-003

(3) Disconnect the actuator [15] from the support [18] as follows:

(a) Support the actuator [15] and remove the nut [12], washer [9], bolt [17], washer [16], and bushings [28].

SUBTASK 21-51-21-020-008

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE DEFLECTOR DOOR. THE DEFLECTOR DOOR WILL RETRACT QUICKLY WHEN THE ACTUATOR DISCONNECTS FROM THE ARM. DAMAGE TO EQUIPMENT AND INJURY TO PERSONS CAN OCCUR.

(4) Disconnect the rod end of the actuator [15] from the arm [10] as follows:

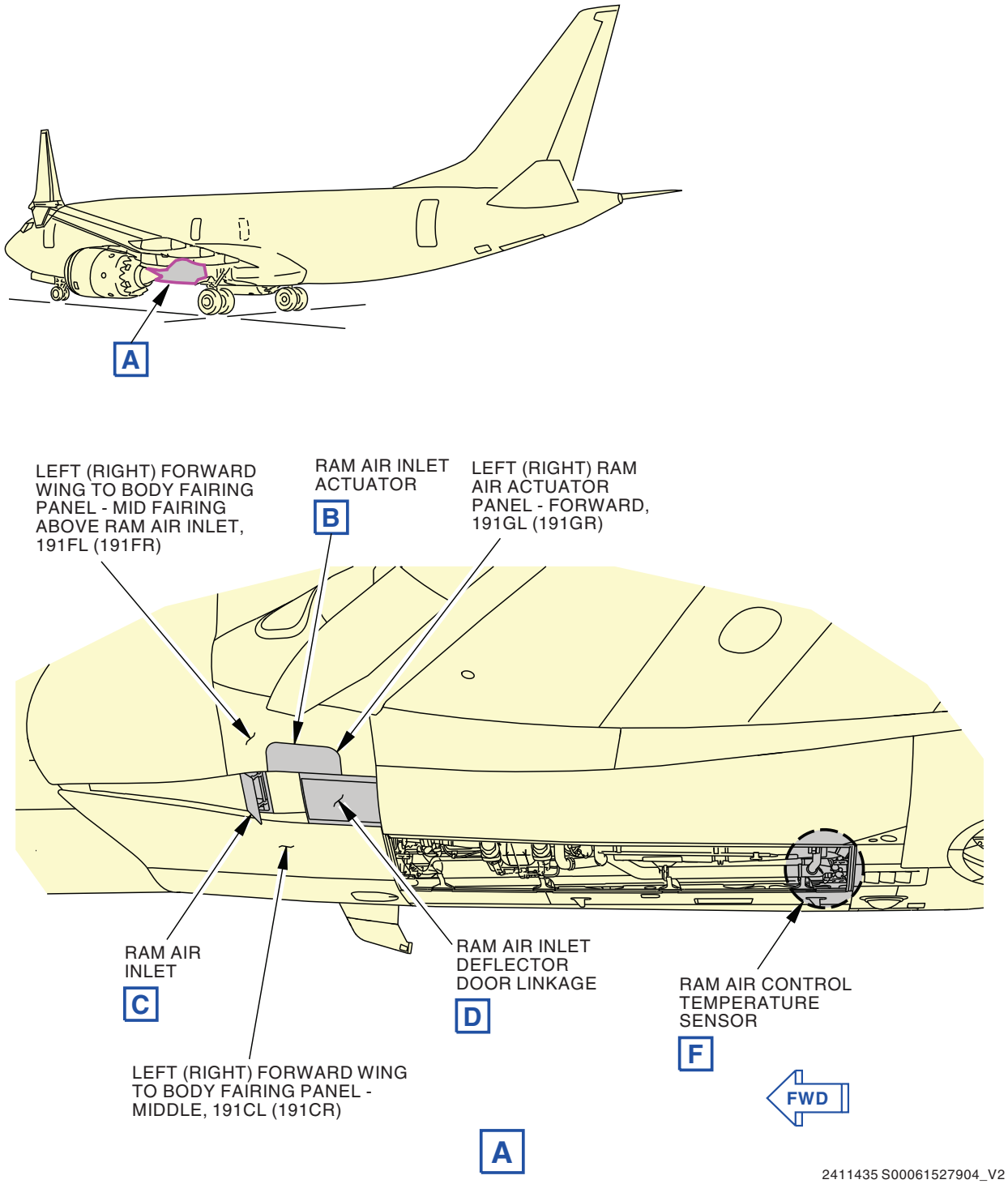
(a) Support the actuator [15] and remove the nut [12], washer [9], bushing [28], washer [16], and bolt [8].

SUBTASK 21-51-21-020-004

(5) Remove the actuator [15].

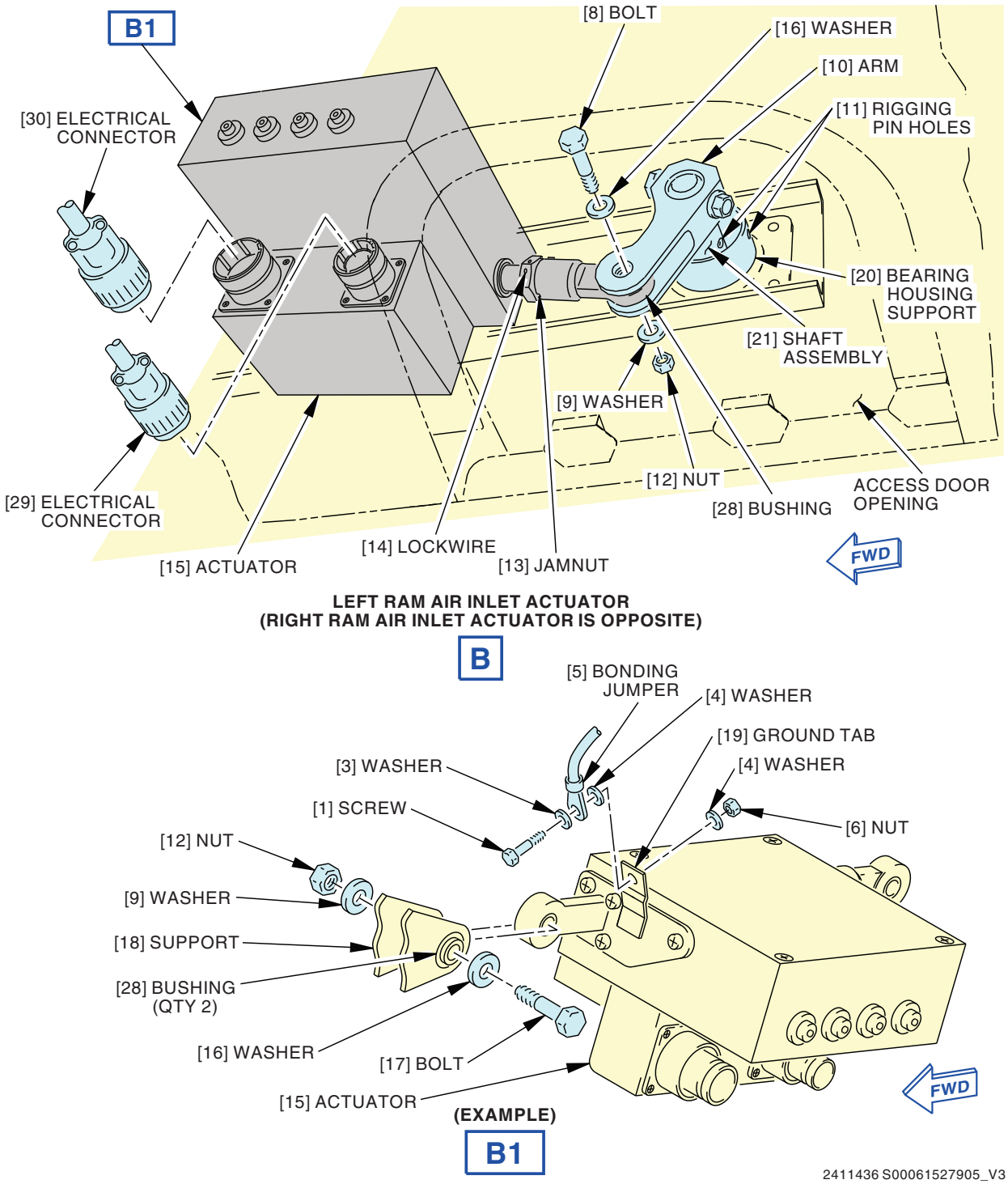
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



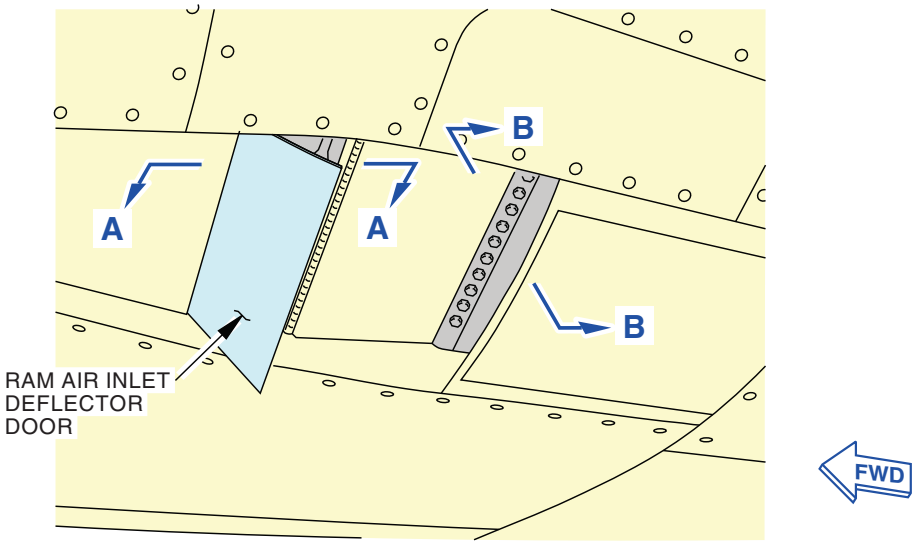
Ram Air Inlet Actuator Installation
Figure 401/21-51-21-990-801 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



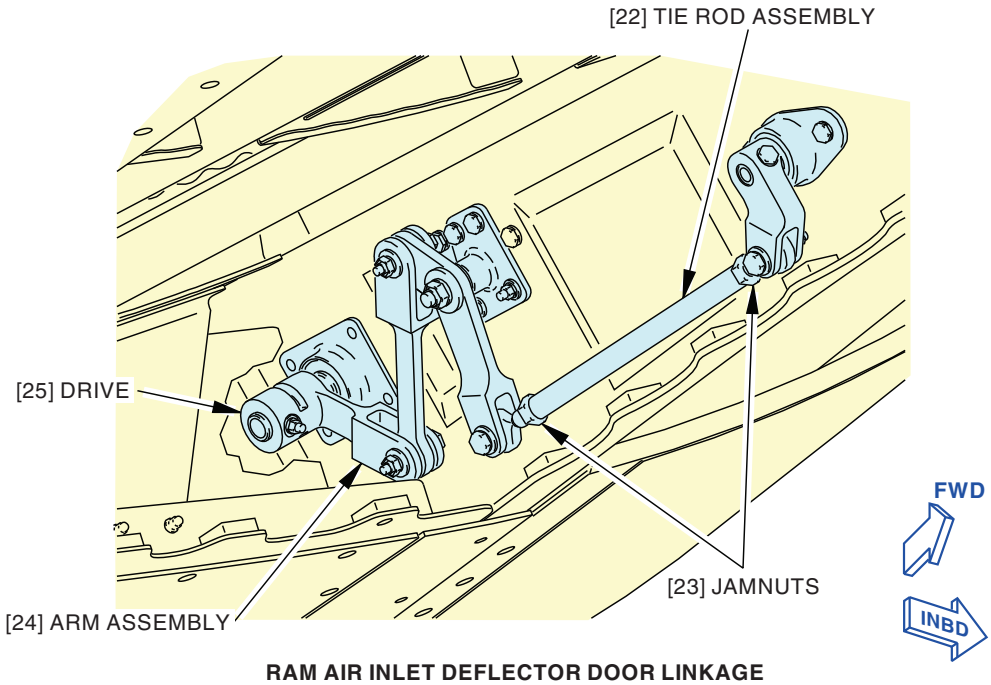
Ram Air Inlet Actuator Installation
Figure 401/21-51-21-990-801 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RAM AIR INLET

C



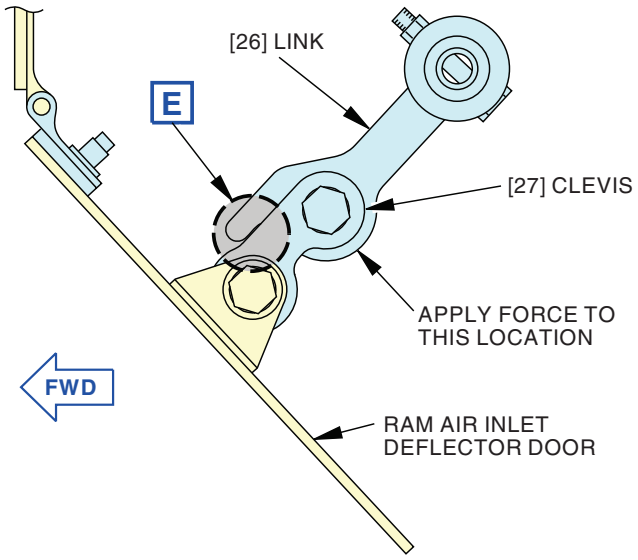
RAM AIR INLET DEFLECTOR DOOR LINKAGE

D

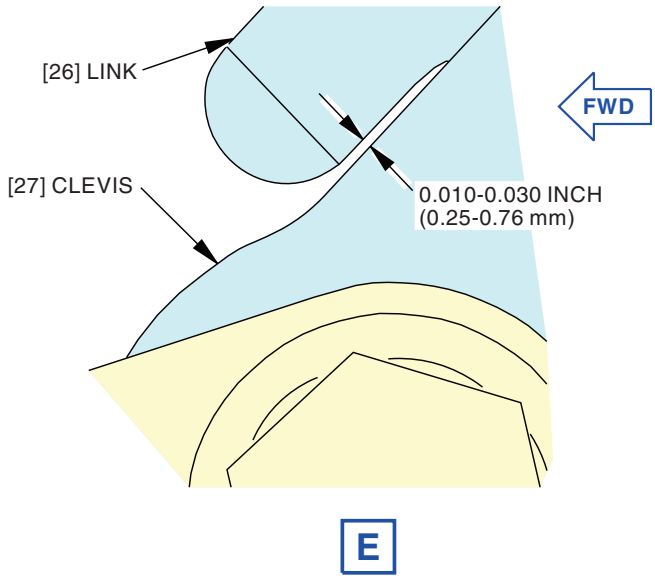
2411437 S00061527906_V1

Ram Air Inlet Actuator Installation
Figure 401/21-51-21-990-801 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A

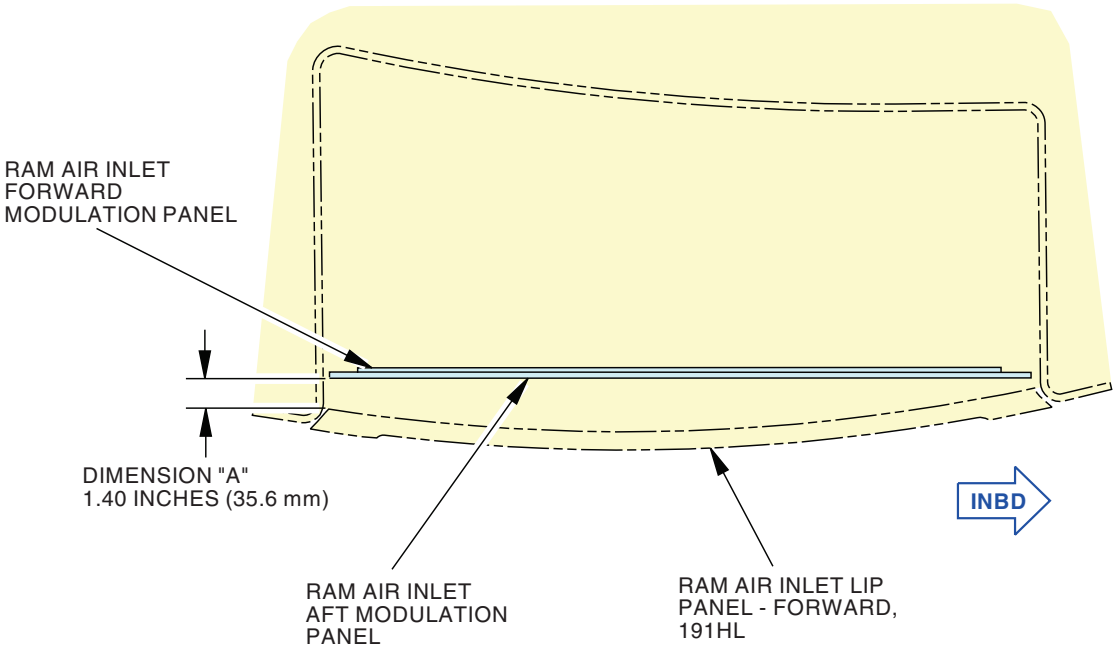


E

2411438 S00061527907_V1

Ram Air Inlet Actuator Installation
Figure 401/21-51-21-990-801 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



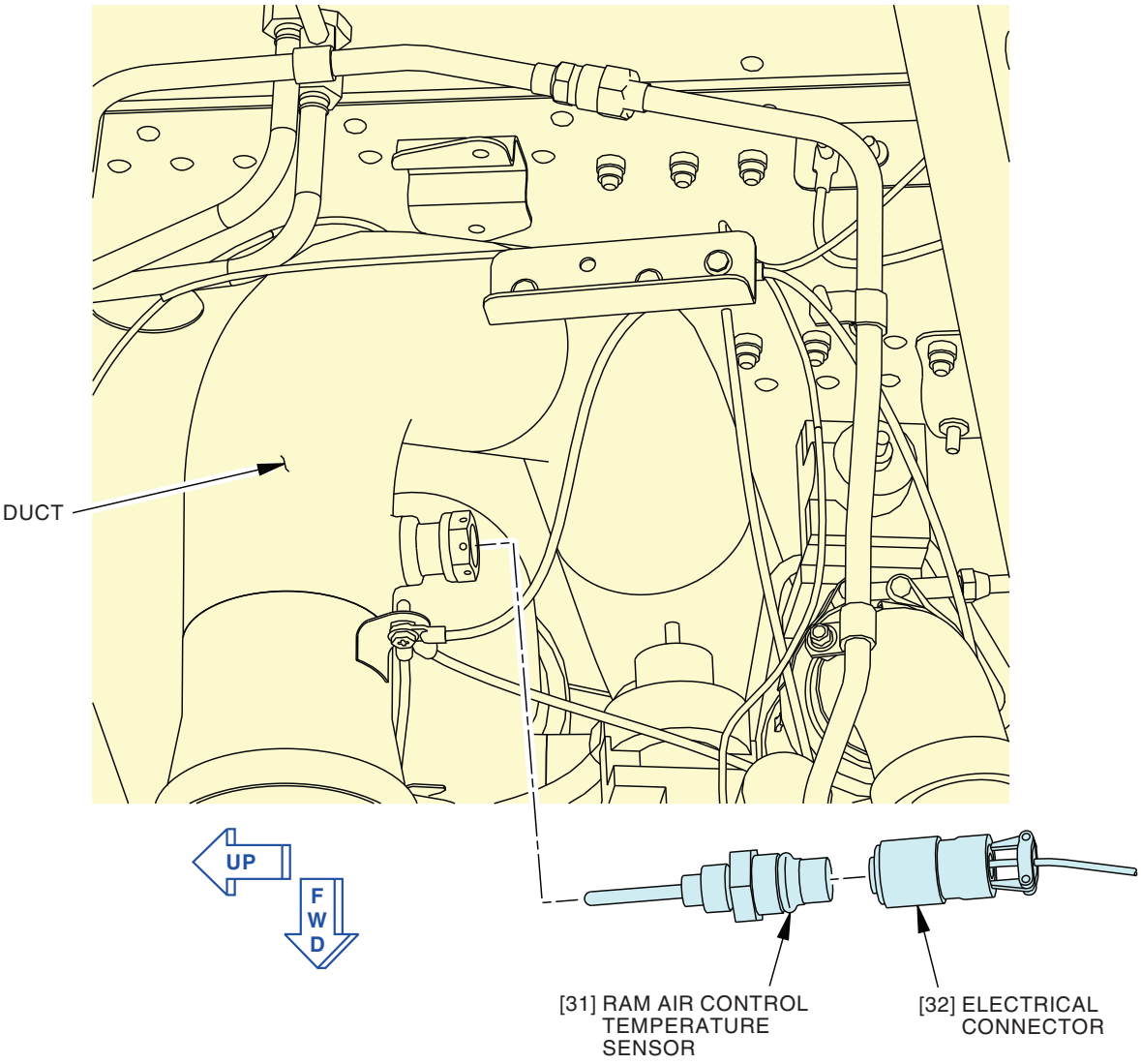
(FORWARD AND AFT MODULATION PANELS SHOWN IN FLIGHT FULL CLOSED POSITION)

B-B

2411439 S00061527908_V2

Ram Air Inlet Actuator Installation
Figure 401/21-51-21-990-801 (Sheet 5 of 6)

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RAM AIR CONTROL TEMPERATURE SENSOR

F

2983857 S0000760636_V1

Ram Air Inlet Actuator Installation
Figure 401/21-51-21-990-801 (Sheet 6 of 6)

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TASK 21-51-21-400-801

3. Ram Air Inlet Actuator Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

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(Continued)

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
COM-4746	Handle - Sealant Removal Tool Part #: 310/1 (for SkyScraper) Supplier: F6892 Part #: SHR0272-T (for PnJ scraper) Supplier: 7RKH2 Part #: ST982LH-F-3 (for ST982LF) Supplier: 81205 Opt Part #: SHW0215 (for PnJ scraper) Supplier: 7RKH2

C. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B00666	Solvent - Methyl Propyl Ketone	BMS11-9
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

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(Continued)

Reference	Description	Specification
G51925	Strap - Plastic, Wire Dressing Tie, Self-Locking, Unpressurized Areas, BACS38AD1	BACS38AD
G51926	Strap - Plastic, Wire Dressing Tie, Self-Locking, Unpressurized Areas, BACS38AD2	BACS38AD
G51927	Spacer - Wire Harness, BACS18BF1 (Use with BACS38AD Plastic Straps)	BACS18BF

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
15	Actuator	21-51-21-03-120	PTF ALL

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
191FL	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Ram Air Inlet Actuator Installation

SUBTASK 21-51-21-420-001

(1) Install the ram air inlet actuator [15] as follows:

(a) Put the ram air inlet actuator [15] in its position.

(b) Attach the actuator [15] to the support [18] as follows:

1) Make sure that the bushings [28] are installed in the support [18].

2) Install the bolt [17], washer [16], washer [9], and nut [12].

a) Put the countersunk side of the washer [16] under the head of the bolt [17].

SUBTASK 21-51-21-420-002

(2) Install the bonding jumper [5] to the ground tab [19] on the actuator [15] as follows:

(a) Clean the mating surfaces of the screw [1], washer [3], washer [4], nut [6], and ground tab [19] as follows:

1) Use sealant removal tool, COM-2481, and sealant removal tool handle, COM-4746, to remove the adhesive.

2) Use cotton wiper, G00034, moist in solvent, B00184, or solvent, B00666, or solvent, B00148, until all the adhesive is removed as follows:

a) Pour the solvent onto the wiper to avoid contamination of the solvent container.

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b) Wipe or scrub the surface to be cleaned with the solvent soaked wiper.

c) Immediately wipe the cleaned surfaces with a clean cotton wiper, G00034.

3) Use clean cotton wiper, G00034, to remove all excess solvent that may remain on the parts.

4) Repeat the cleaning action until no discoloration shows on the wiper.

5) Dry the surface before the solvent evaporates.

(b) Install the screw [1], washer [3], bonding jumper [5], washers [4], and nut [6].

NOTE: It is permissible to install the bonding jumper terminal on both sides of the ground tab with the fastener head and nut in both directions.

1) Make sure that the bonding jumper [5] does not hit wire bundles, ducts, the airplane structure, or other components.

a) If sections of the bonding jumper [5] are closer than 0.25 in. (6.4 mm) to electrical wire bundles or tubing, separate the bonding jumper [5] with spacer, G51927.

b) Install spacer, G51927, with plastic straps, G51925, or plastic straps, G51926.

2) Tighten the nut [6] to 35 in-lb (4.0 N·m) - 45 in-lb (5.1 N·m).

(c) Measure the electrical bonding resistance between the ground tab [19] and terminals of the bonding jumper [5] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.0010 ohm (1 milliohm) or less.

(d) Apply sealant, A50337, or sealant, A50231, or sealant, A50359, so that the bonding installation is totally encapsulated.

SUBTASK 21-51-21-860-003

(3) Install the electrical connector [29] and electrical connector [30] on the ram air actuator [15].

H. Adjustment of the Ram Air Inlet Actuator

SUBTASK 21-51-21-020-005

(1) Disconnect the tie rod assembly [22] from the forward and aft links.

SUBTASK 21-51-21-020-006

(2) Make sure that the rod end of the actuator [15] is not attached to the arm [10].

(a) If the rod end is attached, remove the nut [12], washer [9], bushing [28], washer [16], and bolt [8].

SUBTASK 21-51-21-860-013

(3) To install the ram air actuator [15] for the left pack, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

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(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-21-860-014

(4) To install the ram air actuator [15] for the right pack, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-21-860-005

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Extend the left and right trailing edge flaps to position 1.

(a) To extend the flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

NOTE: The actuator will retract slightly.

SUBTASK 21-51-21-860-006

(6) Simulate the air mode (TASK 32-09-00-860-801).

SUBTASK 21-51-21-860-007

(7) Retract the left and right trailing edge flaps to the full-up position (TASK 27-51-00-860-804).

SUBTASK 21-51-21-020-007

(8) Disconnect the electrical connector [32], D3916 from the left ram air control temperature sensor [31], T481 and electrical connector [32], D3918 from the right ram air control temperature sensor [31], T482.

SUBTASK 21-51-21-700-001

(9) Make sure that the ram air actuator [15] retracts.

NOTE: This position is approximately 0.9 in. (22.9 mm) from the fully retracted position.

SUBTASK 21-51-21-820-001

(10) Adjust the ram air actuator [15] as follows:

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(a) Rotate the shaft assembly [21] until the rigging pin holes [11] in the shaft assembly [21] and bearing housing support [20] align.

(b) Install a 3/16-inch diameter rigging pin through the rigging pin holes [11].

(c) Attach the actuator rod end to the arm [10].

1) Install the bolt [8], washer [16], bushing [28], washer [9], and nut [12] to attach the actuator rod end to the arm [10].

a) Put the countersunk side of the washer [16] under the head of the bolt [8].

2) If the hole in the rod end of the actuator [15] does not align with the holes in the arm [10], do these steps:

a) Remove the lockwire [14] from the jamnut [13] on the actuator rod end.

b) Adjust the length of the rod end by maximum ±0.125 in. (3.18 mm) (or 4 turns in either direction) until the holes in the rod end and actuator drive arm align.

c) Make sure that the jamnut [13] is rotated so that when the actuator is fully extended (door closed) the jamnut [13] does not interfere with the arm [10].

d) After adjustment of the actuator rod end, tighten the jamnut [13] as follows:

<1> Hold the actuator rod end from turning and tighten the jamnut [13] until it is seated on the rod.

<2> Tighten the jamnut [13] to 110 ±10 in-lb (12.43 ±1.13 N·m).

e) Install lockwire [14] (MS20995C32 lockwire, G01048) to hold the jamnut [13] in place (TASK 20-10-44-400-801).

(d) Remove the 3/16-inch rigging pin.

SUBTASK 21-51-21-710-001

(11) Adjust the tie rod assembly [22] as follows:

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Extend the trailing edge flaps to position 1.

1) To extend the flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

NOTE: The actuator will retract slightly.

(b) Put the airplane in the ground mode (TASK 32-09-00-860-802).

(c) Install the tie rod assembly [22] as follows:

1) Adjust the length of the tie rod assembly [22] until contact is just made between the first set of over travel stops (Views A-A and E, Figure 401).

NOTE: The over travel stops are the extensions on the links [26]. One of the over travel stops will hit the clevis [27] before the other travel stop does. It is considered to be the first set of over travel stops.

2) Once contact is made, adjust the tie rod assembly [22] so that it is one full turn shorter.

NOTE: The tie rod can be adjusted without removing it by loosening both jamnuts [23] and rotating the shaft to shorten or lengthen it.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Do a check that the tie rod assembly [22] is properly adjusted as follows:

1) Push with 8 lbf (36 N) - 12 lbf (53 N) at the joint location in the direction indicated by the arrow.

2) Make sure that the smaller of the two gaps is 0.010 in. (0.254 mm) - 0.03 in. (0.76 mm) with the force applied (Views A-A and E, Figure 401).

NOTE: The inboard and outboard gaps on the deflector door linkage may or may not be the same.

3) If the gap is not within the acceptable limits, adjust the tie rod assembly [22] in 1/4 turn increments until the gap is within the 0.010 in. (0.254 mm) - 0.03 in. (0.76 mm) specification.

4) If the gap is within the acceptable limits, tighten both jamnuts [23] on the tie rod assembly [22] to 50 ±10 in-lb (5.65 ±1.13 N·m).

5) Install MS20995C32 lockwire, G01048, to retain the jamnuts [23] in position (TASK 20-10-44-400-801).

SUBTASK 21-51-21-710-002

(12) Do this check of the deflection door and ram air inlet doors adjustment.

(a) Simulate the air mode (TASK 32-09-00-860-801).

(b) Make sure that the inlet doors move towards the closed position smoothly without hanging or binding.

(c) Make sure that the deflector door retracts (closes) to the faired position.

(d) Retract the left and right trailing edge flaps to the full-up position (TASK 27-51-00-860-804).

(e) Connect the electrical connector [32], D3916 to the left ram air control temperature sensor [31], T481 and electrical connector [32], D3918 to the right ram air control temperature sensor [31], T482.

(f) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(g) Wait for a minimum of 10 seconds and then continue.

NOTE: Opening the above circuit breakers for 10 seconds minimum and then closing will clear the Smart Ram Air Door Actuator (SRADA) internal fault memory and allow actuator movement during subsequent steps.

(h) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

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(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(i) Make sure that the actuator drives to the fully extended position (flight fully closed).

NOTE: Dimension “A” should be approximately 1.4 in. (35.6 mm).

(j) Put the airplane in the ground mode (TASK 32-09-00-860-802).

(k) Make sure that the ram air inlet doors return to the fully open position smoothly without any hanging or binding.

(l) Make sure that the deflector door returns to the fully extended position.

I. Ram Air Inlet Actuator Test

SUBTASK 21-51-21-740-001

(1) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

(a) When the test is completed, make sure that PASS shows adjacent to TEST CONDITION.

(b) If FAILED shows, then do the fault isolation procedure for the maintenance message.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-21-860-008

(1) For the ram air inlet actuator for the left pack, install these access panels:

Number	Name/Location
191FL	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GL	Ram Air Actuator Panel - Forward

SUBTASK 21-51-21-410-003

(2) For the left ram air inlet actuator, close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-21-860-009

(3) For the ram air inlet actuator for the right pack, install these access panels:

Number	Name/Location
191FR	Forward Wing-To-Body Fairing Panel - Mid Fairing, Above Ram Air Inlet
191GR	Ram Air Actuator Panel - Forward

SUBTASK 21-51-21-410-004

(4) For the right ram air inlet actuator, close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR INLET DEFLECTOR DOOR AND SHAFT ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ram air inlet deflector door

(2) An installation of the ram air inlet deflector door

(3) A removal of the shaft assembly for the ram air inlet deflector door

(4) An installation of the shaft assembly for the ram air inlet deflector door.

TASK 21-51-22-000-801

2. Ram Air Inlet Deflector Door Removal

(Figure 401)

A. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 21-51-22-860-001

(1) For the left pack ram air inlet deflector door, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-22-860-002

(2) For the right pack ram air inlet deflector door, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

C. Ram Air Inlet Deflector Door Removal

SUBTASK 21-51-22-020-001

(1) Remove the nuts [11] and washers [8] from the ends of the bolts [9] that attach the ram air inlet deflector door [1] to the links.

SUBTASK 21-51-22-020-002

(2) Remove the bolts [9] and washers [10] from the links.

(a) Hold the links while you remove the bolts [9].

NOTE: The spring on the deflector door shaft pulls the links.

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SUBTASK 21-51-22-020-003

(3) Remove the screw [4] and washers [6] that attach the bonding jumper [7] to the ram air inlet deflector door [1].

SUBTASK 21-51-22-020-004

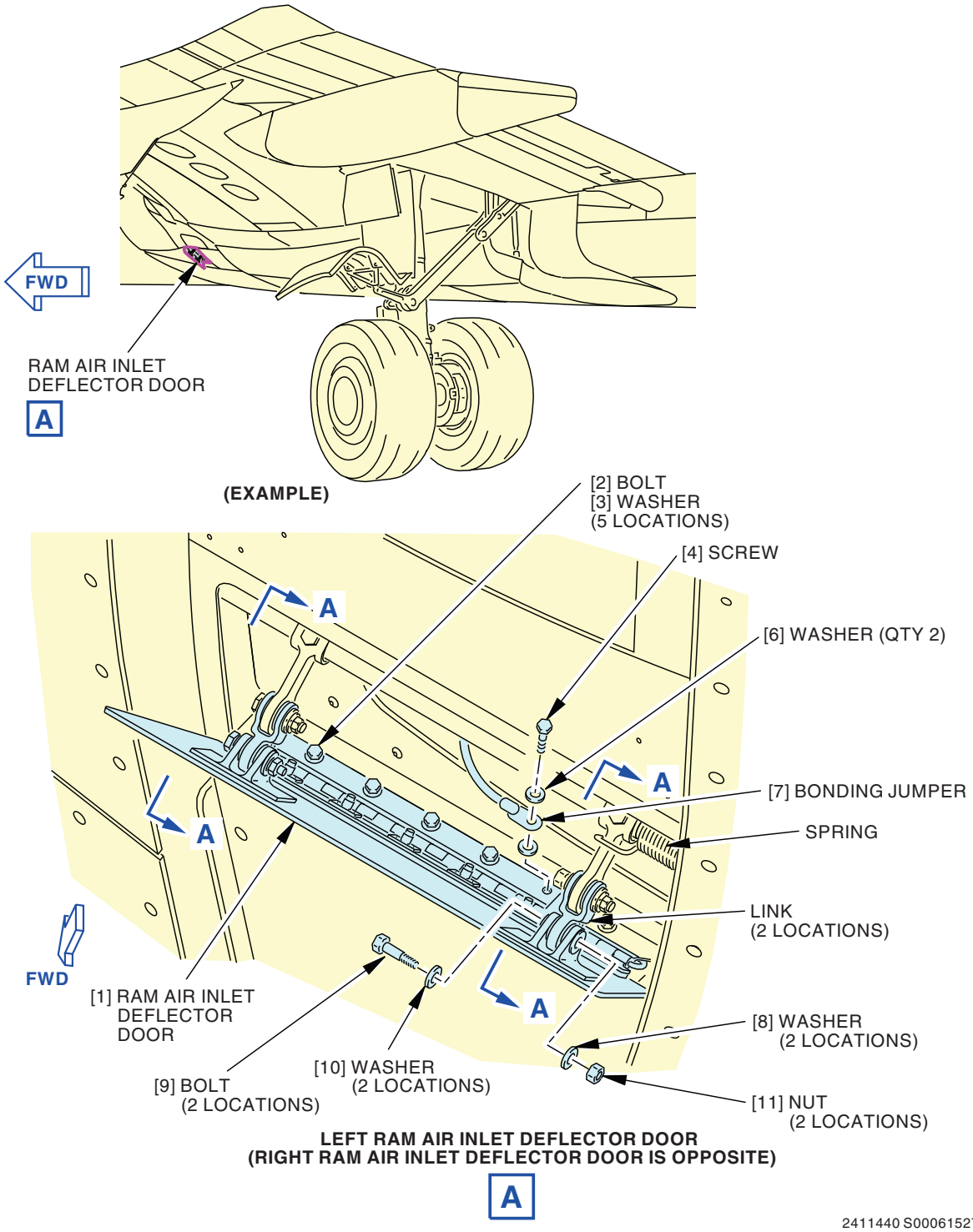
(4) Remove the bolts [2] and washers [3] that attach hinge of the ram air inlet deflector door [1] to the airplane.

SUBTASK 21-51-22-020-005

(5) Carefully lower the ram air inlet deflector door [1] from the airplane.

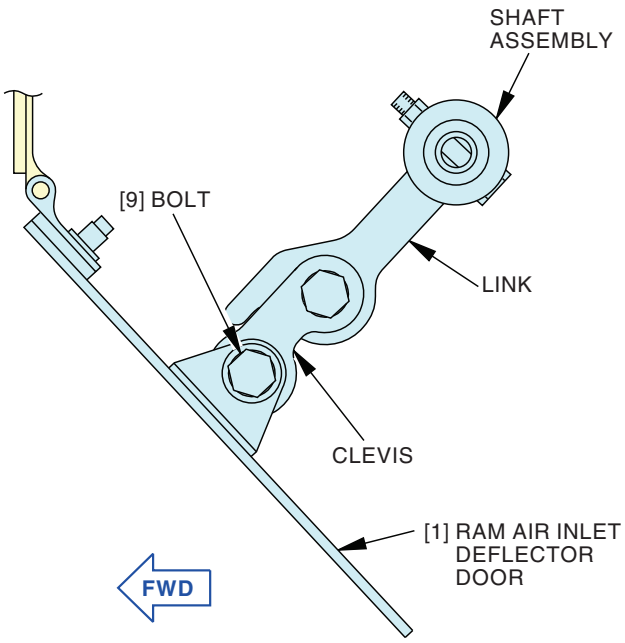
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ram Air Inlet Deflector Door Installation
Figure 401/21-51-22-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

A-A

3026782 S0000797712_V1

Ram Air Inlet Deflector Door Installation
Figure 401/21-51-22-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-22-400-801

3. Ram Air Inlet Deflector Door Installation

(Figure 401)

A. References

Reference	Title
21-51-22-400-803	Ram Air Inlet Deflector Door and Modulation Panels Adjustment (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Deflector door	21-51-22-02-060	PTF ALL

E. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
212	Flight Compartment - Right

F. Ram Air Inlet Deflector Door Installation

SUBTASK 21-51-22-420-001

(1) Put the ram air inlet deflector door [1] in its position.

SUBTASK 21-51-22-420-002

(2) Install the bolts [2] and washers [3] that attach the hinge of ram air inlet deflector door [1] to the airplane.

SUBTASK 21-51-22-100-001

(3) Prepare the mating surfaces (SWPM 20-20-00).

(a) Do abrasive cleaning procedure for mating surface of the airplane structure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do solvent cleaning procedure for mating surfaces of the washers [6] and bonding jumper [7].

SUBTASK 21-51-22-420-003

(4) Install the bonding jumper [7].

(a) Make sure that the bonding jumper [7], screw [4], and washers [6] are clean.

(b) Put the bonding jumper [7] in its position.

(c) Install the screw [4] and washers [6].

(d) Tighten the screw [4] to 32 in-lb (3.6 N·m) - 38 in-lb (4.3 N·m).

SUBTASK 21-51-22-760-001

(5) Measure the electrical bonding resistance between the bonding jumper [7] and the structure (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

SUBTASK 21-51-22-390-001

(6) Apply sealant, A50337, or sealant, A50231, or sealant, A50359, on the terminal of bonding jumper [7], washers [6], and screw [4].

SUBTASK 21-51-22-420-004

(7) Install the bolts [9], washers [10], washers [8], and nuts [11] that attach the ram air inlet deflector door [1] to the links.

(a) Make sure that the spring is in its position on the support assembly and around the inboard link (Figure 401).

(b) Make sure that the links are in the correct position (View A-A, Figure 401).

(c) If it is necessary, adjust the deflector door [1] (TASK 21-51-22-400-803).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-22-860-003

(1) For the left pack ram air inlet door deflector door, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-22-860-004

(2) For the right pack ram air inlet deflector door, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-22-000-802

4. Ram Air Inlet Deflector Door Shaft Assembly Removal

(Figure 402)

A. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
212	Flight Compartment - Right

B. Access Panels

Number	Name/Location
191CL	Forward Wing-To-Body Fairing Panel - Middle
191CR	Forward Wing-To-Body Fairing Panel - Middle

C. Prepare for the Removal

SUBTASK 21-51-22-860-005

(1) For the left pack ram air inlet deflector door shaft assembly, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Remove this access panel:

Number	Name/Location
191CL	Forward Wing-To-Body Fairing Panel - Middle

SUBTASK 21-51-22-860-006

(2) For the right pack ram air inlet deflector shaft assembly, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Remove this access panel:

Number	Name/Location
191CR	Forward Wing-To-Body Fairing Panel - Middle

D. Ram Air Inlet Deflector Door Shaft Assembly Removal

SUBTASK 21-51-22-020-006

(1) Do these steps to remove the arm assembly [40]:

(a) Remove the nut [46], washer [45], washer [41], and bolt [42] to disengage the link extension [37] from the arm assembly [40].

(b) Remove the nut [44], washer [43], washer [39], and bolt [38].

(c) Remove the arm assembly [40] from the end of the shaft [48].

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SUBTASK 21-51-22-020-007

(2) Do these steps to remove the shaft [48]:

(a) Remove the nuts [33], washers [32], and bolts [31] that attach the links [49] to the shaft [48].

(b) Remove the nuts [51], washers [52], and bolts [54] that attach the bearing support assembly [53] to the support assembly.

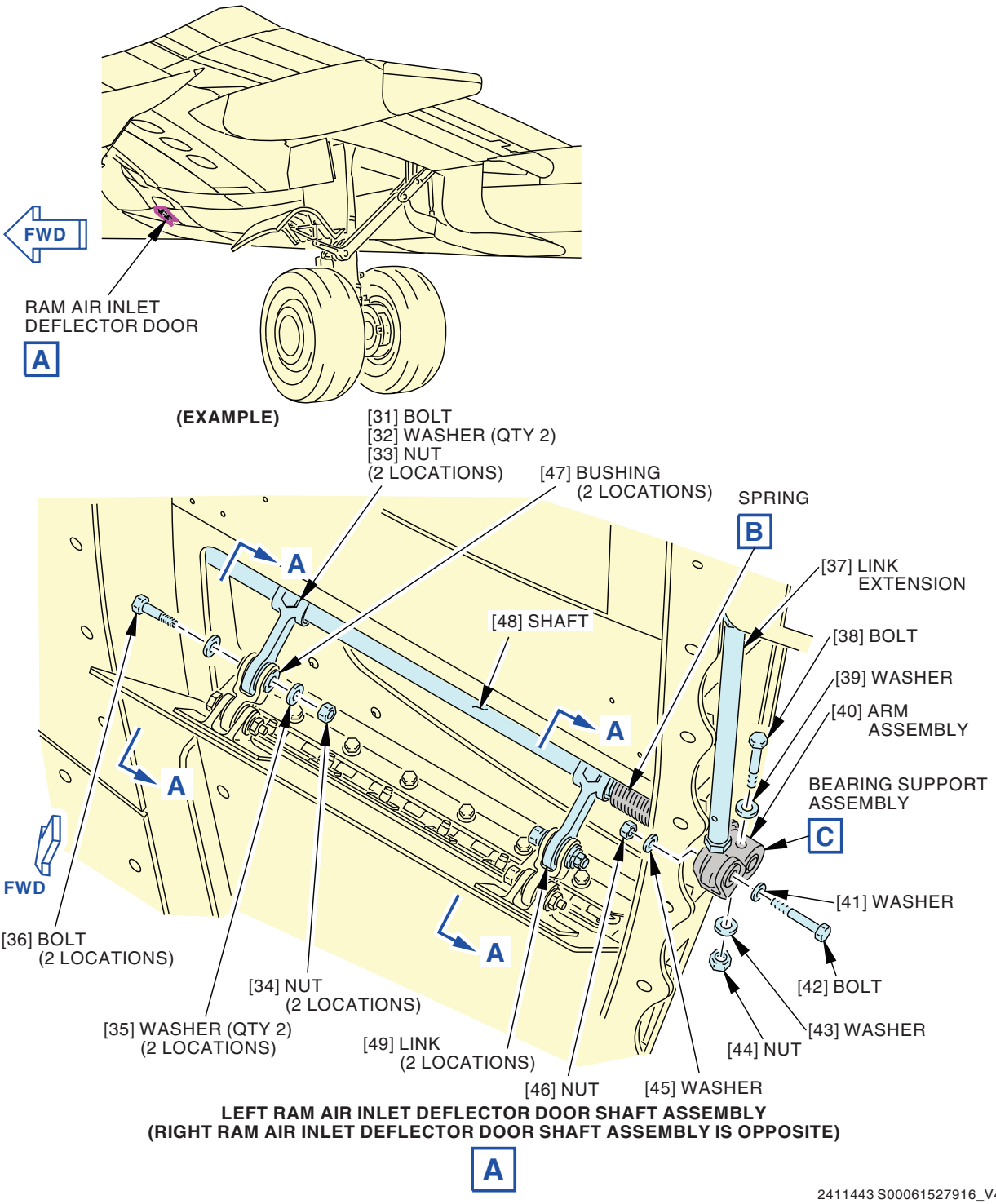
(c) While you hold the links [49], carefully pull the shaft [48] in the inboard direction to remove it.

(d) Remove the nuts [34], washers [35], bolts [36], and bushings [47] from the bottom end of the links [49].

(e) Remove the links [49].

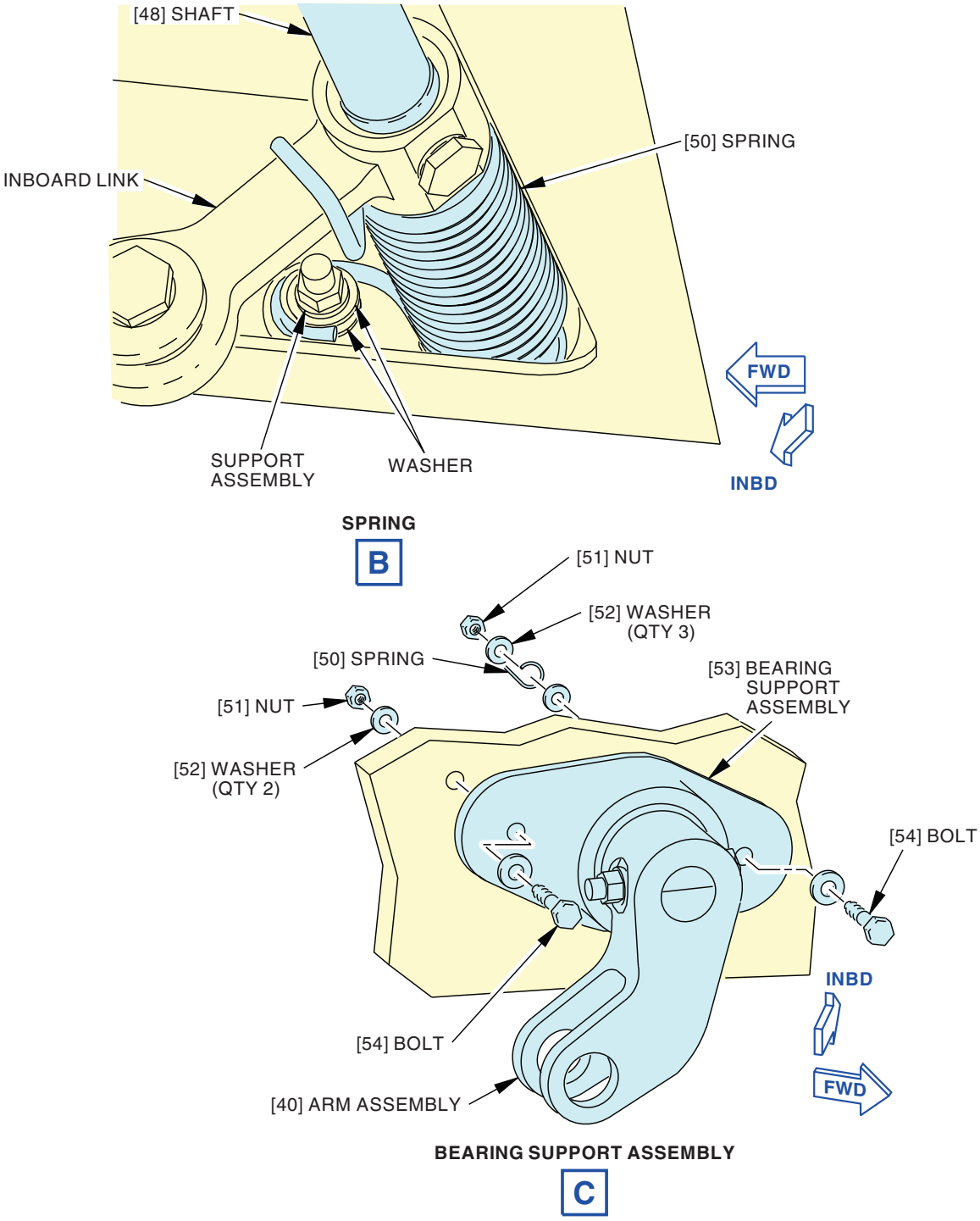
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ram Air Inlet Deflector Door Shaft Assembly Installation
Figure 402/21-51-22-990-805 (Sheet 1 of 3)

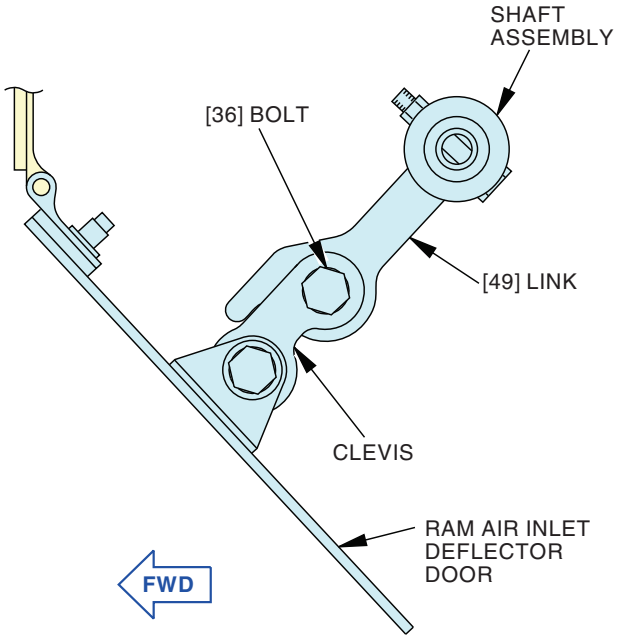
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411444 S00061527917_V1

Ram Air Inlet Deflector Door Shaft Assembly Installation
Figure 402/21-51-22-990-805 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

A-A

3026783 S0000798129_V1

Ram Air Inlet Deflector Door Shaft Assembly Installation
Figure 402/21-51-22-990-805 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-22-400-802

5. Ram Air Inlet Deflector Door Shaft Assembly Installation

(Figure 402)

A. References

Reference	Title
21-51-22-400-803	Ram Air Inlet Deflector Door and Modulation Panels Adjustment (P/B 501)

B. Consumable Materials

Reference	Description	Specification
A50086	Primer - Loctite 7471 Primer T Activator	
A50463	Compound - Retaining - Loctite 641	
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
48	Shaft	21-51-22-02-060	PTF ALL
49	Link	21-51-22-01-295	PTF ALL

D. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
191CL	Forward Wing-To-Body Fairing Panel - Middle
191CR	Forward Wing-To-Body Fairing Panel - Middle

F. Ram Air Inlet Deflector Door Shaft Assembly Installation

SUBTASK 21-51-22-420-005

(1) Do these steps to install the shaft [48]:

NOTE: You must remove the link [49] from the shaft assembly before you can install the shaft [48] into the ram air inlet.

(a) Remove the nut [33], washer [32], and bolt [31], that attach the links [49] to the shaft [48].

(b) Remove the links [49] from the shaft [48].

(c) Put the links [49] in their positions.

1) Make sure that the links [49] are in the correct position (View A-A, Figure 402).

2) Make sure that the inboard and outboard links are opposite for the left and right side.

NOTE: The circular edge of each link must point down after the installation.

(d) Apply Loctite 7471 Primer T Activator, A50086, and Loctite 641 compound, A50463, to surfaces contacting mating parts, shafts, and crankarms or collars.

(e) Apply thin coating of grease, D00633, to mating surfaces of the bushings [47].

(f) Install the bushings [47], washers [35], bolts [36], and nuts [34].

(g) While you hold the links [49], insert the shaft [48] into the ram air inlet in an outboard direction, through the spring [50] and the holes in the top ends of the links [49].

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1) Make sure that you insert the shaft [48] so that the end without the hole goes in first.

(h) Install the bolts [31], washers [32], and nuts [33] that attach the links [49] to the shaft [48].

CAUTION DO NOT LUBRICATE THE BEARING WITH GREASE OR WD-40. IF YOU LUBRICATE, THE BEARING LINER CAN SWELL, DECREASING THE CLEARANCE AND LEADING TO INCREASED ROTATIONAL BREAK-AWAY TORQUE.

(i) Install the bolt [54], washer [52], and nut [51] that attach the bearing support assembly [53] to the support assembly.

NOTE: The bearing is self lubricating.

(j) Make sure that the spring [50] is in its position on the support assembly and around the inboard link.

1) Make sure that the shaft assembly rotates freely and not bind.

SUBTASK 21-51-22-420-006

(2) Do these steps to install the arm assembly [40]:

(a) Put the arm assembly [40] in its position on the end of the shaft [48] with the free end facing aft.

(b) Install the bolt [38], washer [39], washer [43], and nut [44].

(c) Put the link extension [37] in its position and install the bolt [42], washer [41], washer [45], and nut [46].

SUBTASK 21-51-22-820-004

(3) If it is necessary, adjust the air inlet deflector door (TASK 21-51-22-400-803).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-22-860-007

(1) For the left pack ram air inlet actuator, do these steps:

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Close this access panel:

Number	Name/Location
191CL	Forward Wing-To-Body Fairing Panel - Middle

SUBTASK 21-51-22-860-008

(2) For the right pack ram air inlet actuator, do these steps:

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Close this access panel:

Number	Name/Location
191CR	Forward Wing-To-Body Fairing Panel - Middle

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR INLET SYSTEM - ADJUSTMENT

1. General

A. This procedure has this task:

(1) An adjustment of the ram air inlet deflector door and the ram air inlet modulation panels.

B. The left and right ram air inlet systems may be adjusted at the same time or adjusted separately.

TASK 21-51-22-400-803

2. Ram Air Inlet Deflector Door and Modulation Panels Adjustment

(Figure 501)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
53-51-21-000-801	Forward Wing-To-Body Fairing Panel Removal (P/B 401)
53-51-21-400-801	Forward Wing-To-Body Fairing Panel - Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
191CL	Forward Wing-To-Body Fairing Panel - Middle
191CR	Forward Wing-To-Body Fairing Panel - Middle
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

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E. Preparation for the Adjustment

SUBTASK 21-51-22-010-001

(1) For the adjustment of the left ram air inlet system, remove these panels:

(TASK 53-51-21-000-801)

Number	Name/Location
191CL	Forward Wing-To-Body Fairing Panel - Middle
191GL	Ram Air Actuator Panel - Forward

SUBTASK 21-51-22-010-002

(2) To get access to the left ram air temperature sensor, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-22-010-003

(3) For the adjustment of the right ram air inlet system, remove these panels:

(TASK 53-51-21-000-801)

Number	Name/Location
191CR	Forward Wing-To-Body Fairing Panel - Middle
191GR	Ram Air Actuator Panel - Forward

SUBTASK 21-51-22-010-004

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access the right ram air temperature sensor, open the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-22-010-005

(5) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Ram Air Inlet System Adjustment/Rigging

SUBTASK 21-51-22-020-008

(1) Do these steps to disconnect the forward end of the tie rod assembly [1]:

(a) Remove the nut [5], washer [30], bolt [3], and washer [4].

(b) Move the end of the tie rod assembly [1] away from the adjacent arm [2].

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SUBTASK 21-51-22-860-009

(2) Do these steps to put the actuator in the flight position:

(a) If not already done, attach the electrical connector [28] and electrical connector [29] to the actuator [21].

(b) For the adjustment of the actuator for the left ram air system, do these steps:

1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

(c) For the adjustment of the actuator for the right ram air system, do these steps:

1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(d) Extend the trailing edge flaps to position 1.

1) To extend the flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

NOTE: The actuator will retract slightly.

(e) To prevent the standby hydraulic pump from operating when the airplane is in air mode, do this step:

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

1) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

(f) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

(g) Retract the trailing edge flaps to full up position.

1) To retract the flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

(h) Wait for the ram air actuator to fully extend.

(i) Remove the electrical connector [22] from the ram air sensor.

(j) Make sure that the ram air actuator retracts.

NOTE: This position is approximately 0.90 in. (22.86 mm) from the fully retracted position.

SUBTASK 21-51-22-420-007

(3) Do these steps to adjust the rod end of actuator [21] to the arm [13]:

(a) Turn the shaft assembly [16] to align the rigging pin holes [14] in the shaft assembly [16] and the bearing housing support [15].

(b) Install a 1-3/4 inch long, 3/16-inch diameter rigging pin (part of rigging pin kit, SPL-1585) through the rigging pin holes [14].

(c) If the hole in the rod end of the actuator [21] does not align with the holes in the arm [13], do these steps:

1) Remove the lockwire [20] from the jam nut [19] on the actuator rod end.

2) Adjust the length of the rod end plus or minus 0.25 in. (0.64 cm) until the holes align.

(d) Tighten the jam nut [19] on the rod end of the actuator [21] as follows:

1) Rotate the jam nut [19] so that it does not interfere with the arm [13] when the actuator is fully extended.

2) Restrain the actuator rod end until the jam nut [19] is seated on the rod and torque the jam nut [19] to 100 in-lb (11 N·m) - 120 in-lb (14 N·m).

(e) Install lockwire [20] between the jam nut [19] on the actuator rod end and the rod end lockwasher next to it (TASK 20-10-44-400-801).

(f) Remove the rigging pin.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-22-860-010

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Extend the trailing edge flaps to position 1.

(a) To extend the flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

NOTE: The actuator will retract slightly.

SUBTASK 21-51-22-860-011

(5) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

(a) Make sure that the actuator retracts to the fully retracted position (ground fully open).

SUBTASK 21-51-22-820-001

(6) Do the steps that follow to adjust the tie rod assembly [1]:

(a) Remove the lockwire [6] from the jam nuts [7] on the tie rod assembly [1].

(b) Loosen the jam nuts [7].

(c) Install the bolt [3], washer [4], washer [30], and nut [5] that attach the tie rod assembly [1] to the arm [2] on the deflector door pivot shaft.

(d) Adjust the length of the tie rod assembly [1] until the over-travel stop on either the inboard or outboard link [25] just touches the applicable inboard or outboard clevis [26].

(e) Adjust the tie rod assembly [1] so that it is one full turn shorter as follows:

1) Rotate the shaft one full turn.

2) Tighten the jam nuts [7].

(f) Apply 8 lbf (36 N) - 12 lbf (53 N) to the link [25] as follows (View A-A, Figure 501):

1) Apply the force through the axis of the bolt in a plane approximately parallel to the surface of the ram air inlet deflector door.

(g) As you apply the force, make sure that the smaller of the two gaps (either inboard or outboard) measures 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm) between the over-travel arm of the link [25] and the clevis [26] (View A-A, Figure 501).

NOTE: The gaps at the inboard and outboard links may or may not be the same.

(h) If the smaller gap is not within 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm), adjust the tie rod assembly [1] in 1/4 turn increments until you get the specified gap.

(i) Torque the jam nuts [7] on the tie rod assembly [1] to 40 in-lb (5 N·m) - 60 in-lb (7 N·m).

(j) Install the lockwire [6] on the jam nuts [7] (TASK 20-10-44-400-801).

SUBTASK 21-51-22-480-001

(7) Install the electrical connector [22] on the ram air sensor.

SUBTASK 21-51-22-860-017

(8) Do the following steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Wait for a minimum of 10 seconds and then continue.

NOTE: Opening the above circuit breakers for 10 seconds minimum and then closing will clear the Smart Ram Air Door Actuator (SRADA) internal fault memory and allow actuator movement during subsequent steps.

(c) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-22-860-012

(9) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

(a) Make sure that the inlet doors move to a more closed position smoothly without binding.

(b) Make sure that the deflector door closes and is faired with the ram air inlet ramp.

SUBTASK 21-51-22-860-013

(10) Retract the trailing edge flaps to full up position.

(a) To retract the flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 21-51-22-700-001

(11) Do a check of the dimension shown in View B-B, Figure 501.

(a) The dimension “A” should be approximately 1.40 in. (35.6 mm).

(b) This is a reference check only as no action is required at this point in the adjustment of the system.

SUBTASK 21-51-22-820-002

(12) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

(a) Make sure that the deflector door opens to the fully deployed position.

(b) Make sure that the modulation panels move smoothly to the fully open position without any binding.

(c) Apply 8 lbf (36 N) - 12 lbf (53 N) to the link [25] as follows (View A-A, Figure 501):

1) Apply the force through the axis of the bolt in a plane approximately parallel to the surface of the ram air inlet deflector door.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) As you apply the force, make sure that the smaller of the two gaps (either inboard or outboard) measures 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm) between the over-travel arm of the link [25] and the clevis [26] (View A-A of Figure 501).

NOTE: The gaps at the inboard and outboard links may or may not be the same.

(e) If the smaller gap is within 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm), do these steps:

1) Torque the jam nuts [7] on the tie rod assembly [1] to 40 in-lb (5 N·m) - 60 in-lb (7 N·m).

2) Install the lockwire [6] on the jam nuts [7] (TASK 20-10-44-400-801).

(f) If the smaller gap is not within 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm), adjust the tie rod assembly [1] in 1/4 turn increments until you get the specified gap.

(g) Torque the jam nuts [7] on the tie rod assembly [1] to 40 in-lb (5 N·m) - 60 in-lb (7 N·m).

(h) Install the lockwire [6] on the jam nuts [7] (TASK 20-10-44-400-801).

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(i) Make sure that this circuit breaker is closed:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 21-51-22-740-001

(13) Do a Built-In-Test Equipment (BITE) test on the left Integrated Air Systems Controller (M2939) and the right Integrated Air Systems Controller (M2957):

NOTE: This will clear any fault introduced during maintenance when power is interrupted.

(a) Do the applicable ground test in this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

G. Ram Air Inlet Deflector Door Adjustment Check

SUBTASK 21-51-22-420-009

(1) Do these steps to secure the tie rod assembly [1].

(a) Torque the jam nuts [7] on the tie rod assembly [1] to 40 in-lb (5 N·m) - 60 in-lb (7 N·m).

(b) Install the lockwire [6] on the jam nuts [7] (TASK 20-10-44-400-801).

SUBTASK 21-51-22-480-002

(2) Install the electrical connector D3916 (D3918) on the ram air sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-22-860-018

(3) Do the following steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Wait for a minimum of 10 seconds and then continue.

NOTE: Opening the above circuit breakers for 10 seconds minimum and then closing will clear the SRADA internal fault memory and allow actuator movement during subsequent steps.

(c) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-22-860-019

(4) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

(a) Make sure that the inlet doors move to a more closed position smoothly without binding.

(b) Make sure that the deflector door closes and is faired with the ram air inlet ramp.

SUBTASK 21-51-22-860-020

(5) Retract the trailing edge flaps to full up position.

(a) To retract the flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 21-51-22-700-002

(6) Do a check of the dimension shown in View B-B, Figure 501.

(a) The dimension “A” should be approximately 1.40 in. (35.6 mm).

(b) This is a reference check only as no action is required at this point in the adjustment of the system.

SUBTASK 21-51-22-820-003

(7) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

(a) Make sure that the deflector door opens to the fully deployed position.

(b) Make sure that the modulation panels move smoothly to the fully open position without any binding.

(c) Apply 8 lbf (36 N) - 12 lbf (53 N) to the link [25] as follows (View A-A, Figure 501):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Apply the force through the axis of the bolt in a plane approximately parallel to the surface of the ram air inlet deflector door.

(d) As you apply the force, make sure that the smaller of the two gaps (either inboard or outboard) measures 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm) between the over-travel arm of the link [25] and the clevis [26] (View A-A, Figure 501).

NOTE: The gaps at the inboard and outboard links may or may not be the same.

(e) If the smaller gap is within 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm), do these steps:

1) Torque the jam nuts [7] on the tie rod assembly [1] to 40 in-lb (5 N·m) - 60 in-lb (7 N·m).

2) Install the lockwire [6] on the jam nuts [7] (TASK 20-10-44-400-801).

(f) If the smaller gap is not within 0.010 in. (0.254 mm) - 0.030 in. (0.762 mm), do these steps:

1) Adjust the tie rod assembly [1] in 1/4 turn increments until you get the specified gap (See: Ram Air Inlet System Adjustment/Rigging).

2) Do a check if the gap remains within 0.010 in. (0.25 mm) - 0.03 in. (0.76 mm).

a) Do the steps in Ram Air Inlet Deflector Door Adjustment Check again.

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(g) Make sure that this circuit breaker is closed:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 21-51-22-740-002

(8) Do a BITE test on the left Integrated Air Systems Controller (M2939) and the right Integrated Air Systems Controller (M2957):

NOTE: This will clear any fault introduced during maintenance when power is interrupted.

(a) Do the applicable ground test in this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-22-410-003

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-51-22-860-014

(2) When the left ram air inlet system was adjusted, install these access panels:

(TASK 53-51-21-400-801)

Number	Name/Location
191CL	Forward Wing-To-Body Fairing Panel - Middle
191GL	Ram Air Actuator Panel - Forward

SUBTASK 21-51-22-410-001

(3) For the left ram air inlet system, close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-22-860-015

(4) When the right ram air inlet system was adjusted, install these access panels:

(TASK 53-51-21-400-801)

Number	Name/Location
191CR	Forward Wing-To-Body Fairing Panel - Middle
191GR	Ram Air Actuator Panel - Forward

SUBTASK 21-51-22-410-002

(5) For the right ram air inlet system, close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

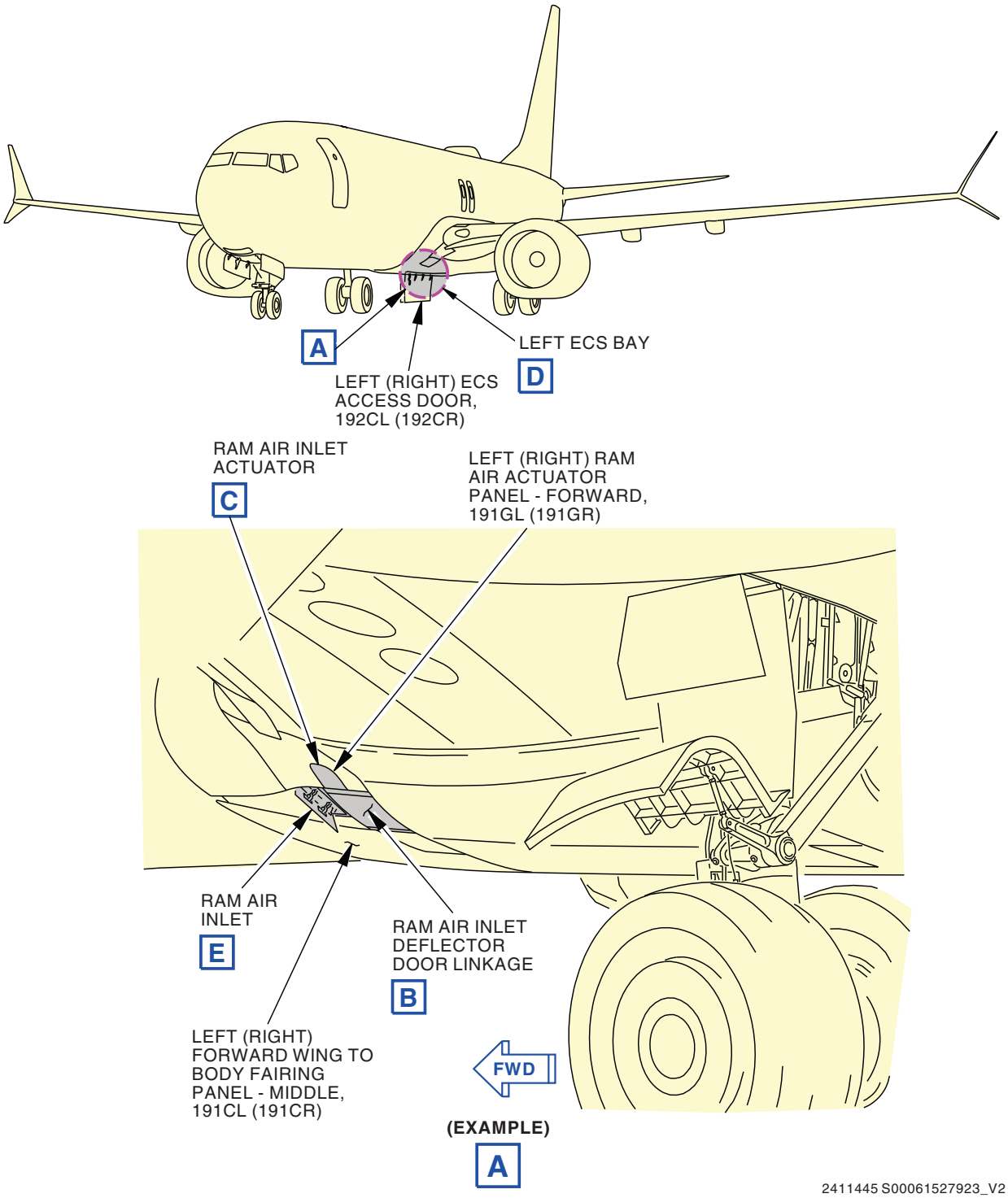
(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

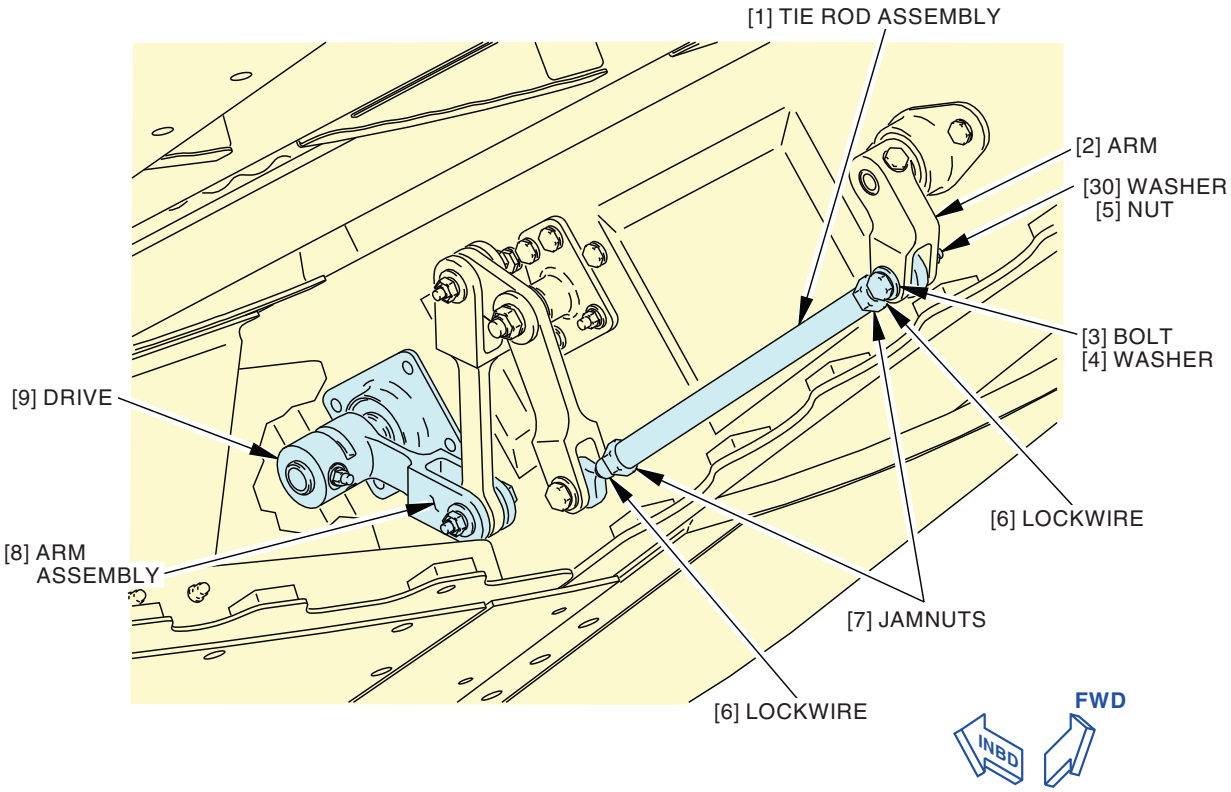
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ram Air Inlet System
Figure 501/21-51-22-990-803 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



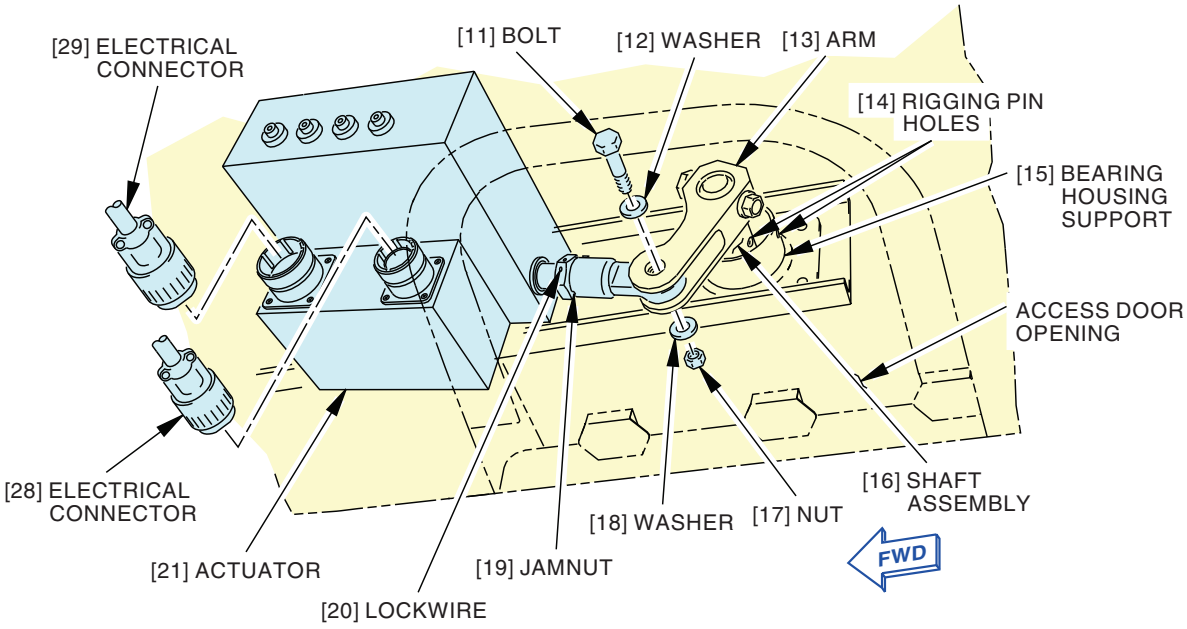
RAM AIR INLET DEFLECTOR DOOR LINKAGE

B

2411446 S00061527924_V2

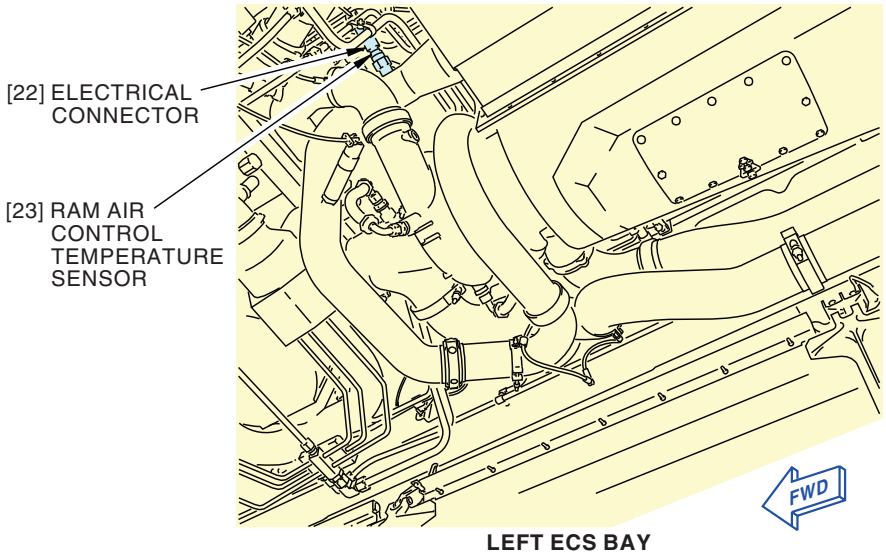
Ram Air Inlet System
Figure 501/21-51-22-990-803 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT RAM AIR INLET ACTUATOR
(RIGHT RAM AIR INLET ACTUATOR IS OPPOSITE)

C



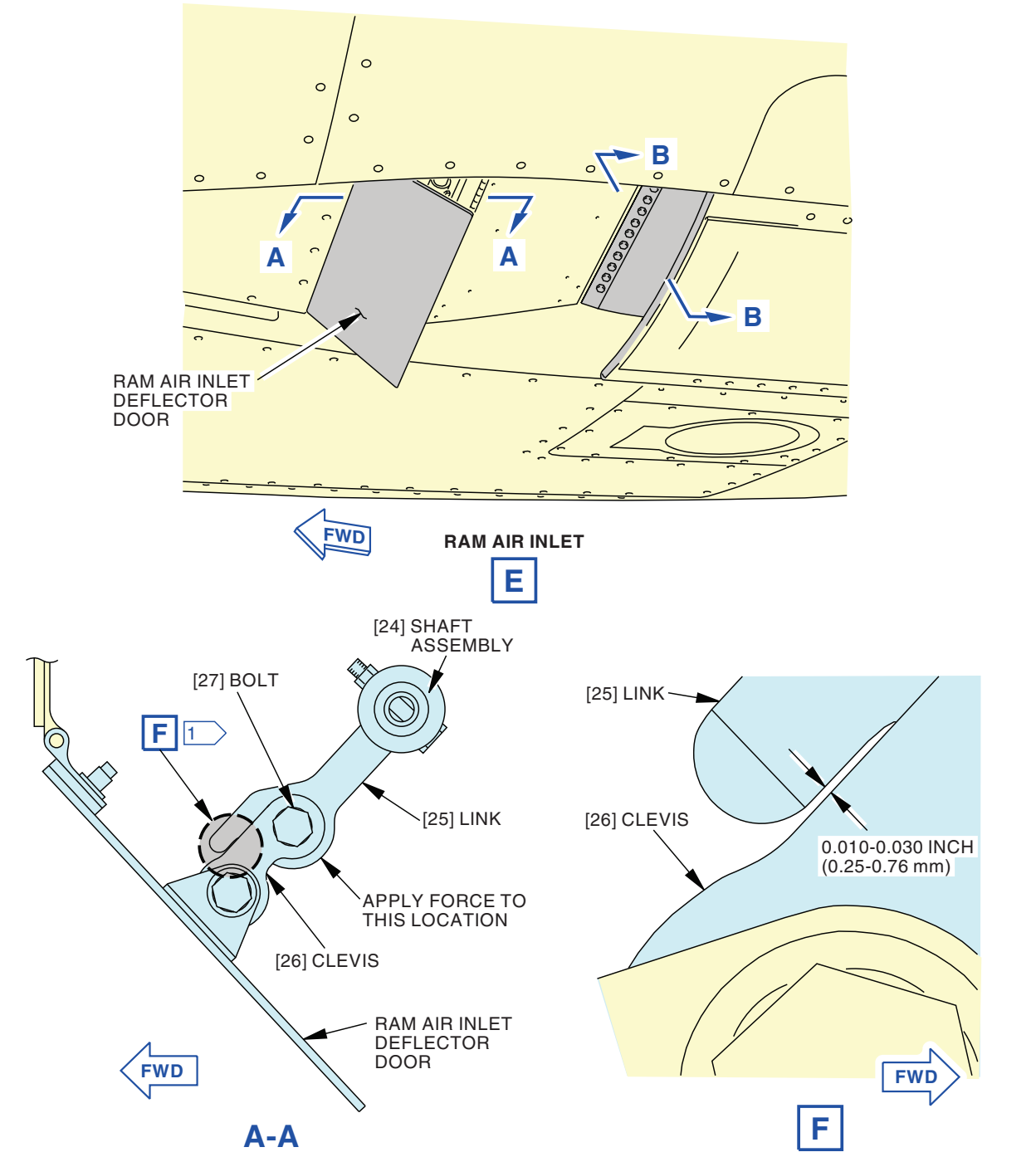
LEFT ECS BAY
(RIGHT ECS BAY IS OPPOSITE)

D

2411448 S00061527926_V1

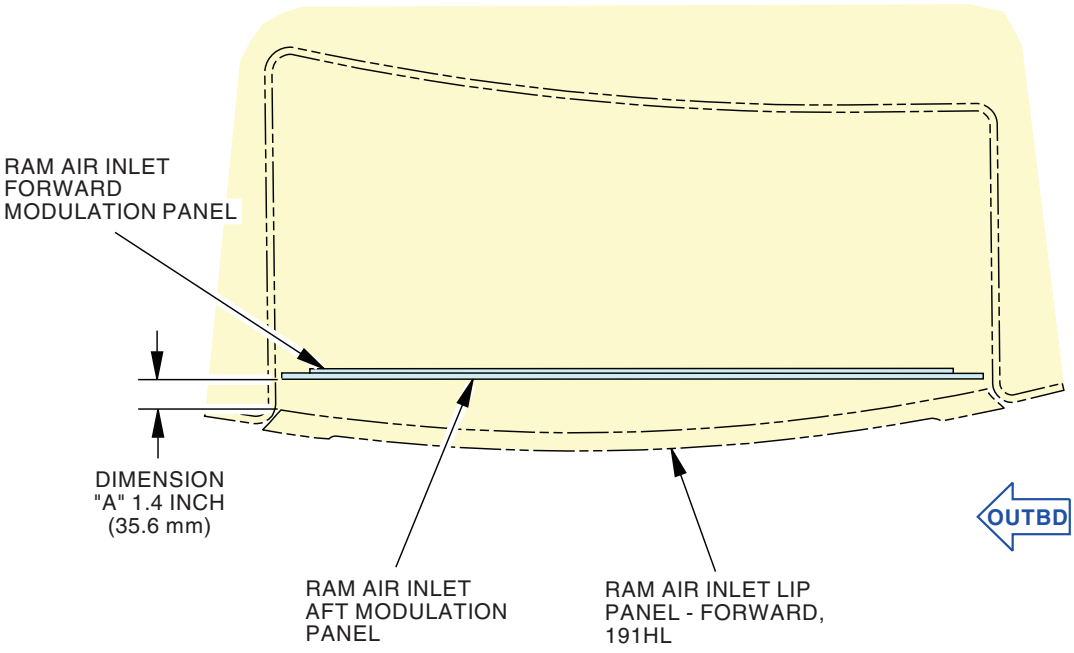
Ram Air Inlet System
Figure 501/21-51-22-990-803 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ram Air Inlet System
Figure 501/21-51-22-990-803 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(FORWARD AND AFT MODULATION PANELS SHOWN IN FLIGHT FULL CLOSED POSITION)

B-B

2411450 S00061527928_V2

Ram Air Inlet System
Figure 501/21-51-22-990-803 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR SHAFT ASSEMBLY AND ARM ASSEMBLIES INSPECTION

1. General

A. This procedure contains one task:

(1) Ram air inlet door linkage inspection.

TASK 21-51-22-200-801

2. Ram Air Inlet Door Linkage Inspection

(Figure 601)

A. General

(1) This task has the steps to inspect the ram air inlet door linkage inspection.

B. References

Reference	Title
21-51-21-000-801	Ram Air Inlet Actuator Removal (P/B 401)
21-51-21-400-801	Ram Air Inlet Actuator Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-7173	Gage - Force, 0-50 lbs. capacity

D. Prepare for the Inspection

SUBTASK 21-51-22-020-009

(1) Disconnect the actuator from the arm assembly [1] (TASK 21-51-21-000-801).

E. Ram Air Inlet Door Linkage Inspection

SUBTASK 21-51-22-710-001

(1) Use a force gage, STD-7173, to check the bearings in the ram air inlet door linkages assembly.

(a) Connect the force gage, STD-7173, to the arm assembly [1].

(b) Pull the force gage, STD-7173, to rotate the arm assembly [1].

1) Make sure that the direction of the force is perpendicular to the arm assembly [1].

(c) Record the initial force (lbf or N) required to move the ram air inlet linkage assembly and share the results with Boeing.

NOTE: Although a maximum force required has not yet been determined, preliminary data shows that an initial force over 20 lbf (89 N), may indicate that the self-lubricating bearings within the ram air inlet linkage assembly may be degraded.

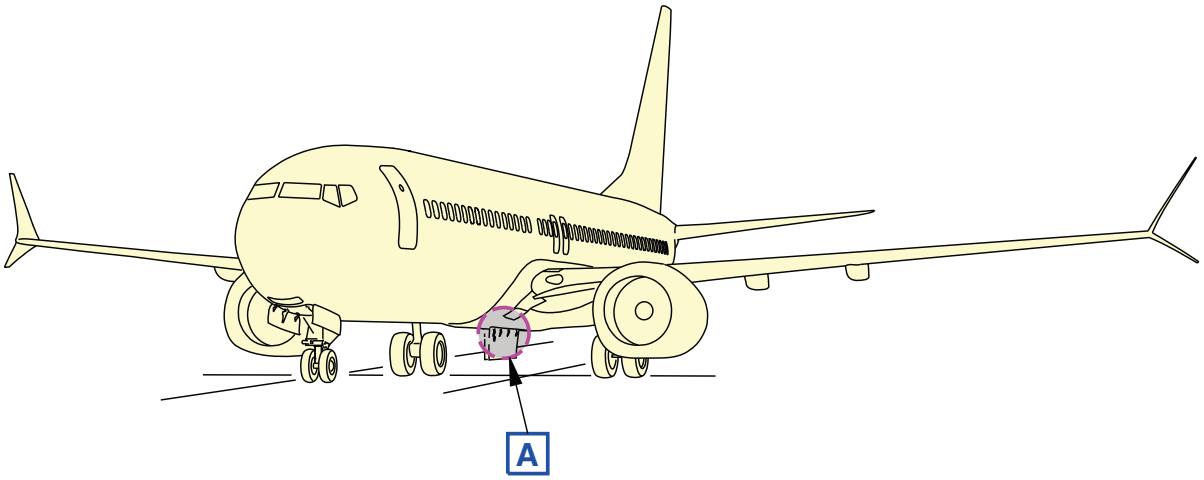
F. Put the Airplane Back to Its Usual Condition

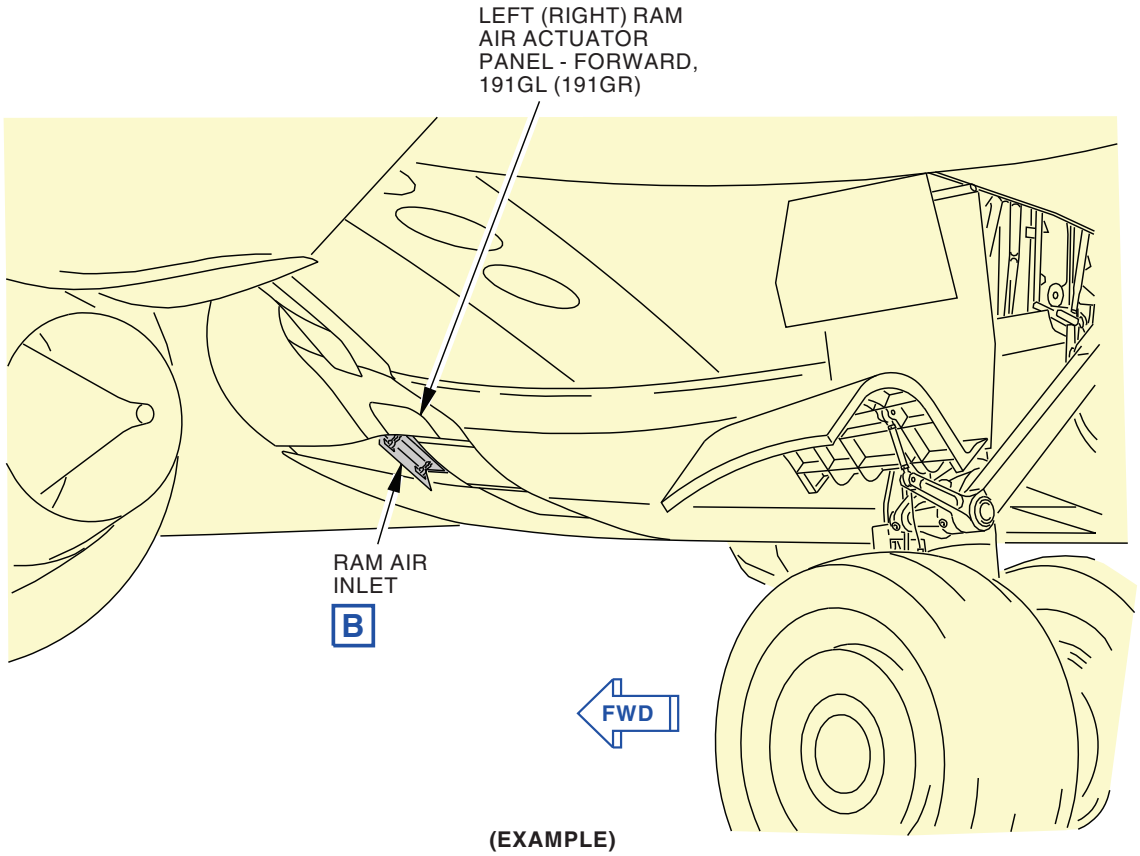
SUBTASK 21-51-22-420-010

(1) Connect the actuator to the arm assembly [1] (TASK 21-51-21-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

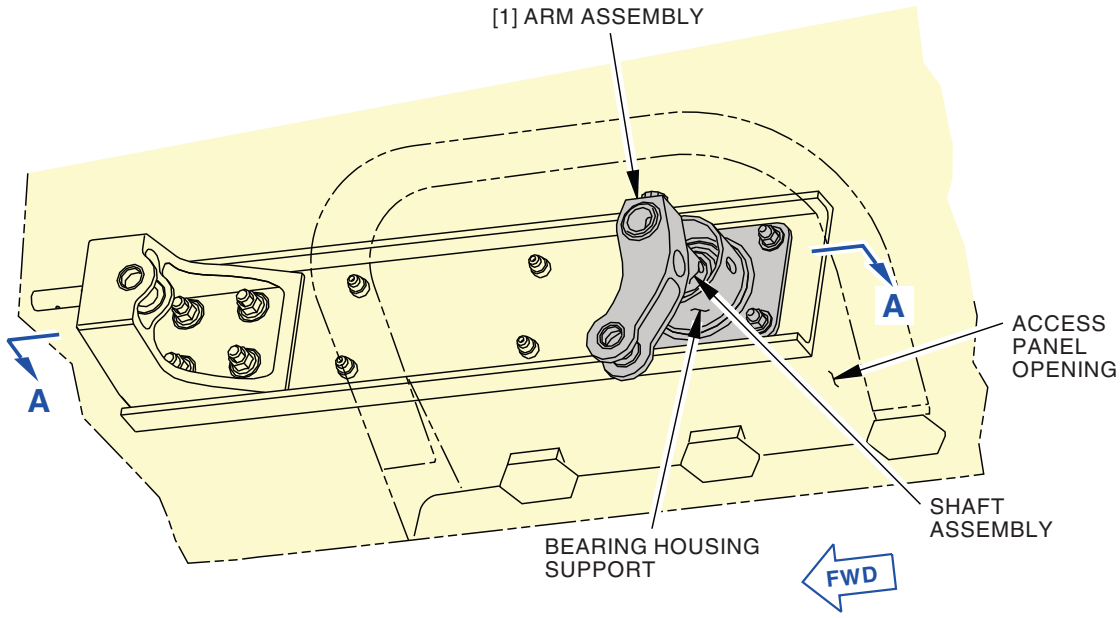




2927756 S0000707247_V1

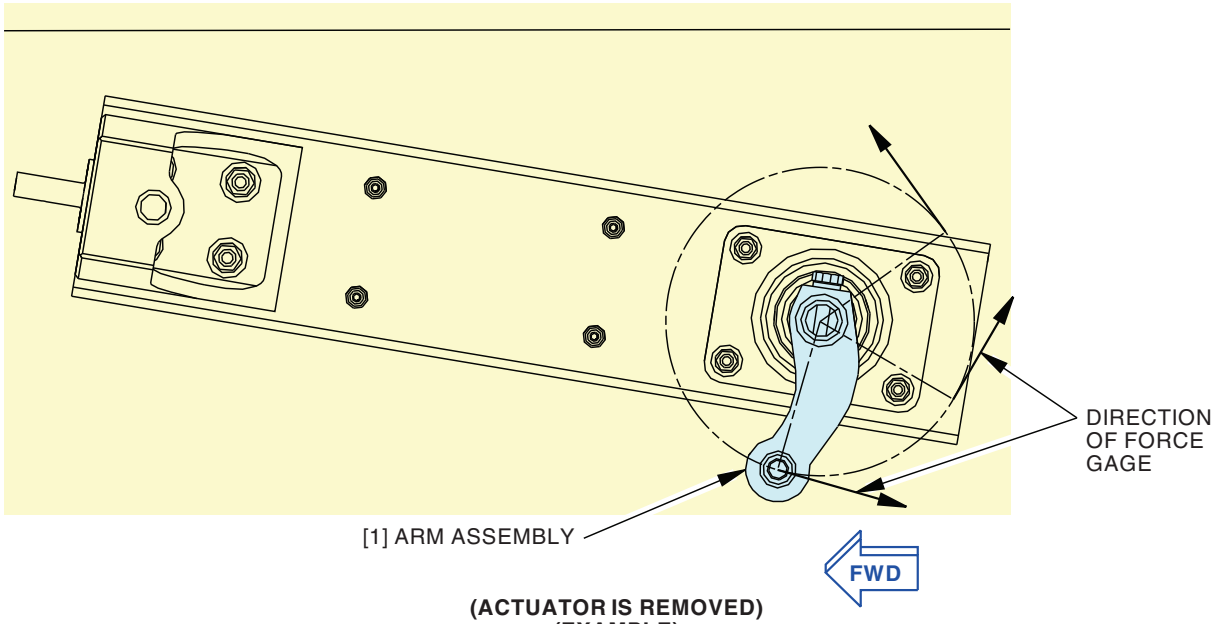
Ram Air Inlet Door Linkage Inspection
Figure 601/21-51-22-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT RAM AIR INLET
(RIGHT RAM AIR INLET IS OPPOSITE)

B



(ACTUATOR IS REMOVED)
(EXAMPLE)

A-A

2927757 S0000707248_V1

Ram Air Inlet Door Linkage Inspection
Figure 601/21-51-22-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR INLET MODULATION PANELS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ram air inlet modulation panels

(2) An installation of the ram air inlet modulation panels.

B. The ram air inlet modulation panels have a forward panel and an aft panel.

C. The modulation panels are installed at the forward end of the wing-to-body fairings below the wings.

TASK 21-51-23-000-801

2. Ram Air Inlet Modulation Panels Removal

(Figure 401)

A. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
21-51-21-000-801	Ram Air Inlet Actuator Removal (P/B 401)

B. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box

C. Prepare for the Removal

SUBTASK 21-51-23-860-001

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

SUBTASK 21-51-23-860-002

(2) For the modulation panels for the left ram air inlet, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT

SUBTASK 21-51-23-860-003

(3) For the modulation panels for the right ram air inlet, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

D. Ram Air Inlet Modulation Panels Removal

NOTE: The ram air inlet modulation panels have a forward panel and an aft panel. It is necessary to remove the forward panel before you can remove the aft panel.

SUBTASK 21-51-23-020-001

(1) Remove the forward modulation panel [1] as follows:

(a) Remove the bolts [3] that attach the forward end of the aft modulation panel [2] to the aft end of the forward modulation panel [1].

NOTE: Let the forward modulation panel come down. This will permit access to the bolts and washers that attach the forward end of the forward modulation panel to the ram air inlet forward fairing support.

(b) Remove the bolts [4] and washers [5] from the forward hinge on the forward modulation panel [1].

(c) Remove the forward modulation panel [1] from the ram air inlet.

SUBTASK 21-51-23-020-002

(2) Remove the aft modulation panel [2] as follows:

(a) If it is necessary, disconnect the ram air inlet actuator arm from the shaft of the aft modulation panel [2] (TASK 21-51-21-000-801).

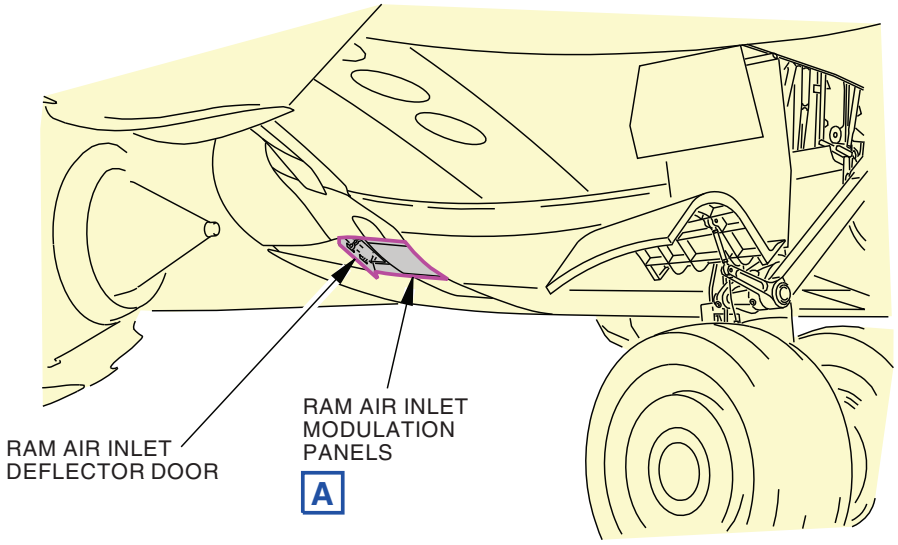
(b) Disconnect the two clevis assemblies [8] from the aft modulation panel [2]:

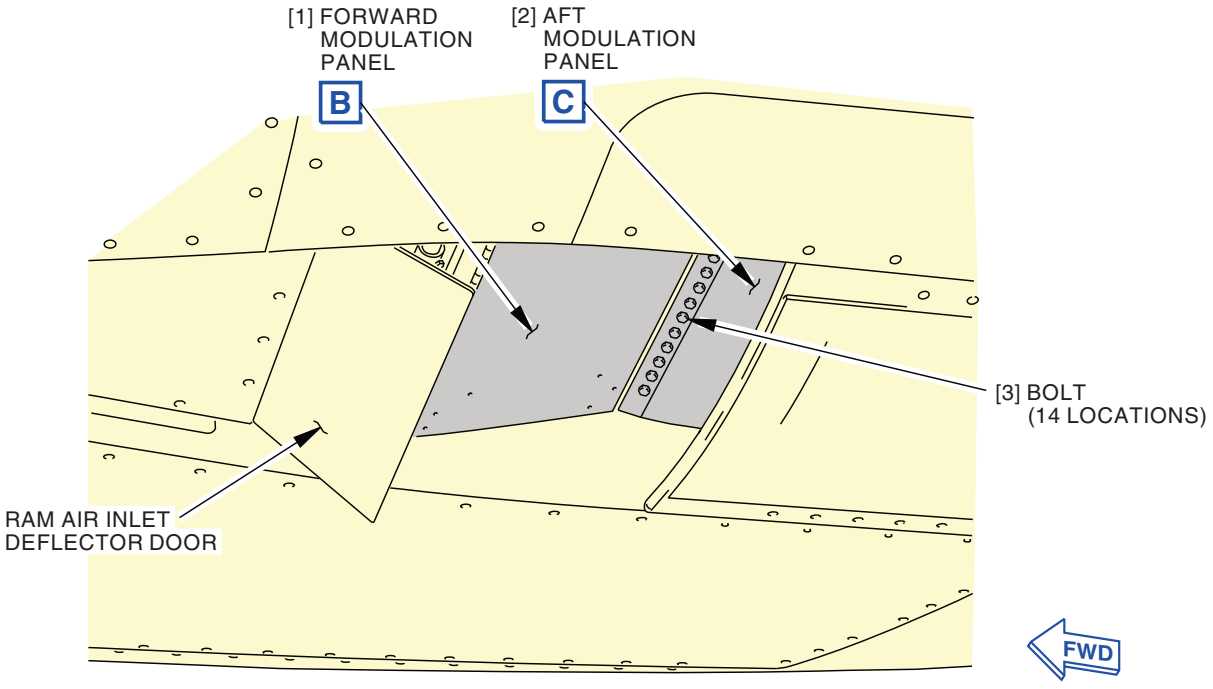
1) Remove the nut [12], washers [10], bushing [11], and bolt [9] that connect each clevis assembly [8] to the aft modulation panel [2].

(c) Pull the aft modulation panel [2] out of the ram air inlet.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





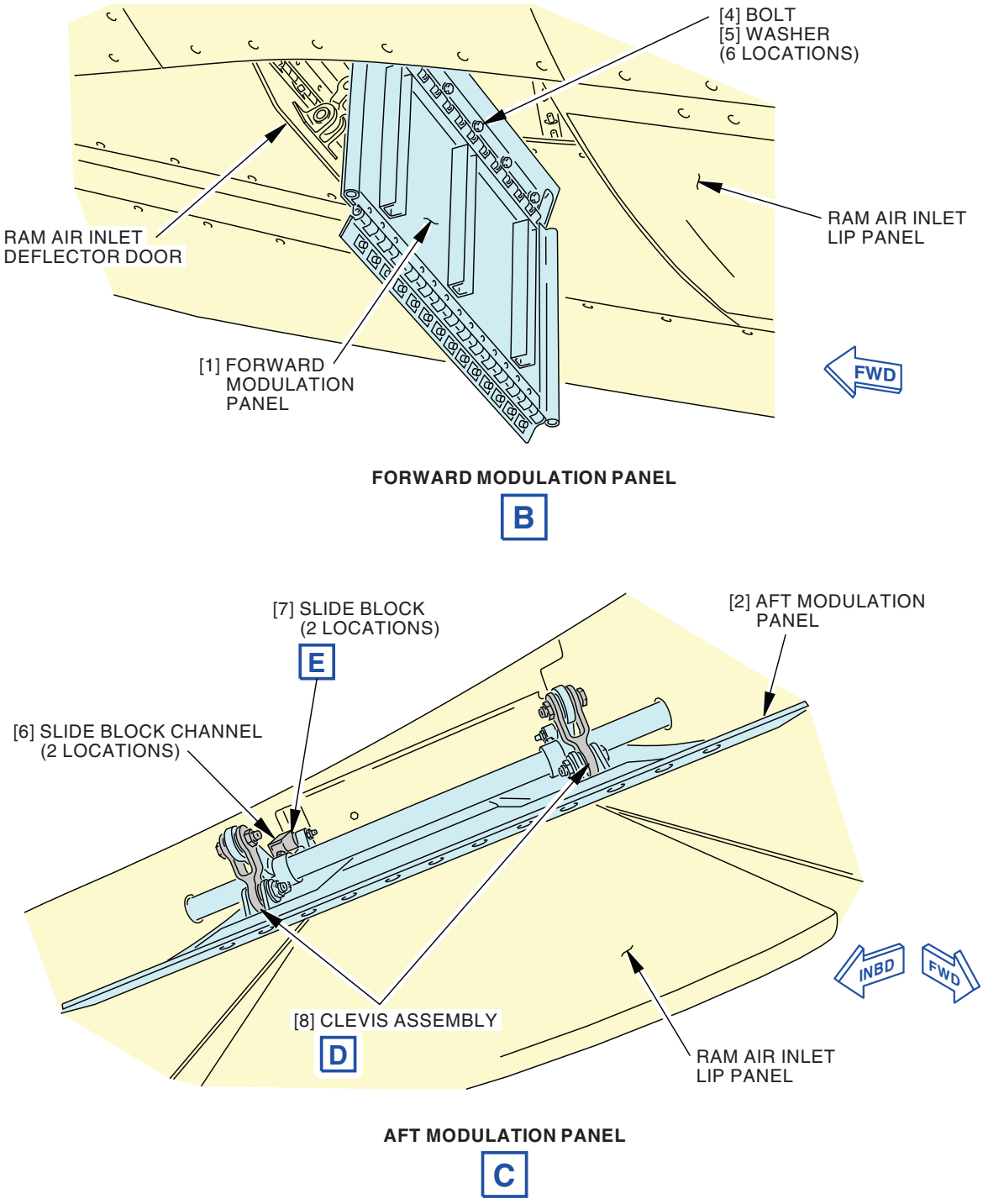
RAM AIR INLET MODULATION PANELS
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

2411451 S00061527932_V1

Ram Air Inlet Modulation Panels Installation
Figure 401/21-51-23-990-801 (Sheet 1 of 4)

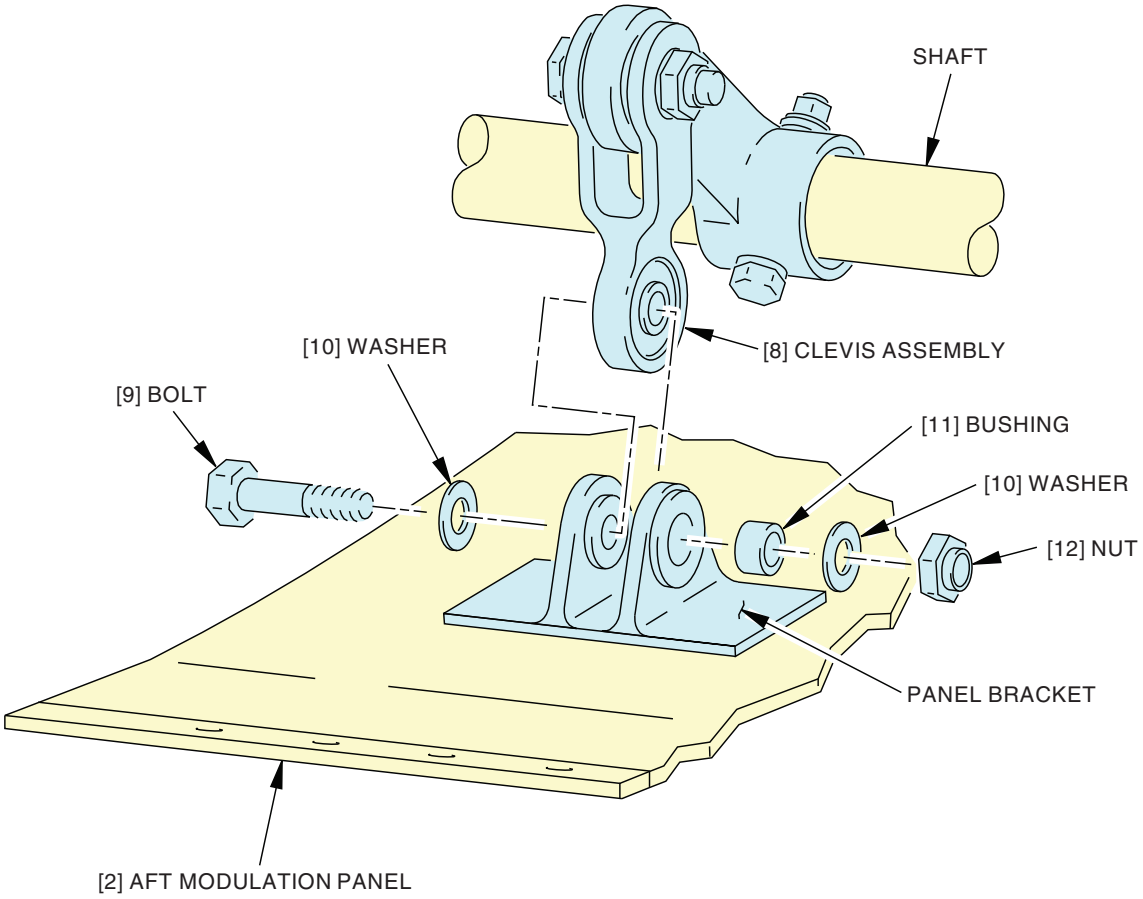
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411452 S00061527933_V1

Ram Air Inlet Modulation Panels Installation
Figure 401/21-51-23-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



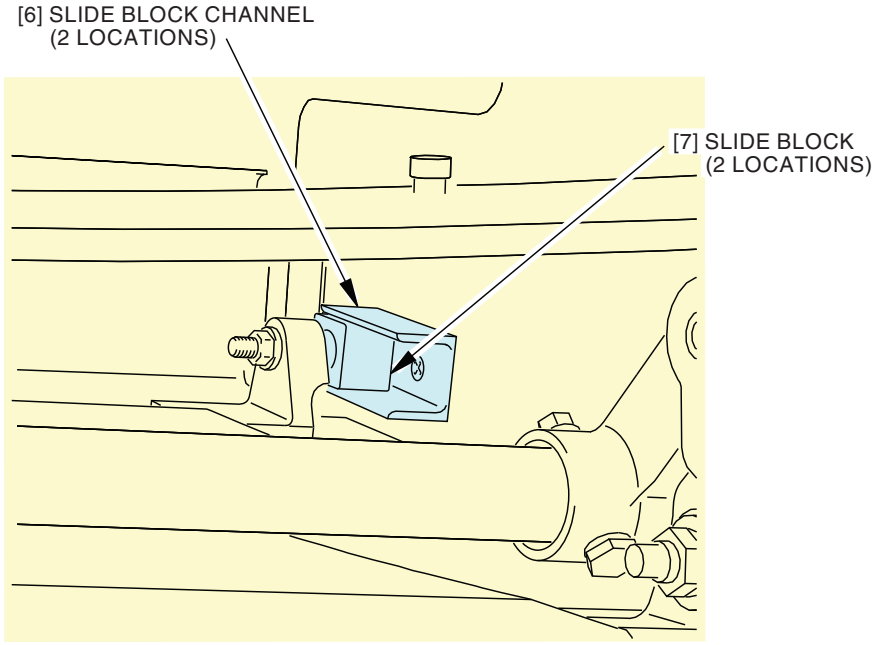
CLEVIS ASSEMBLY (EXAMPLE)

D

2411453 S00061527934_V1

Ram Air Inlet Modulation Panels Installation
Figure 401/21-51-23-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SLIDE BLOCK
(LEFT SIDE)
(RIGHT SIDE IS EQUIVALENT)

E

2411454 S00061527935_V1

Ram Air Inlet Modulation Panels Installation
Figure 401/21-51-23-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-23-400-801

3. Ram Air Inlet Modulation Panels Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
21-51-21-400-801	Ram Air Inlet Actuator Installation (P/B 401)
21-51-22-400-803	Ram Air Inlet Deflector Door and Modulation Panels Adjustment (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward modulation panel	21-51-23-02-125	PTF ALL
		21-51-23-02-130	PTF ALL
2	Aft modulation panel	21-51-23-02-175	PTF ALL
		21-51-23-02-180	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Ram Air Inlet Modulation Panels Installation

SUBTASK 21-51-23-420-001

(1) Install the aft modulation panel [2] in the ram air inlet as follows:

(a) Move the aft modulation panel [2] into its position in the ram air inlet.

(b) Make sure that the slide blocks [7] are in the slide block channels [6].

(c) Install the bolts [9], washers [10], bushings [11], and nuts [12] to connect the clevis assemblies [8] to the aft modulation panel [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If disconnected, attach the ram air inlet actuator arm to the shaft of the aft modulation panel [2] (TASK 21-51-21-400-801).

SUBTASK 21-51-23-420-002

(2) Install the forward modulation panel [1] in the ram air inlet as follows:

(a) Put the forward modulation panel [1] in its position in the ram air inlet with the forward hinge against the ram air inlet forward fairing support.

(b) Install the bolts [4] and washers [5] that attach the forward hinge on the forward modulation panel [1] to the ram air inlet forward fairing support.

(c) Adjust the forward modulation panel [1] and aft modulation panel [2] until the aft hinge on the forward modulation panel [1] is against the forward edge of the aft modulation panel [2].

(d) Make sure that the slide blocks [7] are still in the slide block channels [6].

(e) Install the bolts [3] that attach the forward modulation panel [1] and aft modulation panel [2].

SUBTASK 21-51-23-860-004

WARNING MAKE SURE THAT PERSONNEL ARE CLEAR OF THE RAM AIR INLET MODULATION PANELS, LINKAGES AND ACTUATORS BEFORE YOU SUPPLY POWER TO THE RAM AIR ACTUATOR. POSSIBLE INJURY TO PERSONNEL CAN OCCUR.

(3) If you replaced a modulation panel for the left pack, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-23-860-005

WARNING MAKE SURE THAT PERSONNEL ARE CLEAR OF THE RAM AIR INLET MODULATION PANELS, LINKAGES AND ACTUATORS BEFORE YOU SUPPLY POWER TO THE RAM AIR ACTUATOR. POSSIBLE INJURY TO PERSONNEL CAN OCCUR.

(4) If you replaced a modulation panel for the right pack, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

F. Ram Air Inlet Modulation Panels Installation Test

SUBTASK 21-51-23-860-006

WARNING MAKE SURE THAT PERSONNEL ARE CLEAR OF THE RAM AIR INLET AND EXIT DOORS, LINKAGES AND ACTUATORS BEFORE YOU DO THIS TEST. POSSIBLE INJURY TO PERSONNEL CAN OCCUR.

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-51-23-860-007

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
D	15	C01401	LANDING GEAR AIR/GND RELAY
E	11	C00313	INDICATOR MASTER DIM SECT 1
E	12	C00314	INDICATOR MASTER DIM SECT 2
E	13	C00315	INDICATOR MASTER DIM SECT 3
E	14	C00316	INDICATOR MASTER DIM SECT 4
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 21-51-23-860-008

(3) If you replaced the modulation panels for the left pack, do these steps:

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	5	C00259	AIR CONDITIONING BLEED AIR VALVE ISLN
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

(b) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-23-860-009

(4) If you replaced the modulation panels for the right pack, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	5	C00259	AIR CONDITIONING BLEED AIR VALVE ISLN
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(b) To open the right ECS access door do these steps:

(TASK 06-41-00-800-801)

1) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access door:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-23-860-010

(5) Set these switches, on the P5-10 Bleed Air Control Module, to the OFF position:

(a) BLEED 1

(b) BLEED 2

(c) APU BLEED

(d) L PACK

(e) R PACK.

SUBTASK 21-51-23-730-001

(6) Disconnect the electrical connector D15832 (D15836) from the left (right) pack flow control and shutoff valve.

SUBTASK 21-51-23-010-001

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, DRIVE MECHANISMS AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPERATE THEM WITH ELECTRICAL OR HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Extend the trailing edge flaps to Flaps Position 1 (TASK 27-51-00-860-803).

SUBTASK 21-51-23-860-011

(8) Put the airplane in the air mode (TASK 32-09-00-860-801).

SUBTASK 21-51-23-700-001

(9) Make sure that the ram air inlet deflector door closes and is faired with the ram air inlet ramp.

SUBTASK 21-51-23-860-012

(10) Make sure that the left (right) modulation panels move in the direction of the closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-23-730-002

(11) Retract the trailing edge flaps to the full up position (TASK 27-51-00-860-804).

SUBTASK 21-51-23-730-003

(12) Make sure that the left (right) modulation panels move in the direction of the closed position.

SUBTASK 21-51-23-010-002

(13) Disconnect the electrical connector D3916 (D3918) from the left (right) ram air sensor.

SUBTASK 21-51-23-730-004

(14) Make sure that the ram air inlet modulation panels move to a more open position.

SUBTASK 21-51-23-410-001

(15) Connect the electrical connector D3916 (D3918) to the left (right) ram air sensor.

SUBTASK 21-51-23-860-017

(16) Do the following steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Wait for a minimum of 10 seconds and then continue.

NOTE: Opening the above circuit breakers for 10 seconds minimum and then closing will clear the SRADA internal fault memory and allow actuator movement during subsequent steps.

(c) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-23-410-002

(17) Make sure that the ram air inlet modulation panels move to a more closed position.

SUBTASK 21-51-23-860-013

(18) Extend the trailing edge flaps to Flaps Position 1 (TASK 27-51-00-860-803).

SUBTASK 21-51-23-860-014

(19) Make sure that the left (right) ram air inlet modulation panels move to a more open position.

SUBTASK 21-51-23-860-015

(20) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-23-730-006

(21) Retract the trailing edge flaps to the full up position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 21-51-23-730-005

(22) Make sure that the ram air inlet modulation panels move to a more open position.

SUBTASK 21-51-23-700-002

(23) Make sure that the ram air inlet deflector door opens to the fully deployed position.

SUBTASK 21-51-23-860-016

(24) Connect the electrical connector D15832 (D15836) to the left (right) pack flow control and shutoff valve.

SUBTASK 21-51-23-220-001

(25) If the modulation panels do not move smoothly, do this task: Ram Air Inlet Deflector Door and Modulation Panels Adjustment, TASK 21-51-22-400-803.

SUBTASK 21-51-23-740-001

(26) Do this ground test: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

(a) When the test is completed, make sure that PASS shows adjacent to TEST CONDITION.

(b) If FAILED shows, then do the fault isolation procedure for the maintenance message.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-23-410-003

(1) Close the applicable access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-23-410-004

(2) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR DUCTS - REMOVAL/INSTALLATION

1. General

A. This procedure contains two tasks:

(1) A removal of the ram air ducts

(2) An installation of the ram air ducts.

B. Each pack has a ram air inlet duct and a ram air exhaust duct.

(1) The ram air inlet duct has an anti-static coating that prevents the build up of electric charge on the surface of the duct.

(2) A bonding check is required after the installation of the ram air inlet duct to verify that the ground path is satisfactory.

TASK 21-51-24-000-801

2. Ram Air Ducts Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-23-000-801	Ram Air Inlet Modulation Panels Removal (P/B 401)
21-51-27-000-801	Ram Air Exhaust Actuator - Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
47-11-01-000-802	Air Separation Module Assembly - Removal (P/B 401)
53-51-21-000-801	Forward Wing-To-Body Fairing Panel Removal (P/B 401)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
191HL	Ram Air Inlet Lip Panel - Forward
191HR	Ram Air Inlet Lip Panel - Forward
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 21-51-24-860-001

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-24-860-002

(2) Do these steps, on the P5-10 air conditioning panel (installed on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach the DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-24-010-001

(3) To remove the left ram air inlet duct, do these steps:

WARNING OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Remove this access panel:

(TASK 53-51-21-000-801)

Number	Name/Location
191HL	Ram Air Inlet Lip Panel - Forward

SUBTASK 21-51-24-010-002

(4) To remove the right ram air inlet duct, do these steps:

(a) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

(b) Remove this access panel:

(TASK 53-51-21-000-801)

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Number	Name/Location
191HR	Ram Air Inlet Lip Panel - Forward

SUBTASK 21-51-24-010-003

(5) To get access to the bolts [8], bolts [45], and bolts [43] on the front of the transition duct [2] inlet, remove the applicable ram air inlet modulation panels (TASK 21-51-23-000-801).

SUBTASK 21-51-24-010-004

(6) To get access to the left ram air exhaust duct [26], open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-24-010-005

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(7) To get access to the right ram air exhaust duct [26], do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Ram Air Inlet Duct Removal

SUBTASK 21-51-24-030-001

(1) Do these steps to disconnect the flexible hose [17] between the NGS heat exchanger [19] and the aft, top portion of the left heat exchanger inlet duct [1]:

(a) Loosen the hose clamps [18] that attach the flexible hose [17] to the NGS heat exchanger [19] and heat exchanger inlet duct [1].

(b) Move the end of the flexible hose [17] off of the end of heat exchanger inlet duct [1].

SUBTASK 21-51-24-020-001

(2) Do these steps to remove the heat exchanger inlet duct [1]:

(a) If it is necessary to get more access in the left ram air duct compartment, remove the Nitrogen-Enriched Air Distribution System (NEADS) tube (TASK 47-11-01-000-802).

NOTE: It is possible to disconnect the ends and loosen the two clamps for the applicable section of the NEADS tube assembly. This is to rotate the NEADS tube outwards to get more access.

(b) Remove the bolts [20] and washers [21] that hold each bonding jumper [22] to the airplane structure.

NOTE: These bolts are longer to accommodate the bonding jumper. Keep them separate from the other bolts for the installation.

(c) Remove the bolts [4], washers [5], bolt [41], and washer [42].

(d) Loosen the clamps [6].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Move the end of the flexible hose [3] off the heat exchanger inlet duct [1].

(f) Move the heat exchanger inlet duct [1] down to disengage the upper flange from the retainer on the lower wing beam.

SUBTASK 21-51-24-020-002

(3) Remove the clamps [6].

SUBTASK 21-51-24-020-003

(4) Remove the flexible hose [3] from the transition duct [2].

SUBTASK 21-51-24-020-004

(5) Do these steps to remove the transition duct [2]:

(a) Open the clamp [7] that holds the transition duct [2].

(b) Remove the bolts [8], washers [9], bolts [45], washers [46], bolts [43], and washers [44].

(c) Remove the transition duct [2].

SUBTASK 21-51-24-480-001

(6) Put covers on the duct openings to keep out unwanted materials.

G. Ram Air Exit Duct Removal

SUBTASK 21-51-24-020-005

(1) Remove the ram air exhaust actuator [34] (TASK 21-51-27-000-801).

SUBTASK 21-51-24-020-006

(2) Do these steps to remove the ram air exhaust duct [26]:

(a) Loosen the aft clamp [32].

(b) Move the aft end of the flexible duct [40] forward off of the ram air exhaust duct [26].

(c) Disconnect the bonding jumper [36] and bonding jumper [37] from the rear spar [35] as follows:

1) Remove the nut [39] and washer [38].

2) Remove the bonding jumper [36] and bonding jumper [37].

(d) Support the ram air exhaust duct [26] while you remove the bolts [16] and bolts [27].

(e) Lower the ram air exhaust duct [26] out of the aircraft.

(f) Do these steps to remove the elbow [10]:

1) Remove the bolts [12] and washers [13] at the 7 locations without the bonding jumper.

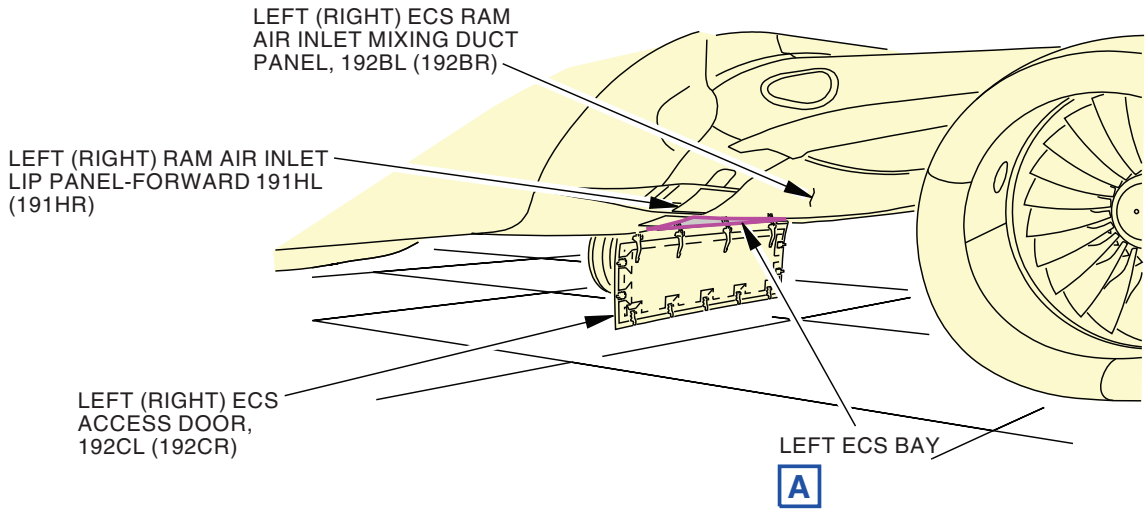
2) At the flange location with the bonding jumper, remove the bolt [28], washer [29], bonding jumper [30], and washer [31].

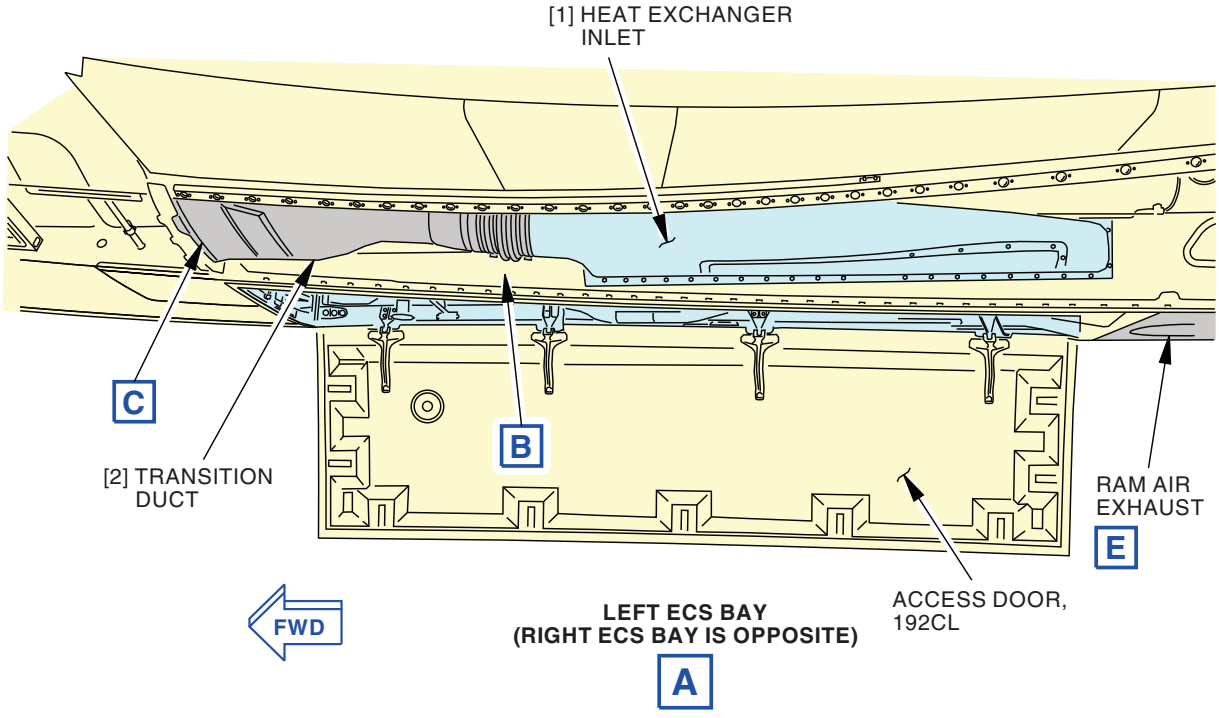
3) Remove the elbow [10] along with the flexible duct [40].

(g) Put a cover on the opening to the plenum/diffuser to keep out unwanted material.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

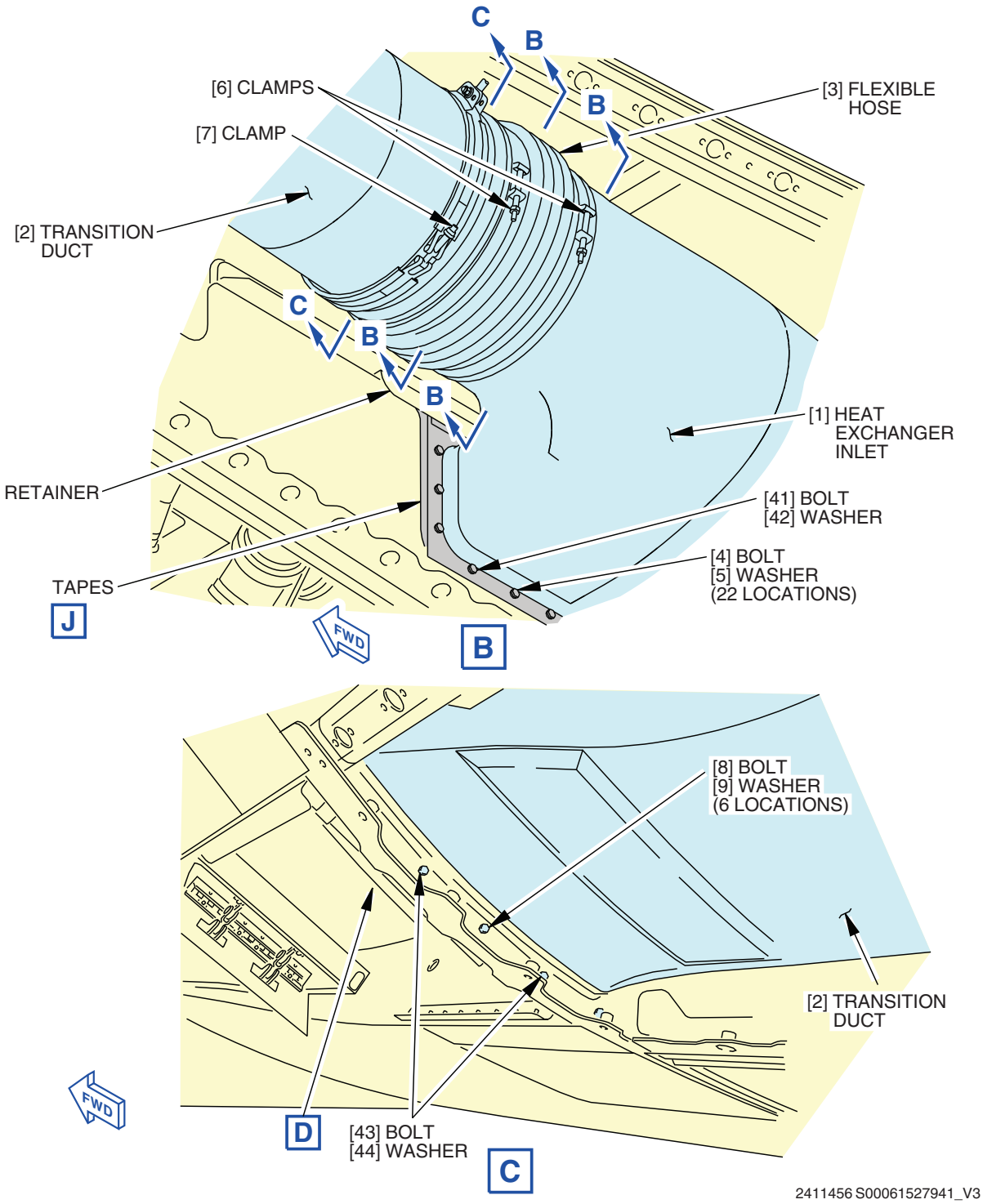




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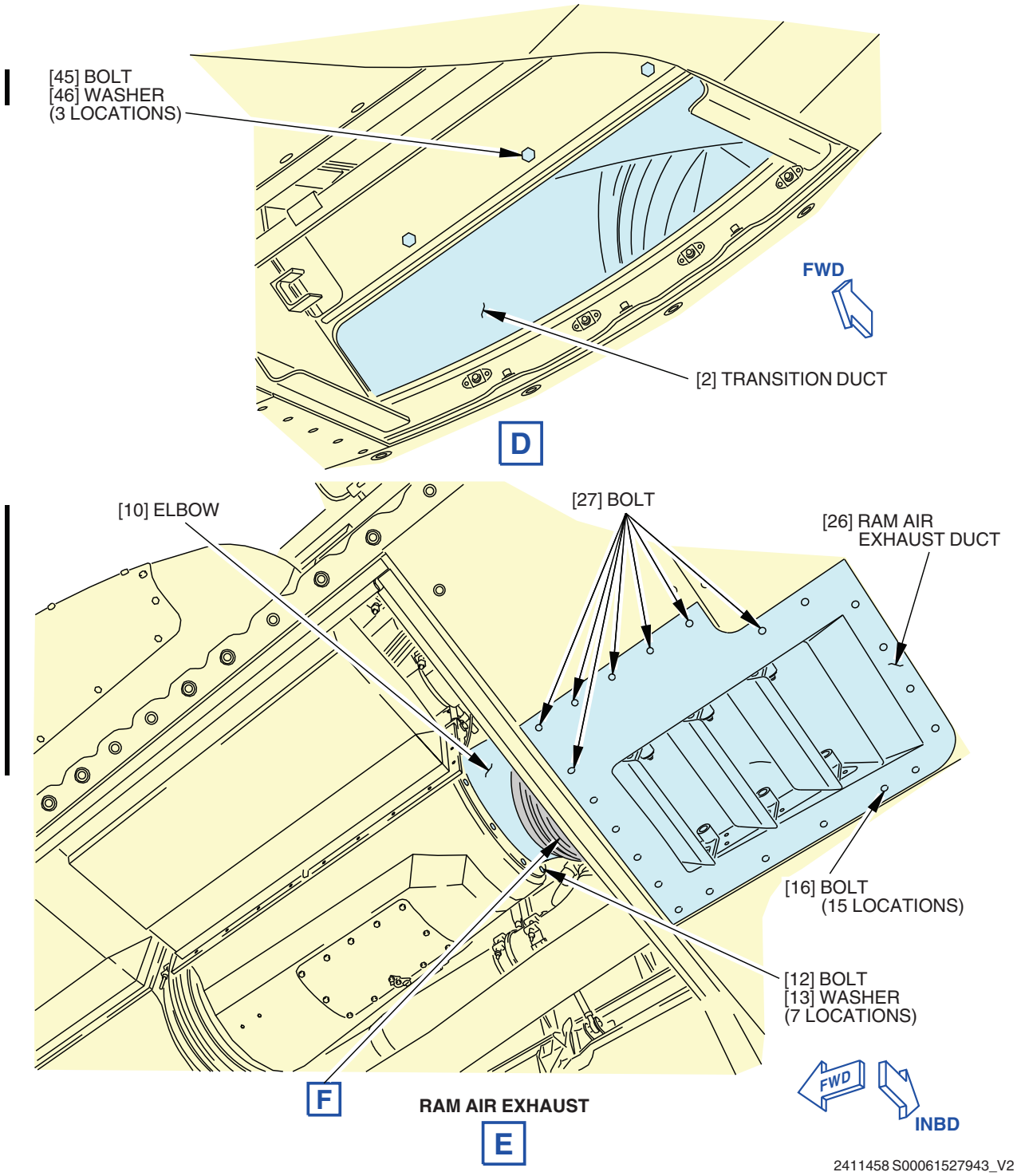
Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



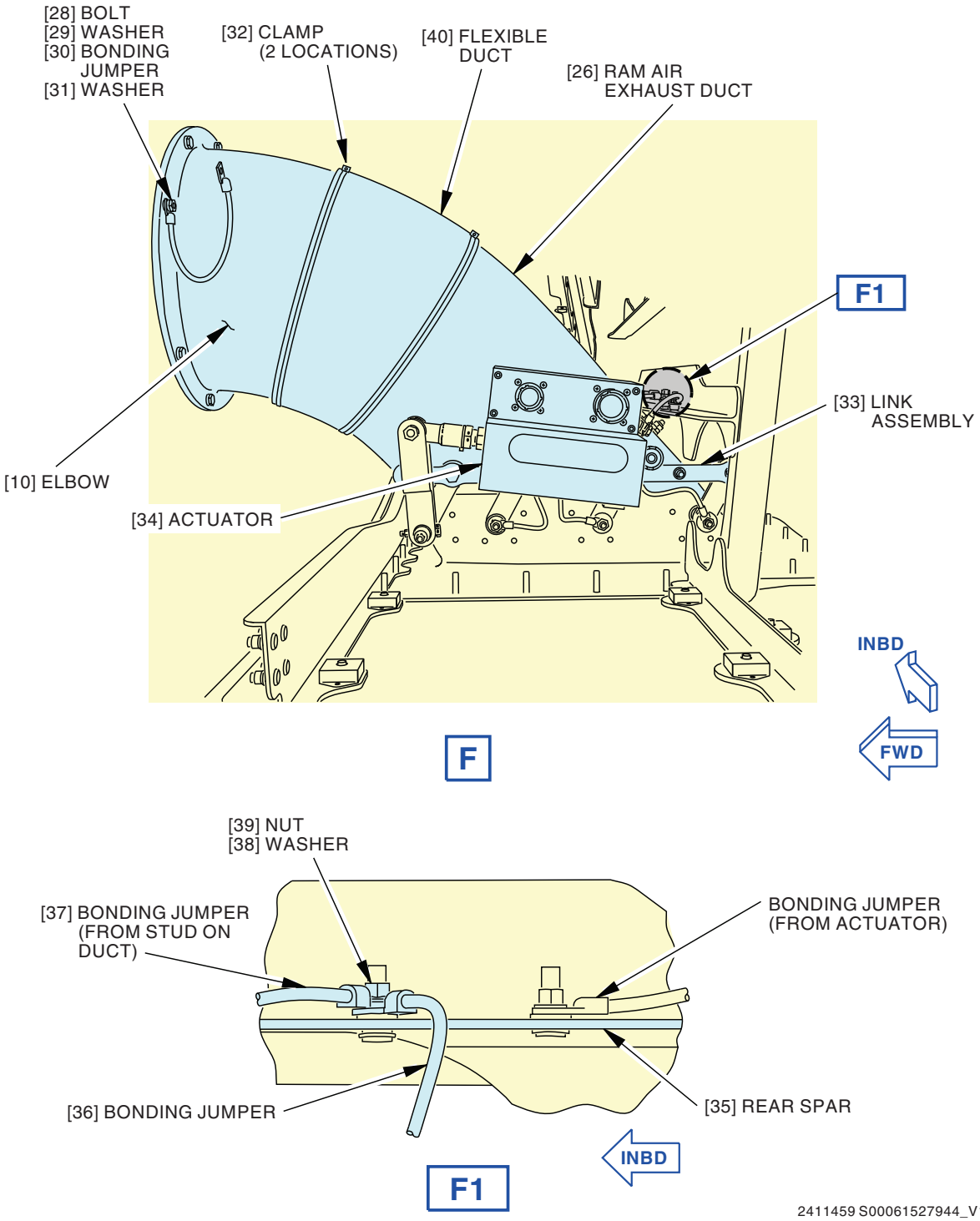
Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 2 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



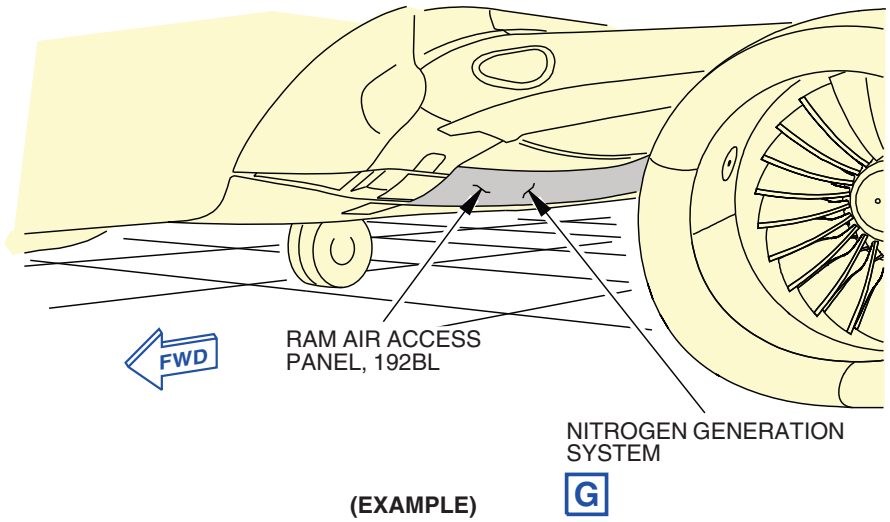
Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

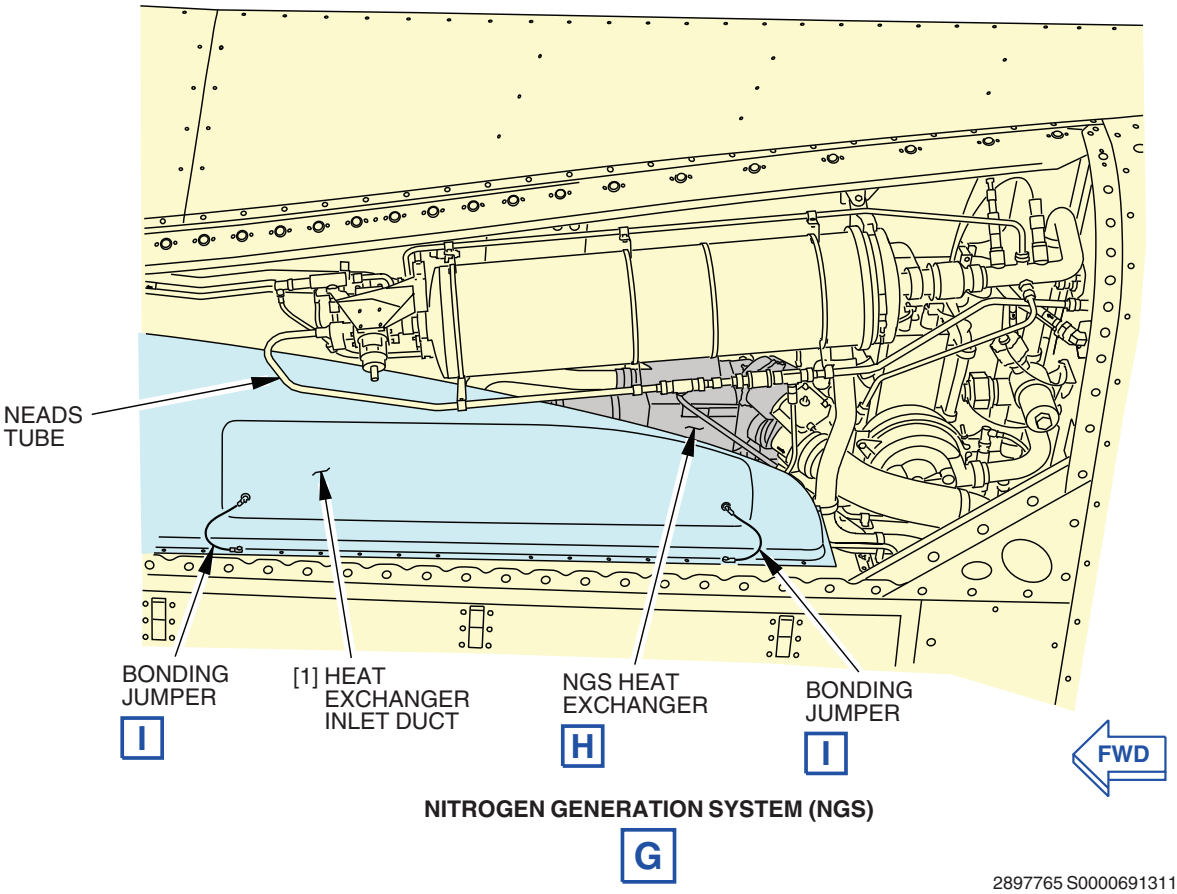


Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



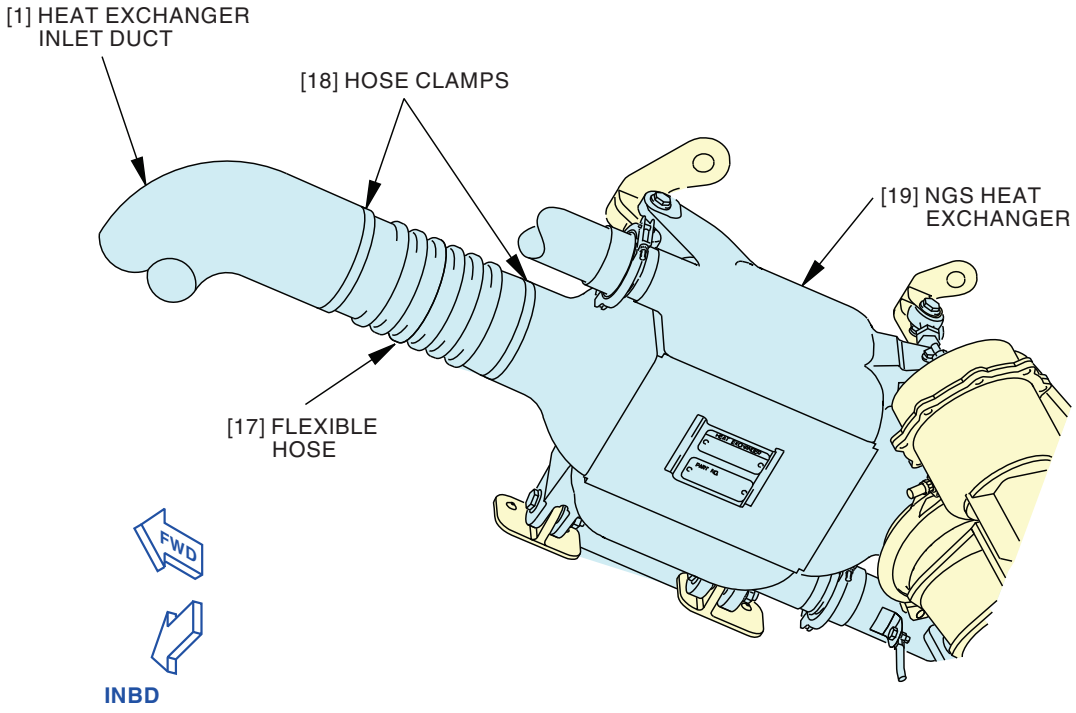
(EXAMPLE)



2897765 S0000691311_V2

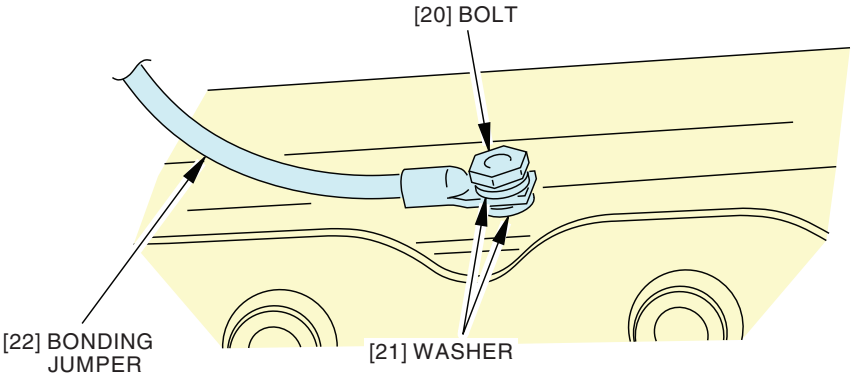
Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NGS HEAT EXCHANGER

H



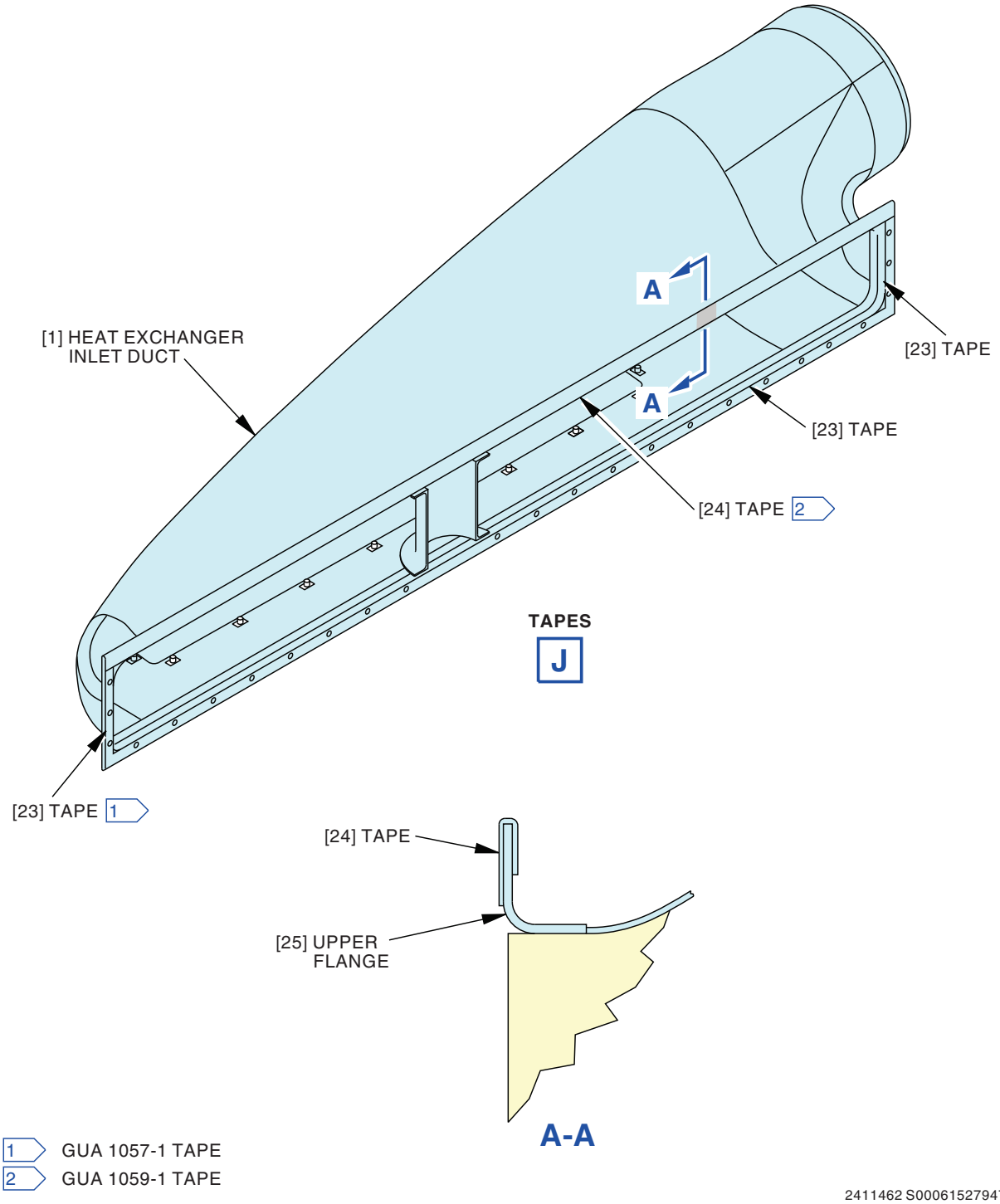
BONDING JUMPER (EXAMPLE)

I

2411461 S00061527946_V1

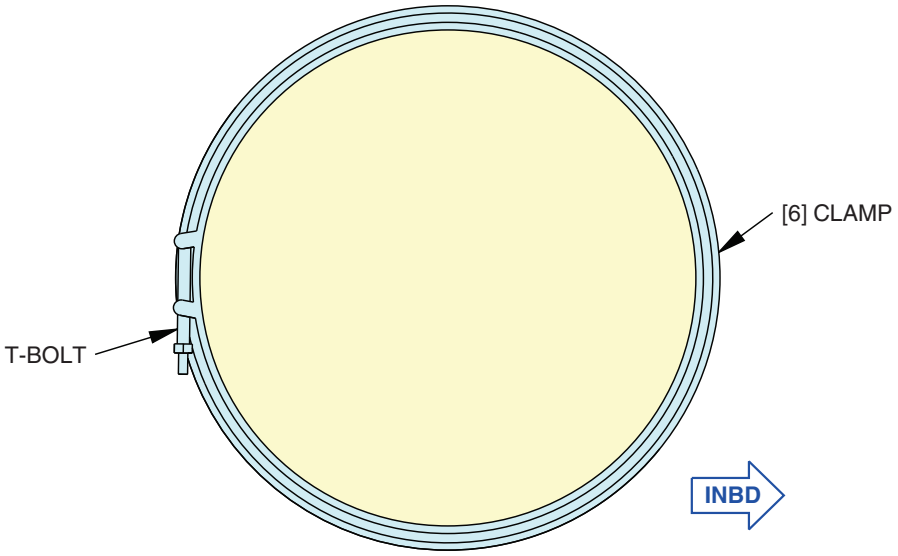
Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



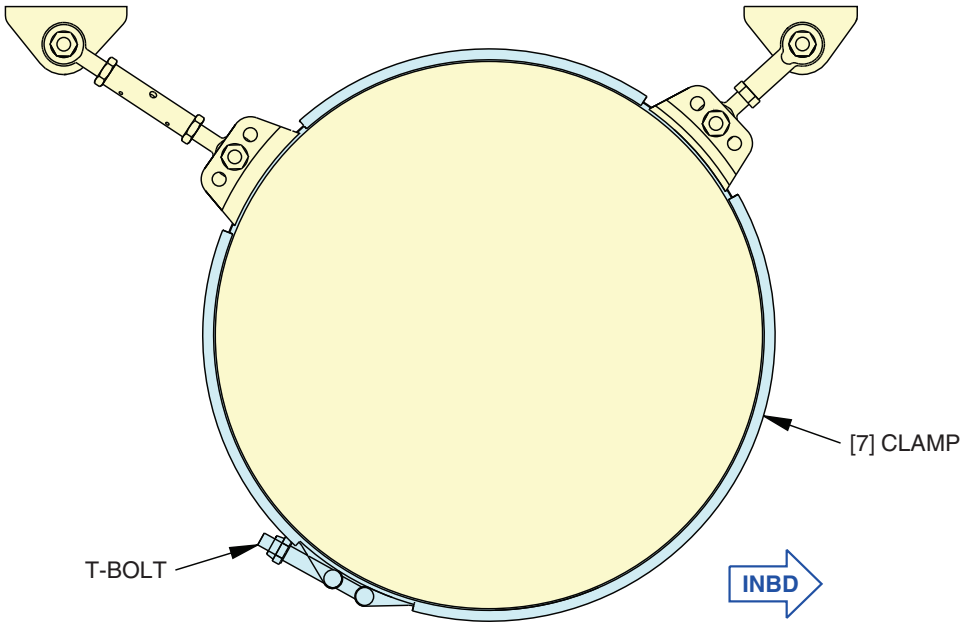
Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(CLAMP T-BOLT ORIENTATION IN THE LEFT ECS BAY IS SHOWN, CLAMP T-BOLT ORIENTATION IN THE RIGHT ECS BAY IS OPPOSITE)

B-B



(CLAMP T-BOLT ORIENTATION IN THE LEFT ECS BAY IS SHOWN, CLAMP T-BOLT ORIENTATION IN THE RIGHT ECS BAY IS OPPOSITE)

C-C

3092401 S0000842521_V1

Ram Air Ducts Installation
Figure 401/21-51-24-990-801 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-24-400-801

3. Ram Air Ducts Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
21-51-23-400-801	Ram Air Inlet Modulation Panels Installation (P/B 401)
21-51-27-400-801	Ram Air Exhaust Actuator - Installation (P/B 401)
47-00-00-790-801	Leak Check of the Nitrogen Generation System (P/B 601)
47-00-00-800-801	Nitrogen Generation System Ground Operation (P/B 201)
47-11-01-400-801	Air Separation Module Assembly - Installation (P/B 401)
51-21-71-370-802	Apply BMS 10-21 Type III Conductive Coating to Specified External Surfaces (P/B 701)
53-51-21-400-801	Forward Wing-To-Body Fairing Panel - Installation (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536

C. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
C00766	Primer - Nonchromated Primer For Composites	BMS10-103 Type I
C00767	Coating - Anti-Static Coating	BMS10-21 Type III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02424	Tape - Teflon, Skyflex Noise Reduction - GUA1057-1, GUA1059-1, GUA1057-144	
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Heat exchanger inlet duct	21-51-24-04A-160	PTF ALL
		21-51-24-04B-165	PTF ALL
2	Transition duct	21-51-24-04A-115	PTF ALL
		21-51-24-04B-120	PTF ALL

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
191HL	Ram Air Inlet Lip Panel - Forward
191HR	Ram Air Inlet Lip Panel - Forward
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Ram Air Inlet Duct Inspection

SUBTASK 21-51-24-210-001

(1) Inspect the tape [23] and tape [24] on the flange of the heat exchanger inlet duct [1] for these conditions:

(a) Missing or damaged sections

(b) Presence of oil or grease

(c) Dirt or debris.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-24-360-001

(2) Remove any damaged or unserviceable section of tape.

SUBTASK 21-51-24-160-001

(3) Clean the section of the duct flange where tape is to be installed as follows:

(a) Remove any fragments of tape from the flange surface where the new tape will be applied.

(b) Clean the flange surface with a cotton wiper, G00034, wet with solvent, B00083, until the surface is clean.

(c) Dry the cleaned surfaces with a cotton wiper, G00034, until all traces of the solvent are removed.

SUBTASK 21-51-24-420-001

(4) Install a new section of GUA1057-1, GUA1059-1, GUA1057-144 tape, G02424, on the duct flange.

(a) GUA 1057-1 for tape [23].

(b) GUA 1059-1 for tape [24].

(c) Cut the tape to the required length.

(d) Install the tape on the flange surface.

1) Do not stretch or wrinkle the tape as you place it on the flange.

2) Make sure that you install the tape [24] on the upper flange [25] so that it wraps over the top of the flange and onto the back surface of the flange.

(e) If it is necessary, cut holes in the tape at the bolt locations.

H. Ram Air Inlet Duct Installation

SUBTASK 21-51-24-080-001

(1) Remove the covers from the duct openings.

SUBTASK 21-51-24-420-002

(2) Do these steps to install the transition duct [2]:

(a) Apply primer, C00259, to all areas of the fastener holes for the bolts [45] and let it dry.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin coat of Cor-Ban 27L Compound, G50237, to the shank and thread of the bolts [8] and bolts [45].

(c) Install the transition duct [2] with the bolts [8] and washers [9], and the bolts [45] and washers [46].

(d) Tighten the bolts [8] and bolts [45] to 30.0 in-lb (3.4 N·m).

(e) Install the clamp [7] on the transition duct [2].

1) Tighten the clamp [7] to 40 ±5 in-lb (4.5 ±0.6 N·m).

2) Make sure that the T-bolt of the clamp [7] points in the outboard direction (View C-C, Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Install the flexible hose [3] and clamp [6] on the transition duct [2].

1) Make sure that the clamp [6] is not installed on the duct bead.

2) Keep a 0.12 in. (3.0 mm) minimum distance between the clamp and edge of the flexible hose [3].

3) Make sure that the T-bolt of the clamp [6] points in the outboard direction (View B-B, Figure 401).

(g) To install the bolts [43] and washers [44], do these steps:

1) Use a cotton wiper, G00034, moist in solvent, B00148, to clean the mating surfaces of the transition duct [2], bolts [43], and washers [44].

2) Make sure that the anti-static coating is continuous from the fastener holes to the surface coat.

a) If necessary, apply a continuous layer of the coating, C00767, on the panel from the fastener holes to the surface coat (Apply BMS 10-21 Type III Conductive Coating to Specified External Surfaces, TASK 51-21-71-370-802).

3) Install the bolts [43] and washer [44].

4) Measure the electrical bonding resistance between the bolts [43] and anti-static coating of the transition duct [2] (SWPM 20-20-00).

a) Use a digital/analog multimeter, COM-1793.

b) Make sure that the electrical bonding resistance is 300,000 ohm or less.

c) If it is necessary, remove a small area of the surface finish to get access to the anti-static coating of the transition duct [2].

d) If the resistance measurement is not satisfactory, disassemble the bonding connection, clean the mating surfaces, and repeat the bonding connection installation.

e) If it is necessary, put a thin coating of primer, C00766, on the surface of the transition duct [2] after the measurement is completed.

SUBTASK 21-51-24-420-003

(3) Do these steps to install the heat exchanger inlet duct [1]:

(a) Loosely put the other clamp [6] on the flexible hose [3].

(b) Put the heat exchanger inlet duct [1] in its position on the lower wing beam.

1) Make sure that the top flange is in the retainer on the lower wing beam.

2) Make sure that the flexible hose [3] is on the forward end of the heat exchanger inlet duct [1].

(c) Install the bolts [4] and washers [5] that attach the heat exchanger inlet duct [1] to the lower wing beam.

(d) To install the bolt [41] and washer [42], do these steps:

1) Use cotton wiper, G00034, moist in solvent, B00148, to clean the mating surfaces of the heat exchanger inlet duct [1], bolt [41], and washer [42].

2) Make sure that the anti-static coating is continuous from the fastener holes to the surface coat.

a) If necessary, apply a continuous layer of the coating, C00767, on the panel from the fastener holes to the surface coat (Apply BMS 10-21 Type III Conductive Coating to Specified External Surfaces, TASK 51-21-71-370-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Install the bolt [41] and washer [42].

4) Measure the electrical bonding resistance between the bolt [41] and anti-static coating of the heat exchanger inlet duct [1] (SWPM 20-20-00).

a) Use a digital/analog multimeter, COM-1793.

b) Make sure that the electrical bonding resistance is 300,000 ohm or less.

c) If it is necessary, remove a small area of the surface finish to get access to the anti-static coating of the heat exchanger inlet duct [1].

d) If the resistance measurement is not satisfactory, disassemble the bonding connection, clean the mating surfaces, and repeat the bonding connection installation.

e) If it is necessary, put a thin coating of primer, C00766, on the surface of the heat exchanger inlet duct [1] after the measurement is completed.

(e) Install the other clamp [6] on the flexible hose [3].

1) Make sure that the clamp [6] is not installed on the duct bead.

2) Keep a 0.12 in. (3.0 mm) minimum distance between the clamp [6] and edge of the flexible hose [3].

3) Make sure that the T-bolt of the clamp [6] points in the outboard direction (View B-B, Figure 401).

(f) Connect the heat exchanger inlet duct [1] to the NGS heat exchanger [19] at the area above the aft, top portion of the left heat exchanger inlet duct [1].

1) Move the end of the flexible hose [17] over the end of the cooling duct from the heat exchanger inlet duct [1].

2) Tighten the hose clamp [18] to 13 in-lb (1.5 N·m) - 18 in-lb (2.0 N·m).

(g) Clean the mating surfaces of the bolts [20], washers [21], and bonding jumpers [22].

1) Use a cotton wiper, G00034, moist with solvent, B00148, to clean the mating surfaces.

(h) Install the bolts [20], washers [21], and bonding jumpers [22] that hold the heat exchanger inlet duct [1] to the airplane structure.

1) Make sure that the bolts [20] are the same length as the ones that were removed.

2) Install the washer [21] between the bonding jumper [22] and bolt [20] head.

a) Make sure that the washer [21] is installed with the countersink next to the bolt [20] head.

3) Install the washer [21] between the heat exchanger inlet duct [1] and bonding jumper [22].

(i) Measure the electrical bonding resistance between the bonding jumper [22] terminal and anti-static coating of the heat exchanger inlet duct [1] (SWPM 20-20-00).

1) Use a digital/analog multimeter, COM-1793.

2) Make sure that the electrical bonding resistance is 300,000 ohms or less.

a) If it is necessary, remove a small area of the surface finish to gain access to the anti-static coating of the heat exchanger inlet duct [1].

b) If the resistance measurement is not satisfactory, then disassemble the bonding connection, clean the mating surfaces, and repeat the bonding connection installation.

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3) If it is necessary, put a thin coating of primer, C00766, on the surface of the heat exchanger inlet duct [1] after the measurement is completed.

SUBTASK 21-51-24-410-009

(4) If it was removed, install or connect the Nitrogen-Enriched Air Distribution System (NEADS) tube (TASK 47-11-01-400-801).

(a) Do this task: Nitrogen Generation System Ground Operation, TASK 47-00-00-800-801.

(b) Do this task: Leak Check of the Nitrogen Generation System, TASK 47-00-00-790-801.

SUBTASK 21-51-24-410-001

(5) Install the ram air inlet modulation panels, do this task: Ram Air Inlet Modulation Panels Installation, TASK 21-51-23-400-801.

I. Ram Air Exit Duct Installation

SUBTASK 21-51-24-010-006

(1) Remove the cover from the duct opening.

SUBTASK 21-51-24-420-004

(2) Do these steps to install the elbow [10]:

(a) Put the elbow [10] and flexible duct [40] in position.

(b) Install the bolts [12] and washers [13] without the bonding jumper [30].

(c) Clean the mating surfaces of the bolt [28], washer [29], bonding jumper [30], washer [31] and the elbow [10].

1) Use a cotton wiper, G00034, moist with solvent, B00148, to clean the mating surfaces.

(d) Put the bonding jumper [30] in position and install the bolt [28], washer [29], and washer [31].

1) Install the bonding jumper [30] between the washer [29] and washer [31] with the countersunk side of the washer [29] under the head of the bolt [28].

(e) Measure the electrical bonding resistance between the bonding jumper [30] terminal and anti-static coating on the elbow [10] (SWPM 20-20-00).

1) Use a digital/analog multimeter, COM-1793.

2) Make sure that the electrical bonding resistance is less than 300,000 ohms.

(f) After the resistance check, apply primer, C00766, to the exposed area around the ground stud.

(g) Install the forward clamp [32] on the ram air exhaust duct [26] as follows:

1) Make sure that the forward clamp [32] is not installed on the duct bead.

2) Tighten the forward clamp [32] to 13 in-lb (1.5 N·m) - 17 in-lb (1.9 N·m).

SUBTASK 21-51-24-420-005

(3) Do these steps to install the ram air exhaust duct [26]:

(a) Apply compound, C00528, to the fastener holes in the airplane structure.

(b) Put the ram air exhaust duct [26] into its installed position.

(c) Install the bolts [16] and bolts [27] that attach the duct to the airplane structure.

1) Make sure that the longer bolts [27] are installed in the locations around the ram air exhaust actuator [34].

(d) Install the aft end of the flexible duct [40] onto the ram air exhaust duct [26].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Install the aft clamp [32] onto the ram air exhaust duct [26] as follows:

1) Make sure that the aft clamp [32] is not installed on the duct bead.

2) Tighten the aft clamp [32] to 13 in-lb (1.5 N·m) - 17 in-lb (1.9 N·m).

SUBTASK 21-51-24-420-006

(4) Install the bonding jumper [36] and bonding jumper [37].

(a) Put the bonding jumper [36] and bonding jumper [37] in their position on the rear spar.

(b) Install the washer [38] and nut [39].

(c) Keep the ground stud from turning and tighten the nut [39] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

(d) Measure the electrical bonding resistance between the bonding jumper terminal and structure where the ground stud is installed (SWPM 20-20-00).

1) Use the digital/analog multimeter, COM-1793.

2) Put one probe on the top bonding jumper terminal and other probe on the structure close to the ground stud.

3) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

SUBTASK 21-51-24-410-002

(5) Install the ram air exhaust actuator, do this task: Ram Air Exhaust Actuator - Installation, TASK 21-51-27-400-801.

J. Ram Air Ducts Installation Test

SUBTASK 21-51-24-860-003

(1) Remove the DO-NOT-OPERATE tags from these switches, on the P5-10 air conditioning panel:

(a) The L PACK and R PACK switches

(b) The BLEED 1 and BLEED 2 switches

(c) The BLEED APU switch.

SUBTASK 21-51-24-860-005

(2) Operate the applicable air conditioning pack, do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

SUBTASK 21-51-24-790-001

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(3) Make sure that there is no air leakage at the applicable ram air duct connections.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-24-860-006

(1) Shut down the applicable air conditioning pack, do this task: Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804.

SUBTASK 21-51-24-410-003

(2) Install this access panel:

(TASK 53-51-21-400-802)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-24-410-004

(3) Install this access panel:

(TASK 53-51-21-400-801)

Number	Name/Location
191HL	Ram Air Inlet Lip Panel - Forward

SUBTASK 21-51-24-410-005

(4) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BR	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 21-51-24-410-006

(5) Install this access panel:

(TASK 53-51-21-400-801)

Number	Name/Location
191HR	Ram Air Inlet Lip Panel - Forward

SUBTASK 21-51-24-410-007

(6) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-24-410-008

(7) Do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR DUCTS - REPAIRS

1. General

A. This procedure gives instructions for the repair of the ram air inlet and ram air exhaust ducts.

TASK 21-51-24-340-801

2. Resin Damage - Fiberglass Ducts

A. General

(1) This procedure gives instructions for the repair of a fiberglass ram air duct with an application of resin.

(2) A resin damage repair is applicable when there are no holes or punctures in the ram air duct but one of these defects are present:

(a) Exposed fibers

(b) Soft areas from impact damage

(c) Delamination

(d) Loss of resin or voids

B. References

Reference	Title
21-51-24-000-801	Ram Air Ducts Removal (P/B 401)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-122	Brush
STD-854	Syringe - Hypodermic

D. Consumable Materials

Reference	Description	Specification
B00094	Solvent - Toluene	A-A-59107
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
C00766	Primer - Nonchromated Primer For Composites	BMS10-103 Type I
C00767	Coating - Anti-Static Coating	BMS10-21 Type III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02428	Abrasive - Aluminum Oxide Paper, 150 grit	A-A-1048
G02451	Resin - Composite Structural Repair, 2 Part (150F Cure)	BMS8-301 Class 2
G50104	Film, FEP or TFE Solid Parting Film, 0.001 inch Thick Minimum	
G50381	Abrasive - Aluminum Oxide Paper, 180 Grit	
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)

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E. Prepare for the Repair

SUBTASK 21-51-24-860-008

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-51-24-840-001

(2) Remove the damaged duct from the airplane. To remove the duct, do this task: Ram Air Ducts Removal, TASK 21-51-24-000-801

SUBTASK 21-51-24-840-002

(3) Do these steps to prepare the damaged area of the duct:

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct repair area.

(b) Sand the damaged area to remove any unwanted contamination from the duct.

(c) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct joint area.

SUBTASK 21-51-24-840-003

(4) Prepare the applicable resin:

(a) Ram air inlet duct repair - resin, G50400

(b) Ram air exhaust duct repair - resin, G02451

F. Repair the Air Conditioning Duct

SUBTASK 21-51-24-340-001

(1) Use resin, G50400 to repair the damaged area of the ram air inlet duct:

(a) Use a brush, STD-122 or hypodermic syringe, STD-854 to apply the resin to the damaged area of the duct.

(b) Apply a thin, smooth layer of the resin to the damaged area of the duct.

(c) Put a piece of solid parting film, G50104 on the duct and press lightly to remove any bubbles or any wrinkles and to remove unwanted resin.

(d) Remove the solid parting film, G50104 from the duct and remove any unwanted resin with a wiping cloth.

(e) Do one of these steps to cure the resin:

1) Permit the resin, G50400 to cure for a minimum of 30 minutes at 150°, ± 10° F.

2) Permit the resin, G50400 to cure for 24 hours at room temperature (70° F minimum).

SUBTASK 21-51-24-340-002

(2) Use resin, G02451 to repair the damaged area of the ram air exhaust duct.

(a) Use a brush, STD-122 or hypodermic syringe, STD-854 to apply the resin to the damaged area of the duct.

(b) Apply a thin, smooth layer of the resin to the damaged area of the duct.

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(c) Put a piece of solid parting film, G50104 on the duct and press lightly to remove any bubbles or any wrinkles and to remove unwanted resin.

(d) Remove the solid parting film, G50104 from the duct and remove any unwanted resin with a wiping cloth.

(e) Permit the resin, G02451 to cure for 3 hours, ± 10 minutes at 150°, ± 10° F.

SUBTASK 21-51-24-340-003

(3) If the repair was made to the outer surface of the ram air inlet duct access panel, do the steps that follow:

(a) Hand sand the surface with 180 grit abrasive paper, G50381 or finer abrasive paper to remove the glossy finish.

(b) Remove the dust with a cotton wiper, G00034.

(c) Prior to the application of the coating, C00767, clean the surface with solvent, B00094 and a cotton wiper, G00034.

(d) Apply the coating, C00767 by spray to a dry film thickness of 0.4 mil (0.010 mm) to 0.8 mil (0.020 mm).

(e) Allow the coating, C00767 to cure a minimum of 2 hours at 60°F (16°C) or 30 minutes at 140°F (60°C).

(f) Hand sand the surface with 150 grit abrasive paper, G02428 or finer abrasive paper.

(g) Remove the dust with a cotton wiper, G00034.

(h) Apply the primer, C00766 by spray to a dry film thickness of 0.5 mil (0.013 mm) to 0.8 mil (0.020 mm).

(i) Allow the primer, C00766 to cure a minimum of 6 hours at 60°F (16°C) or 40 minutes at 140°F (60°C).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-24-840-004

(1) Install the repaired duct on the airplane. To install the duct, do this task: Ram Air Ducts Installation, TASK 21-51-24-400-801

——— END OF TASK ———

TASK 21-51-24-340-802

3. External Patch - Fiberglass Ducts

(Figure 801)

A. General

(1) This procedure gives instructions for the repair of a crack or cut in a fiberglass ram air duct with an external patch.

(2) An external patch is applicable when the cut or the crack is less than 0.25 in. (6.35 mm) wide and is less than 3 in. (76 mm) long.

B. References

Reference	Title
21-51-24-000-801	Ram Air Ducts Removal (P/B 401)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

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C. Tools/Equipment

Reference	Description
STD-94	Bag - Vacuum
STD-821	Squeegee - Plastic

D. Consumable Materials

Reference	Description	Specification
B00094	Solvent - Toluene	A-A-59107
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
C00766	Primer - Nonchromated Primer For Composites	BMS10-103 Type I
C00767	Coating - Anti-Static Coating	BMS10-21 Type III
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02428	Abrasive - Aluminum Oxide Paper, 150 grit	A-A-1048
G02451	Resin - Composite Structural Repair, 2 Part (150F Cure)	BMS8-301 Class 2
G50104	Film, FEP or TFE Solid Parting Film, 0.001 inch Thick Minimum	
G50173	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type H-3 Style 7781
G50381	Abrasive - Aluminum Oxide Paper, 180 Grit	
G50399	Resin - Fiberglass Layup, Long Worklife, Non-Brominated	BMS8-201 Type III (Supersedes BMS8-201 Type I)
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)
G51266	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type H-2 Style 1581

E. Prepare for the Repair

SUBTASK 21-51-24-840-005

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-51-24-840-006

(2) Remove the damaged duct from the airplane. Do this task: Ram Air Ducts Removal, TASK 21-51-24-000-801.

SUBTASK 21-51-24-840-007

(3) Do these steps to prepare the damaged area of the duct.

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WARNING

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(a) Use the solvent, B00148, and a cotton wiper, G00034, to clean the duct repair area.

(b) Drill a hole at each end of the cut or crack to prevent the cut or crack from getting larger.

1) Make sure that the holes are a maximum of 0.062 in. (1.57 mm) diameter.

(c) Sand the damaged area to remove any unwanted contamination from the duct.

(d) Use the solvent, B00148, and a cotton wiper, G00034, to clean the duct joint area.

SUBTASK 21-51-24-340-004

(4) Cut sufficient pieces of the glass fabric, G00031, fabric, G50173, or fabric, G51266, to be equal to the thickness of the duct.

(a) Cut the pieces of fiberglass fabric larger than 0.5 in. (12.7 mm) on all sides of the damage.

(b) Make sure that the corners of the pieces of the fiberglass fabric are rounded.

SUBTASK 21-51-24-340-005

(5) Prepare the applicable resin:

(a) Ram air inlet duct repair - resin, G50400, or resin, G50399

(b) Ram air exhaust duct repair - resin, G02451

F. Repair the Air Conditioning Duct

SUBTASK 21-51-24-340-006

(1) Apply a thin, smooth layer of resin, G50400, resin, G50399, or resin, G02451, to the duct where the fiberglass fabric will be applied.

(a) Make sure that the area of the resin is at the minimum of the dimension and shape of the fiberglass fabric pieces.

SUBTASK 21-51-24-340-007

(2) Put a piece of the fiberglass fabric on the resin so that it extends beyond the cut or crack a minimum of 0.5 in. (12.7 mm) on all sides.

SUBTASK 21-51-24-340-008

(3) Soak the fiberglass fabric with the applicable resin.

SUBTASK 21-51-24-340-009

(4) Put a piece of solid parting film, G50104, on the fiberglass fabric.

(a) Apply light pressure with a plastic squeegee, STD-821, or with your hand.

(b) From the center of the fabric to the edges, remove all bubbles, wrinkles, and unwanted resin.

SUBTASK 21-51-24-340-010

(5) Remove the solid parting film, G50104, from the fiberglass fabric piece.

SUBTASK 21-51-24-340-045

(6) Remove unwanted resin with a cotton wiper, G00034.

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SUBTASK 21-51-24-340-011

(7) Put another piece of the fiberglass fabric on the first piece.

(a) Soak the fiberglass fabric with resin.

1) Make sure that the resin is smooth.

(b) Continue with the same procedure until the thickness of the patch is equal to the thickness of the duct.

SUBTASK 21-51-24-340-050

WARNING DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(8) Wipe the duct around the patch with a cotton wiper, G00034, and solvent, B00148, to remove unwanted resin.

SUBTASK 21-51-24-340-012

(9) To cure resin, G50399, or resin, G50400, do these steps (Structural Repair Manual (SRM) 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Allow the resin to cure for 24 hours at 75 ±5°F (24 ±3°C).

2) Cure for a minimum of 2 - 2.5 hours under a minimum of 20 in/Hg (68 kPa) at 170 ±10°F (77 ±6°C).

NOTE: Moisture protection is not necessary during 170 ±10°F (77 ±6°C) cure.

SUBTASK 21-51-24-340-044

(10) To cure resin, G02451, do these steps (SRM 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Cure for a minimum of 3 hours under a minimum of 22 in/Hg (75 kPa) at 150 ±10°F (66 ±6°C).

2) Decrease the temperature at a maximum rate of 5°F (3°C) for each minute.

3) Below 125°F (52°C), remove the vacuum bag, STD-94, without damage to the cured layers.

SUBTASK 21-51-24-340-013

(11) If the repair was made to the outer surface of the ram air inlet duct access panel, do the steps that follow:

(a) Hand sand the surface with 180 grit abrasive paper, G50381, or finer abrasive paper to remove the glossy finish.

(b) Remove the dust with a cotton wiper, G00034.

(c) Prior to the application of the coating, C00767, clean the surface with solvent, B00094, and a cotton wiper, G00034.

(d) Apply the coating, C00767, by spray to a dry film thickness of 0.4 mil (0.010 mm) to 0.8 mil (0.020 mm).

(e) Allow the coating, C00767, to cure a minimum of 2 hours at 60°F (16°C) or 30 minutes at 140°F (60°C).

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(f) Hand sand the surface with 150 grit abrasive paper, G02428 or finer abrasive paper.

(g) Remove the dust with a cotton wiper, G00034.

(h) Apply the primer, C00766, by spray to a dry film thickness of 0.3 mil (0.008 mm) to 0.8 mil (0.020 mm).

(i) Allow the primer, C00766, to cure a minimum of 6 hours at 60°F (16°C) or 40 minutes at 140°F (60°C).

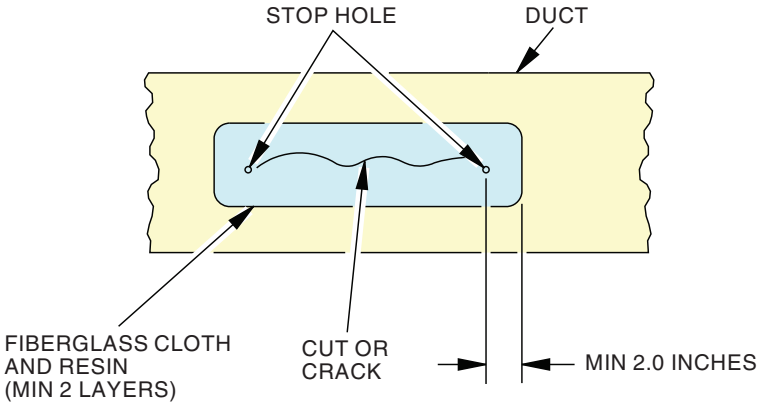
G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-24-840-008

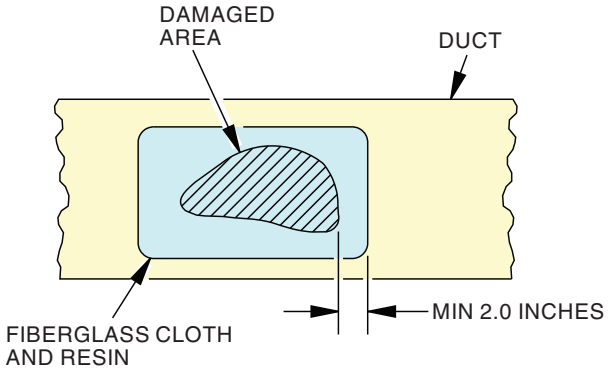
(1) Install the repaired duct on the airplane. Do this task: Ram Air Ducts Installation, TASK 21-51-24-400-801.

——— END OF TASK ———

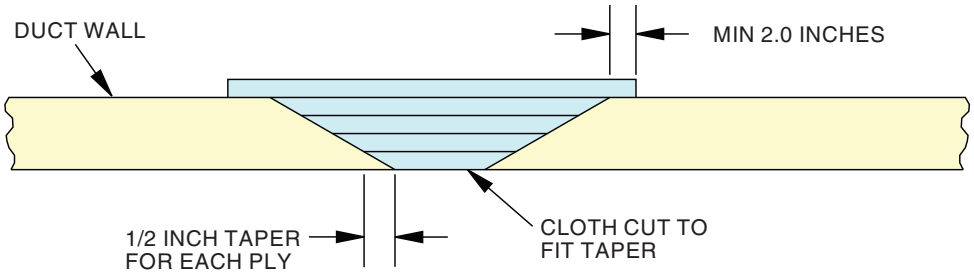
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EXTERNAL PATCH - FIBERGLASS DUCT



STRUCTURAL REPAIR - FIBERGLASS - OVERHEAD VIEW



STRUCTURAL REPAIR - FIBERGLASS - CROSS-SECTION VIEW

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Ram Air Ducts Repair
Figure 801/21-51-24-990-802

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TASK 21-51-24-340-803

4. Structural Repair - Fiberglass Ducts

(Figure 801)

A. General

(1) This procedure gives instructions for making a structural repair to a fiberglass air conditioning duct.

(2) A structural repair is applicable when the damage to the air conditioning duct is a cut or a hole that is more than 0.25 in. (6.4 mm) wide and more than 3 in. (76.2 mm) long but less than 5 percent of the total surface area of the duct. If the damage is more than 5 percent of the total surface area of the duct, then replace the duct section.

B. References

Reference	Title
21-51-24-000-801	Ram Air Ducts Removal (P/B 401)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-94	Bag - Vacuum
STD-122	Brush
STD-821	Squeegee - Plastic

D. Consumable Materials

Reference	Description	Specification
B00094	Solvent - Toluene	A-A-59107
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
C00766	Primer - Nonchromated Primer For Composites	BMS10-103 Type I
C00767	Coating - Anti-Static Coating	BMS10-21 Type III
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02428	Abrasive - Aluminum Oxide Paper, 150 grit	A-A-1048
G02451	Resin - Composite Structural Repair, 2 Part (150F Cure)	BMS8-301 Class 2
G50104	Film, FEP or TFE Solid Parting Film, 0.001 inch Thick Minimum	
G50173	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type H-3 Style 7781
G50381	Abrasive - Aluminum Oxide Paper, 180 Grit	
G50399	Resin - Fiberglass Layup, Long Worklife, Non-Brominated	BMS8-201 Type III (Supersedes BMS8-201 Type I)

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(Continued)

Reference	Description	Specification
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)
G50797	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type D Style 120
G50798	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type D-1 Style 220
G51266	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type H-2 Style 1581

E. Prepare for the Repair

SUBTASK 21-51-24-840-009

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-51-24-840-010

(2) Remove the damaged duct from the airplane. Do this task: Ram Air Ducts Removal, TASK 21-51-24-000-801.

SUBTASK 21-51-24-840-011

(3) Do these steps to prepare the damaged area of the duct.

(a) Cut away the damaged area to make a rounded opening with a smooth shape.

1) Cut the sides of the opening at an angle to make a taper of 0.5 in. (12.7 mm) between each ply of material.

(b) Sand the edge of the opening to make a smooth surface.

(c) Sand the surface of the duct around the opening a distance of 0.5 in. (12.7 mm) for each ply in the duct.

NOTE: For example, if there are 6 plies, sand 3 in. (76.2 mm) from the opening on all sides (1/2 x 6 = 3 inches) or (12.7 x 6 = 76.2 millimeters).

WARNING

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(d) Use the solvent, B00148, and a cotton wiper, G00034, to clean the edge of the opening and the sanded surface of the duct.

SUBTASK 21-51-24-340-014

(4) Cut sufficient pieces of the glass fabric, G00031, fabric, G50173, or fabric, G51266, to be equal to the thickness of the duct.

(a) Cut each piece of the fiberglass fabric 0.5 in. (12.7 mm) larger to fit the taper of the opening.

(b) Cut one piece of the fiberglass fabric a minimum of 2 in. (50.8 mm) larger on all sides than the opening in the duct.

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SUBTASK 21-51-24-340-015

(5) Cut a piece of solid parting film, G50104, 2 in. (50.8 mm) to 3 in. (76.2 mm) larger on all sides than the opening.

SUBTASK 21-51-24-340-048

(6) Cut a piece of cardboard or aluminum 2 in. (50.8 mm) to 3 in. (76.2 mm) larger than the opening on all sides.

SUBTASK 21-51-24-340-016

(7) Attach the solid parting film, G50104, to the inside of the opening with adhesive tape so that it covers the opening.

SUBTASK 21-51-24-340-049

(8) Attach the cardboard or aluminum to the inside of the opening with adhesive tape so that it covers the opening and provides support for the solid parting film, G50104.

SUBTASK 21-51-24-340-017

(9) Prepare the applicable resin:

(a) Ram air inlet duct repair - resin, G50399, or resin, G50400

(b) Ram air exhaust duct repair - resin, G02451

F. Repair the Air Conditioning Duct

SUBTASK 21-51-24-340-018

(1) Apply a thin, smooth layer of resin to the solid parting film, G50104, in the opening in the duct.

SUBTASK 21-51-24-340-019

(2) Put a piece of the fiberglass fabric in the opening over the solid parting film, G50104, so that it covers the opening completely.

SUBTASK 21-51-24-340-020

(3) Soak the fiberglass fabric with the applicable resin.

SUBTASK 21-51-24-340-021

(4) Put a piece of solid parting film, G50104, on the fiberglass fabric.

(a) Apply light pressure with a plastic squeegee, STD-821, or with your hand.

(b) From the center of the fabric to the edges, remove all bubbles, wrinkles, and unwanted resin.

SUBTASK 21-51-24-340-022

(5) Remove the solid parting film, G50104, from the fiberglass fabric piece.

SUBTASK 21-51-24-340-046

(6) Remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-51-24-340-023

(7) Put the next piece of the fiberglass fabric on the first piece of fabric.

(a) Make sure that the fabric is on all of the opening.

(b) If it is necessary, cut the fiberglass fabric.

(c) Soak the fiberglass fabric with resin.

1) Make sure that the resin is smooth.

(d) Continue with the same procedure until the thickness of the patch is equal to the thickness of the duct.

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SUBTASK 21-51-24-340-024

(8) Put the last piece of the fiberglass fabric on top of the patch so it extends beyond the opening a minimum of 2 in. (50.8 mm) on all sides.

(a) Make the fiberglass fabric smooth to remove bubbles or wrinkles.

SUBTASK 21-51-24-340-025

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(9) Wipe the duct around the patch with a cotton wiper, G00034, and solvent, B00148, to remove unwanted resin.

SUBTASK 21-51-24-340-026

(10) To cure resin, G50399, or resin, G50400, do these steps (Structural Repair Manual (SRM) 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Allow the resin to cure for 24 hours at 75 ±5°F (24 ±3°C).

2) Cure for a minimum of 2 - 2.5 hours under a minimum of 20 in/Hg (68 kPa) at 170 ±10°F (77 ±6°C).

NOTE: Moisture protection is not necessary during 170 ±10°F (77 ±6°C) cure.

SUBTASK 21-51-24-340-043

(11) To cure resin, G02451, do these steps (SRM 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Cure for a minimum of 3 hours under a minimum of 22 in/Hg (75 kPa) at 150 ±10°F (66 ±6°C).

2) Decrease the temperature at a maximum rate of 5°F (3°C) for each minute.

3) Below 125°F (52°C), remove the vacuum bag, STD-94, without damage to the cured layers.

SUBTASK 21-51-24-340-027

(12) If the repair was made to the outer surface of the ram air inlet duct access panel, do the steps that follow:

(a) If it is necessary, remove the solid parting film, G50104, the cardboard or aluminum, and the adhesive tape you used to support the repair.

(b) Hand sand the surface with 180 grit abrasive paper, G50381, or finer abrasive paper to remove the glossy finish.

(c) Remove the dust with a cotton wiper, G00034.

(d) Prior to the application of the coating, C00767, clean the surface with solvent, B00094, and a cotton wiper, G00034.

(e) Apply the coating, C00767, by spray to a dry film thickness of 0.4 mil (0.010 mm) to 0.8 mil (0.020 mm).

(f) Allow the coating, C00767, to cure a minimum of 2 hours at 60°F (16°C) or 30 minutes at 140°F (60°C).

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(g) Hand sand the surface with 150 grit abrasive paper, G02428, or finer abrasive paper.

(h) Remove the dust with a cotton wiper, G00034.

(i) Apply the primer, C00766, by spray to a dry film thickness of 0.3 mil (0.008 mm) to 0.8 mil (0.020 mm).

(j) Allow the primer, C00766, to cure a minimum of 6 hours at 60°F (16°C) or 40 minutes at 140°F (60°C).

G. Repair the Beaded End

NOTE: More repair plies are necessary on the external and internal surfaces. If no damage is found on the duct plies (the plies below the bead), do the bead repairs and not the duct plies repairs. Only repair the ply that attaches to the bead.

SUBTASK 21-51-24-340-041

(1) Prepare the applicable resin:

(a) Duct repair of the ram air inlet - resin, G50399, or resin, G50400.

(b) Duct repair of the ram air exhaust - resin, G02451.

SUBTASK 21-51-24-340-029

(2) Apply a thin, smooth layer of resin to the solid parting film, G50104, in the opening in the duct.

SUBTASK 21-51-24-340-030

(3) Put a piece of fabric in the opening over the solid parting film, G50104.

(a) Use fabric, G50797, fabric, G50798, fabric, G50173, glass fabric, G00031, fabric, G51266, or fabric, G50173, to repair the beaded end.

(b) Make sure that the fabric is on all of the opening.

SUBTASK 21-51-24-340-031

(4) Soak the fabric with resin.

SUBTASK 21-51-24-340-032

(5) Put a piece of the solid parting film, G50104, on the fabric.

(a) Apply light pressure to the solid parting film, G50104, with a plastic squeegee, STD-821, or with your hand.

(b) From the center of the fabric to the edges, remove all bubbles, wrinkles, and unwanted resin.

SUBTASK 21-51-24-340-033

(6) Remove the solid parting film, G50104, from the fiberglass fabric piece.

SUBTASK 21-51-24-340-047

(7) Remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-51-24-340-034

(8) Put the next piece of fabric on the first piece of fabric.

(a) Make sure that the fabric is on all of the opening.

(b) If it is necessary, cut the fabric.

(c) Soak the fabric with resin.

1) Make sure that the resin is smooth.

(d) Continue the same procedure with the fabric until the bead agrees with the necessary duct end dimensions.

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SUBTASK 21-51-24-340-035

(9) Cut the last piece of fabric to the necessary dimensions.

NOTE: The dimensions can include the repaired bead and the internal and external surfaces of the full diameter of the duct.

(a) Make sure that the external layer of the Outer Mold Line (OML) is on all of the repaired bead and is around the part end.

(b) Make sure that the external layer extends more than 0.5 in. (12.7 mm) from the repair layer of the Inner Mold Line (IML).

(c) From the center of the fabric to the edges, remove all bubbles or wrinkles.

SUBTASK 21-51-24-340-036

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(10) Clean the duct around the patch with a cotton wiper, G00034, and solvent, B00148, to remove unwanted resin.

SUBTASK 21-51-24-340-037

(11) To cure resin, G50399, or resin, G50400 (SRM 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Allow the resin to cure for 24 hours at 75 ±5°F (24 ±3°C).

2) Cure for a minimum of 2 - 2.5 hours under a minimum of 20 in/Hg (68 kPa) at 170 ±10°F (77 ±6°C).

NOTE: Moisture protection is not necessary during 170 ±10°F (77 ±6°C) cure.

SUBTASK 21-51-24-340-042

(12) To cure resin, G02451, do these steps (SRM 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Cure for a minimum of 3 hours under a minimum of 22 in/Hg (75 kPa) at 150 ±10°F (66 ±6°C).

2) Decrease the temperature at a maximum rate of 5°F (3°C) for each minute.

3) Below 125°F (52°C), remove the vacuum bag, STD-94, without damage to the cured layers.

SUBTASK 21-51-24-340-038

(13) If it is necessary, sand the patch on the outer duct surface to get a correct fit.

(a) Keep a wall thickness of 0.045 in. (1.143 mm) - 0.065 in. (1.651 mm) in this area.

NOTE: Wrinkles, creases, and unwanted resin are not permitted in this area. Sand these items flush where necessary.

SUBTASK 21-51-24-340-039

(14) Use a brush, STD-122, to seal exposed fabric with a layer of resin.

SUBTASK 21-51-24-340-040

(15) Let the resin cure.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-24-840-014

(1) Remove the solid parting film, G50104, the cardboard or aluminum, and the adhesive tape you used to support the repair.

SUBTASK 21-51-24-840-013

(2) Install the repaired duct on the airplane. Do this task: Ram Air Ducts Installation, TASK 21-51-24-400-801.

——— END OF TASK ———

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RAM AIR EXHAUST ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the ram air exhaust actuator

(2) Installation of the ram air exhaust actuator.

B. There are two ram air exhaust actuators (left and right).

(1) They are located aft of each ECS bay and adjacent to the air exhaust duct.

(2) The removal and installation procedures are the same for the two actuators.

TASK 21-51-27-000-801

2. Ram Air Exhaust Actuator - Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
192	Lower Wing-To-Body Fairing - Under Wing Box

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

E. General

SUBTASK 21-51-27-800-001

(1) This procedure removes the aft Smart Ram Air Door Actuator (SRADA) from the ram air exit duct assembly.

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F. Prepare for the Removal

SUBTASK 21-51-27-860-001

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM BEFORE YOU DO MAINTENANCE. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-27-860-002

(2) Set these switches on the P5-10 air conditioning panel as follows:

(a) Set the BLEED 1 and BLEED 2 switches to OFF and attach DO NOT OPERATE tags, STD-858.

(b) Set the APU BLEED switch to OFF and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-27-010-001

(3) To get access to the left ram air exhaust actuator [2], open these access panels:

(a) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

SUBTASK 21-51-27-010-002

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the right ram air exhaust actuator [2], do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

(b) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-27-020-001

(5) To remove the ram air exhaust actuator [2] for the left pack, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

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(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-51-27-020-002

(6) To remove the ram air exhaust actuator [2] for the right pack, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

G. Ram Air Exhaust Actuator Removal

SUBTASK 21-51-27-020-003

(1) Disconnect the two electrical connectors [8] on the ram air exhaust actuator [2].

SUBTASK 21-51-27-020-004

(2) Disconnect the bonding jumper [3] from the bonding tab on the ram air exhaust actuator [2].

(a) Remove the adhesive from the bonding jumper [3] fasteners.

(b) Remove the nut [13], three washers [14], and screw [15].

(c) Remove the bonding jumper [3].

SUBTASK 21-51-27-020-005

(3) Disconnect the rod end [17] of the actuator from the actuator drive arm [16]:

(a) Remove the nut [12], washer [10], washer [11], and bolt [9] that retain the rod end [17] to the actuator drive arm [16].

SUBTASK 21-51-27-020-006

(4) Disconnect the aft end of the ram air exhaust actuator [2] from the ram air exhaust duct [1] assembly.

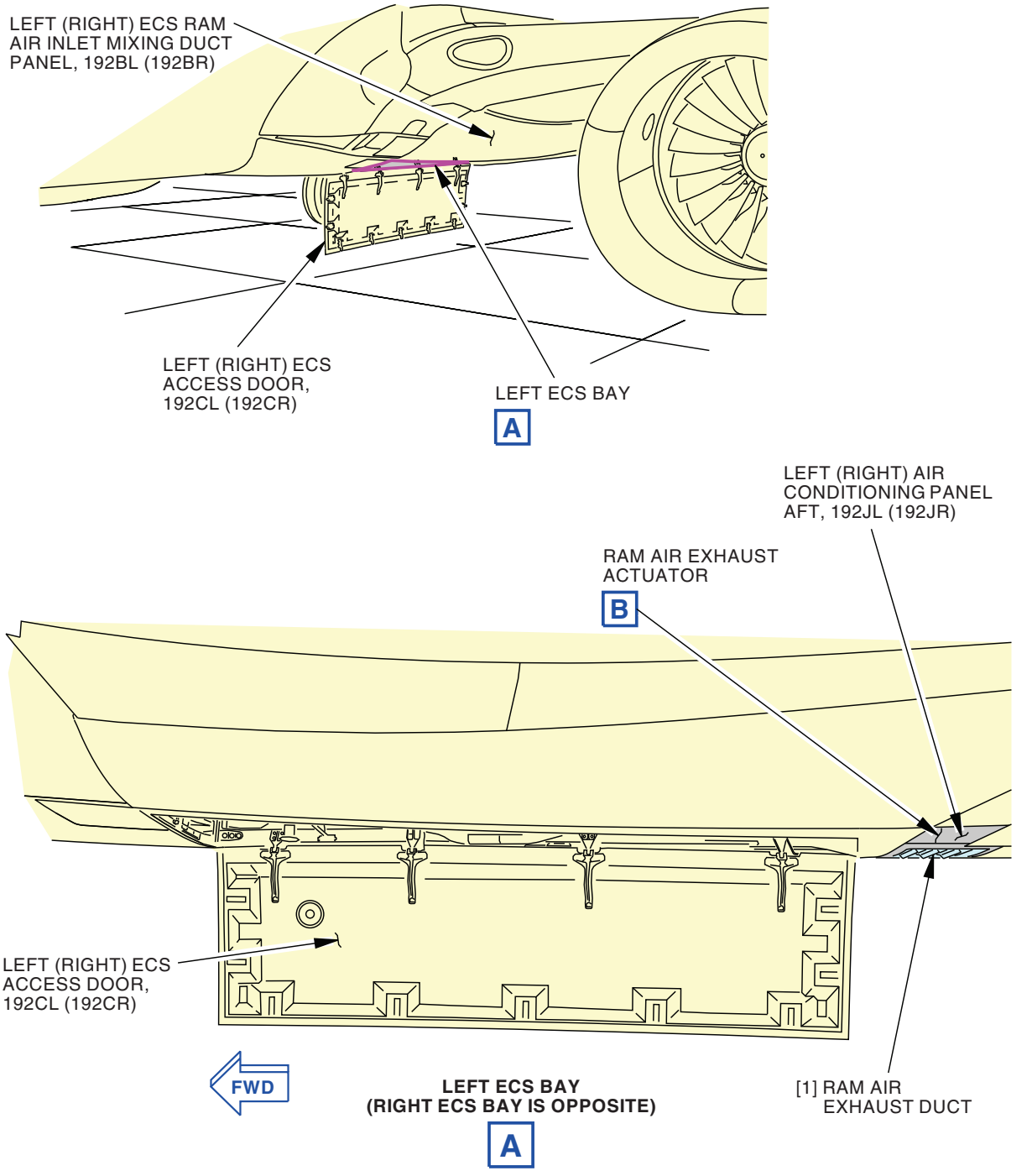
(a) Remove the nut [7], washer [5], washer [6], and bolt [4].

SUBTASK 21-51-27-020-007

(5) Remove the ram air exhaust actuator [2].

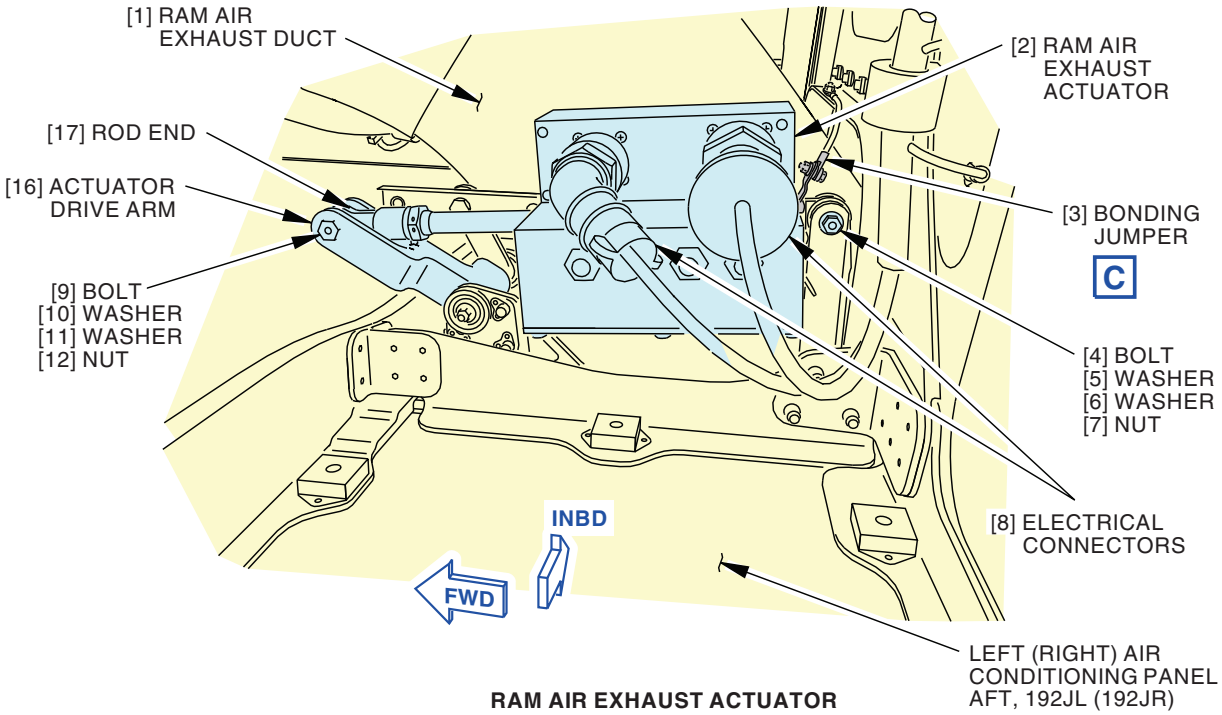
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



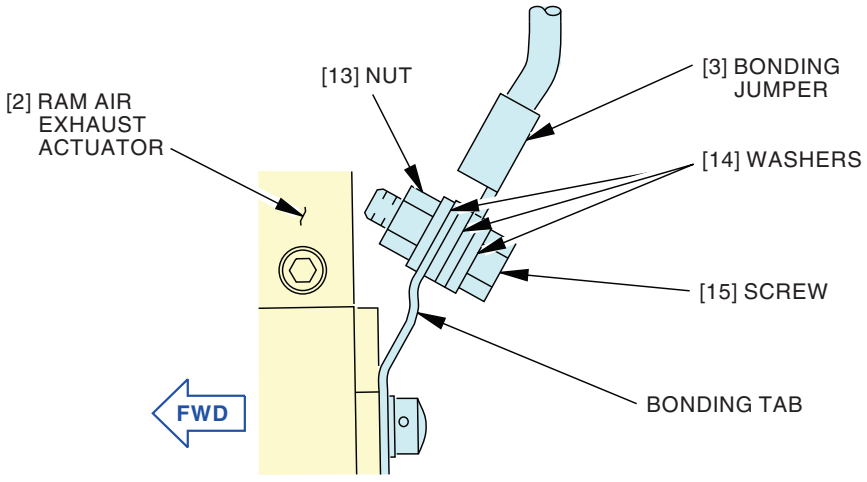
Ram Air Exhaust Actuator Installation
Figure 401/21-51-27-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RAM AIR EXHAUST ACTUATOR (ACCESS PANEL IS REMOVED)

B



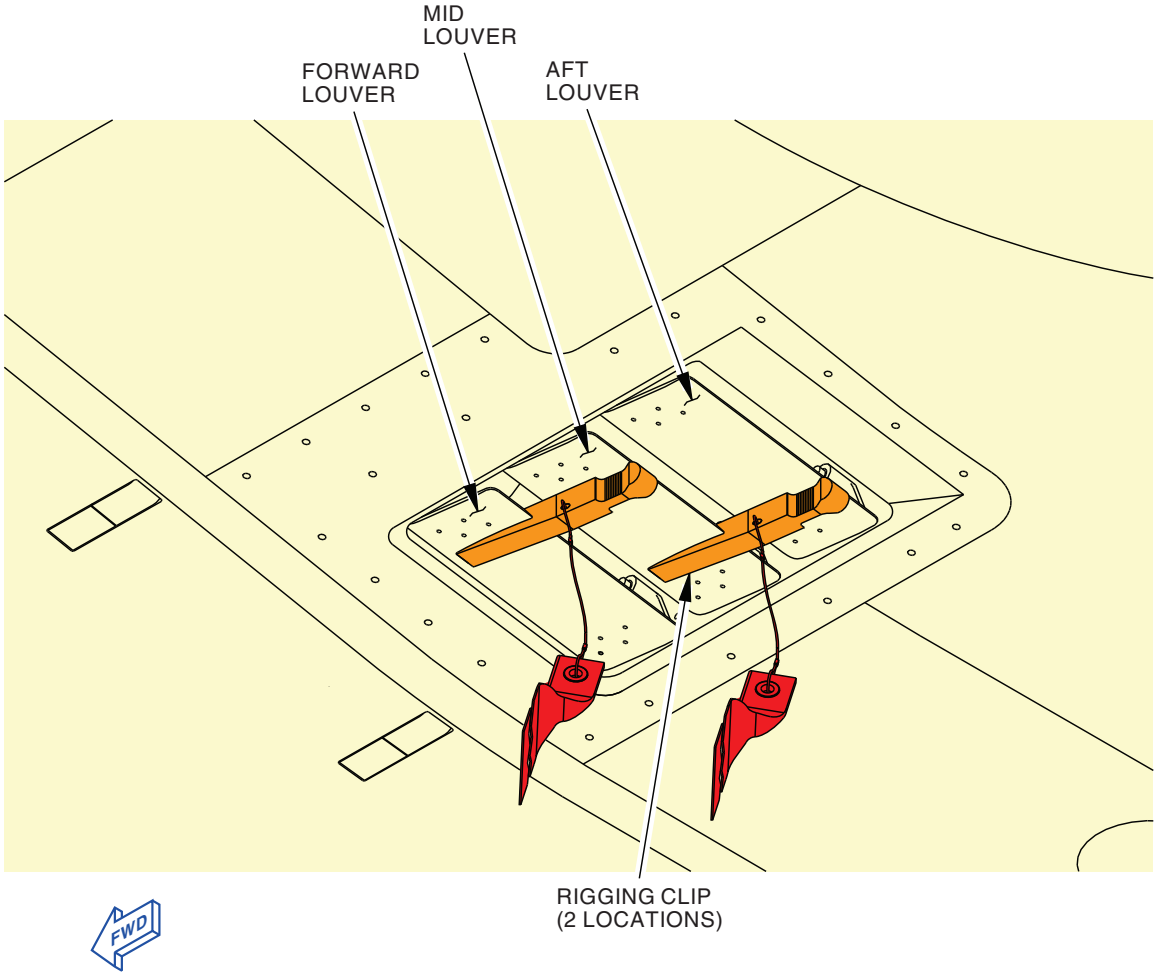
BONDING JUMPER

C

2411465 S00061527958_V1

Ram Air Exhaust Actuator Installation
Figure 401/21-51-27-990-801 (Sheet 2 of 2)

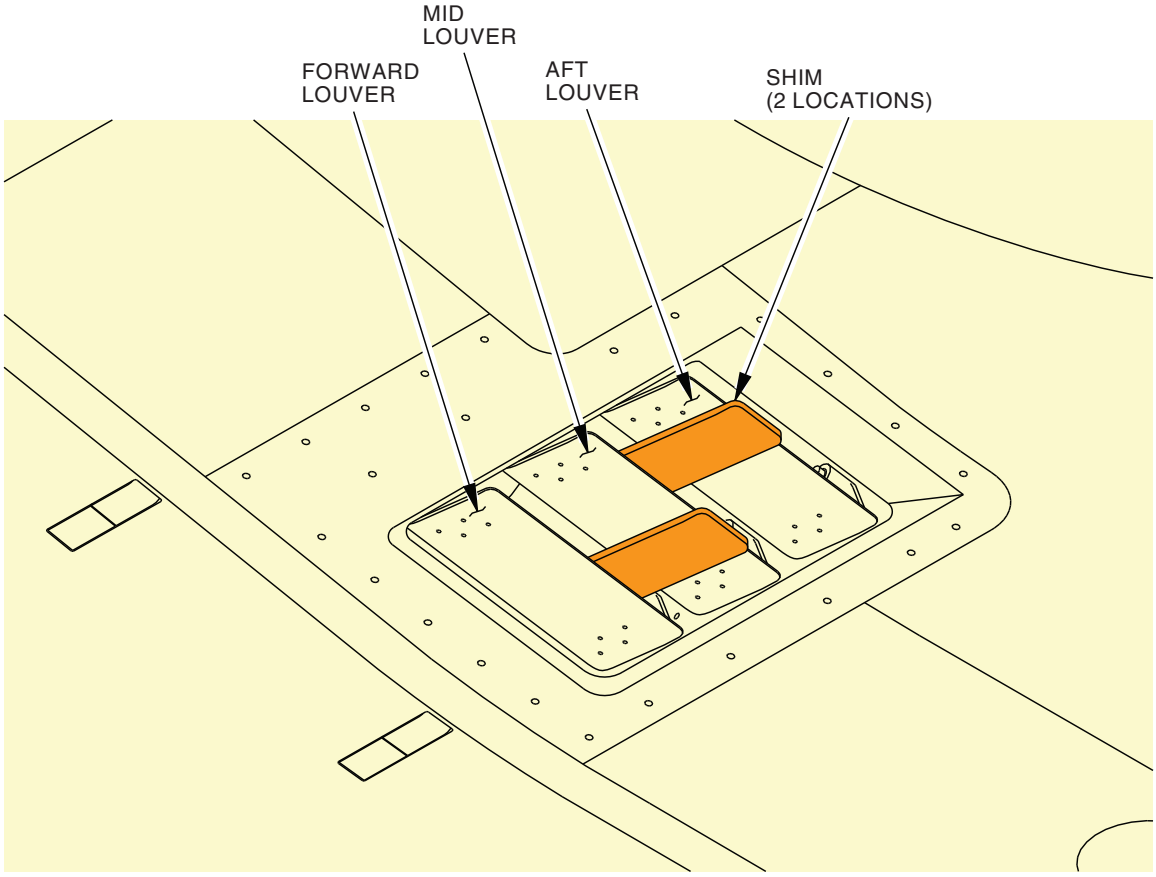
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2974250 S0000753367_V1

Ram Air Exhaust Actuator Adjustment Using Rigging Clips (Preferred Method)
Figure 402/21-51-27-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2967728 S0000747162_V1

Ram Air Exhaust Actuator Adjustment Using Shims (Alternate Method)
Figure 403/21-51-27-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-27-400-801

3. Ram Air Exhaust Actuator - Installation

(Figure 401 or Figure 402)

A. General

(1) This task gives instructions for the installation, adjustment and a post-installation test of the aft Smart Ram Air Door Actuator (SRADA).

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-50-11-910-801	Standard Torque Values (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
SPL-20203	Gap Tool - SRADA Exit Louvers Part #: 68SHIMBLOCK Supplier: 053H3 Part #: C21012-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

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(Continued)

Reference	Description	Specification
G51239	Shim - Resin Bonded, Laminated, Polyimide	BMS7-335 Type VI Grade B

E. Location Zones

Zone	Area
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
192	Lower Wing-To-Body Fairing - Under Wing Box

F. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

G. Ram Air Exhaust Actuator Installation

SUBTASK 21-51-27-420-001

(1) Put the ram air exhaust actuator [2] in its position on the ram air exhaust duct [1] (Figure 401).

SUBTASK 21-51-27-420-002

(2) Attach the aft end of the ram air exhaust actuator [2] to the ram air exhaust duct [1] assembly.

(a) Put the countersunk side of the washer [5] under the head of the bolt [4].

(b) Put the bolt [4] through the ram air exhaust duct [1] assembly and boss on the ram air exhaust actuator [2].

(c) Install the washer [5], washer [6], and nut [7].

(d) Tighten the nut [7] (TASK 20-50-11-910-801).

SUBTASK 21-51-27-420-003

(3) Do not attach the forward end of the ram air exhaust actuator [2] to the actuator drive arm [16] at this time.

SUBTASK 21-51-27-420-004

(4) Do these steps to install bonding jumper [3]:

(a) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the washers [14], screw [15], nut [13], bonding jumper [3], and bonding tab.

(b) Install the washers [14], screw [15], nut [13] and bonding jumper [3] on the bonding tab.

(c) Tighten the nut [13] to 36.5 ±3.5 in-lb (4.1 ±0.4 N·m).

(d) Measure the electrical resistance between the bonding tab and bonding jumper [3].

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1 milliohm) or less.

(e) Apply a fillet seal to the fasteners with sealant, A50337, sealant, A50231, or sealant, A50359.

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H. Ram Air Exhaust Actuator Adjustment

SUBTASK 21-51-27-820-001

(1) Make sure that the rod end [17] of the ram air exhaust actuator [2] is not attached to the actuator drive arm [16].

SUBTASK 21-51-27-420-006

(2) Connect the two electrical connectors [8] to the left ram air exhaust actuator [2].

SUBTASK 21-51-27-420-007

(3) Connect the two electrical connectors [8] to the right ram air exhaust actuator [2].

SUBTASK 21-51-27-860-010

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-51-27-860-014

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-27-860-011

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-51-27-860-013

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-27-860-003

(8) Extend the trailing edge flaps to position 1 (TASK 27-51-00-860-803).

SUBTASK 21-51-27-860-004

(9) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-51-27-860-005

(10) Retract the left and right trailing edge flaps to the flaps fully up position (TASK 27-51-00-860-804).

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SUBTASK 21-51-27-820-002

CAUTION MAKE SURE TO FOLLOW THE STEPS TO ADJUST THE LOUVERS TO THE FULLY CLOSED POSITION CORRECTLY. IF YOU DO NOT OBEY, THIS CAN CAUSE AN INCORRECT RIGGING CONDITION AND POSSIBLE DAMAGE TO THE AIR CYCLE MACHINE.

(11) Do one of the following steps to set the louvers to the flight fully closed position:

NOTE: The preferred procedure uses rigging clips. The alternate procedure uses non-metallic shims.

(a) Use rigging clips of gap tool, SPL-20203, to set the louvers to the flight fully closed position as follows (Figure 402):

1) Move the actuator drive arm [16] to turn the exhaust louvers until a rigging clip can be inserted on the forward and mid louvers.

2) Insert one rigging clip on the forward louver, and insert another rigging clip on the mid louver.

3) Position the rigging clips with slight separation between them.

4) Manually turn the louvers to the closed position until one of the rigging clips touches the adjacent louver.

NOTE: This position of the exhaust louvers is the flight fully closed position with the louvers set at approximately 12 degrees.

5) Make sure that the rigging clip that touched the adjacent louver does not become loose or fall out during the connection of the actuator rod end [17] to the actuator drive arm [16].

(b) Use non-metallic shims of gap tool, SPL-20203, or non-metallic shims, G51239, to set the louvers to the flight fully closed position as follows (Figure 403):

NOTE: Non-metallic shims can be machined or sanded to get a completed dimension of 0.680 ±0.010 in. (17.272 ±0.254 mm) thick x 7 in. (18 cm) long x 2 in. (5 cm) wide. Multiple thinner laminations can be bonded together to get the correct thickness.

1) Move the actuator drive arm [16] to turn the exhaust louvers until a shim (0.680 ±0.010 in. (17.272 ±0.254 mm) thick x 7 in. (18 cm) long x 2 in. (5 cm) wide) can be inserted between each pair of louvers.

2) Insert one shim between the forward and mid louvers, and insert another shim between the mid and aft louvers.

3) Position the shims with a slight separation between them.

4) Manually turn the louvers to the closed position until one of the shims is caught between the louvers.

NOTE: This position of the exhaust louvers is the flight fully closed position with the louvers set at approximately 12 degrees.

5) Make sure that the shim that was caught between the louvers does not become loose or fall out during the connection of the actuator rod end [17] to the actuator drive arm [16].

SUBTASK 21-51-27-420-005

(12) Connect the rod end [17] of the ram air exhaust actuator [2] to the actuator drive arm [16] (Figure 401):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If the hole in the rod end [17] does not align with the holes in the actuator drive arm [16], do these steps:

1) Remove the lockwire from the jam nut on the rod end [17].
2) Adjust the length of the rod end [17] by maximum of ±0.125 in. (3.175 mm) (or 4 turns in either direction) until the holes in the rod end [17] and actuator drive arm [16] align.
3) Hold the rod end [17] and tighten the jam nut until it is seated against the rod end.
4) Tighten the jam nut to 110 ±10 in-lb (12.4 ±1.1 N·m).
5) Install lockwire to keep the jam nut in position.

(b) Put the countersunk side of the washer [10] under the head of the bolt [9].

(c) Put the bolt [9] through the actuator drive arm [16] and rod end [17] of the actuator drive shaft.

(d) Install the washer [11] and nut [12].

(e) Tighten the nut [12] (TASK 20-50-11-910-801).

SUBTASK 21-51-27-080-001

(13) Remove the rigging clips of gap tool, SPL-20203, or shims of gap tool, SPL-20203, or shims, G51239, from the louvers.

I. Ram Air Exhaust Actuator Adjustment Check

SUBTASK 21-51-27-860-008

(1) Extend the left and right trailing edge flaps to position 1 (TASK 27-51-00-860-803).

SUBTASK 21-51-27-710-001

(2) Make sure that the inlet doors move to the open position smoothly without hanging or binding.

SUBTASK 21-51-27-860-012

(3) Retract the trailing edge flaps to the fully up position (TASK 27-51-00-860-804).

SUBTASK 21-51-27-710-004

(4) Make sure that the left and right ram air inlets doors move towards the closed position.

SUBTASK 21-51-27-860-009

(5) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-51-27-710-002

(6) Make sure that the deflector door extends.

SUBTASK 21-51-27-710-003

(7) Make sure that the ram air exit louvers move to the most open position without hanging or binding.

J. Ram Air Exhaust Actuator Test

SUBTASK 21-51-27-740-001

(1) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

(a) When the test is completed, make sure that PASS shows adjacent to TEST CONDITION.

(b) If FAILED shows, then do the fault isolation procedure for the maintenance message.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-27-410-001

(1) If you replaced the left ram air exhaust actuator [2], close these access panels:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) These are the panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

SUBTASK 21-51-27-410-002

(2) If you replaced the right ram air exhaust actuator [2], do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Close this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMPRESSOR DISCHARGE OVERHEAT SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the compressor discharge overheat switch

(2) An installation of the compressor discharge overheat switch.

TASK 21-51-40-000-801

2. Compressor Discharge Overheat Switch Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

D. Prepare for the Removal

SUBTASK 21-51-40-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-40-860-002

(2) To replace the compressor discharge overheat switch [1] for the left cooling pack, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-40-860-003

(3) To replace the compressor discharge overheat switch [1] for the right cooling pack, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-40-010-001

(4) To get access to the compressor discharge overheat switch [1] on the left cooling pack, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-40-010-002

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To get access to the compressor discharge overheat switch [1] on the right cooling pack, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

E. Overheat Switch Removal

SUBTASK 21-51-40-020-001

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the electrical connector [2] from the compressor discharge overheat switch [1].

(a) If it is necessary to remove the turbine inlet overheat switch and compressor discharge overheat switch [1], make a mark on the electrical connector [2].

NOTE: The electrical connector of the compressor discharge overheat switch is near the connector of the turbine inlet overheat switch.

NOTE: You can make a mark on the electrical connectors to identify the connectors in the installation task.

SUBTASK 21-51-40-020-002

(2) Remove and discard the lockwire from the compressor discharge overheat switch [1].

SUBTASK 21-51-40-020-003

CAUTION USE TWO WRENCHES TO INSTALL THE OVERHEAT SWITCH IN THE BOSS. IF YOU USE ONLY ONE WRENCH TO INSTALL THE OVERHEAT SWITCH, IT CAN CAUSE DAMAGE TO THE DUCT.

(3) Use two wrenches to remove the compressor discharge overheat switch [1].

(a) Use one wrench to hold the boss and the other wrench to turn the compressor discharge overheat switch [1].

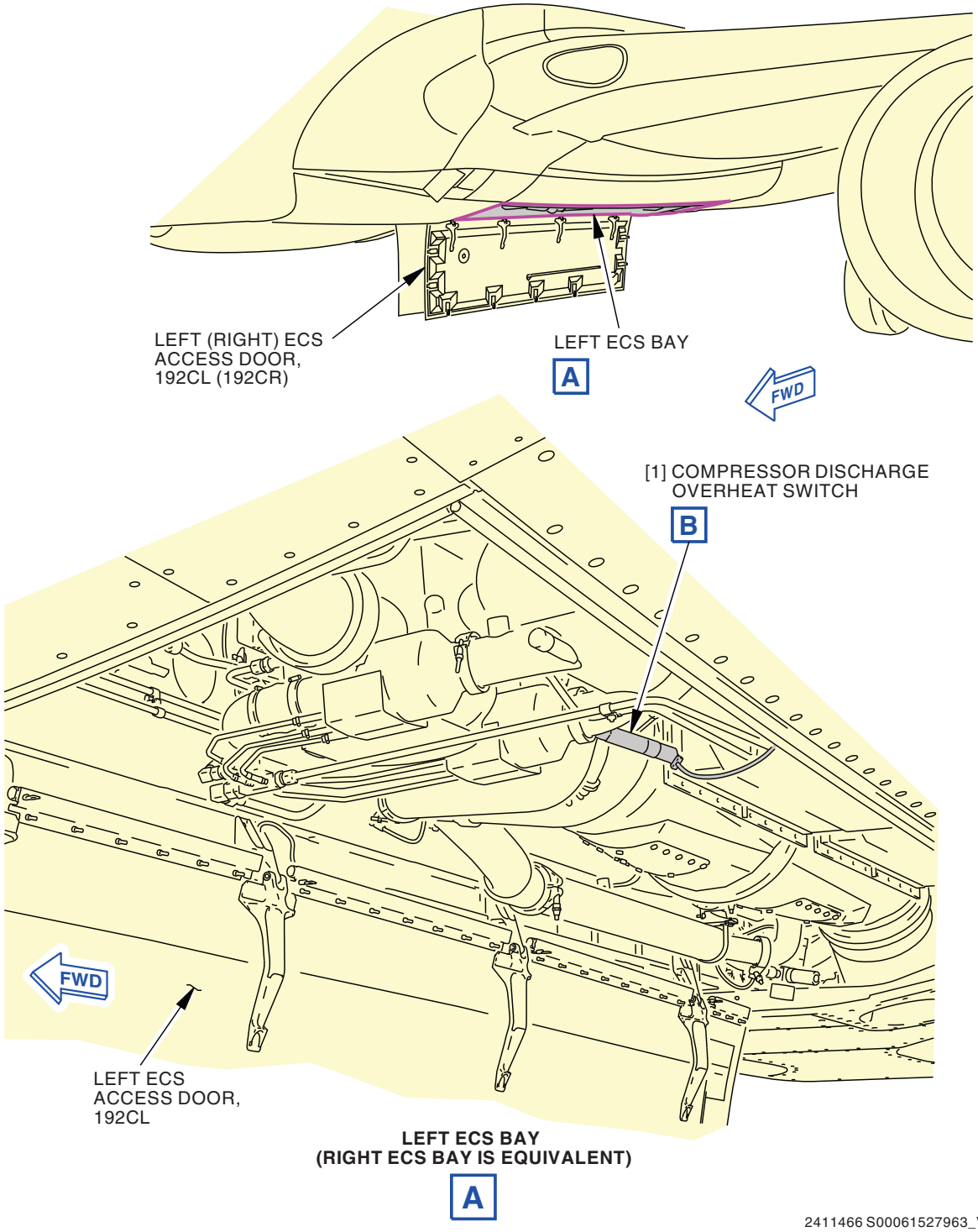
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-40-020-004

(4) Remove and discard the O-ring [3] from the compressor discharge overheat switch [1].

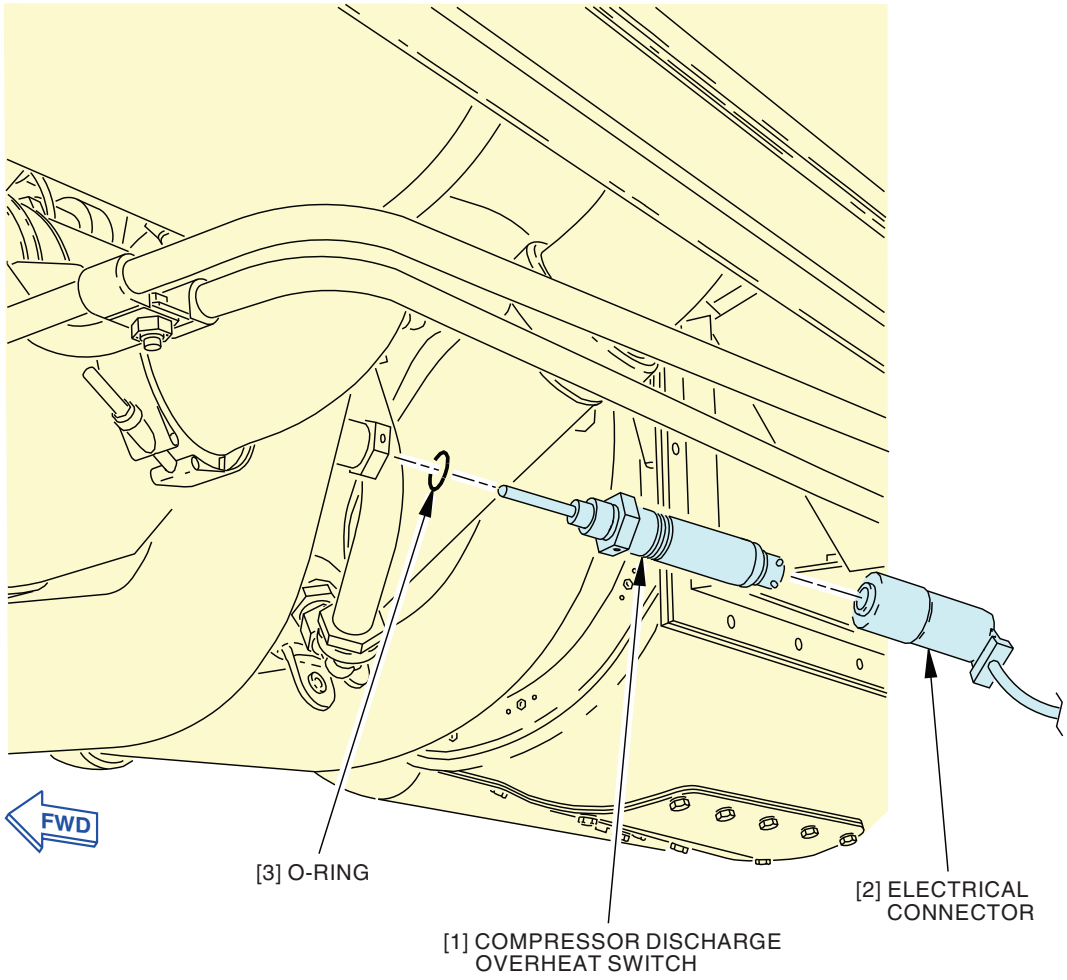
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Compressor Discharge Overheat Switch Installation
Figure 401/21-51-40-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Compressor Discharge Overheat Switch Installation
Figure 401/21-51-40-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-40-400-801

3. Compressor Discharge Overheat Switch Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Compressor discharge overheat switch	21-51-40-01-015	PTF ALL
		21-51-40-02-005	PTF ALL
3	O-ring	21-51-40-01-020	PTF ALL
		21-51-40-02-010	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Overheat Switch Installation

SUBTASK 21-51-40-640-001

(1) Apply a light coat of Molykote 44 grease, medium, D50104, to a new O-ring [3].

NOTE: The new compressor discharge overheat switch can be supplied with an O-ring already installed from the manufacturer. In this case it is also necessary to apply grease on the O-ring.

SUBTASK 21-51-40-420-001

(2) Install the lubricated O-ring [3] on the compressor discharge overheat switch [1].

SUBTASK 21-51-40-160-001

(3) Make sure that the mating surfaces of the boss and compressor discharge overheat switch [1] nut are clean.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-40-640-002

(4) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the compressor discharge overheat switch [1].

SUBTASK 21-51-40-420-002

(5) Put the compressor discharge overheat switch [1] in the boss on the duct.

SUBTASK 21-51-40-420-003

CAUTION

USE TWO WRENCHES TO INSTALL THE OVERHEAT SWITCH IN THE BOSS. IF YOU USE ONLY ONE WRENCH TO INSTALL THE OVERHEAT SWITCH, IT CAN CAUSE DAMAGE TO THE DUCT.

(6) Use two wrenches to install the compressor discharge overheat switch [1]:

(a) Use one wrench to hold the boss and the other wrench to turn the compressor discharge overheat switch [1] into the boss.

(b) Tighten the compressor discharge overheat switch [1] to 133 in-lb (15.0 N·m) - 147 in-lb (16.6 N·m).

SUBTASK 21-51-40-420-004

(7) Install new MS20995C32 lockwire, G01048, on the compressor discharge overheat switch [1].

SUBTASK 21-51-40-420-005

CAUTION

MAKE SURE THAT YOU CONNECT THE CORRECT ELECTRICAL CONNECTOR TO THE OVERHEAT SWITCH FOR COMPRESSOR DISCHARGE. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(8) Connect the electrical connector [2] to the compressor discharge overheat switch [1].

(a) If there is a mark on the electrical connector [2], remove the mark.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-40-860-004

(1) If the compressor discharge overheat switch [1] for the left cooling pack was replaced, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-40-860-005

(2) If the compressor discharge overheat switch [1] for the right cooling pack was replaced, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-40-410-001

(3) If the compressor discharge overheat switch [1] on the left cooling pack was replaced, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-40-410-002

(4) If the compressor discharge overheat switch [1] on the right cooling pack was replaced, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMPRESSOR DISCHARGE OVERHEAT SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Compressor discharge overheat switch functional test.

TASK 21-51-40-710-801

2. Compressor Discharge Overheat Switch Functional Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the functional test of the compressor discharge overheat switch.

B. References

Reference	Title
21-51-40-000-801	Compressor Discharge Overheat Switch Removal (P/B 401)
21-51-40-400-801	Compressor Discharge Overheat Switch Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-3924	Tester - Thermo Switch, K-Type Thermocouple Switches Part #: H394R Supplier: 98869 Opt Part #: H294 Supplier: 98869 Opt Part #: H394 SERIES Supplier: 98869
SPL-3952	Kit - Heater Probes, TEMPCAL Part #: BH24944-13 Supplier: 98869 Part #: BH24944-14 Supplier: 98869 Part #: BH24944-16 Supplier: 98869

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Prepare for the Test

SUBTASK 21-51-40-020-005

(1) Remove the compressor discharge overheat switch that you will test (TASK 21-51-40-000-801).

F. Compressor Discharge Overheat Switch Functional Test (Method I)

SUBTASK 21-51-40-420-006

(1) Connect the electrical connector back to the compressor discharge overheat switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-40-860-007

(2) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-40-480-001

(3) Prepare to use the K-Type thermocouple switch tester, SPL-3924, and TEMPCAL heater probe kit, SPL-3952, for the test.

NOTE: Use the applicable heater probe to test the overheat switch.

SUBTASK 21-51-40-710-001

(4) Use the test equipment to increase the temperature of the compressor discharge overheat switch as follows:

(a) Set the test temperature on the tester to 370 ±5°F (188 ±3°C).

(b) Increase the test temperature on the tester at a rate of not more than 2°F (1°C) per minute to 390 ±10°F (199 ±6°C).

(c) Move the tester POWER switch to the ON position.

SUBTASK 21-51-40-210-001

(5) When the temperature reaches 390 ±10°F (199 ±6°C), make sure that the PACK light, on the P5 air conditioning panel, and MASTER CAUTION lights come on.

SUBTASK 21-51-40-860-008

(6) Push one of the MASTER CAUTION lights to reset.

SUBTASK 21-51-40-210-002

(7) Make sure that both MASTER CAUTION lights go off.

SUBTASK 21-51-40-710-002

(8) Move the tester POWER switch to the OFF position.

SUBTASK 21-51-40-710-003

(9) Let the compressor discharge overheat switch probe cool down at a rate of not more than 2°F (1°C) per minute.

NOTE: The compressor discharge overheat switch will reset when the probe temperature is 350°F (177°C).

SUBTASK 21-51-40-860-009

(10) Push the TRIP RESET button, on the P5 air conditioning panel.

SUBTASK 21-51-40-210-003

(11) Make sure that the PACK light on the air conditioning panel goes off.

SUBTASK 21-51-40-020-006

(12) Disconnect the electrical connector from the compressor discharge overheat switch.

SUBTASK 21-51-40-410-003

(13) Do this task: Compressor Discharge Overheat Switch Installation, TASK 21-51-40-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Compressor Discharge Overheat Switch Functional Test (Method II)

SUBTASK 21-51-40-420-007

(1) Install a new or serviceable compressor discharge overheat switch (TASK 21-51-40-400-801).

NOTE: The compressor discharge overheat switch that was removed, should be tested according to the vendor Component Maintenance Manual (CMM) before it is put back into service.

SUBTASK 21-51-40-020-007

(2) Remove the electrical connector from the compressor discharge overheat switch.

SUBTASK 21-51-40-480-002

(3) Connect a jumper wire between pin 1 and pin 2 of the electrical connector.

SUBTASK 21-51-40-210-004

(4) Make sure that these lights come on:

(a) PACK light on the P5-10 panel.

(b) MASTER CAUTION light on the pilots' glareshield.

SUBTASK 21-51-40-080-001

(5) Disconnect the jumper wire.

SUBTASK 21-51-40-860-010

(6) Push the TRIP RESET button on the P5-10 panel.

SUBTASK 21-51-40-860-011

(7) Push the MASTER CAUTION reset button on the P7 glareshield panel.

SUBTASK 21-51-40-210-005

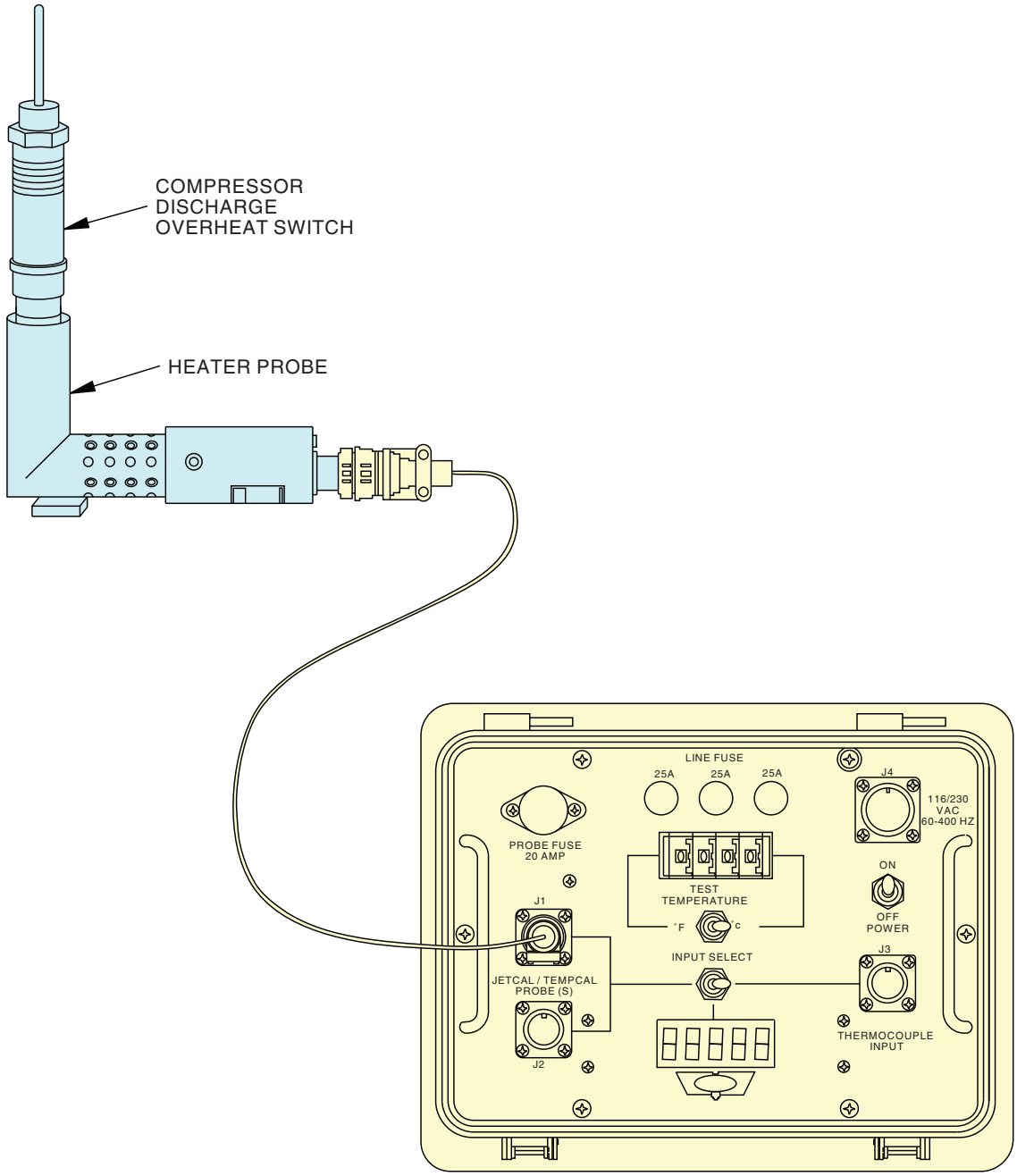
(8) Make sure that the MASTER CAUTION light goes off.

SUBTASK 21-51-40-420-008

(9) Connect the electrical connector to the compressor discharge overheat switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

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Compressor Discharge Overheat Switch Functional Test
Figure 501/21-51-40-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TURBINE INLET OVERHEAT SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the turbine inlet overheat switch

(2) An installation of the turbine inlet overheat switch.

B. There is a turbine inlet overheat switch installed on each of the two cooling packs.

TASK 21-51-41-000-801

2. Turbine Inlet Overheat Switch Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-41-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-41-860-002

(2) To replace the turbine inlet overheat switch [1] for the left cooling pack, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-41-860-003

(3) To replace the turbine inlet overheat switch [1] for the right cooling pack, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-41-010-001

(4) To get access to the turbine inlet overheat switch [1] for the left cooling pack, do this step:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-41-010-002

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To get access to the turbine inlet overheat switch [1] for the right cooling pack, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Overheat Switch Removal

SUBTASK 21-51-41-020-001

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the electrical connector from the turbine inlet overheat switch [1].

(a) If it is necessary to remove the turbine inlet overheat switch [1] and compressor discharge overheat switch, make a mark on the electrical connector.

NOTE: The electrical connector of the compressor discharge overheat switch is near the connector of the turbine inlet overheat switch.

NOTE: You can make a mark on the electrical connectors to identify the connectors in the installation task.

SUBTASK 21-51-41-020-002

(2) Remove and discard MS20995C32 lockwire, G01048, from the turbine inlet overheat switch [1].

SUBTASK 21-51-41-020-003

(3) Remove the turbine inlet overheat switch [1] from the duct.

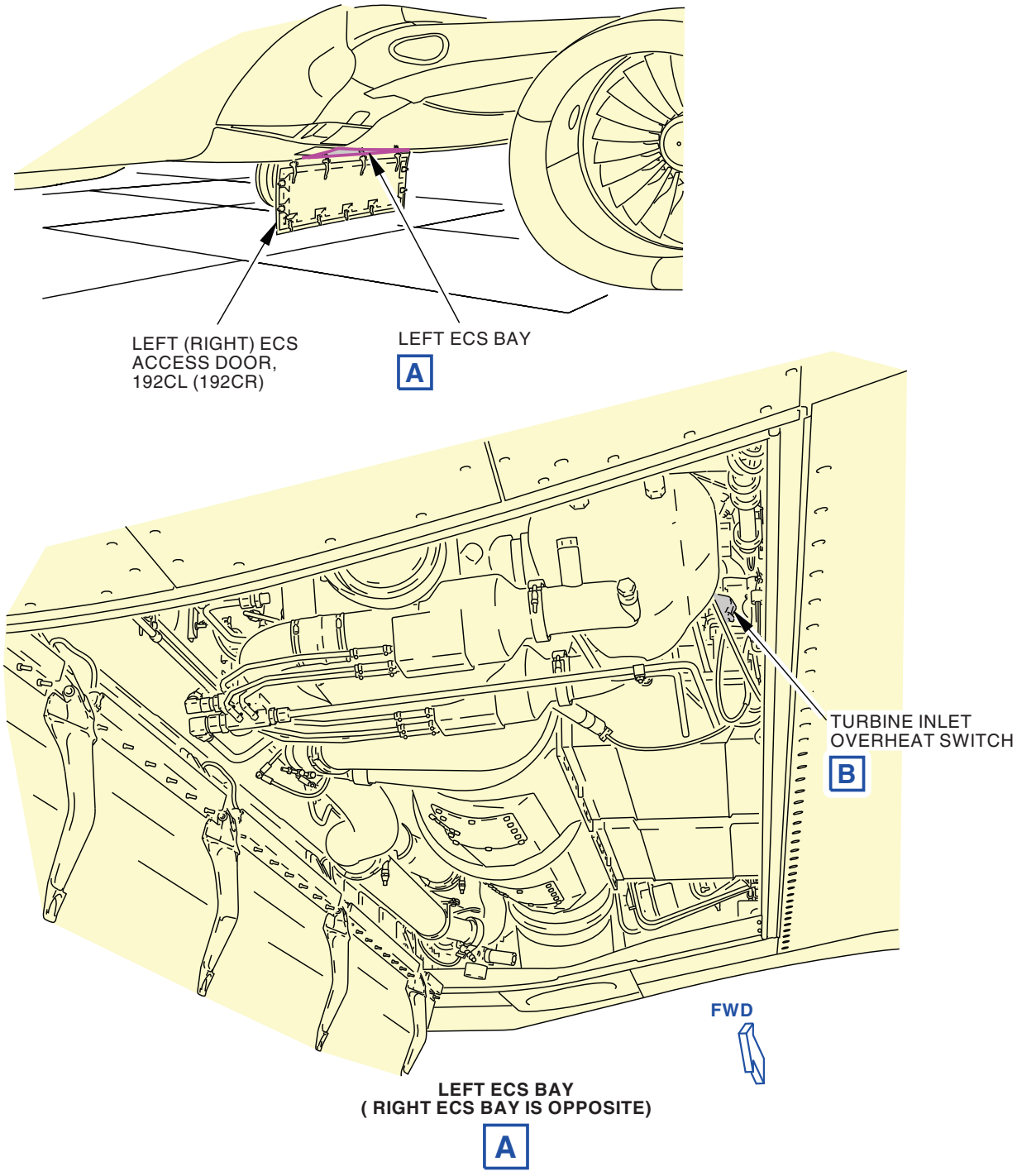
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-41-020-004

(4) Remove and discard the O-ring [2] from the turbine inlet overheat switch [1].

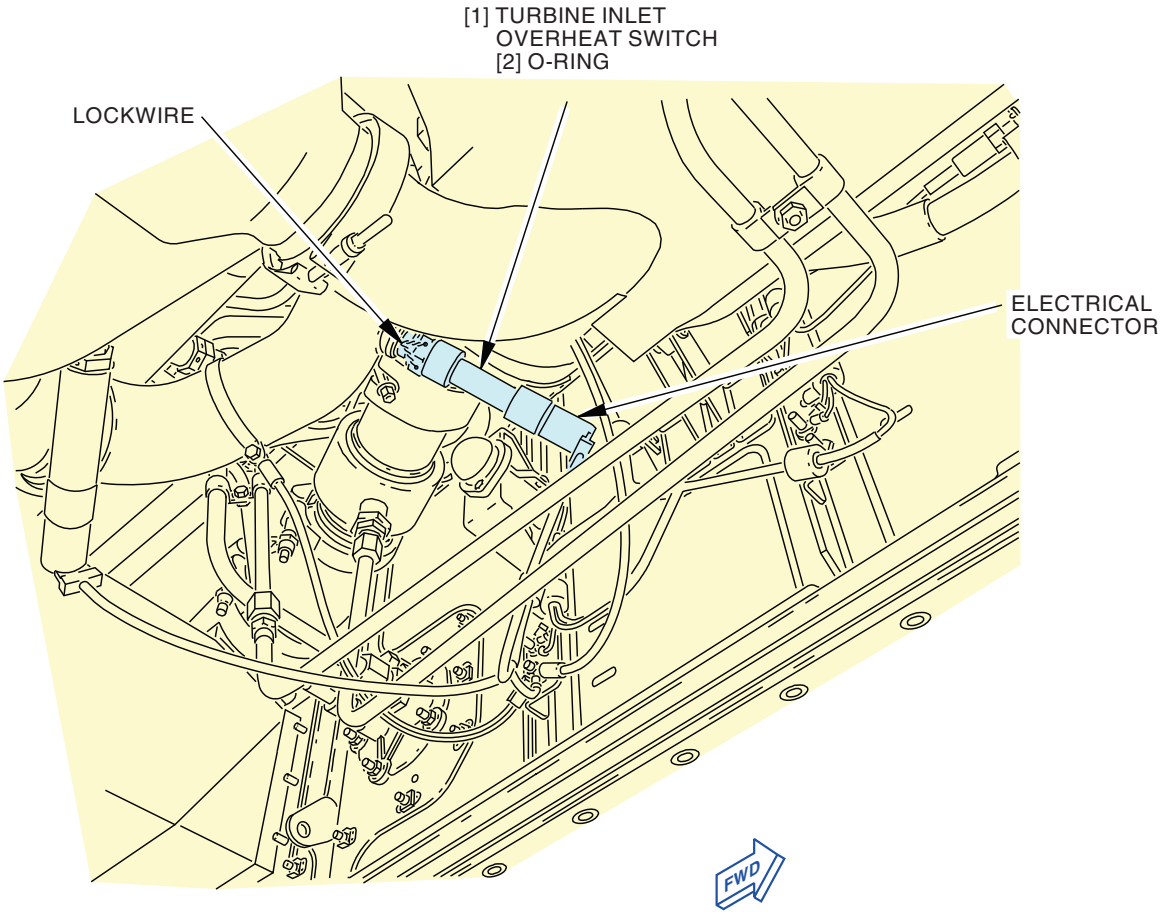
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Turbine Inlet Overheat Switch Installation
Figure 401/21-51-41-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Turbine Inlet Overheat Switch Installation
Figure 401/21-51-41-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-41-400-801

3. Turbine Inlet Overheat Switch Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00649	Lubricant - O-Ring - Krytox GPL206	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Turbine inlet overheat switch	21-51-41-09-080	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Overheat Switch Installation

SUBTASK 21-51-41-160-001

(1) Make sure that the mating surfaces of the boss and overheat switch nut are clean.

SUBTASK 21-51-41-640-001

(2) Lightly lubricate a new O-ring [2] with Krytox GPL206 lubricant, D00649.

SUBTASK 21-51-41-420-001

(3) Install the O-ring [2] on the turbine inlet overheat switch [1].

SUBTASK 21-51-41-640-002

(4) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the turbine inlet overheat switch [1].

SUBTASK 21-51-41-420-002

(5) Install the turbine inlet overheat switch [1] on the duct.

SUBTASK 21-51-41-420-003

(6) Tighten the turbine inlet overheat switch [1] to 225 ±10 in-lb (25.4 ±1.1 N·m).

SUBTASK 21-51-41-420-004

(7) Install new MS20995C32 lockwire, G01048, on the turbine inlet overheat switch [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-41-420-005

CAUTION

MAKE SURE THAT YOU CONNECT THE CORRECT ELECTRICAL CONNECTOR TO THE OVERHEAT SWITCH FOR THE TURBINE INLET. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(8) Connect the electrical connector to the turbine inlet overheat switch [1].

(a) If there is a mark on the electrical connector, remove the mark.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-41-860-004

(1) If the turbine inlet overheat switch [1] for the left cooling pack was replaced, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-41-860-005

(2) If the turbine inlet overheat switch [1] for the right cooling pack was replaced, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-41-410-001

(3) If the turbine inlet overheat switch [1] for the left cooling pack was replaced, do this step:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-41-410-002

(4) If the turbine inlet overheat switch [1] for the right cooling pack was replaced, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TURBINE INLET THERMAL SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Turbine inlet thermal switch functional test.

TASK 21-51-41-710-801

2. Turbine Inlet Thermal Switch - Functional Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
21-51-41-000-801	Turbine Inlet Overheat Switch Removal (P/B 401)
21-51-41-400-801	Turbine Inlet Overheat Switch Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1552	Kit - Heater Probes, TEMPCAL Part #: BH24944-7 Supplier: 98869
SPL-3924	Tester - Thermo Switch, K-Type Thermocouple Switches Part #: H394R Supplier: 98869 Opt Part #: H294 Supplier: 98869 Opt Part #: H394 SERIES Supplier: 98869

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

D. Turbine Inlet Thermal Switch - Functional Test (Method I)

SUBTASK 21-51-41-020-005

(1) Do this task: Turbine Inlet Overheat Switch Removal, TASK 21-51-41-000-801.

(a) Re-connect the electrical connector to the turbine inlet thermal switch.

SUBTASK 21-51-41-480-001

(2) Prepare the K-Type thermocouple switch tester, SPL-3924, and TEMPCAL heater probe kit, SPL-1552, to test the turbine inlet thermal switch.

NOTE: Use the correct heater probe to test overheat switch part number 572757-1.

SUBTASK 21-51-41-710-001

(3) Use the test equipment to increase the temperature of the turbine inlet thermal switch as follows:

(a) Set the test temperature on the tester to 190 ±5°F (88 ±3°C).

(b) Increase the test temperature on the tester at a rate of not more than 2°F (1°C) per minute to 210 ±10°F (99 ±6°C).

(c) Move the tester POWER switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-41-710-002

(4) Make sure that these lights come on when the temperature of the turbine inlet thermal switch probe get to 210 ±10°F (99 ±6°C).

(a) PACK light on the P5-10 panel.

(b) MASTER CAUTION light on the pilots' glareshield.

SUBTASK 21-51-41-710-003

(5) Remove the heat from the turbine inlet thermal switch probe.

SUBTASK 21-51-41-710-004

(6) Let the turbine inlet thermal switch probe become cool down at a rate of not more than 2°F (1°C) per minute.

NOTE: The reset temperature for the turbine inlet thermal switch is 170°F (77°C).

SUBTASK 21-51-41-710-005

(7) Push the TRIP RESET light on the P5-10 panel.

SUBTASK 21-51-41-710-006

(8) Push the MASTER CAUTION reset button on the P7 glareshield panel.

SUBTASK 21-51-41-710-007

(9) Make sure that the MASTER CAUTION light goes off.

SUBTASK 21-51-41-420-006

(10) Do this task: Turbine Inlet Overheat Switch Installation, TASK 21-51-41-400-801.

E. Turbine Inlet Thermal Switch - Functional Test (Method II)

SUBTASK 21-51-41-020-006

(1) Do this task: Turbine Inlet Overheat Switch Removal, TASK 21-51-41-000-801.

SUBTASK 21-51-41-420-007

(2) Install a new or serviceable turbine inlet thermal switch (TASK 21-51-41-400-801).

NOTE: The turbine inlet thermal switch that was removed should be tested according to the vendor CMM before it is put back into service.

SUBTASK 21-51-41-020-007

(3) Remove the electrical connector from the turbine inlet thermal switch.

SUBTASK 21-51-41-480-002

(4) Connect a jumper wire between pin 1 and pin 2 of the electrical connector.

SUBTASK 21-51-41-210-001

(5) Make sure that these lights come on:

(a) PACK light on the P5-10 panel.

(b) MASTER CAUTION light on the pilots' glareshield.

SUBTASK 21-51-41-080-001

(6) Disconnect the jumper wire.

SUBTASK 21-51-41-860-007

(7) Push the TRIP RESET button on the P5-10 panel.

SUBTASK 21-51-41-860-008

(8) Push the MASTER CAUTION reset button on the P7 glareshield panel.

SUBTASK 21-51-41-210-002

(9) Make sure that the MASTER CAUTION light goes off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-41-420-008

(10) Connect the electrical connector to the turbine inlet thermal switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK DISCHARGE OVERHEAT SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the pack discharge overheat switch

(2) An installation of the pack discharge overheat switch.

B. There is a pack discharge overheat switch for each of the two packs. The pack discharge overheat switches are found in the mix bay.

TASK 21-51-42-000-801

2. Pack Discharge Overheat Switch Removal

(Figure 401)

A. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

D. Prepare for the Removal

SUBTASK 21-51-42-860-001

(1) Do this step for the left pack discharge overheat switch:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-42-860-002

(2) Do this step for the right pack discharge overheat switch:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-42-860-003

(3) Make sure that the L PACK and R PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

SUBTASK 21-51-42-010-001

(4) Open the forward cargo door.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-42-010-002

(5) Remove the aft center bulkhead liner in the forward cargo compartment, do this task: Forward Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-11-02-000-801.

E. Pack Discharge Overheat Switch Removal

SUBTASK 21-51-42-020-001

(1) Disconnect the electrical connector [1] from the overheat switch [2].

SUBTASK 21-51-42-020-002

(2) Remove MS20995C32 lockwire, G01048, from the overheat switch [2].

SUBTASK 21-51-42-020-003

CAUTION

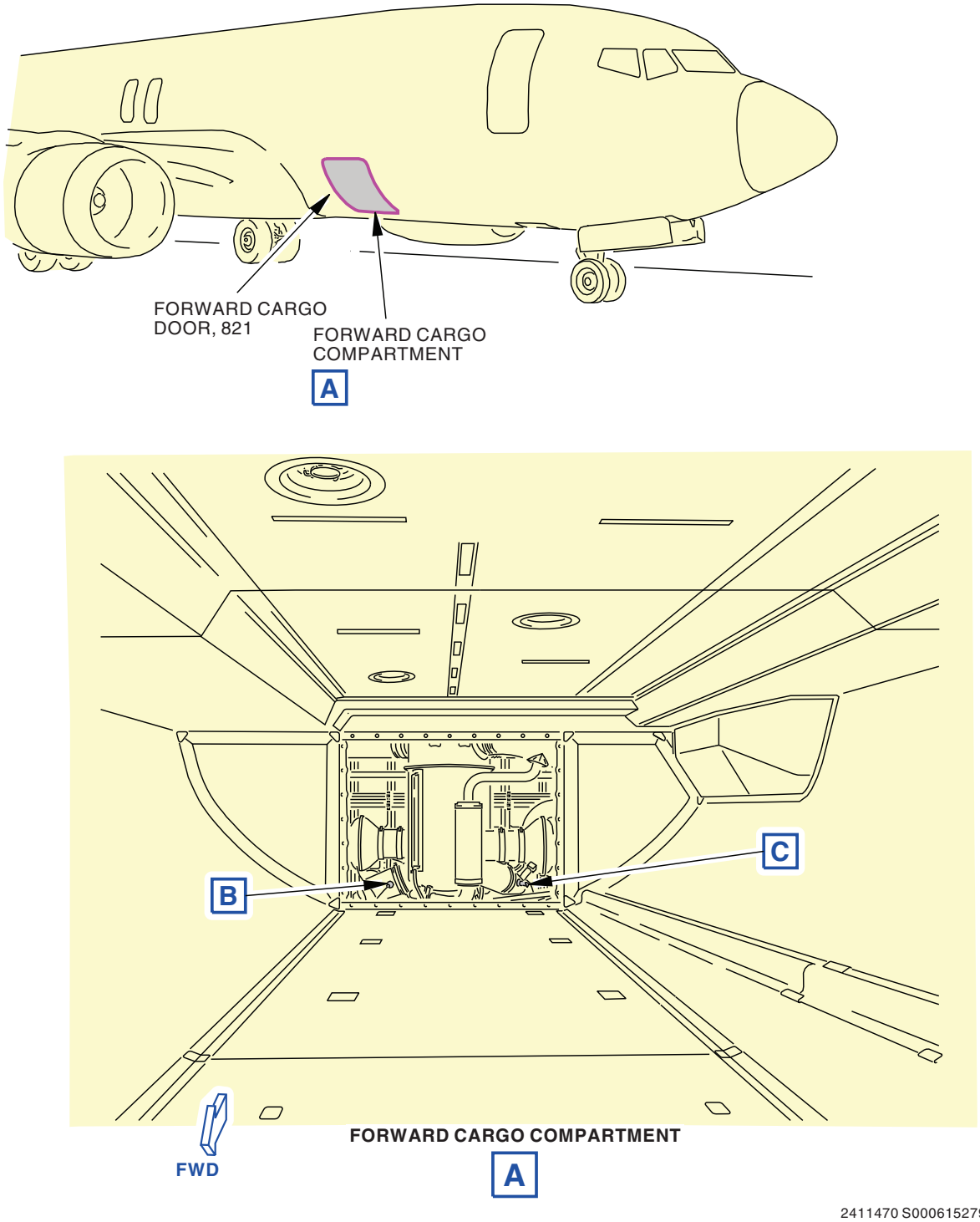
USE TWO WRENCHES TO REMOVE THE OVERHEAT SWITCH FROM THE BOSS. IF YOU USE ONLY ONE WRENCH TO REMOVE THE OVERHEAT SWITCH, IT CAN CAUSE DAMAGE TO THE DUCT.

(3) Use two wrenches to remove the pack discharge overheat switch [2].

NOTE: Use one wrench to hold the boss. Use the other wrench to turn the overheat switch.

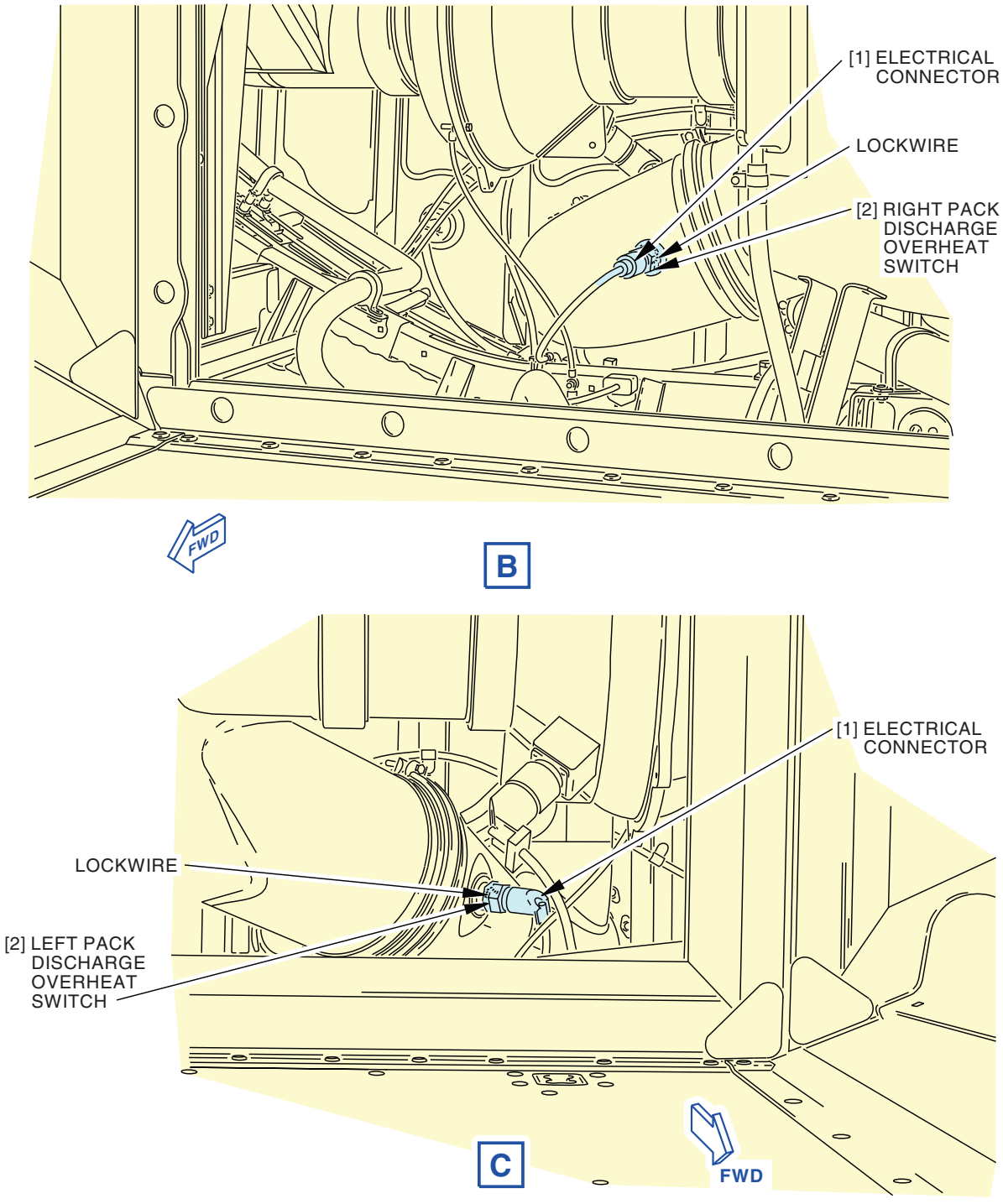
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pack Discharge Overheat Switch Installation
Figure 401/21-51-42-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pack Discharge Overheat Switch Installation
Figure 401/21-51-42-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-42-400-801

3. Pack Discharge Overheat Switch Installation

(Figure 401)

A. References

Reference	Title
21-51-42-710-801	Pack Discharge Overheat Switch - Functional Test (P/B 501)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Switch	21-21-01-02-305	PTF ALL
		21-21-01-02-380	PTF ALL

D. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

E. Pack Discharge Overheat Switch Installation Test

SUBTASK 21-51-42-710-006

(1) Do this task: Pack Discharge Overheat Switch - Functional Test, TASK 21-51-42-710-801.

F. Pack Discharge Overheat Switch Installation

SUBTASK 21-51-42-160-001

(1) Make sure that the mating surfaces of the boss and overheat switch nut are clean.

SUBTASK 21-51-42-640-001

(2) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the overheat switch [2].

SUBTASK 21-51-42-420-001

CAUTION

USE TWO WRENCHES TO INSTALL THE OVERHEAT SWITCH IN THE BOSS. IF YOU USE ONLY ONE WRENCH TO INSTALL THE OVERHEAT SWITCH, IT CAN CAUSE DAMAGE TO THE DUCT.

(3) Use two wrenches to install the pack discharge overheat switch [2].

NOTE: Use one wrench to hold the boss. Use the other wrench to turn the overheat switch.

SUBTASK 21-51-42-420-002

(4) Install new MS20995C32 lockwire, G01048, to the overheat switch [2] and boss.

SUBTASK 21-51-42-420-003

(5) Connect the electrical connector [1] to the overheat switch [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-42-860-004

(1) If replaced the left pack discharge overheat switch, do this step:

(a) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-42-860-005

(2) If replaced the right pack discharge overheat switch, do this step:

(a) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT

SUBTASK 21-51-42-010-003

(3) Install the aft center bulkhead liner in the forward cargo compartment, do this task: Forward Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-11-02-400-801.

SUBTASK 21-51-42-010-004

(4) Close the forward cargo door.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK DISCHARGE OVERHEAT SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) A functional test of the pack discharge overheat switch.

TASK 21-51-42-710-801

2. Pack Discharge Overheat Switch - Functional Test

A. References

Reference	Title
21-51-42-000-801	Pack Discharge Overheat Switch Removal (P/B 401)
21-51-42-400-801	Pack Discharge Overheat Switch Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-3924	Tester - Thermo Switch, K-Type Thermocouple Switches Part #: H394R Supplier: 98869 Opt Part #: H294 Supplier: 98869 Opt Part #: H394 SERIES Supplier: 98869
SPL-3952	Kit - Heater Probes, TEMPCAL Part #: BH24944-13 Supplier: 98869 Part #: BH24944-14 Supplier: 98869 Part #: BH24944-16 Supplier: 98869

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

D. Pack Discharge Overheat Switch - Functional Test (Method I)

SUBTASK 21-51-42-020-004

(1) If installed, remove the pack overheat discharge switch (TASK 21-51-42-000-801).

(a) Re-connect the electrical connector to the pack discharge overheat switch.

SUBTASK 21-51-42-480-001

(2) Prepare the K-Type thermocouple switch tester, SPL-3924, and TEMPCAL heater probe kit, SPL-3952, to test the pack discharge overheat switch.

NOTE: Use the correct heater probe to test the pack discharge overheat switch.

SUBTASK 21-51-42-710-001

(3) Use the test equipment to increase the temperature of the turbine inlet thermal switch as follows:

(a) Set the test temperature on the tester to 230°F (110°C).

(b) Wait for a minimum of five minutes.

(c) Increase the test temperature on the tester at a rate of not more than 5°F (2.8°C) per minute to 250 ±10°F (121 ±6°C).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-42-710-002

(4) Make sure that the PACK light, on the P5-10 panel, comes on when the temperature of the pack discharge overheat switch probe gets to 250 ±10°F (121 ±6°C).

SUBTASK 21-51-42-710-003

(5) Remove the heat from the turbine inlet thermal switch probe.

SUBTASK 21-51-42-710-004

(6) Let the turbine inlet thermal switch probe cool down at a rate of not more than 5°F (2.8°C) per minute.

NOTE: The reset temperature for the pack discharge overheat switch is 150°F (66°C).

SUBTASK 21-51-42-710-005

(7) Push the TRIP RESET light on the P5-10 panel.

SUBTASK 21-51-42-420-004

(8) Do this task: Pack Discharge Overheat Switch Installation, TASK 21-51-42-400-801.

E. Pack Discharge Overheat Switch - Functional Test (Method II)

SUBTASK 21-51-42-020-005

(1) If installed, remove the pack overheat discharge switch (TASK 21-51-42-000-801).

SUBTASK 21-51-42-420-005

(2) Install a new or serviceable pack overheat discharge switch (TASK 21-51-42-400-801).

NOTE: The pack discharge overheat switch that was removed should be tested according to the vendor CMM before it is put back into service.

SUBTASK 21-51-42-000-001

(3) Remove the electrical connector from the pack discharge overheat switch.

SUBTASK 21-51-42-480-002

(4) Connect a jumper wire between pin 1 and pin 2 of the electrical connector.

SUBTASK 21-51-42-210-001

(5) Make sure that the PACK light, on the P5-10 panel, comes on.

SUBTASK 21-51-42-080-001

(6) Disconnect the jumper wire.

SUBTASK 21-51-42-860-006

(7) Push the TRIP RESET button on the P5-10 panel.

SUBTASK 21-51-42-420-006

(8) Connect the electrical connector to the pack discharge overheat switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RAM AIR CONTROL TEMPERATURE SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ram air control temperature sensor

(2) An installation of the ram air control temperature sensor.

B. There is a ram air control temperature sensor installed on each of the two cooling packs.

TASK 21-51-50-000-801

2. Ram Air Control Temperature Sensor Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

D. Prepare for the Removal

SUBTASK 21-51-50-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-50-010-001

(2) For the ram air control temperature sensor [1] on the left cooling pack, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) To get access to the ram air control temperature sensor [1] on the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-50-010-002

(3) For the ram air control temperature sensor [1] on the right cooling pack, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(c) To get access to the ram air control temperature sensor [1] on the right cooling pack, do these steps:

(TASK 06-41-00-800-801)

1) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

E. Temperature Sensor Removal

SUBTASK 21-51-50-020-001

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the electrical connector [2] from the ram air control temperature sensor [1].

SUBTASK 21-51-50-020-002

(2) Remove and discard the safety cable from the ram air control temperature sensor [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-50-020-003

CAUTION

USE TWO WRENCHES TO REMOVE THE TEMPERATURE SENSOR FROM THE BOSS. IF YOU USE ONLY ONE WRENCH TO REMOVE THE TEMPERATURE SENSOR, IT CAN CAUSE DAMAGE TO THE DUCT.

(3) Use two wrenches to remove the ram air control temperature sensor [1].

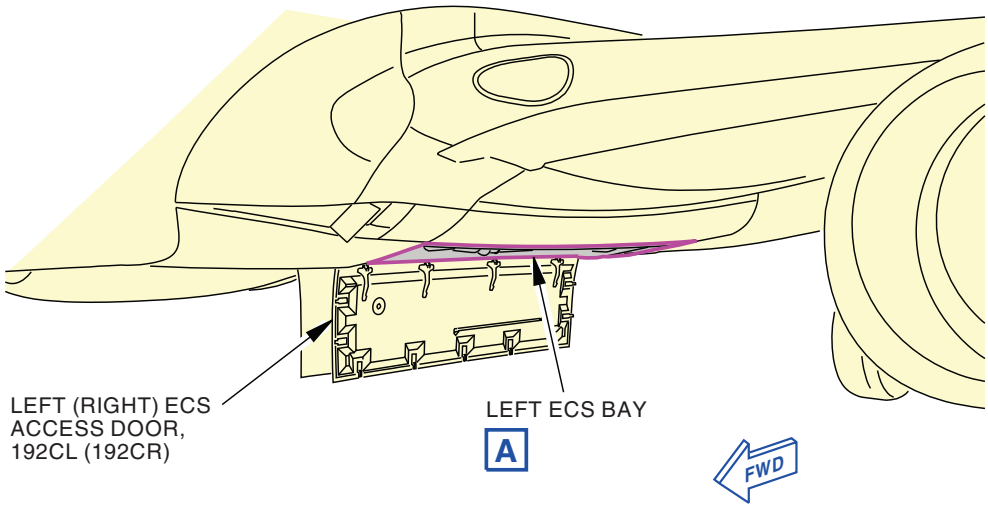
(a) Use one wrench to hold the boss and the other wrench to turn the ram air control temperature sensor [1].

SUBTASK 21-51-50-020-004

(4) Remove and discard the O-ring [3] from the ram air control temperature sensor [1].

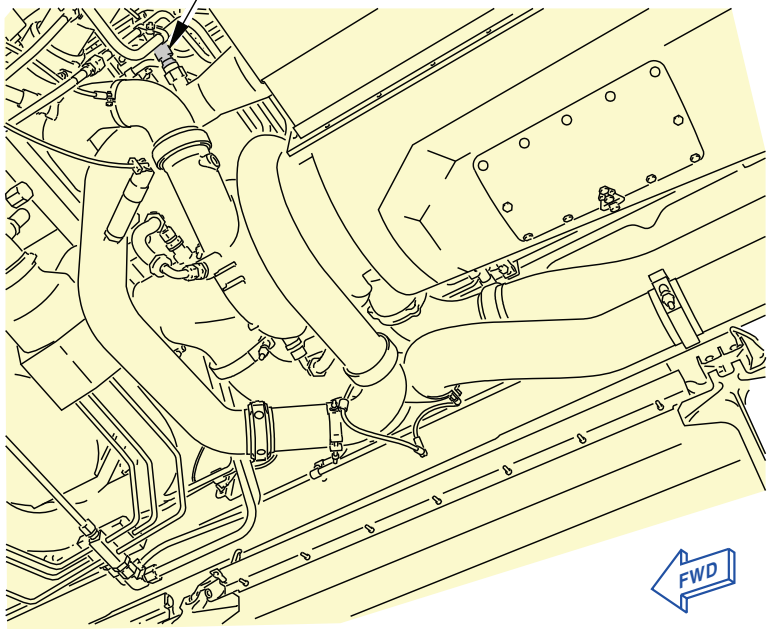
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



[1] RAM AIR CONTROL
TEMPERATURE SENSOR

B



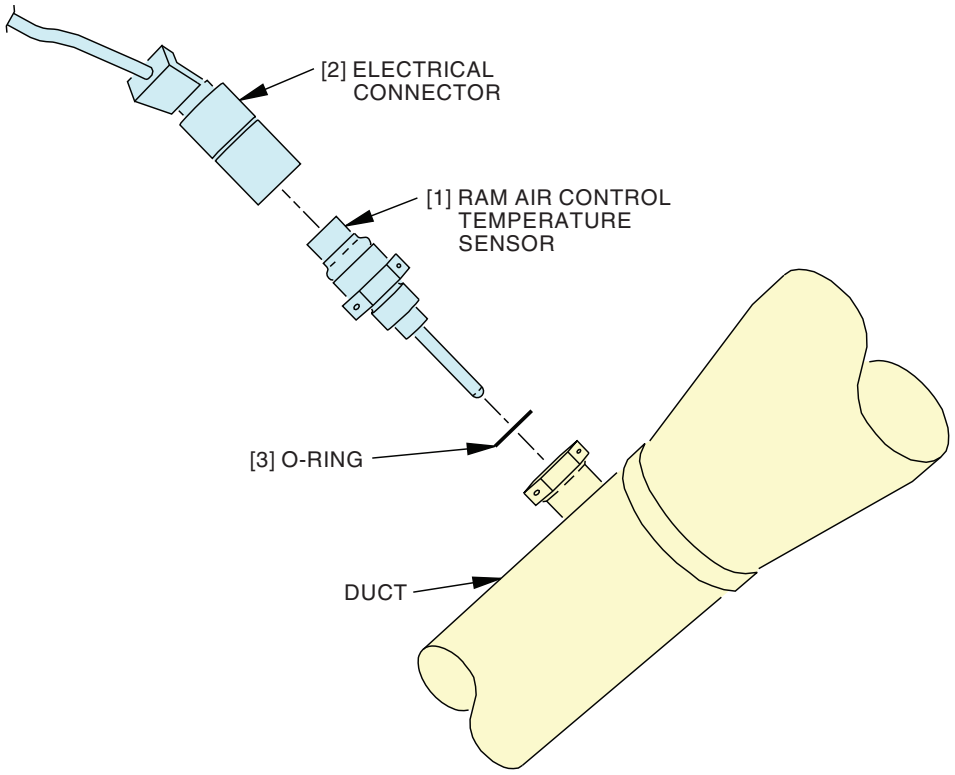
LEFT ECS BAY
(RIGHT ECS BAY IS OPPOSITE)

A

2411472 S00061527981_V1

Ram Air Control Temperature Sensor Installation
Figure 401/21-51-50-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RAM AIR CONTROL TEMPERATURE SENSOR

B

2411473 S00061527982_V1

Ram Air Control Temperature Sensor Installation
Figure 401/21-51-50-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-50-400-801

3. Ram Air Control Temperature Sensor Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
21-00-00-730-802	Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00504	Grease - Petrolatum	VV-P-236
G50375	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule)	BACC13AT3K, AMS 5689

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Ram air control temperature sensor	21-51-52-01-015	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Temperature Sensor Installation

SUBTASK 21-51-50-600-001

(1) Apply a light coat of grease, D00504, to a new O-ring [3].

SUBTASK 21-51-50-420-001

(2) Install the lubricated O-ring [3] on the ram air control temperature sensor [1].

SUBTASK 21-51-50-160-001

(3) Make sure that the mating surfaces of the boss and temperature sensor nut are clean.

SUBTASK 21-51-50-640-001

(4) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the ram air control temperature sensor [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-50-420-002

CAUTION USE TWO WRENCHES TO INSTALL THE TEMPERATURE SENSOR FROM THE BOSS. IF YOU USE ONLY ONE WRENCH TO INSTALL THE TEMPERATURE SENSOR, IT CAN CAUSE DAMAGE TO THE DUCT.

(5) Use two wrenches to install the ram air control temperature sensor [1].

(a) Use one wrench to hold the boss and the other wrench to turn the ram air control temperature sensor [1].

(b) Tighten the ram air control temperature sensor [1] to 220 in-lb (24.9 N·m) to 230 in-lb (26.0 N·m).

SUBTASK 21-51-50-420-003

(6) Install a new safety cable kit, G50375, on the ram air control temperature sensor [1] (TASK 20-10-44-400-801).

SUBTASK 21-51-50-420-004

(7) Connect the electrical connector [2] to the ram air control temperature sensor [1].

SUBTASK 21-51-50-860-002

(8) For the left cooling pack, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-50-860-003

(9) For the right cooling pack, do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-50-740-001

(10) Do this ground test: Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test, TASK 21-00-00-730-802.

(a) When the test is completed, make sure that PASS shows adjacent to TEST CONDITION.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If FAILED shows, do the fault isolation procedure for the maintenance message.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-50-840-001

(1) If the ram air control temperature sensor [1] for the left cooling pack was replaced, close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-50-840-002

(2) If the ram air control temperature sensor [1] for the right cooling pack was replaced, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MIX MANIFOLD TEMPERATURE SENSOR - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the mix manifold temperature sensor

(2) A cleaning of the mix manifold temperature sensor

(3) An installation of the mix manifold temperature sensor.

B. If it is necessary, the mix manifold temperature sensors should be cleaned before the check, to determine if the resistance of the sensors has the drifted out of the specification.

C. There is a mix manifold temperature sensor for each of the two cooling packs. The mix manifold temperature sensors are installed behind the liner at the aft end of the forward cargo compartment.

TASK 21-51-51-000-801

2. Mix Manifold Temperature Sensor Removal

(Figure 201)

A. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 21-51-51-860-001

(1) To remove the mix manifold temperature sensor [1] for the left cooling pack, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-51-860-002

(2) To remove the mix manifold temperature sensor [1] for the right cooling pack, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-51-860-003

(3) Put these switches, on the P5-10 air conditioning panel, to the OFF position:

(a) L PACK

(b) R PACK

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(c) L RECIRC FAN

(d) R RECIRC FAN.

SUBTASK 21-51-51-010-001

(4) Open the forward cargo door.

SUBTASK 21-51-51-010-002

(5) Remove the aft center bulkhead liner in the forward cargo compartment, do this task: Forward Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-11-02-000-801.

D. Mix Manifold Temperature Sensor Removal

SUBTASK 21-51-51-020-001

(1) Disconnect the elecrical connector [2] from the mix manifold temperature sensor [1].

SUBTASK 21-51-51-020-002

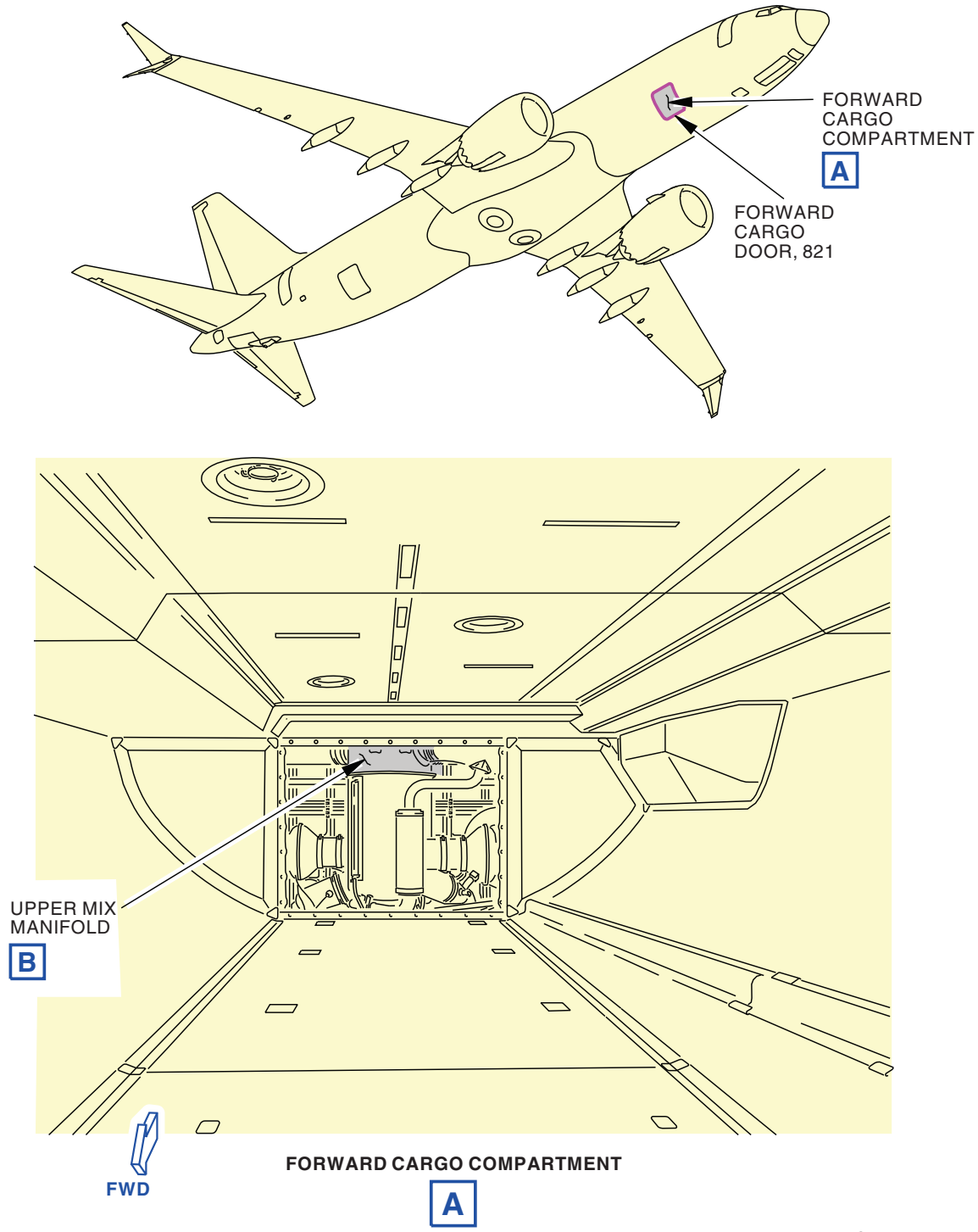
(2) Remove the screws [3] that hold the mix manifold temperature sensor [1] to the upper mix manifold.

SUBTASK 21-51-51-020-003

(3) Remove the mix manifold temperature sensors [1] from the upper mix manifold.

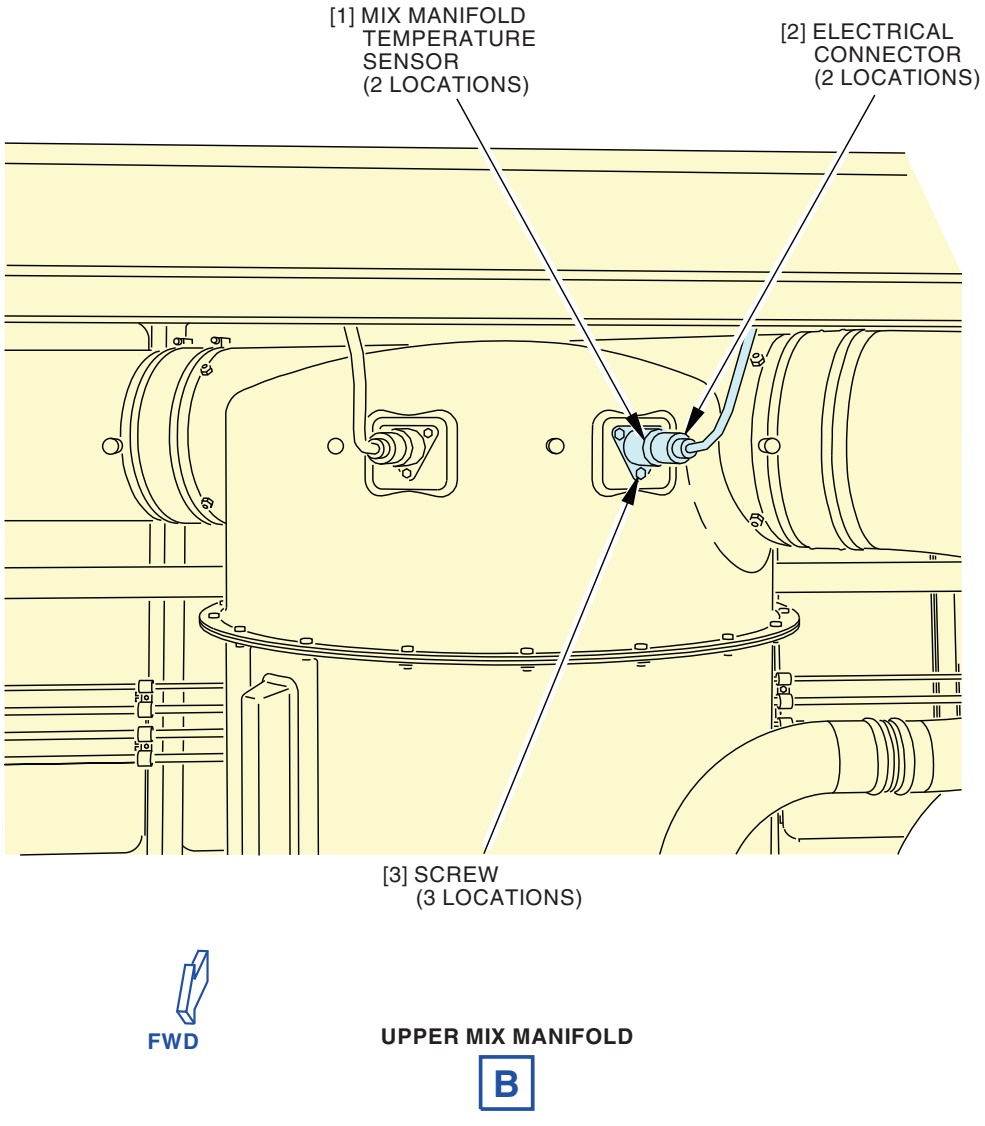
——— END OF TASK ———

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Mix Manifold Temperature Sensor Installation
Figure 201/21-51-51-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Mix Manifold Temperature Sensor Installation
Figure 201/21-51-51-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-51-110-801

3. Mix Manifold Temperature Sensor Cleaning

(Figure 201)

A. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G00215	Brush - Soft Bristle Fiber	
G50637	Wiper - MicroCare Fiber Wipes	

B. Prepare for the Cleaning

SUBTASK 21-51-51-020-004

(1) Remove the mix manifold temperature sensor [1] (TASK 21-51-51-000-801).

SUBTASK 21-51-51-960-001

(2) Replace the mix manifold temperature sensor [1] if you find:

(a) Bent, broken or missing pins

(b) Damaged, stripped or missing threads

(c) Broken or dirty insulation

(d) Broken or cracked thermistor beads.

C. Mix Manifold Temperature Sensor Cleaning

SUBTASK 21-51-51-160-001

(1) Remove all unwanted material from between the connector pins and around the thermistor beads:

(a) Use soft-bristle brush, G00215, moistened with isopropyl alcohol, B00130.

(b) Use MicroCare fiber wipes, G50637, to remove all traces of the isopropyl alcohol, B00130.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-51-420-001

(1) Install the mix manifold temperature sensor [1] (TASK 21-51-51-400-801).

——— END OF TASK ———

TASK 21-51-51-400-801

4. Mix Manifold Temperature Sensor Installation

(Figure 201)

A. References

Reference	Title
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Mix manifold temperature sensor	21-21-05-01-015	PTF ALL

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C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

D. Mix Manifold Temperature Sensor Installation

SUBTASK 21-51-51-080-001

(1) Remove the protective cap from the mix manifold temperature sensor [1].

SUBTASK 21-51-51-420-002

(2) Put the mix manifold temperature sensor [1] in its position in the upper mix manifold.

SUBTASK 21-51-51-420-003

(3) Install the screws [3] that attach the mix manifold temperature sensor [1] to the upper mix manifold.

SUBTASK 21-51-51-420-004

(4) Connect the elecrical connector [2] to the mix manifold temperature sensor [1].

SUBTASK 21-51-51-860-004

(5) If the mix manifold temperature sensor [1] for the left cooling pack was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-51-860-005

(6) If the mix manifold temperature sensor [1] for the right cooling pack was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-51-51-740-001

(7) Do a BITE test on the left and right Integrated Air Systems Controllers (M2939 and M2957).

NOTE: This will clear any fault introduced during maintenance when power is interrupted.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-51-410-001

(1) Install the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 21-51-51-410-002

(2) Close the forward cargo door.

——— END OF TASK ———

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MIX MANIFOLD TEMPERATURE SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) A test of the mix manifold temperature sensors

B. There are two mix manifold temperature sensors installed.

TASK 21-51-51-710-801

2. Mix Manifold Temperature Sensor Test

(Figure 501)

A. References

Reference	Title
21-51-51 P/B 201	MIX MANIFOLD TEMPERATURE SENSOR - MAINTENANCE PRACTICES
21-61-22-000-801	Integrated Air Systems Controller (IASC) Removal (P/B 401)
21-61-22-400-801	Integrated Air Systems Controller (IASC) Installation (P/B 401)
50-12-01-000-801	Aft Cargo Compartment Forward Bulkhead Liner Removal (P/B 401)
50-12-01-400-801	Aft Cargo Compartment Forward Bulkhead Liner Installation (P/B 401)
WDM 21-51-12	Wiring Diagram Manual
WDM 21-51-22	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-3955	Thermometer - Digital (Type K) Part #: 51 2B Supplier: 89536 Opt Part #: 51 SERIES II Supplier: 89536

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C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

D. Prepare For The Resistance Test

SUBTASK 21-51-51-860-006

(1) Do this task: Aft Cargo Compartment Forward Bulkhead Liner Removal, TASK 50-12-01-000-801.

SUBTASK 21-51-51-860-007

(2) To do a test of the left mix manifold sensor, remove the left Integrated Air Systems Controller (IASC) (TASK 21-61-22-000-801).

SUBTASK 21-51-51-860-008

(3) To do a test of the right mix manifold sensor, remove the right Integrated Air Systems Controller (IASC) (TASK 21-61-22-000-801).

SUBTASK 21-51-51-710-001

(4) Make sure the pack has been off for 20 minutes or until it has reached ambient temperature.

E. Resistance Check Procedure

SUBTASK 21-51-51-700-001

(1) Use a digital thermometer, COM-3955, or equivalent, to measure the temperature adjacent to the applicable mix manifold temperature sensor.

(a) Write down the temperature.

SUBTASK 21-51-51-760-001

(2) Use the digital/analog multimeter, COM-1793, (preferred) to measure resistance between the following pin connections.

NOTE: An equivalent meter can be used as long as the digital multimeter minimizes self-heating of the sensor element. The instrument must limit the current through the sensor to less than 100 microamperes when set to the 10K ohms scale. If you use resistance measuring instruments without this current-limiting feature you will get an error in resistance value due to self-heating of the sensor element.

SUBTASK 21-51-51-700-002

(3) For the right mix manifold temperature sensor, measure the resistance from pin, F9, to pin, F10, on connector, D15840B, for the right Integrated Air Systems Controller (IASC) (WDM 21-51-22).

SUBTASK 21-51-51-700-003

(4) For the left mix manifold temperature sensor, measure the resistance from pin, F9, to pin, F10, on connector, D15838B, for the left Integrated Air Systems Controller (IASC) (WDM 21-51-12).

SUBTASK 21-51-51-700-004

(5) Write down the resistance in Ohms that you measure.

(a) For the mix manifold temperature sensors, find the temperature that is nearest to the temperature that you wrote down (Figure 501).

(b) Compare the permitted range of resistance for that temperature to the resistance you wrote down.

(c) If the resistance is out of range, replace the sensor (PAGEBLOCK 21-51-51/201).

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F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-51-420-005

(1) Install the applicable Integrated Air Systems Controller (IASC) (TASK 21-61-22-400-801).

SUBTASK 21-51-51-410-003

(2) Do this task: Aft Cargo Compartment Forward Bulkhead Liner Installation, TASK 50-12-01-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

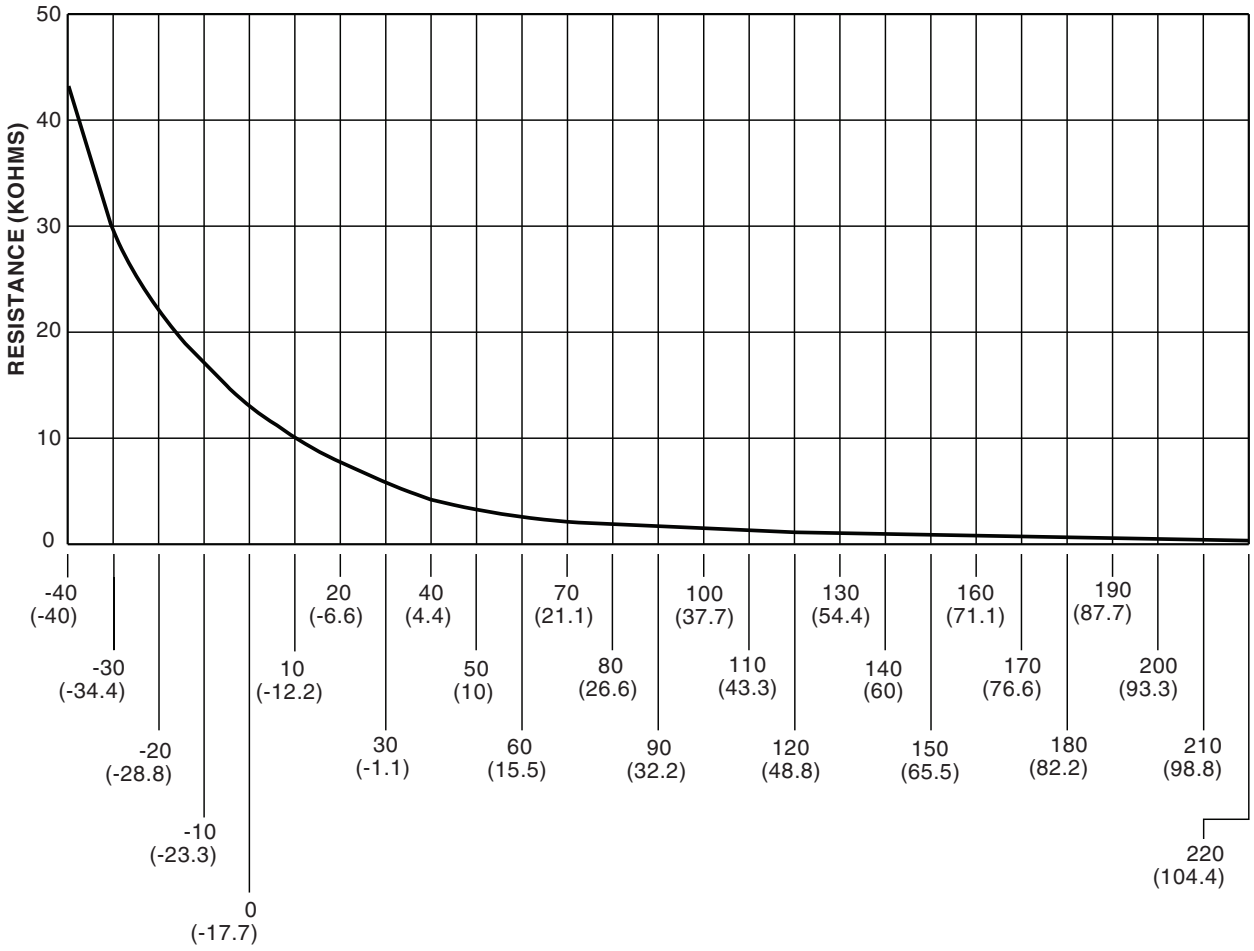
TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)	TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)
-40 (-40)	43,030-45,210	95 (35)	1,340-1,400
-35 (-37.2)	36,690-38,550	100 (37.7)	1,210-1,270
-30 (-34.4)	31,360-32,960	105 (40.5)	1,100-1,150
-25 (-31.6)	26,930-28,230	110 (43.3)	997-1,140
-20 (-28.8)	23,140-24,260	115 (46.1)	905-944
-15 (-26.1)	19,940-20,900	120 (48.8)	822-859
-10 (-23.3)	17,220-18,050	125 (51.6)	748-782
-5 (-20.5)	14,890-15,580	130 (54.4)	682-713
0 (-17.7)	12,950-13,550	135 (57.2)	623-652
5 (-15)	11,310-11,780	140 (60)	568-598
10 (-12.2)	9,870-10,290	145 (62.7)	520-547
15 (-9.4)	8,640-9,000	150 (65.5)	477-502
20 (-6.6)	7,520-7,900	155 (68.3)	438-461
25 (-3.8)	6,660-6,940	160 (71.1)	413-434
30 (-1.1)	5,870-6,120	165 (73.8)	371-390
35 (1.6)	5,180-5,400	170 (76.6)	342-360
40 (4.4)	4,580-4,770	175 (79.4)	315-332
45 (7.2)	4,060-4,230	180 (82.2)	291-307
50 (10)	3,600-3,750	185 (85)	269-284
55 (12.7)	3,200-3,340	190 (87.7)	249-263
60 (15.5)	2,850-2,980	195 (90.5)	231-245
65 (18.3)	2,550-2,660	200 (93.3)	214-226
70 (21.1)	2,293-2,390	205 (96.1)	198-211
75 (23.8)	2,040-2,130	210 (98.8)	184-196
80 (26.6)	1,830-1,920	215 (101.6)	172-183
85 (29.4)	1,650-1,720	220 (104.4)	160-170
90 (32.2)	1,480-1,550		

RESISTANCE TABLE

2411476 S00061527993_V1

Mix Manifold Temperature Sensor - Temperature and Resistance Data
Figure 501/21-51-51-990-802 (Sheet 1 of 2)

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**MIX MANIFOLD TEMPERATURE SENSOR
RESISTANCE VS TEMPERATURE GRAPH**

2411477 S00061527994_V1

**Mix Manifold Temperature Sensor - Temperature and Resistance Data
Figure 501/21-51-51-990-802 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK TEMPERATURE SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the pack temperature sensor

(2) An installation of the pack temperature sensor.

B. There is a pack temperature sensor installed on the left and right cooling pack.

TASK 21-51-52-000-801

2. Pack Temperature Sensor Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-52-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-52-010-001

(2) To remove the pack temperature sensor [1] on the left cooling pack, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

(b) To get access to the pack temperature sensor [1] on the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-52-010-002

(3) To remove the pack temperature sensor [1] on the right cooling pack, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(b) To get access to the pack temperature sensor [1] for the right cooling pack, do these steps:

(TASK 06-41-00-800-801)

1) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Pack Temperature Sensor Removal

SUBTASK 21-51-52-020-001

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the electrical connector from the pack temperature sensor [1].

SUBTASK 21-51-52-020-002

(2) Remove and discard MS20995C32 lockwire, G01048, from the pack temperature sensor [1].

SUBTASK 21-51-52-020-003

CAUTION

USE TWO WRENCHES TO REMOVE THE TEMPERATURE SENSOR FROM THE BOSS. IF YOU USE ONLY ONE WRENCH TO REMOVE THE TEMPERATURE SENSOR, IT CAN CAUSE DAMAGE TO THE DUCT.

(3) Use two wrenches to remove the pack temperature sensor [1].

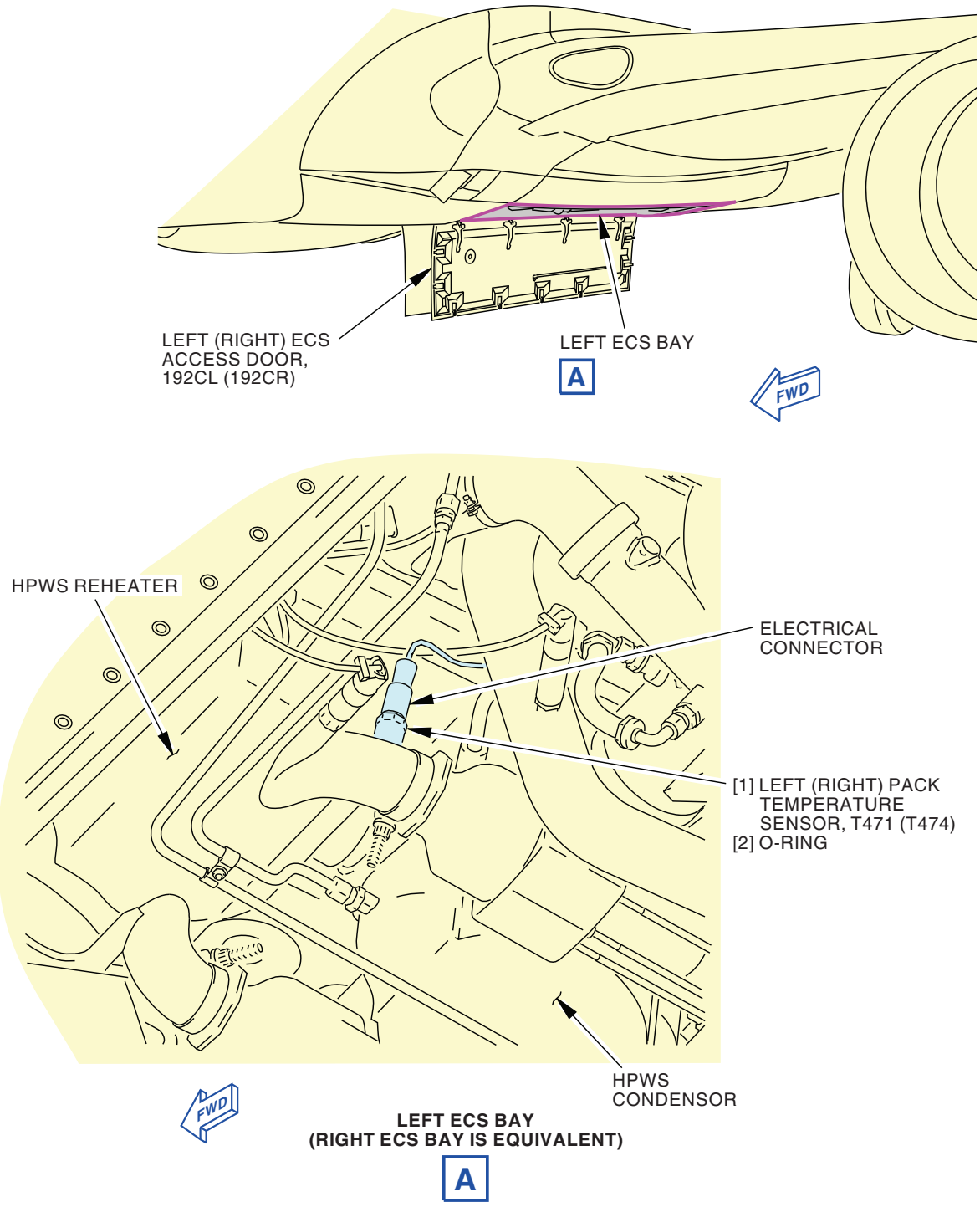
NOTE: Use one wrench to hold the boss. Use the other wrench to turn the pack temperature sensor.

SUBTASK 21-51-52-020-004

(4) Remove and discard the O-ring [2] from the pack temperature sensor [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pack Temperature Sensor Installation
Figure 401/21-51-52-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-52-400-801

3. Pack Temperature Sensor Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Pack temperature sensor	21-51-52-04-025	PTF ALL
2	O-ring	21-51-52-04-020	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Pack Temperature Sensor Installation

SUBTASK 21-51-52-420-001

(1) Install a new O-ring [2] on the pack temperature sensor [1].

SUBTASK 21-51-52-160-001

(2) Make sure that the mating surfaces of the boss and pack temperature sensor nut are clean.

SUBTASK 21-51-52-640-001

(3) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the pack temperature sensor [1].

SUBTASK 21-51-52-420-002

CAUTION

USE TWO WRENCHES TO INSTALL THE TEMPERATURE SENSOR FROM THE BOSS. IF YOU USE ONLY ONE WRENCH TO INSTALL THE TEMPERATURE SENSOR, IT CAN CAUSE DAMAGE TO THE DUCT.

(4) Use two wrenches to install the pack temperature sensor [1].

NOTE: Use one wrench to hold the boss. Use the other wrench to turn the pack temperature sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-52-420-003

(5) Install new MS20995C32 lockwire, G01048, on the pack temperature sensor [1].

SUBTASK 21-51-52-420-004

(6) Connect the electrical connector to the pack temperature sensor [1].

SUBTASK 21-51-52-010-003

(7) If the temperature pack temperature sensor [1] on the left cooling pack was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-52-010-004

(8) If the pack temperature sensor [1] on the right cooling pack was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-52-010-005

(1) If the pack temperature sensor [1] on the left cooling pack was replaced, do this step:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-52-010-006

(2) If the pack temperature sensor [1] on the right cooling pack was replaced, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK TEMPERATURE SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) A test of the pack temperature sensors.

(2) The test consists of a resistance measurements to make sure that the sensor meets specifications.

B. There are two pack temperature sensors, T471 and T474.

TASK 21-51-52-720-801

2. Pack Temperature Sensor Test

A. References

Reference	Title
21-51-52 P/B 401	PACK TEMPERATURE SENSOR - REMOVAL/INSTALLATION
21-61-22-000-801	Integrated Air Systems Controller (IASC) Removal (P/B 401)
21-61-22-400-801	Integrated Air Systems Controller (IASC) Installation (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-3955	Thermometer - Digital (Type K) Part #: 51 2B Supplier: 89536 Opt Part #: 51 SERIES II Supplier: 89536

C. Preparation for the Resistance Test

SUBTASK 21-51-52-864-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-52-020-005

(2) To do a test of the left pack temperature sensor, remove the left IASC (TASK 21-61-22-000-801).

SUBTASK 21-51-52-020-006

(3) To do a test of the right pack temperature sensor, remove the right IASC (TASK 21-61-22-000-801).

SUBTASK 21-51-52-800-001

(4) Allow the pack bay to cool until the pack bay interior has reached the outside ambient temperature.

D. Sensor Resistance Check Procedure

SUBTASK 21-51-52-700-001

(1) Use a digital thermometer, COM-3955, or equivalent, to measure the temperature adjacent to the applicable pack temperature sensor in the pack bay.

(a) Record the temperature for later use.

SUBTASK 21-51-52-700-002

(2) Use the digital/analog multimeter, COM-1793, (preferred) to measure resistance between connector pins.

NOTE: An equivalent meter can be used as long as the digital multimeter minimizes self-heating of the sensor element. The instrument must limit the current through the sensor to less than 100 microamperes when set to the 10K ohms scale. If you use resistance measuring instruments without this current-limiting feature you will get an error in resistance value due to self-heating of the sensor element.

SUBTASK 21-51-52-700-003

(3) For the left pack temperature sensor, T471, make these resistance measurements:

(a) Measure the resistance from pin, J9, to pin, J10, on the electrical connector, D15838B, for the left Integrated Air Systems Controller (IASC) (WDM 21-51-12).

(b) Measure the resistance from pin, 16, to pin, 22, on the electrical connector, D15840C, for the right Integrated Air Systems Controller (IASC) (WDM 21-51-13).

SUBTASK 21-51-52-700-004

(4) For the right pack temperature sensor, T474, make these resistance measurements:

(a) Measure the resistance from pin, J9, to pin, J10, on the electrical connector, D15840B, for the right Integrated Air Systems Controller (IASC) (WDM 21-51-22).

(b) Measure the resistance from pin, 16, to pin, 22, on the electrical connector, D15838C, for the left Integrated Air Systems Controller (IASC) (WDM 21-51-23).

SUBTASK 21-51-52-700-005

(5) Make a record of the resistances for each sensor that you measured.

SUBTASK 21-51-52-700-006

(6) Find the temperature that is nearest to the temperature that you recorded (Figure 501).

SUBTASK 21-51-52-700-007

(7) Compare the serviceable range of resistance for that temperature to the resistance you recorded.

NOTE: The serviceable range of resistance is between the two parallel lines on the graph.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-52-960-001

(8) If the resistance is out of the serviceable range, replace the sensor (PAGEBLOCK 21-51-52/401).

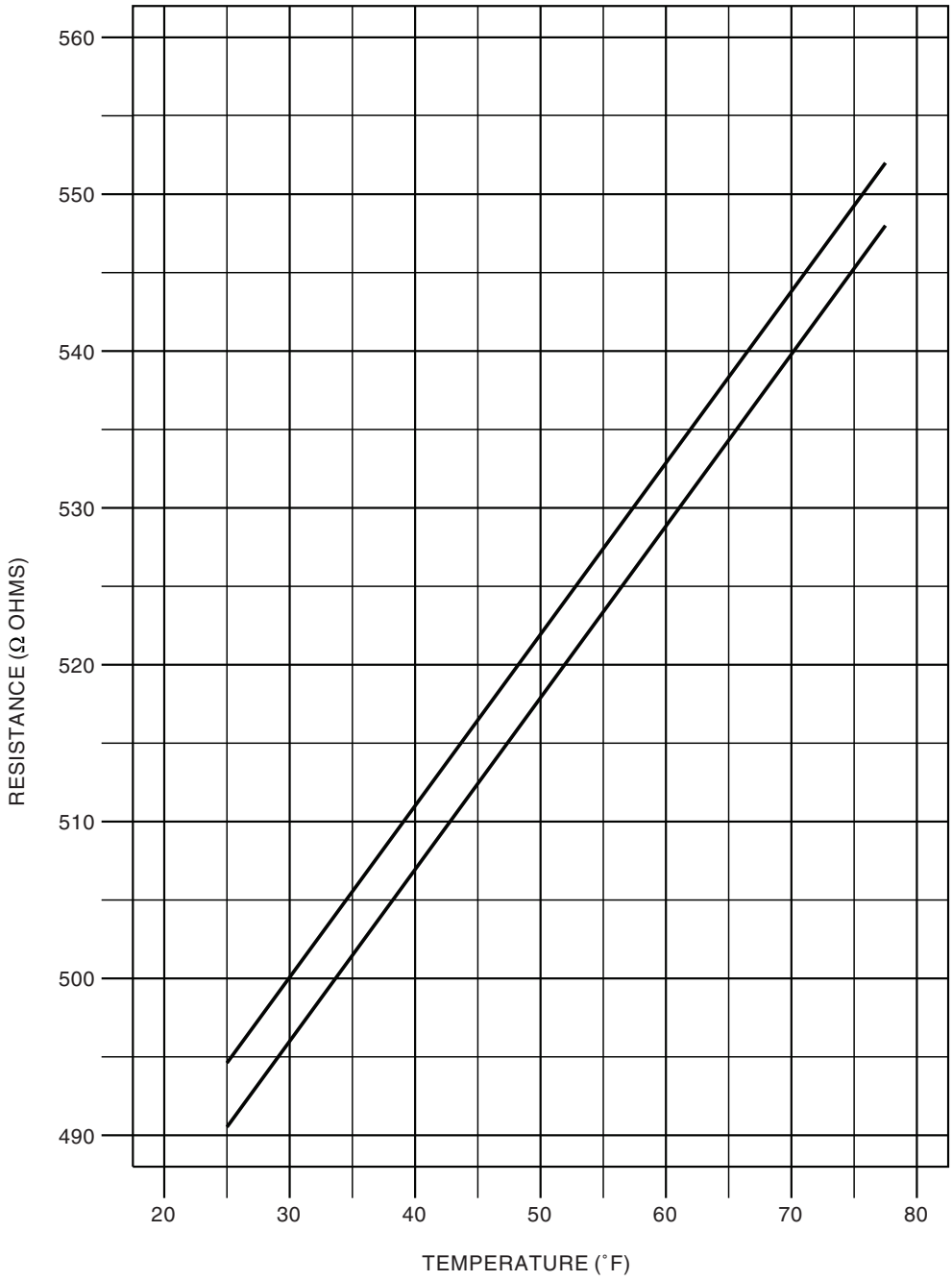
E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-52-420-005

(1) Install the applicable Integrated Air Systems Controller (IASC) (TASK 21-61-22-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PACK TEMPERATURE SENSOR RESISTANCE VS TEMPERATURE GRAPH

2411479 S00061528002_V1

Pack Temperature Sensor Test
Figure 501/21-51-52-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK TEMPERATURE BULB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the pack temperature bulb

(2) An installation of the pack temperature bulb.

B. There is a pack temperature bulb installed on the left and right cooling pack.

TASK 21-51-60-000-801

2. Pack Temperature Bulb Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-51-60-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-60-860-002

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

SUBTASK 21-51-60-010-001

(3) To get access to the pack temperature bulb [1] on the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-60-010-002

CAUTION: DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the pack temperature bulb [1] on the right cooling pack, do these steps:
(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Pack Temperature Bulb Removal

SUBTASK 21-51-60-020-001

WARNING: DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the electrical connector from the pack temperature bulb [1].

SUBTASK 21-51-60-020-002

(2) Remove and discard MS20995C32 lockwire, G01048, from the overheat pack temperature bulb [1].

SUBTASK 21-51-60-020-003

CAUTION: USE TWO WRENCHES TO REMOVE THE PACK TEMPERATURE BULB FROM THE BOSS. IF YOU USE ONLY ONE WRENCH TO REMOVE THE PACK TEMPERATURE BULB, IT CAN CAUSE DAMAGE TO THE DUCT.

(3) Use two wrenches to remove the pack temperature bulb [1].

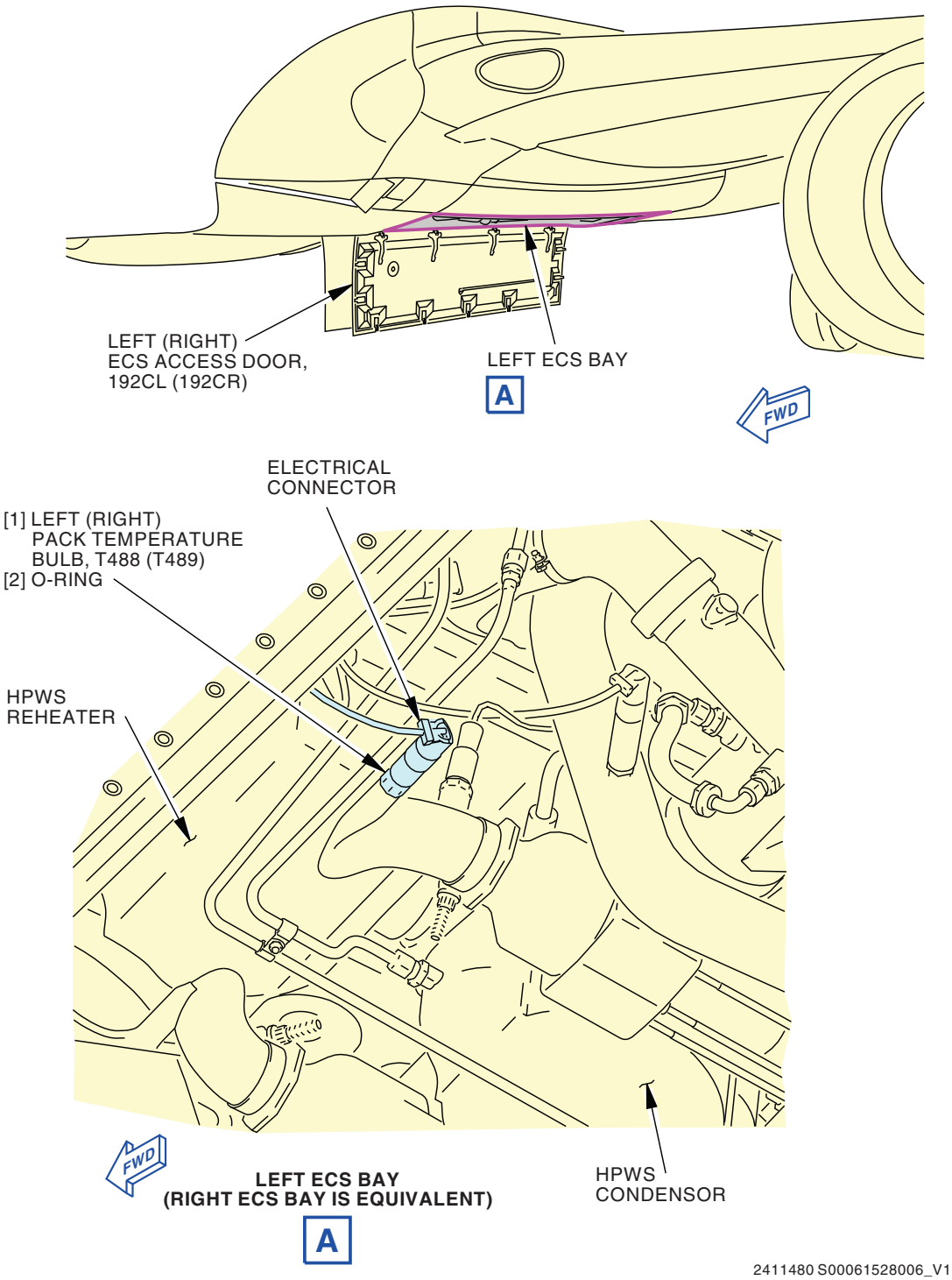
NOTE: Use one wrench to hold the boss. Use the other wrench to turn the pack temperature bulb.

SUBTASK 21-51-60-020-004

(4) Remove and discard the O-ring [2] from the pack temperature bulb [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411480 S00061528006_V1

Pack Temperature Bulb Installation
Figure 401/21-51-60-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-60-400-801

3. Pack Temperature Bulb Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Pack temperature bulb	21-51-60-02-025	PTF ALL
2	O-ring	21-51-60-02-020	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Pack Temperature Bulb Installation

SUBTASK 21-51-60-020-005

(1) If it is installed, remove the copper gasket from the pack temperature bulb [1].

SUBTASK 21-51-60-420-001

(2) Install a new O-ring [2] on the pack temperature bulb [1].

SUBTASK 21-51-60-160-001

(3) Make sure that the mating surfaces of the boss and pack temperature bulb nut are clean.

SUBTASK 21-51-60-640-001

(4) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the pack temperature bulb [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-60-420-002

CAUTION: USE TWO WRENCHES TO INSTALL THE PACK TEMPERATURE BULB IN THE BOSS. IF YOU USE ONLY ONE WRENCH TO INSTALL THE PACK TEMPERATURE BULB, IT CAN CAUSE DAMAGE TO THE DUCT.

(5) Use two wrenches to install the pack temperature bulb [1].

NOTE: Use one wrench to hold the boss. Use the other wrench to turn the pack temperature bulb.

(a) Tighten the pack temperature bulb [1] to 225 ±11 in-lb (25.4 ±1.2 N·m).

SUBTASK 21-51-60-420-003

(6) Install new MS20995C32 lockwire, G01048, on the pack temperature bulb [1].

SUBTASK 21-51-60-420-004

(7) Connect the electrical connector to the pack temperature bulb [1].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-60-410-001

(1) If the pack temperature bulb [1] for the left cooling pack was replaced, close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-60-410-002

(2) If the pack temperature bulb [1] for the right cooling pack was replaced, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CONDITIONING MODULE - REMOVAL/INSTALLATION

1. General

A. The air conditioning module is located on the upper right corner of the P5 forward overhead panel in the flight deck.

B. This procedure has these tasks:

(1) A removal of the air conditioning module.

(2) An installation of the air conditioning module.

TASK 21-51-65-000-801

2. Air Conditioning Module Removal

(Figure 401)

A. References

Reference	Title
36-21-02-600-801	Dual Duct Pressure Indicator Removal (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Air Conditioning Module Removal

SUBTASK 21-51-65-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	8	C00326	OVERHEAD 5V SECONDARY
E	9	C00329	OVERHEAD 5V SECONDARY
E	10	C00328	OVERHEAD 5V SECONDARY
F	8	C00327	OVERHEAD 5V SECONDARY
F	9	C00330	OVERHEAD 5V SECONDARY
F	10	C00390	OVERHEAD 5V SECONDARY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-51-65-860-002

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	3	C01613	AIR COND PACK CONT VALVES L ALT
D	4	C01614	AIR COND PACK CONT VALVES R ALT

SUBTASK 21-51-65-010-001

(3) Get access to the electrical connectors on the back of the P5-10 air conditioning module [1] on the P5 forward overhead panel:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Loosen the 1/4-turn fasteners that hold the P5 forward overhead panel in position and let the panel rotate downward.

SUBTASK 21-51-65-020-001

(4) Remove the P5-10 air conditioning module [1] from the P5 forward overhead panel as follows:

(a) Remove the duct pressure indicator (TASK 36-21-02-600-801).

(b) Disconnect the electrical connectors at the back of the module.

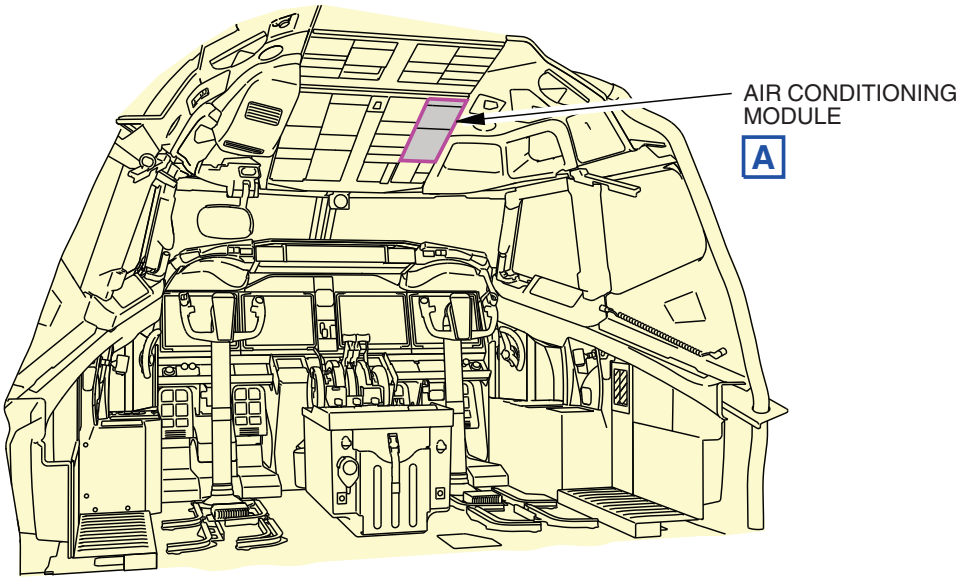
WARNING HOLD THE AIR CONDITIONING MODULE WHEN YOU LOOSEN THE 1/4-TURN FASTENERS. IF THE MODULE FALLS, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Hold the air conditioning module [1] in place and loosen the 1/4-turn fasteners that secure it to the P5 forward overhead panel.

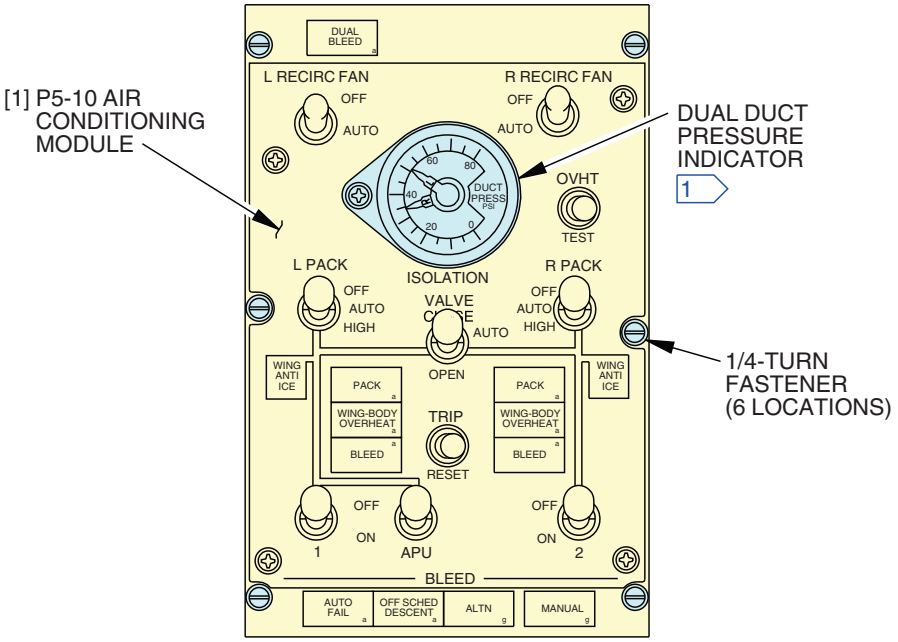
(d) Remove the air conditioning module [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



AIR CONDITIONING
MODULE
(EXAMPLE)

A

1 IF APPLICABLE

2411481 S00061528011_V3

Air Conditioning Module Installation
Figure 401/21-51-65-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-65-400-801

3. Air Conditioning Module Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-25-00-710-801	Recirculation System - Operational Test (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-804	Supply Pressure to the Pneumatic System with One or Both Engines (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
36-21-02-600-802	Dual Duct Pressure Indicator Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-3907	Mirror - Dental

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Module	21-51-65-01-050	PTF ALL
		31-11-94-01-205	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Air Conditioning Module Installation

SUBTASK 21-51-65-010-002

(1) If it has not already been done, lower the P5 forward overhead panel.

SUBTASK 21-51-65-420-001

(2) Install the P5-10 air conditioning module [1] into the P5 forward overhead panel as follows:

(a) Put the air conditioning module [1] into the P5 forward overhead panel and hold it in position.

(b) Turn the 1/4-turn fasteners on the air conditioning module [1] to secure it to the P5 forward overhead panel.

(c) Install the dual duct pressure indicator (TASK 36-21-02-600-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Connect the two electrical connectors to the air conditioning module [1].

SUBTASK 21-51-65-010-003

(3) Lift the P5 forward overhead panel to the closed position and turn the 1/4-turn fasteners.

SUBTASK 21-51-65-860-003

(4) Close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	8	C00326	OVERHEAD 5V SECONDARY
E	9	C00329	OVERHEAD 5V SECONDARY
E	10	C00328	OVERHEAD 5V SECONDARY
F	8	C00327	OVERHEAD 5V SECONDARY
F	9	C00330	OVERHEAD 5V SECONDARY
F	10	C00390	OVERHEAD 5V SECONDARY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-51-65-860-004

(5) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	3	C01613	AIR COND PACK CONT VALVES L ALT
D	4	C01614	AIR COND PACK CONT VALVES R ALT

G. Air Conditioning Module Installation Test

SUBTASK 21-51-65-860-005

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-51-65-860-006

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-51-65-860-007

(3) Make sure that all circuit breakers for the air conditioning and the pneumatic systems are closed.

SUBTASK 21-51-65-860-008

(4) Make sure that all the circuit breakers for these systems are closed:

(a) Master caution

(b) Air/ground systems (nose gear and main gear)

(c) Indicator lights.

SUBTASK 21-51-65-860-009

(5) Make sure the L and R PACK switches on the P5-10 panel are set to OFF.

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SUBTASK 21-51-65-710-001

(6) Set the switch for the master dim and test lights, on the P1 captain's main instrument panel, to the TEST position and make sure that these lights come on:

(a) AUTO FAIL

(b) OFF SCHED DESCENT

(c) ALTN

(d) MANUAL

(e) L and R PACK

(f) L and R BLEED

(g) DUAL BLEED

SUBTASK 21-51-65-710-006

(7) Set the switch for the master dim and test lights, on the P1 panel, to the usual mode.

SUBTASK 21-51-65-710-002

(8) Do a test of the overheat indication system:

(a) Press the OVHT TEST button on the P5-10 panel.

(b) Make sure the WING-BODY OVERHEAT lights come on.

SUBTASK 21-51-65-710-003

(9) Do this task to do a test of the recirculation system, Recirculation System - Operational Test, TASK 21-25-00-710-801.

SUBTASK 21-51-65-710-004

(10) Do a test of the pneumatic system pressurization:

(a) Open the left access panel:

Number	Name/Location
192CL	ECS Access Door

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(b) To open the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

1) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access door:

Number	Name/Location
192CR	ECS Access Door

(c) Do this task: Supply Pressure to the Pneumatic System with the APU, TASK 36-00-00-860-803.

(d) Make sure the ISOLATION VALVE switch is in the CLOSED position.

(e) Position the APU BLEED switch to ON.

(f) Make sure there is pressure indication for the left pneumatic system.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Position the L PACK switch to AUTO.

(h) Make sure the indication on the pressure gauge increases.

(i) Make sure the left pack flow control valve opens.

1) Use a dental mirror, STD-3907 if necessary to view the valve position.

(j) Position the L PACK switch to OFF.

(k) Position the ISOLATION VALVE switch to OPEN.

(l) Make sure there is pressure indication for the right pneumatic system.

(m) Position the R PACK switch to AUTO.

(n) Make sure the indication on the pressure gauge increases.

(o) Make sure the right pack flow control valve opens.

1) Use a dental mirror, STD-3907 if necessary to view the valve position.

(p) Set the R PACK switch to OFF.

(q) Remove pressure from the pneumatic system, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-65-710-005

(11) If testing of the engine bleed switches functionality is desired, then do the following test:

NOTE: It is recommended to test the engine bleed switches functionality after P5-10 panel replacement due to bleed switch failure.

(a) Supply pressure to the pneumatic system with the engine on the applicable side, do this task: Supply Pressure to the Pneumatic System with One or Both Engines, TASK 36-00-00-860-804.

(b) Set these pneumatic system control switches to the positions shown:

1) APU BLEED switch to OFF

2) ISOLATION VALVE switch to CLOSE

(c) If applicable, remove any external pneumatic source.

(d) Set the pack switches to AUTO.

(e) Do these steps to do a check of the operation of the engine BLEED switch/PRSOV:

NOTE: The engine BLEED switch will be cycled in these steps to make sure that there are no intermittent malfunctions in the system operation.

1) Set the applicable engine BLEED 1 or 2 switch to the OFF position.

2) Make sure that the Pointer for the applicable system on the Dual Duct Pressure Indicator decreases to less than 10 psi (68.9 kPa).

3) Set the applicable engine BLEED switch to the ON position.

4) Make sure that the Manifold Pressure Pointer for the applicable system increases to 10 psi (68.9 kPa) - 25 psi (172.4 kPa) with the engine at steady idle without user systems in operation.

NOTE: The engine operation at high power is not necessary.

NOTE: The Duct Pressure Pointers on the Dual Duct Pressure Indicator may fluctuate without user systems in operation.

5) Set the applicable engine BLEED switch to the OFF position.

6) Make sure that the applicable pointer on the Dual Duct Pressure Indicator decreases to less than 10 psi (68.9 kPa).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Set the pack switches to OFF.

(g) Remove pressure from the pneumatic system, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-65-860-010

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-65-410-001

(2) Close the left access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-65-410-002

(3) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-65-410-003

(4) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ZONE TEMPERATURE CONTROL AND INDICATION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Cabin Temperature Control and Indication Test.

(2) Duct Overheat Switch - Operational Test.

TASK 21-61-00-700-801

2. Cabin Temperature Control and Indication - Operational Test

A. General

(1) This task does an operational test of the cabin temperature control and indication components.

B. References

Reference	Title
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1122	Thermometer - Alcohol/Mercury (or equivalent meter meets task requirements)

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Procedure

SUBTASK 21-61-00-710-001

(1) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

SUBTASK 21-61-00-860-002

(2) Do these steps at the P5-17 Cabin Temperature Panel:

(a) Put the CONT CAB temperature selector to the full COLD position.

(b) Put the FWD CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

(c) Let the pack operate for 1 to 3 minutes to let the output temperature become stable.

SUBTASK 21-61-00-710-002

(3) Do these steps to check temperature indication for the forward passenger cabin:

(a) Use a thermometer, STD-1122 to measure the ambient temperature in the forward passenger cabin near the cabin temperature sensor assembly.

(b) Put the AIR TEMP selector on the P5-17 Cabin Temperature Panel to the FWD PASS CABIN position.

(c) Make sure that the temperature indicator shows approximately the same temperature as the ambient temperature in the forward passenger cabin.

(d) Move the FWD CAB temperature selector to the fully WARM position.

(e) Make sure that the temperature indicator shows an increase in temperature.

(f) Move the FWD CAB temperature selector to the AUTO (12:00) position.

(g) Let the pack operate for 1 to 3 minutes to let the output temperature become stable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-00-710-003

(4) Do these steps to check temperature indication for the aft passenger cabin:

(a) Use a thermometer, STD-1122 to measure the ambient temperature in the aft passenger cabin near the cabin temperature sensor assembly.

(b) Put the AIR TEMP selector on the P5-17 Cabin Temperature Panel to the AFT PASS CABIN position.

(c) Make sure that the temperature indicator shows approximately the same temperature as the ambient temperature in the aft passenger cabin.

(d) Move the AFT CAB temperature selector to the fully WARM position.

(e) Make sure that the temperature indicator shows an increase in temperature.

(f) Move the AFT CAB temperature selector to the full COLD position.

(g) Let the pack operate for 1 to 3 minutes to let the output temperature become stable.

SUBTASK 21-61-00-710-004

(5) Do these steps to check temperature control for the control cabin:

(a) Put the CONT CABIN temperature selector to the AUTO (12:00) position.

(b) Let the pack operate for 1 to 3 minutes to let the output temperature become stable.

(c) Put the AIR TEMP selector on the P5-17 Cabin Temperature Panel to the SUPPLY DUCT CONT CAB position.

(d) Make a record of the temperature shown on the indicator.

(e) Move the CONT CAB temperature selector to the fully WARM position.

(f) Make sure that the temperature indicator shows an increase in temperature.

(g) Move the CONT CAB and AFT CAB temperature selectors to the AUTO (12:00) position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-00-710-005

(1) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804.

——— END OF TASK ———

TASK 21-61-00-700-802

3. Duct Overheat Switch - Operational Test

A. References

Reference	Title
21-61-05-000-801	Duct Overheat Switch Removal (P/B 401)
21-61-05-400-801	Duct Overheat Switch Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-3924	Tester - Thermo Switch, K-Type Thermocouple Switches Part #: H394R Supplier: 98869 Opt Part #: H294 Supplier: 98869 Opt Part #: H394 SERIES Supplier: 98869

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-3952	Kit - Heater Probes, TEMPCAL Part #: BH24944-13 Supplier: 98869 Part #: BH24944-14 Supplier: 98869 Part #: BH24944-16 Supplier: 98869

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Preparation for the Operational Test

SUBTASK 21-61-00-010-001

(1) To get access to the duct overheat switch for the control cabin, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-00-010-002

(2) To get access to the duct overheat switches for the passenger cabin, remove the applicable ceiling panel that is forward of the overwing escape hatches.

F. Operational Test of the Control Cabin Duct Overheat Switch (Method I)

SUBTASK 21-61-00-020-001

(1) Remove the duct overheat switch (TASK 21-61-05-000-801).

(a) Re-connect the electrical connector to the duct overheat switch.

SUBTASK 21-61-00-480-001

(2) Prepare the K-Type thermocouple switch tester, SPL-3924 and the TEMPCAL heater probe kit, SPL-3952 to test the overheat switch.

NOTE: Use the correct heater probe to test the overheat switch.

SUBTASK 21-61-00-710-006

CAUTION

DO NOT LET THE TEMPERATURE OF THE OVERHEAT SWITCH PROBE BECOME MORE THAN 200 DEGREES F (93 DEGREES C). THIS CAN CAUSE DAMAGE TO THE OVERHEAT SWITCH.

(3) Use the test equipment to increase the temperature of the overheat switch as follows:

(a) Set the TEST TEMPERATURE on the tester to 170°F (77°C).

(b) Connect the duct overheat switch to the K-Type thermocouple switch tester, SPL-3924.

(c) Wait for a minimum of five minutes to find the switch closing temperature.

(d) Increase the test temperature on the tester at a rate of not more than 2°F (1°C) per minute to 190 ±10°F (88 ±6°C).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-00-710-007

(4) Make sure that these lights come on when the temperature of the overheat switch probe gets to 190 ±10°F (88 ±6°C):

(a) ZONE TEMP light on the P5-17 panel

(b) MASTER CAUTION and AIR COND lights on the pilots' glareshield.

SUBTASK 21-61-00-710-008

(5) Remove the heat from the overheat switch probe.

SUBTASK 21-61-00-710-009

(6) Let the overheat switch probe become cool down at a rate of not more than 2°F (1°C) per minute.

NOTE: The reset temperature for the duct overheat switch is 150°F (66°C).

SUBTASK 21-61-00-710-010

(7) Push the TRIP RESET button on the P5-10 panel.

SUBTASK 21-61-00-710-011

(8) Push the MASTER CAUTION reset button on the P7 glareshield panel.

SUBTASK 21-61-00-710-012

(9) Make sure that the ZONE TEMP, MASTER CAUTION and AIR COND lights go off.

SUBTASK 21-61-00-420-001

(10) Install the duct overheat switch (TASK 21-61-05-400-801).

G. Operational Test of the Control Cabin Duct Overheat Switch (Method II)

SUBTASK 21-61-00-020-002

(1) Remove the duct overheat switch (TASK 21-61-05-000-801).

SUBTASK 21-61-00-420-002

(2) Install a new or serviceable duct overheat switch (TASK 21-61-05-400-801).

NOTE: The duct overheat switch that was removed should be tested according to the vendor CMM before it is put back into service.

SUBTASK 21-61-00-020-003

(3) Remove the electrical connector from the duct overheat switch.

SUBTASK 21-61-00-480-002

(4) Connect a jumper wire between pin 1 and pin 2 of the electrical connector.

SUBTASK 21-61-00-210-001

(5) Make sure that these lights come on:

(a) ZONE TEMP light on the P5-17 panel.

(b) MASTER CAUTION and AIR COND lights on the pilots' glareshield.

SUBTASK 21-61-00-080-001

(6) Disconnect the jumper wire.

SUBTASK 21-61-00-860-005

(7) Push the TRIP RESET button on the P5-10 panel.

SUBTASK 21-61-00-860-006

(8) Push the MASTER CAUTION reset button on the P7 glareshield panel.

SUBTASK 21-61-00-210-002

(9) Make sure that the ZONE TEMP, MASTER CAUTION, and AIR COND lights go off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-00-420-003

(10) Connect the electrical connector to the duct overheat switch.

H. Operational Test of the Passenger Cabin Duct Overheat Switch (Method I)

SUBTASK 21-61-00-020-004

(1) Remove the forward passenger cabin duct overheat switch (TASK 21-61-05-000-801).

(a) Re-connect the electrical connector to the duct overheat switch.

SUBTASK 21-61-00-480-003

(2) Prepare the K-Type thermocouple switch tester, SPL-3924 and the TEMPCAL heater probe kit, SPL-3952 to test the overheat switch.

NOTE: Use the correct heater probe to test overheat switch.

SUBTASK 21-61-00-710-013

CAUTION

DO NOT LET THE TEMPERATURE OF THE OVERHEAT SWITCH PROBE BECOME MORE THAN 200 DEGREES F (93 DEGREES C). THIS CAN CAUSE DAMAGE TO THE OVERHEAT SWITCH.

(3) Use the test equipment to increase the temperature of the overheat switch as follows:

(a) Set the TEST TEMPERATURE on the tester to 170°F (77°C).

(b) Connect the duct overheat switch to the K-Type thermocouple switch tester, SPL-3924.

(c) Wait for a minimum of five minutes to find the switch closing temperature.

(d) Increase the test temperature on the tester at a rate of not more than 2°F (1°C) per minute to 190 ±10°F (88 ±6°C).

SUBTASK 21-61-00-710-014

(4) Make sure that these lights come on when the temperature of the overheat switch probe gets to 190 ±10°F (88 ±6°C):

(a) ZONE TEMP light on the P5-17 panel

(b) MASTER CAUTION and AIR COND lights on the pilots' glareshield.

SUBTASK 21-61-00-710-015

(5) Remove the heat from the overheat switch probe.

SUBTASK 21-61-00-710-016

(6) Let the overheat switch probe become cool down at a rate of not more than 2°F (1°C) per minute.

NOTE: The reset temperature for the duct overheat switch is 150°F (66°C).

SUBTASK 21-61-00-710-017

(7) Push the TRIP RESET button on the P5-10 panel.

SUBTASK 21-61-00-710-018

(8) Push the MASTER CAUTION reset button on the P7 glareshield panel.

SUBTASK 21-61-00-710-019

(9) Make sure that the ZONE TEMP, MASTER CAUTION and AIR COND lights go off.

SUBTASK 21-61-00-420-004

(10) Install the duct overheat switch (TASK 21-61-05-400-801).

SUBTASK 21-61-00-710-020

(11) Repeat the test for the aft passenger cabin duct overheat switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Operational Test of the Passenger Cabin Duct Overheat Switch (Method II)

SUBTASK 21-61-00-020-005

(1) Remove the duct overheat switch (TASK 21-61-05-000-801).

SUBTASK 21-61-00-420-005

(2) Install a new or serviceable duct overheat switch (TASK 21-61-05-400-801).

NOTE: The duct overheat switch that was removed should be tested according to the vendor CMM before it is put back into service.

SUBTASK 21-61-00-020-006

(3) Remove the electrical connector from the duct overheat switch.

SUBTASK 21-61-00-480-004

(4) Connect a jumper wire between pin 1 and pin 2 of the electrical connector.

SUBTASK 21-61-00-210-003

(5) Make sure that these lights come on:

(a) ZONE TEMP light on the P5-17 panel.

(b) MASTER CAUTION and AIR COND lights on the pilots' glareshield.

SUBTASK 21-61-00-080-002

(6) Disconnect the jumper wire.

SUBTASK 21-61-00-860-007

(7) Push the TRIP RESET button on the P5-10 panel.

SUBTASK 21-61-00-860-008

(8) Push the MASTER CAUTION reset button on the P7 glareshield panel.

SUBTASK 21-61-00-210-004

(9) Make sure that the ZONE TEMP, MASTER CAUTION, and AIR COND lights go off.

SUBTASK 21-61-00-420-006

(10) Connect the electrical connector to the duct overheat switch.

SUBTASK 21-61-00-710-021

(11) Repeat the test for the aft passenger cabin duct overheat switch.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-00-410-001

(1) Install the ceiling panels that you removed to do the test.

SUBTASK 21-61-00-410-002

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DUCT TEMPERATURE LIMIT SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of these components:

(a) Primary control cabin duct temperature limit sensor

(b) Backup control cabin duct temperature limit sensor

(c) Forward passenger cabin duct temperature limit sensor

(d) Aft passenger cabin duct temperature limit sensor.

(2) An installation of these components:

(a) Primary control cabin duct temperature limit sensor

(b) Backup control cabin duct temperature limit sensor

(c) Forward passenger cabin duct temperature limit sensor

(d) Aft passenger cabin duct temperature limit sensor.

TASK 21-61-04-000-801

2. Duct Temperature Limit Sensor Removal

(Figure 401, Figure 402)

A. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)

B. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 21-61-04-860-001

(1) Make sure that these switches, on the P5-10 air conditioning panel, are set to the positions that follow:

(a) Make sure that the L PACK and R PACK switches are in the OFF position.

(b) Make sure that the BLEED 1, BLEED 2, and BLEED APU switches are in the OFF position.

SUBTASK 21-61-04-860-003

(2) For the forward and primary duct temperature limit sensor, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-04-860-004

(3) For the aft and backup duct temperature limit sensor, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-61-04-010-001

(4) To get access to the temperature sensors for the control cabin, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-04-010-002

(5) To get access to the temperature sensors for the passenger cabin, remove the applicable ceiling panel that is forward of the overwing escape hatches, do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

E. Passenger Cabin Duct Temperature Limit Sensor Removal

SUBTASK 21-61-04-020-001

(1) Disconnect the electrical connector [2] from the passenger cabin duct temperature limit sensor [1].

SUBTASK 21-61-04-020-002

(2) Remove the screws [3] from the passenger cabin duct temperature limit sensor [1].

SUBTASK 21-61-04-020-003

(3) Remove the passenger cabin duct temperature limit sensor [1].

F. Control Cabin Duct Temperature Limit Sensor Removal

SUBTASK 21-61-04-020-004

(1) Disconnect the electrical connector [22] from the control cabin duct temperature limit sensor [21].

SUBTASK 21-61-04-020-005

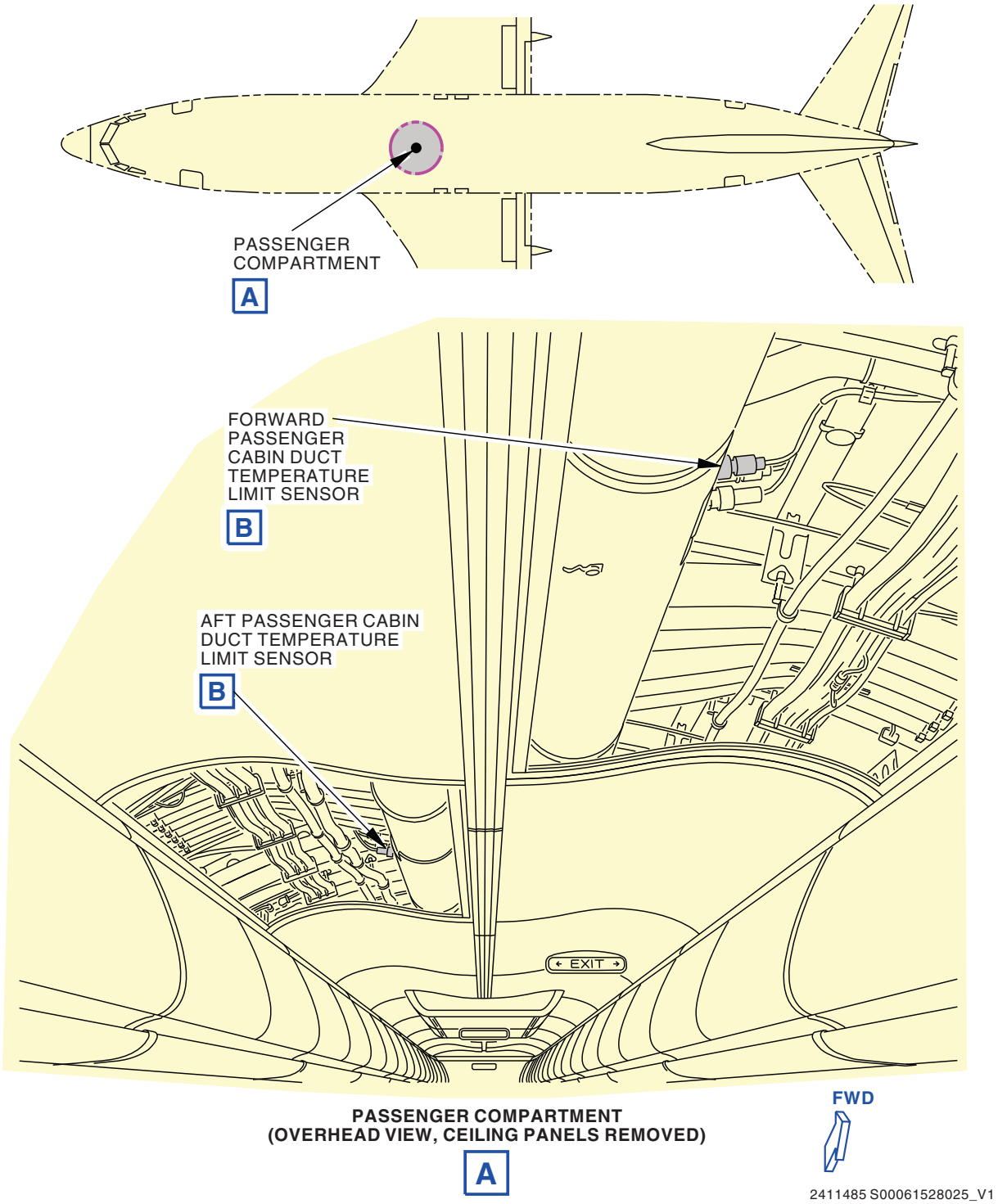
(2) Remove the screw [23] from the control cabin duct temperature limit sensor [21].

SUBTASK 21-61-04-020-006

(3) Remove the control cabin duct temperature limit sensor [21].

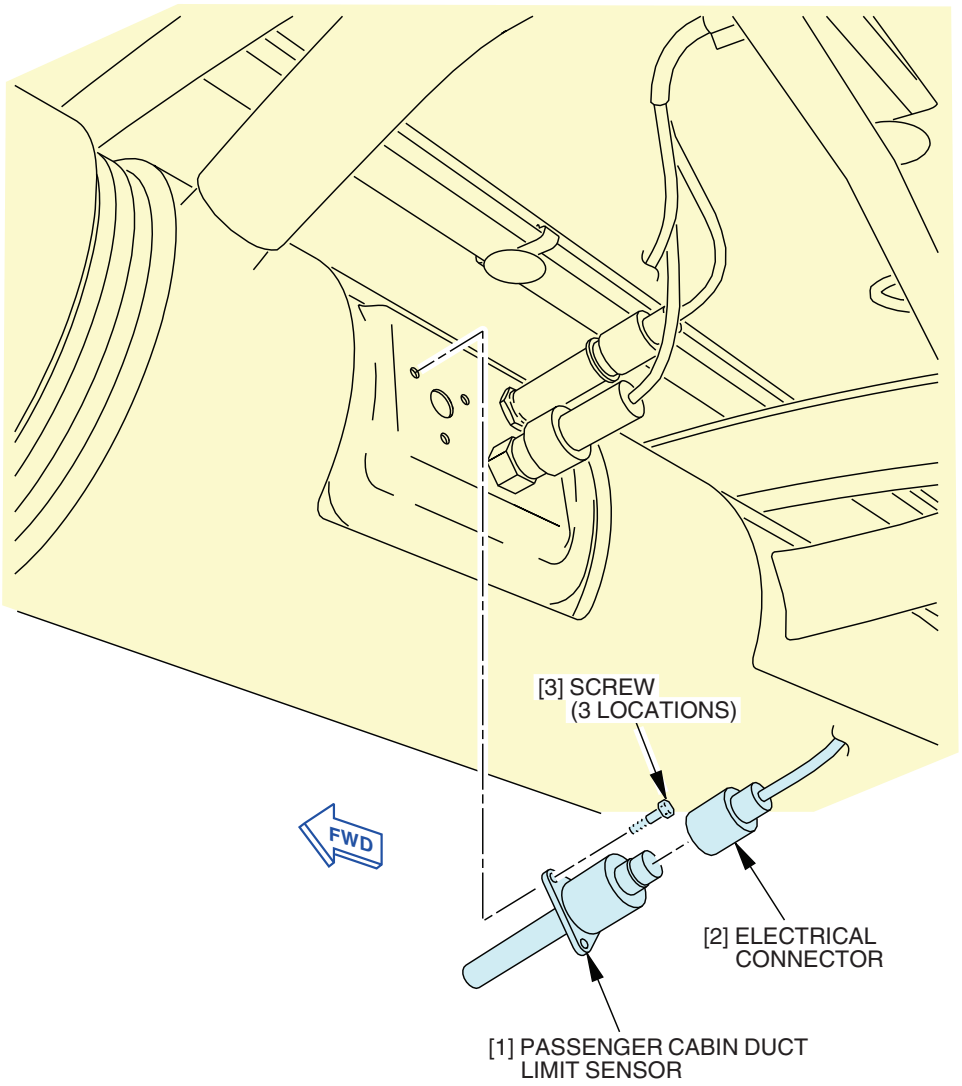
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Passenger Cabin Duct Temperature Limit Sensor Installation
Figure 401/21-61-04-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



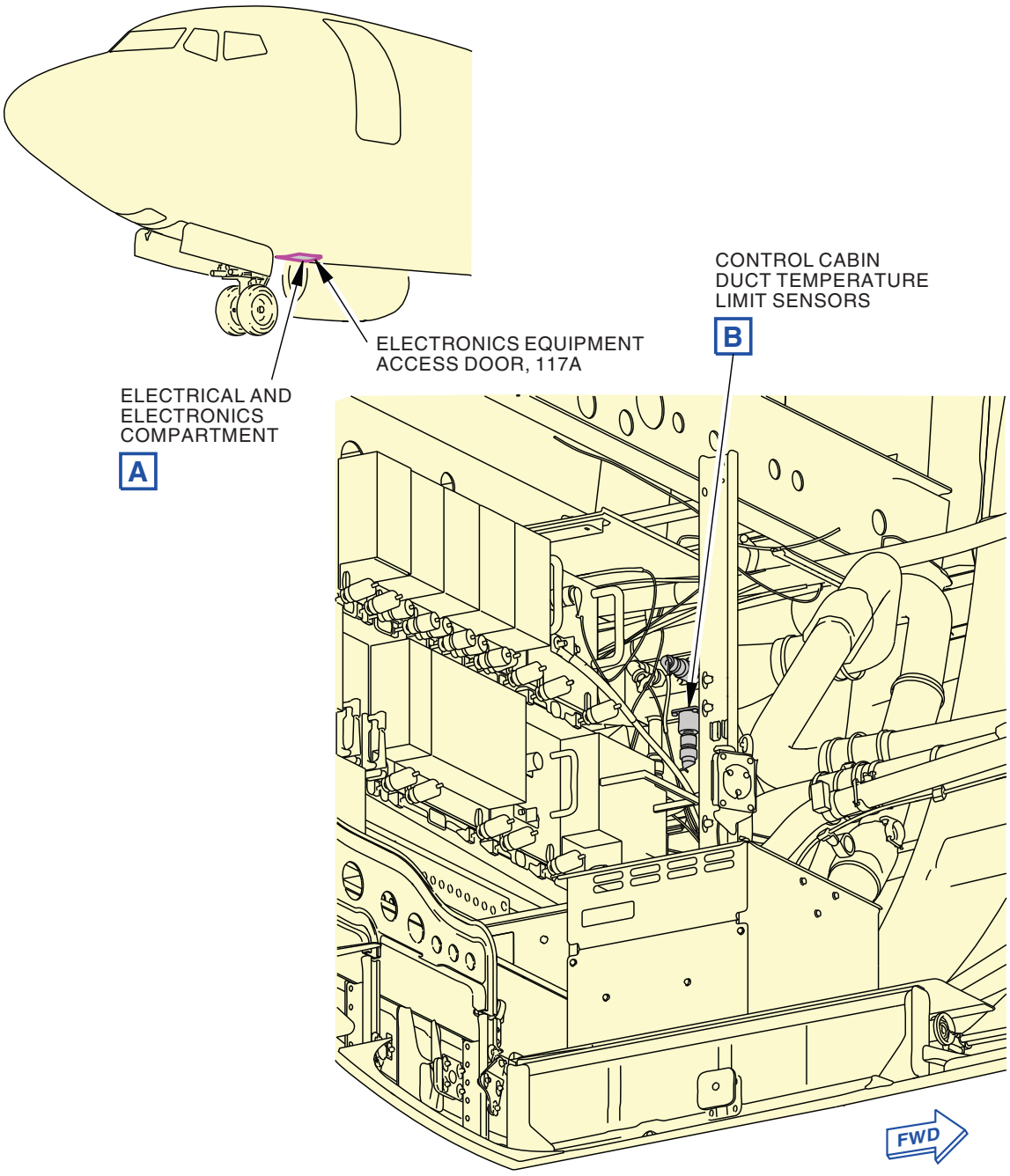
**PASSENGER CABIN DUCT TEMPERATURE LIMIT SENSOR
(EXAMPLE)**

B

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**Passenger Cabin Duct Temperature Limit Sensor Installation
Figure 401/21-61-04-990-801 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



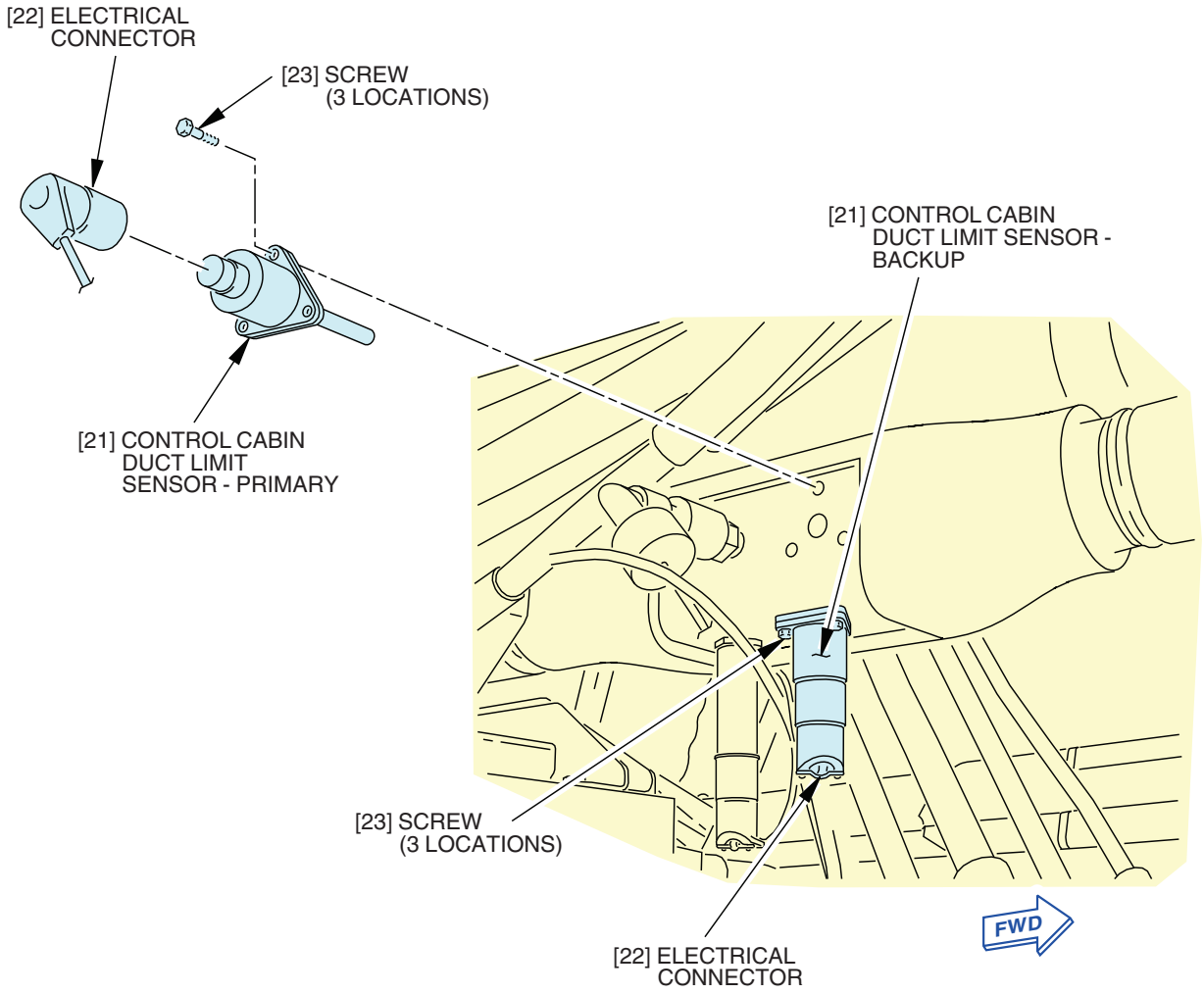
ELECTRICAL AND ELECTRONICS COMPARTMENT
(VIEW THROUGH ACCESS DOOR, 117A)

A

2411487 S00061528027_V2

Control Cabin Duct Temperature Limit Sensor Installation
Figure 402/21-61-04-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONTROL CABIN DUCT TEMPERATURE LIMIT SENSORS (EXAMPLE)

B

2411488 S00061528028_V3

Control Cabin Duct Temperature Limit Sensor Installation
Figure 402/21-61-04-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-04-400-801

3. Duct Temperature Limit Sensor Installation

(Figure 401, Figure 402)

A. References

Reference	Title
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Control cabin duct temperature limit sensor	21-61-04-04-020	PTF ALL

C. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Passenger Cabin Duct Temperature Limit Sensor Installation

SUBTASK 21-61-04-420-001

(1) Put the passenger cabin duct temperature limit sensor [1] in its position on the duct.

SUBTASK 21-61-04-420-002

(2) Install the screws [3].

SUBTASK 21-61-04-420-003

(3) Connect the electrical connector [2] to the passenger cabin duct temperature limit sensor [1].

F. Control Cabin Duct Temperature Limit Sensor Installation

SUBTASK 21-61-04-420-005

(1) Put the control cabin duct temperature limit sensor [21] in its position on the duct.

SUBTASK 21-61-04-420-006

(2) Install the screws [23].

SUBTASK 21-61-04-420-007

(3) Connect the electrical connector [22] to the control cabin duct temperature limit sensor [21].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-04-860-005

(1) If the forward or primary duct temperature limit sensor is replaced, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-04-860-006

(2) If the aft or backup duct temperature limit sensor is replaced, remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-61-04-010-003

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-04-010-004

(4) If you replaced a temperature sensor for the passenger cabin, install the applicable ceiling panel, do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DUCT TEMPERATURE LIMIT SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) A test of the duct temperature limit sensors

B. There are four duct temperature limit sensors installed. There is one primary and one backup duct temperature limit sensor for the control cabin and one forward and one aft duct temperature limit sensor for the passenger cabin.

TASK 21-61-04-000-802

2. Duct Temperature Limit Sensor Test

(Figure 501, Figure 502, Figure 503, Figure 504)

A. References

Reference	Title
21-61-04 P/B 401	DUCT TEMPERATURE LIMIT SENSOR - REMOVAL/INSTALLATION
21-61-22-000-801	Integrated Air Systems Controller (IASC) Removal (P/B 401)
21-61-22-400-801	Integrated Air Systems Controller (IASC) Installation (P/B 401)
WDM 21-61-11	Wiring Diagram Manual
WDM 21-61-12	Wiring Diagram Manual
WDM 21-61-13	Wiring Diagram Manual
WDM 21-61-14	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-3955	Thermometer - Digital (Type K) Part #: 51 2B Supplier: 89536 Opt Part #: 51 SERIES II Supplier: 89536

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare For The Resistance Test

SUBTASK 21-61-04-860-002

(1) To get access to the temperature limit sensor for the passenger cabin, remove the applicable ceiling panels (Figure 501).

SUBTASK 21-61-04-010-005

(2) To get access to the temperature limit sensor for the control cabin, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-04-020-008

(3) To do a test of the aft passenger duct temperature sensor or the backup control cabin temperature sensor, remove the left integrated air systems controller (TASK 21-61-22-000-801).

SUBTASK 21-61-04-020-009

(4) To do a test of the forward passenger duct temperature sensor or the primary control cabin temperature sensor, remove the right integrated air systems controller (TASK 21-61-22-000-801).

SUBTASK 21-61-04-710-001

(5) Make sure the pack has been off for 20 minutes or until it has reached ambient temperature.

F. Resistance Check Procedure

SUBTASK 21-61-04-700-001

(1) Use a digital thermometer, COM-3955, or equivalent, to measure the temperature adjacent to the applicable duct temperature sensor.

(a) Write down the temperature.

SUBTASK 21-61-04-760-001

(2) Use a digital/analog multimeter, COM-1793, to measure resistance between the following pin connections.

NOTE: The digital multimeter must be specifically selected to minimize self-heating of the sensor element. The instrument must limit the current through the sensor to less than 100 microamperes. If you use resistance measuring instruments without this current-limiting feature you will get an error in resistance value due to self-heating of the sensor element.

SUBTASK 21-61-04-700-002

(3) For the primary control cabin duct temperature sensor, measure the resistance from pin E9 to pin E10 on connector, D15840B, for the right integrated air systems controller (WDM 21-61-11).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-04-700-003

(4) For the forward passenger cabin duct temperature sensor, measure the resistance from pin B8 to pin B9 on connector, D15840B, for the right integrated air systems controller(WDM 21-61-13).

SUBTASK 21-61-04-700-004

(5) For the backup control cabin backup duct temperature sensor, measure the resistance between pin E9 and pin E10 on connector, D15838B, for the left integrated air systems controller (WDM 21-61-12).

SUBTASK 21-61-04-700-005

(6) For the aft passenger cabin duct temperature sensor, measure the resistance from pin B8 to pin B9 on connector, D15838B, for the left integrated air systems controller (WDM 21-61-14).

SUBTASK 21-61-04-700-006

(7) Write down the resistance in Ohms that you measure.

(a) For the duct temperature sensors for the control cabin or the passenger cabin, find the temperature that is nearest to the temperature that you wrote down (Figure 504).

(b) Compare the permitted range of resistance for that temperature to the resistance you wrote down.

(c) If the resistance is out of range, replace the sensor (PAGEBLOCK 21-61-04/401).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-04-420-008

(1) Install the applicable integrated air systems controller (TASK 21-61-22-400-801).

SUBTASK 21-61-04-410-001

(2) Install the passenger cabin ceiling panels.

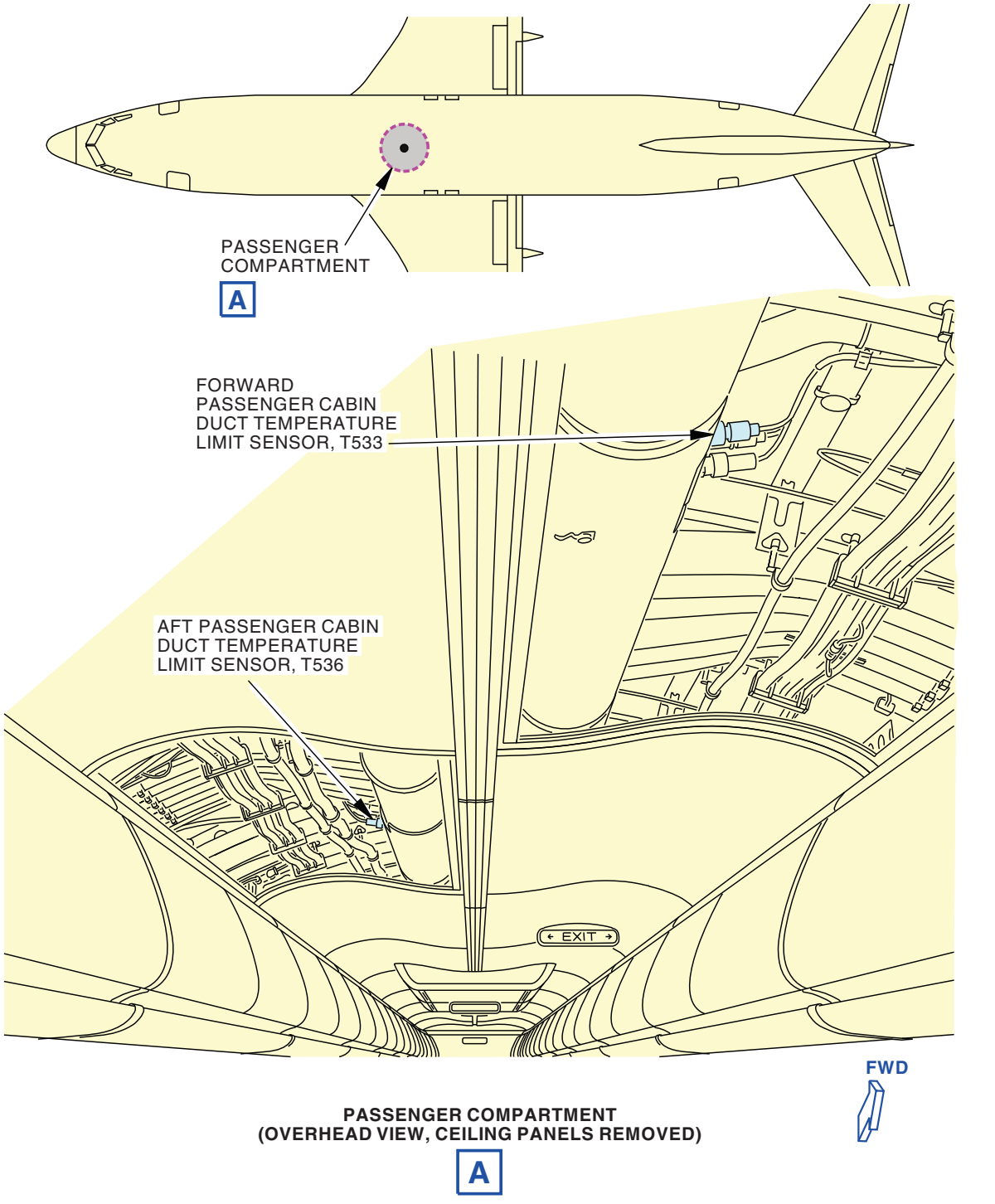
SUBTASK 21-61-04-410-002

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



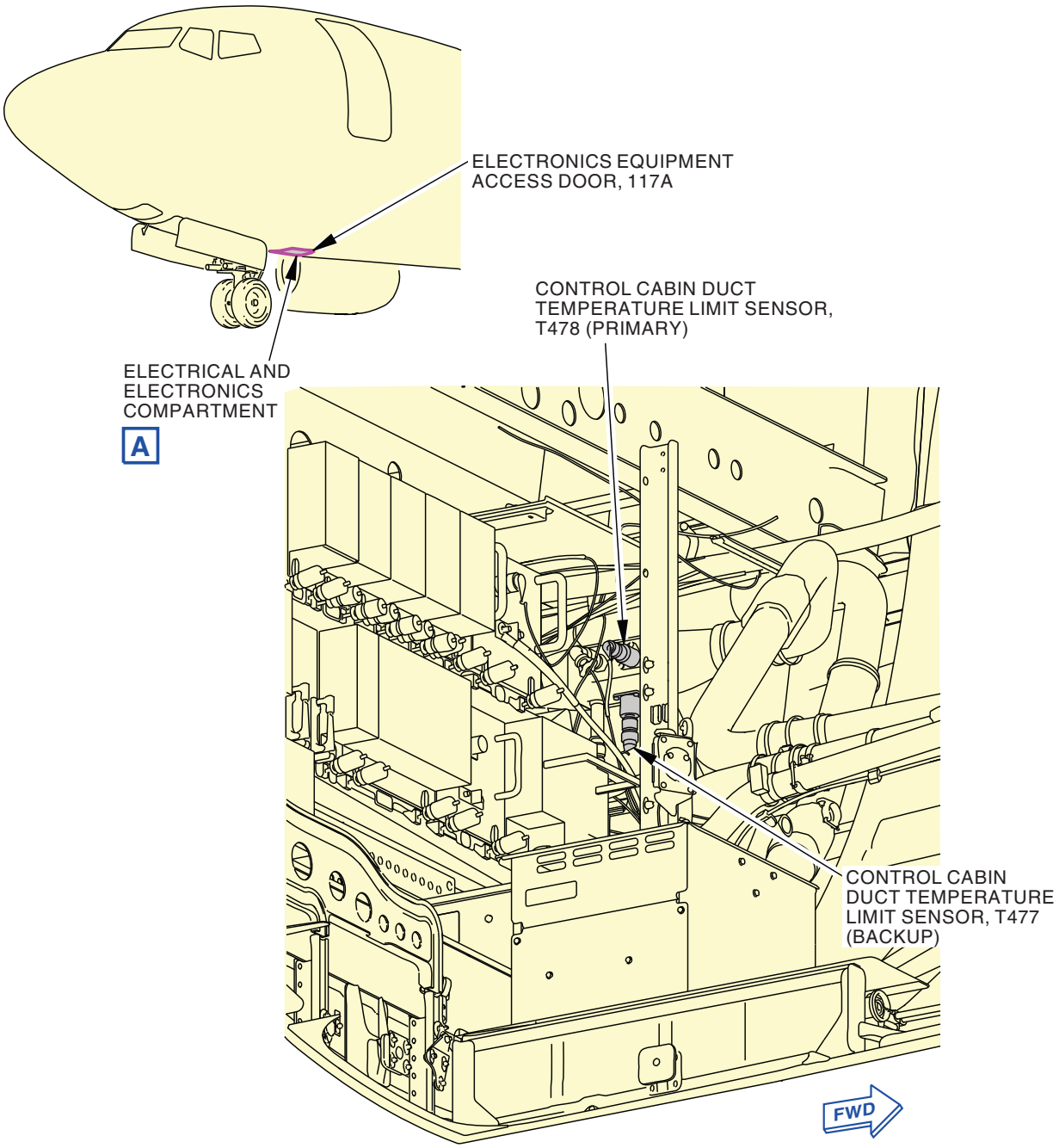
PASSENGER COMPARTMENT
(OVERHEAD VIEW, CEILING PANELS REMOVED)

A

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Passenger Cabin Duct Temperature Limit Sensor Location
Figure 501/21-61-04-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



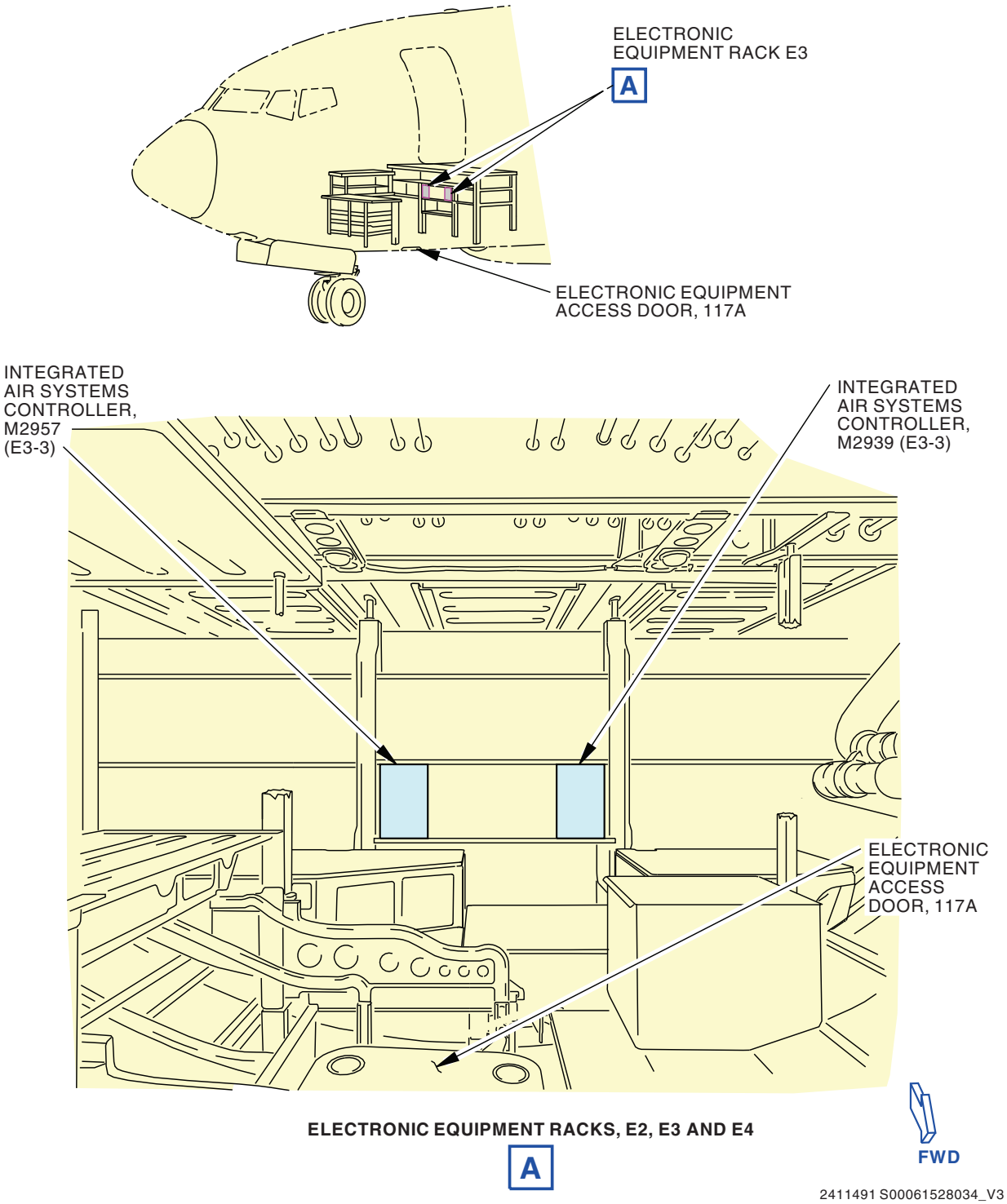
ELECTRICAL AND ELECTRONICS COMPARTMENT (VIEW THROUGH ACCESS DOOR, 117A)

A

2411490 S00061528033_V1

Control Cabin Duct Temperature Limit Sensor Location
Figure 502/21-61-04-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Integrated Air Systems Controller Location
Figure 503/21-61-04-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

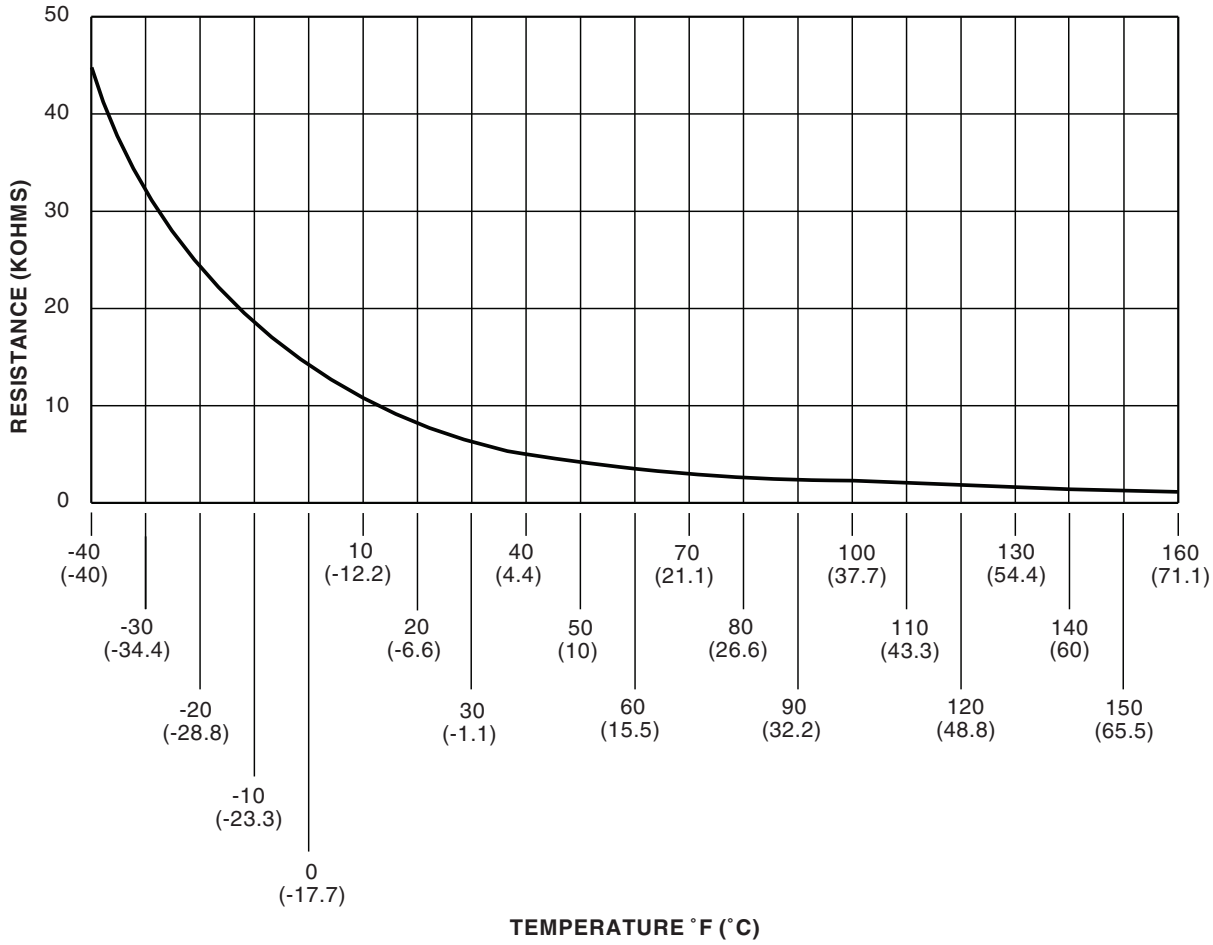
TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)	TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)
-40 (-40)	37,260-51,890	65 (18.3)	2,560-2,680
-35 (-37.2)	32,030-43,880	70 (21.1)	2,300-2,390
-30 (-34.4)	27,610-37,220	75 (23.8)	2,070-2,140
-25 (-31.6)	23,850-31,690	80 (26.6)	1,860-1,920
-20 (-28.8)	20,680-27,030	85 (29.4)	1,660-1,730
-15 (-26.1)	17,960-23,150	90 (32.2)	1,490-1,570
-10 (-23.3)	15,640-19,870	95 (35)	1,330-1,430
-5 (-20.5)	13,650-17,110	100 (37.7)	1,200-1,300
0 (-17.7)	11,960-14,750	105 (40.5)	1,070-1,180
5 (-15)	10,500-12,760	110 (43.3)	974-1,080
10 (-12.2)	9,230-11,070	115 (46.1)	879-977
15 (-9.4)	8,140-9,620	120 (48.8)	795-893
20 (-6.6)	7,190-8,380	125 (51.6)	720-816
25 (-3.8)	6,350-7,340	130 (54.4)	653-747
30 (-1.1)	5,800-6,175	135 (57.2)	593-685
35 (1.6)	5,125-5,425	140 (60)	540-628
40 (4.4)	4,460-4,950	145 (62.7)	492-578
45 (7.2)	3,980-4,360	150 (65.5)	448-532
50 (10)	3,560-3,850	155 (68.3)	435-465
55 (12.7)	3,190-3,400	160 (71.1)	400-428
60 (15.5)	2,860-3,020		

DUCT TEMPERATURE SENSOR P/N 622878-2 (10-62091-14)
RESISTANCE VS TEMPERATURE TABLE

2411492 S00061528035_V2

Duct Temperature Sensor - Temperature and Resistance Data
Figure 504/21-61-04-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**DUCT TEMPERATURE SENSOR
RESISTANCE VS TEMPERATURE GRAPH**

2411493 S00061528036_V1

**Duct Temperature Sensor - Temperature and Resistance Data
Figure 504/21-61-04-990-806 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DUCT OVERHEAT SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the duct overheat switches for these zones:

(a) Control Cabin

(b) Forward Passenger Cabin

(c) Aft Passenger Cabin.

(2) An installation of the duct overheat switches for these zones:

(a) Control Cabin

TASK 21-61-05-000-801

2. Duct Overheat Switch Removal

(Figure 401, Figure 402)

A. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)

B. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 21-61-05-010-001

(1) To get access to the duct overheat switch for the control cabin, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-05-010-002

(2) To get access to the passenger cabin duct overheat switches [1], remove the applicable ceiling panel that is forward of the overwing escape hatches (Main Ceiling Panel - Removal, TASK 25-21-45-000-801).

E. Passenger Cabin Duct Overheat Switch Removal

(Figure 401)

SUBTASK 21-61-05-020-002

(1) Disconnect the electrical connector [2] from the passenger cabin duct overheat switch [1].

SUBTASK 21-61-05-020-003

(2) Remove the passenger cabin duct overheat switch [1].

SUBTASK 21-61-05-020-004

(3) Remove and discard the packing [3] from the passenger cabin duct overheat switch [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Control Cabin Duct Overheat Switch Removal

(Figure 402)

SUBTASK 21-61-05-020-005

(1) Disconnect the electrical connector [23] from the control cabin duct overheat switch [21].

SUBTASK 21-61-05-020-006

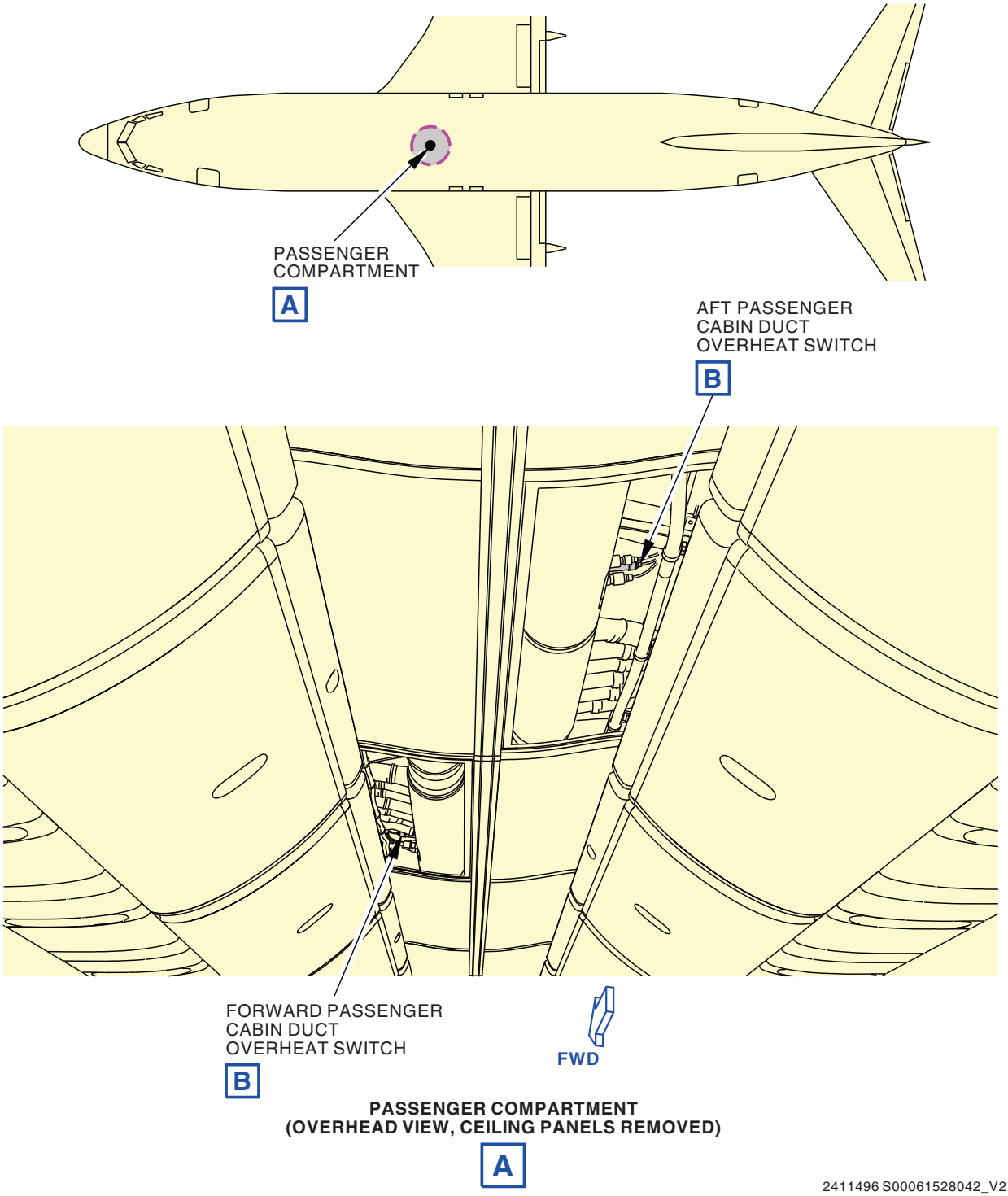
(2) Remove the control cabin duct overheat switch [21].

SUBTASK 21-61-05-020-007

(3) Remove and discard the packing [22] from the control cabin duct overheat switch [21].

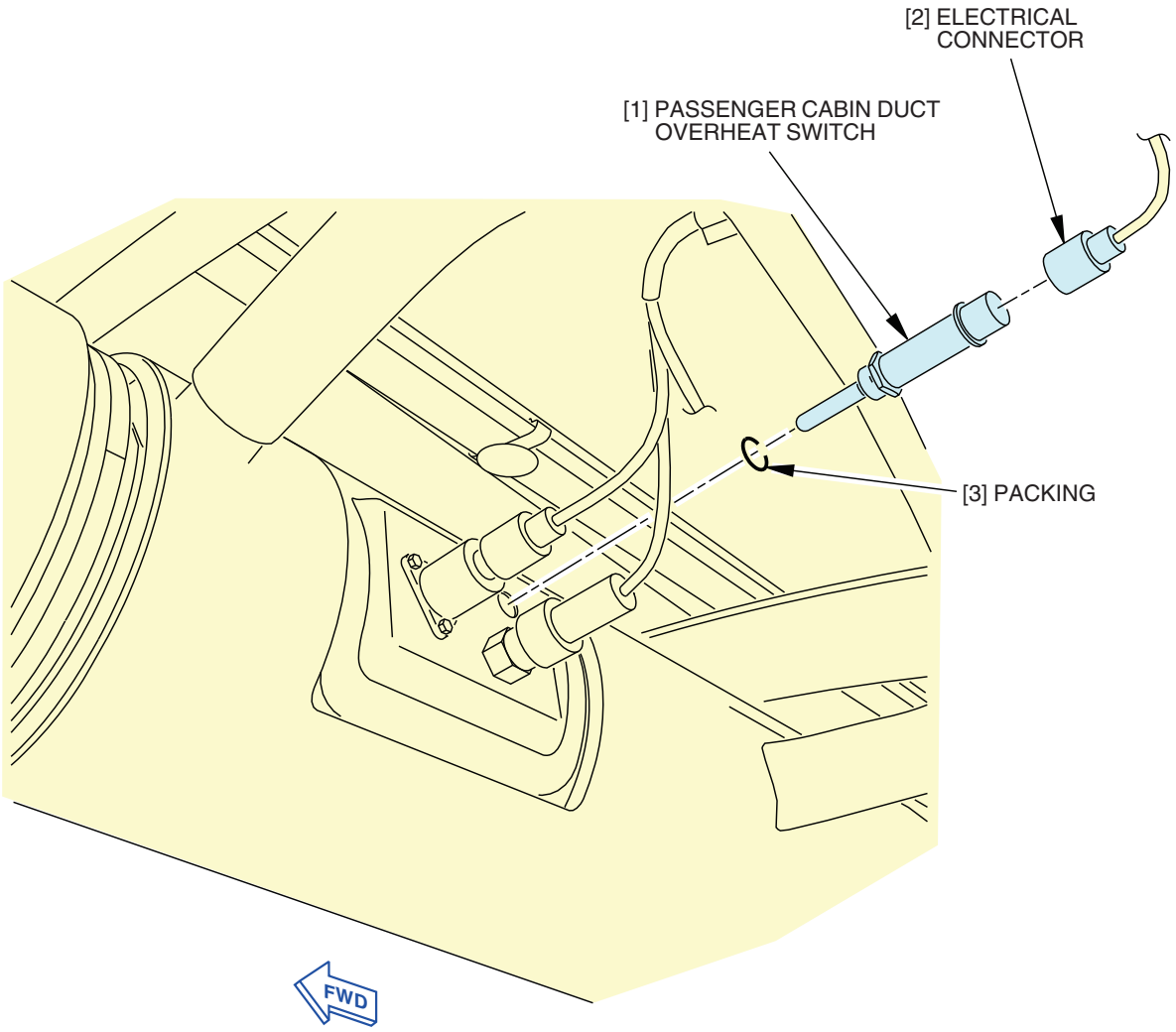
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Passenger Cabin Duct Overheat Switch Installation
Figure 401/21-61-05-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



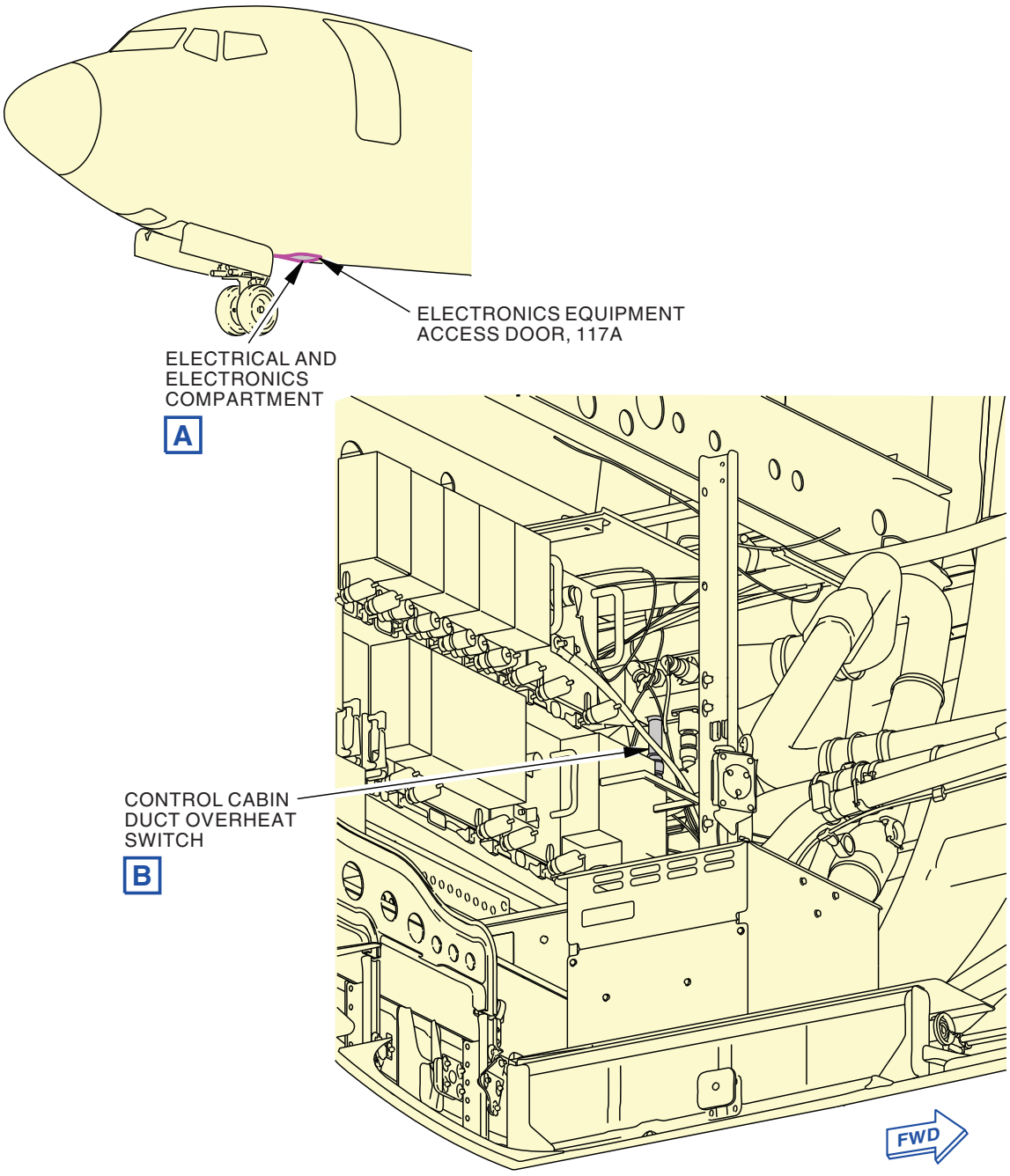
PASSENGER CABIN DUCT OVERHEAT SWITCH (EXAMPLE)

B

2411497 S00061528043_V2

Passenger Cabin Duct Overheat Switch Installation
Figure 401/21-61-05-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



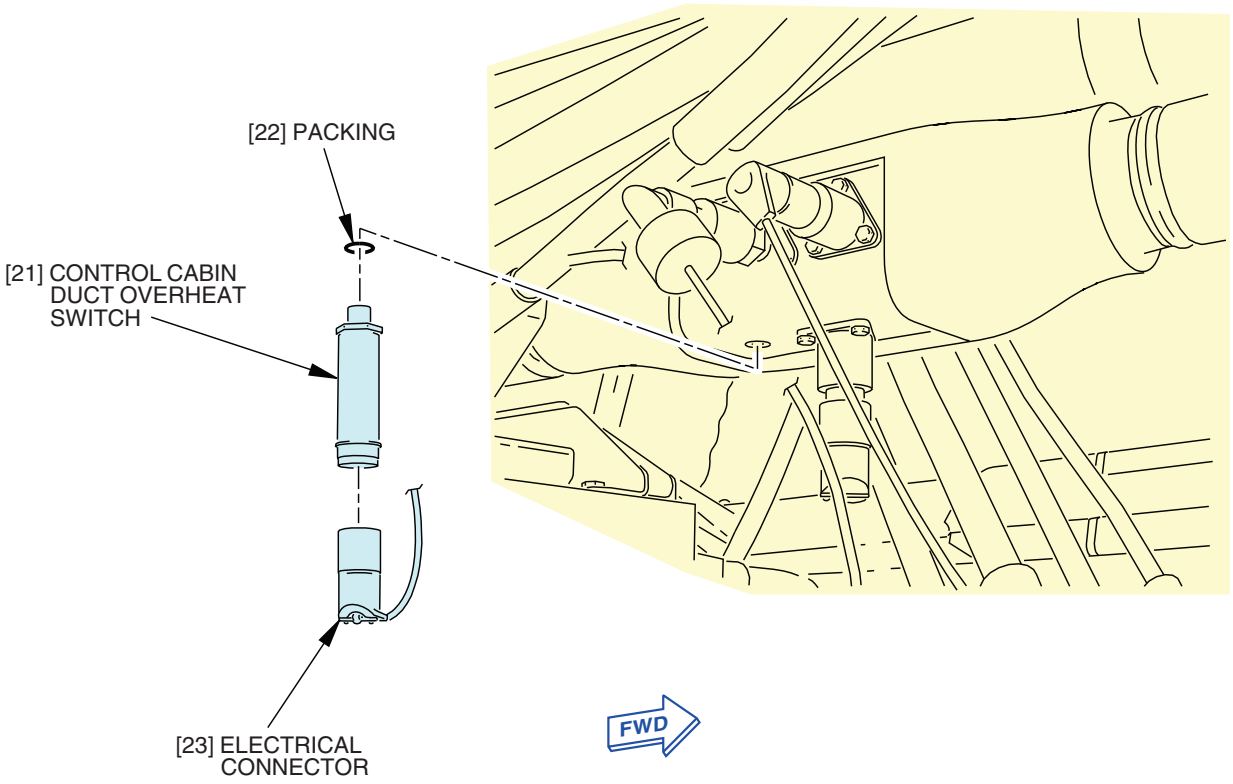
ELECTRICAL AND ELECTRONICS COMPARTMENT (VIEW THROUGH ACCESS DOOR, 117A)

A

2411498 S00061528044_V1

Control Cabin Duct Overheat Switch Installation
Figure 402/21-61-05-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONTROL CABIN DUCT OVERHEAT SWITCH (EXAMPLE)

B

2411499 S00061528045_V2

Control Cabin Duct Overheat Switch Installation
Figure 402/21-61-05-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-05-400-801

3. Duct Overheat Switch Installation

A. References

Reference	Title
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	Packing	21-61-05-09-020	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
120	Subzone - Body Station 396 to Body Station 540
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Passenger Cabin Duct Overheat Switch Installation

(Figure 401)

SUBTASK 21-61-05-420-001

(1) When it is not supplied, install a new packing [3] on the passenger cabin duct overheat switch [1].

SUBTASK 21-61-05-640-001

(2) Put compound, D50004 on the threads of the passenger cabin duct overheat switch [1].

SUBTASK 21-61-05-420-002

(3) Do these steps to install the passenger cabin duct overheat switch [1]:

(a) Put the passenger cabin duct overheat switch [1] in its position in the duct.

(b) Tighten the passenger cabin duct overheat switch [1] until the packing [3] makes contact with the duct.

(c) Tighten the passenger cabin duct overheat switch [1] an additional 3/4 turn.

SUBTASK 21-61-05-420-003

(4) Connect the electrical connector [2] to the passenger cabin duct overheat switch [1].

G. Control Cabin Duct Overheat Switch Installation

(Figure 402)

SUBTASK 21-61-05-420-004

(1) Install a new packing [22] on the control cabin duct overheat switch [21].

SUBTASK 21-61-05-640-002

(2) Put compound, D50004 on the threads of the control cabin duct overheat switch [21].

SUBTASK 21-61-05-420-005

(3) Do these steps to install the control cabin duct overheat switch [21]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put the control cabin duct overheat switch [21] in its position in the duct.

(b) Tighten the control cabin duct overheat switch [21] until the packing [22] makes contact with the duct.

(c) Tighten the control cabin duct overheat switch [21] an additional 3/4 turn.

SUBTASK 21-61-05-420-006

(4) Connect the electrical connector [23] to the control cabin duct overheat switch [21].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-05-010-003

(1) If you replaced the control cabin duct overheat switch [21], do this step:

(a) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-05-010-004

(2) If you replaced a passenger cabin duct overheat switch [1], install the applicable ceiling panel (Main Ceiling Panel - Installation, TASK 25-21-45-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN TEMPERATURE SENSOR ASSEMBLY FILTER - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Cabin Temperature Sensor Assembly Filter Removal

(2) Cabin Temperature Sensor Assembly Filter Cleaning

(3) Cabin Temperature Sensor Assembly Filter Installation.

TASK 21-61-06-000-801

2. Cabin Temperature Sensor Assembly Filter Removal

(Figure 201, Figure 202)

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

Reference	Description
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

B. Consumable Materials

Reference	Description	Specification
B00039	Cleaner - Brulin Formula 815MX	
B00130	Alcohol - Isopropyl	TT-I-735
B00541	Cleaner - General Purpose Household Detergent	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00624	Bag - Plastic, General Purpose	
G50140	Gloves - Protective, Latex or Nitrile	

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
232	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Right
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Prepare for the Removal

SUBTASK 21-61-06-860-001

(1) Do this step for the removal of the filter [1] for the flight compartment zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-06-860-002

(2) Do the step that follows to remove the filter [21] for the forward passenger compartment zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-06-860-003

(3) Do the step that follows to remove the filter [21] for the aft passenger compartment zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

SUBTASK 21-61-06-010-001

(4) Find the filter for the flight compartment cabin temperature sensor assembly on the aft ceiling in the flight compartment (Figure 201).

SUBTASK 21-61-06-860-004

(5) Locate the filter for the passenger compartment directly above the passenger seats on the right side of the aisle (Figure 202).

SUBTASK 21-61-06-940-001

WARNING

PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(6) Before you touch the cabin temperature sensor filter [1] or filter [21], put on the personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140.

NOTE: Nitrile gloves are preferred when in contact with alcohol.

SUBTASK 21-61-06-500-001

(7) Select a plastic bag, G00624, larger than the filter(s) and tape suitable for sealing the bag closed.

E. Flight Compartment Zone - Cabin Temperature Sensor Assembly Filter Removal

(Figure 201)

SUBTASK 21-61-06-020-001

(1) Do the step that follows to remove the filter [1] in the flight compartment:

(a) Remove the four screws [2] that attach the grille [3] to the aft ceiling in the flight compartment.

(b) Remove the grille [3] and filter [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT LET THE FILTER TOUCH YOUR SKIN. DO NOT SHAKE OR HIT THE FILTER. DO NOT LET THE FILTER FALL. DO NOT USE COMPRESSED AIR TO CLEAN THE FILTER OR FILTER HOUSING. THIS CAN CAUSE THE INFECTIOUS MATERIAL TO BECOME AIRBORNE. DISCARD THE FILTER IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(c) If possible, slide the plastic bag, G00624, over the filter when removing the filter [1] from the grille [3] to minimize contamination.

(d) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(e) Refer to the airline, local health, safety and regulatory procedures for the disposal of material.

F. Passenger Compartment Zone - Cabin Temperature Sensor Assembly Filter Removal

(Figure 202)

SUBTASK 21-61-06-020-002

(1) Do the step that follows to remove the filter [21] in the passenger compartment:

CAUTION DO NOT APPLY TOO MUCH FORCE TO REMOVE THE GRILLE. THE GRILLE WILL COME OUT EASILY IF YOU PUSH ON THE TAB LIGHTLY WITH A TOOL OF THE CORRECT DIMENSION. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE GRILLE OR THE PANEL.

(a) Put a metal rod with a blunt end, approximately 1/16 in. (1.6 mm) diameter, into one of the slots on the edge of the grille [22].

(b) Apply pressure to the rod end to release the tab that holds the grille [22] onto the ceiling panel.

(c) Remove the grille [22] and filter [21].

WARNING DO NOT LET THE FILTER TOUCH YOUR SKIN. DO NOT SHAKE OR HIT THE FILTER. DO NOT LET THE FILTER FALL. DO NOT USE COMPRESSED AIR TO CLEAN THE FILTER OR FILTER HOUSING. THIS CAN CAUSE THE INFECTIOUS MATERIAL TO BECOME AIRBORNE. DISCARD THE FILTER IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(d) If possible, slide the plastic bag, G00624, over the filter when removing the filter [21] from the grille [22] to minimize contamination.

(e) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(f) Refer to the airline, local health, safety and regulatory procedures for the disposal of material.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Cabin Temperature Sensor Assembly Grille Cleaning

SUBTASK 21-61-06-110-003

(1) Prepare a cleaning solution of water and general purpose household detergent cleaner, B00541, or Brulin Formula 815MX cleaner, B00039.

(a) Follow the mixing instructions for the cleaner.

SUBTASK 21-61-06-110-004

(2) Clean the grille with the cleaning solution.

SUBTASK 21-61-06-100-005

(3) If the cleaning solution did not remove all dirt or contamination from the grille, then clean the grille with a cotton wiper, G00034, that is moist with alcohol, B00130.

SUBTASK 21-61-06-100-006

(4) Flush the grille with clean water until all of the cleaning solution or alcohol has been removed.

SUBTASK 21-61-06-100-007

(5) Dry the grille with cotton wiper, G00034.

H. Post-Filter Removal Cleanup

SUBTASK 21-61-06-670-002

WARNING

ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(1) Disinfect the air grille [3] and air grilles [22] with a solution of 70% alcohol, B00130.

NOTE: Follow the manufacturer's instruction and refer to the local health authority guidance for required contact/dwell time.

(a) Use pre-moistened cotton wiper, G00034.

1) Make sure that the wiper is not over-saturated not to let the alcohol drips and soaks into crevices or porous surfaces.

SUBTASK 21-61-06-670-001

(2) Disinfect all tools used for filter replacement.

NOTE: Refer to the airline procedure and policies for the disinfecting of tools.

SUBTASK 21-61-06-020-005

WARNING

DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(3) Remove the personal protective equipment, put them in a plastic bag, G00624, with any used wipers and seal the bag closed with tape.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(a) Discard the bag with the personal protective equipment.

NOTE: Refer to the airline, local health, safety and regulatory procedures for disposal of material.

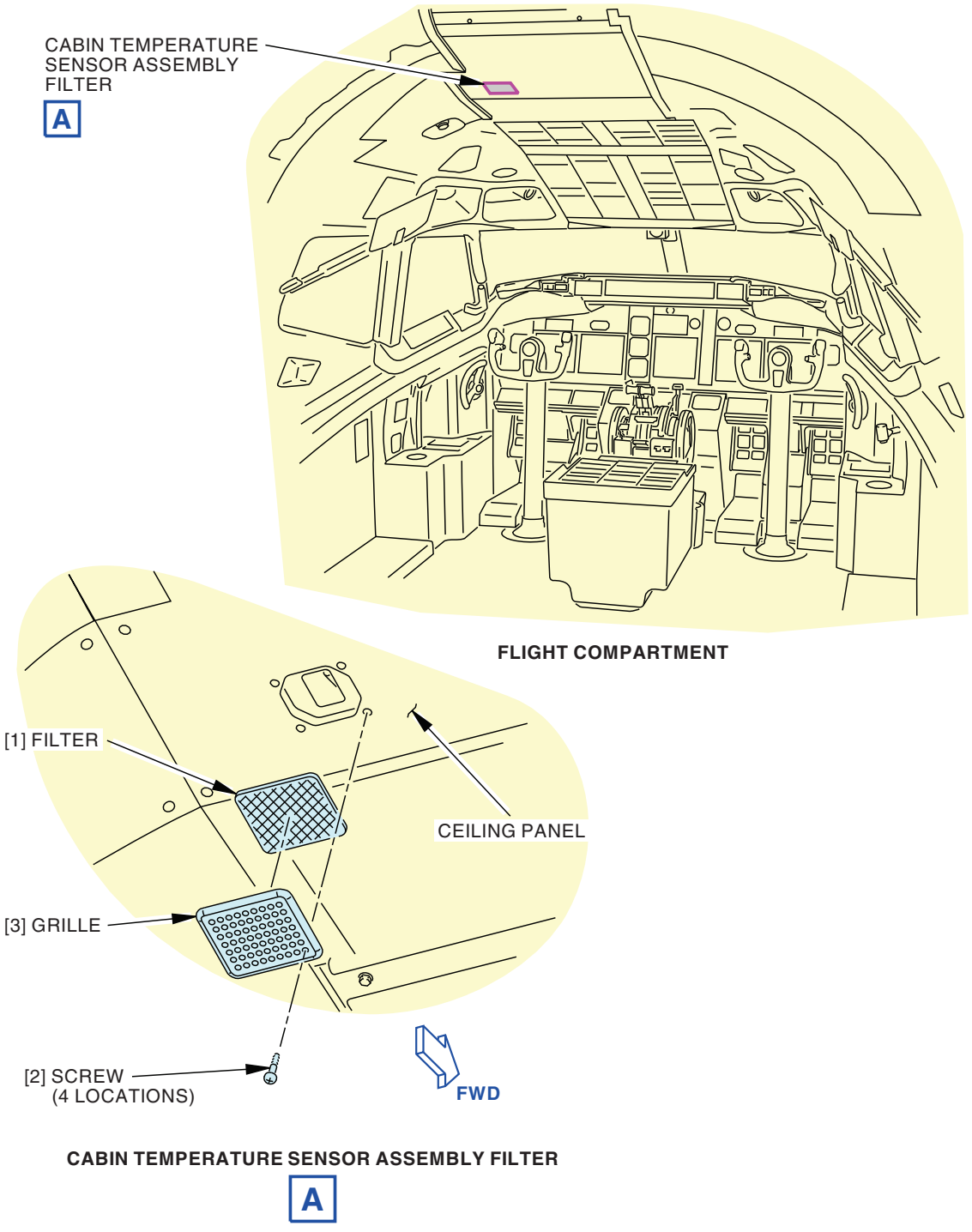
SUBTASK 21-61-06-110-005

WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(4) Clean your hands with soap and running hot water.

——— END OF TASK ———

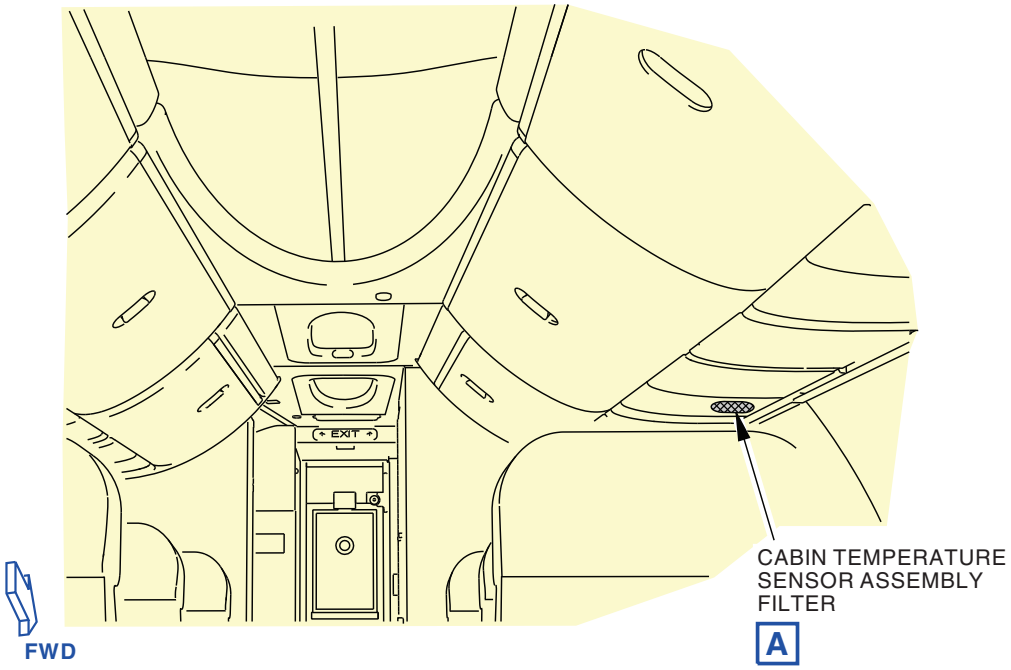
ECCN 9E991 BOEING PROPRIETARY - See title page for details



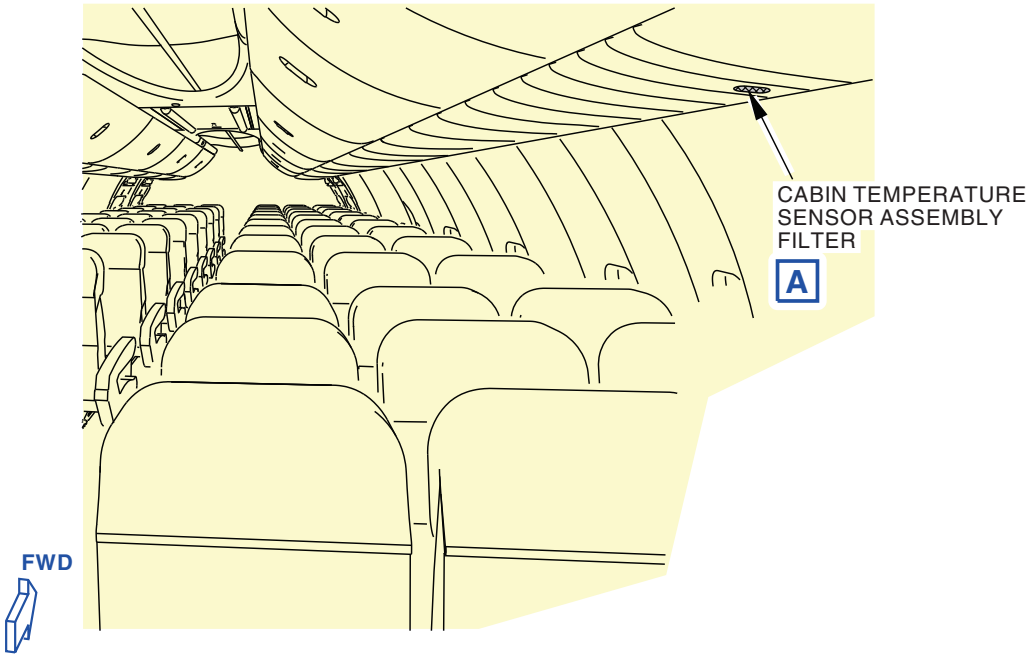
2411500 S00061528050_V1

Flight Compartment - Cabin Temperature Sensor Assembly Filter Installation
Figure 201/21-61-06-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD PASSENGER COMPARTMENT

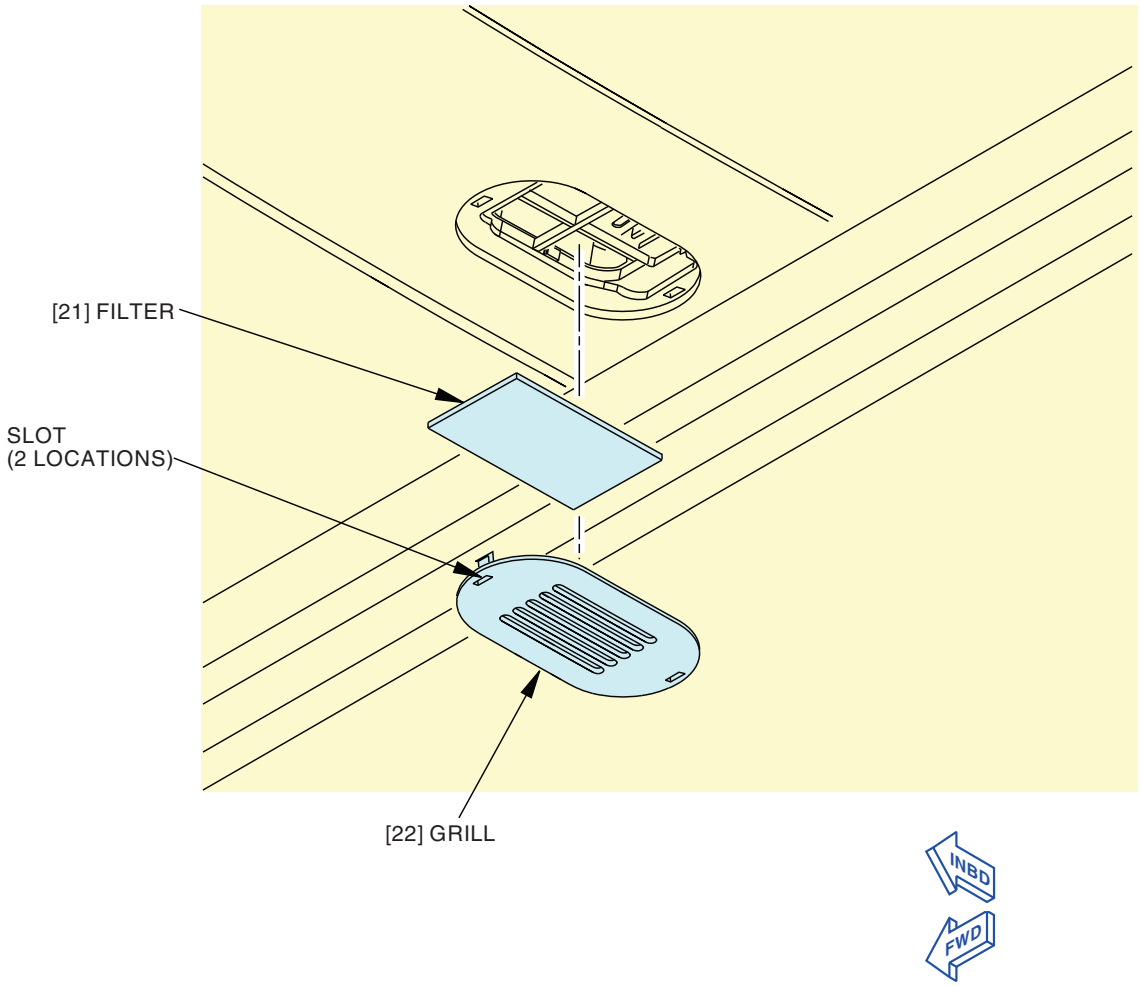


AFT PASSENGER COMPARTMENT

2411501 S00061528051_V2

Cabin Temperature Sensor Assembly Filter Installation
Figure 202/21-61-06-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE SENSOR ASSEMBLY FILTER (EXAMPLE)

A

2411502 S00061528052_V1

Cabin Temperature Sensor Assembly Filter Installation
Figure 202/21-61-06-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-06-100-801

3. Cabin Temperature Sensor Assembly Filter Cleaning

(Figure 201, Figure 202)

A. Tools/Equipment

Reference	Description
STD-1129	Coveralls - Tyvek/Saranex 23-p (Approved Cotton Substitute), No Pockets
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented

B. Consumable Materials

Reference	Description	Specification
B00039	Cleaner - Brulin Formula 815MX	
B00130	Alcohol - Isopropyl	TT-I-735
B00541	Cleaner - General Purpose Household Detergent	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00624	Bag - Plastic, General Purpose	
G02472	Filter - 3/16 inch, Porosity Grade 45 Ppi, Color-Gray, Flame Retardent Foam	
G50140	Gloves - Protective, Latex or Nitrile	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Filter	21-61-06-01-020	PTF ALL
21	Filter	21-61-06-02-005	PTF ALL

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
232	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Right
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

E. Prepare for the Cleaning

SUBTASK 21-61-06-100-008

WARNING PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(1) Make sure that personal protection equipment is used when handling the barrier filter.

(a) particulate respirator, STD-6435

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) coverall (approved cotton substitute), STD-1129

(c) safety goggles, STD-6437

(d) protective gloves, G50140

SUBTASK 21-61-06-020-003

(2) Remove the filter [1] for the temperature sensor assembly in the flight compartment (TASK 21-61-06-000-801).

SUBTASK 21-61-06-020-004

(3) Remove the filter [21] for the temperature sensor assembly in the passenger compartment (TASK 21-61-06-000-801).

SUBTASK 21-61-06-210-001

(4) If the filter is damaged, discard the filter.

SUBTASK 21-61-06-350-001

(5) If necessary, cut a new filter from foam filter, G02472, to the same dimensions as the filter you removed.

NOTE: The old filter can be used as a template to prepare a new filter.

F. Cabin Temperature Sensor Assembly Filter Cleaning

SUBTASK 21-61-06-110-001

(1) Prepare a cleaning solution of water and general purpose household detergent cleaner, B00541, or Brulin Formula 815MX cleaner, B00039.

(a) Follow the mixing instructions for the cleaner.

SUBTASK 21-61-06-110-002

(2) Clean the grille and the filter with the cleaning solution.

NOTE: It is not necessary to clean a new filter.

SUBTASK 21-61-06-100-001

(3) If the cleaning solution did not remove all dirt or contamination from the grille, then clean the grille with a cotton wiper, G00034, that is moist with alcohol, B00130.

SUBTASK 21-61-06-100-002

(4) Flush the grille and the filter with clean water until you remove all traces of the cleaning solution or alcohol, B00130.

SUBTASK 21-61-06-100-003

(5) Dry the grille with cotton wiper, G00034.

SUBTASK 21-61-06-100-004

(6) Dry the filter with compressed air or a cotton wiper, G00034.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-06-420-001

(1) Install the filter [1] for the temperature sensor assembly in the flight compartment (TASK 21-61-06-400-801).

SUBTASK 21-61-06-420-002

(2) Install the filter [21] for the temperature sensor assembly in the passenger compartment (TASK 21-61-06-400-801).

H. Post-Filter Maintenance Cleanup

SUBTASK 21-61-06-100-009

(1) Make sure that you fully clean around the area in which you cleaned the filter.

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SUBTASK 21-61-06-670-003

(2) Disinfect all tools used for filter maintenance.

NOTE: Refer to the airline procedure and policies for the disinfecting of tools.

SUBTASK 21-61-06-100-010

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(3) Remove the personal protective equipment and put them in a plastic bag, G00624.

(a) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(b) Discard the bag with the personal protective equipment.

NOTE: Refer to the airline, local health, safety and regulatory procedures for disposal of material.

(4) Clean you hands with soap and water.

END OF TASK

TASK 21-61-06-400-801

4. Cabin Temperature Sensor Assembly Filter Installation

NOTE: This procedure is a scheduled maintenance task.

A. Consumable Materials

Reference	Description	Specification
G02472	Filter - 3/16 inch, Porosity Grade 45 Ppi, Color-Gray, Flame Retardent Foam	

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Filter	21-61-06-01-020	PTF ALL
21	Filter	21-61-06-02-005	PTF ALL

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
232	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Right
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Flight Compartment Zone - Cabin Temperature Sensor Assembly Filter Installation

(Figure 201)

SUBTASK 21-61-06-350-002

(1) If applicable, cut the new filter from the foam filter, G02472, the same size as the removed filter.

NOTE: Use brand new filter as a template because using bagged filter as a template may lead to incorrect dimension or risk contact with infectious material.

SUBTASK 21-61-06-420-003

(2) Do the steps that follow to install the filter [1] for the cabin temperature sensor assembly in the flight compartment:

(a) Put a new filter [1] in its position in the aft ceiling panel.

(b) Put the grille [3] in its position.

(c) Install the four screws [2] that attach grille [3] to the aft ceiling panel.

E. Passenger Compartment Zone - Cabin Temperature Sensor Assembly Filter Installation

(Figure 202)

SUBTASK 21-61-06-350-003

(1) If applicable, cut the new filter from the foam filter, G02472, the same size as the removed filter.

NOTE: Use brand new filter as a template because using bagged filter as a template may lead to incorrect dimension or risk contact with infectious material.

SUBTASK 21-61-06-420-004

(2) Do the steps that follow to install the filter [21] for the temperature sensor assembly in the passenger compartment:

(a) Put a new filter [21] in its position in the assembly.

(b) Align index tabs on the grille [22] with the notches on the temperature sensor assembly.

(c) Push the grille [22] into the temperature sensor assembly to engage the two tabs.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-06-860-005

(1) If you installed the filter [1] for the flight compartment zone, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

SUBTASK 21-61-06-860-006

(2) If you installed the filter [21] for the forward passenger compartment zone, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-06-860-007

(3) If you installed the filter [21] for the aft passenger compartment zone, do this step:

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(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN TEMPERATURE SELECTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Cabin temperature selector removal.

(2) Cabin temperature selector installation.

B. There are three cabin temperature selectors installed in the P5-17 Cabin Temperature Panel. The removal and installation procedure is the same for the three cabin temperature selectors.

C. The P5-17 Cabin Temperature Panel is found in the P5 Forward Overhead Panel.

TASK 21-61-07-000-801

2. Cabin Temperature Selector Removal

(Figure 401)

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 21-61-07-840-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-61-07-860-001

(2) Put the L and R PACK switches on the P5-10 Air Conditioning Panel to the OFF position.

C. Cabin Temperature Selector Removal

SUBTASK 21-61-07-010-001

(1) Do these steps to get access to the electrical connectors [1] on the back of the cabin temperature panel [3]:

(a) Loosen the 1/4-turn fasteners that hold the aft end of the P5 forward overhead panel in place.

NOTE: The forward end of the P5 panel is hinged. The aft end of the P5 panel will rotate downward when the fasteners are loose.

(b) Release the safety latch and carefully lower the P5 forward overhead panel to the open position.

SUBTASK 21-61-07-020-001

(2) Do the steps that follow to remove the cabin temperature selector [2] from the cabin temperature panel [3] (P5-17):

(a) Disconnect the electrical connector [1] from the cabin temperature selector [2].

(b) Loosen the setscrews that hold the knob [6] to the cabin temperature selector [2].

(c) Remove the knob [6] from the cabin temperature selector [2].

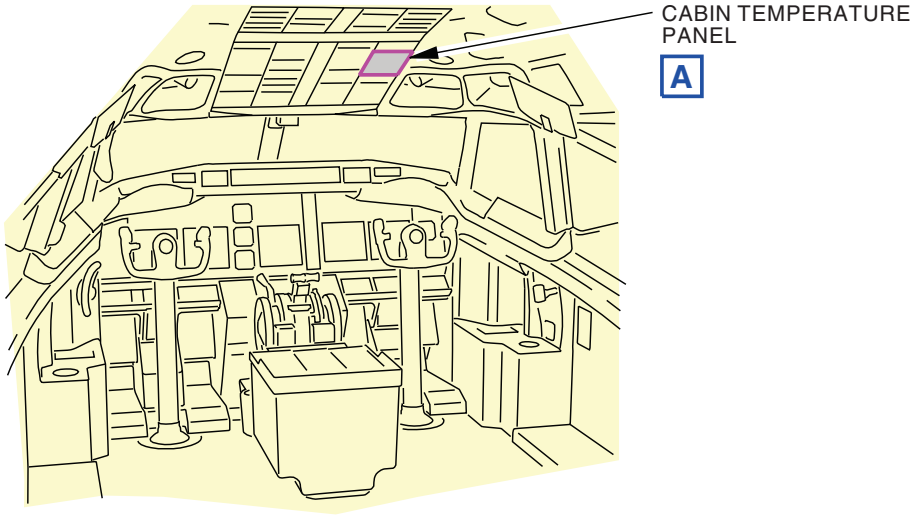
(d) Remove the nut [5] and washer [4] from the cabin temperature selector [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

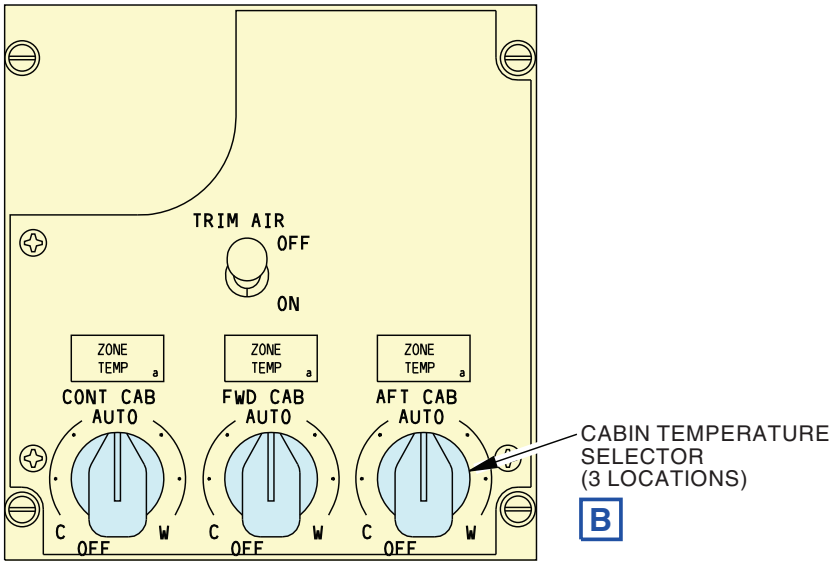
(e) Carefully pull the cabin temperature selector [2] out of the cabin temperature panel [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



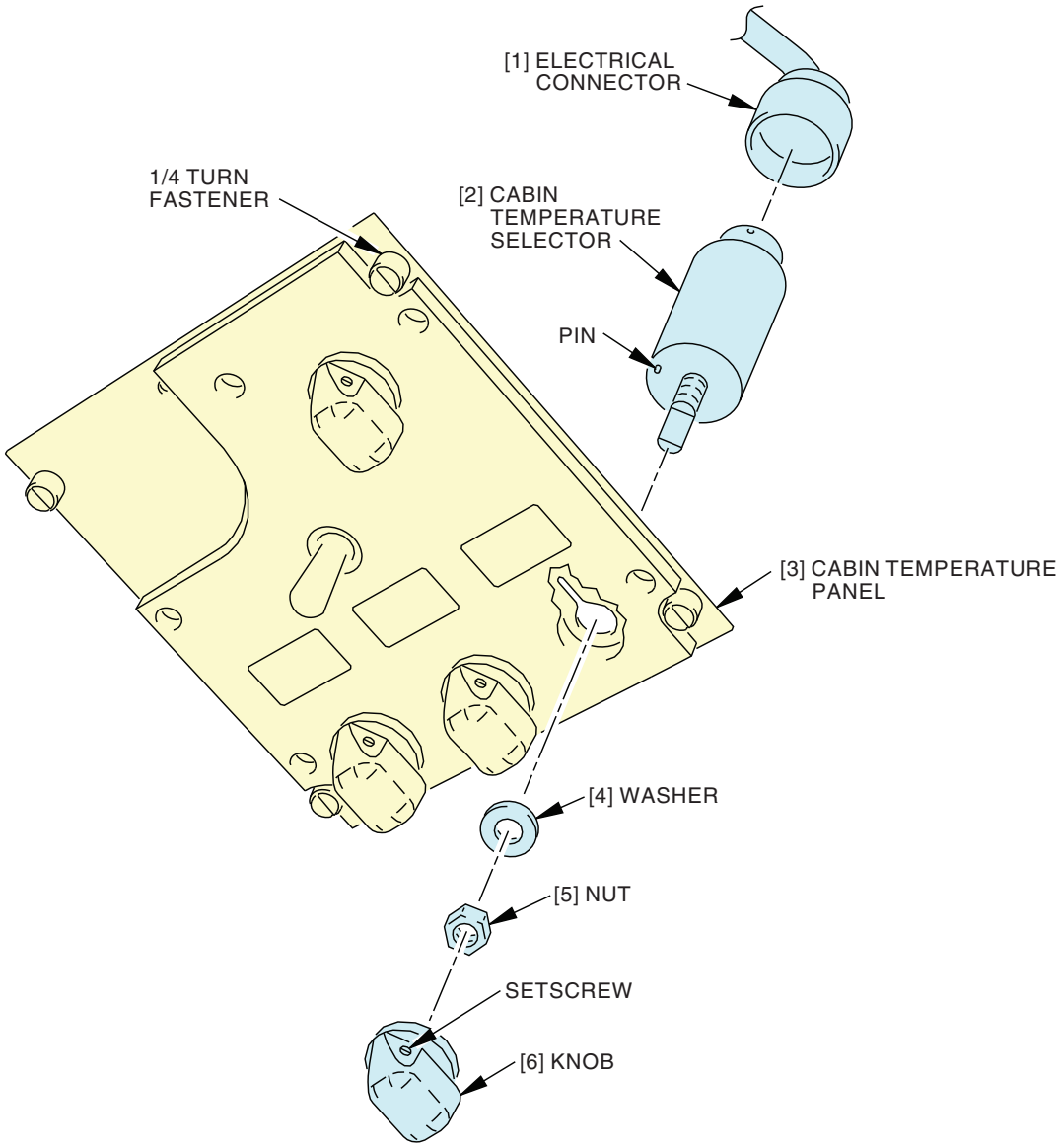
CABIN TEMPERATURE PANEL
(EXAMPLE)

A

2411503 S00061528058_V2

Cabin Temperature Selector Installation
Figure 401/21-61-07-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE SELECTOR (EXAMPLE)

B

2411504 S00061528059_V2

Cabin Temperature Selector Installation
Figure 401/21-61-07-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-07-400-801

3. Cabin Temperature Selector Installation

(Figure 401)

A. References

Reference	Title
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Cabin Temperature Selector Installation

SUBTASK 21-61-07-420-001

(1) To install the cabin temperature selector [2] in the cabin temperature panel [3] (P5-17), do these steps:

(a) Carefully push the cabin temperature selector [2] into its position in the cabin temperature panel [3].

1) Make sure that the pin on the front of the cabin temperature selector [2] is in the slot on the cabin temperature panel [3].

(b) Install the washer [4] and nut [5] on the cabin temperature selector [2].

(c) Put the knob [6] on the cabin temperature selector [2].

(d) Tighten the setscrews that hold the knob [6] to the cabin temperature selector [2].

(e) Connect the electrical connector [1] to the cabin temperature selector [2].

SUBTASK 21-61-07-410-001

(2) Carefully lift the P5 forward overhead panel to the closed position.

SUBTASK 21-61-07-410-002

(3) Tighten the 1/4-turn fasteners that hold the aft end of the P5 forward overhead panel in position.

D. Cabin Temperature Selector Installation Test

SUBTASK 21-61-07-840-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-61-07-860-002

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-61-07-860-003

(3) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

SUBTASK 21-61-07-710-001

(4) Put the R PACK switch and the L PACK switch on the P5-10 Air Conditioning Panel in the AUTO position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-07-710-002

(5) Put the TRIM AIR switch, on the P5-17 Cabin Temperature Panel, to the ON position.

SUBTASK 21-61-07-710-003

(6) Put the cabin temperature selector [2] in the AUTO position.

(a) Make sure that the airflow from the conditioned air outlets in the applicable cabin/zone becomes warm.

SUBTASK 21-61-07-710-004

(7) Put the cabin temperature selector [2] in the fully Cool position.

(a) Make sure that the airflow from the conditioned air outlets in the applicable cabin/zone becomes cool.

SUBTASK 21-61-07-710-005

(8) Hold the cabin temperature selector [2] in the fully Warm position.

(a) Make sure that the airflow from the conditioned air outlets in the applicable cabin/zone becomes warm.

SUBTASK 21-61-07-810-001

(9) If you find a problem with the cabin temperature selector [2], refer to the IFIM and do the applicable procedure(s) to correct the problem.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-07-710-006

(1) Put the cabin temperature selector [2] in the NORMAL position.

SUBTASK 21-61-07-860-004

(2) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRIM AIR CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the trim air check valve

(2) An installation of the trim air check valve.

(3) There is a trim air check valve for each air conditioning pack. The trim air check valves are installed in the aft inboard area of the Environmental Control System (ECS) bays.

TASK 21-61-09-000-801

2. Trim Air Check Valve Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-24-000-801	Ram Air Ducts Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-61-09-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-61-09-010-001

(2) To get access to the left trim air check valve, open this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-61-09-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(3) To get access to the right trim air check valve, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-61-09-860-002

(4) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach DO NOT OPERATE tag, STD-858.

SUBTASK 21-61-09-020-001

(5) To get access to the trim air check valve [3], remove the applicable ram air exhaust duct [5] (TASK 21-51-24-000-801), if it is necessary.

F. Trim Air Check Valve Removal

SUBTASK 21-61-09-020-004

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

CAUTION

MAKE SURE THAT YOU DO NOT BEND OR TWIST THE ADJACENT DUCTS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(1) If it is necessary, to get more access to the trim air check valve [3], do these steps:

(a) Remove the duct assembly [9]:

1) Remove the two couplings [6].

2) Remove the coupling [14].

3) Remove the clamp [8].

4) Remove the duct assembly [9].

(b) Adjust the duct assembly [12]:

1) Remove the coupling [10].

2) Remove the clamp [11].

(c) Put protective covers or blanks on all open ends of the ducts and valves.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-09-020-002

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(2) To remove the trim air check valve [3], do these steps:

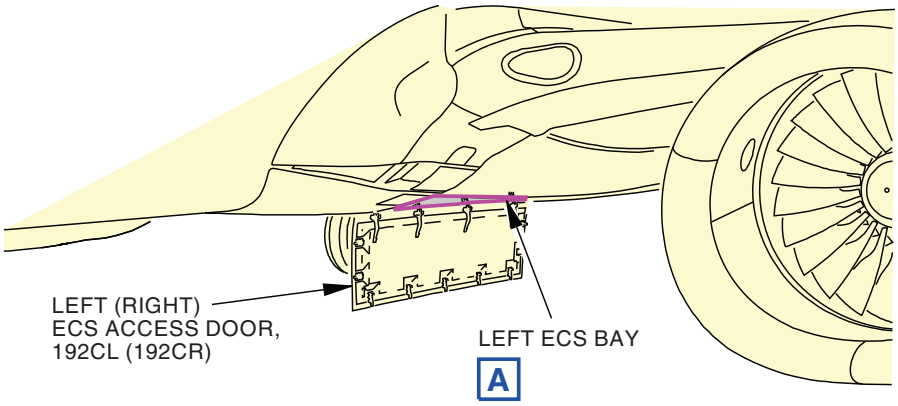
(a) Loosen the two clamps [4].

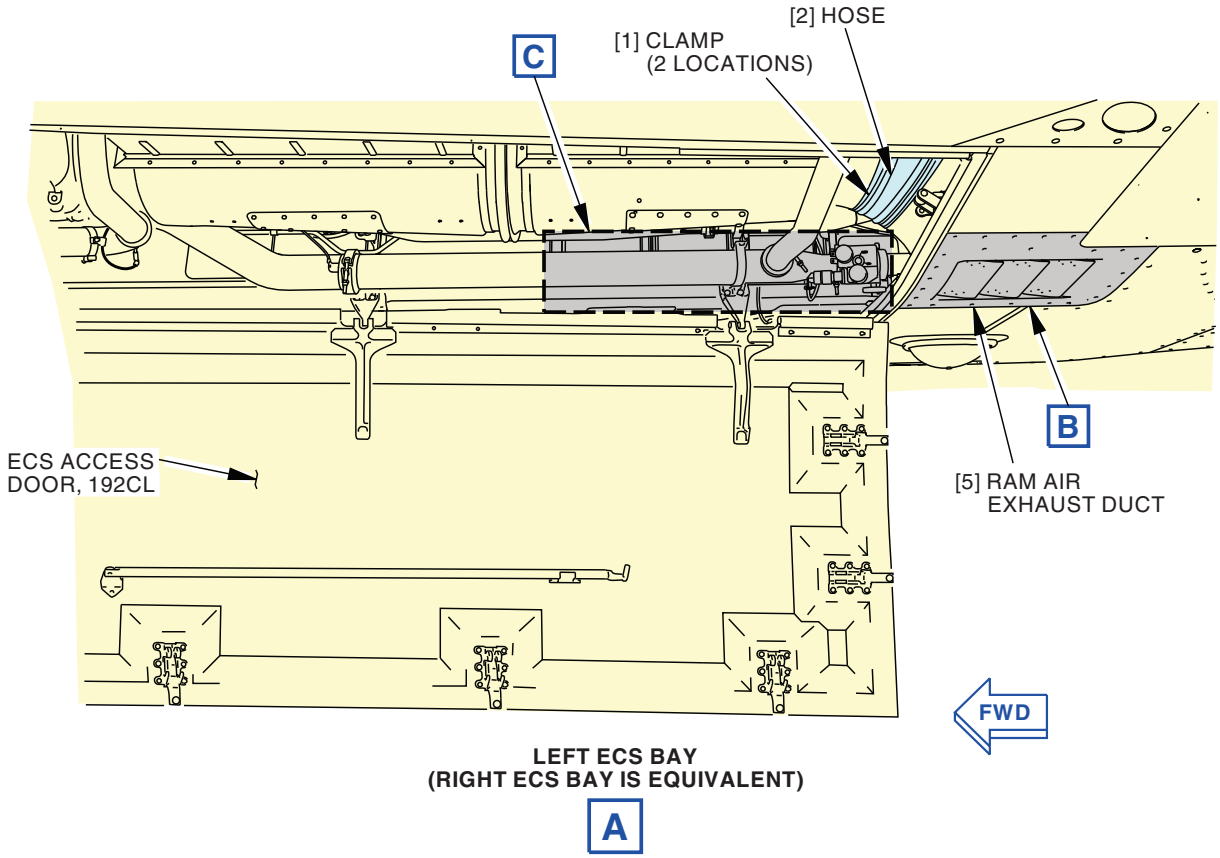
(b) Move the clamps [4] to the adjacent ducts.

(c) Remove the trim air check valve [3] from the ECS bay.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

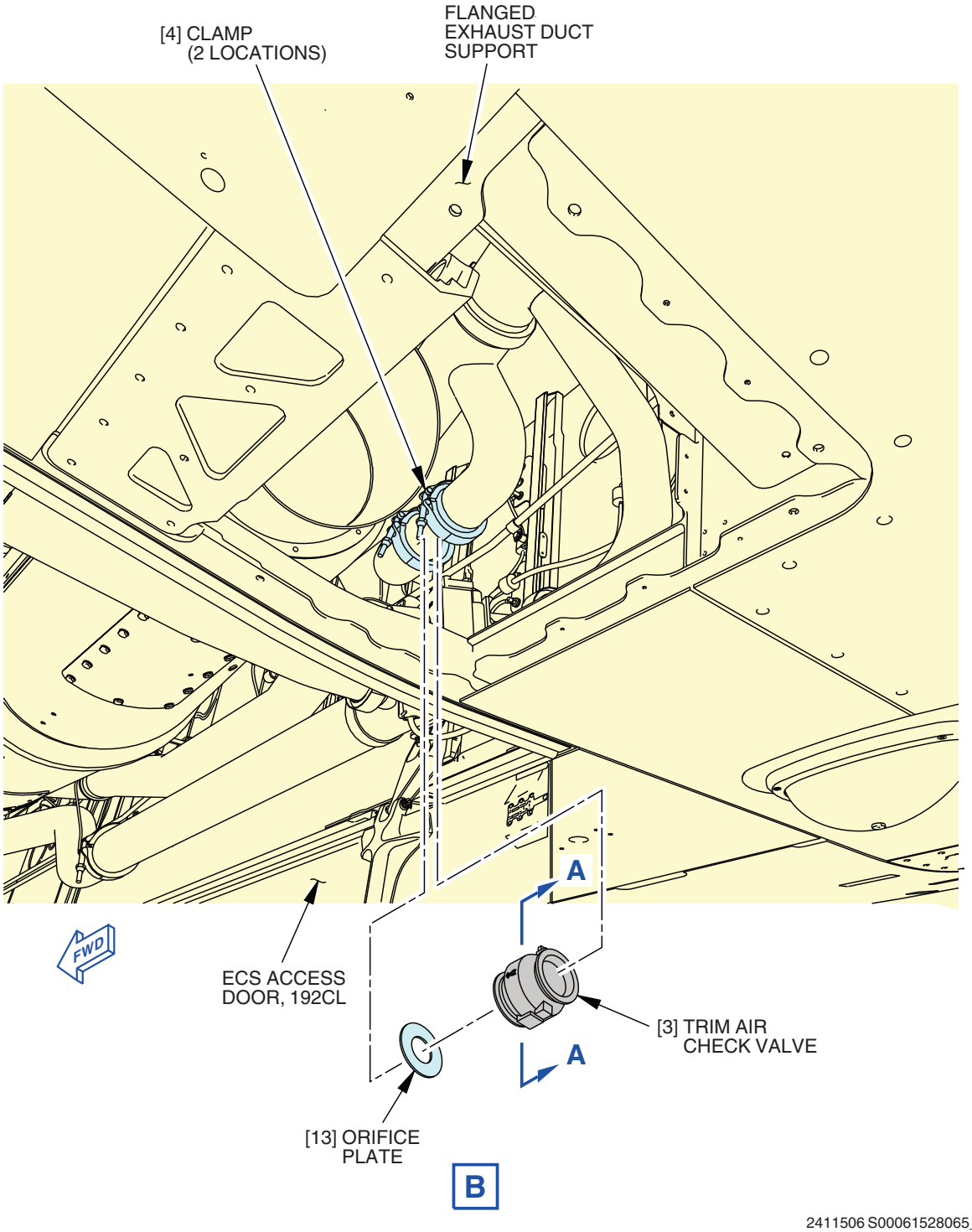




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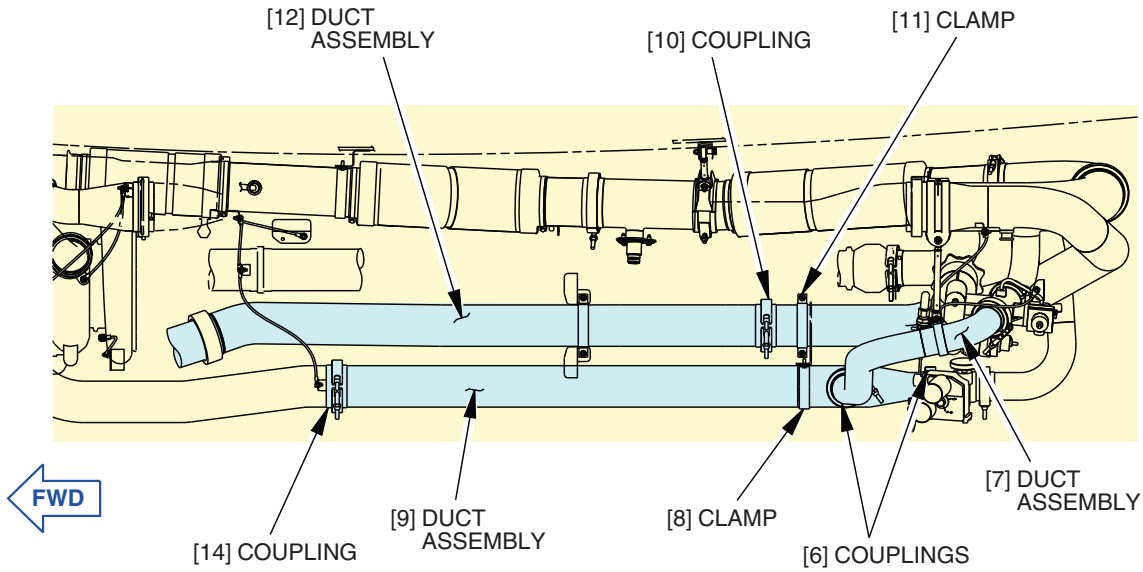
Trim Air Check Valve Installation
Figure 401/21-61-09-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



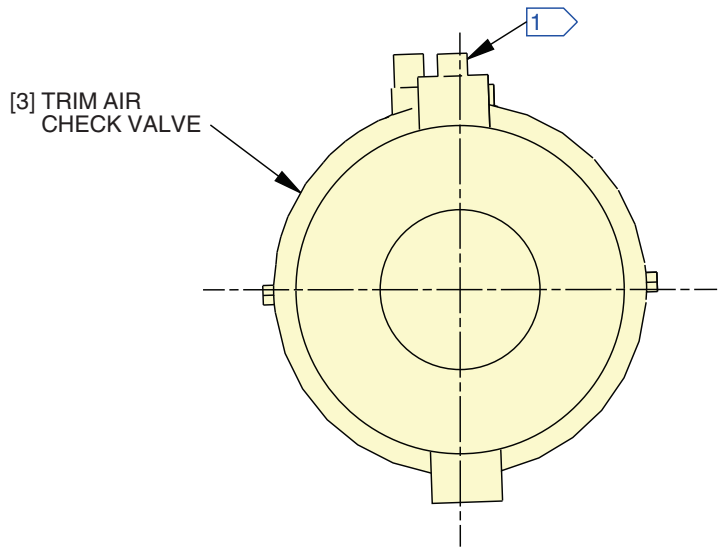
Trim Air Check Valve Installation
Figure 401/21-61-09-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT ECS BAY
(RIGHT ECS BAY IS EQUIVALENT)

C



(EXAMPLE)

A-A

1 ORIENT CHECK VALVE PORTS UP

3018972 S0000791312_V2

Trim Air Check Valve Installation
Figure 401/21-61-09-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-09-400-801

3. Trim Air Check Valve Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00281	Adhesive - Dow Corning 3145 RTV	MIL-A-46146 (BAC5010 Type 79)
A50126	Primer - Adhesive Bonding (Formerly Dow Corning 1204 Clear)	
B00130	Alcohol - Isopropyl	TT-I-735
C50242	Primer - Dow Corning PR-1204 RTV Prime Coat	
G02428	Abrasive - Aluminum Oxide Paper, 150 grit	A-A-1048
G50262	Wiper - Cleaning	BMS15-5

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Trim air check valve	21-61-09-01-200	PTF ALL
		21-61-09-01-210	PTF ALL
13	Orifice plate	21-61-09-01-205	PTF ALL

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Trim Air Check Valve Installation

SUBTASK 21-61-09-420-001

(1) Install the trim air check valve [3] as follows:

(a) Install the orifice plate [13] using one of the methods below:

1) Method 1, install the orifice plate [13] un-bonded to the outlet side of the trim air check valve [3].

2) Method 2, install the orifice plate [13] bonded to the outlet side of the trim air check valve [3]. Bonding is optional and not required for proper operation.

NOTE: Bonding is optional and not required for proper operation.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) If bonding is preferred, bond the orifice plate [13] to the trim air check valve [3] as follows:

<1> Use wiper, G50262, moist in alcohol, B00130, to clean the mating surfaces of the orifice plate [13] and trim air check valve [3].

<a> If it is necessary, use clean wiper, G50262, to remove remaining solvent.

<2> Clean the mating surface of the orifice plate [13] with 150 grit abrasive paper, G02428.

<a> Use wiper, G50262, moist in alcohol, B00130, to clean the mating surface of the orifice plate [13].

<b> If it is necessary, use clean wiper, G50262, to remove remaining solvent.

<3> Apply PR 1204 primer, A50126, or Dow Corning PR-1204 prime coat, C50242, to mating surfaces of the orifice plate [13] and trim air check valve [3].

<a> Let the primer dry for a minimum of 60 minutes at 65°F (18.3°C) - 100°F (37.8°C).

<4> Apply Dow Corning 3145 RTV adhesive, A00281, to the mating surfaces of the orifice plate [13] and trim air check valve [3].

<5> Attach the orifice plate [13] to the outlet flange of the trim air check valve [3].

<a> Make sure that the orifice plate [13] is centered on the outlet flange of trim air check valve [3].

<6> Apply the pressure to the bonding surfaces.

<a> Make sure that there is no squeeze out.

<7> Let adhesive dry under pressure for a minimum of 24 hours at 65°F (18.3°C) - 100°F (37.8°C) and at more than 30 percent relative humidity.

(b) Put the trim air check valve [3] in its position.

1) Make sure that the flow arrow on the trim air check valve [3] points in the forward direction.

2) Make sure that the ports of the trim air check valve [3] point up.

(c) Move the clamp [4] to the duct connections.

(d) Tighten the nut on the two clamps [4] to 55 in-lb (6.2 N·m) - 60 in-lb (6.8 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-61-09-420-004

CAUTION

MAKE SURE THAT YOU DO NOT BEND OR TWIST THE ADJACENT DUCTS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(2) If it is necessary, install the duct assembly [12], do these steps:

(a) Remove any protective covers or blanks.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Install the coupling [10] to the duct assembly [12] connection.

1) Tighten the nut on the coupling [10] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

2) Make sure that the loop bands do not touch.

3) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

(c) Install the clamp [11].

SUBTASK 21-61-09-420-005

(3) If it is necessary, install the duct assembly [9], do these steps:

(a) Remove any protective covers or blanks.

(b) Put duct assembly [9] into its position.

(c) Install the clamp [8].

(d) Install the two couplings [6] to the duct assembly [9] connections.

(e) Install the coupling [14] to the duct assembly [9] connection.

(f) Tighten the nut on the two couplings [6] to 55 in-lb (6.2 N·m) - 60 in-lb (6.8 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

(g) Tighten the nut on the coupling [14] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-61-09-420-002

(4) Do the steps in this task to install the applicable ram air exhaust duct [5], if it is removed: Ram Air Ducts Installation, TASK 21-51-24-400-801.

G. Trim Air Check Valve Leakage Check

SUBTASK 21-61-09-860-003

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-61-09-860-004

(2) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the AUTO position and remove the DO NOT OPERATE tag.

(b) Set the BLEED 1 and 2 switches to the ON position and remove the DO NOT OPERATE tag.

(c) Set the BLEED APU switch to the ON position and remove the DO NOT OPERATE tag.

SUBTASK 21-61-09-790-001

(3) Do a soap bubble test of the duct joints at the trim air check valve [3] and hose [2] to the ram air exhaust duct [5].

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the applicable clamps [1] or clamps [4] as necessary.
3) Make sure that the ducts are aligned at the joints.
4) Tighten the applicable clamps [1] or clamps [4].
   a) For the clamps [4], make sure that the loop bands do not touch.
   b) For the clamps [4], make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).
5) Put the L PACK and R PACK switches to the AUTO position.
6) Make sure that the leak has been repaired.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-09-010-003

(1) If you replaced the left trim air check valve, close this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-61-09-010-004

(2) If you replaced the right trim air check valve, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRIM AIR CHECK VALVE - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) An inspection of the trim air check valve.

TASK 21-61-09-200-801

2. Trim Air Check Valve Inspection

(Figure 601)

A. References

Reference	Title
21-61-09-000-801	Trim Air Check Valve Removal (P/B 401)
21-61-09-400-801	Trim Air Check Valve Installation (P/B 401)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left

C. Prepare for the Inspection

SUBTASK 21-61-09-010-005

(1) Do this task: Trim Air Check Valve Removal, TASK 21-61-09-000-801.

D. Trim Air Check Valve Inspection

SUBTASK 21-61-09-200-001

(1) Inspect the trim air check valve for these defects:

(a) The valve body [1] has cracks or corrosion.

(b) The valve flappers [2] do not move freely.

(c) The valve flappers [2] do not touch the valve body equally.

(d) There are carbon particles at the valve flappers [2] and body interface.

(e) The valve flappers [2] are bent.

(f) The stop pin tube [3] has cracks or deterioration.

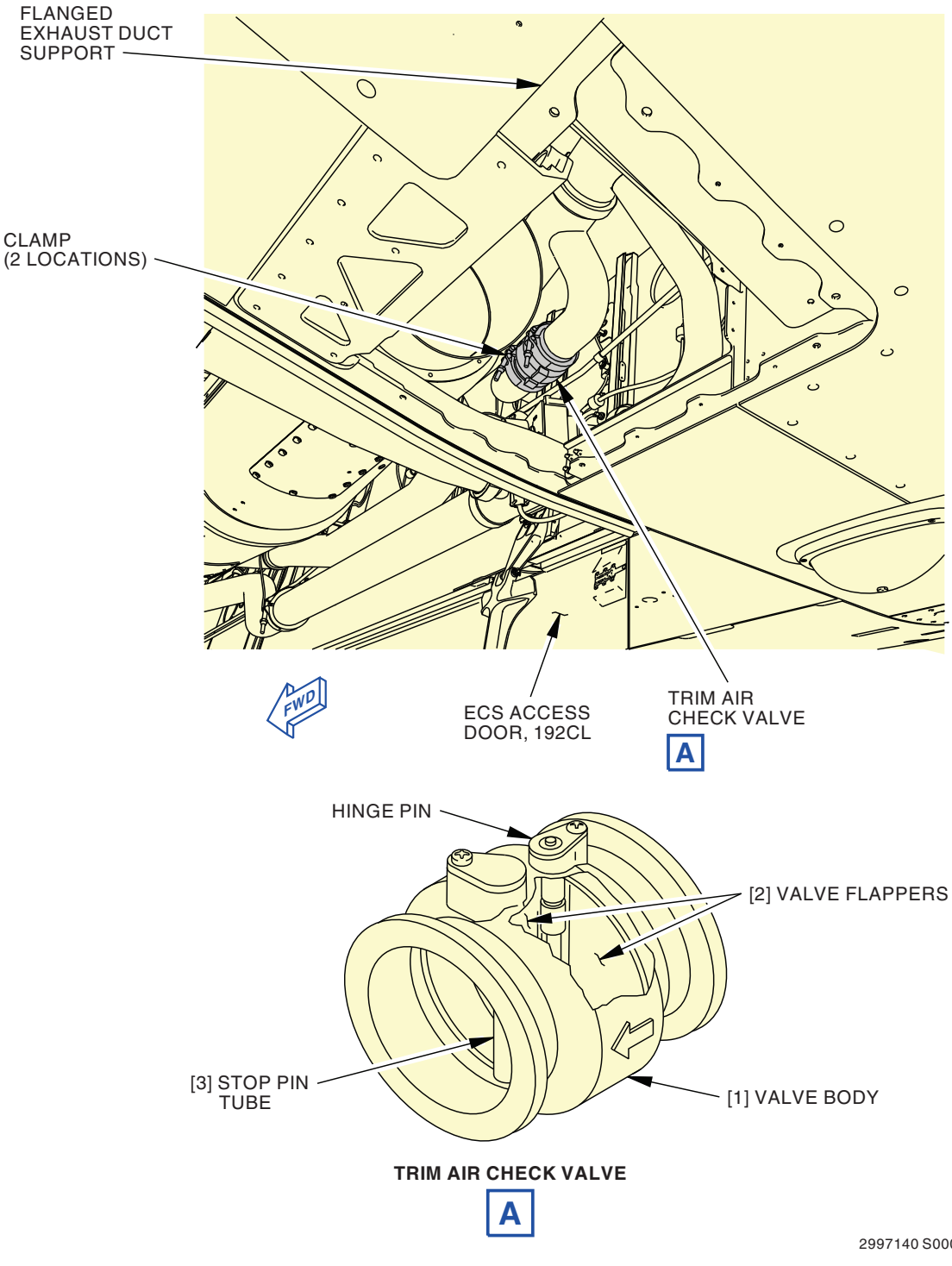
(g) If the trim air check valve has one or more of the defects, replace the trim air check valve.

SUBTASK 21-61-09-420-003

(2) Do this task: Trim Air Check Valve Installation, TASK 21-61-09-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trim Air Check Valve Inspection
Figure 601/21-61-09-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN TEMPERATURE SENSOR - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Cabin Temperature Sensor Removal

(2) Cabin Temperature Sensor Cleaning

(3) Cabin Temperature Sensor Installation.

(4) There is a cabin temperature sensor for the flight compartment zone, the forward passenger compartment zone and the aft passenger compartment zone.

TASK 21-61-10-000-801

2. Cabin Temperature Sensor Removal

A. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

C. Prepare for the Removal

SUBTASK 21-61-10-860-004

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-61-10-860-005

(2) On the P5 panel, set the L PACK, R PACK and BLEED switches to OFF.

SUBTASK 21-61-10-020-001

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

D. Cabin Temperature Sensor Removal

SUBTASK 21-61-10-860-001

(1) To remove the cabin temperature sensor [1] in the flight compartment, do these steps (Figure 201 or Figure 202):

(a) Loosen the access panel on the aft ceiling lining in the flight compartment and let it hang on the straps.

NOTE: The access panel on the aft ceiling lining is held in position by hook and loop strips.

(b) Disconnect the electrical connector [7] from the sensor [1].

(c) Remove the screws [2] and washers [3] that attach the sensor [1] to the housing [6].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Remove the sensor [1].

SUBTASK 21-61-10-860-002

(2) To remove the cabin temperature sensor [35] for the passenger compartment zone, do these steps:

(a) Lower the cabin temperature sensor panel.

1) Put a small Allen wrench or other applicable tool in the access holes to release the latches.

2) Partially lower the cabin temperature sensor panel and disconnect the lanyard.

3) Move the cabin temperature sensor panel to the fully lowered position.

(b) Disconnect the electrical connector [36] from the sensor [35].

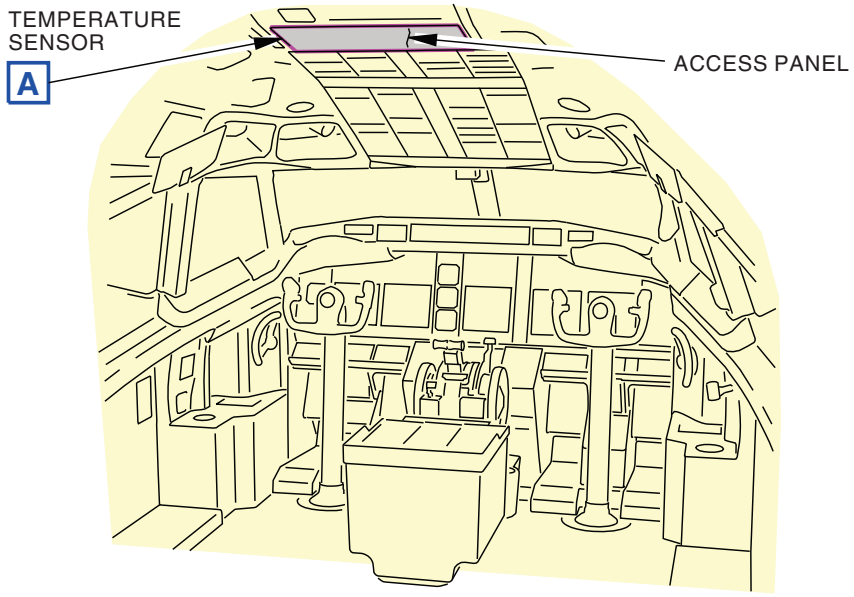
(c) Remove the lockwire [42].

(d) Remove the screws [39], lanyard [43] (if installed), and washers [40].

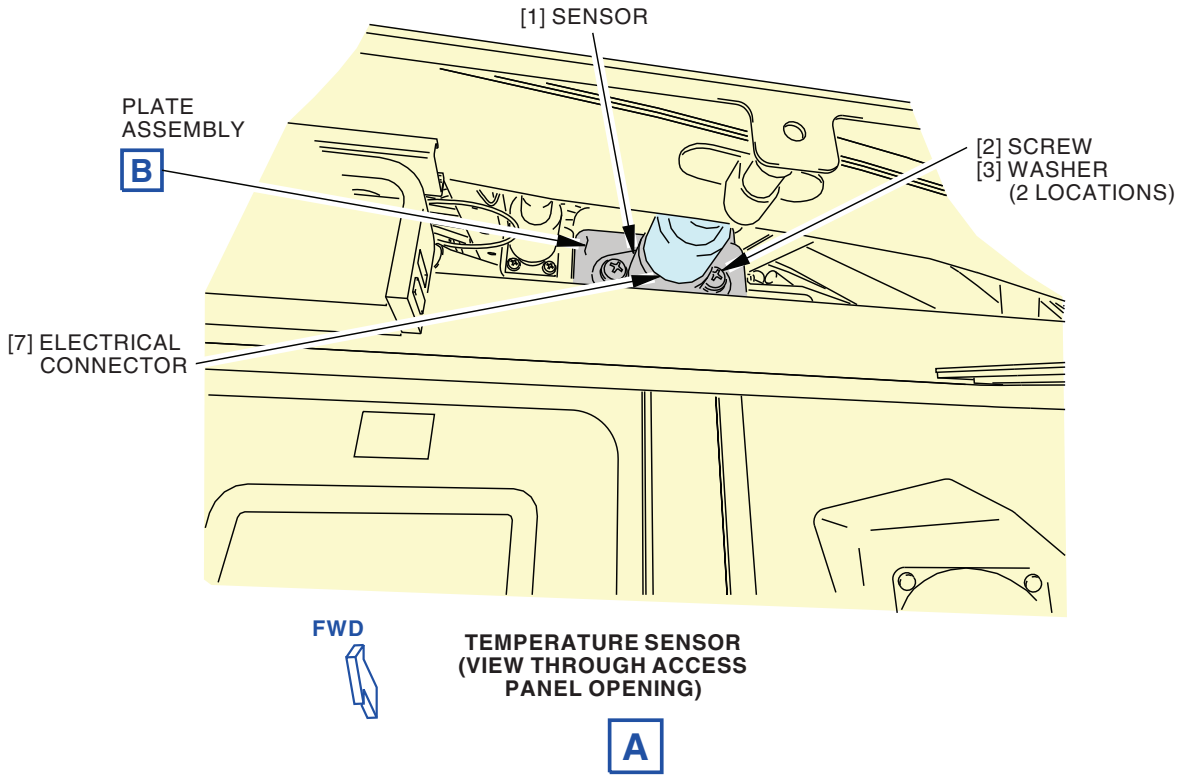
(e) Remove the sensor [35].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



2411507 S00061528072_V1

Flight Compartment - Cabin Temperature Sensor Installation
Figure 201/21-61-10-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

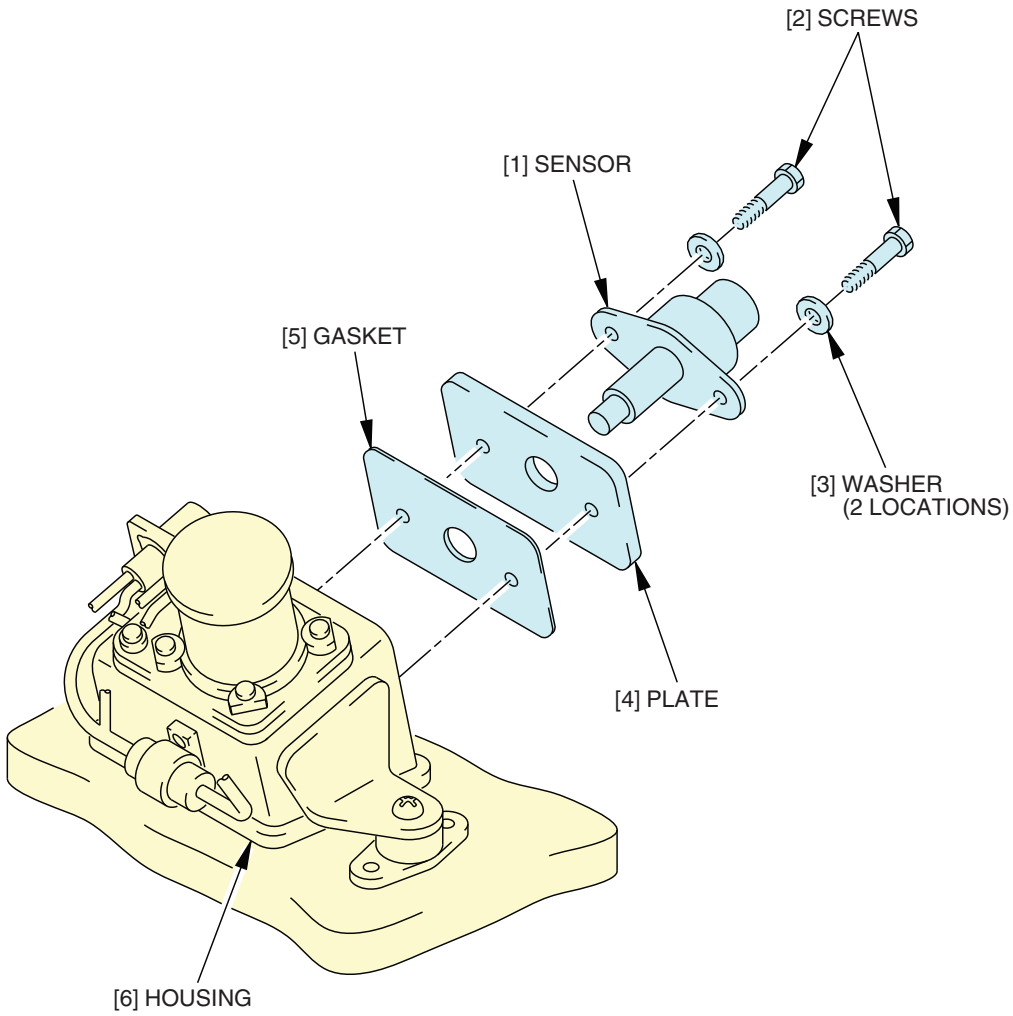


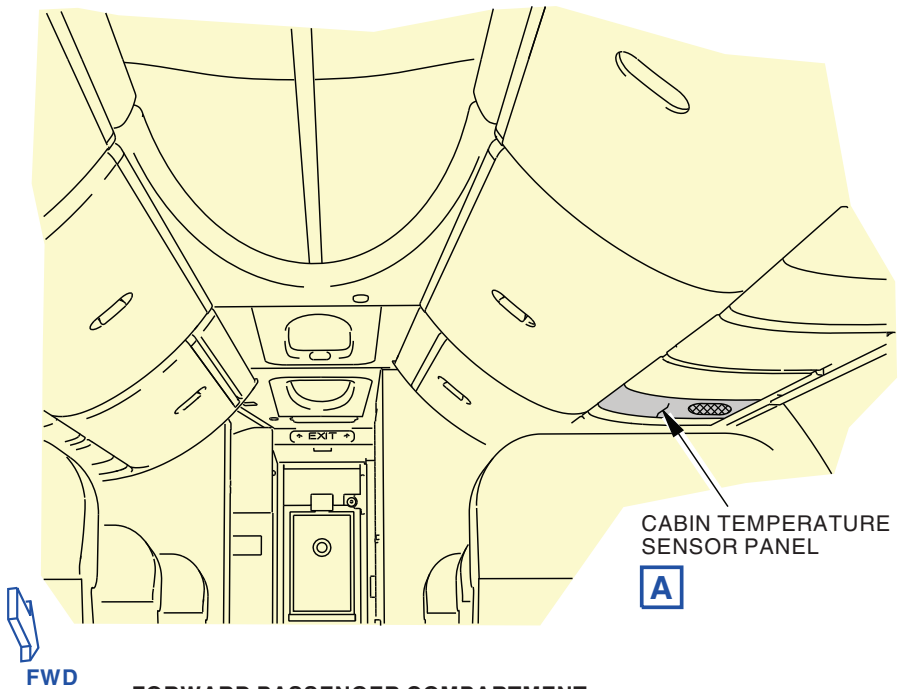
PLATE ASSEMBLY

B

2411508 S00061528073_V2

Flight Compartment - Cabin Temperature Sensor Installation
Figure 201/21-61-10-990-801 (Sheet 2 of 2)

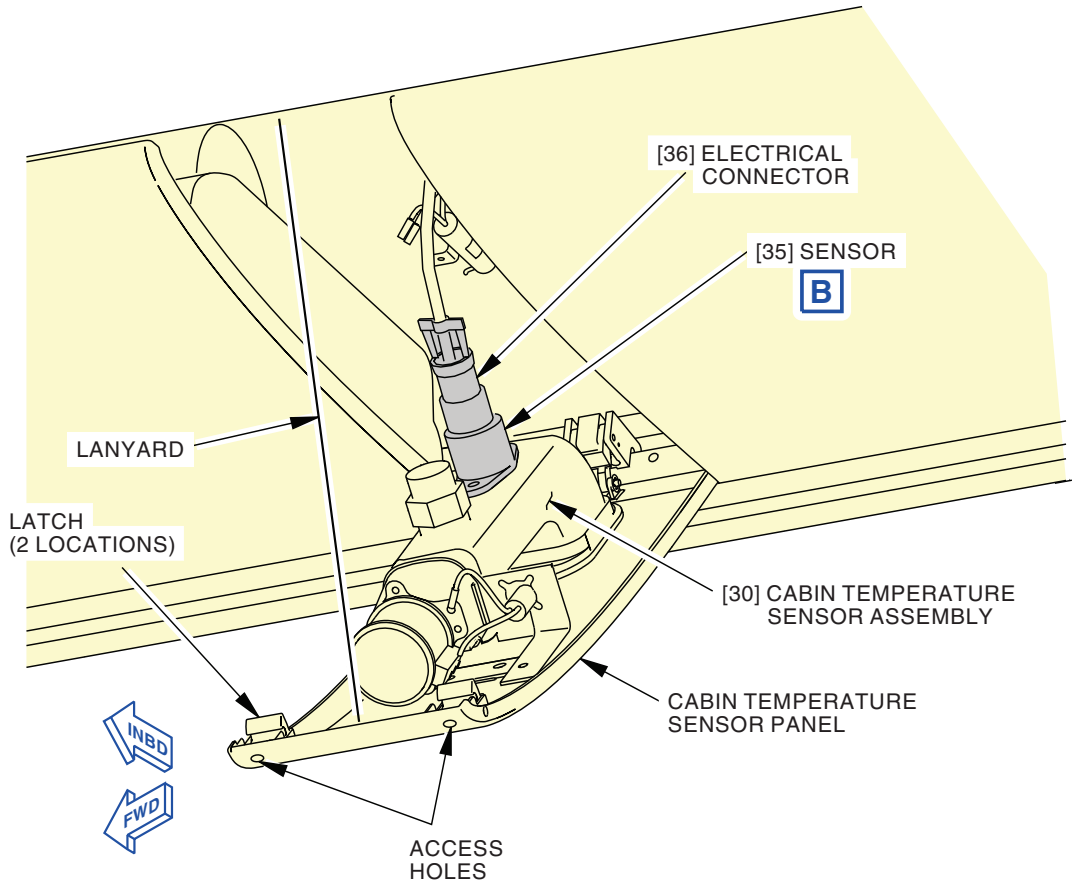
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411509 S00061528074_V1

Passenger Compartment - Cabin Temperature Sensor Installation
Figure 202/21-61-10-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE SENSOR PANEL (EXAMPLE)

A

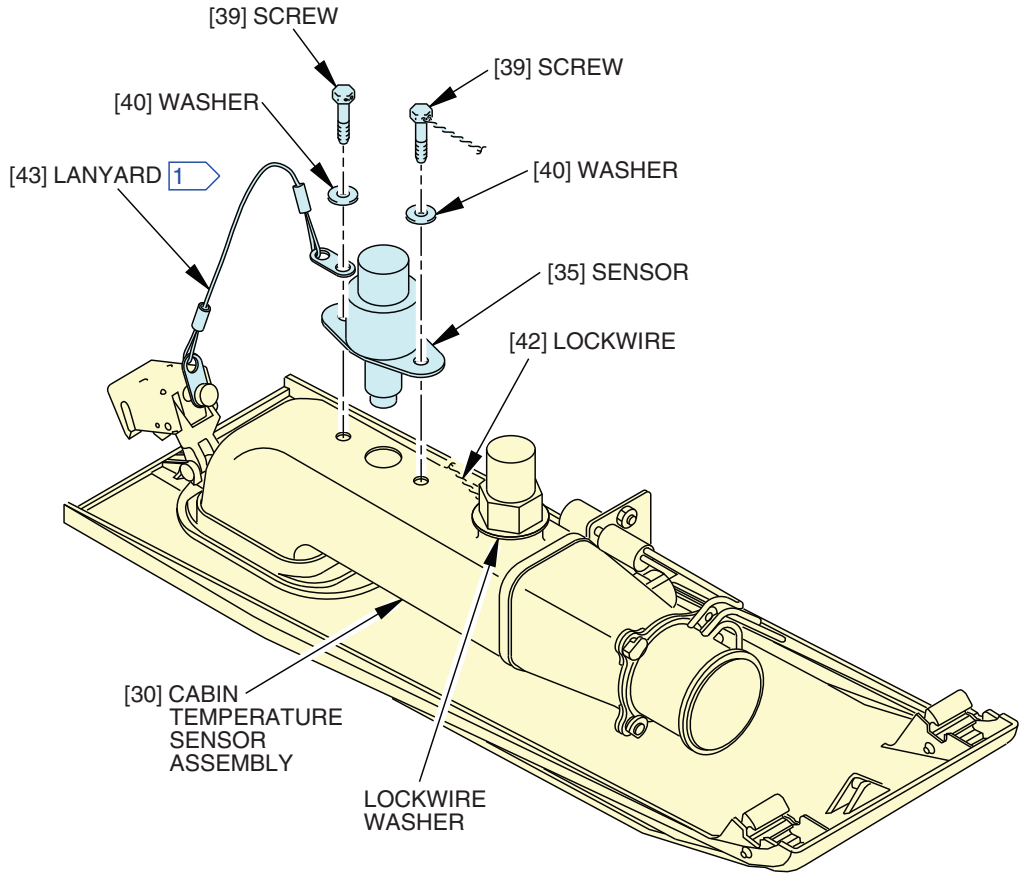
NOTE:

THE CABIN TEMPERATURE SENSOR PANEL IS SHOWN IN THE OPEN POSITION.

2411510 S00061528075_V2

Passenger Compartment - Cabin Temperature Sensor Installation
Figure 202/21-61-10-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SENSOR (EXAMPLE)

B

1 IF INSTALLED.

2411511 S00061528076_V4

Passenger Compartment - Cabin Temperature Sensor Installation
Figure 202/21-61-10-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-10-100-801

3. Cabin Temperature Sensor Cleaning

A. Consumable Materials

Reference	Description	Specification
B00039	Cleaner - Brulin Formula 815MX	
B00130	Alcohol - Isopropyl	TT-I-735
B00541	Cleaner - General Purpose Household Detergent	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

B. Prepare for the Cabin Temperature Sensor Cleaning

SUBTASK 21-61-10-020-002

(1) Remove the temperature sensor for the flight compartment (TASK 21-61-10-000-801).

SUBTASK 21-61-10-020-003

(2) Remove the temperature sensors for the passenger compartment (TASK 21-61-10-000-801).

C. Cabin Temperature Sensor Cleaning

SUBTASK 21-61-10-100-001

(1) Do these steps to clean the temperature sensor:

(a) Prepare a cleaning solution of water and general purpose household detergent cleaner, B00541, or Brulin Formula 815MX cleaner, B00039.

1) Follow the mixing instructions for the cleaner.

(b) Clean the temperature sensors with the cleaning solution and a cotton wiper, G00034.

(c) If the cleaning solution did not remove all dirt deposits or contamination from the sensor, then clean the temperature sensor with a cotton wiper, G00034, that is moist with alcohol, B00130.

(d) Flush the temperature sensor with clean water until all cleaning solution or alcohol, B00130, has been removed.

(e) Dry the temperature sensor with a clean cotton wiper, G00034.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-10-420-001

(1) Install the temperature sensor for the flight compartment (TASK 21-61-10-400-801).

SUBTASK 21-61-10-420-002

(2) Install the temperature sensors for the passenger compartment (TASK 21-61-10-400-801).

——— END OF TASK ———

TASK 21-61-10-400-801

4. Cabin Temperature Sensor Installation

(Figure 201 or Figure 202)

A. References

Reference	Title
46-13-02-700-804	Present Leg Faults Check (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Cabin Temperature Sensor Installation

SUBTASK 21-61-10-420-003

(1) To install the cabin temperature sensor [1] for the flight compartment, do these steps:

(a) Make sure that the gasket [5] and plate [4] are in position on the housing [6] of the temperature sensor assembly.

(b) Put the sensor [1] in its position on the temperature sensor assembly.

(c) Install the washers [3] and screws [2].

(d) Connect the electrical connector [7] to the sensor [1].

(e) To install the access panel, lift it to its closed position and make sure the hook and loop strips are fully engaged.

SUBTASK 21-61-10-420-004

(2) To install the cabin temperature sensor [35] for the passenger compartment, do these steps:

(a) Put the sensor [35] in position on the cabin temperature sensor assembly [30].

(b) Install the washers [40], lanyard [43] (if installed), and screws [39].

(c) Install the lockwire [42] (MS20995C32 lockwire, G01048).

(d) Connect the electrical connector [36].

(e) Partially lift the panel for the cabin temperature sensor assembly [30] and install the lanyard.

(f) Lift the cabin temperature sensor panel for the cabin temperature sensor assembly [30] to its closed position.

SUBTASK 21-61-10-860-003

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Cabin Temperature Sensor Installation Test

SUBTASK 21-61-10-740-001

(1) Look for maintenance messages that are related to the cabin temperature sensor, do this task: Present Leg Faults Check, TASK 46-13-02-700-804.

(a) Make sure that maintenance messages that are related to the temperature sensor are NOT ACTIVE.

(b) If the maintenance message is ACTIVE, refer to the IFIM and do the applicable procedure(s) to repair the fault indicated by the maintenance message.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-10-740-002

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN TEMPERATURE SENSOR - INSPECTION/CHECK

1. General

A. This procedure does an electrical resistance check of the cabin temperature sensors.

TASK 21-61-10-200-801

2. Cabin Temperature Sensor Resistance Check

(Figure 601, Figure 602, Figure 603)

A. General

(1) This procedure does a temperature versus electrical resistance check of the cabin temperature sensors to do a check for sensor drift.

B. References

Reference	Title
21-61-10-000-801	Cabin Temperature Sensor Removal (P/B 201)
21-61-10-400-801	Cabin Temperature Sensor Installation (P/B 201)
21-61-22-000-801	Integrated Air Systems Controller (IASC) Removal (P/B 401)
21-61-22-400-801	Integrated Air Systems Controller (IASC) Installation (P/B 401)
WDM 21-61-11	Wiring Diagram Manual
WDM 21-61-12	Wiring Diagram Manual
WDM 21-61-13	Wiring Diagram Manual
WDM 21-61-14	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
COM-3955	Thermometer - Digital (Type K) Part #: 51 2B Supplier: 89536 Opt Part #: 51 SERIES II Supplier: 89536

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Cabin Temperature Sensor Resistance Check

SUBTASK 21-61-10-210-001

(1) Find the inlet grilles of the plenums that contain the cabin temperature sensors in the flight compartment and the forward and aft passenger compartment (Figure 601).

SUBTASK 21-61-10-970-001

(2) Use a digital thermometer, COM-3955, or equivalent, to measure the ambient temperature near the inlet grille of the cabin temperature sensors for the flight compartment and forward and aft passenger compartment.

SUBTASK 21-61-10-970-002

(3) Open this access door:

(Figure 602)

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-10-010-001

(4) To remove the left and right integrated air systems controller, do this task: Integrated Air Systems Controller (IASC) Removal, TASK 21-61-22-000-801.

SUBTASK 21-61-10-020-004

(5) Use a digital/analog multimeter, COM-1793, or equivalent, to measure the electrical resistance of the cabin temperature sensor at the electrical connector for the left or right integrated air systems controller (M2939 and M2957):

(a) Cabin temperature sensor for the flight compartment:

1) For the main element, measure the resistance across pin E11 and pin E12 of connector D15840B for the right integrated air systems controller (M2957) (WDM 21-61-11).

2) For the backup element, measure the resistance across pin E11 and pin E12 of connector D15838B for the left integrated air systems controller (M2939) (WDM 21-61-12).

(b) Cabin temperature sensor for the forward passenger compartment:

1) For the main element, measure the resistance across pin C8 and pin C9 of connector D15840B for the right integrated air systems controller (M2957) (WDM 21-61-13).

2) For the backup element, measure the resistance across pin 4 and pin 10 of connector D15838C for the left integrated air systems controller (M2939) (WDM 21-61-13).

(c) Cabin temperature sensor for the aft passenger compartment:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) For the main element, measure the resistance across pin C8 and pin C9 of connector D15838B for the left integrated air systems controller (M2939) (WDM 21-61-14).

2) For the backup element, measure the resistance across pin 4 and pin 10 of connector D15840C for the right integrated air systems controller (M2957)(WDM 21-61-14).

SUBTASK 21-61-10-210-002

(6) Compare the measured resistance to the resistance given in Figure 603 for the ambient temperature measured at the inlet grille.

(a) If the measured resistance is out of range of the electrical resistance given in (Figure 603, replace the cabin temperature sensor.

Do these tasks:

- Cabin Temperature Sensor Removal, TASK 21-61-10-000-801,
- Cabin Temperature Sensor Installation, TASK 21-61-10-400-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-10-420-005

(1) To install the left and right integrated air systems controllers, do this task: Integrated Air Systems Controller (IASC) Installation, TASK 21-61-22-400-801.

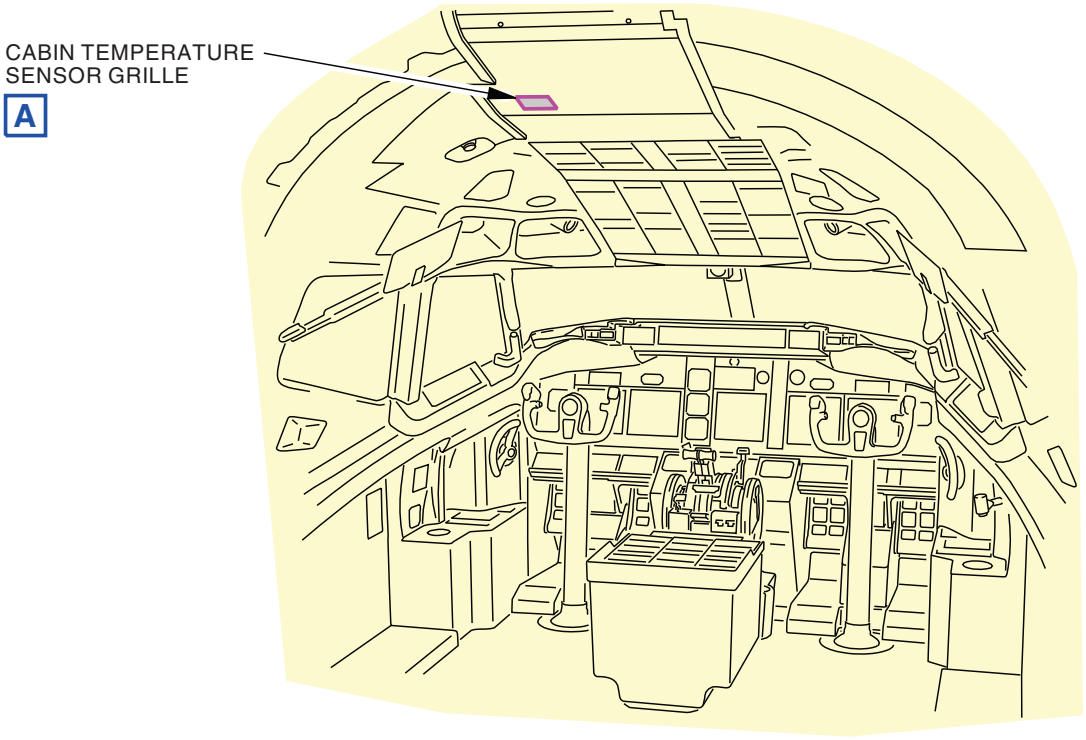
SUBTASK 21-61-10-760-001

(2) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

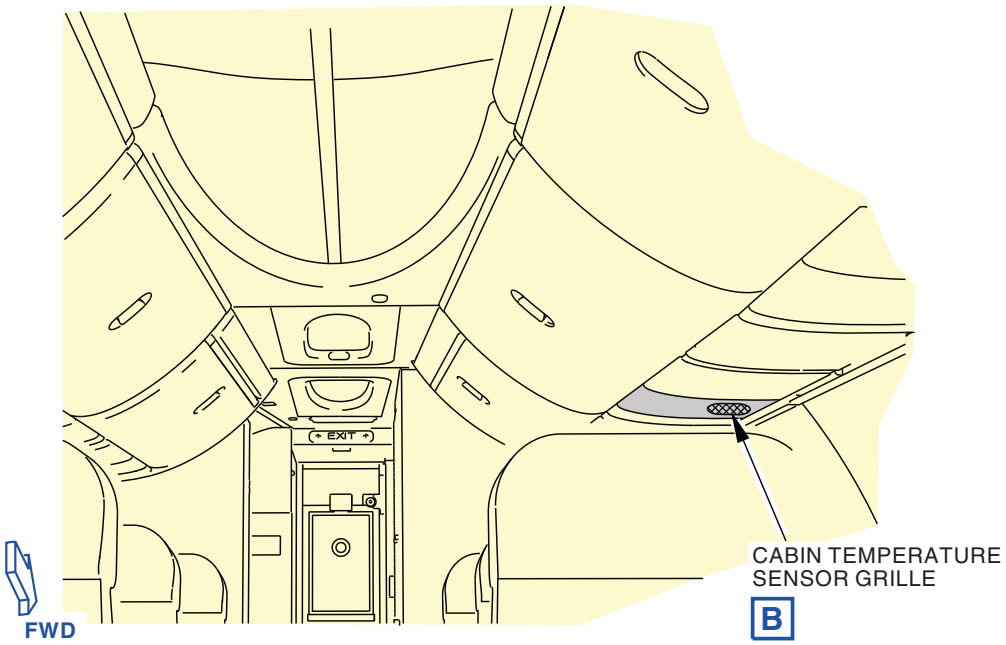
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

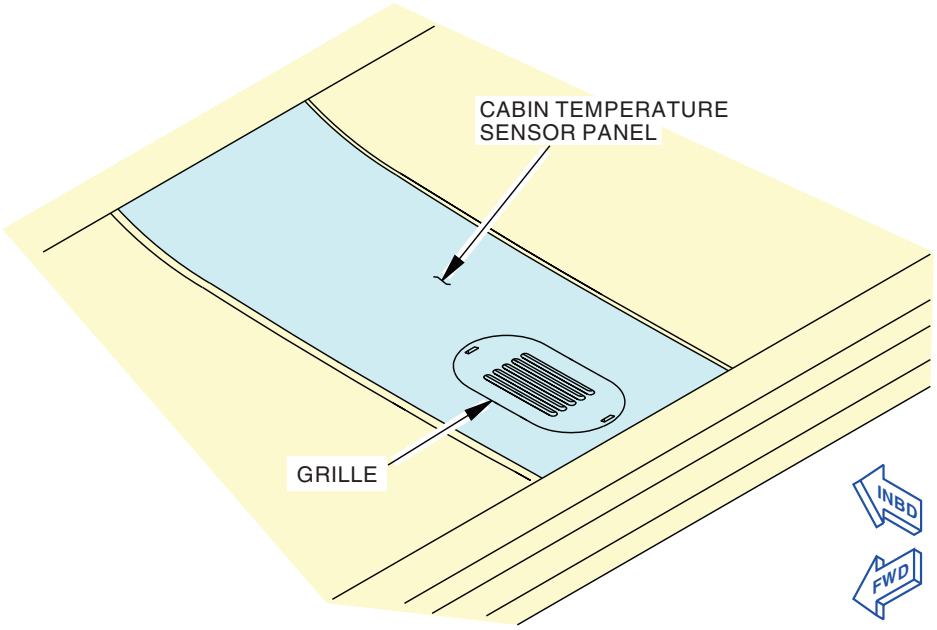


Cabin Temperature Sensor Inspection
Figure 601/21-61-10-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD PASSENGER COMPARTMENT



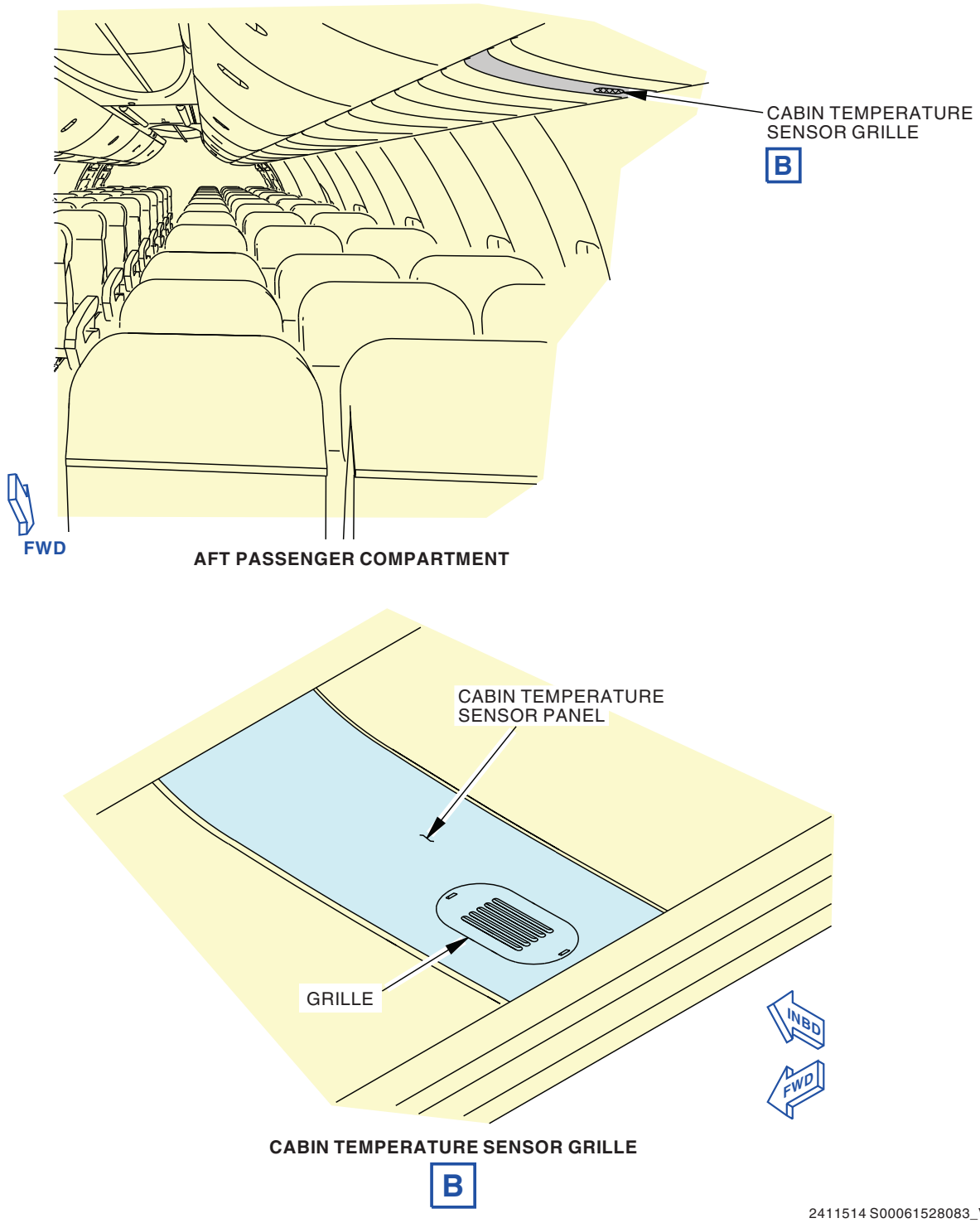
CABIN TEMPERATURE SENSOR GRILLE

B

2411513 S00061528082_V1

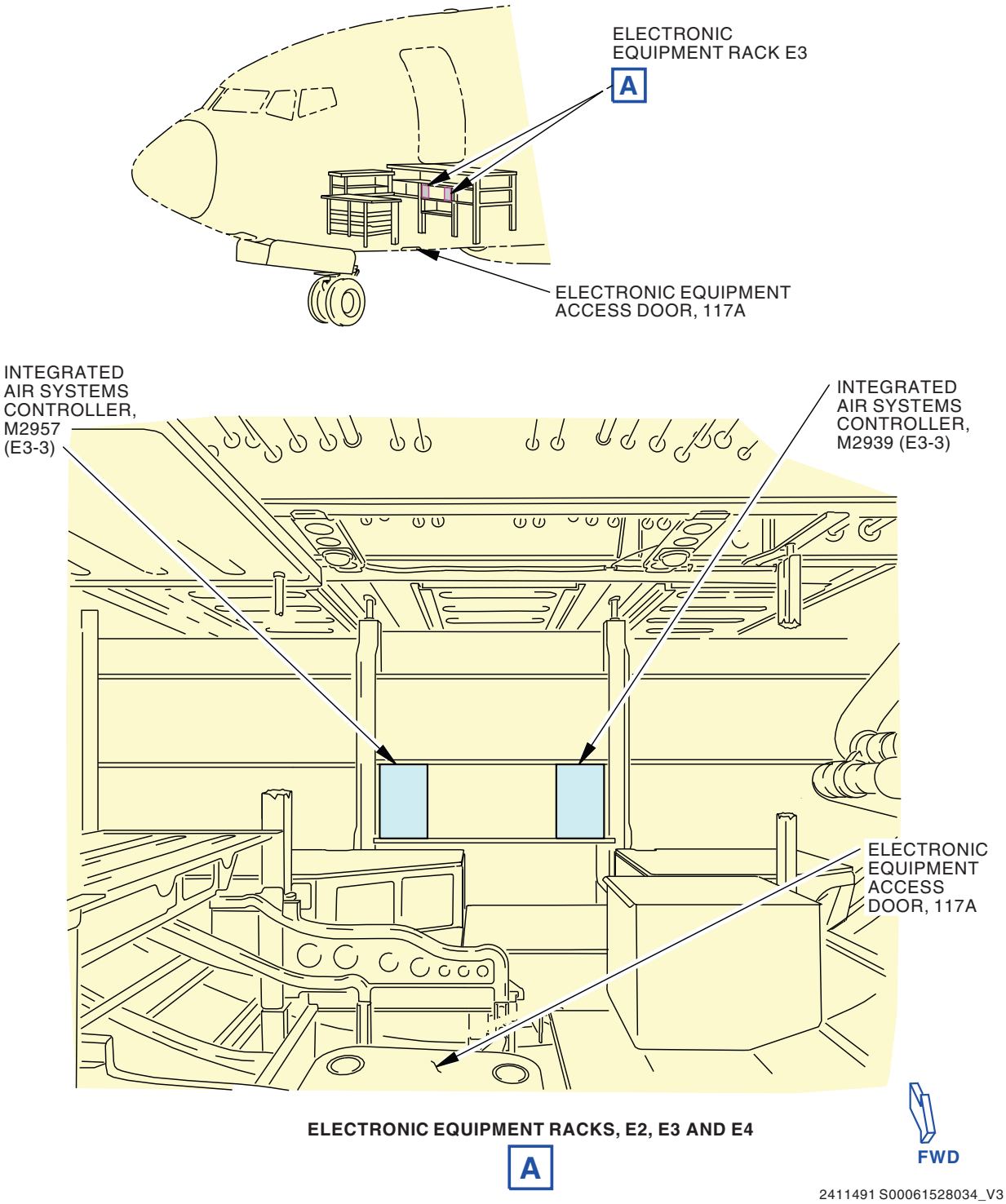
Cabin Temperature Sensor Inspection
Figure 601/21-61-10-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Temperature Sensor Inspection
Figure 601/21-61-10-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Integrated Air Systems Controller Location]
Figure 602/21-61-10-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN ELEMENT

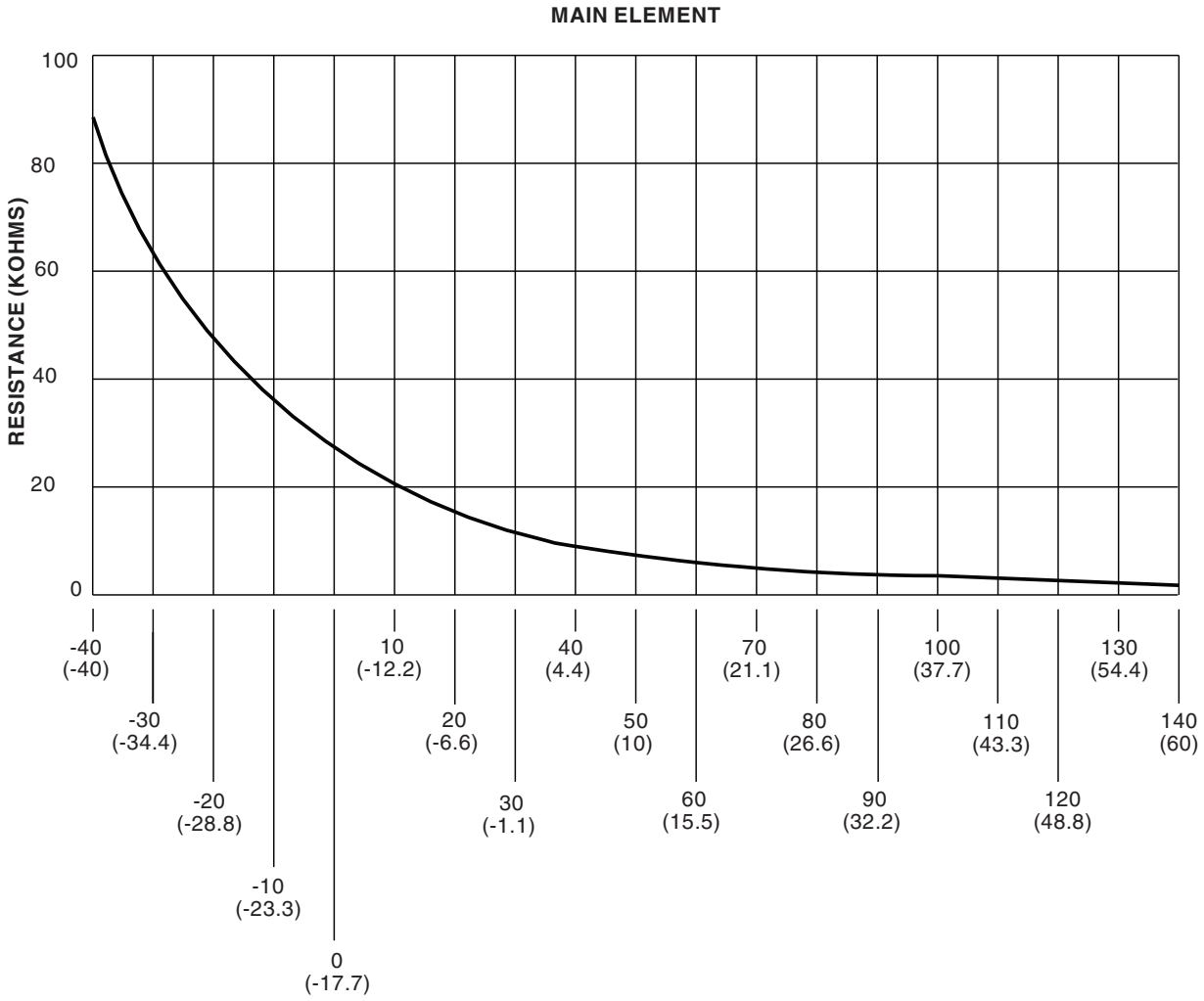
TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)	TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)
-40 (-40)	87350-89130	55 (12.7)	6470-6610
-35 (-37.2)	74480-76000	60 (15.5)	5770-5900
-30 (-34.4)	63670-64970	65 (18.3)	5150-5270
-25 (-31.6)	54600-55720	70 (21.1)	4610-4710
-20 (-28.8)	46920-47880	75 (23.8)	4130-4220
-15 (-26.1)	40430-41250	80 (26.6)	3710-3790
-10 (-23.3)	34910-35630	85 (29.4)	3330-3410
-5 (-20.5)	30160-30780	90 (32.2)	3000-3070
0 (-17.7)	26240-26780	95 (35)	2710-2780
5 (-15)	22850-23320	100 (37.7)	2450-2510
10 (-12.2)	19950-20370	105 (40.5)	2220-2270
15 (-9.4)	17460-17820	110 (43.3)	2010-2060
20 (-6.6)	15310-15630	115 (46.1)	1830-1870
25 (-3.8)	13460-13750	120 (48.8)	1660-1700
30 (-1.1)	11860-12110	125 (51.6)	1510-1550
35 (1.6)	10470-10690	130 (54.4)	1380-1410
40 (4.4)	9250-9450	135 (57.2)	1260-1290
45 (7.2)	8200-8380	140 (60)	1150-1180
50 (10)	7270-7430	145 (62.7)	1050-1080

CABIN TEMPERATURE SENSOR P/N 624018-1 (S210T130-21)
RESISTANCE VS TEMPERATURE TABLE

2411516 S00061528085_V1

Cabin Temperature Sensor - Temperature and Resistance Data
Figure 603/21-61-10-990-805 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE SENSOR P/N 624018-1 (S210T130-21)
RESISTANCE VS TEMPERATURE GRAPH

2411517 S00061528086_V1

Cabin Temperature Sensor - Temperature and Resistance Data
Figure 603/21-61-10-990-805 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BACKUP ELEMENT

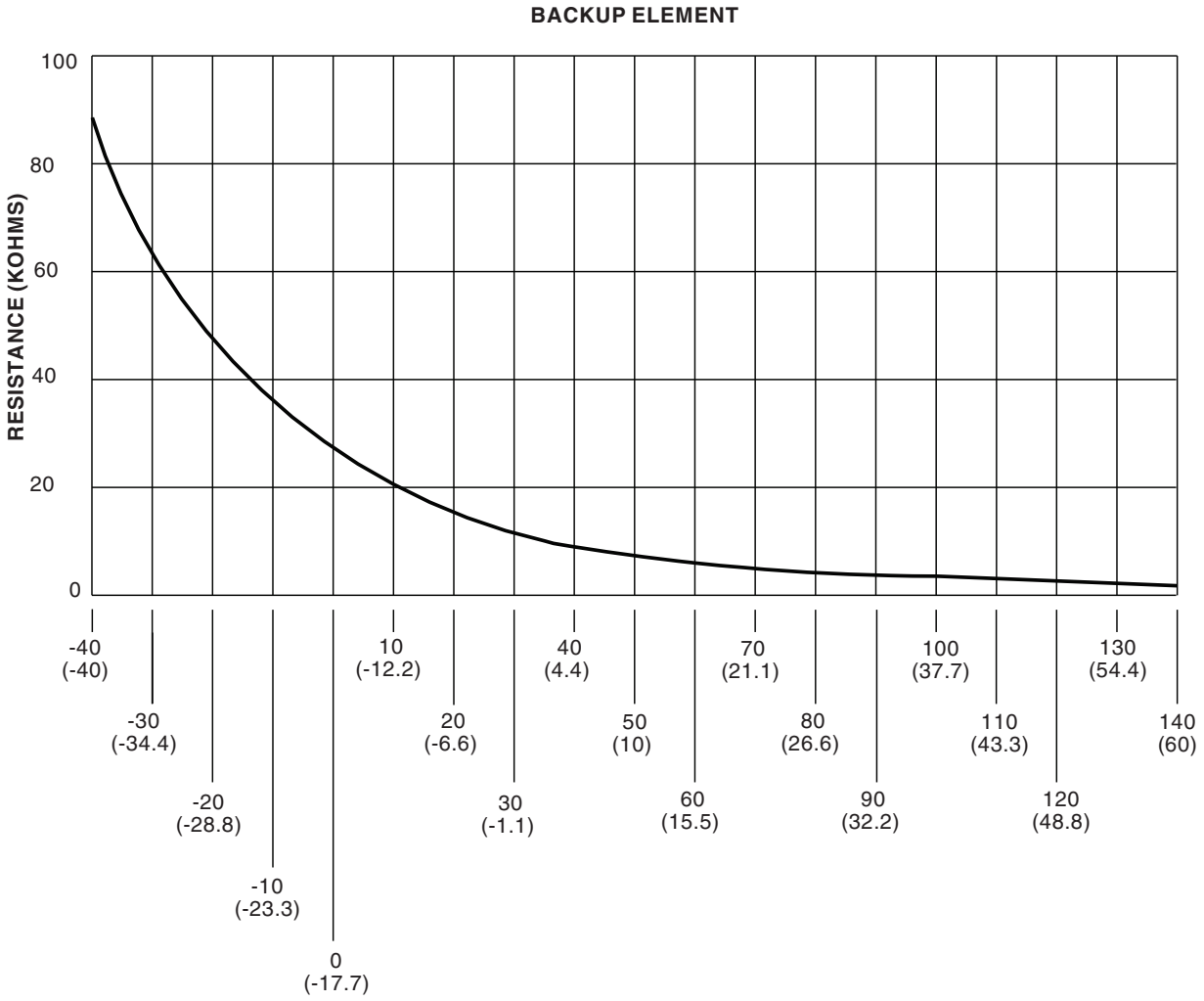
TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)	TEMPERATURE ˚F (˚C)	RESISTANCE (OHMS)
-40 (-40)	85850-90630	55 (12.7)	6360-6720
-35 (-37.2)	73200-77280	60 (15.5)	5670-5990
-30 (-34.4)	62580-66060	65 (18.3)	5060-5350
-25 (-31.6)	53670-56650	70 (21.1)	4530-4790
-20 (-28.8)	46120-48680	75 (23.8)	4060-4290
-15 (-26.1)	39730-41950	80 (26.6)	3640-3860
-10 (-23.3)	34310-36230	85 (29.4)	3280-3470
-5 (-20.5)	29640-31300	90 (32.2)	2950-3130
0 (-17.7)	25790-27230	95 (35)	2660-2820
5 (-15)	22460-23720	100 (37.7)	2410-2550
10 (-12.2)	19610-20710	105 (40.5)	2180-2310
15 (-9.4)	17160-18120	110 (43.3)	1980-2100
20 (-6.6)	15050-15890	115 (46.1)	1790-1900
25 (-3.8)	13230-13980	120 (48.8)	1630-1730
30 (-1.1)	11660-12310	125 (51.6)	1480-1580
35 (1.6)	10290-10870	130 (54.4)	1350-1440
40 (4.4)	9090-9610	135 (57.2)	1240-1310
45 (7.2)	8060-8520	140 (60)	1130-1200
50 (10)	7150-7560	145 (62.7)	1030-1100

CABIN TEMPERATURE SENSOR P/N 624018-1 (S210T130-21)
RESISTANCE VS TEMPERATURE TABLE

2411518 S00061528087_V1

Cabin Temperature Sensor - Temperature and Resistance Data
Figure 603/21-61-10-990-805 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE SENSOR P/N 624018-1 (S210T130-21)
RESISTANCE VS TEMPERATURE GRAPH

2411519 S00061528088_V1

Cabin Temperature Sensor - Temperature and Resistance Data
Figure 603/21-61-10-990-805 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN TEMPERATURE SENSOR FAN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cabin temperature sensor fan

(2) An installation of the cabin temperature sensor fan.

B. There is a cabin temperature sensor fan for each of the three temperature control zones, the flight compartment, forward, and aft passenger compartment zones.

TASK 21-61-11-000-801

2. Cabin Temperature Sensor Fan Removal

(Figure 401, Figure 402)

A. References

Reference	Title
25-11-21-000-803	Flight Compartment Aft Ceiling Panel Removal (P/B 201)

B. Consumable Materials

Reference	Description	Specification
B00316	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I, ASTM D-3735 Type I
B50078	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I (Supersedes BMS3-2 Type I)
B50079	Solvent - Aliphatic Naphtha (For Acrylic Plastics)	TT-N-95 Type II (Supersedes BMS3-2 Type II)
G50262	Wiper - Cleaning	BMS15-5

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Preparation for the Removal

SUBTASK 21-61-11-860-001

(1) Do this step for the removal of the fan for the flight compartment zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

SUBTASK 21-61-11-860-002

(2) Do this step for the removal of the fan for the forward passenger compartment zone:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-11-860-003

(3) Do this step for the removal of the fan for the aft passenger compartment zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

E. Flight Compartment Cabin Temperature Sensor Fan Removal

SUBTASK 21-61-11-020-001

(1) Do these steps to remove the fan [3] in the flight compartment (Figure 401):

(a) Lower the access panel on the forward end of the aft ceiling panel [7] in the flight compartment.

NOTE: The access panel is held in position with hook and loop strips. This panel will give the access to the electrical connectors on the cabin temperature sensor assembly.

(b) Disconnect the electrical connector [1] and electrical connector [2] from the cabin temperature sensor assembly.

(c) To remove the aft ceiling panel [7] in the flight compartment, do this task: Flight Compartment Aft Ceiling Panel Removal, TASK 25-11-21-000-803.

(d) Attach tags to each of the fan wires [4] to show its correct location for later installation.

(e) Remove the compound from the screws [5] with wiper, G50262, soaked in aliphatic naphtha, B50079, or aliphatic naphtha, B50078, or solvent, B00316.

(f) Remove the screws [5] and washers [6].

(g) Disconnect the fan wires [4] from the terminal board [13] on the fan [3].

1) Do not remove the ground wire from the metal flange [9].

(h) Remove the four clamps [10], washers [11], and screws [12] that attach the fan [3] to the cabin temperature sensor assembly.

(i) Remove the fan [3] from the cabin temperature sensor assembly.

F. Passenger Compartment Cabin Temperature Sensor Fan Removal

SUBTASK 21-61-11-020-003

CAUTION

THE FAN, P/N 041510000, CONTAINS SENSITIVE BEARINGS AND YOU MUST HANDLE IT CAREFULLY. DAMAGE TO THE FAN CAN OCCUR IF THIS CAUTION IS NOT OBEYED.

(1) Remove the fan [14] in the passenger compartment as follows (Figure 402):

(a) Lower the cabin temperature sensor panel.

(b) Disconnect the electrical connector [15].

1) Remove the nuts [16], washers [17], and screws [18].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the screws [19], washers [20], and clamps [21] that attach the fan [14] to the cabin temperature sensor assembly.

(d) Remove the fan [14] from the cabin temperature sensor assembly.

NOTE: Fan P/N 041510000 does not have a terminal board, fan includes the capacitor.

(e) Attach a tag to each fan wire [24] to show its correct location for later installation.

CAUTION

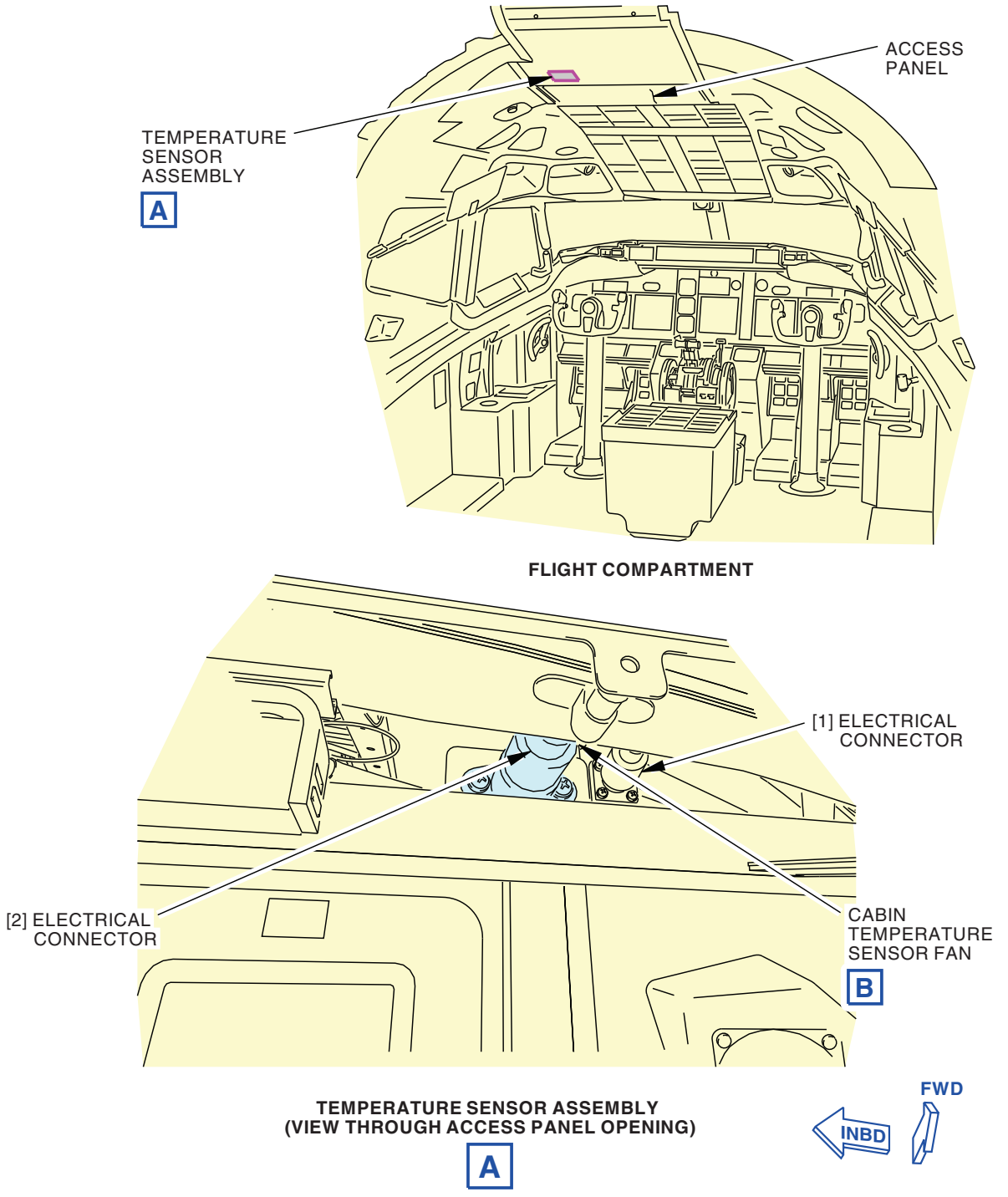
THE FAN, P/N 041510000, CONTAINS SENSITIVE BEARINGS AND YOU MUST HANDLE IT CAREFULLY. DAMAGE TO THE FAN CAN OCCUR IF THIS CAUTION IS NOT OBEYED.

1) Remove the strap [22] from the capacitor [23] and bracket assembly.

2) Use an extraction tool to remove the contacts and fan wires [24] from the electrical connector [15].

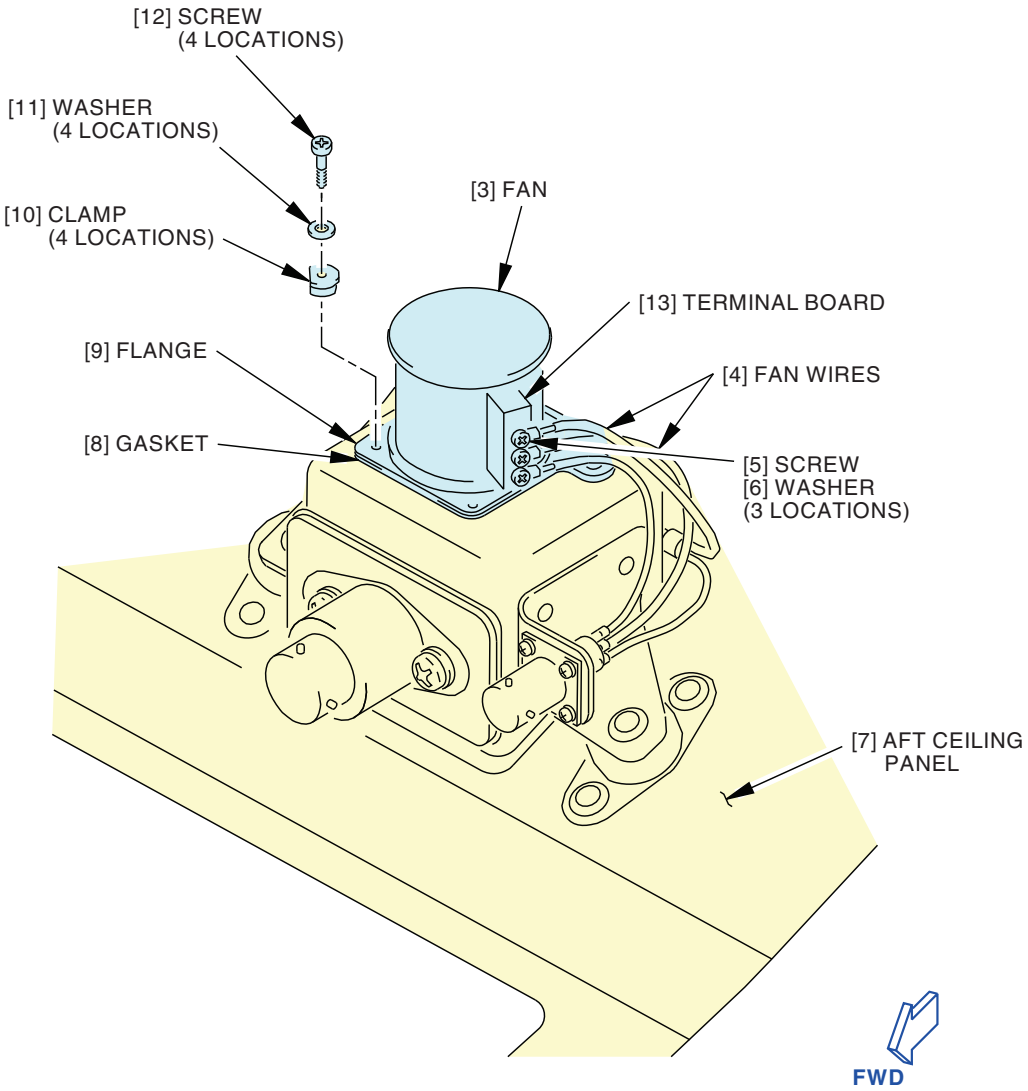
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Compartment Cabin Temperature Sensor Fan Installation
Figure 401/21-61-11-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



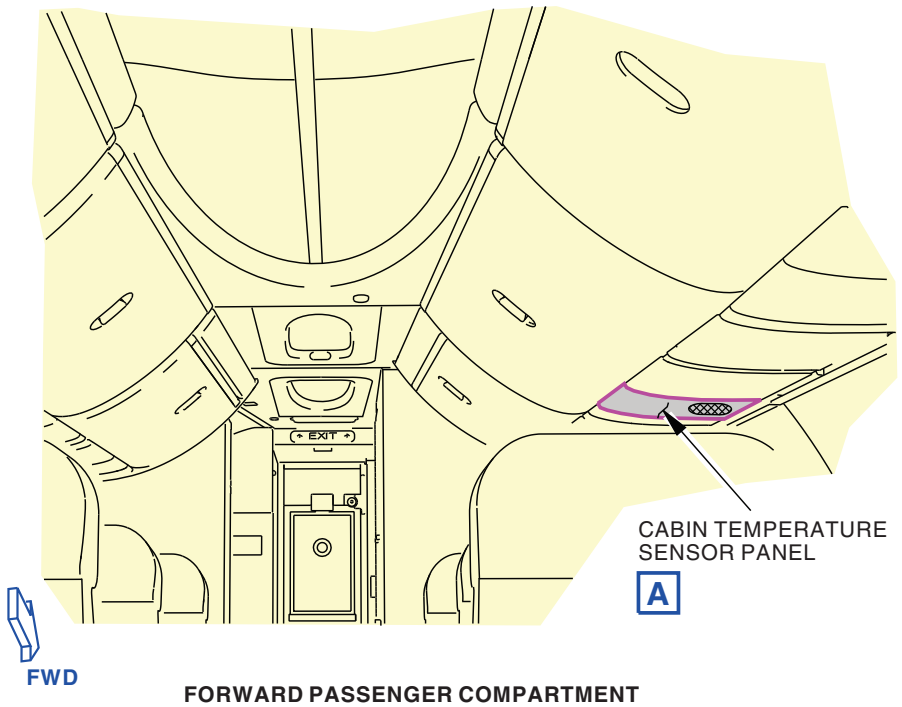
CABIN TEMPERATURE SENSOR FAN
(EXAMPLE)

B

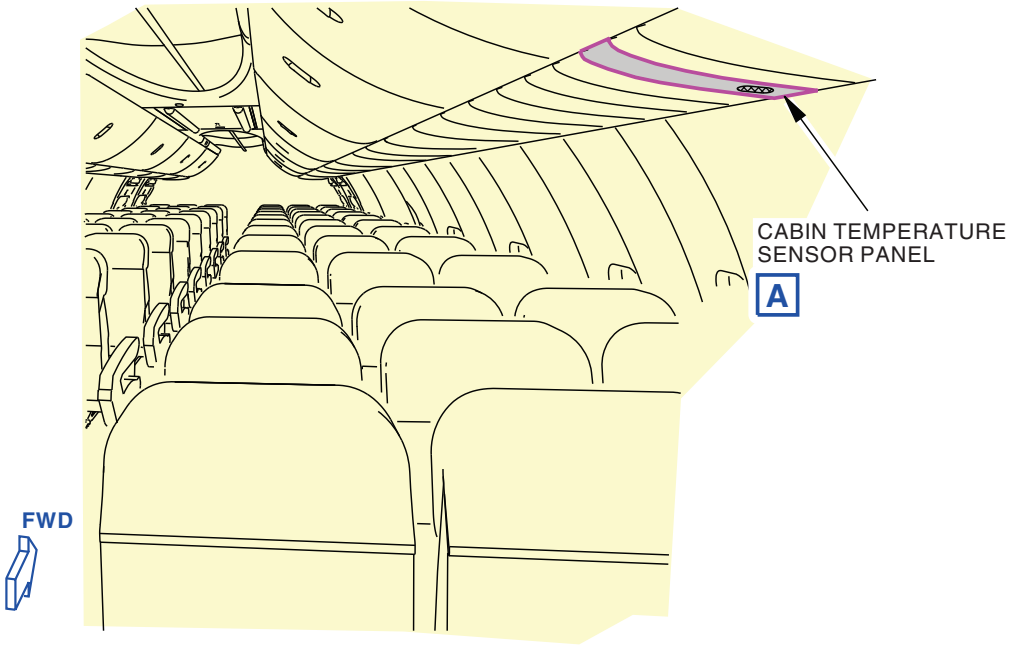
2411521 S00061528093_V2

Flight Compartment Cabin Temperature Sensor Fan Installation
Figure 401/21-61-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD PASSENGER COMPARTMENT

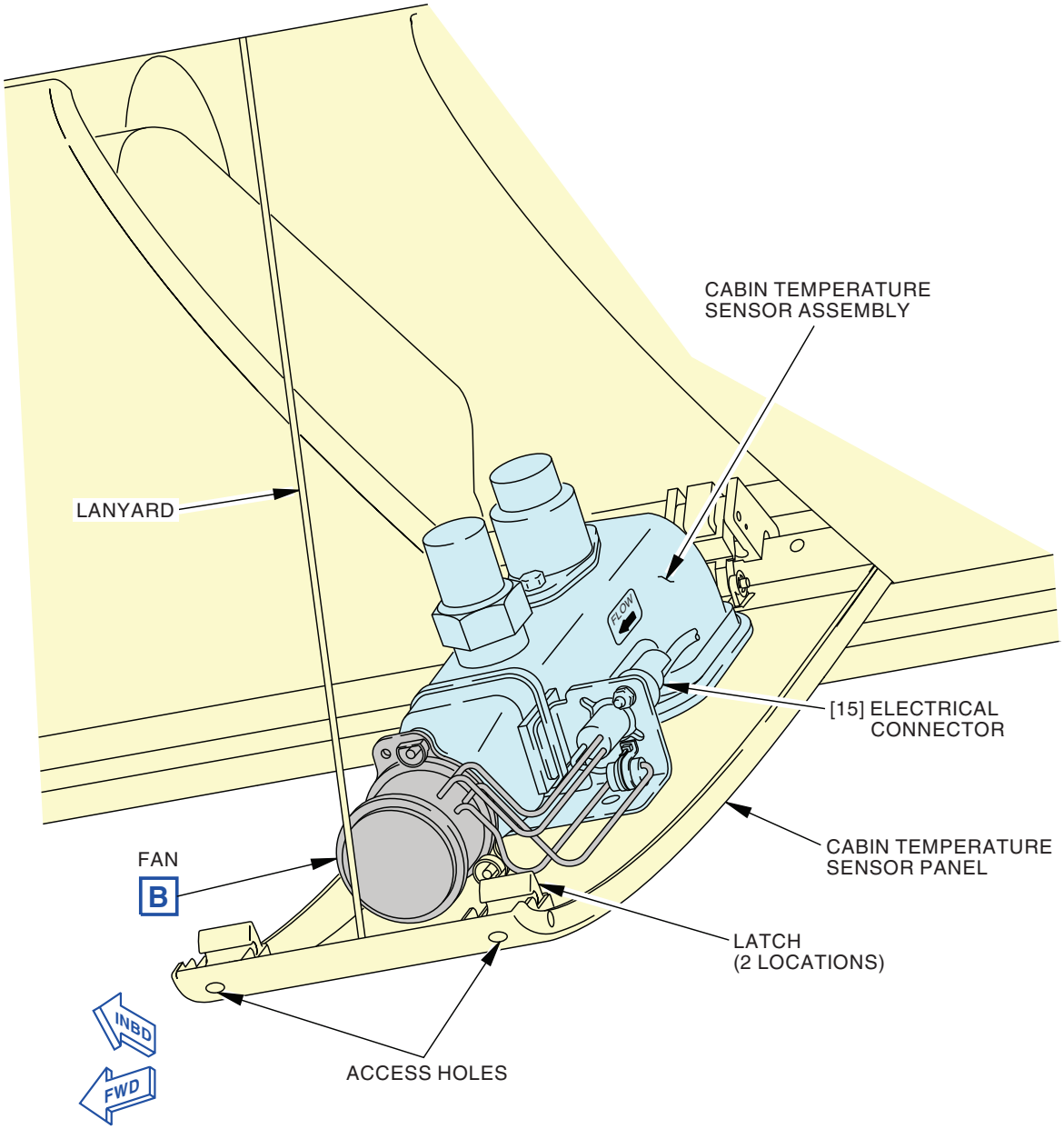


AFT PASSENGER COMPARTMENT

2794662 S0000637169_V1

Passenger Cabin Temperature Sensor Fan Installation
Figure 402/21-61-11-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE SENSOR PANEL (EXAMPLE)

A

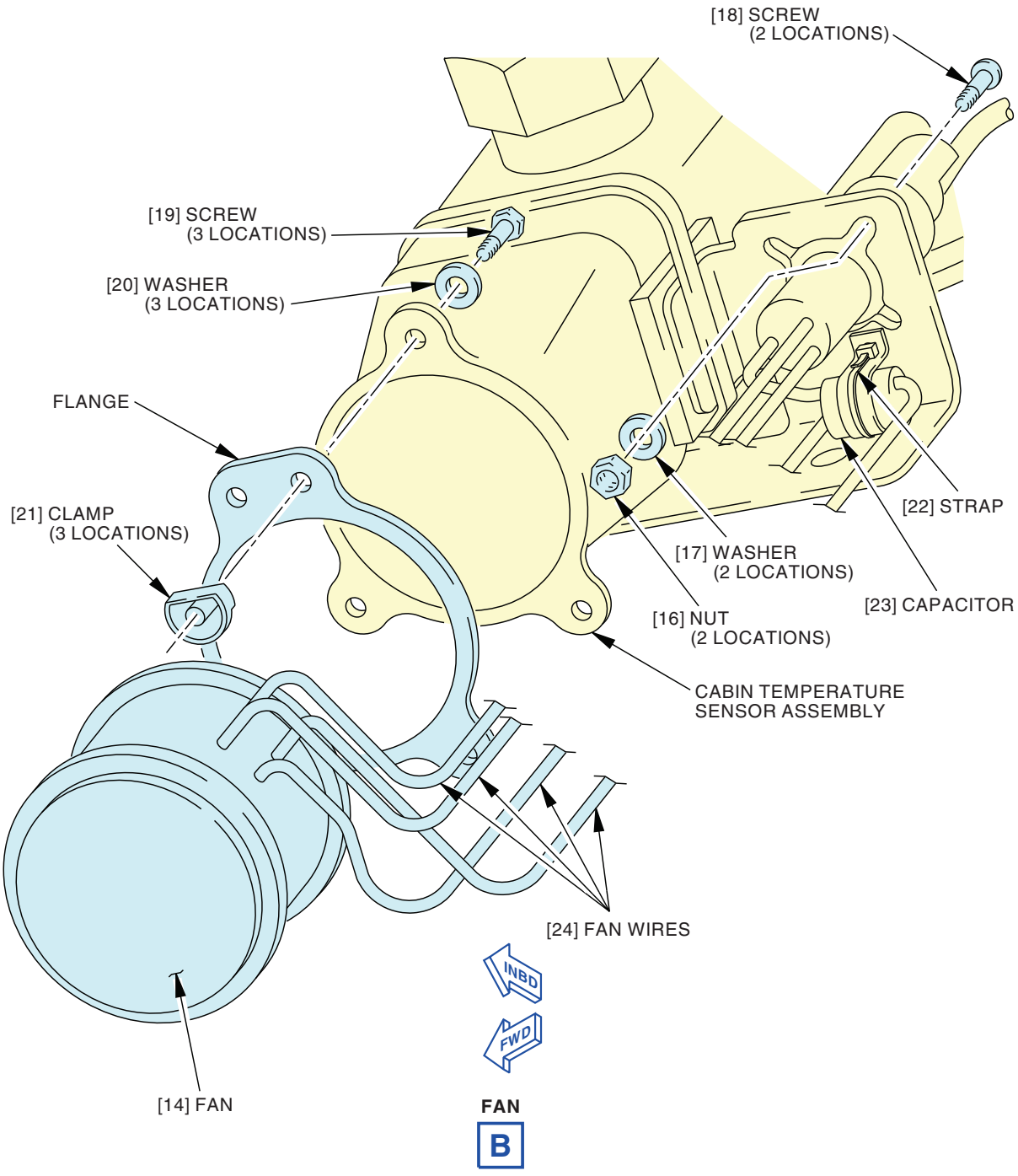
NOTE:

THE CABIN TEMPERATURE SENSOR PANEL IS SHOWN IN THE OPEN POSITION.

2794663 S0000637168_V2

Passenger Cabin Temperature Sensor Fan Installation
Figure 402/21-61-11-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Passenger Cabin Temperature Sensor Fan Installation
Figure 402/21-61-11-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-11-400-801

3. Cabin Temperature Sensor Fan Installation

(Figure 401, Figure 402)

A. References

Reference	Title
25-11-21-400-803	Flight Compartment Aft Ceiling Panel Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A50589	Compound - Electrical Insulating Coating	BMS5-37 Type I Class B-2
B00316	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I, ASTM D-3735 Type I
B50078	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I (Supersedes BMS3-2 Type I)
B50079	Solvent - Aliphatic Naphtha (For Acrylic Plastics)	TT-N-95 Type II (Supersedes BMS3-2 Type II)
G50262	Wiper - Cleaning	BMS15-5

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Fan	21-61-11-01-055	PTF ALL

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

E. Flight Compartment Cabin Temperature Sensor Fan Installation

SUBTASK 21-61-11-420-001

(1) Do these steps to install the fan [3] in the flight compartment (Figure 401):

(a) Put the fan [3] in its position with the other components of the cabin temperature sensor assembly.

(b) Install the clamps [10], washers [11], and screws [12] to attach the fan [3] to the cabin temperature sensor assembly.

NOTE: The screws head direction can be with the 180 degrees turn. Cabin temperature sensor assembly part numbers P/N 213A6101-14 and P/N 213A6101-15 have fastener head outside the housing.

(c) Install the fan wires [4] on the fan terminal board [13] with the screws [5] and washers [6].

NOTE: Use the tags on the fan wires to put the fan wires in the correct locations.

1) Clean the mating surfaces on the terminal board [41] with wiper, G50262, soaked in aliphatic naphtha, B50079, or aliphatic naphtha, B50078, or solvent, B00316.

2) Wipe dry with wiper, G50262, before solvent evaporates.

a) If it is necessary repeat steps to completely remove contaminants.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Coat 0.05 ±0.03 in. (1.3 ±0.8 mm) thick of compound, A50589, on fan wires [4].

NOTE: Coatings that are thinner than 0.04 in. (1.0 mm) are difficult to remove in-service and should be avoided.

(d) To install the aft ceiling panel [7] in the flight compartment, do this task: Flight Compartment Aft Ceiling Panel Installation, TASK 25-11-21-400-803.

(e) Connect the electrical connector [1] and electrical connector [2] to the cabin temperature sensor assembly.

(f) To install the access panel, lift it to its position and push firmly to make sure that the hook and loop strips are fully engaged.

F. Passenger Compartment Cabin Temperature Sensor Fan Installation

SUBTASK 21-61-11-420-003

(1) Install the fan [14] in the passenger compartment as follows:

CAUTION

THE FAN, P/N 041510000, CONTAINS SENSITIVE BEARINGS AND YOU MUST HANDLE IT CAREFULLY. DAMAGE TO THE FAN CAN OCCUR IF THIS CAUTION IS NOT OBEYED.

(a) Put the fan [14] in its position with the other components of the cabin temperature sensor panel assembly.

1) Safety the capacitor [23] to the bracket assembly with strap [22].

a) Use the tags on the fan wires [24] to put the fan wire [24] in the correct locations.

NOTE: Fan P/N 041510000 does not have a terminal board. Fan includes the capacitor.

NOTE: The capacitor and contacts can be replaced with the supplier Component Maintenance Manual (CMM).

(b) Install the screws [19], washers [20], and clamps [21] to attach the fan [14] to the cabin temperature sensor assembly.

(c) Install the nuts [16], washers [17], and screws [18].

(d) Install the contacts in the electrical connector [15].

(e) Install the electrical connector [15].

(f) Partially lift the panel for the cabin temperature sensor and install the lanyard.

(g) Lift the panel for the cabin temperature sensor to its closed position.

G. Sensor Fan Installation Test

SUBTASK 21-61-11-860-008

(1) Do this step for the fan in the flight compartment zone:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

SUBTASK 21-61-11-860-009

(2) Do this step for the fan in the forward passenger compartment zone:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-11-860-010

(3) Do this step for the fan in the aft passenger compartment zone:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

SUBTASK 21-61-11-710-002

(4) Do this test:

(a) Hold a 3 in. (76 mm) by 4 in. (102 mm) sheet of lightweight paper over the air inlet grille of the cabin temperature sensor assembly.

1) Make sure that the fan holds the paper in its position.

2) Remove the paper.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRIM AIR DUCT PRESSURE SEAL - REMOVAL/INSTALLATION

1. General

A. This procedure has the following tasks:

(1) Trim Air Duct Pressure Seal Removal.

(2) Trim Air Duct Pressure Seal Installation.

B. The Trim Air Duct Pressure Seals and support rings are used to seal ECS (Environmental Control System) ducting penetration through the pressure bulkhead of the Airplane.

TASK 21-61-12-000-801

2. Trim Air Duct Pressure Seal Removal

(Figure 401)(Figure 402)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-61-21-000-801	Zone Trim Air Modulating Valve Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box

C. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
821	Forward Cargo Door

D. Prepare for Removal

SUBTASK 21-61-12-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806

SUBTASK 21-61-12-860-002

(2) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach DO-NOT-OPERATE tags.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach DO-NOT-OPERATE tags.

(c) Set the BLEED APU switch to the OFF position and attach a DO-NOT-OPERATE tag.

SUBTASK 21-61-12-860-003

(3) Do this step on the P5-17 Cabin Temperature Panel (located on the P5 forward overhead panel):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Set the TRIM AIR switch to the OFF position and attach a DO-NOT-OPERATE tag.

SUBTASK 21-61-12-010-001

(4) To get access to the trim air duct pressure seals [6] in the ECS pack bay, do these steps:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(b) To open the right ECS access door do these steps:

(TASK 06-41-00-800-801)

1) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

2) Open this access door:

Number	Name/Location
192CR	ECS Access Door

E. Trim Air Duct Pressure Seal Removal

SUBTASK 21-61-12-000-001

(1) To remove the trim air duct pressure seal [6], first remove the trim air modulating valve [1]: (TASK 21-61-21-000-801).

SUBTASK 21-61-12-020-004

(2) To remove trim air duct pressure seal [6], do these steps:

(a) Remove the 12 bolts [2], washers [3], and the two seal ring halvess [5], to disconnect the trim air duct pressure seal [6] from the pressure bulkhead.

NOTE: The 2 bolts at the 11 and 12 o'clock positions on the flight deck and aft zone pressure seals attach the duct overheat sensor to the pressure bulkhead. They are longer than the other bolts.

(b) Remove the clamp [7] from the trim air duct pressure seal [6].

(c) Remove the trim air duct pressure seal [6] from the trim air duct [8].

SUBTASK 21-61-12-020-005

(3) If it is necessary to remove the trim air duct [8] that penetrates through the pressure bulkhead, then do these steps: (Figure 402)

(a) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

(b) Remove the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

(c) Support the trim air duct [12].

(d) Remove the aft coupling [11] that connects trim air duct [12] to trim air duct [8] that penetrates the pressure bulkhead in the mix manifold bay.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

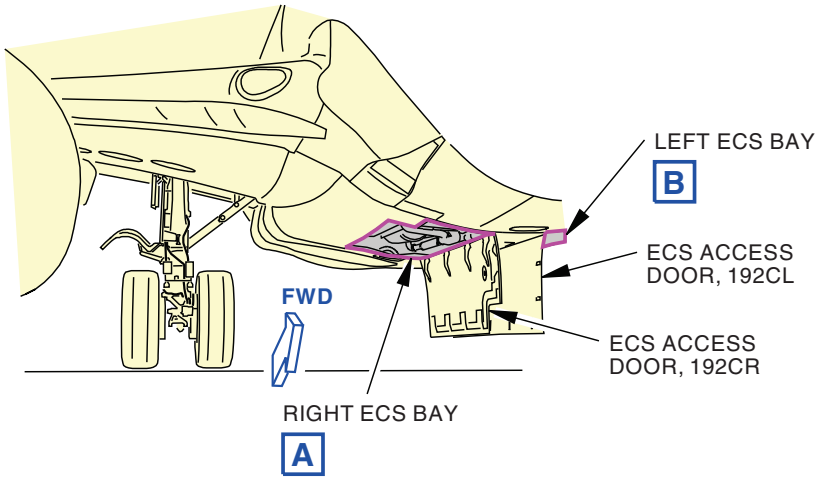
(e) Remove trim air duct [8].

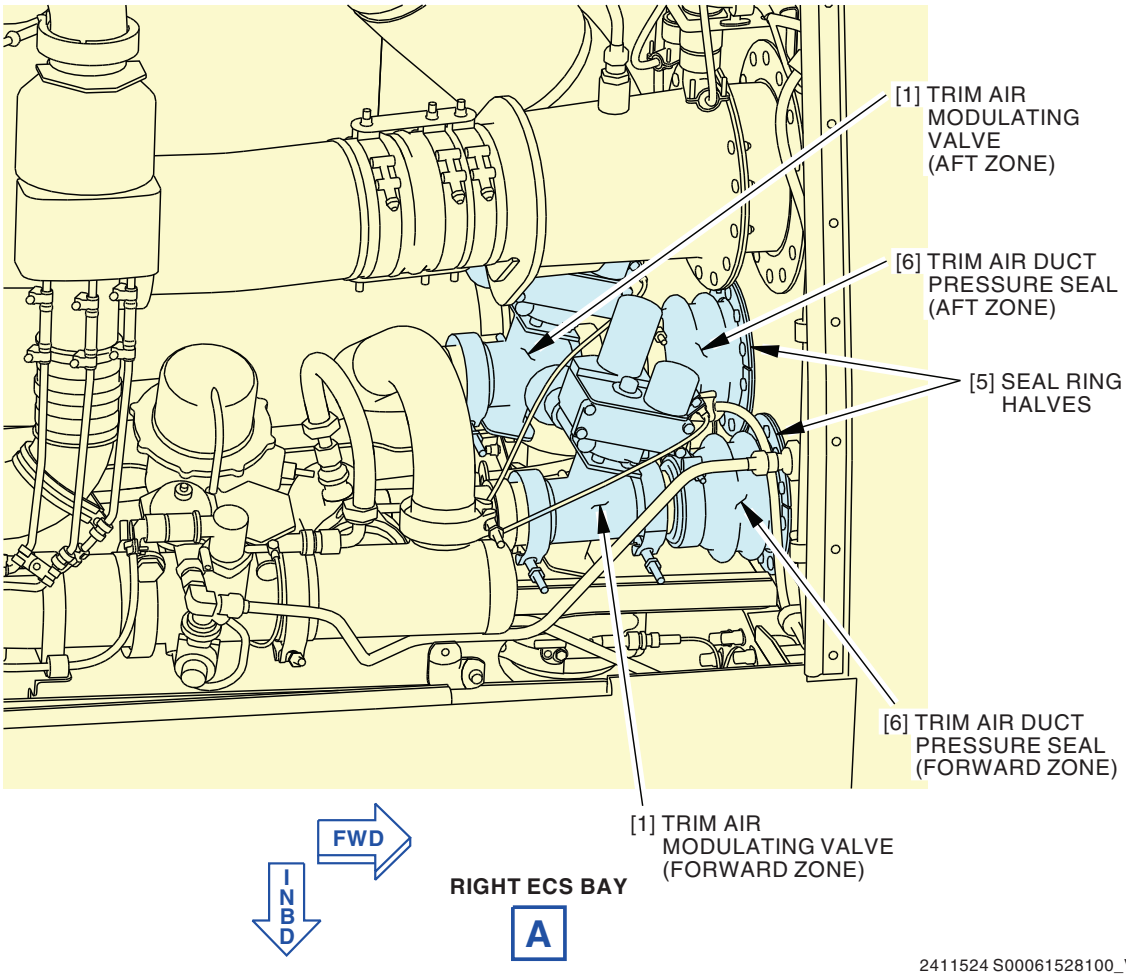
SUBTASK 21-61-12-420-001

(4) Install covers on all duct openings to keep out unwanted materials.

——— END OF TASK ———

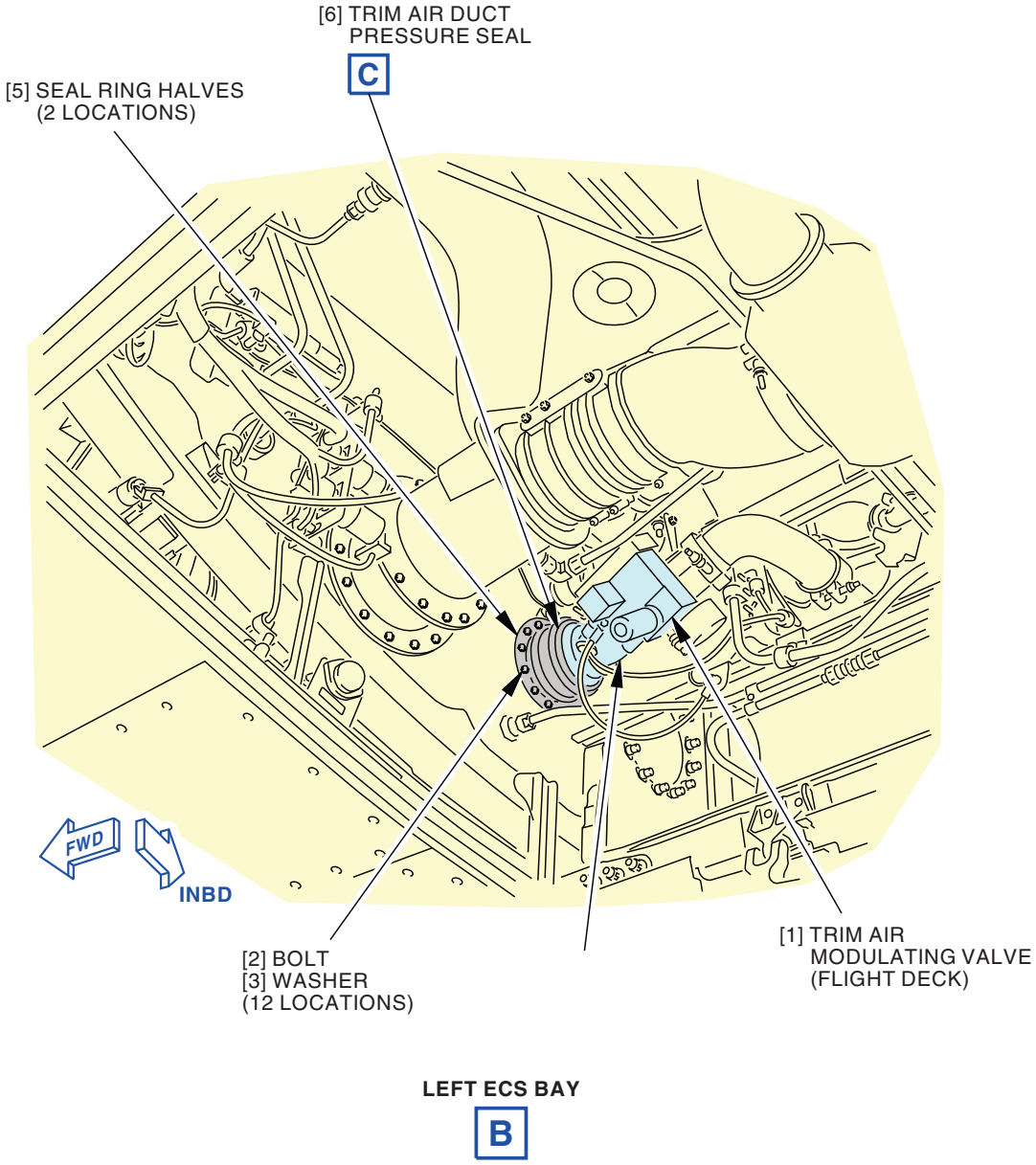
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Trim Air Duct Pressure Seal Installation
Figure 401/21-61-12-990-801 (Sheet 1 of 3)

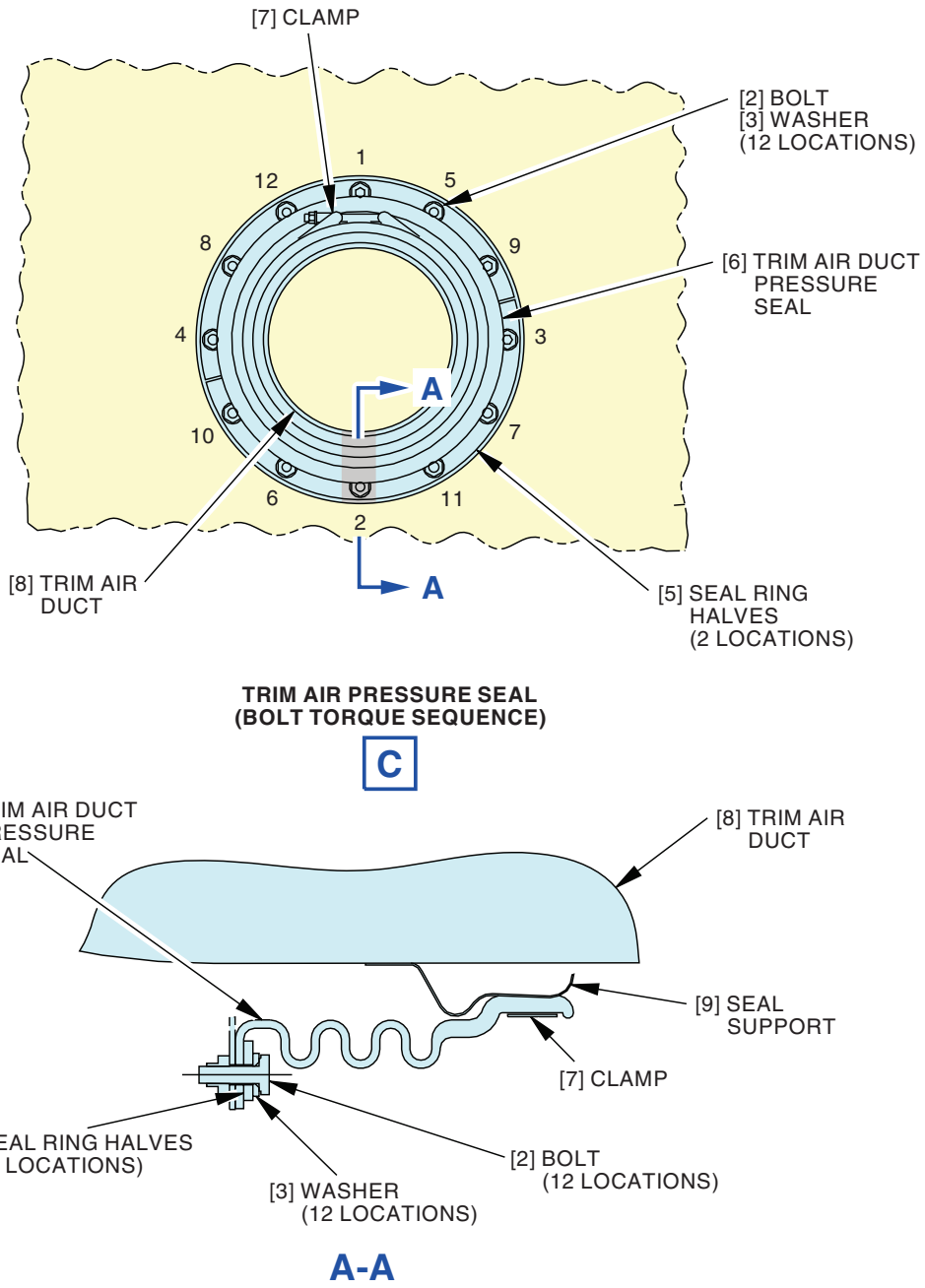
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411525 S00061528101_V1

Trim Air Duct Pressure Seal Installation
Figure 401/21-61-12-990-801 (Sheet 2 of 3)

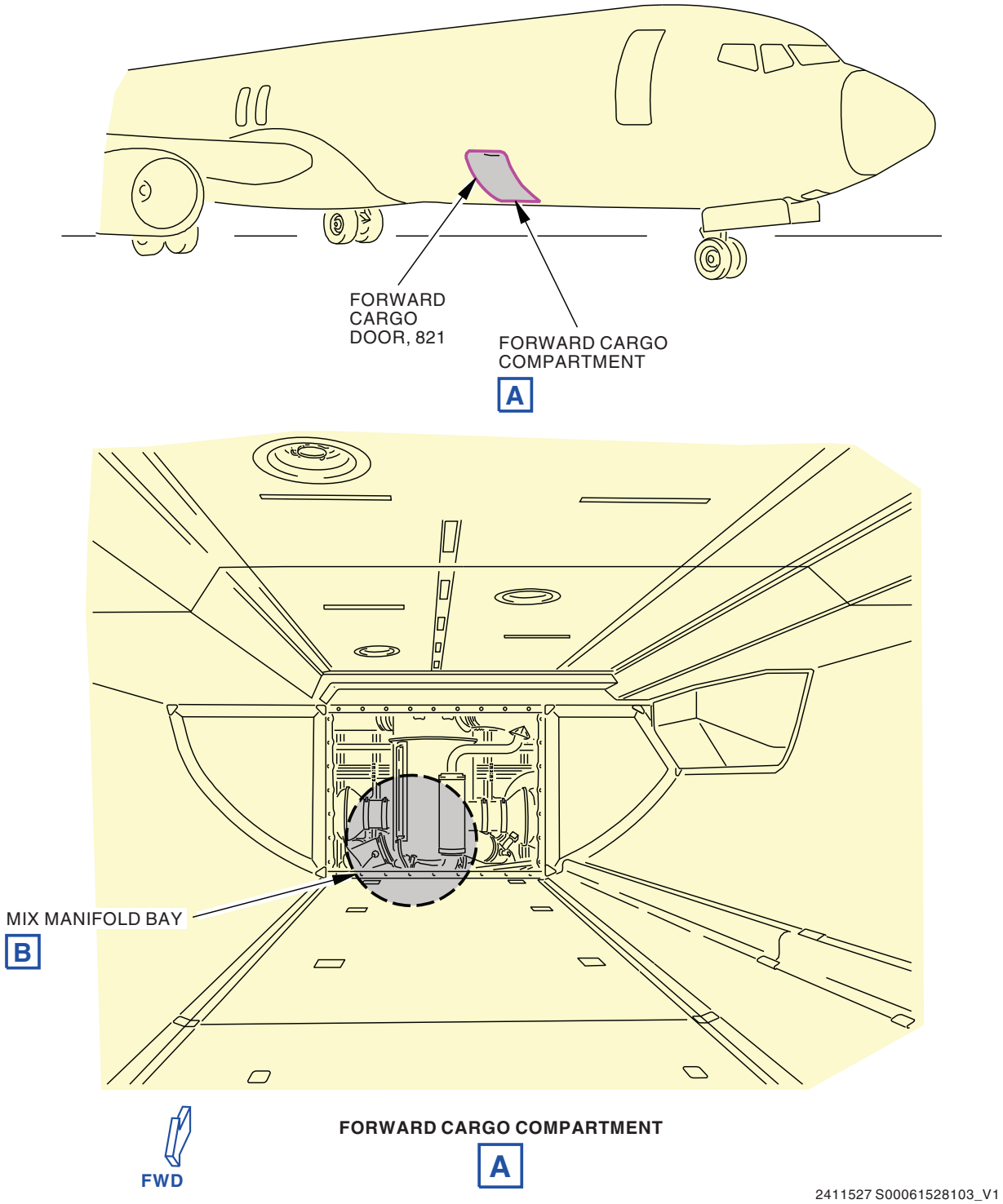
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2411526 S00061528102_V1

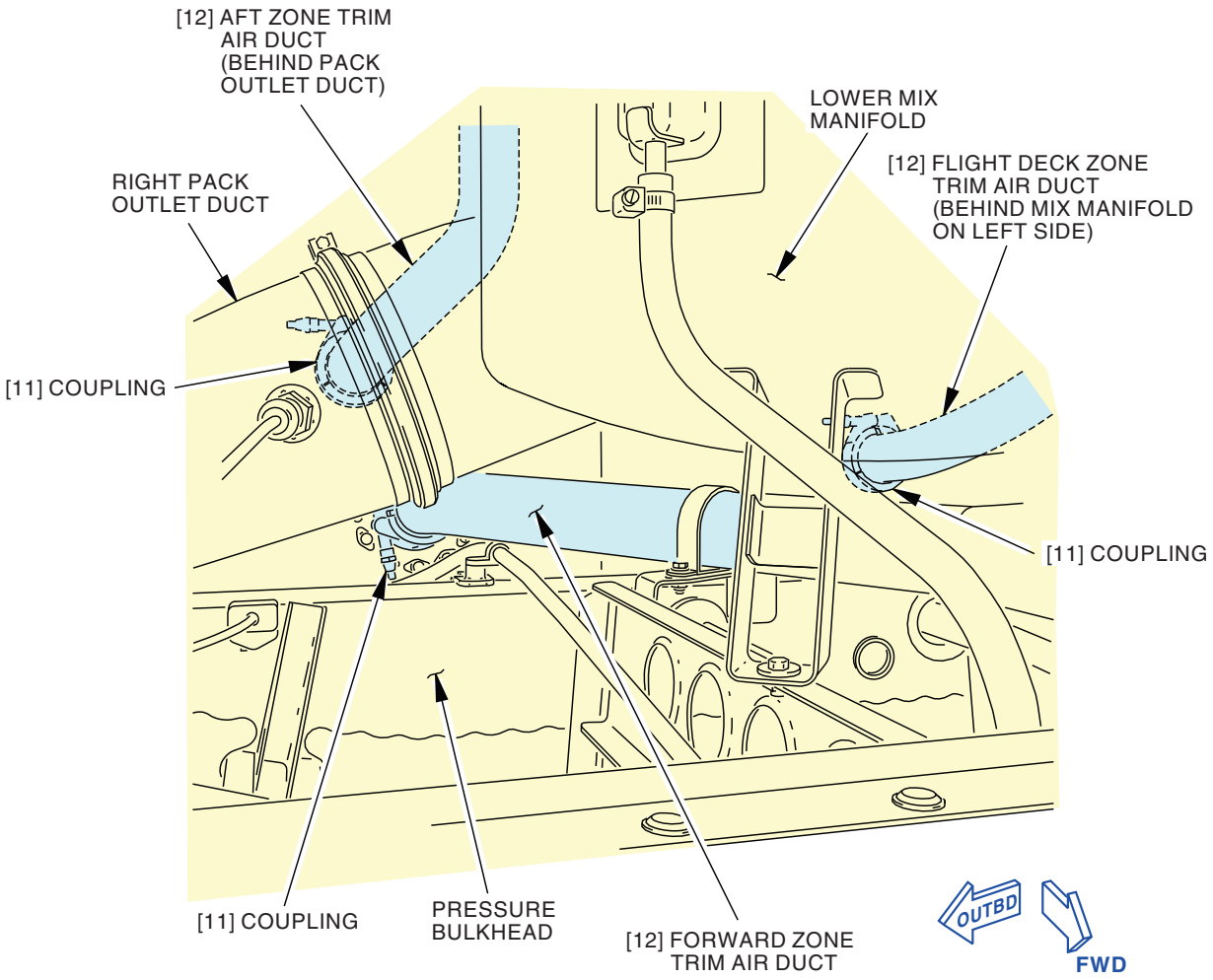
Trim Air Duct Pressure Seal Installation
Figure 401/21-61-12-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trim Air Duct Installation
Figure 402/21-61-12-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411528 S00061528104_V1

Trim Air Duct Installation
Figure 402/21-61-12-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-12-400-801

3. Trim Air Duct Pressure Seal Installation

(Figure 401, Figure 402)

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
21-61-21-400-801	Zone Trim Air Modulating Valve Installation (P/B 401)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
D00010	Compound - Thread Antiseize, High Temperature	MIL-PRF-907
G00091	Compound - Oxygen System Leak Detection - Snoop Leak Detector	MIL-PRF-25567

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
821	Forward Cargo Door

E. Trim Air Duct Pressure Seal Installation

SUBTASK 21-61-12-630-001

(1) Remove the covers from the duct openings and make sure there is no unwanted materials inside the duct.

SUBTASK 21-61-12-420-002

(2) If you removed the trim air duct [8] that penetrates through the pressure bulkhead, then do these steps to install it (Figure 401):

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(a) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

(b) Remove the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

(c) Slide the trim air duct [8] through the pressure bulkhead hole.

(d) Support the trim air duct [8].

(e) Loosely install the coupling [11] in the mix manifold bay to connect the trim air duct [8] to trim air duct [12].

SUBTASK 21-61-12-100-001

(3) Use a clean wiper and solvent, B00083 to clean the pressure seal support [9] area on the trim air duct [8] that penetrates through the pressure bulkhead (Figure 401).

SUBTASK 21-61-12-420-003

(4) Slide the seal support [9] onto the trim air duct [8].

SUBTASK 21-61-12-420-004

(5) For installation of the trim air duct pressure seal [6], do these steps (Figure 401):

(a) Push the trim air duct pressure seal [6] to the mating surface of the pressure bulkhead.

(b) Apply compound, D00010 to the threads of the bolts [2].

(c) Install the two seal ring halvess [5] on the trim air duct pressure seal [6] with the 12 bolts [2] and washers [3].

NOTE: The 2 bolts at the 11 and 12 o'clock positions on the flight deck and aft zone pressure seals attach the duct overheat sensor to the pressure bulkhead. They are longer than the other bolts. The duct overheat sensor support bracket must be installed on top of the seal ring halves [5].

(d) Tighten the bolts [2] to 32.5 ±2.5 in-lb (3.7 ±0.3 N·m) in the sequence shown in Figure 401.

(e) Install the clamp [7] loosely on the trim air duct pressure seal [6].

(f) Do these steps to tighten the clamp [7]:

1) Turn the torque wrench to tighten the clamp [7]. Observe the torque needed to turn the nut on the clamp's T-bolt before the clamp becomes tight.

NOTE: The observed torque should be 2 to 15 inch-pounds (0.23-1.7 newton-meters). This torque is referred to as run-on torque.

2) Add the observed run-on torque to 20 ±2 in-lb (2.3 ±0.3 N·m). For example, if the run-on torque needed to turn the T-bolt nut was 10 in-lb (1 N·m), then the torque needed to tighten the clamp is 30 ±2 in-lb (3 ±1 N·m). This is the calculated torque needed to tighten the clamp.

3) Tighten the clamp [7] to the torque calculated in the previous step.

4) Wait a minimum of 2 hours, then retighten the clamp [7] to the torque calculated in the previous step.

SUBTASK 21-61-12-420-005

(6) Install the trim air modulating valve [1] (TASK 21-61-21-400-801).

SUBTASK 21-61-12-420-006

(7) Align the trim air duct [8] and trim air duct [12].

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SUBTASK 21-61-12-420-007

(8) Tighten the coupling [11] to 55 +5 / -0 in-lb (6.2 +0.6 / -0.0 N·m).

(a) Make sure that the loop bands do not touch.

(b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.310 in. (7.874 mm).

F. Trim Air Duct Leak Test

SUBTASK 21-61-12-860-004

(1) Do this step on the P5-17 Cabin Temperature Panel:

(a) Remove the DO-NOT-OPERATE tag from the TRIM AIR switch.

SUBTASK 21-61-12-860-005

(2) Do these steps on the P5-10 air conditioning panel:

(a) Remove the DO-NOT-OPERATE tags from the L and R PACK switches.

(b) Remove the DO-NOT-OPERATE tags from the BLEED 1 and 2 switches.

(c) Remove the DO-NOT-OPERATE tag from the BLEED APU switch.

SUBTASK 21-61-12-860-006

(3) Do this task: Supply Pressure to the Pneumatic System with the APU, TASK 36-00-00-860-803.

SUBTASK 21-61-12-860-007

(4) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

(a) Make sure the TRIM AIR switch is set to ON position.

SUBTASK 21-61-12-790-001

(5) Do a leak test:

(a) Apply a Snoop Leak Detector compound, G00091 to the trim air duct joint areas.

(b) Make sure there is no leakage at the trim air duct joint areas.

(c) If there is leakage, do these steps:

1) Do this task : Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804.

a) Make sure the TRIM AIR switch is set to OFF position.

2) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

3) Loosen the couplings at the trim air duct joint areas.

4) Make sure the ducts are aligned at the joints.

5) Tighten the couplings to 55 +5 / -0 in-lb (6.2 +0.6 / -0.0 N·m).

a) Make sure that the loop bands do not touch.

b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.310 in. (7.874 mm).

6) Do the trim air leak test again.

G. Cabin Pressure Leak Test

SUBTASK 21-61-12-790-002

(1) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane to its Usual Condition

SUBTASK 21-61-12-860-008

(1) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804

(a) Make sure the TRIM AIR switch is set to OFF position.

SUBTASK 21-61-12-860-009

(2) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806

SUBTASK 21-61-12-410-001

(3) Install the aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 21-61-12-410-002

(4) Close these access panels:

Number	Name/Location
192CL	ECS Access Door
821	Forward Cargo Door

SUBTASK 21-61-12-410-003

(5) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

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TEMPERATURE SENSOR ASSEMBLY DUCT/HOUSING - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Flight Compartment Temperature Sensor Assembly Housing Removal

(2) Flight Compartment Temperature Sensor Assembly Housing Installation

(3) Passenger Cabin Temperature Sensor Assembly Duct Removal

(4) Passenger Cabin Temperature Sensor Assembly Duct Installation

(5) Flight Compartment Temperature Sensor Assembly Housing Cleaning

(6) Passenger Cabin Temperature Sensor Assembly Duct Cleaning.

B. There is a temperature sensor assembly housing in the flight compartment temperature control zone.

C. There are two temperature sensor assembly ducts in the passenger cabin temperature control zone.

(1) In the forward passenger cabin temperature control zone.

(2) In the aft passenger cabin temperature control zone.

TASK 21-61-14-000-801

2. Flight Compartment Temperature Sensor Assembly Housing Removal

Figure 201

A. References

Reference	Title
25-11-21-000-803	Flight Compartment Aft Ceiling Panel Removal (P/B 201)

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Prepare for the Removal

SUBTASK 21-61-14-860-001

(1) Do this step for the removal of the temperature sensor assembly housing:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

D. Temperature Sensor Assembly Housing Removal

SUBTASK 21-61-14-010-001

(1) Loosen the access panel on the aft ceiling lining in the flight compartment and let it hang on the straps.

NOTE: The access panel on the aft ceiling lining is held in position by hook and loop strips.

SUBTASK 21-61-14-020-001

(2) Disconnect the electrical connector [2] for the fan.

SUBTASK 21-61-14-020-002

(3) Disconnect the electrical connector [3] from the temperature sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-14-020-003

(4) Remove the aft ceiling panel [10] in the flight compartment. To remove the panel, do this task: Flight Compartment Aft Ceiling Panel Removal, TASK 25-11-21-000-803.

SUBTASK 21-61-14-020-004

(5) Remove the two screws [9] and the temperature sensor assembly housing [1] from the aft ceiling panel [10].

SUBTASK 21-61-14-020-005

(6) Remove the screws [11], washers [12], temperature sensor assembly [4] and the mounting plate [13] from the temperature sensor assembly housing [1].

SUBTASK 21-61-14-020-006

(7) Remove the screws [7], washers [6] and clamps [5] that attach the fan [8] to the temperature sensor assembly housing [1].

SUBTASK 21-61-14-010-002

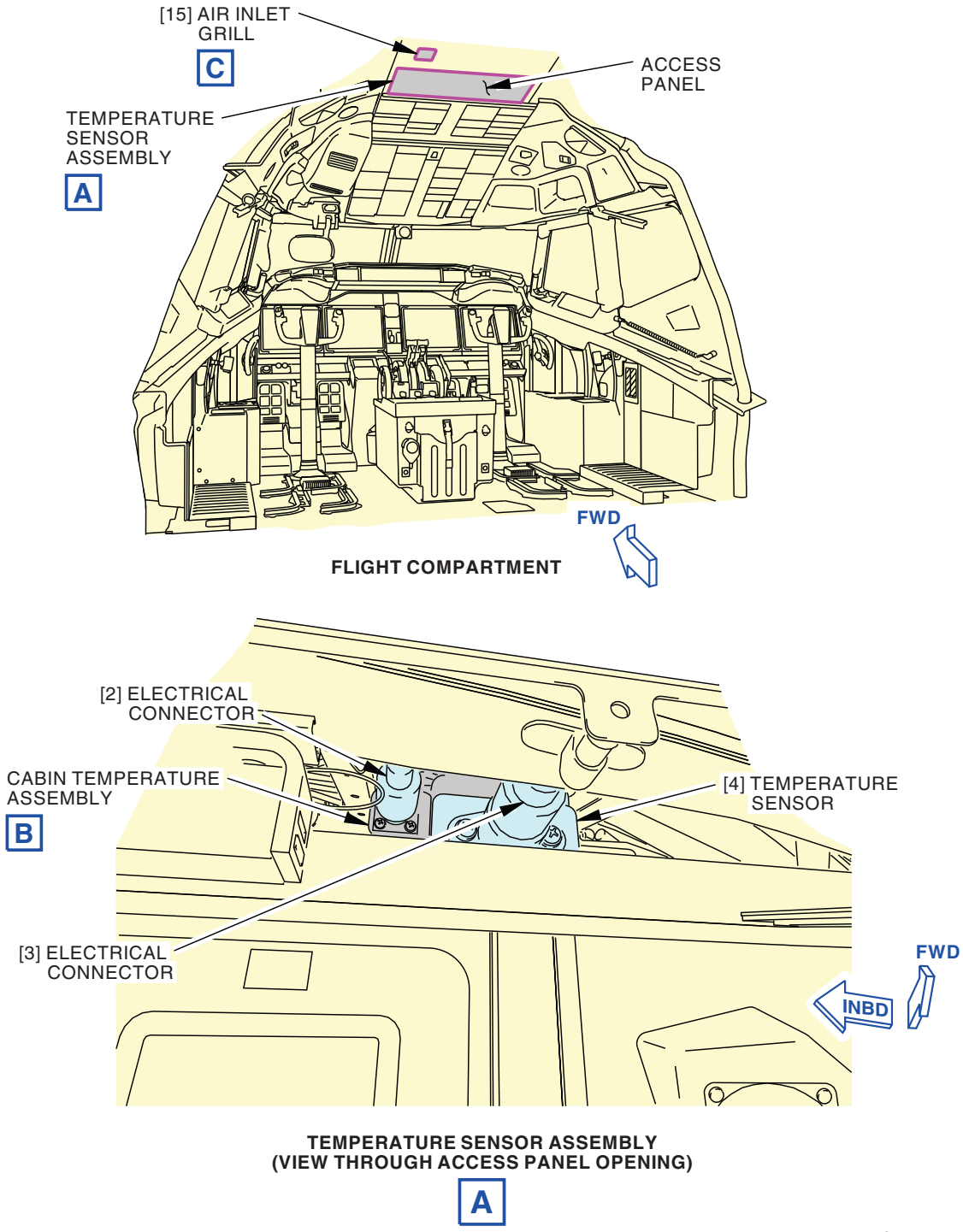
(8) Move the fan [8] away from the opening in the temperature sensor assembly housing [1].

SUBTASK 21-61-14-000-001

(9) Remove the four screws [16] that attach the air inlet grill [15] and the temperature sensor assembly housing [1] to the aft ceiling panel [10].

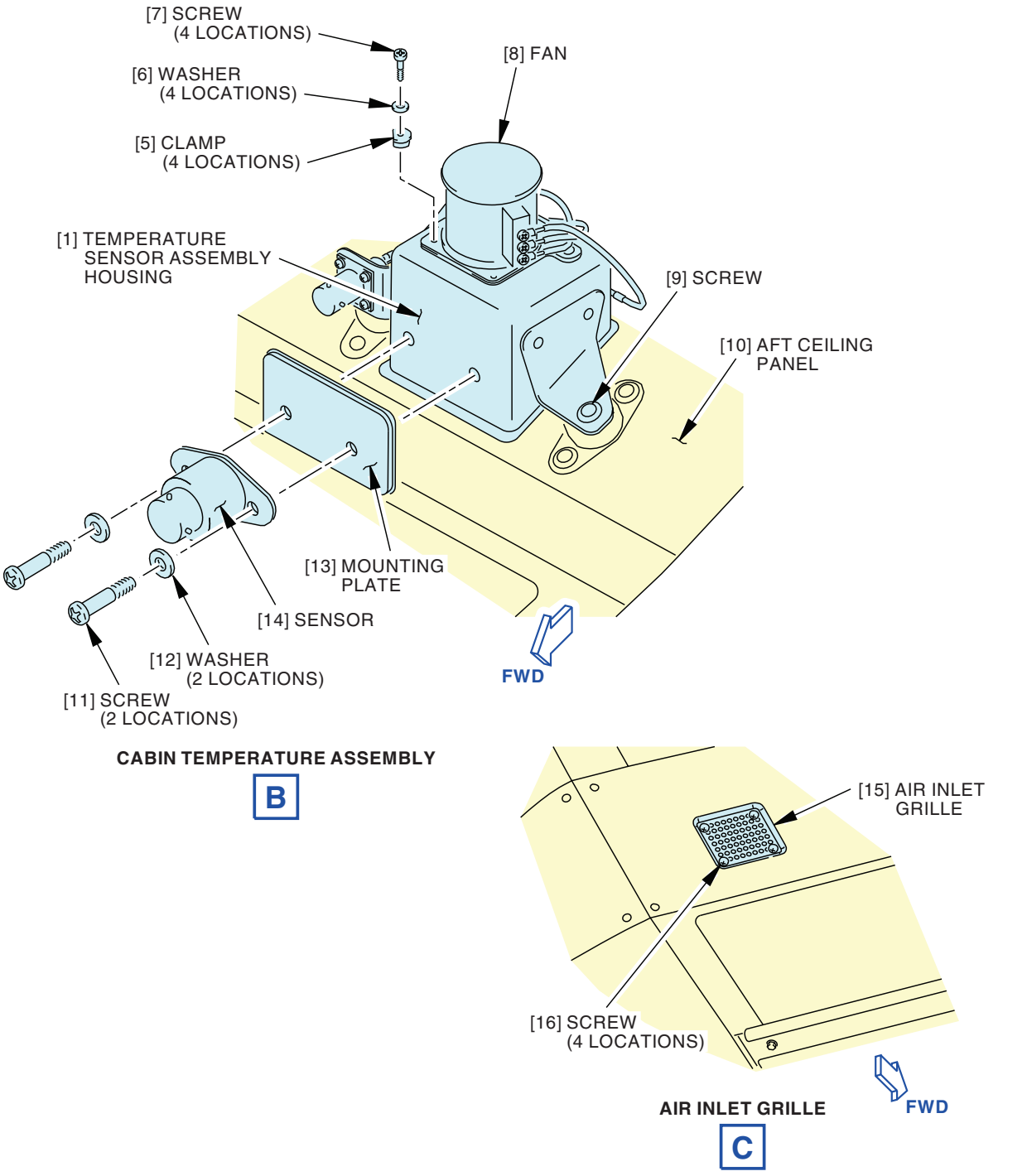
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Compartment Cabin Temperature Housing Installation
Figure 201/21-61-14-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Compartment Cabin Temperature Housing Installation
Figure 201/21-61-14-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-14-400-801

3. Flight Compartment Temperature Sensor Assembly Housing Installation

Figure 201

A. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
25-11-21-400-803	Flight Compartment Aft Ceiling Panel Installation (P/B 201)
46-13-02-700-804	Present Leg Faults Check (P/B 201)

B. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Flight Compartment Temperature Sensor Assembly Housing Installation

SUBTASK 21-61-14-420-001

(1) Put the fan [8] in its position on the temperature sensor assembly housing [1].

(a) Install the four clamps [5], washers [6] and screws [7] to attach the fan [8] to the temperature sensor assembly housing [1].

SUBTASK 21-61-14-420-002

(2) Put the mounting plate [13] and the sensor [14] in their position on the temperature sensor assembly housing [1].

(a) Make sure that the foam gasket of the mounting plate assembly is not damaged.

(b) Install the two screws [11] and washers [12].

SUBTASK 21-61-14-420-003

(3) Install the temperature sensor assembly housing [1].

(a) Put the temperature sensor assembly housing [1] in its position on the aft ceiling panel [10].

(b) Install the two screws [9] to attach the temperature sensor assembly housing [1] to the aft ceiling panel [10].

SUBTASK 21-61-14-420-017

(4) Put the air inlet grill [15] in position on the aft ceiling panel [10].

(a) Install the four screws [16] to attach the air inlet grill [15] to the aft ceiling panel [10].

SUBTASK 21-61-14-420-004

(5) Install the aft ceiling panel [10] in the flight compartment. To install the panel, do this task: Flight Compartment Aft Ceiling Panel Installation, TASK 25-11-21-400-803.

SUBTASK 21-61-14-420-005

(6) Connect the electrical connector [3] to the sensor [14].

SUBTASK 21-61-14-420-006

(7) Connect the electrical connector [2] for the fan [8].

SUBTASK 21-61-14-420-007

(8) Install the access panel.

(a) Lift the panel to its position and push firmly to make sure that the hook and loop strips are fully engaged.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Flight Compartment Temperature Sensor Assembly Housing Installation Test

SUBTASK 21-61-14-860-004

(1) Do this step for the installation of the fan:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

SUBTASK 21-61-14-710-001

(2) Do this test of the fan operation:

(a) Hold a 3 inch by 4 inch sheet of lightweight paper over the air inlet grill [15]of the temperature sensor assembly housing [1].

1) Make sure that the fan holds the paper in its position.

2) Remove the paper.

SUBTASK 21-61-14-700-001

(3) Look for maintenance messages that are related to the cabin temperature sensor Present Leg Faults Check, TASK 46-13-02-700-804.

(a) Make sure that maintenance messages that are related to the temperature sensor are NOT ACTIVE.

(b) If a maintenance message is ACTIVE, refer to the IFIM and do the applicable procedure(s) to repair the fault indicated by the maintenance message.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-14-860-005

(1) Remove electrical power from the airplane if it is not necessary. To remove electrical power, do this task: Remove Electrical Power, TASK 24-22-00-860-802.

——— END OF TASK ———

TASK 21-61-14-000-802

4. Passenger Cabin Temperature Sensor Assembly Duct Removal

(Figure 202)

A. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

B. Prepare for the Removal

SUBTASK 21-61-14-860-002

(1) Do this step for the removal of the passenger cabin temperature assembly duct [61] for the forward passenger cabin zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-14-860-003

(2) Do this step for the removal of the passenger cabin temperature assembly duct [61] for the aft passenger cabin zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

C. Passenger Cabin Temperature Sensor Assembly Duct Removal

SUBTASK 21-61-14-010-004

(1) Lower the panel for the cabin temperature assembly duct [61].

(a) Put a small allen wrench or other applicable tool in the access holes to release the latches.

(b) Lower the panel and disconnect the lanyard.

(c) Move the panel to the fully lowered position.

SUBTASK 21-61-14-020-007

(2) Disconnect the electrical connector [30] for the fan [43].

SUBTASK 21-61-14-020-008

(3) Disconnect the electrical connector [33] for the temperature sensor [49].

SUBTASK 21-61-14-020-009

(4) Disconnect the electrical connector [32] for the temperature bulb [45].

SUBTASK 21-61-14-020-010

(5) Remove the nuts [64], washer [63], and screw [62].

SUBTASK 21-61-14-020-011

(6) Remove the three screws [39], washers [40], clamps [42], flange [41], and fan [43] from the temperature assembly duct [61].

SUBTASK 21-61-14-020-012

(7) Remove the sensor [49] and temperature bulb [45] from the temperature assembly duct [61]:

(a) Remove the lockwire [47].

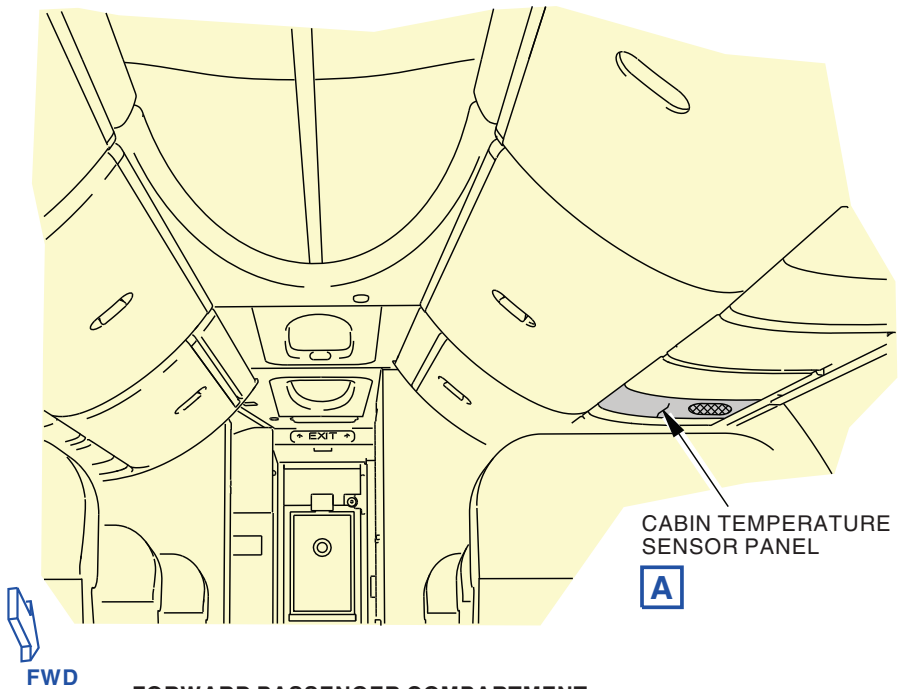
I (b) Remove the screws [46], lanyard [51] (if installed), and washers [48].

(c) Remove the sensor [49] and gasket [50].

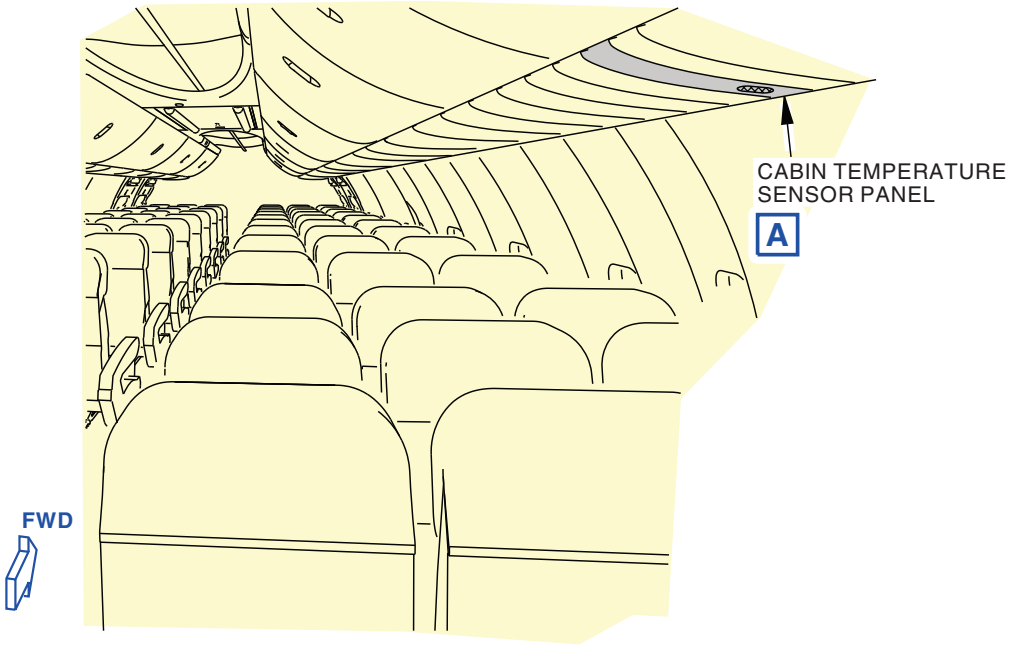
(d) Remove the temperature bulb [45] and lockwire washer [44].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD PASSENGER COMPARTMENT

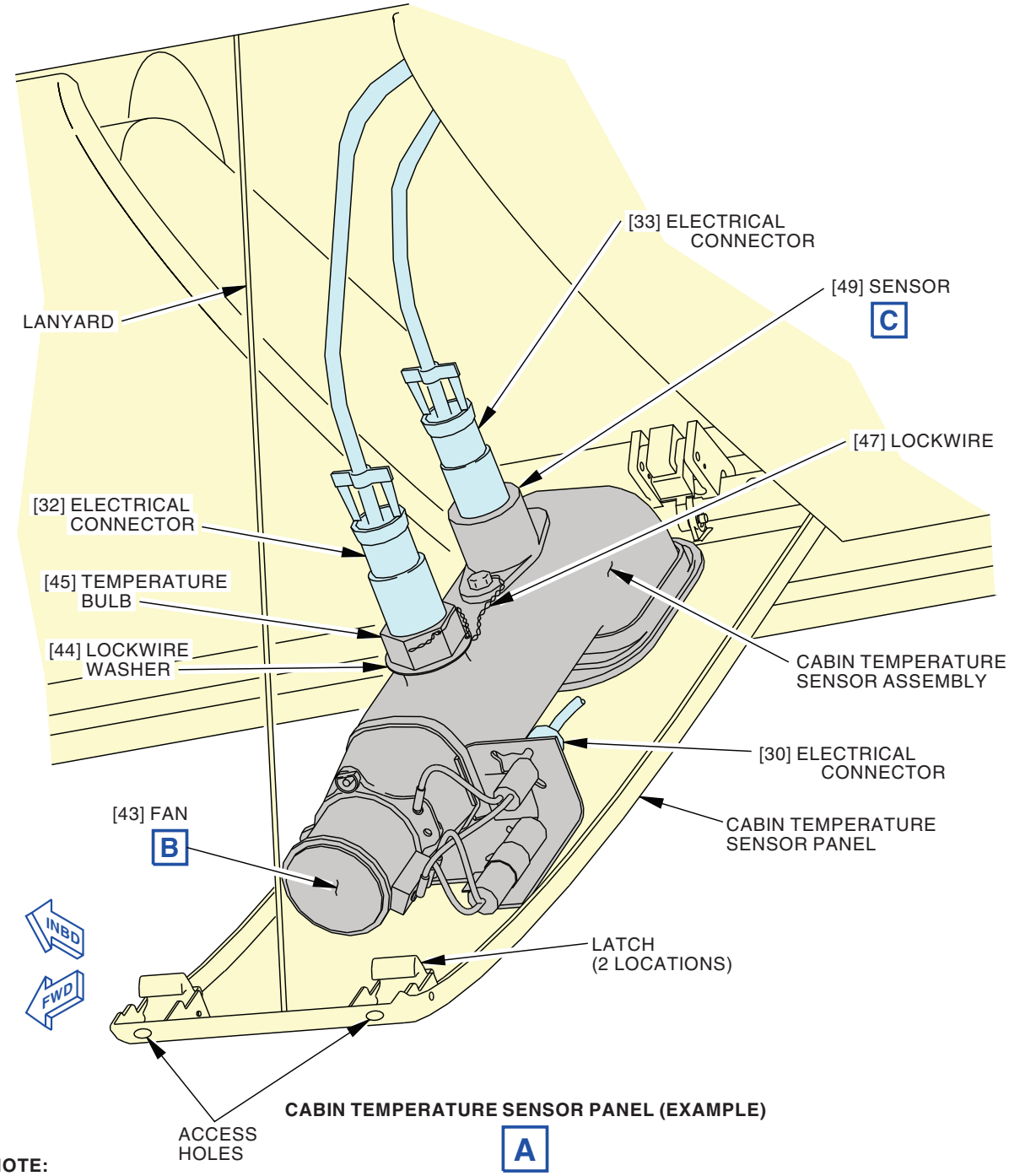


AFT PASSENGER COMPARTMENT

2411509 S00061528074_V1

Passenger Compartment - Cabin Temperature Duct Installation
Figure 202/21-61-14-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE SENSOR PANEL (EXAMPLE)

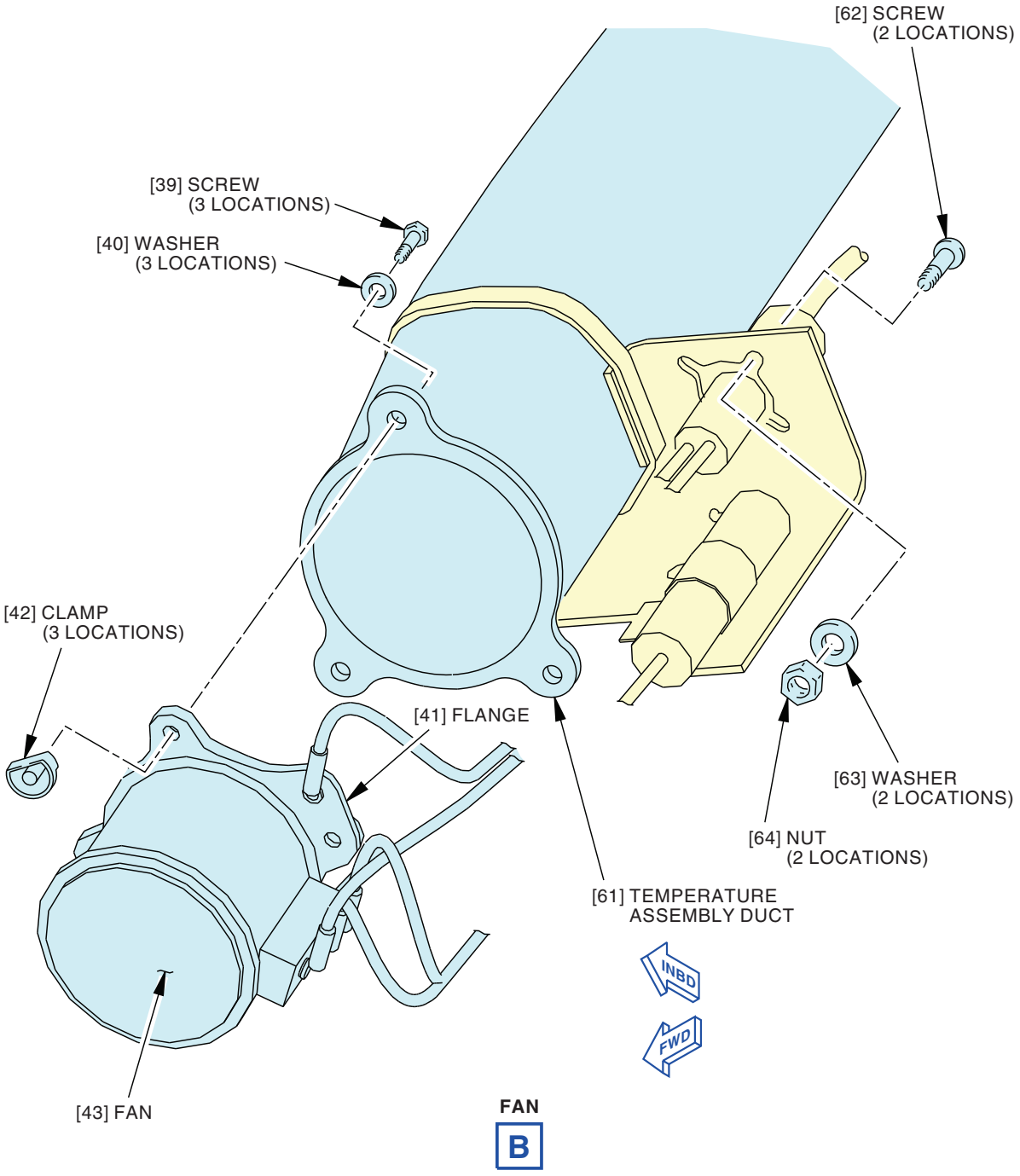
NOTE:

THE CABIN TEMPERATURE SENSOR PANEL IS SHOWN IN THE OPEN POSITION.

2411531 S00061528112_V3

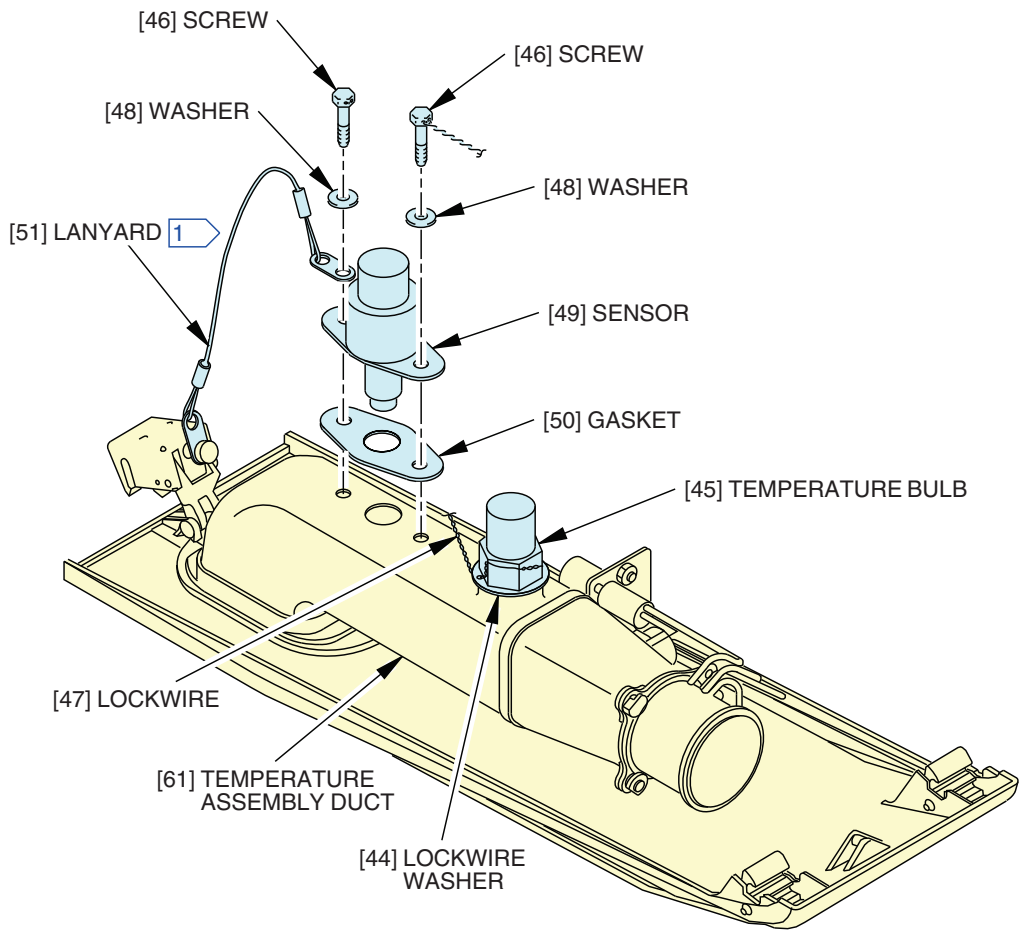
Passenger Compartment - Cabin Temperature Duct Installation
Figure 202/21-61-14-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Passenger Compartment - Cabin Temperature Duct Installation
Figure 202/21-61-14-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SENSOR (EXAMPLE)

C

1 IF INSTALLED.

2411533 S00061528114_V3

Passenger Compartment - Cabin Temperature Duct Installation
Figure 202/21-61-14-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-14-400-802

5. Passenger Cabin Temperature Sensor Assembly Duct Installation

Figure 202

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
46-13-02-700-804	Present Leg Faults Check (P/B 201)

B. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

C. Passenger Cabin Temperature Assembly Duct Installation

SUBTASK 21-61-14-420-018

(1) Do these steps to install the cabin temperature sensor fan [43]:

(a) Put the fan [43] and flange [41] in their position with the other components of the temperature sensor assembly.

(b) Install the three clamps [42], washers [40], and screws [39] to attach the fan [43] to the temperature sensor assembly.

(c) Install the screws [62], washers [63], and nuts [64].

SUBTASK 21-61-14-420-010

(2) Put the gasket [50] and sensor [49] in their position on the temperature assembly duct [61].

(a) Install the washers [48], lanyard [51] (if installed), and screws [46].

SUBTASK 21-61-14-420-009

(3) To install the cabin temperature bulb [45], do these steps:

(a) Install the lockwire washer [44] and temperature bulb [45] on the temperature sensor assembly.

SUBTASK 21-61-14-420-019

(4) Install the lockwire [47] trough the screw [46], lockwire washer [44], and temperature bulb [45] (TASK 20-10-44-400-801).

SUBTASK 21-61-14-420-012

(5) Connect the electrical connector [30] for the fan [43].

SUBTASK 21-61-14-420-013

(6) Connect the electrical connector [33] for the temperature sensor [49].

SUBTASK 21-61-14-420-014

(7) Connect the electrical connector [32] for the temperature bulb [45].

SUBTASK 21-61-14-410-001

(8) Partially lift the panel for the cabin temperature sensor and install the lanyard.

SUBTASK 21-61-14-410-002

(9) Lift the panel for the cabin temperature sensor to its closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Passenger Cabin Temperature Sensor Assembly Duct Installation Test

SUBTASK 21-61-14-860-006

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-14-710-002

(2) Do this test of the fan operation:

(a) Hold a 3 inch by 4 inch sheet of lightweight paper over the air inlet grille of the temperature sensor assembly.

1) Make sure that the fan holds the paper in its position.

2) Remove the paper.

SUBTASK 21-61-14-700-002

(3) Look for maintenance messages that are related to the cabin temperature sensor Present Leg Faults Check, TASK 46-13-02-700-804.

(a) Make sure that maintenance messages that are related to the temperature sensor are NOT ACTIVE.

(b) If the maintenance message is ACTIVE, refer to the IFIM and do the applicable procedure(s) to repair the fault indicated by the maintenance message.

SUBTASK 21-61-14-710-003

(4) Do this test of the cabin temperature bulb that was replaced:

(a) Use a thermometer and make a record of the ambient temperature in the passenger compartment.

(b) Put the AIR TEMP selecto,r on the P5-17 temperature control panel, to PASS CABIN.

(c) Make sure that the temperature record with the thermometer and PASS CABIN TEMP gage indication are approximately the same.

——— END OF TASK ———

TASK 21-61-14-100-801

6. Flight Compartment Temperature Sensor Assembly Housing Cleaning

Figure 201

A. References

Reference	Title
21-61-10-100-801	Cabin Temperature Sensor Cleaning (P/B 201)

B. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00541	Cleaner - General Purpose Household Detergent	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Prepare for the Flight Compartment Temperature Sensor Assembly Housing Cleaning

SUBTASK 21-61-14-020-013

(1) Remove the temperature sensor assembly housing [1] for the flight compartment. To remove the housing, do this task: Flight Compartment Temperature Sensor Assembly Housing Removal, TASK 21-61-14-000-801.

E. Procedure

SUBTASK 21-61-14-160-001

(1) Prepare a cleaning solution of warm water and general purpose household detergent cleaner, B00541.

(a) Follow the mixing instructions for the cleaner.

SUBTASK 21-61-14-160-002

(2) Clean the inner and external surfaces of the housing with a cotton wiper, G00034 that is moist with the cleaning solution.

SUBTASK 21-61-14-160-003

(3) If the cotton wiper, G00034 with the cleaning solution did not remove all dirt deposits or contamination, clean the housing with a cotton wiper, G00034 that is moist with alcohol, B00130.

SUBTASK 21-61-14-160-004

(4) Wipe the housing with a cotton wiper, G00034 that is moist with clean water until all of the cleaning solution or alcohol are removed.

SUBTASK 21-61-14-160-005

(5) Dry the housing with a dry, clean cotton wiper, G00034.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-14-210-001

(1) Before the installation of the temperature sensor assembly housing [1], do these steps:

(a) Examine the sensor [14] for unwanted material.

(b) If unwanted material is found, clean the sensor: Cabin Temperature Sensor Cleaning, TASK 21-61-10-100-801.

SUBTASK 21-61-14-420-015

(2) Install the temperature sensor assembly housing [1], do this task: Passenger Cabin Temperature Sensor Assembly Duct Installation, TASK 21-61-14-400-802.

——— END OF TASK ———

TASK 21-61-14-100-802

7. Passenger Cabin Temperature Sensor Assembly Duct Cleaning

Figure 202

A. References

Reference	Title
21-61-10-100-801	Cabin Temperature Sensor Cleaning (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00541	Cleaner - General Purpose Household Detergent	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

C. Prepare for Cabin Temperature Assembly Duct Cleaning

SUBTASK 21-61-14-020-014

(1) Remove the temperature sensor assembly duct. To remove the duct, do this task: Passenger Cabin Temperature Sensor Assembly Duct Removal, TASK 21-61-14-000-802.

D. Procedure

SUBTASK 21-61-14-160-006

(1) Prepare a cleaning solution of warm water and general purpose household detergent cleaner, B00541.

(a) Follow the mixing instructions for the cleaner.

SUBTASK 21-61-14-160-007

(2) Clean the inner and external surfaces of the duct with a cotton wiper, G00034 that is moist with the cleaning solution.

SUBTASK 21-61-14-160-008

(3) If the cotton wiper, G00034 with the cleaning solution did not remove all dirt deposits or contamination, clean the duct with a cotton wiper, G00034 that is moist with alcohol, B00130.

SUBTASK 21-61-14-160-009

(4) Wipe the duct with a cotton wiper, G00034 that is moist with clean water until all of the cleaning solution or alcohol are removed.

SUBTASK 21-61-14-160-010

(5) Dry the duct with a dry, clean cotton wiper, G00034.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-14-210-002

(1) Before the installation of the temperature assembly duct in the passenger compartment, do the steps that follow:

(a) Examine the temperature sensor [49] for unwanted material.

(b) If unwanted material is found, clean the sensor [49]: Cabin Temperature Sensor Cleaning, TASK 21-61-10-100-801.

SUBTASK 21-61-14-420-016

(2) Install the cabin temperature assembly duct for the passenger compartment, do this task: Passenger Cabin Temperature Sensor Assembly Duct Installation, TASK 21-61-14-400-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TEMPERATURE CONTROL MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the temperature control module

(2) An installation of the temperature control module.

B. The temperature control module is located on the upper right corner of the P5 forward overhead panel in the flight deck.

TASK 21-61-15-000-801

2. Temperature Control Module Removal

(Figure 401)

A. References

Reference	Title
21-61-32-000-801	Cabin Temperature Indicator Removal (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Temperature Control Module Removal

SUBTASK 21-61-15-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	9	C00329	OVERHEAD 5V SECONDARY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	8	C00076	AIR CONDITIONING TEMP IND

SUBTASK 21-61-15-010-001

(2) Get access to the electrical connectors on the back of the P5-17 temperature control module [1], on the P5 forward overhead panel.

(a) Loosen the 1/4-turn fasteners that hold the P5 forward overhead panel in position and let the panel turn downward.

SUBTASK 21-61-15-020-001

(3) Remove the temperature control module [1] from the P5 forward overhead panel:

(a) Disconnect the electrical connectors at the back of the temperature control module [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING HOLD THE TEMPERATURE CONTROL MODULE WHEN YOU LOOSEN THE 1/4-TURN FASTENERS. IF THE MODULE FALLS, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Hold the temperature control module [1] in position and loosen the 1/4-turn fasteners that hold the temperature control module [1] to the P5 forward overhead panel.

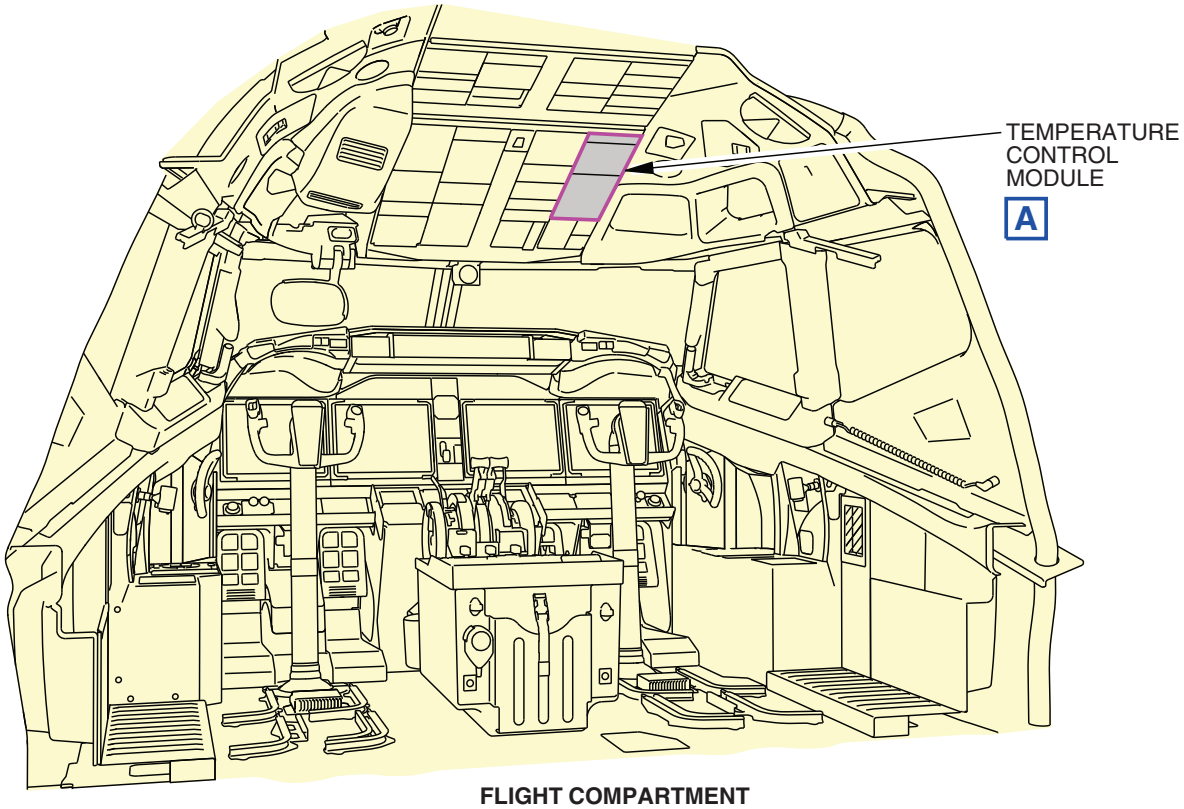
(c) Remove the temperature control module [1].

SUBTASK 21-61-15-020-002

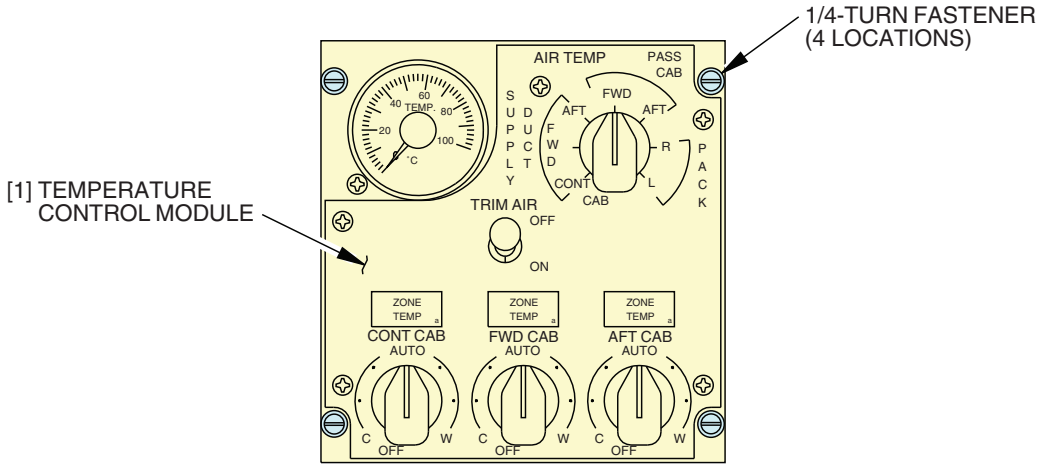
(4) Remove the cabin temperature indicator from the temperature control module [1] (TASK 21-61-32-000-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



TEMPERATURE CONTROL MODULE (EXAMPLE)

A

2411534 S00061528122_V3

Temperature Control Module Installation
Figure 401/21-61-15-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-15-400-801

3. Temperature Control Module Installation

(Figure 401)

A. References

Reference	Title
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
21-61-00-700-801	Cabin Temperature Control and Indication - Operational Test (P/B 501)
21-61-32-400-801	Cabin Temperature Indicator Installation (P/B 401)
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Temperature Control Module Installation

SUBTASK 21-61-15-010-002

(1) If necessary, lower the P5 forward overhead panel.

SUBTASK 21-61-15-420-002

(2) Install the cabin temperature indicator in the temperature control module [1] (TASK 21-61-32-400-801).

SUBTASK 21-61-15-420-001

(3) Install the P5-17 temperature control module [1] into the P5 forward overhead panel as follows:

(a) Put the temperature control module [1] into the P5 forward overhead panel and hold it in position.

(b) Turn the 1/4-turn fasteners on the temperature control module [1] to hold the module to the P5 forward overhead panel.

(c) Connect the two electrical connectors to the temperature control module [1].

SUBTASK 21-61-15-010-003

(4) Lift the P5 forward overhead panel to the closed position and turn the 1/4-turn fasteners.

SUBTASK 21-61-15-860-002

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	9	C00329	OVERHEAD 5V SECONDARY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	8	C00076	AIR CONDITIONING TEMP IND

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Temperature Control Module Test

SUBTASK 21-61-15-860-003

(1) Make sure that the electrical power is supplied (TASK 24-22-00-860-801).

SUBTASK 21-61-15-860-004

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-61-15-710-001

(3) Do this test of the temperature control module:

(a) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

(b) Do this task: Cabin Temperature Control and Indication - Operational Test, TASK 21-61-00-700-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-15-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ZONE TRIM AIR MODULATING VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the zone trim air modulating valve.

(2) An installation of the zone trim air modulating valve.

(3) There are three zone trim air modulating valves. Two of the zone trim air modulating valves are installed in the right air conditioning bay. One zone trim air modulating valve is installed in the left air conditioning bay.

TASK 21-61-21-000-801

2. Zone Trim Air Modulating Valve Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-61-21-860-001

(1) Make sure that the pneumatic power is off (TASK 36-00-00-860-806).

SUBTASK 21-61-21-860-002

(2) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach the DO NOT OPERATE tag, STD-858.

(d) Set the L and R RECIRC FAN switches to the OFF position and attach the DO NOT OPERATE tags, STD-858.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-21-860-003

(3) Do these steps on the P5-17 cabin temperature panel (located on the P5 forward overhead panel):

(a) Set the TRIM AIR switch to the OFF position and attach a DO NOT OPERATE tag, STD-858.

(b) Set the applicable cabin temperature selector (CONT CAB, FWD CAB, or AFT CAB) to the OFF position and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-61-21-860-004

(4) Do this step to remove the trim air modulating valve [1] for the control cabin:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

SUBTASK 21-61-21-860-005

(5) Do this step to remove the trim air modulating valve [1] for the forward cabin:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-21-860-006

(6) Do this step to remove the trim air modulating valve [1] for the aft cabin:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

SUBTASK 21-61-21-010-001

(7) To get access to the control compartment trim air modulating valve [1], do this step:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-61-21-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(8) To get access to the aft and forward passenger compartment trim air modulating valves [1], do these steps in this sequence:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Zone Trim Air Modulating Valve Removal

SUBTASK 21-61-21-020-001

(1) Disconnect the electrical connector [6] from the trim air modulating valve [1].

SUBTASK 21-61-21-020-006

(2) Do these steps to remove the bonding jumper [2] from the trim air modulating valve [1] for the aft passenger compartment:

(a) Remove the screw [3], washers [4], and nut [5] that hold the bonding jumper [2] to the trim air modulating valve [1].

(b) Move the bonding jumper [2] away from the trim air modulating valve [1].

SUBTASK 21-61-21-020-007

(3) Do these steps to remove the bonding jumper [2] and bonding jumper [8] from the trim air modulating valve [1] for the forward passenger compartment:

(a) Remove the screw [10], washers [4], and nut [5] that hold the bonding jumper [2] and bonding jumper [8] to the trim air modulating valve [1].

(b) Move the bonding jumper [2] and bonding jumper [8] away from the trim air modulating valve [1].

SUBTASK 21-61-21-020-008

(4) Do these steps to remove the bonding jumper [9] from the trim air modulating valve [1] for the control compartment:

(a) Remove the screw [3], washers [4], and nut [5] that hold the bonding jumper [9] to the trim air modulating valve [1].

(b) Move the bonding jumper [9] away from the trim air modulating valve [1].

SUBTASK 21-61-21-020-003

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(5) Do these steps to remove the trim air modulating valve [1]:

(a) Loosen the clamps [7].

(b) Move the clamp [7] to the adjacent ducts.

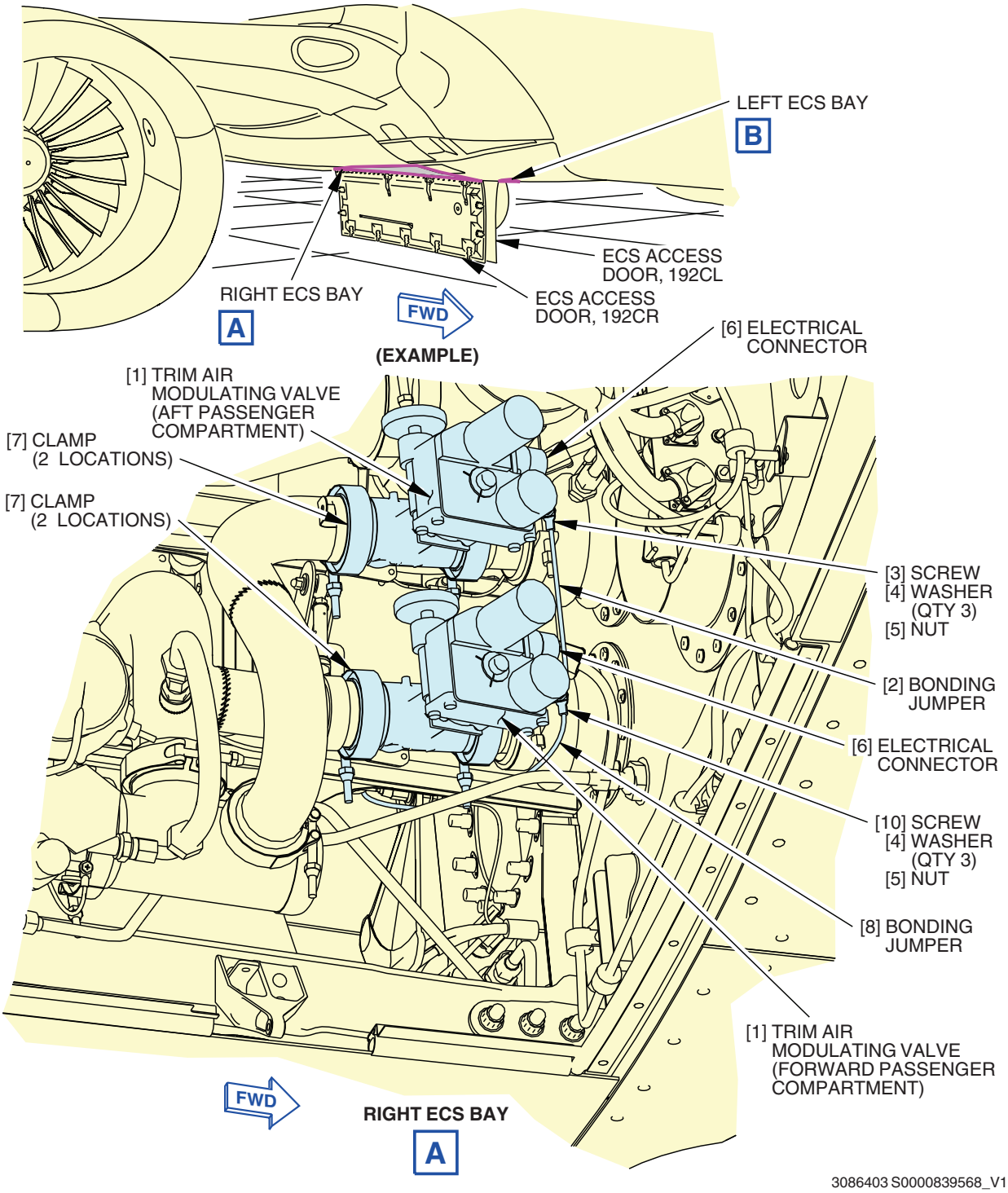
(c) Remove the trim air modulating valve [1] from the ECS bay.

SUBTASK 21-61-21-420-003

(6) Install the covers on the openings of the removed trim air modulating valve [1] and on the duct openings to keep out unwanted materials.

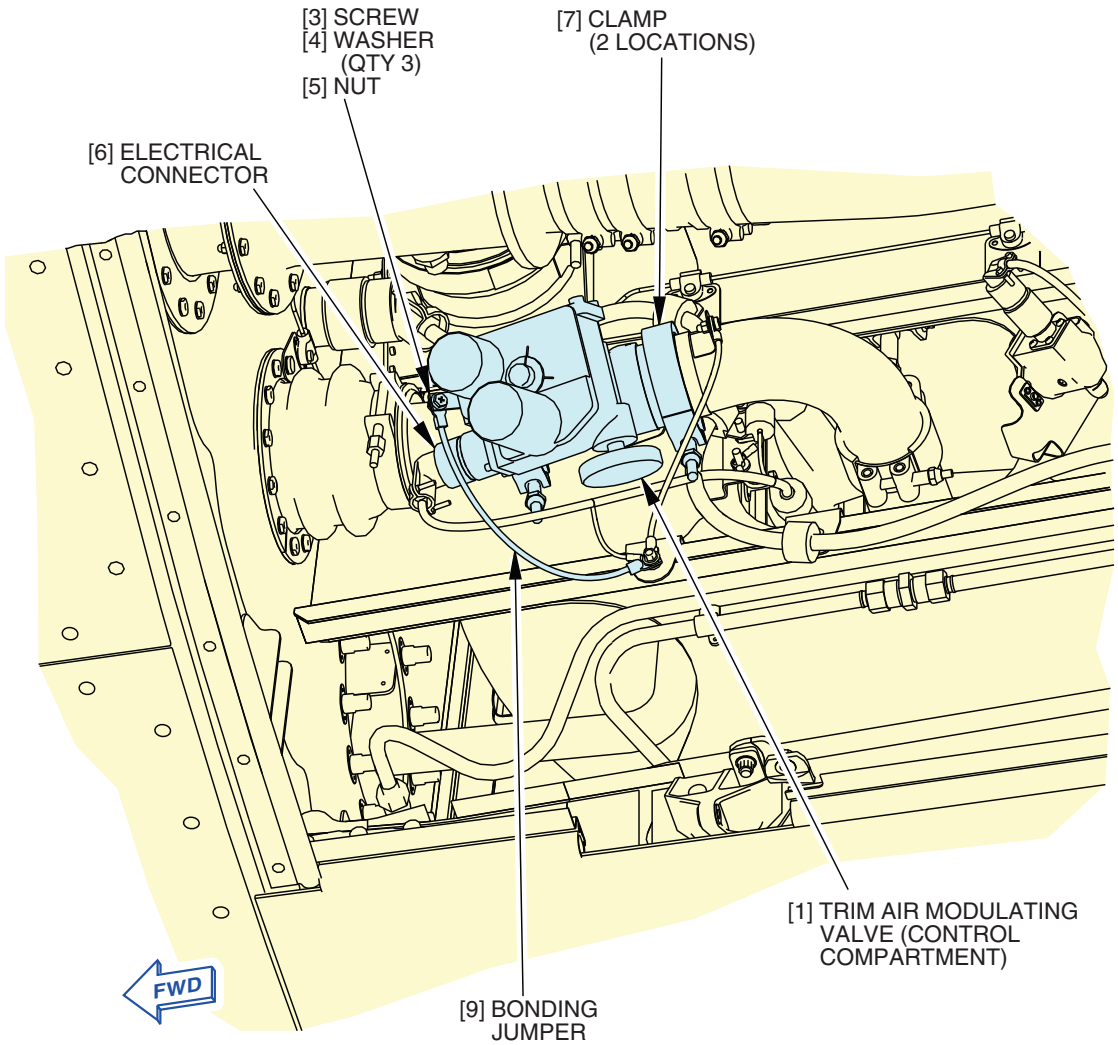
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Zone Trim Air Modulating Valve Installation
Figure 401/21-61-21-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Zone Trim Air Modulating Valve Installation
Figure 401/21-61-21-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-21-400-801

3. Zone Trim Air Modulating Valve Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Zone Trim Air Modulating Valve Installation

SUBTASK 21-61-21-020-005

(1) Remove the covers from the openings of trim air modulating valve [1] and duct openings.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that there is no unwanted materials inside.

SUBTASK 21-61-21-420-001

(2) Do these steps to install the trim air modulating valve [1]:

(a) Put the trim air modulating valve [1] in its position between the ducts.

1) Make sure that the flow arrow on the trim air modulating valve [1] points in the forward direction.

(b) Make sure that there is a minimum of 0.15 in. (3.81 mm) between the trim air modulating valve [1] and access door structure, tubes, and clamps.

(c) Move the clamps [7] to the duct connections.

(d) Tighten the clamps [7] to 55 +5 / -0 in-lb (6.2 +0.6 / -0.0 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

SUBTASK 21-61-21-420-004

(3) Do these steps to install the bonding jumper [2] on the trim air modulating valve [1] for the aft passenger compartment:

(a) Use cotton wiper, G00034, moist in solvent, B00184, or alcohol, B00130, to clean the mating surfaces of the trim air modulating valve [1], washers [4], and bonding jumper [2].

(b) Put the bonding jumper [2] in its position on the trim air modulating valve [1].

(c) Install the screw [3], washers [4], and nut [5] that hold the bonding jumper [2] to the trim air modulating valve [1].

(d) Tighten the nut [5] to 33 in-lb (3.7 N·m) - 40 in-lb (4.5 N·m).

(e) Measure the electrical bonding resistance between the bonding jumper [2] and trim air modulating valve [1] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.002 ohm (2.0 milliohm) or less.

SUBTASK 21-61-21-420-005

(4) Do these steps to install the bonding jumper [2] and bonding jumper [8] on the trim air modulating valve [1] for the forward passenger compartment:

(a) Use cotton wiper, G00034, moist in solvent, B00184, or alcohol, B00130, to clean the mating surfaces of the trim air modulating valve [1], washers [4], bonding jumper [2], and bonding jumper [8].

(b) Put the bonding jumper [2] and bonding jumper [8] in its position on the trim air modulating valve [1].

(c) Install the screw [10], washers [4], and nut [5] that hold the bonding jumper [2] and bonding jumper [8] to the trim air modulating valve [1].

(d) Tighten the nut [5] to 33 in-lb (3.7 N·m) - 40 in-lb (4.5 N·m).

(e) Measure the electrical bonding resistance between the bonding jumper [2], bonding jumper [8], and trim air modulating valve [1] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.002 ohm (2.0 milliohm) or less.

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SUBTASK 21-61-21-420-006

(5) Do these steps to install the bonding jumper [9] on the trim air modulating valve [1] for the control compartment:

(a) Use cotton wiper, G00034, moist in solvent, B00184, or alcohol, B00130, to clean the mating surfaces of the trim air modulating valve [1], washers [4], and bonding jumper [9].

(b) Put the bonding jumper [9] in its position on the trim air modulating valve [1].

(c) Install the screw [3], washers [4], and nut [5] that hold the bonding jumper [9] to the trim air modulating valve [1].

(d) Tighten the nut [5] to 33 in-lb (3.7 N·m) - 40 in-lb (4.5 N·m).

(e) Measure the electrical bonding resistance between the bonding jumper [9] and trim air modulating valve [1] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.002 ohm (2.0 milliohm) or less.

SUBTASK 21-61-21-020-004

(6) Connect the electrical connector [6] to the trim air modulating valve [1].

G. Zone Trim Air Modulating Valve Installation Test

SUBTASK 21-61-21-860-007

(1) If the trim air modulating valve [1] for the control cabin was replaced, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK

SUBTASK 21-61-21-710-001

(2) Do the ground test for the right air conditioning system (TASK 21-00-00-730-801).

SUBTASK 21-61-21-860-008

(3) If the trim air modulating valve [1] for the forward cabin was replaced, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-21-710-002

(4) Do the ground test for the right air conditioning system (TASK 21-00-00-730-801).

SUBTASK 21-61-21-860-009

(5) If the trim air modulating valve [1] for the aft cabin was replaced, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

SUBTASK 21-61-21-710-003

(6) Do the ground test for the left air conditioning system (TASK 21-00-00-730-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-21-860-010

(7) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-61-21-860-011

(8) Do these steps on the P5-10 air conditioning panel:

(a) Set the L and R PACK switches to the AUTO position and remove the DO NOT OPERATE tags.

(b) Set the BLEED 1 and 2 switches to the ON position and remove the DO NOT OPERATE tags.

(c) Set the BLEED APU switch to the ON position and remove the DO NOT OPERATE tag.

(d) Set the L and R RECIRC FAN switches to the AUTO position and remove the DO NOT OPERATE tags.

SUBTASK 21-61-21-860-012

(9) Do these steps on the P5-17 cabin temperature panel:

(a) Set the TRIM AIR switch to the ON position and remove the DO NOT OPERATE tag.

(b) Set the applicable cabin temperature selector (CONT CAB, FWD CAB, or AFT CAB) to the AUTO position and remove the DO NOT OPERATE tag.

SUBTASK 21-61-21-860-015

(10) Make sure that the trim air modulating valve [1] is open:

(a) If necessary, turn the applicable P5 temperature selector to a warmer position.

NOTE: It is necessary for the trim air modulating valve [1] to be in the open position when you do the soap bubble test.

SUBTASK 21-61-21-790-001

(11) Do a soap bubble test of the duct joints at the trim valve.

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches to the OFF position.

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

2) Loosen the clamps [7].

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps [7] to 55 +5 / -0 in-lb (6.2 +0.6 / -0.0 N·m).

a) Make sure that the loop bands do not touch.

b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

5) Put the L PACK and R PACK switches to the AUTO position.

6) Make sure that the leak has been repaired.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-21-860-014

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-21-010-003

(2) If the trim air modulating valve [1] was replaced for the left pack, do this step:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-61-21-010-004

(3) If the trim air modulating valve [1] was replaced for the right pack, do these steps in this sequence:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INTEGRATED AIR SYSTEMS CONTROLLER - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A configuration check for the Integrated Air System Controller

TASK 21-61-22-700-801

2. Integrated Air System Controller Configuration Check

A. General

(1) This configuration check makes sure that the Integrated Air System Controller (IASC) has the correct hardware.

(2) This configuration check is performed on the Multi Function Display (MFD).

B. References

Reference	Title
21-61-22 P/B 401	INTEGRATED AIR SYSTEMS CONTROLLER - REMOVAL/INSTALLATION

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Integrated Air System Controller Configuration Check

SUBTASK 21-61-22-860-010

(1) Set the NAVIGATION FMC switch, on the P5-28 panel, to the NORMAL position.

E. Integrated Air System Controller Configuration Check

SUBTASK 21-61-22-700-001

(1) Do these steps to make sure the IASC has the correct configuration:

(a) Push the MFD SYS button on the P9-82 panel.

(b) Select MAINT on the MFD.

(c) Select ONBD MAINT.

(d) Select LINE MAINT.

(e) Select SYSTEM CONFIGURATION.

(f) Select 21 - Air Conditioning or 36 - Pneumatics.

(g) Select CONTINUE.

1) Make sure that the part number on the MFD matches the physical part number of the IASC.

a) If the IASC hardware part number is not correct, replace the IASC (PAGEBLOCK 21-61-22/401).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INTEGRATED AIR SYSTEMS CONTROLLER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the integrated air systems controller.

(2) An installation of the integrated air systems controller.

TASK 21-61-22-000-801

2. Integrated Air Systems Controller (IASC) Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for Removal

SUBTASK 21-61-22-860-014

(1) Do this task:Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-61-22-860-015

(2) On the P5 panel, set the L PACK, R PACK and BLEED switches to OFF.

SUBTASK 21-61-22-040-001

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	6	C04035	AIR COND BLEED AIR CONTROL PRIMARY LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	6	C04036	AIR COND BLEED AIR CONTROL PRIMARY RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
D	5	C04037	AIR COND BLEED AIR CONTROL BACKUP LEFT
D	6	C04038	AIR COND BLEED AIR CONTROL BACKUP RIGHT

SUBTASK 21-61-22-860-003

(4) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

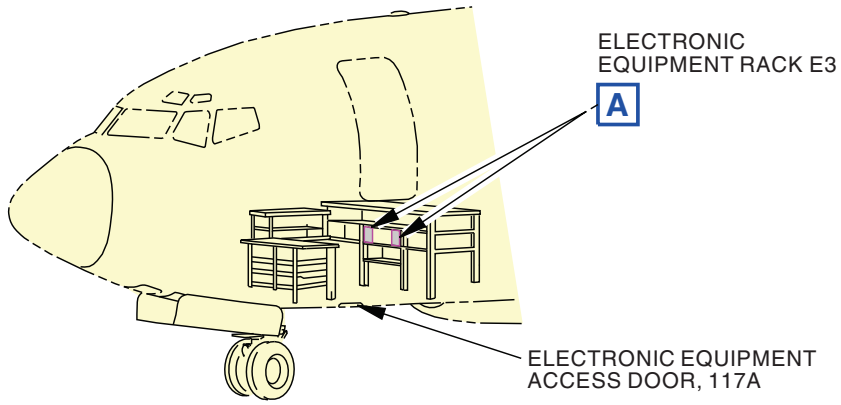
E. Integrated Air Systems Controller Removal

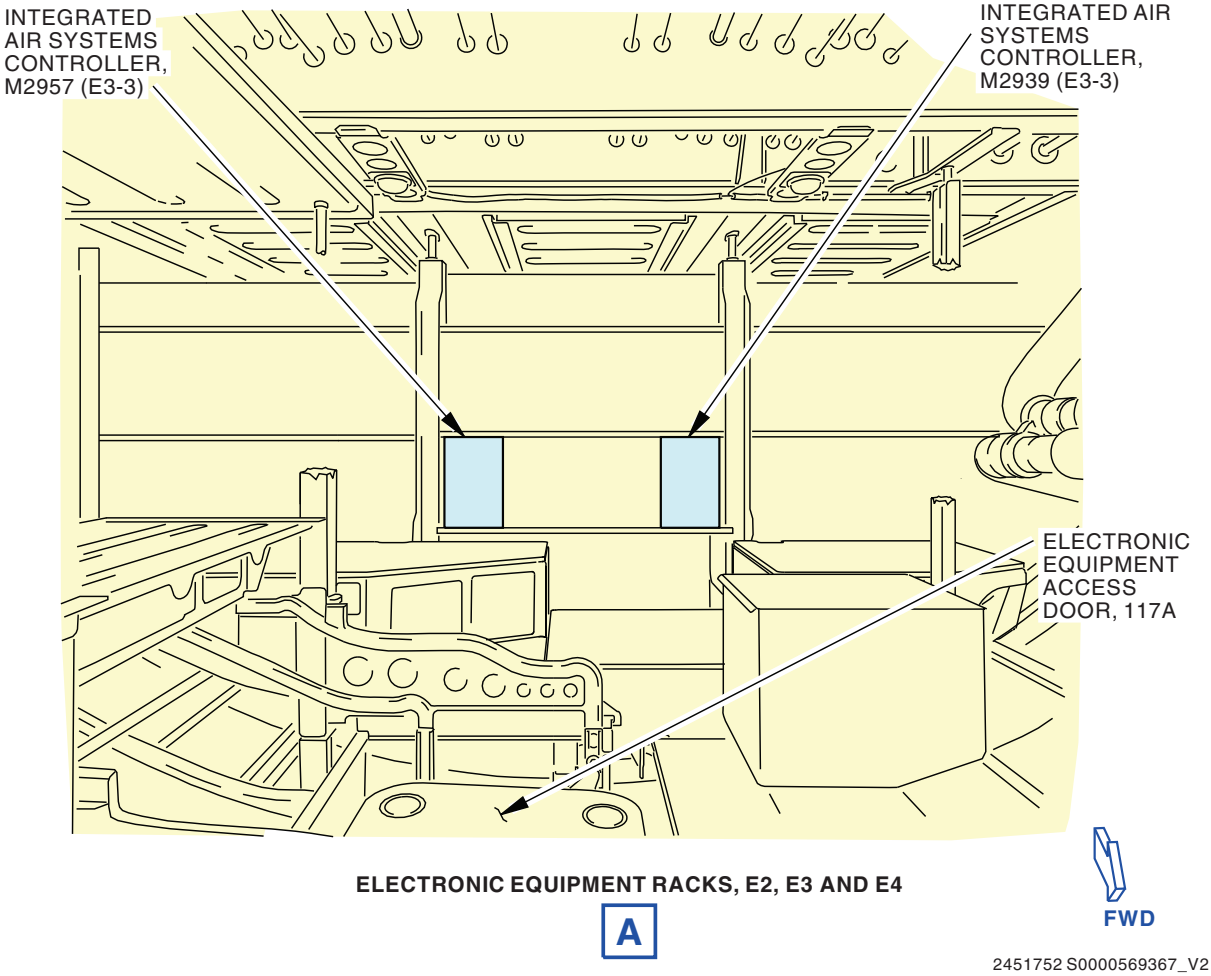
SUBTASK 21-61-22-000-001

(1) Remove the integrated air systems controller [1] (TASK 20-10-07-000-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





Integrated Air Systems Controller Installation
Figure 401/21-61-22-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-22-400-801

3. Integrated Air Systems Controller (IASC) Installation

(Figure 401)

A. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
36-00-00-710-801	Electrical LRU - Replacement Test (P/B 501)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Integrated Air Systems Controller Installation

SUBTASK 21-61-22-860-004

(1) Install the integrated air systems controller [1] (TASK 20-10-07-400-801).

SUBTASK 21-61-22-860-009

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	6	C04035	AIR COND BLEED AIR CONTROL PRIMARY LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	6	C04036	AIR COND BLEED AIR CONTROL PRIMARY RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
D	5	C04037	AIR COND BLEED AIR CONTROL BACKUP LEFT
D	6	C04038	AIR COND BLEED AIR CONTROL BACKUP RIGHT

E. Integrated Air Systems Controller Installation Test

SUBTASK 21-61-22-730-001

(1) Do the applicable ground test in this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

SUBTASK 21-61-22-710-001

(2) Do both of these ground tests in this task: Electrical LRU - Replacement Test, TASK 36-00-00-710-801.

(a) 36 Pneumatics, LRU Replacement Test, Electrical LRU - Left.

(b) 36 Pneumatics, LRU Replacement Test, Electrical LRU - Right.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-22-710-002

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-22-860-008

(2) If all maintenance on the airplane is completed, set the GND TEST switch, on the P61-4 panel, to the NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INLET PRESSURE SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the inlet pressure sensor.

(2) An installation of the inlet pressure sensor.

TASK 21-61-23-000-801

2. Inlet Pressure Sensor Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the left and right pack inlet pressure sensors.

(2) The pack inlet pressure sensor are installed on the center wing section of the wing front spar in the keel beam area.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
129	Keel Beam (Part) - Section 43
192	Lower Wing-To-Body Fairing - Under Wing Box
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-61-23-040-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806

SUBTASK 21-61-23-040-002

(2) Put the L PACK and R PACK switches, on the P5-10 air conditioning panel, to the OFF positions.

SUBTASK 21-61-23-010-001

(3) To remove the left pack inlet pressure sensor [5], open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-23-040-003

(4) To remove the right pack inlet pressure sensor [5], open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC

SUBTASK 21-61-23-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To get access to the right pack inlet pressure sensor [5], open these access doors in this sequence:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-61-23-010-003

(6) To get access to the left pack inlet pressure sensor [5], open this access door:

Number	Name/Location
192CL	ECS Access Door

F. Pack Inlet Pressure Sensor Removal

SUBTASK 21-61-23-020-001

CAUTION

THIS EQUIPMENT IS SUSCEPTIBLE TO DAMAGE BY ELECTRICAL DISCHARGE. PROTECTIVE CAPS MUST STAY ON THE ELECTRICAL CONNECTION ENDS OF THE SENSOR UNTIL FINAL INSTALLATION.

(1) Disconnect the electrical connector [4] from the inlet pressure sensor [5].

(a) Install a protective cap on the electrical connector [4].

SUBTASK 21-61-23-020-002

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(2) Disconnect the pressure sense line [6A] or pressure sense line [6B] from the inlet pressure sensor [5].

SUBTASK 21-61-23-000-001

(3) Remove the screws [1] and washers [2] that secure the clamp [3] that attach the inlet pressure sensor [5] to the bracket on the front spar.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-23-000-002

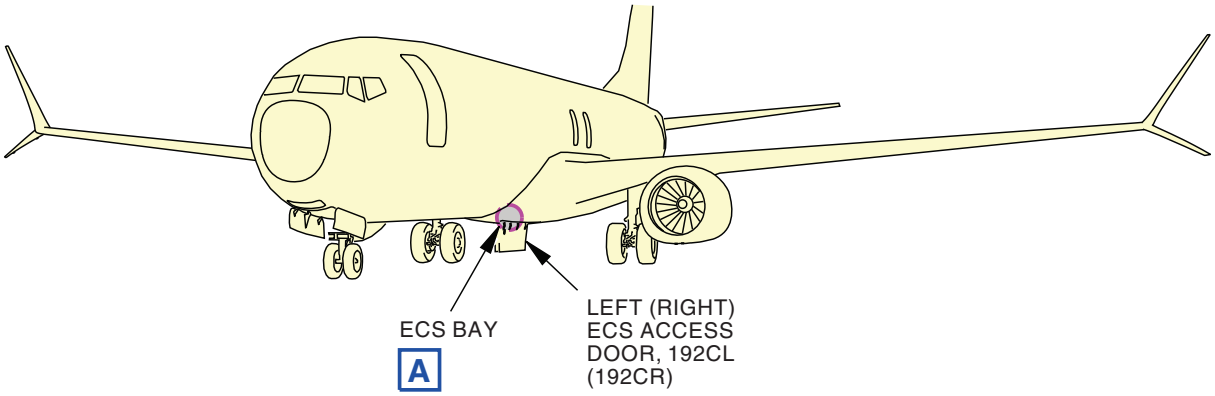
(4) Remove the inlet pressure sensor [5].

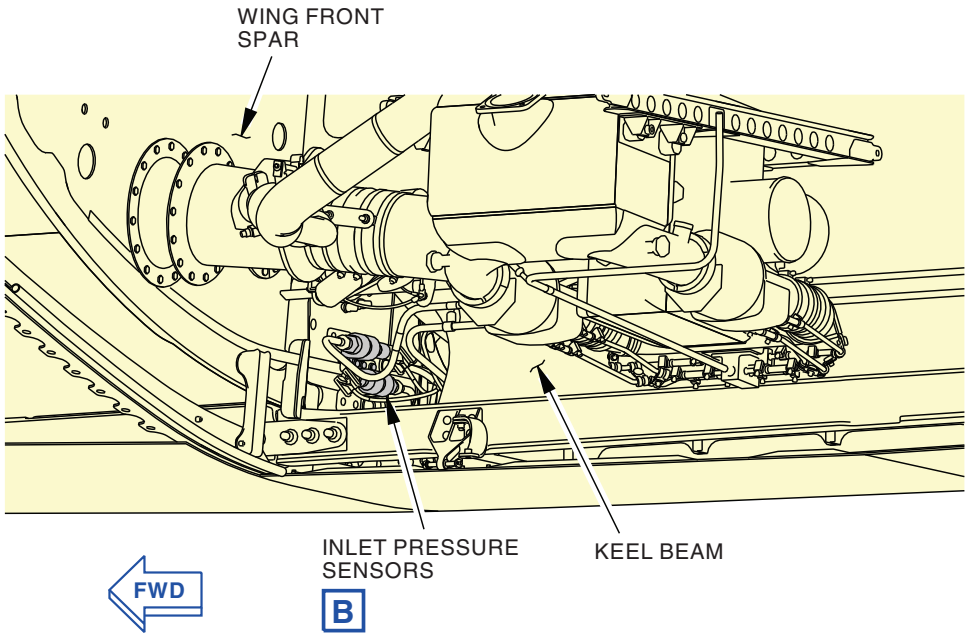
SUBTASK 21-61-23-420-001

(5) Put a protective cap on the pressure sense line [6A] or pressure sense line [6B] to keep out unwanted material.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

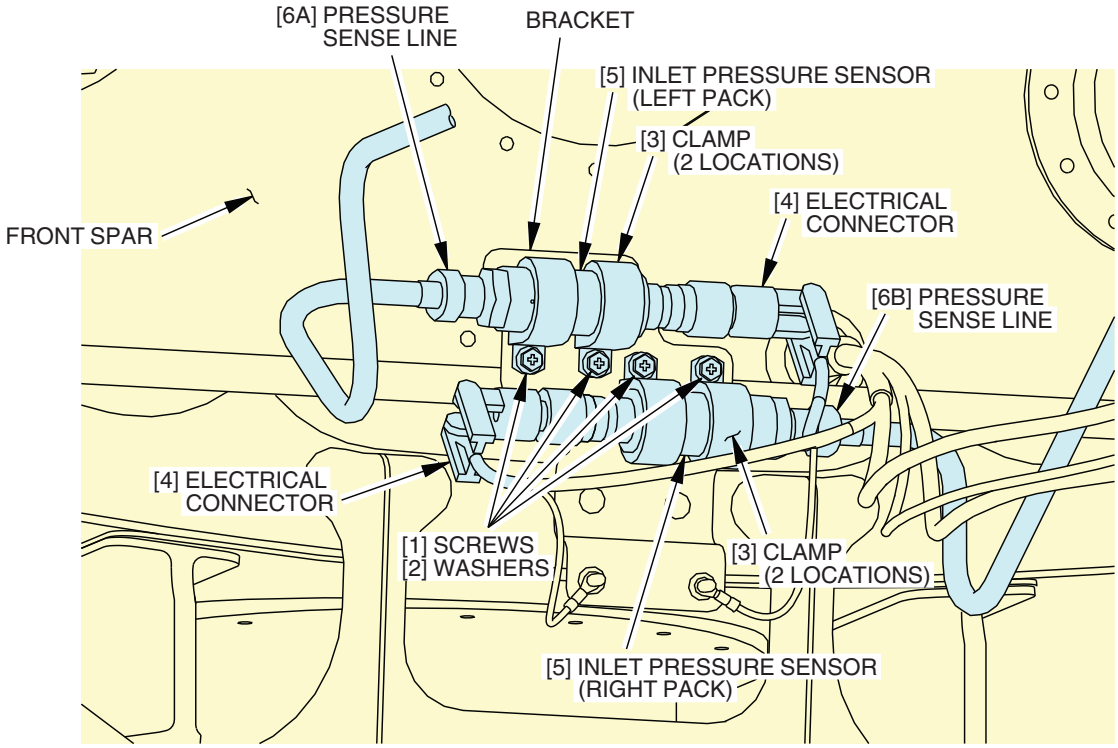




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Inlet Pressure Sensor Installation
Figure 401/21-61-23-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



INLET PRESSURE SENSORS

B

2531244 S0000597229_V2

Inlet Pressure Sensor Installation
Figure 401/21-61-23-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-23-400-801

3. Inlet Pressure Sensor Installation

(Figure 401)

A. General

(1) This task includes the steps to install the left and right pack inlet pressure sensors.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50135	Leak Detector - Liquid, Non-Corrosive Soap Compound	MIL-PRF-25567
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Location Zones

Zone	Area
129	Keel Beam (Part) - Section 43
192	Lower Wing-To-Body Fairing - Under Wing Box
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for the Installation

SUBTASK 21-61-23-000-003

(1) Remove the protective cap from the pressure sense line [6A] or pressure sense line [6B].

G. Pack Inlet Pressure Sensor Installation

SUBTASK 21-61-23-980-001

(1) Put the two clamps [3] on the inlet pressure sensor [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-23-980-002

(2) Position the inlet pressure sensor [5] with the clamps [3] on the bracket so that the pressure sense line [6A] or pressure sense line [6B] can be easily connected to the inlet pressure sensor [5].

(a) Make sure that you make the correct connections.

NOTE: The left and right inlet pressure sensors can be connected to the opposite pack wiring.

SUBTASK 21-61-23-420-002

(3) Install the screws [1] and washers [2] to attach the clamp [3] to the bracket.

SUBTASK 21-61-23-420-003

(4) Connect the pressure sense line [6A] or pressure sense line [6B] to the inlet pressure sensor [5] as follows:

(a) Align the pressure sense line [6A] or pressure sense line [6B] with the inlet pressure sensor [5].

(b) Apply corrosion inhibiting material, G50136, to the threads on the inlet pressure sensor [5].

(c) Connect the pressure sense line [6A] or pressure sense line [6B] to the inlet pressure sensor [5]:

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

1) Tighten the B-nut on the pressure sense line [6A] or pressure sense line [6B] to 90 in-lb (10.2 N·m) - 100 in-lb (11.3 N·m).

SUBTASK 21-61-23-000-004

(5) Remove the protective cap from the electrical connector of the inlet pressure sensor [5].

SUBTASK 21-61-23-420-004

(6) Connect the electrical connector [4] to the inlet pressure sensor [5] (TASK 20-60-07-910-802).

H. Pack Inlet Pressure Sensor Installation Test

SUBTASK 21-61-23-440-001

(1) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC

SUBTASK 21-61-23-720-001

(2) If both inlet pressure sensors [5] have been disconnected and reconnected, then do a check that the electrical connectors [4] are correctly installed:

(a) Look for maintenance message 21-55230 “LEFT AIR CONDITIONING PACK INLET PRESSURE SENSOR IS OUT OF RANGE” and 21-55231 “RIGHT AIR CONDITIONING PACK INLET PRESSURE SENSOR IS OUT OF RANGE” (TASK 46-13-02-700-805).

1) Make sure that the maintenance message is NOT ACTIVE or that the message does not show.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If the maintenance message is ACTIVE, refer to the IFIM and do the applicable procedure(s) to repair the fault indicated by the maintenance message.

(b) Disconnect the electrical connector [4] for the right pack inlet pressure sensor [5].

NOTE: The right pack inlet pressure sensors electrical connector is most easily connected through the left pack bay door. The right pack inlet pressure sensors is connected to the pneumatic pressure sense line that goes to the right pack.

1) Look for maintenance message 21-55231 “RIGHT AIR CONDITIONING PACK INLET PRESSURE SENSOR IS OUT OF RANGE” (TASK 46-13-02-700-805).

a) Make sure that the maintenance message is ACTIVE.

b) If maintenance message 21-55231 is not active and maintenance message 21-55230 “LEFT AIR CONDITIONING PACK INLET PRESSURE SENSOR IS OUT OF RANGE” is ACTIVE, then connect the electrical connectors to the correct sensors.

<1> Disconnect the electrical connector [4] on the left inlet pressure sensor [5] and connect that electrical connector [4] to the right inlet pressure sensor [5].

<2> Connect the electrical connector [4] to the left inlet pressure sensor [5].

c) Reconnect the electrical connector [4] to the right pack inlet pressure sensor [5].

d) Make sure that the maintenance message is NOT ACTIVE or that the message does not show.

NOTE: You have verified that the electrical connectors are connected to the correct sensors.

SUBTASK 21-61-23-730-001

(3) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

SUBTASK 21-61-23-440-002

(4) Supply conditioned air to the airplane (TASK 21-00-00-800-801).

SUBTASK 21-61-23-700-001

(5) Use leak detector, G50135, to do a leak check at the pressure sense line [6A] or pressure sense line [6B] connection.

(a) Repair leaks that show.

SUBTASK 21-61-23-100-001

(6) Use a clean cotton wiper, G00034, to remove the remaining leak detector.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-23-040-004

(1) Remove conditioned air from the airplane (TASK 21-00-00-800-802).

SUBTASK 21-61-23-410-001

(2) If the inlet pressure sensor [5] for the left cooling pack was removed, close this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-61-23-410-002

(3) If the inlet pressure sensor [5] for the right cooling pack was removed, close these access doors in this sequence:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK PRESSURE SENSOR SENSE LINE - CLEANING/PAINTING

1. General

A. This procedure has this task:

(1) A Cleaning of the Pack Inlet Pressure Sensor Sense Line.

B. This procedure provides instructions to clean the left pack pressure sensor, right pack pressure sensor and sense lines.

TASK 21-61-23-100-801

2. Pack Inlet Pressure Sensor Sense Line Cleaning

(Figure 701)

A. References

Reference	Title
21-61-23-000-801	Inlet Pressure Sensor Removal (P/B 401)
21-61-23-400-801	Inlet Pressure Sensor Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
G02314	Air - Compressed, Breathing	BB-A-1034 Source I Grade A
G50256	Water, Regular	

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Prepare for the Cleaning

SUBTASK 21-61-23-000-005

(1) Remove the pack pressure sensors [5].

(a) Do this task: Inlet Pressure Sensor Removal, TASK 21-61-23-000-801.

SUBTASK 21-61-23-000-006

(2) Remove the sense line [6] as follows:

(a) Remove the clamps [7].

1) Remove the screw [8] and washers [9] that secure each clamp [7] to the structure.

2) Remove the clamps [7].

(b) Disconnect the sense line [6] from the duct connector [10] as follows:

1) Use a second wrench to hold the duct connector [10] when you disconnect the sense line [6].

(c) Remove the sense line [6].

SUBTASK 21-61-23-000-007

(3) Remove the sense line [12] and sense line [18] as follows:

(a) Remove the clamp [15] and clamps [21].

1) Remove the screw [13], washer [14], screw [19], and washer [20] that secure clamp [15] and each clamp [21] to the structure.

2) Remove the clamp [15] and clamps [21].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Disconnect the sense line [18] from the duct connector [10] as follows:

1) Use a second wrench to hold the duct connector [10] when you disconnect the sense line [18].

(c) Remove the sense line [12] and sense line [18].

(d) If it is necessary, disconnect the nut [17] from the union [16] to disconnect sense line [18] from sense line [12].

E. Cleaning Procedure

SUBTASK 21-61-23-100-002

(1) If it is necessary, remove any ice buildup in the pack pressure sensors [5].

SUBTASK 21-61-23-100-003

(2) If it is necessary, remove any ice buildup in the sense line with warm water, G50256.

SUBTASK 21-61-23-100-004

(3) Blow air, G02314, through the sense line weep hole [11] with compressed air.

(a) Make sure that there is no debris, dirt or unwanted materials in the sense line.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-23-400-002

(1) Install the sense line [6] as follows:

(a) Connect the sense line [6] to the duct connector [10] as follows:

1) Use a second wrench to hold duct connector [10] when you tighten the sense line [6].

2) Tighten the B-nut on the sense line [6] to 170 ±10 in-lb (19 ±1 N·m).

(b) Install the clamps [7].

1) Install the clamps [7] with the screws [8] and washers [9].

SUBTASK 21-61-23-400-003

(2) Install the sense line [12] and sense line [18] as follows:

(a) If it was disconnected, reconnect the nut [17] to the union [16].

(b) Connect the sense line [18] to the duct connector [10] as follows:

1) Use a second wrench to hold duct connector [10] when you tighten the sense line [18].

2) Tighten the B-nut on the sense line [18] to 170 ±10 in-lb (19 ±1 N·m).

(c) Install the clamp [15] and clamps [21].

1) Install the clamp [15] with the screw [13], washer [14].

2) Install the clamps [21] with the screw [19], and washer [20].

SUBTASK 21-61-23-400-001

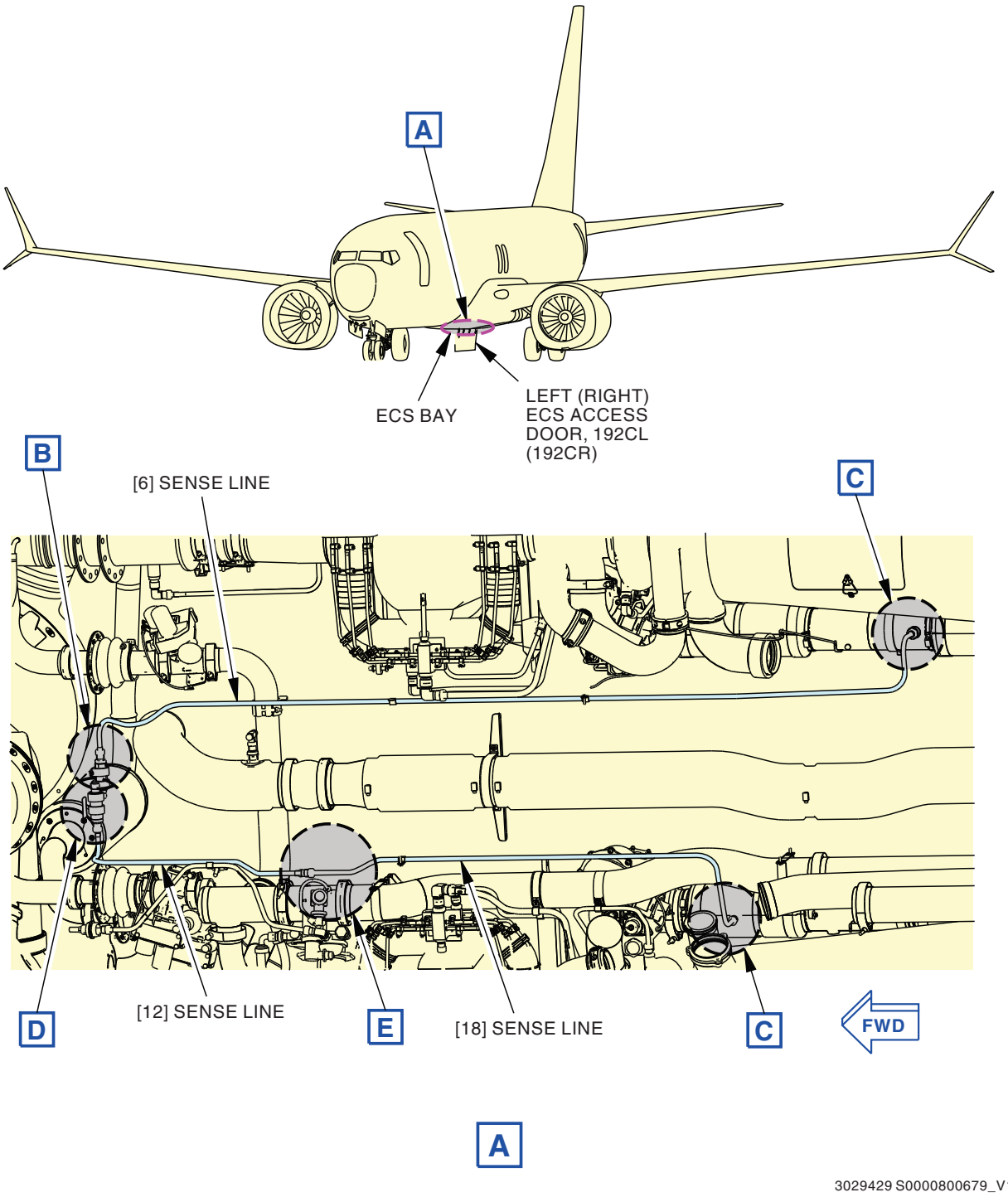
(3) Install the pack pressure sensors [5].

(a) Do this task: Inlet Pressure Sensor Installation, TASK 21-61-23-400-801.

(b) Tighten the B-nuts on the sense line [6] and sense line [12] to 90 in-lb (10 N·m) - 100 in-lb (11 N·m).

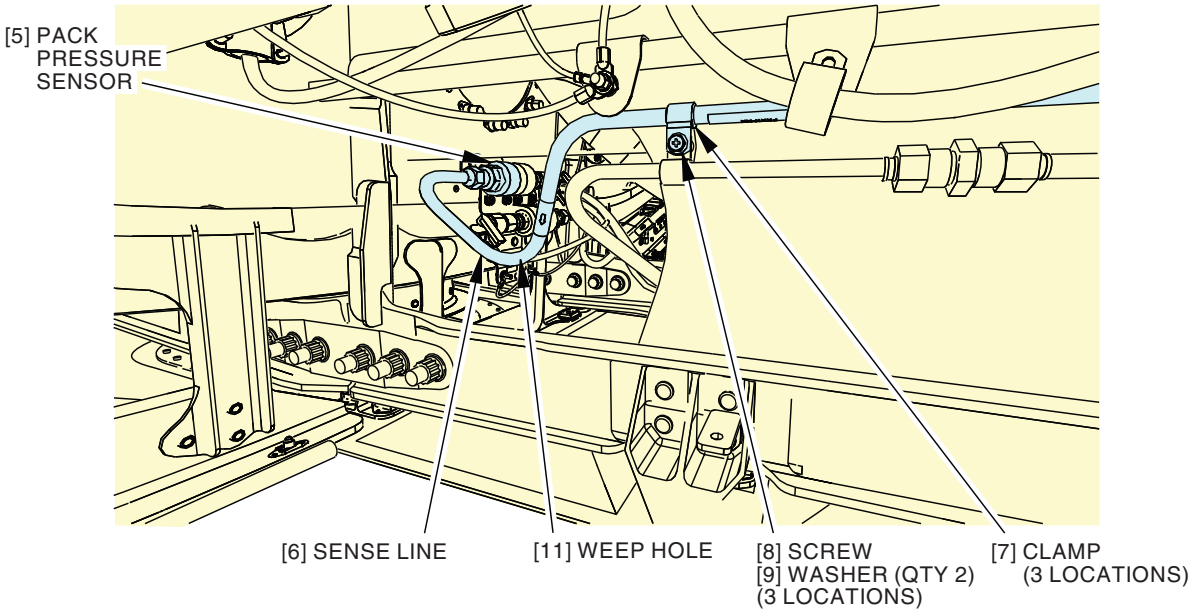
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

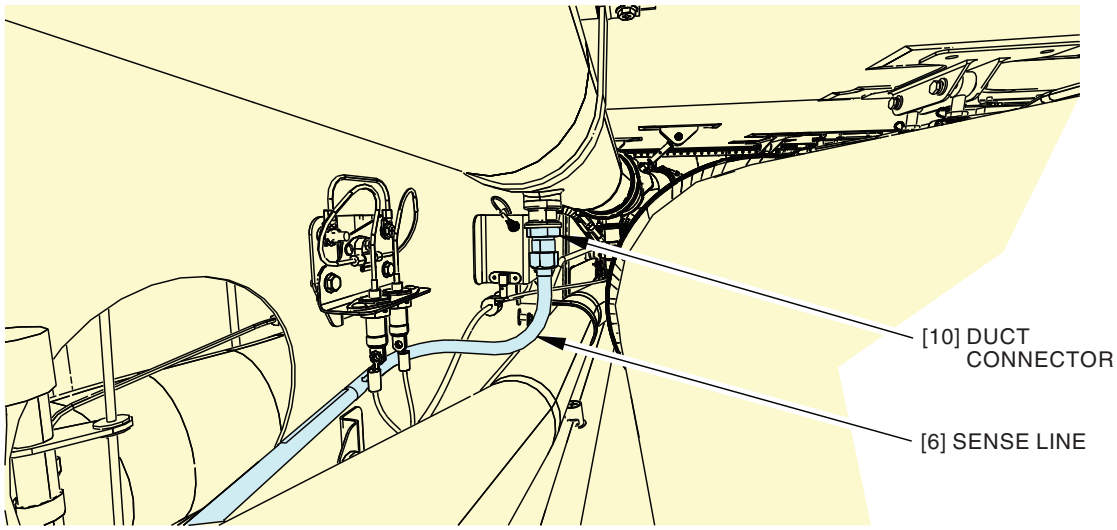


Pack Inlet Pressure Sensor Sense Line Cleaning
Figure 701/21-61-23-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B



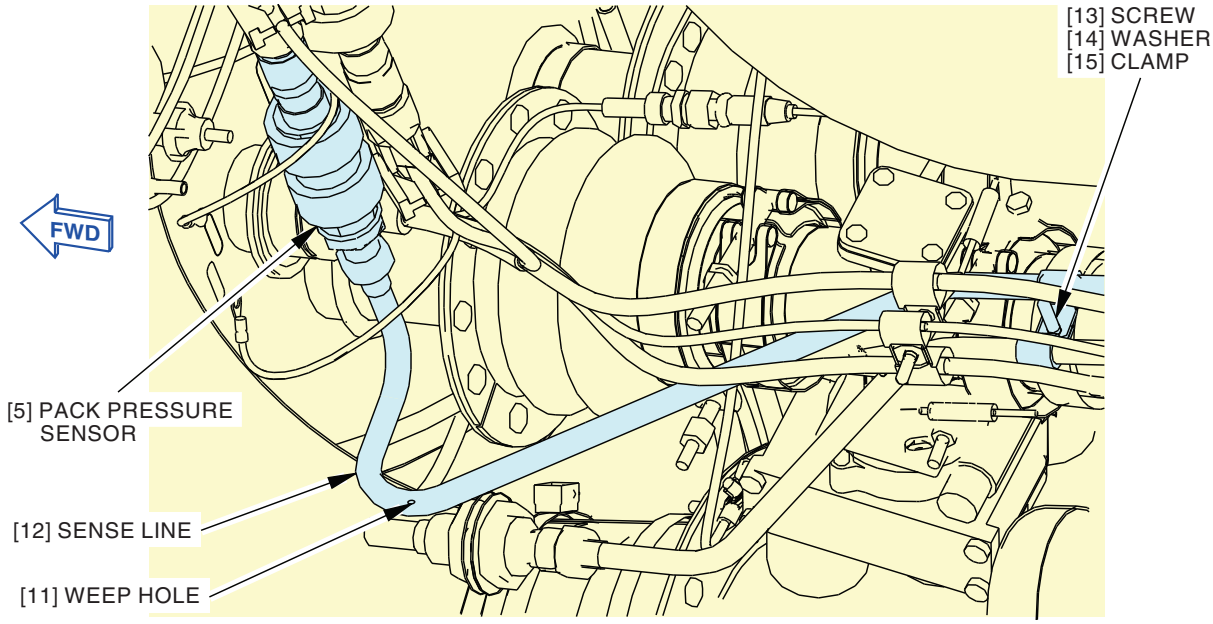
(EXAMPLE)

C

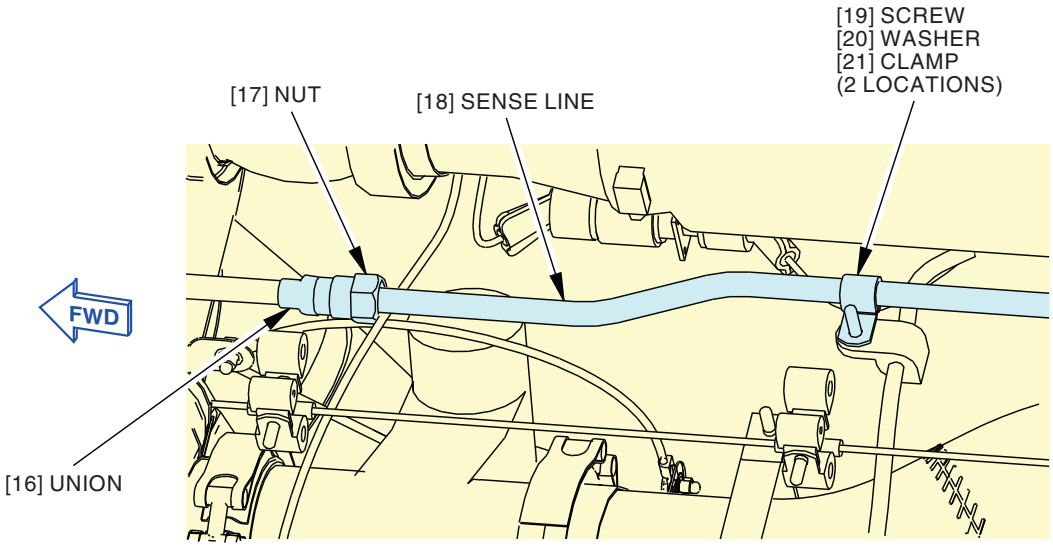
3029430 S0000800680_V1

Pack Inlet Pressure Sensor Sense Line Cleaning
Figure 701/21-61-23-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



D



E

3030675 S0000801296_V1

Pack Inlet Pressure Sensor Sense Line Cleaning
Figure 701/21-61-23-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLOW SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flow sensor.

(2) An installation of the flow sensor.

TASK 21-61-24-000-801

2. Flow Sensor Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the left and right pack flow sensors.

(2) The pack flow sensor is located in the duct that goes to the pack and trim air system. It is between the keel beam and the primary heat exchanger.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-61-24-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806

SUBTASK 21-61-24-040-001

(2) To remove the left pack flow sensor [1], do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-61-24-040-002

(3) To remove the right pack flow sensor [1], do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-61-24-010-001

(4) To get access to the left pack flow sensor [1], do this step:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-61-24-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To get access to the right pack flow sensor [1], do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Pack Flow Sensor Removal

SUBTASK 21-61-24-000-002

(1) Disconnect the electrical connector [2] from the flow sensor [1].

SUBTASK 21-61-24-000-003

(2) Remove the three screws [3] and washers [4].

SUBTASK 21-61-24-000-004

(3) Remove the flow sensor [1].

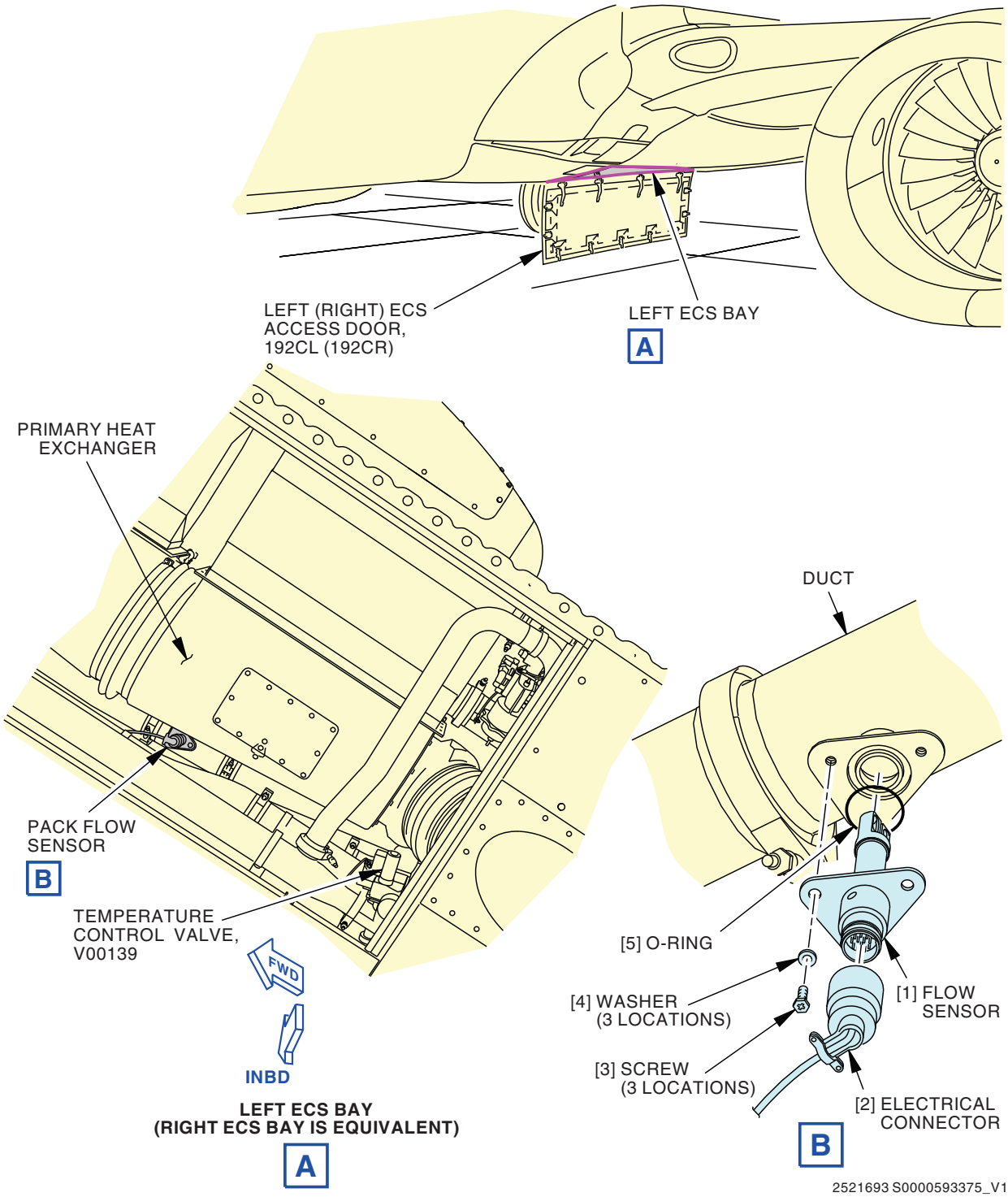
(a) Discard the o-ring [5].

SUBTASK 21-61-24-860-002

(4) Install a cover over the opening in the duct to keep out unwanted material.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pack Flow Sensor Installation
Figure 401/21-61-24-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-24-400-801

3. Flow Sensor Installation

(Figure 401)

A. General

(1) This task includes the steps to install the left and right pack flow sensor.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-730-802	Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
B00065	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002 (Supersedes O-A-396)
B50073	Alcohol - Isopropyl	ASTM D 770
D00504	Grease - Petrolatum	VV-P-236
G50262	Wiper - Cleaning	BMS15-5

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Prepare for the Installation

SUBTASK 21-61-24-000-001

(1) Remove the cover from the flow sensor [1] opening in the duct.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Pack Flow Sensor Installation

SUBTASK 21-61-24-100-001

(1) Do these steps to clean the faying surfaces of the flow sensor [1] and the duct boss:

(a) Clean the surfaces of the flow sensor [1] and the duct boss using wiper, G50262 that are soaked with alcohol, B00065 or alcohol, B50073.

(b) Remove all of the alcohol with a clean wiper, G50262.

SUBTASK 21-61-24-640-001

(2) Lubricate a new o-ring [5] with grease, D00504.

SUBTASK 21-61-24-400-001

(3) Install the new o-ring [5] on the flow sensor [1].

SUBTASK 21-61-24-400-002

(4) Position the flow sensor [1] on the boss on the duct.

SUBTASK 21-61-24-400-003

(5) Install the three screws [3] and washers [4] to attach the flow sensor [1] to the boss.

(a) Tighten screws [3] to 30 ±1 in-lb (3.4 ±0.1 N·m).

SUBTASK 21-61-24-700-001

(6) Use an intrinsically safe approved bonding meter, COM-1550, to do a check of the electrical bond resistance between the flow sensor [1] and the boss (SWPM 20-20-00).

(a) Make sure the resistance is not more than 0.001 ohm (1.0 milliohm).

SUBTASK 21-61-24-420-001

(7) Connect the electrical connector [2] to the flow sensor [1].

SUBTASK 21-61-24-410-003

(8) For the left pack flow sensor [1] do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-61-24-410-004

(9) For the right pack flow sensor [1] do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

I. Pack Flow Sensor Installation Test

SUBTASK 21-61-24-730-001

(1) Do this task: Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test, TASK 21-00-00-730-802.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-24-410-001

(1) If the flow sensor [1] for the left cooling pack was replaced, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-61-24-410-002

(2) If the flow sensor [1] for the right cooling pack was replaced, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DUCT TEMPERATURE BULB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the control cabin duct temperature bulb.

(2) A removal of the passenger cabin duct temperature bulbs.

(3) An installation of the control cabin duct temperature bulb.

(4) An installation of the passenger duct temperature bulbs.

(5) There is a duct temperature bulb installed for each of the three cabin temperature zones.

TASK 21-61-30-000-801

2. Duct Temperature Bulb Removal

(Figure 401, Figure 402)

A. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)

B. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 21-61-30-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

SUBTASK 21-61-30-010-001

(2) To get access the control cabin duct temperature sensor open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-30-010-002

(3) To get access the passenger cabin duct temperature sensors remove the applicable ceiling panel that is forward of the overwing escape hatches. To remove the panel, do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

E. Passenger Cabin Duct Temperature Bulb Removal

SUBTASK 21-61-30-020-001

(1) Disconnect the electrical connector [1] from the passenger cabin duct temperature bulb [2].

SUBTASK 21-61-30-020-002

(2) Remove the passenger cabin duct temperature bulb [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-30-020-003

(3) Remove and discard the O-ring [3] from the passenger cabin duct temperature bulb [2].

F. Control Cabin Duct Temperature Bulb Removal

SUBTASK 21-61-30-020-004

(1) Disconnect the electrical connector [22] from the control cabin duct temperature bulb [21].

SUBTASK 21-61-30-020-005

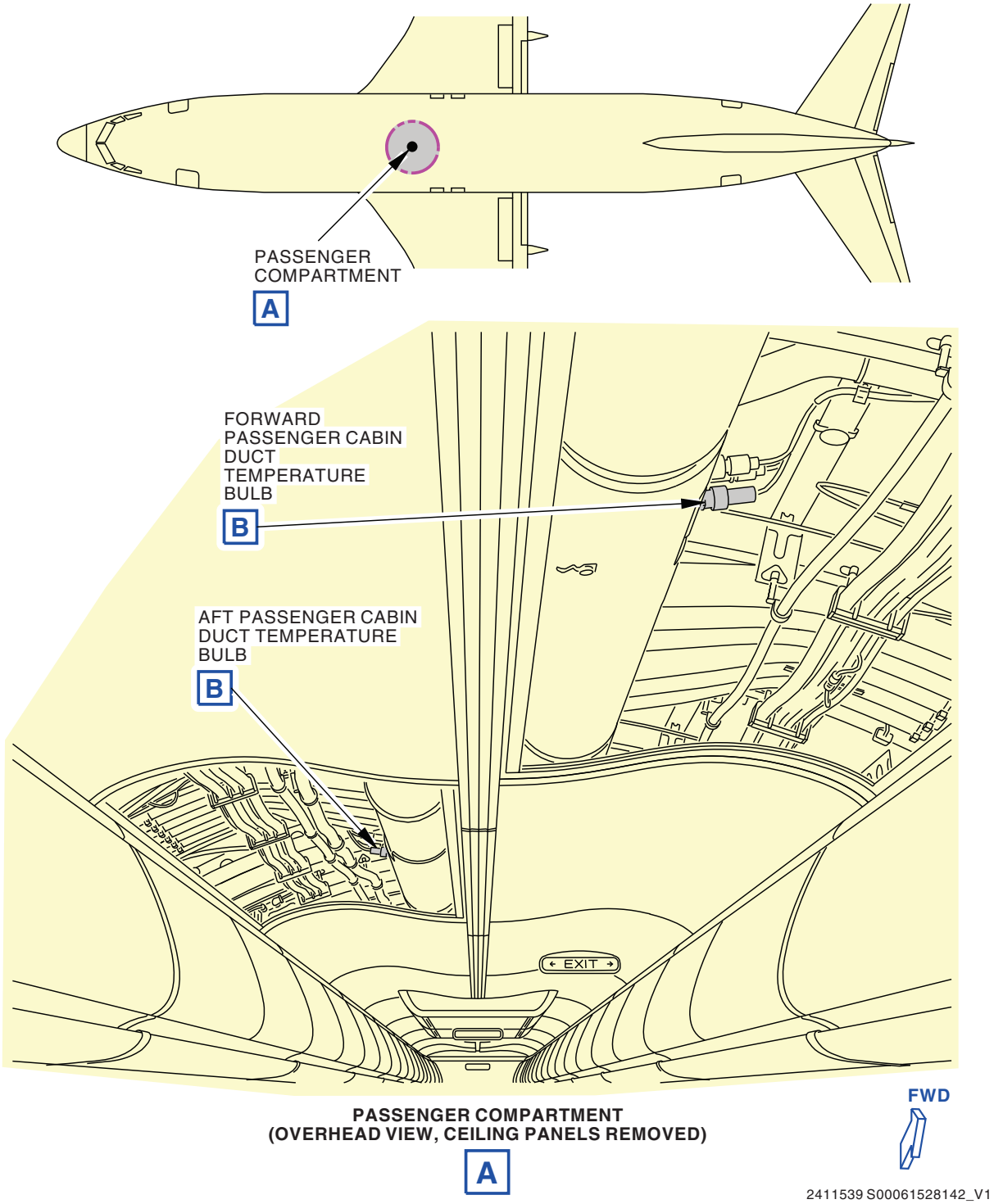
(2) Remove the control cabin duct temperature bulb [21].

SUBTASK 21-61-30-020-006

(3) Remove and discard the O-ring [23] from the control cabin duct temperature bulb [21].

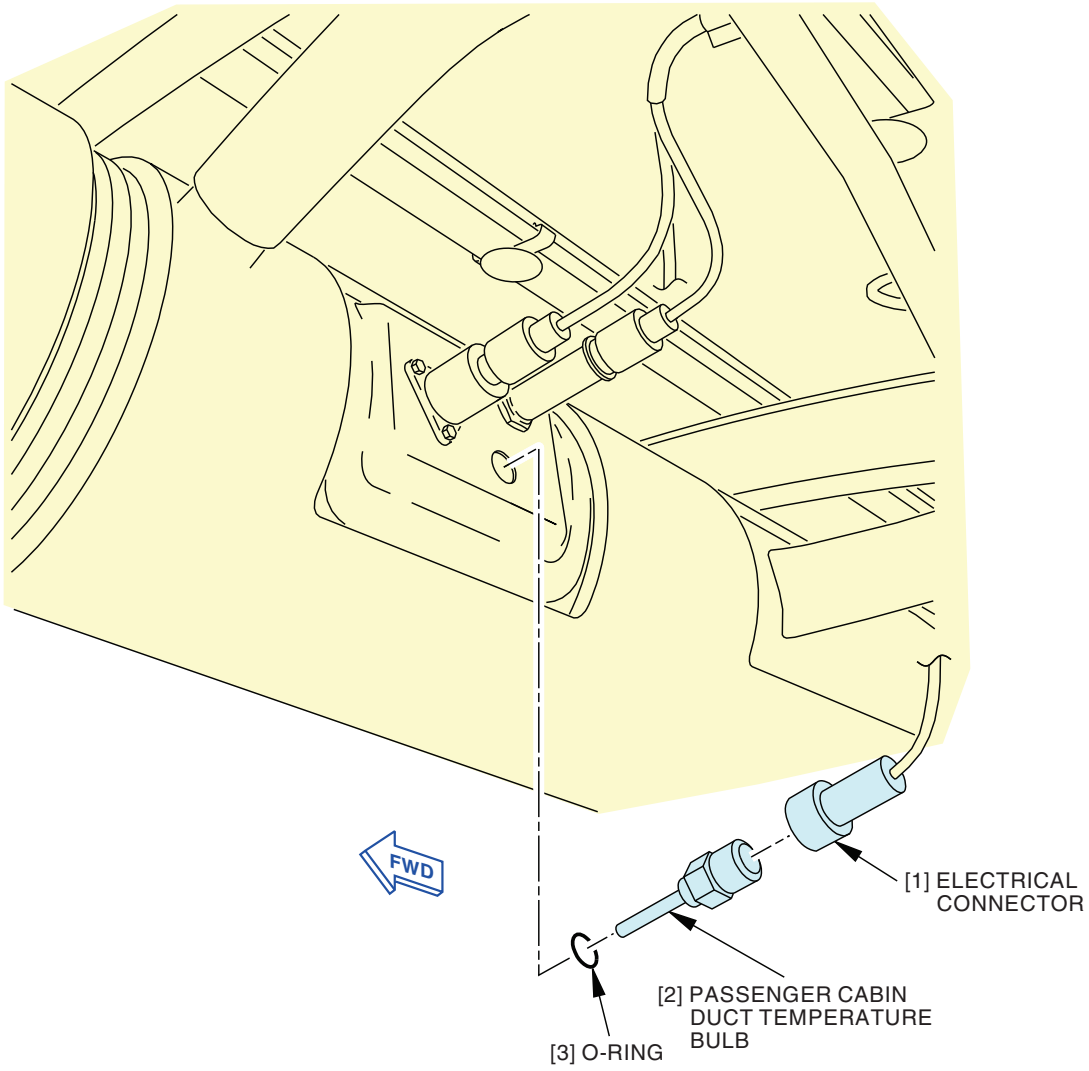
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Passenger Cabin Duct Temperature Bulb Installation
Figure 401/21-61-30-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



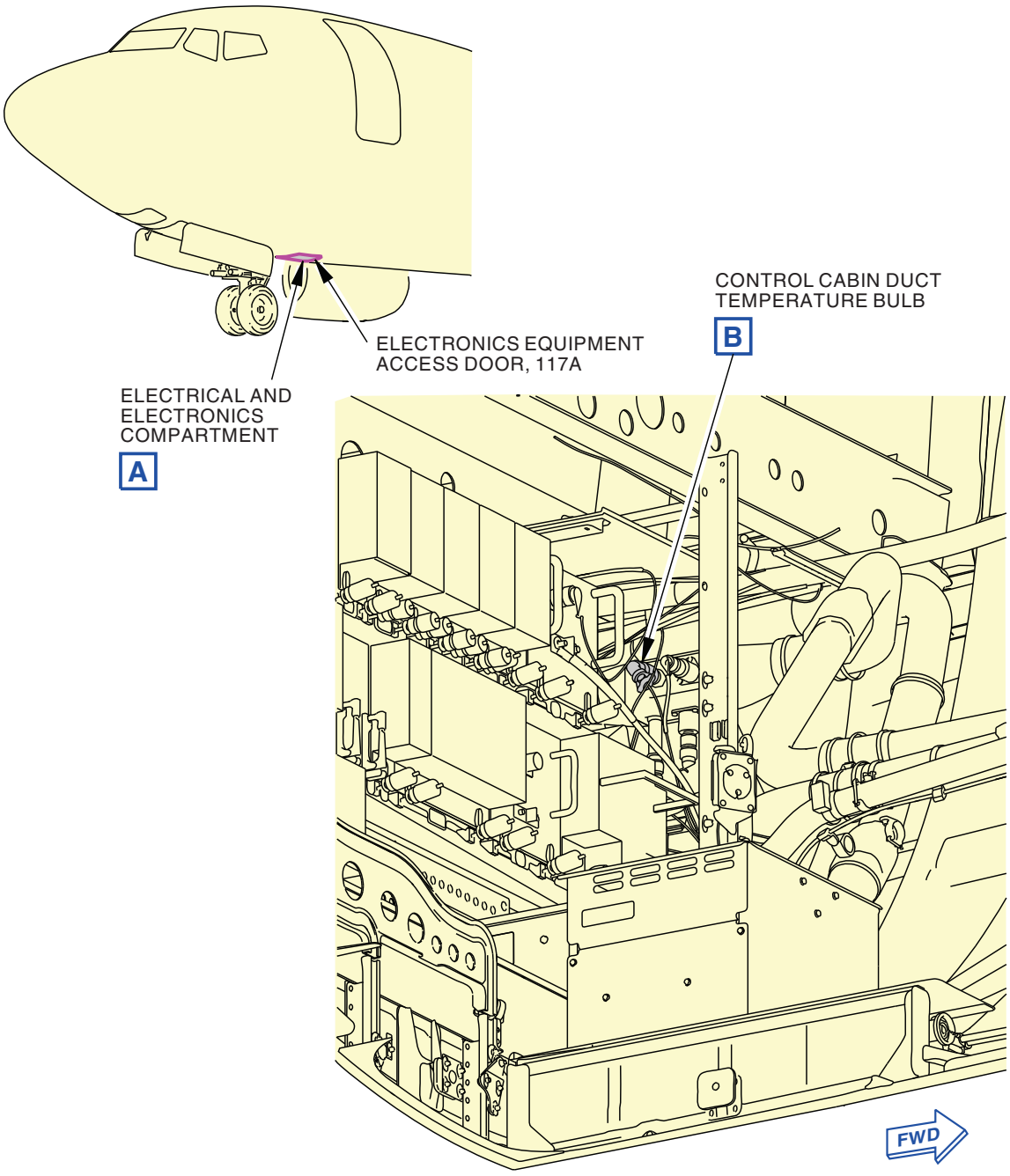
PASSENGER CABIN DUCT TEMPERATURE BULB
(FORWARD CABIN BULB IS SHOWN, AFT CABIN BULB IS OPPOSITE)

B

2411540 S00061528143_V1

Passenger Cabin Duct Temperature Bulb Installation
Figure 401/21-61-30-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



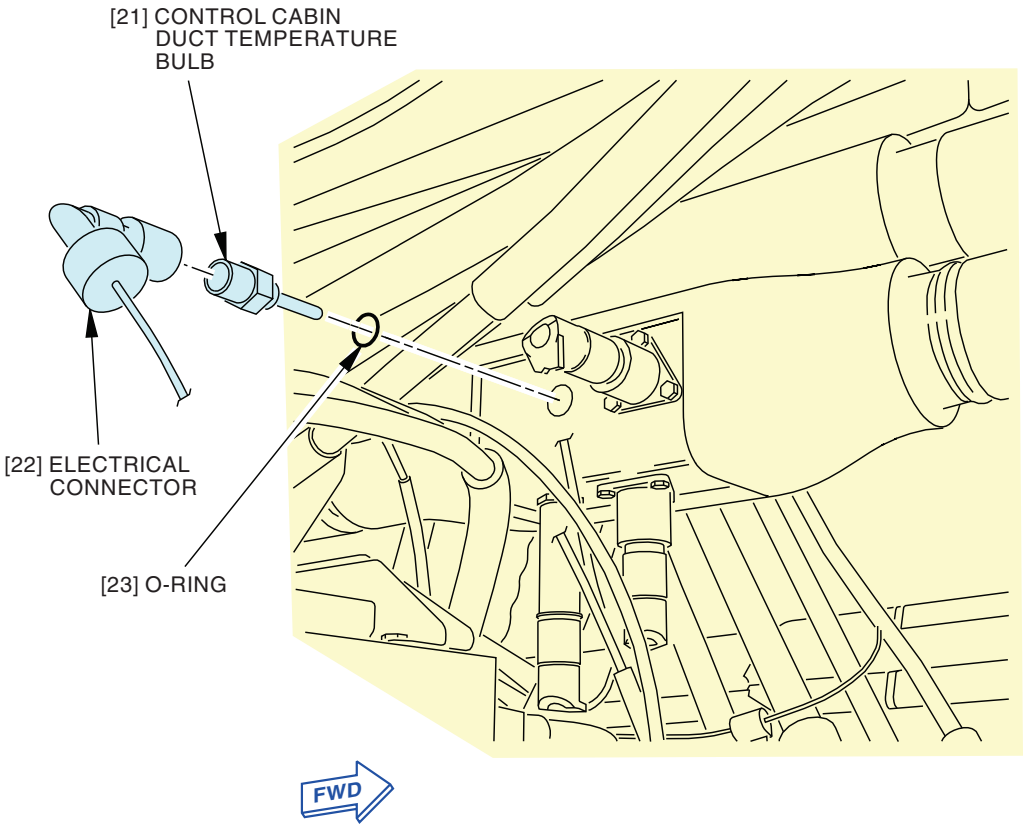
ELECTRICAL AND ELECTRONICS COMPARTMENT (VIEW THROUGH ACCESS DOOR, 117A)

A

2411541 S00061528144_V1

Control Cabin Duct Temperature Bulb Installation
Figure 402/21-61-30-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONTROL CABIN DUCT TEMPERATURE BULB

B

2411542 S00061528145_V1

Control Cabin Duct Temperature Bulb Installation
Figure 402/21-61-30-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-30-400-801

3. Duct Temperature Bulb Installation

(Figure 401, Figure 402)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Passenger cabin duct temperature bulb	21-61-30-01-005	PTF ALL
		21-61-30-04-015	PTF ALL
		21-61-30-09-005	PTF ALL
3	O-ring	21-61-30-01-010	PTF ALL
		21-61-30-09-010	PTF ALL
21	Control cabin duct temperature bulb	21-61-30-04-015	PTF ALL

D. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Passenger Cabin Duct Temperature Bulb Installation

SUBTASK 21-61-30-020-007

(1) Remove the copper gasket, if installed, from the new passenger cabin duct temperature bulb [2].

SUBTASK 21-61-30-420-001

(2) Install a new O-ring [3] on the passenger cabin duct temperature bulb [2].

SUBTASK 21-61-30-640-001

(3) Put compound, D50004 on the threads of the passenger cabin duct temperature bulb [2].

SUBTASK 21-61-30-420-002

(4) Do these steps to install the passenger cabin duct temperature bulb [2]:

(a) Put the passenger cabin duct temperature bulb [2] in its position in the duct.

(b) Tighten the passenger cabin duct temperature bulb [2] until the O-ring [3] touches the duct.

(c) Tighten the duct temperature passenger cabin duct temperature bulb [2] 3/4 turn more.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-30-420-003

(5) Connect the electrical connector [1] to the passenger cabin duct temperature bulb [2].

G. Control Cabin Duct Temperature Bulb Installation

SUBTASK 21-61-30-020-008

(1) Remove the copper gasket, if installed, from the new control cabin duct temperature bulb [21].

SUBTASK 21-61-30-420-004

(2) Install a new O-ring [23] on the control cabin duct temperature bulb [21].

SUBTASK 21-61-30-640-002

(3) Put compound, D50004 on the threads of the control cabin duct temperature bulb [21].

SUBTASK 21-61-30-420-005

(4) Do these steps to install the control cabin duct temperature bulb [21]:

(a) Put the control cabin duct temperature bulb [21] in its position in the duct.

(b) Tighten the control cabin duct temperature bulb [21] until the O-ring [23] touches the duct.

(c) Tighten the control cabin duct temperature bulb [21] an additional 3/4 turn.

SUBTASK 21-61-30-420-006

(5) Connect the electrical connector [22] to the control cabin duct temperature bulb [21].

H. Do a Test of the Duct Temperature Bulb

SUBTASK 21-61-30-860-002

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-61-30-860-003

(2) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

SUBTASK 21-61-30-860-004

(3) Put the AIR TEMP selector on the temperature control module on the overhead P5 panel to the SUPPLY DUCT position for the applicable zone.

SUBTASK 21-61-30-710-001

(4) Use a thermometer to find the temperature of the applicable zone air in the cabin.

SUBTASK 21-61-30-710-002

(5) Make sure the temperature shown on the indicator on the P5 panel is approximately the same as the temperature recorded in the cabin.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-61-30-010-003

(1) If the control cabin duct temperature sensor was replaced close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-61-30-010-004

(2) Install the ceiling panel if a passenger cabin duct temperature bulb was replaced. To install the panel, do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN TEMPERATURE BULB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Cabin Temperature Bulb Removal

(2) Cabin Temperature Bulb Installation.

(3) There is a cabin temperature bulb for each of the forward and aft passenger compartment zones.

TASK 21-61-31-000-801

2. Cabin Temperature Bulb Removal

(Figure 401)

A. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

B. Prepare for the Removal

SUBTASK 21-61-31-860-001

(1) Do this step for the removal of the temperature bulb for the forward passenger cabin zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-31-860-002

(2) Do this step for the removal of the temperature bulb for the aft passenger compartment zone:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

C. Cabin Temperature Bulb Removal

SUBTASK 21-61-31-010-001

(1) Lower the cabin temperature sensor panel.

(a) Put a small allen wrench or other applicable tool in the access holes to release the latches.

(b) Lower the panel and disconnect the lanyard.

(c) Move the panel to the fully lowered position.

SUBTASK 21-61-31-020-001

(2) Remove the cabin temperature bulb [18] from the panel:

(a) Disconnect the electrical connector [10] from the temperature bulb [18].

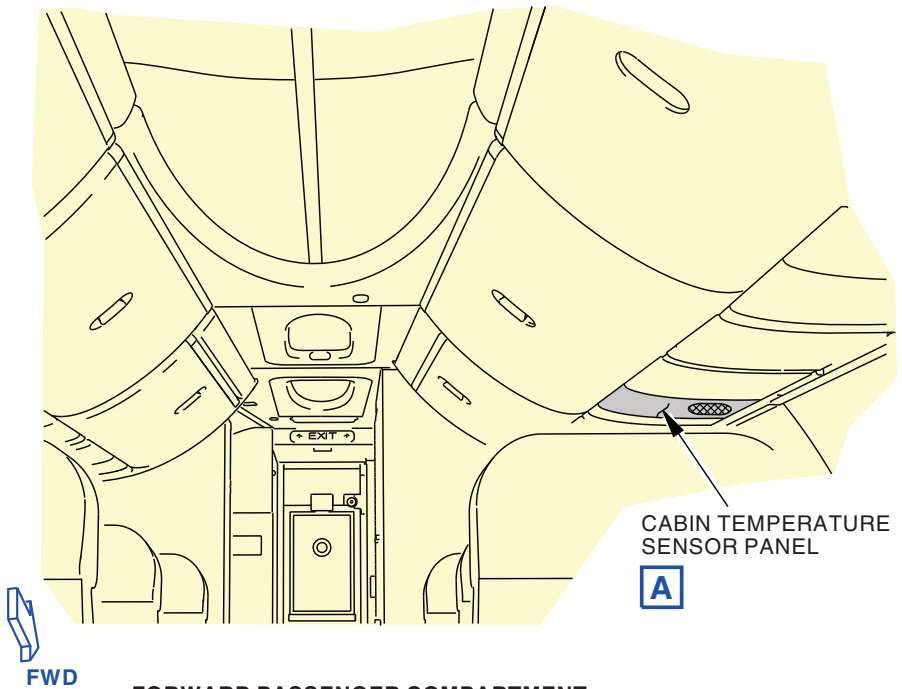
(b) Remove the lockwire [19].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the temperature bulb [18] and the lockwire washer [20].

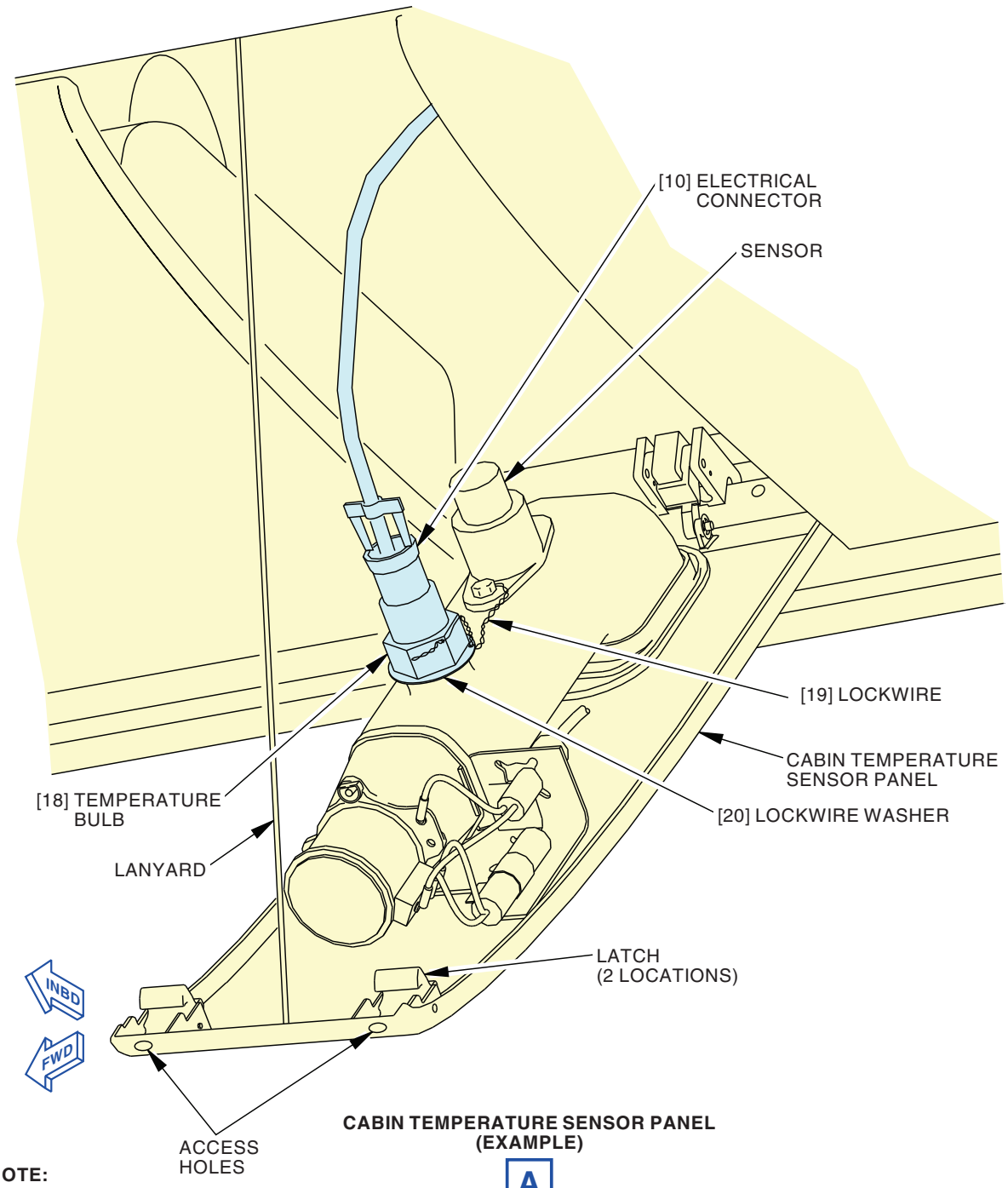
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Temperature Bulb Installation
Figure 401/21-61-31-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Temperature Bulb Installation
Figure 401/21-61-31-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-31-400-801

3. Cabin Temperature Bulb Installation

(Figure 401)

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-1122	Thermometer - Alcohol/Mercury (or equivalent meter meets task requirements)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
18	Temperature bulb	21-61-10-02-105	PTF ALL
		21-61-30-01-005	PTF ALL
		21-61-30-04-015	PTF ALL
		21-61-30-09-005	PTF ALL

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

E. Cabin Temperature Bulb Installation

SUBTASK 21-61-31-420-001

(1) Install the cabin temperature bulb [18]:

(a) Install the lockwire washer [20] and the temperature bulb [18] on the temperature sensor assembly.

(b) Install the lockwire [19] (TASK 20-10-44-400-801).

(c) Install the electrical connector [10].

(d) Partially lift the panel for the cabin temperature bulb and install the lanyard.

(e) Lift the panel for the cabin temperature bulb to its closed position.

SUBTASK 21-61-31-860-004

(2) Do this step for the installation of the cabin temperature bulb [18] for the forward passenger compartment zone:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS

SUBTASK 21-61-31-860-005

(3) Do this step for the installation of the cabin temperature bulb [18] for the aft passenger compartment zone:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS

F. Cabin Temperature Bulb Installation Test

SUBTASK 21-61-31-740-001

(1) Do this test of the cabin temperature bulb [18] that was replaced:

(a) Use a thermometer, STD-1122, and make a record of the ambient temperature in the passenger compartment.

(b) Put the AIR TEMP selector on the P5-17 temperature control panel to PASS CABIN.

(c) Make sure that the temperature record with the thermometer, STD-1122, and the PASS CABIN TEMP gauge indication are approximately the same.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN TEMPERATURE INDICATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Cabin temperature indicator removal

(2) Cabin temperature indicator installation.

B. The cabin temperature indicator is installed in the P5-17 Cabin Temperature Panel.

C. The P5-17 Cabin Temperature Panel is installed in the P5 Forward Overhead Panel.

TASK 21-61-32-000-801

2. Cabin Temperature Indicator Removal

(Figure 401)

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 21-61-32-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	9	C00329	OVERHEAD 5V SECONDARY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

SUBTASK 21-61-32-010-001

(2) Do these steps to get access to the electrical connector [3] on the back of the P5-17 cabin temperature module:

(a) Loosen the 1/4-turn fasteners that hold the aft end of the P5 forward overhead panel in the closed position.

NOTE: The forward end of the P5 panel is hinged. The aft end of the P5 panel will turn downward when the fasteners are loose.

(b) Release the safety latch and carefully lower the P5 forward overhead panel to the open position.

C. Cabin Temperature Indicator Removal

SUBTASK 21-61-32-020-001

(1) Do these steps to remove cabin temperature indicator [1] from the P5-17 module:

(a) Turn the clamp tension screw [4] in the counterclockwise direction to loosen the clamp that holds the indicator to the cabin temperature panel [2].

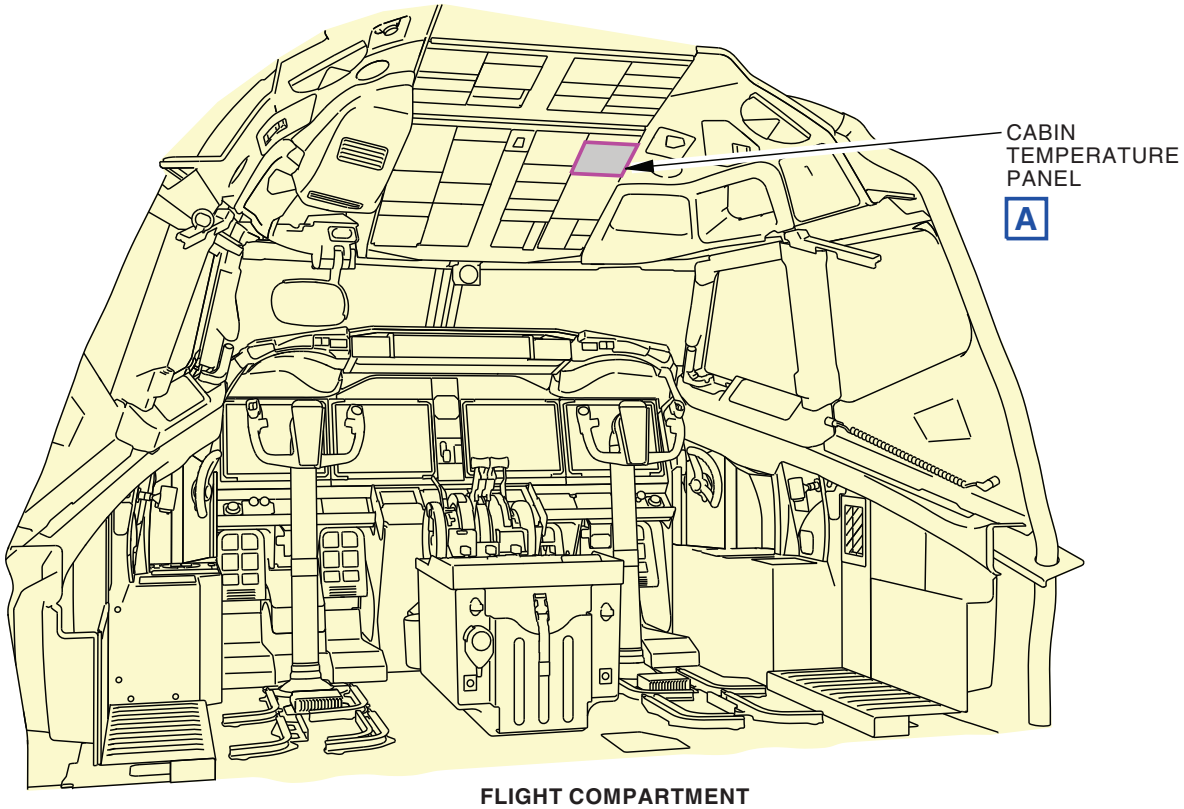
1) Do not remove the clamp tension screw [4].

(b) Carefully pull the cabin temperature indicator [1] out of the cabin temperature panel [2].

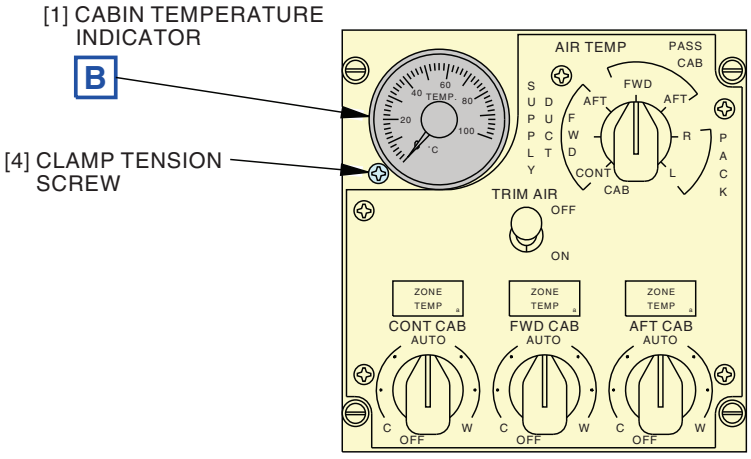
(c) Disconnect the electrical connector [3] from the cabin temperature indicator [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



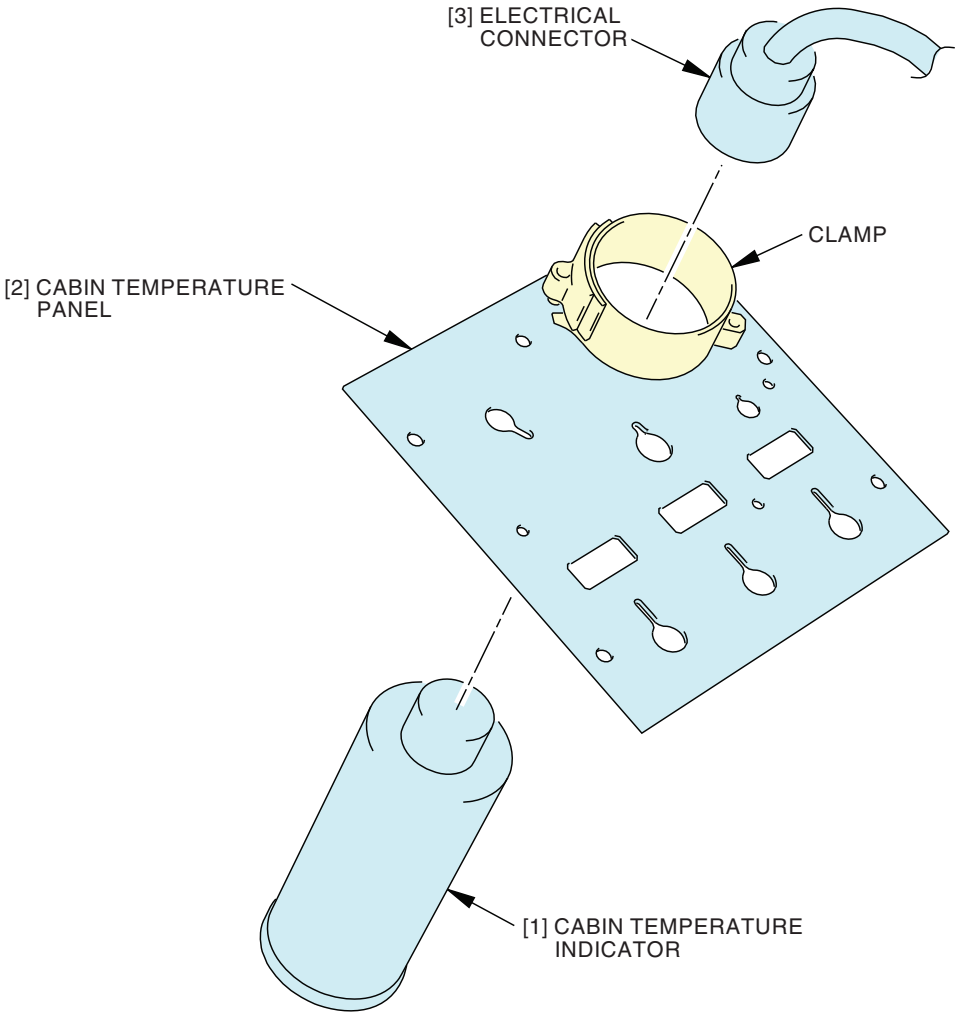
CABIN TEMPERATURE PANEL

A

2411544 S00061528155_V2

Cabin Temperature Indicator Installation
Figure 401/21-61-32-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN TEMPERATURE INDICATOR (EXAMPLE)

B

2411545 S00061528156_V2

Cabin Temperature Indicator Installation
Figure 401/21-61-32-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-61-32-400-801

3. Cabin Temperature Indicator Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cabin temperature indicator	21-61-32-01-050	PTF ALL

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Cabin Temperature Indicator Installation

SUBTASK 21-61-32-420-001

(1) Do these steps to install the cabin temperature indicator [1] in the cabin temperature panel [2]:

(a) Connect the electrical connector [3] to the cabin temperature indicator [1].

(b) Carefully push the cabin temperature indicator [1] into its position in the cabin temperature panel [2].

(c) Tighten the clamp tension screw [4] to 6.5 ±1.5 in-lb (0.7 ±0.2 N·m).

1) Hold the cabin temperature indicator [1] against the cabin temperature panel [2] while you tighten the clamp tension screw [4].

SUBTASK 21-61-32-410-001

(2) Carefully lift the P5 forward overhead panel to the closed position.

SUBTASK 21-61-32-410-002

(3) Tighten the 1/4-turn fasteners that hold the aft end of the P5 forward overhead panel in the closed position.

SUBTASK 21-61-32-860-002

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	9	C00329	OVERHEAD 5V SECONDARY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	8	C00076	AIR CONDITIONING TEMP IND

D. Cabin Temperature Indicator Test

SUBTASK 21-61-32-710-001

(1) Put the AIR TEMP switch on the P5-17 panel in the FWD PASS CABIN position.

SUBTASK 21-61-32-710-002

(2) Hold a thermometer near the inlet of the forward passenger cabin temperature sensor assembly for one minute.

SUBTASK 21-61-32-970-001

(3) Make a note of the temperature that shows on the thermometer.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-61-32-710-003

(4) Make sure that the cabin temperature indicator shows the same temperature as the thermometer.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRIM AIR PRESSURE REGULATION AND SHUTOFF CONTROL - ADJUSTMENT/TEST

1. General

A. This procedure has one task. The task is an operational test of the trim air pressure regulating and shutoff valve.

TASK 21-62-00-710-801

2. Trim Air Pressure Regulating and Shutoff Valve - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left

C. Trim Air Pressure Regulating and Shutoff Valve Operational Test

SUBTASK 21-62-00-860-001

(1) Do this task: Supply Pressure to the Pneumatic System with the APU, TASK 36-00-00-860-803.

SUBTASK 21-62-00-860-002

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

SUBTASK 21-62-00-860-003

(3) Do these steps, on the P5-10 air conditioning panel:

(a) Set the L RECIRC switch to the AUTO position.

(b) Set the R RECIRC switch to the AUTO position.

(c) Set the L PACK switch to the AUTO position.

(d) Set the R PACK switch to the OFF position.

(e) Set the BLEED APU switch to the ON position.

(f) Set the BLEED 1 switch to the OFF position.

(g) Set the BLEED 2 switch to the OFF position.

(h) Set the ISOLATION VALVE switch to the OPEN position.

SUBTASK 21-62-00-860-004

(4) Do these steps, on the P5-17 cabin temperature panel:

(a) Set the TRIM AIR switch to the OFF position.

(b) Set the AIR TEMP selector to the L PACK position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Set the CONT CAB selector to the full WARM position.

(d) Set the FWD CAB selector to the full WARM position.

(e) Set the AFT CAB selector to the full WARM position.

SUBTASK 21-62-00-860-005

(5) Operate the pack for 3 to 5 minutes to let the output temperature become stable.

SUBTASK 21-62-00-970-001

(6) Make a record of the temperature for the L PACK.

NOTE: The temperature indication is shown on the AIR TEMP gage (P5-17 panel).

SUBTASK 21-62-00-860-006

(7) Do these steps, on the P5-17 cabin temperature panel:

(a) Set the CONT CAB selector to the full COLD position.

(b) Set the FWD CAB selector to the full COLD position.

(c) Set the AFT CAB selector to the full COLD position.

SUBTASK 21-62-00-970-002

(8) Make sure that the temperature for the L PACK decreases.

SUBTASK 21-62-00-860-007

(9) Set the TRIM AIR switch to the ON position.

SUBTASK 21-62-00-860-008

(10) Operate the pack for 3 to 5 minutes to let the output temperature become stable.

SUBTASK 21-62-00-970-003

(11) Make a record of the temperature for the L PACK.

NOTE: The temperature indication is shown on the AIR TEMP gage (P5-17 panel).

SUBTASK 21-62-00-860-009

(12) Set the AIR TEMP selector to the FWD SUPPLY DUCT position.

SUBTASK 21-62-00-970-004

(13) Make a record of the temperature for the FWD SUPPLY DUCT.

NOTE: The temperature indication is shown on the AIR TEMP gage (P5-17 panel).

SUBTASK 21-62-00-860-010

(14) Move the FWD CAB temperature selector to the full WARM position.

SUBTASK 21-62-00-970-005

(15) Make sure that the temperature for the FWD SUPPLY DUCT increases.

SUBTASK 21-62-00-860-011

(16) Move the AIR TEMP selector to the L PACK position.

SUBTASK 21-62-00-970-006

(17) Make sure that the L PACK temperature stays approximately the same.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 21-62-00-860-012

(1) Do these steps, on the P5-10 air conditioning panel:

(a) Set the L and R PACK switches to the OFF position.

(b) Set the BLEED APU switch to the OFF position.

(c) Make sure that the BLEED 1 and BLEED 2 switches are in the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-62-00-860-013

(2) Do these steps, on the P5-17 cabin temperature panel:

(a) Set the TRIM AIR switch to the ON position.

(b) Set the CONT CAB selector to the AUTO position.

(c) Set the FWD CAB selector to the AUTO position.

(d) Set the AFT CAB selector to the AUTO position.

SUBTASK 21-62-00-860-014

(3) Remove pneumatic power, if it is not necessary (TASK 36-00-00-860-806).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TRIM AIR PRESSURE REGULATING AND SHUTOFF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Trim Air Pressure Regulating and Shutoff Valve Removal

(2) Trim Air Pressure Regulating and Shutoff Valve Installation.

B. The trim air pressure regulating and shutoff valve is installed in the forward, inboard area of the right air conditioning bay.

TASK 21-62-01-000-801

2. Trim Air Pressure Regulating and Shutoff Valve Removal

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left

D. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for the Removal

SUBTASK 21-62-01-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-62-01-860-002

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

SUBTASK 21-62-01-860-003

(3) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach DO NOT OPERATE tags, STD-858.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Set the BLEED APU switch to the OFF position and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-62-01-860-004

(4) Do these steps on the P5-17 Cabin Temperature Panel (located on the P5 forward overhead panel):

(a) Set the CONT CAB, FWD CAB, and AFT CAB selectors to the OFF position and attach DO NOT OPERATE tags, STD-858.

(b) Set the TRIM AIR switch to the OFF position and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-62-01-010-001

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To get access to the trim air pressure regulating and shutoff valve [1], do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Trim Air Pressure Regulating and Shutoff Valve Removal

SUBTASK 21-62-01-020-001

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the electrical connector [3] from the trim air pressure regulating and shutoff valve [1].

SUBTASK 21-62-01-020-002

(2) Remove the screw [8], washers [5], nut [4], and bonding jumper [9] from the trim air pressure regulating and shutoff valve [1].

SUBTASK 21-62-01-020-003

(3) Do these steps to remove the sense line:

(a) If you decide to replace the sense line, make a record of which end of the sense line goes to the valve and which end goes to the ducting.

NOTE: The end fittings on the sense line are not always the same.

(b) Loosen the B-nut and disconnect the sense line from the union [6].

(c) Remove the union [6] from the trim air pressure regulating and shutoff valve [1].

(d) Remove and discard the packings [7].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-62-01-020-004

(4) Do these steps to remove the hose assembly:

(a) Loosen the B-nut to disconnect the hose assembly from the trim air pressure regulating and shutoff valve [1].

(b) Remove the union [6] from the trim air pressure regulating and shutoff valve [1].

(c) Remove and discard the packing [7].

SUBTASK 21-62-01-020-005

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(5) To remove the trim air pressure regulating and shutoff valve [1], do these steps:

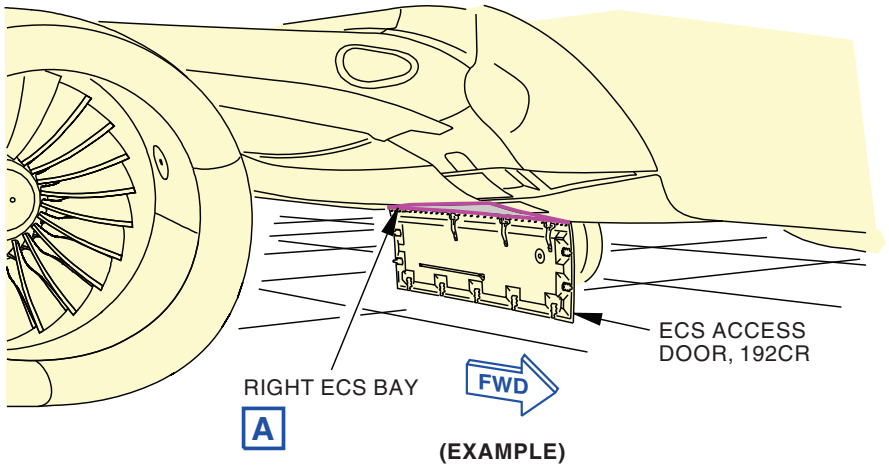
(a) Loosen the two clamps [2].

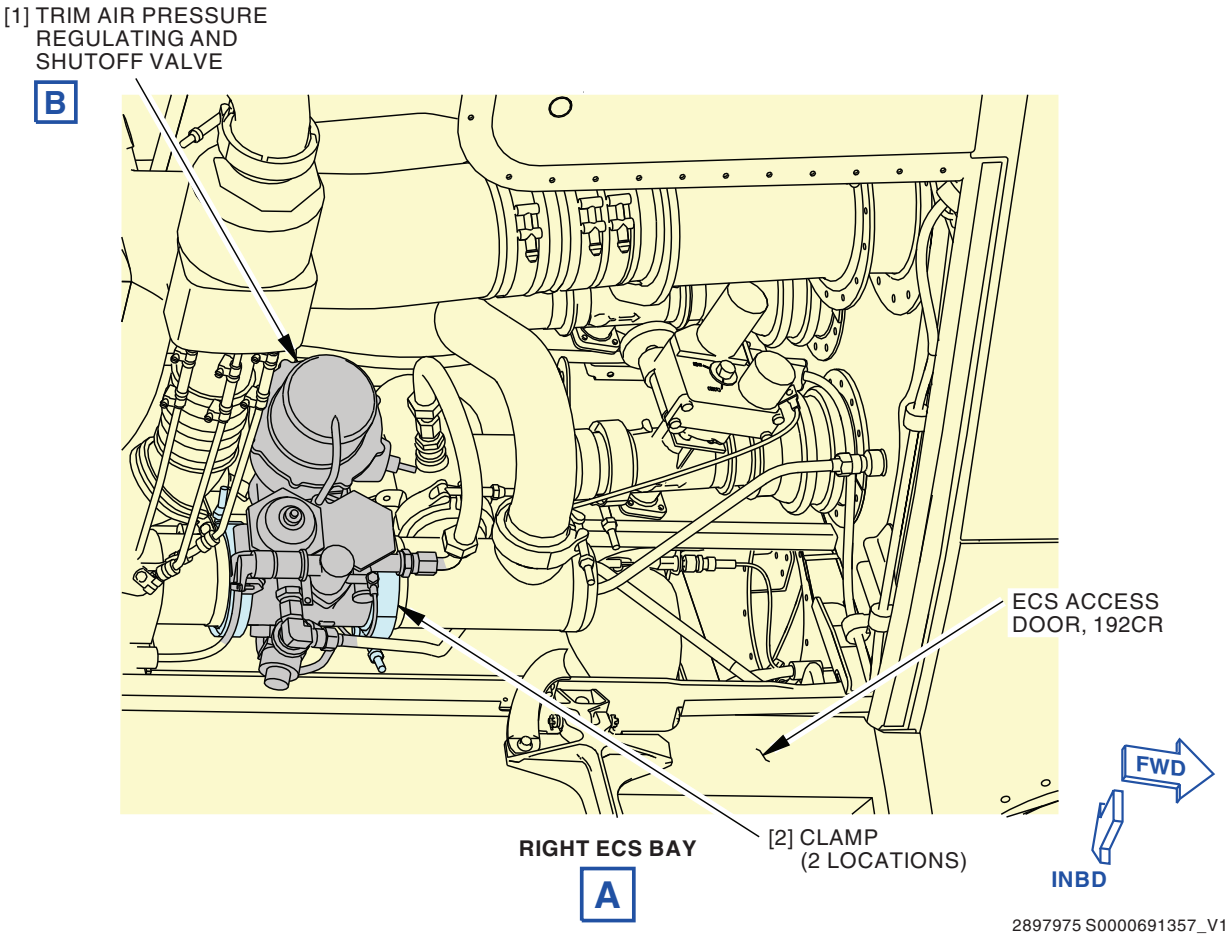
(b) Move the clamps [2] to the adjacent ducts.

(c) Remove the trim air pressure regulating and shutoff valve [1] from the ECS bay.

END OF TASK

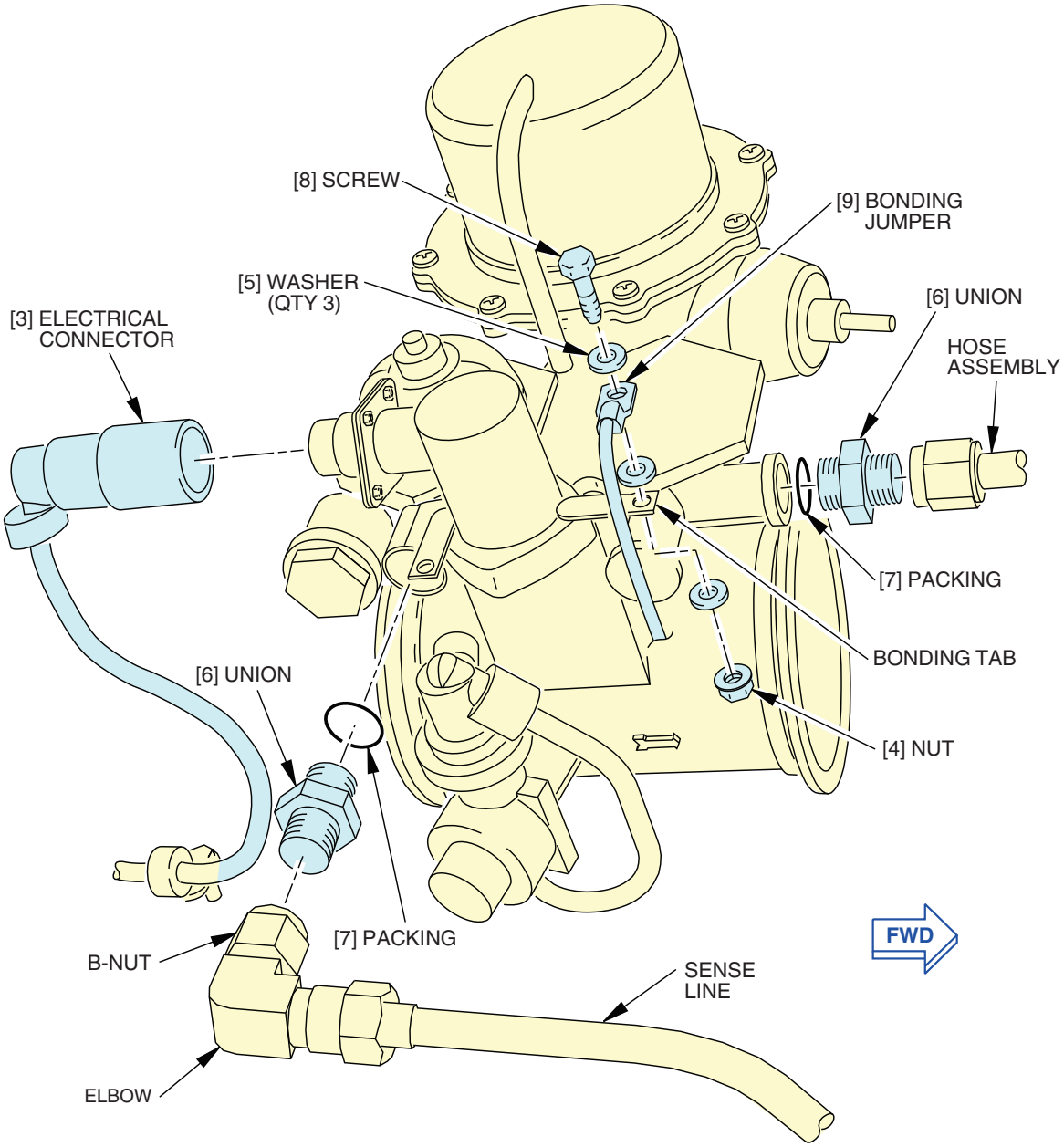
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Trim Air Pressure Regulating and Shutoff Valve Installation
Figure 401/21-62-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TRIM AIR PRESSURE REGULATING AND SHUTOFF VALVE

B

2411548 S00061528167_V2

Trim Air Pressure Regulating and Shutoff Valve Installation
Figure 401/21-62-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-62-01-400-801

3. Trim Air Pressure Regulating and Shutoff Valve Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-34-110-802	Clean Bare, Clad, or Plated Metal with Solvent (P/B 701)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Trim air pressure regulating and shutoff valve	21-62-01-01-010	PTF ALL
7	Packing	21-62-01-01-095	PTF ALL

E. Location Zones

Zone	Area
120	Subzone - Body Station 396 to Body Station 540
192	Lower Wing-To-Body Fairing - Under Wing Box
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

F. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Trim Air Pressure Regulating and Shutoff Valve Installation

SUBTASK 21-62-01-420-001

(1) To install the trim air pressure regulating and shutoff valve [1], do these steps:

(a) Put the trim air pressure regulating and shutoff valve [1] in its position between the ducts.

1) Make sure that the flow arrow on the valve points in the forward direction.

(b) Rotate the trim air pressure regulating and shutoff valve [1] to make sure that there is a minimum distance of 0.75 in. (1.90 cm) between the trim air pressure regulating and shutoff valve [1] and bottom of the keel beam.

1) Make sure that there is a minimum clearance of 0.15 in. (0.38 cm) between the structure, tubes, and clamps [2].

(c) Loosely install the clamps [2].

SUBTASK 21-62-01-420-002

(2) To install the sense line on the trim air pressure regulating and shutoff valve [1], do these steps:

(a) Install a new packing [7] on the union [6].

(b) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the union [6].

(c) Install the union [6] on the trim air pressure regulating and shutoff valve [1].

(d) Install the sense line on the union [6]:

1) Make sure that you install the same end of the sense line to the trim air pressure regulating and shutoff valve [1] as you noted during the removal.

NOTE: Some sense lines have different end fittings.

SUBTASK 21-62-01-020-006

(3) Do these steps to install the hose assembly:

(a) Install a new packing [7] on the union [6].

(b) Apply a thin layer of Pure Nickel Special compound, D00006, to the threads of the union [6].

(c) Install the union [6] on the trim air pressure regulating and shutoff valve [1].

(d) Install the hose assembly on the union [6].

SUBTASK 21-62-01-020-007

(4) Do these steps to install the bonding jumper [9]:

(a) Solvent clean the mating surfaces of the bonding jumper [9], bonding tab, and their fasteners (TASK 20-10-34-110-802).

(b) Put the bonding jumper [9] in its position on the trim air pressure regulating and shutoff valve [1].

(c) Install the screw [8], washers [5], nut [4], and bonding jumper [9] to the trim air pressure regulating and shutoff valve [1].

(d) Measure the electrical bonding resistance between the bonding tab and bonding jumper [9] (SWPM 20-20-00).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the resistance is not more than 0.002 ohms (2.0 milliohms).

SUBTASK 21-62-01-020-008

(5) Install the electrical connector [3] on the trim air pressure regulating and shutoff valve [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-62-01-420-003

(6) Tighten the clamps [2] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

(a) Make sure that the loop bands do not touch.

(b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

H. Trim Air Pressure Regulating and Shutoff Valve Test

SUBTASK 21-62-01-860-005

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

SUBTASK 21-62-01-860-006

(2) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-62-01-860-007

(3) Do these steps on the P5-10 air conditioning panel:

(a) Set the L and R PACK switches to the AUTO position and remove the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and 2 switches to the ON position and remove the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the ON position and remove the DO NOT OPERATE tag, STD-858.

SUBTASK 21-62-01-860-008

(4) Do these steps on the P5-17 cabin temperature panel:

(a) Set the TRIM AIR switch to the ON position and remove the DO NOT OPERATE tag, STD-858.

(b) Set the CONT CAB selector to the fully WARM position and remove the DO NOT OPERATE tag, STD-858.

(c) Set the FWD CAB selector to the fully COLD position and remove the DO NOT OPERATE tag, STD-858.

(d) Set the AFT CAB selector to the fully COLD position and remove the DO NOT OPERATE tag, STD-858.

SUBTASK 21-62-01-860-009

(5) Operate the pack for a minute to let the output temperature become stable.

SUBTASK 21-62-01-860-010

(6) Do these steps on the P5-17 cabin temperature panel:

(a) Set AIR TEMP selector to the each of the three SUPPLY DUCT positions (AFT, FWD, and CONT CAB)

(b) Make a record of the temperature shown on the indicator for each of the three SUPPLY DUCT positions.

(c) Make sure that the supply duct temperature for the CONT CAB is higher than the supply duct temperatures for the FWD and AFT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Trim Air Pressure Regulating and Shutoff Valve Leakage Test

SUBTASK 21-62-01-790-001

(1) Do a soap bubble test of the duct joints at the trim air pressure regulating and shutoff valve [1].

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches to the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

2) Loosen the clamps [2].

3) Make sure that the ducts are aligned at the joints.

4) Tighten the clamps [2] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

a) Make sure that the loop bands do not touch.

b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.410 in. (10.414 mm).

5) Put the L PACK and R PACK switches to the AUTO position.

6) Make sure that the leak is repaired.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-62-01-010-002

(1) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-62-01-860-011

(2) Remove pneumatic power if it is not necessary (TASK 36-00-00-860-806).

END OF TASK

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OZONE CONVERTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ozone converter.

(2) An installation of the ozone converter.

TASK 21-73-01-000-801

2. Ozone Converter Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the removal of the ozone converter.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-07-000-801	Conditioned Air Check Valve Removal (P/B 401)
21-61-21-000-801	Zone Trim Air Modulating Valve Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for the Removal

SUBTASK 21-73-01-860-002

(1) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the OFF position and attach a DO NOT OPERATE tag, STD-858.

(b) Set the BLEED 1 and 2 switches to the OFF position and attach a DO NOT OPERATE tag, STD-858.

(c) Set the BLEED APU switch to the OFF position and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-73-01-010-001

(2) To remove the ozone converter [1] in the left ECS bay, do this step:

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(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-73-01-010-002

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(3) To remove the ozone converter [1] in the right ECS bay, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-73-01-010-003

(4) If it is necessary, do this task: Zone Trim Air Modulating Valve Removal, TASK 21-61-21-000-801.

SUBTASK 21-73-01-010-004

(5) If it is necessary, do this task: Conditioned Air Check Valve Removal, TASK 21-51-07-000-801.

G. Ozone Converter Removal

SUBTASK 21-73-01-020-001

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(1) Remove the coupling [2] from the aft end of the ozone converter [1].

SUBTASK 21-73-01-020-002

(2) Remove the coupling [3] from the forward end of the ozone converter [1].

SUBTASK 21-73-01-020-003

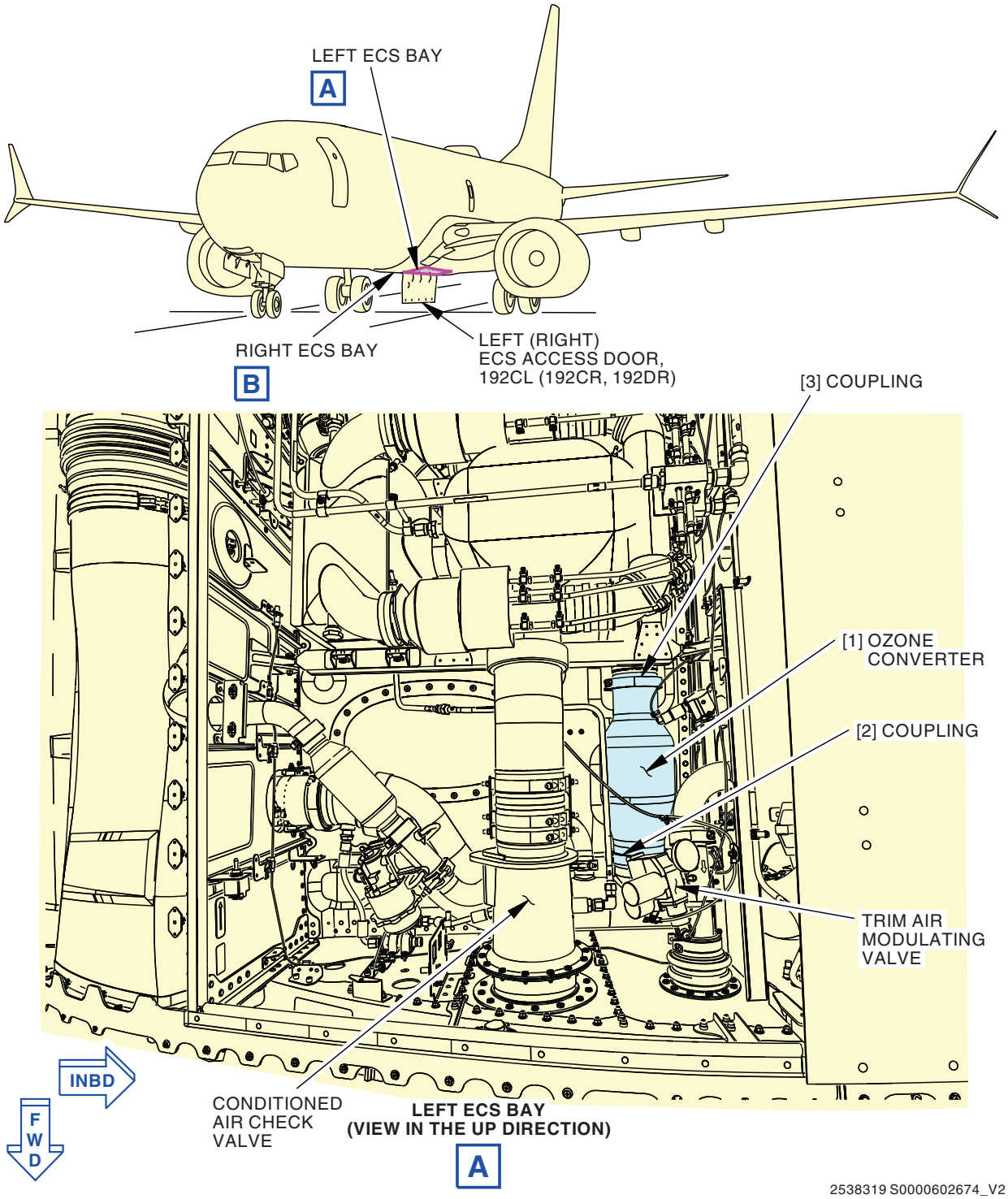
(3) Remove the ozone converter [1].

SUBTASK 21-73-01-420-001

(4) Put the protective covers on the duct openings.

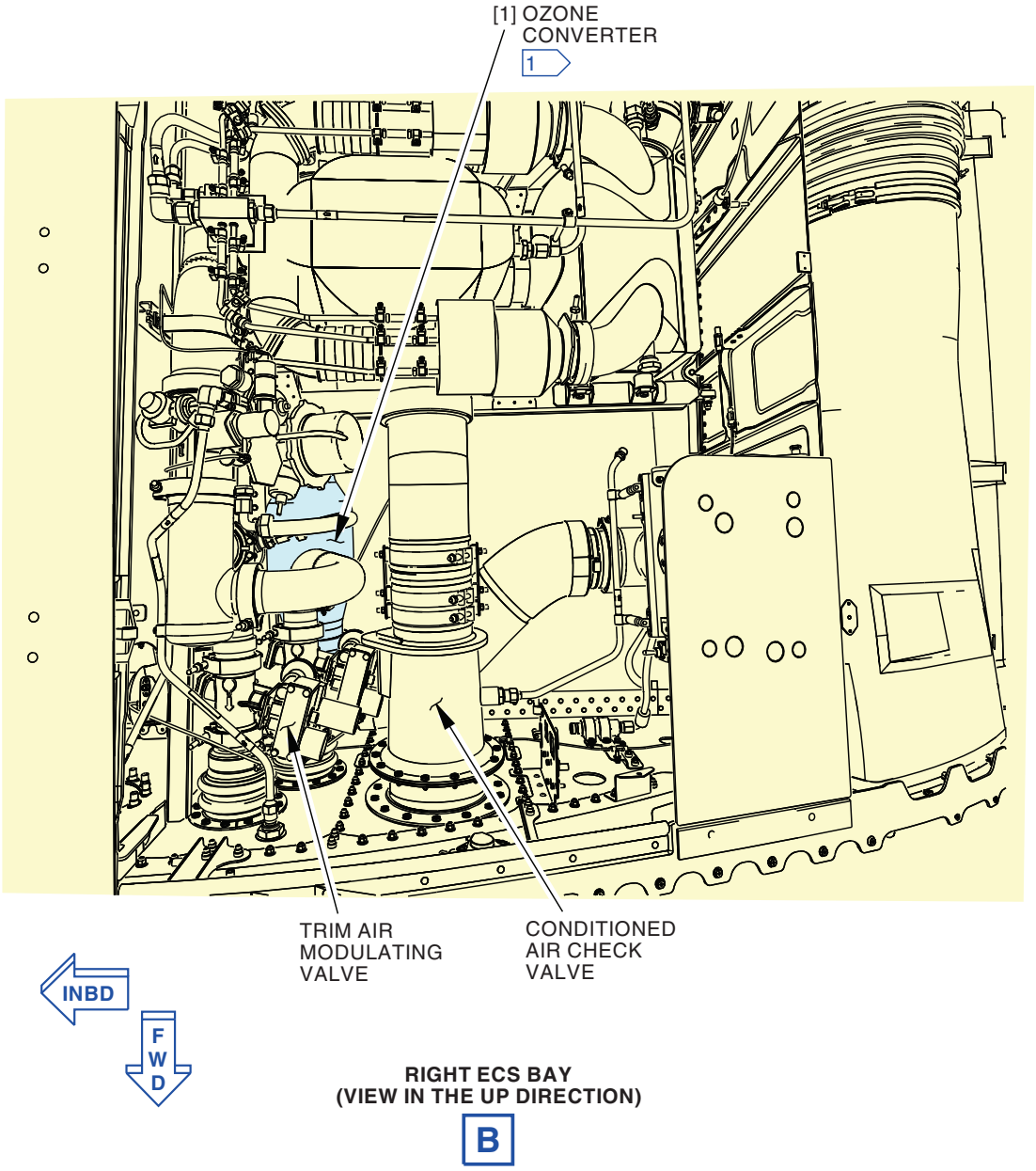
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ozone Converter Installation
Figure 401/21-73-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ozone Converter Installation
Figure 401/21-73-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-73-01-400-801

3. Ozone Converter Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the installation of the ozone converter.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-07-400-801	Conditioned Air Check Valve Installation (P/B 401)
21-61-21-400-801	Zone Trim Air Modulating Valve Installation (P/B 401)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Ozone converter	21-73-01-01-095	PTF ALL

E. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

G. Ozone Converter Installation

SUBTASK 21-73-01-020-004

(1) Remove the protective covers from the duct openings.

SUBTASK 21-73-01-420-002

(2) Put the ozone converter [1] between the air supply ducts.

SUBTASK 21-73-01-420-003

(3) Loosely install the coupling [3] to connect the forward end of the ozone converter [1] to the air supply duct.

SUBTASK 21-73-01-420-004

(4) Loosely install the coupling [2] to connect the aft end of the ozone converter [1] to the air supply duct.

SUBTASK 21-73-01-420-005

(5) Tighten the nut on the coupling [2] and coupling [3] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

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(a) Make sure that the loop bands do not touch.

(b) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.310 in. (7.874 mm).

H. Ozone Converter Leakage Test

SUBTASK 21-73-01-410-003

(1) If it is necessary, do this task: Zone Trim Air Modulating Valve Installation, TASK 21-61-21-400-801.

SUBTASK 21-73-01-410-004

(2) If it is necessary, do this task: Conditioned Air Check Valve Installation, TASK 21-51-07-400-801.

SUBTASK 21-73-01-860-004

(3) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Remove the DO NOT OPERATE tag, STD-858, from the L and R PACK switches.

(b) Remove the DO NOT OPERATE tag, STD-858, from the BLEED 1 and 2 switches.

(c) Remove the DO NOT OPERATE tag, STD-858, from the BLEED APU switch.

SUBTASK 21-73-01-860-005

(4) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-73-01-860-006

(5) Set the applicable BLEED switches, on the P5-10 air conditioning panel, to the ON position.

SUBTASK 21-73-01-790-001

(6) Do a soap bubble test of all the applicable duct joints.

NOTE: No leakage is permitted.

SUBTASK 21-73-01-860-008

(7) If there is leakage, do these steps:

(a) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(b) Loosen the coupling [2] or coupling [3].

(c) Make sure that the ducts are aligned at the joints.

(d) Tighten the nut on the coupling [2] or coupling [3] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

1) Make sure that the loop bands do not touch.

2) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.310 in. (7.874 mm).

(e) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

(f) Make sure that there is no leakage.

SUBTASK 21-73-01-860-007

(8) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-73-01-410-001

(1) Close this access door:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-73-01-410-002

(2) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details